CA3162E Liquid Crystal Display (LCD) Application

Fig. 6 shows the CA3162E in a typical LCD application. LCD's may be used in favor of LED displays in applications requiring lower power dissipation, such as battery-operated equipment, or when visibility in high-ambient-light conditions is desired.

Multiplexing of LCD digits is not practical, since LCD's must be driven by an ac signal and the average voltage across each segment is zero. Three CD4056B liquid-crystal decoder/drivers are therefore used. Each CD4056B contains an input latch so that the BCD data for each digit may be latched into the decoder using the inverted digit-select outputs of the CA3162E as strobes.

Inverters G1 and G2 are used as an astable multivibrator to provide the ac drive to the LCD backplane. Inverters G3, G4, and G5 are the digit-select inverters and require pull-up resistors to interface the open-collector outputs of the CA3162E to COS/MOS logic. The BCD outputs of the CA3162E may be connected directly to the corresponding CD4056B inputs (using pull-up resistors). In this arrangement, the CD4056B decodes the negative sign (–) as an "L" and the positive overload indicator (E) as an "H".

CA3162E Common-Cathode, LED Display Application

Fig. 7 shows the CA3162E connected to a CD4511B decoder/driver to operate a common-cathode LED display. Unlike the CA3161E, the CD4511B remains blank for all BCD codes greater than nine. After 999 mV the display blanks rather than displaying EEE, as with the CA3161E. When displaying negative voltage, the first digit remains blank instead of (–), and during a negative overrange the display blanks.

The additional logic shown within the dotted area of Fig. 7 restores the negative sign (–), allowing the display of negative numbers as low as –99 mV. Negative overrange is indicated by a negative sign (–) in the MSD position. The rest of the display is blanked. During a positive overrange, only segment b of the MSD is displayed.

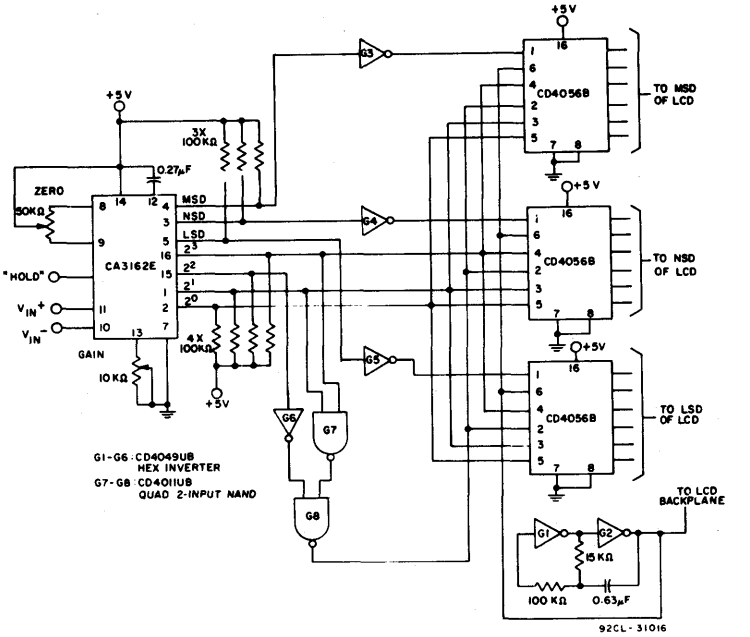


Fig. 6 – Typical LCD application.

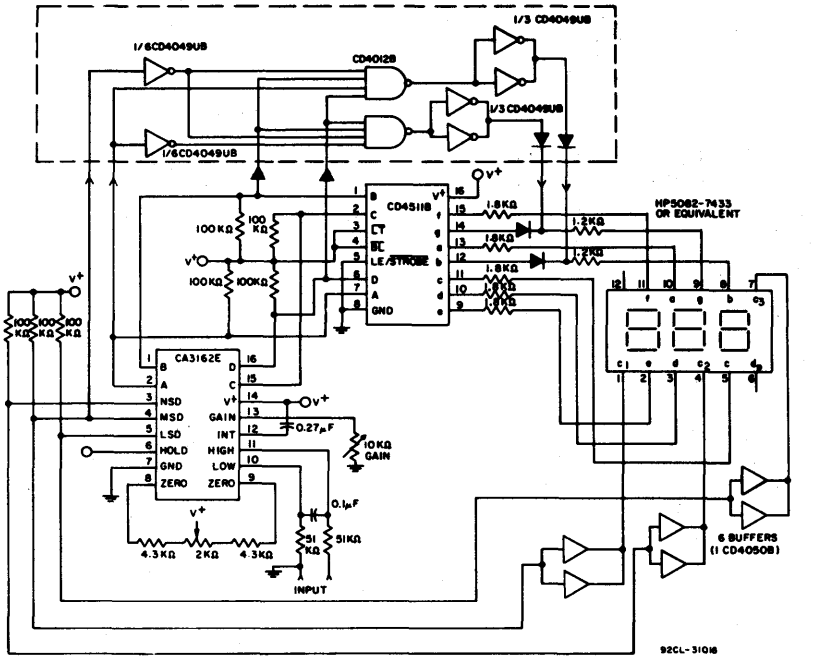


Fig. 7 – Typical common-cathode LED application.

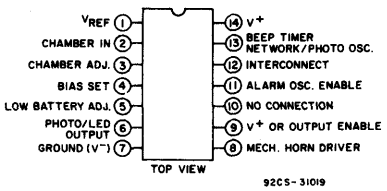
CA3164E

Preliminary Data

BiMOS Single-Chip Smoke Detector

For Fire Detection Systems

TERMINAL ASSIGNMENT



The RCA-CA3164E is a monolithic BiMOS integrated circuit designed to meet the stringent system requirements of a battery- or line-operated smoke detector circuit. When used with an ionization chamber and electro-mechanical horn, it provides a one-chip approach to smoke detection. No external active devices are required to interface with either the chamber input or horn output terminals. The CA3164E can also be used with photoelectric chambers by the addition of several external components.

The CA3164E was designed to comply with U.L. 217 and is supplied in the 14-lead dual-in-line plastic package.

Features:

- **Interfaces directly with ionization chamber — no external buffer FET required**
- **Low input current: 1 pA max.**
- **Gate-protected input terminals**
- **On-chip beep oscillator for low battery indication — requires one external capacitor**
- **Output capable of driving a conventional horn**
- **Self-contained low-battery-voltage detection circuit**
  - **(a) Fixed or adjustable trip point available**
  - **(b) Dynamic battery test when filter capacitor = 2 μF**
- **Chamber trigger voltage independent of battery supply voltage (less than 150 mV over temperature and supply variations)**
- **Designed to comply with U.L. 217**
- **Reference source current available = 5 μA (typ.)**
- **Low standby battery current = 8 μA (typ.)**
- **Can be used with photoelectric sensors by using a minimum of external passive components in combination with the RCA-CA3078 micropower op-amp**
- **Multiple-unit interconnect terminal controls a common annunciator circuit**
  - **(a) A fault to ground doesn't prevent local operation.**
  - **(b) The low battery alarm signal triggers only the local unit.**
- **LED output indicates status of smoke-detector circuit**
- **Operates from 11 V (max.) supply (either battery or line)**
- **Battery reversal protection feature**

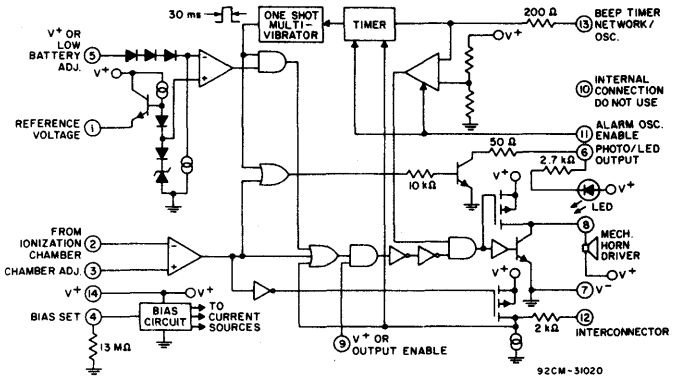


Fig. 1 — Simplified functional diagram for CA3164E.

MAXIMUM RATINGS, *Absolute-Maximum Values:*

DC SUPPLY VOLTAGE, V^+	+11 V
DEVICE DISSIPATION, P_D:	
Up to T_A = 25°C	600 mW
Above T_A = 25°C derate linearly at	6.7 mW/°C
AMBIENT TEMPERATURE RANGE:	
Operating	0 to +50°C
Storage	−65 to +150°C
LEAD TEMPERATURE (DURING SOLDERING):	
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max.	+265°C

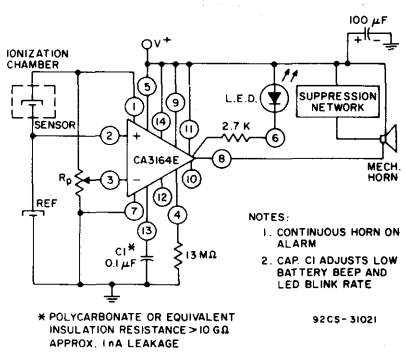


Fig. 2 — Basic ionization detector with electro-mechanical horn.

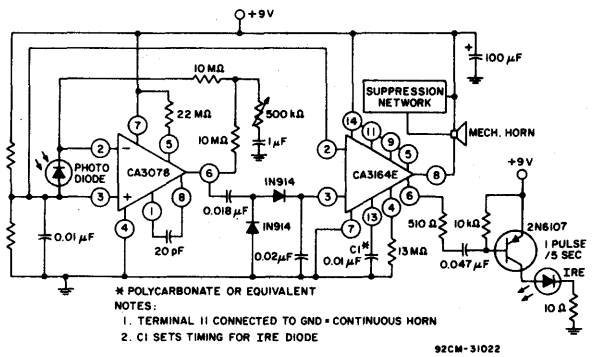


Fig. 3 — Typical photoelectric system using CA3164E.

ELECTRICAL CHARACTERISTICS at $T_A = 25°C$, $V^+ = 9$ V

CHARACTERISTIC	TEST CONDITIONS	LIMITS			UNITS
		Min.	Typ.	Max.	
Operating Voltage		7	9	11	V
Common-Mode Input Voltage Range, V_{ICR}	$(V^+ - 2\ V) = 7$ V	0	–	7	V
Low-Battery Trigger Voltage	External adjust (increase only)	7.3	7.7	7.9	V
Horn Driver $V_{CE(SAT)}$	Term. 8 = 100 mA	–	0.5	–	V
	Term. 8 = 300 mA	–	1	–	
Reference Voltage		5.8	6.2	6.6	V
Input Leakage Current, I_L	Term. 2	–	–	1	pA
	Term. 2 at 50°C	–	–	2.5	
	Term. 3	–	–	50	
Standby Current (13 MΩ from Term. 4 to gnd)	No LED connected	–	8*	12	μA
	LED connected–20 mA for 30 ms every 60s	–	18	–	
	Photoelectric operation – LED photocurrent = 0.6 A (5 sec. rate)	–	13	–	
Reference Source Current		5	–	–	μA
LED Driver Sink Current		40	50	–	mA
Interconnect Current Source	I_{Sink} = 10 μA typ.	–	2.8	–	mA
Sink	I_{Source} = 1.3 mA typ.	–	50	–	μA
Low-Battery Adjust, Term.5 Input Current		50	70	100	nA
Timing Current	Term. 13	10	–	50	nA
LED Blink Period	Adjustable	–	–	1	PPM
LED Pulse Width	Fixed	–	30	–	ms
Remote Fan-Out		20	–	–	
Alarm Pulse Duty Cycle (4.7 MΩ from Term. 11 to gnd	On-time	–	95	–	%
	On-time = 95%	–	0.5	–	sec.
	Off-time = 5%	–	0.026	–	

* Adjustable to 5 μA

OPERATING MODES TRUTH TABLE

Condition	Smoke Ionization Chamber	Low Battery	Led 6	Alarm Horn 8	Alarm Enable Pulser 11	System Interconnect 12	Remote Unit Status
Normal	No	No	Blink	Off	X	Low	Off
Low Battery	No	Yes	Blink	Beep	X	Low	Off
Smoke In Chamber	Yes	X	On	Pulsed*	Resistor to ground	High	On
External Input A1 From Remote Unit	No	No	Blink	On**	High	High	On

** Alarm Horn follows mode programmed for internal system input. For example, if terminal 11 has resistor connected to ground, horn will beep. If terminal 11 is connected to V^+, horn will be "on."

X = Don't Care

Blink & Beep = 30 msec (fixed) every 50 sec (ADJ)

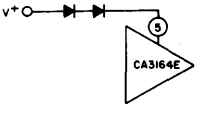
Pulsed = 95% "on" time – Period is determined by resistor from terminal 11 to ground-5% Off Time

* Horn "Continuous" if terminal 11 is connected to V^+

Connections for Optional Functions

1. Low Battery Adjustment – Terminal 5

Add diodes as shown below to increase the the low-battery trigger point.



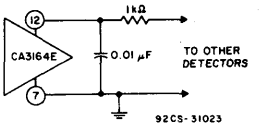
2. Sounder Operating Mode

Continuous sound on alarm – connect terminal 11 to V^+.

Pulsed sound on alarm – connect resistor between terminal and ground.

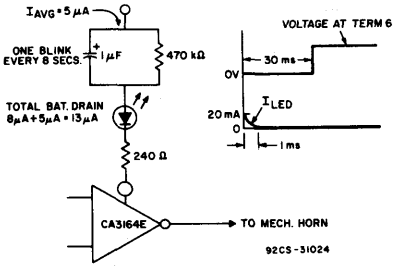
3. Remote (Interconnect)

Connect terminal 12 to same terminal on all other units (fan out = 20 units). When interconnecting units for the remote-alarm function, the extremely low currents involved make it extremely important that a provision be made for limiting externally induced transients into the remote terminal. For example, inadvertent contact with external power sources or electrical storm activity may cause triggering of the remote alarm function. The circuit below will reduce the possibility of such occurences.



4. LED On-Time Adjustment

Option 1: The CA3164E is designed to provide a fixed LED on-time of approximately 30 ms. For applications requiring a reduction in on-time the following circuit is recommended:



This circuit reduces the LED on-time but does not affect the horn on-time of 30 ms. When using this configuration during the continuous-alarm mode (smoke in chamber) the LED will be off instead of on, as shown in the truth table. If the horn is pulsed during the alarm mode, the LED will blink at the pulse rate.

Option 2: A chip design change can be implemented to generate a 10 ms LED on-time. The sounder would also be pulsed for only 10 ms. A further reduction in this time may result in faulty operation of the sounder.

5. Cleaning Procedure

To insure leakage currents of less than 1 pA the following procedure is recommended:

(a) degrease in trichlorethylene
(b) rinse in de-ionized water

Circuit Description

Basic Functions – The CA3164E is designed to interface directly with an ionization-chamber type of smoke detector. Upon being triggered by a decreasing voltage at the ionization-chamber output, the IC operates a mechanical transducer. In addition to this basic smoke-detector function, another circuit monitors and compares the battery voltage to an internal reference-voltage source. Once the battery voltage drops below a defined level, a short 30-ms beep sound is produced in synchronism with an LED indicator every 50 seconds. This rate is determined by a programming resistor connected between terminal 4 and ground and an external 0.1 μF capacitor connected between terminal 13 and ground.

A buffered output voltage is available from the reference supply that may be used to operate the ionization chamber. This voltage helps maintain constant sensitivity with decreasing supply voltage.

There are two alarm modes and two conditions that will sound the alarm. The first alarm condition is the normal smoke in the ionization chamber; the other condition is a high level to the remote input/output terminal of the IC.

The first alarm mode is the customary continuous sound. The second alarm mode is an interrupted or pulsed sound.

Operation – The CA3164E is current programmable by placing a resistor from terminal 4 to ground. This resistor establishes the operating current levels for all the current sources within the IC including the timing circuits.

An operational amplifier configuration is used for the ionization chamber input. P-channel MOS field effect transistors are used on this input in the bootstrap configuration shown in Fig. 4 to drive the protection diodes and maintain the subpioampere input current.

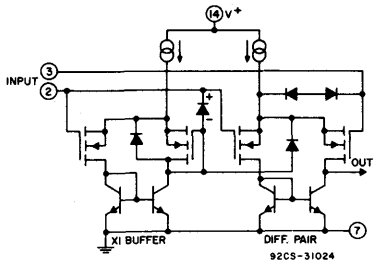


Fig. 4 – Schematic of ionization-chamber amplifier.

A conventional bipolar amplifier is used for the battery monitor circuit. The zener diode is biased at about 3 μA. This zener voltage is raised one V_{AK} and then applied to the base of an emitter-follower transistor to buffer and reflect the zener voltage to the outside reference terminal. By providing an additional input terminal (terminal 5), where three level-shifting diodes are available, an additional external means is provided to raise the voltage level at which the CA3164E goes into the low-voltage alarm mode.

An integrating type of timer is used to generate the one-minute LED power-monitor and battery-function indicator pulse. Fig. 5 shows the system. A constant-current source charges the external 0.1-μF timing amplifier P1, which subsequently triggers the 30-ms one-shot multivibrator composed of n-channel MOS transistor N_3 and n-p-n transistor Q1.

N_3 is then cut off and its drain climbs to the supply rail, linearly charging capacitor CP. When the drain of N_3 reaches the supply rail, the charging current ceases, cutting off the base current of N_3 and discharging the capacitor.

An open-collector n-p-n transistor is used to drive the optional external LED power monitor and battery condition indicator.

When terminal 11 is returned to V^+, the alarm sounds continuously. However, it terminal 11 is returned to ground through a programming resistor as shown in the block diagram, the alarm pulses. The pulse rate is determined by the sum of the current through the programming resistor connected to terminal 11 and the current from the basic timer current source. Thus, when the detector goes into the alarm mode, the nominal 50-second timer is increased to a nominal 0.5-second period. This second 0.5-second rate is a function of the external 4.3-M Ω programming resistor.

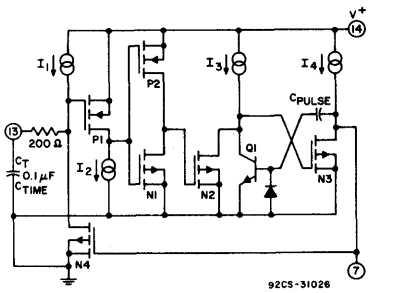


Fig. 5 – Schematic of timer and one-shot multivibrator.

A large n-p-n transistor at terminal 8 is capable of operating the typical mechanical interrupter type sounders. An active pull-up transistor is also incorporated in this circuit. Terminal 9 must be returned directly to V^+.

Terminal 12, the interconnect terminal, is both an input and output for the circuit. When connected by two wires to other units, alarm in any one unit will activate the other units. A small sinking current of only 10 μA keeps the line impedance down while a sourcing current of over 2 mA is available in the alarm mode. This current is more than sufficient to trigger over 20 additional units.

Dual BiMOS Operational Amplifiers

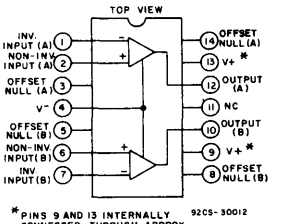
With MOS/FET Input, Bipolar Output

The RCA-CA3240A and CA3240 are dual versions of the popular CA3140-series integrated circuit operational amplifiers. They combine the advantages of MOS and bipolar transistors on the same monolithic chip. The gate-protected MOS/FET (PMOS) input transistors provide high input impedance and a wide common-mode input voltage range (typically to 0.5 V below the negative supply rail). The bipolar output transistors allow a wide output voltage swing and provide a high output current capability.

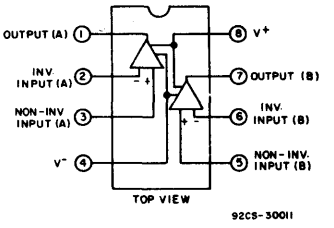
The CA3240A and CA3240 are supplied in the 8-lead dual-in-line plastic package (Mini-DIP, E suffix), and in the 14-lead dual-in-line plastic package (E1 suffix). They are pin-compatible with the industry standard 747 and 1458 operational amplifiers in similar packages. The CA3240A and CA3240 have an operating-temperature range of −40 to +85°C. The offset null feature is available only when these types are supplied in the 14-lead dual-in-line plastic package (E1 suffix).

Features:

- **Dual version of CA3140**
- **Internally compensated**
- **MOS/FET input stage**
  - **(a) Very high input impedance (Z_{IN}) − 1.5 TΩ typ.**
  - **(b) Very low input current (I_I) − 10 pA typ. at ± 15 V**
  - **(c) Wide common-mode input-voltage range (V_{ICR}) − can be swung 0.5 volt below negative supply-voltage rail**
  - **(d) Rugged input stage − bipolar diode protected**
- **Directly replaces industry types 747 and 1458 in most applications**
- **Operation from 4-to-36 volts single or dual supplies**
- **Characterized for ± 15-volt operation and for TTL supply systems with operation down to 4 volts**
- **Wide bandwidth − 4.5 MHz unity gain at ± 15 V or 30 V**
- **High voltage-follower slew rate − 9 V/μs**
- **Output swings to within 0.5 volt of negative supply at $V^+ = 5$ V, $V^- = 0$**



E1 Suffix
Pin compatible with the industry-standard 747



E Suffix
Pin compatible with the industry-standard 1458

Fig. 1 − Functional diagrams.

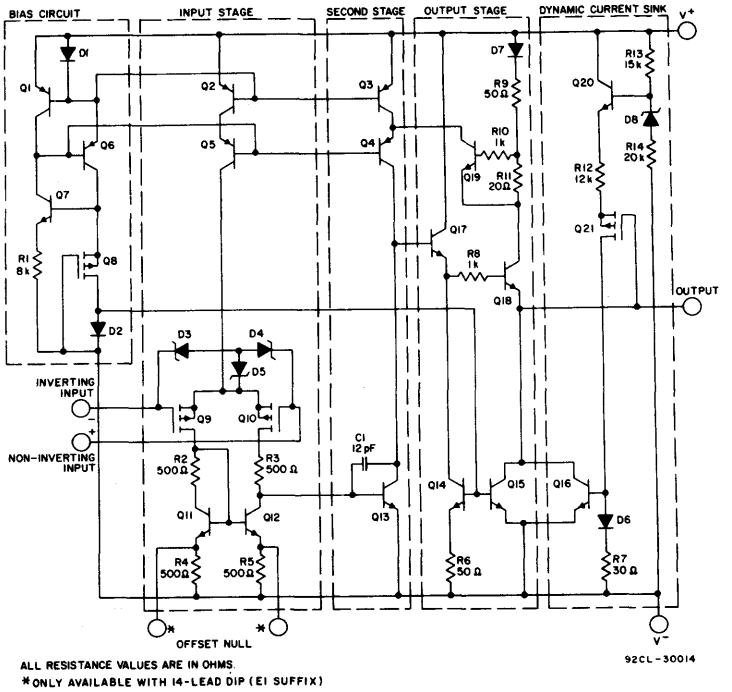


Fig. 2 − Schematic diagram of one-half CA3240 series.

Applications:

- **Ground-referenced single-supply amplifiers in automobile and portable instrumentation**
- **Sample and hold amplifiers**
- **Long-duration timers/multivibrators (microseconds−minutes−hours)**
- **Photocurrent instrumentation**
- **Active filters ■ Intrusion alarm systems**
- **Comparators ■ Instrumentation amplifiers**
- **Function generators ■ Power supplies**

Circuit Description

The schematic diagram of one amplifier section of the CA3240 is shown in Fig. 2. It consists of a differential amplifier stage using PMOS transistors Q9 and Q10 with gate-to-source protection against static discharge damage provided by zener diodes D3, D4, and D5. Constant current bias is applied to the differential amplifier from transistors Q2 and Q5 connected as a constant-current source. This assures a high common-mode rejection ratio. The output of the differential amplifier is coupled to the base of gain stage transistor Q13 by means of an n-p-n current mirror that supplies the required differential-to-single-ended conversion. Provision for offset null for types in the 14-lead plastic package (E1 suffix) is provided through the use of this current mirror.

MAXIMUM RATINGS, *Absolute-Maximum Values:*

DC SUPPLY VOLTAGE
(BETWEEN V^+ AND V^- TERMINALS) . . . . . 36 V
OPERATING VOLTAGE RANGE . . . . . 4 to 36 V
or ±2 to ±18 V
DIFFERENTIAL-MODE INPUT VOLTAGE . . . . . ±8 V
COMMON-MODE DC INPUT VOLTAGE . . . . . (V^+ +8 V) to (V^- −0.5 V)
INPUT-TERMINAL CURRENT . . . . . 1 mA
DEVICE DISSIPATION:
UP TO 55°C . . . . . 630 mW
ABOVE 55°C . . . . . Derate linearly 6.67 mW/°C
TEMPERATURE RANGE:
OPERATING . . . . . −40 to +85°C
STORAGE . . . . . −65 to +150°C
OUTPUT SHORT-CIRCUIT DURATION● . . . . . UNLIMITED
LEAD TEMPERATURE (DURING SOLDERING):
AT DISTANCE 1/16 ±1/32 INCH (1.59 ± 0.79 MM)
FROM CASE FOR 10 SECONDS MAX. . . . . +265°C

● Short circuit may be applied to ground or to either supply. Temperatures and/or supply voltages must be limited to keep dissipation within maximum rating.

ELECTRICAL CHARACTERISTICS FOR EQUIPMENT DESIGN
At V^+ = 15 V, V^- = 15 V, T_A = 25°C Unless Otherwise Specified

CHARACTERISTIC		LIMITS						UNITS
		CA3240A			CA3240			
		Min.	Typ.	Max.	Min.	Typ.	Max.	
Input Offset Voltage,	$\lvert V_{IO}\rvert$	–	2	5	–	5	15	mV
Input Offset Current,	$\lvert I_{IO}\rvert$	–	0.5	20	–	0.5	30	pA
Input Current,	I_I	–	10	40	–	10	50	pA
Large-Signal Voltage Gain,	A_{OL}●	20 k	100 k	–	20 k	100 k	–	V/V
(See Figs. 4, 19)		86	100	–	86	100	–	dB
Common-Mode Rejection Ratio,	CMRR	–	32	320	–	32	320	μV/V
(See Fig. 9)		70	90	–	70	90	–	dB
Common-Mode Input-Voltage Range, (See Fig. 16)	V_{ICR}	−15	−15.5 to +12.5	12	−15	−15.5 to +12.5	11	V
Power-Supply Rejection Ratio, (See Fig. 11)	$\Delta V_{IO}/\Delta V$ PSRR	–	100	150	–	100	150	μV/V
		76	80	–	76	80	–	dB
Maximum Output Voltage,■ (See Figs. 22, 16)	V_{OM}^+	+12	13	–	+12	13	–	V
	V_{OM}^-	−14	−14.4	–	−14	−14.4	–	
Maximum Output Voltage,†	V_{OM}^-	0.4	0.13	–	0.4	0.13	–	V
Supply Current, (See Fig. 7) For Both Amps.	I^+	–	8	12	–	8	12	mA
Total Device Dissipation,	P_D	–	240	360	–	240	360	mW

● At V_O = 26 V_{p-p}, +12 V, −14 V and R_L = 2 kΩ.
■ At R_L = 2 kΩ.
† At V^+ = 5 V, V^- = GND, I_{Sink} = 200 μA.

The gain stage transistor Q13 has a high-impedance active load (Q3 and Q4) to provide maximum open-loop gain. The collector of Q13 directly drives the base of the compound emitter-follower output stage. Pull-down for the output stage is provided by two independent circuits: (1) constant-current-connected transistors Q14 and Q15 and (2) dynamic current-sink transistor Q16 and its associated circuitry. *The level of pull-down current is constant at about 1 mA for Q15 and varies from 0 to 18 mA for Q16 depending on the magnitude of the voltage between the output terminal and V^+.* The dynamic current sink becomes active whenever the output terminal is more negative than V^+ by about 15 V. When this condition exists, transistors Q21 and Q16 are turned on causing Q16 to sink current from the output terminal to V^-. This current always flows when the output is in the linear region, either from the load resistor or from the emitter of Q18 if no load resistor is present. The purpose of this dynamic sink is to permit the output to go within 0.2 V (V_{CE}(sat)) of V^- with a 2-kΩ load to ground. *When the load is returned to V^+, it may be necessary to supplement the 1 mA of current from Q15 in order to turn on the dynamic current sink (Q16).* This may be accomplished by placing a resistor (approx. 2 kΩ) between the output and V^-.

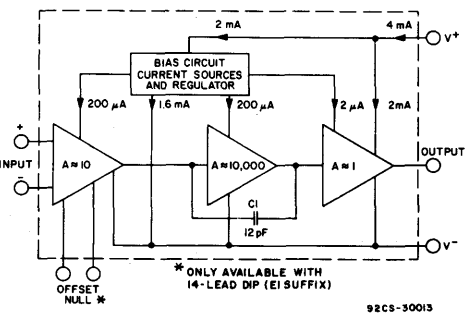


Fig. 3 — Block diagram of one-half CA3240 series.

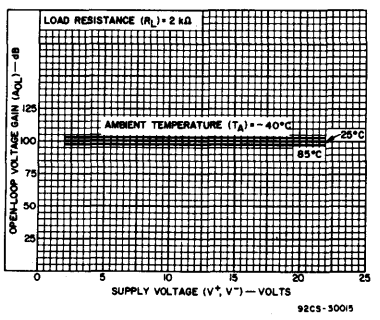


Fig. 4 — Open-loop voltage gain as a function of supply voltage and temperature.

TYPICAL ELECTRICAL CHARACTERISTICS

CHARACTERISTIC		TEST CONDITIONS $V^+ = +15$ V, $V^- = -15$ V, $T_A = 25°C$		TYPICAL VALUES CA3240A	TYPICAL VALUES CA3240	UNITS
Input Offset Voltage Adjustment Resistor (E1 Package Only)		Typ. Value of Resistor Between Terms. 4 and 3(5) or Between 4 and 14(8) to Adjust Max. V_{IO}		18	4.7	kΩ
Input Resistance	R_I			1.5	1.5	TΩ
Input Capacitance	C_I			4	4	pF
Output Resistance	R_O			60	60	Ω
Equivalent Wideband Input Noise Voltage (See Fig. 21)	e_n	BW=140 kHz, R_S = 1 MΩ		48	48	μV
Equivalent Input Noise Voltage (See Fig. 10)	e_n	f= 1 kHz	R_S= 100 Ω	40	40	nV/√Hz
		f=10 kHz		12	12	
Short-Circuit Current to Opposite Supply Source	I_{OM}^+			40	40	mA
Sink	I_{OM}^-			11	11	
Gain-Bandwidth Product (See Figs. 5 and 19)	f_T			4.5	4.5	MHz
Slew Rate (See Fig. 6)	SR			9	9	V/μs
Transient Response: Rise Time	t_r	R_L=2 kΩ, C_L=100 pF		0.08	0.08	μs
Overshoot (See Fig. 20)	t_r			10	10	%
Settling Time at 10 V_{p-p}, (See Fig. 17)	1 mV / 10 mV t_s	R_L=2 kΩ, C_L=100 pF, Voltage Follower		4.5 / 1.4	4.5 / 1.4	μs
Crosstalk		f = 1 kHz		120	120	dB

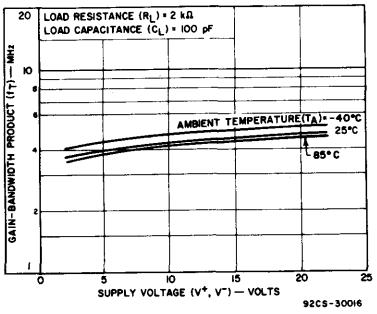


Fig. 5 – Gain-bandwidth product as a function of supply voltage and temperature.

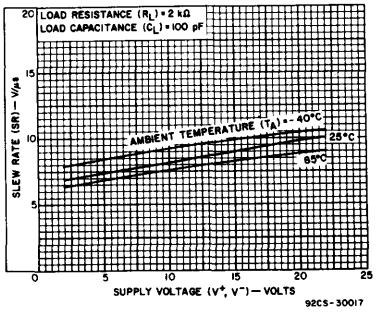


Fig. 6 – Slew rate as a function of supply voltage and temperature.

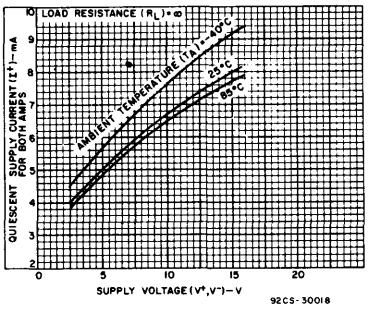


Fig. 7 – Quiescent supply current as a function of supply voltage and temperature.

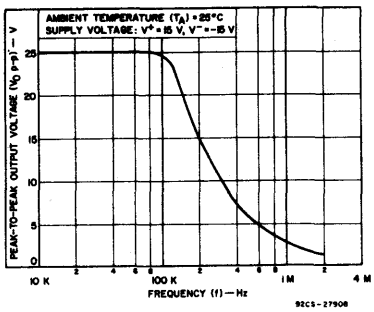


Fig. 8 – Maximum output voltage swing as a function of frequency.

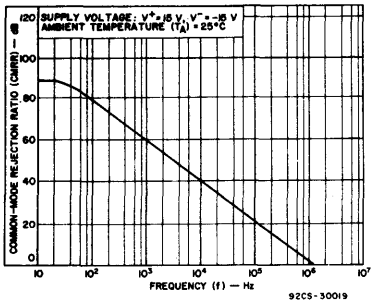


Fig. 9 – Common-mode rejection ratio as a function of frequency.

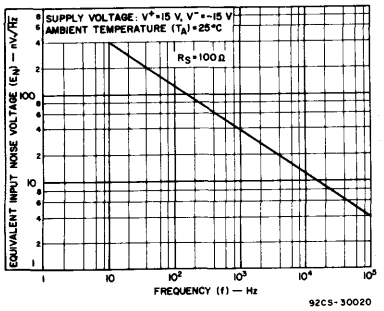


Fig. 10 – Equivalent input noise voltage as a function of frequency.

ELECTRICAL CHARACTERISTICS FOR EQUIPMENT DESIGN
At $V^+ = 15$ V, $V^- = 15$ V, $T_A = -40$ to $+85^\circ$C Unless Otherwise Specified

CHARACTERISTIC		TYPICAL VALUES CA3240A	TYPICAL VALUES CA3240	UNITS
Input Offset Voltage,	$\|V_{IO}\|$	3	10	mV
Input Offset Current,♦	$\|I_{IO}\|$	32	32	pA
Input Current,♦	I_I	640	640	pA
Large-Signal Voltage Gain, (See Figs. 4, 19)	A_{OL}●	63 k	63 k	V/V
		96	96	dB
Common-Mode Rejection Ratio, (See Fig. 9)	CMRR	32	32	μV/V
		90	90	dB
Common-Mode Input-Voltage Range, (See Fig. 16)	V_{ICR}	−15 to +12.3	−15 to +12.3	V
Power-Supply-Rejection Ratio, (See Fig. 11)	$\frac{\Delta V_{IO}/\Delta V}{PSRR}$	150	150	μV/V
		76	76	dB
Maximum Output Voltage,■ (See Figs. 16, 22)	V_{OM}^+	12.4	12.4	V
	V_{OM}^-	−14.2	−14.2	
Supply Current, (See Fig. 7) For Both Amps.	I^+	8.4	8.4	mA
Total Device Dissipation,	P_D	252	252	mW
Temperature Coefficient of Input Offset Voltage,	$\Delta V_{IO}/\Delta T$	15	15	μV/°C

● At $V_O = 26\ V_{p\text{-}p}$, +12 V, −14 V and $R_L = 2$ kΩ.
■ At $R_L = 2$ kΩ.
♦ At $T_A = 85^\circ$C

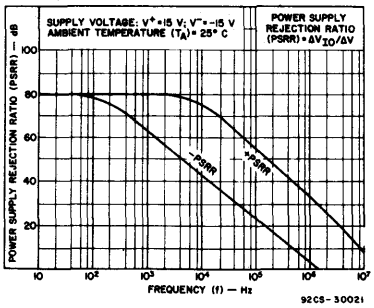


Fig. 11 — Power supply rejection ratio as a function of frequency.

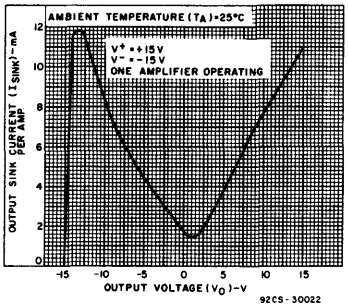


Fig. 12 — Output sink current as a function of output voltage.

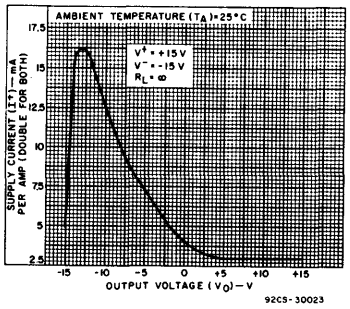


Fig. 13 — Supply current as a function of output voltage.

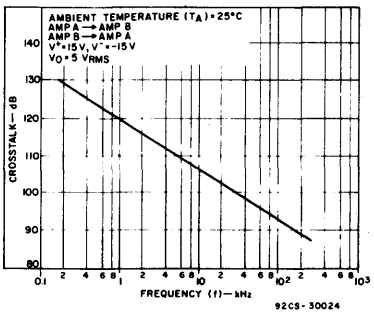


Fig. 14 — Crosstalk as a function of frequency.

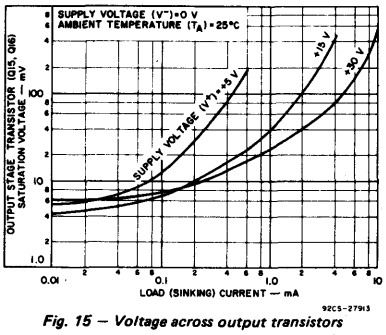


Fig. 15 — Voltage across output transistors Q15 and Q16 as a function of load current.

TYPICAL ELECTRICAL CHARACTERISTICS FOR DESIGN GUIDANCE
At $V^+ = 5$ V, $V^- = 0$ V, $T_A = 25°C$

CHARACTERISTIC		TYPICAL VALUES CA3240A	TYPICAL VALUES CA3240	UNITS
Input Offset Voltage,	$\|V_{IO}\|$	2	5	mV
Input Offset Current,	$\|I_{IO}\|$	0.1	0.1	pA
Input Current,	I_I	2	2	pA
Input Resistance		1	1	TΩ
Large-Signal Voltage Gain, (See Figs. 4, 19)	A_{OL}	100 k	100 k	V/V
		100	100	dB
Common-Mode Rejection Ratio,	CMRR	32	32	μV/V
		90	90	dB
Common-Mode Input-Voltage Range, (See Fig. 22)	V_{ICR}	−0.5	−0.5	V
		2.6	2.6	
Power-Supply Rejection Ratio,	PSRR	31.6	31.6	μV/V
		90	90	dB
Maximum Output Voltage, (See Figs. 16,22)	V_{OM}^+	3	3	V
	V_{OM}^-	0.3	0.3	
Maximum Output Current: Source,	I_{OM}^+	20	20	mA
Sink	I_{OM}^-	1	1	
Slew Rate (See Fig. 6)		7	7	V/μs
Gain-Bandwidth Product, (See Fig. 5)	f_T	4.5	4.5	MHz
Supply Current, (See Fig. 7)	I^+	4	4	mA
Device Dissipation,	P_D	20	20	mW

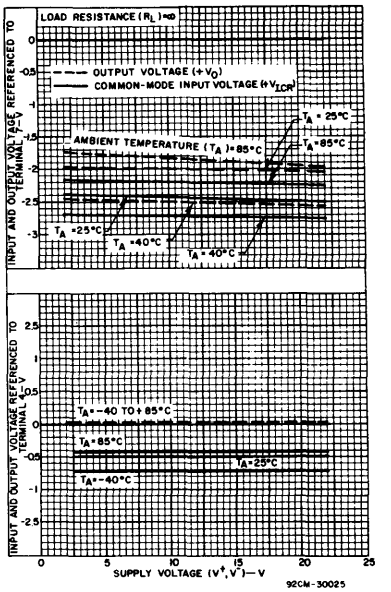


Fig. 16 — Output-voltage-swing capability and common-mode input-voltage range as a function of supply voltage and temperature.

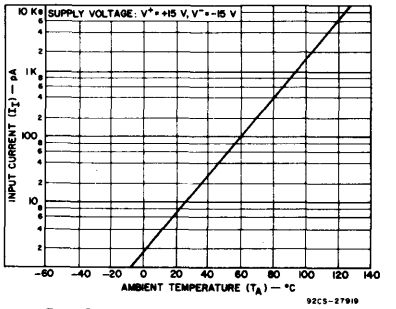
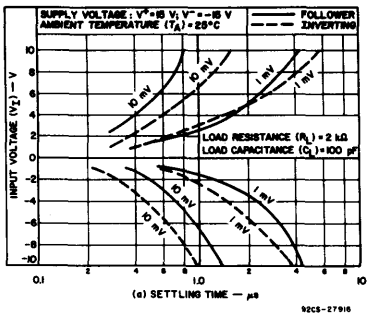


Fig. 18 — Input current as a function of ambient temperature.



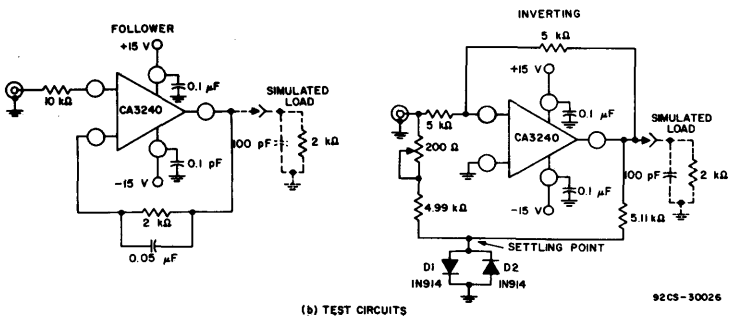


Fig. 17 — Input voltage as a function of settling time.

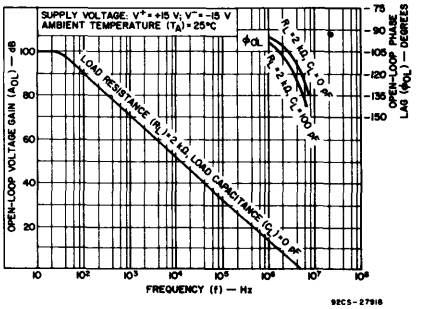


Fig. 19 — Open-loop voltage gain and phase lag as a function of frequency.

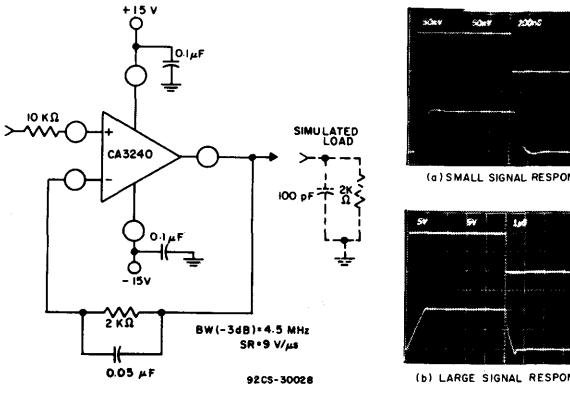


Fig. 20 – Split-supply voltage-follower test circuit and associated waveforms.

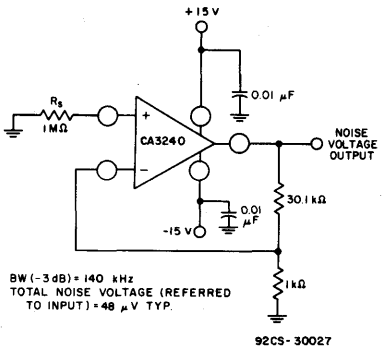


Fig. 21 – Test-circuit amplifier (30-dB gain) used for wideband noise measurement.

APPLICATIONS CONSIDERATIONS

Output Circuit Considerations

Fig. 22 shows output current-sinking capabilities of the CA3240 at various supply voltages. Output voltage swing to the negative supply rail permits this device to operate both power transistors and thyristors directly without the need for level-shifting circuitry usually associated with the 741 series of operational amplifiers.

Fig. 23 shows some typical configurations. Note that a series resistor, R_L, is used in both cases to limit the drive available to the driven device. Moreover, it is recommended that a series diode and shunt diode be used at the thyristor input to prevent large negative transient surges that can appear at the gate of thyristors, from damaging the integrated circuit.

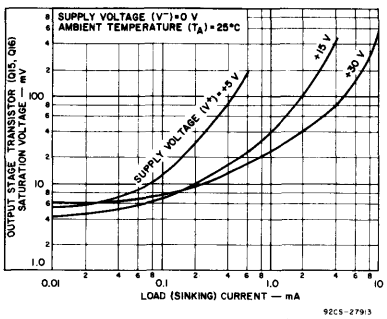


Fig. 22 – Voltage across output transistors Q15 and Q16 as a function of load current.

Input Circuit Considerations

As indicated by the typical VICR, this device will accept inputs as low as 0.5 V below V^-. However, a series current-limiting resistor is recommended to limit the maximum input terminal current to less than 1 mA to prevent damage to the input protection circuitry.

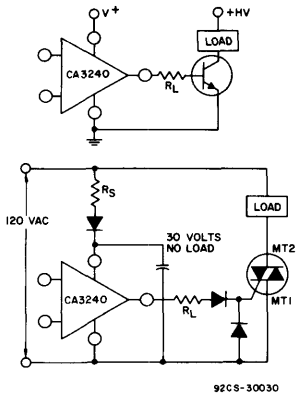


Fig. 23 – Methods of utilizing the $V_{CE}(sat)$ sinking-current capability of the CA3240 series.

Moreover, some current-limiting resistance should be provided between the inverting input and the output when the CA3240 is used as a unity-gain voltage follower. This resistance prevents the possibility of extremely large input-signal transients from forcing a signal through the input-protection network and directly driving the internal constant-current source which could result in positive feedback via the output terminal. A 3.9-kΩ resistor is sufficient.

The typical input current is in the order of 10 pA when the inputs are centered at nominal device dissipation. As the output supplies load current, device dissipation will increase, raising the chip temperature and resulting in increased input current. Fig. 24 shows typical input-terminal current versus ambient temperature for the CA3240.

It is well known that MOS/FET devices can exhibit slight changes in characteristics (for example, small changes in input offset voltage) due to the application of large differential input voltages that are sustained over long periods at elevated temperatures.

Both applied voltage and temperature accelerate these changes. The process is reversible and offset voltage shifts of the opposite polarity reverse the offset. In typical linear applications, where the differential voltage is small and symmetrical, these incremental changes are of about the same magnitude as those encountered in an operational amplifier employing a bipolar transistor input stage.

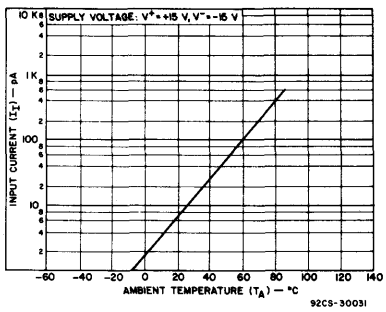


Fig. 24 – Input current as a function of ambient temperature.

Offset-Voltage Nulling

The input-offset voltage of the CA3240AE1 and CA3240E1 can be nulled by connecting a 10-kΩ potentiometer between Terminals 3 and 14 or 5 and 8 and returning its wiper arm to Terminal 4, see Fig. 25a. This technique, however, gives more adjustment range than required and therefore, a considerable portion of the potentiometer rotation is not fully utilized. Typical values of series resistors that may be placed at either end of the potentiometer, see Fig. 25b, to optimize its utilization range are given in the table "Electrical Characteristics For Design Guidance" shown in this bulletin.

An alternate system is shown in Fig. 25c. This circuit uses only one additional resistor of approximately the value shown in the table. For potentiometers, in which the resistance does not drop to zero ohms at either end of rotation, a value of resistance 10% lower than the values shown in the table should be used.

TYPICAL APPLICATIONS

On/Off Touch Switch

The on/off touch switch shown in Fig. 26 uses the CA3240E to sense small currents flowing between two contact points on a touch plate consisting of a PC board metallization "grid". When the "on" plate is touched, current flows between the two halves of the grid causing a positive shift in the output voltage (Term. 7) of the CA3240E. These positive transitions are fed into the CA3059, which is used as a latching circuit and zero-crossing triac driver. When a positive pulse occurs at Terminal 7 of the CA3240E, the triac is turned on and held on by the CA3059 and its associated positive feedback circuitry (51-kΩ resistor and 36-kΩ/42-kΩ voltage divider). When the positive pulse occurs at Terminal 1 (CA3240E), the triac is turned off and held off in a similar manner. Note that power for the CA3240E is supplied by the CA3059 internal power supply.

The advantage of using the CA3240E in this circuit is that it can sense the small currents associated with skin conduction while allowing sufficiently high circuit impedance to provide protection against electrical shock.

Dual Level Detector (window comparator)

Fig. 27 illustrates a simple dual liquid level detector using the CA3240E as the sensing amplifier. This circuit operates on the principle that most liquids contain enough ions in solution to sustain a small amount of current flow between two electrodes submersed in the liquid. The current, induced by an 0.5-V potential applied between two halves of a PC board grid, is converted to a voltage level by the CA3240E in a circuit similar to that of the on/off touch switch shown in Fig. 26. The changes in voltage for both the upper and lower level sensors are processed by the CA3140 to activate an LED whenever the liquid level is above the upper sensor or below the lower sensor.

Constant-Voltage/Constant-Current Power Supply

The constant-voltage/constant-current power supply shown in Fig. 28 uses the CA3240E as a voltage-error and current-sensing amplifier. The CA3240E is ideal for this application because its input common-mode voltage-range includes ground, allowing the supply to adjust from 20 mV to 25 V without requiring an additional negative input voltage. Also, the ground reference capability of the CA3240E allows it to sense the voltage across the 1-Ω current-sensing resistor in the negative output lead of the power supply. The CA3086 transistor array functions as a reference for both constant-voltage and constant-current limiting. The 2N6385 power Darlington is used as the pass element and may be required to dissipate as much as 40 W. Fig. 29 shows the transient response of the supply during a 100-mA to 1-A load transition.

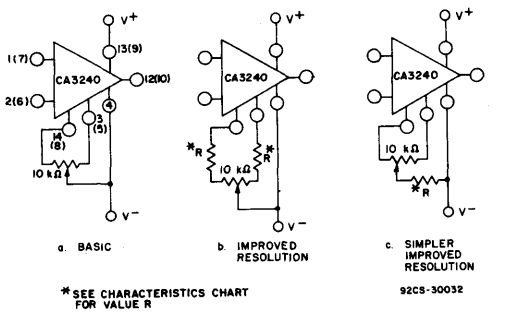


Fig. 25 – Three offset-voltage nulling methods. (CA3240AE1, CA3240E1 only.)

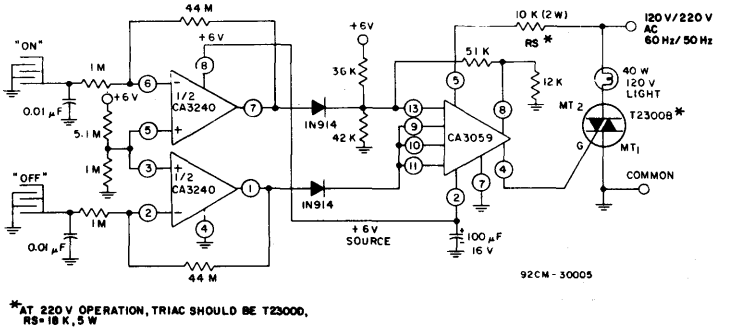


Fig. 26 – On/off touch switch.

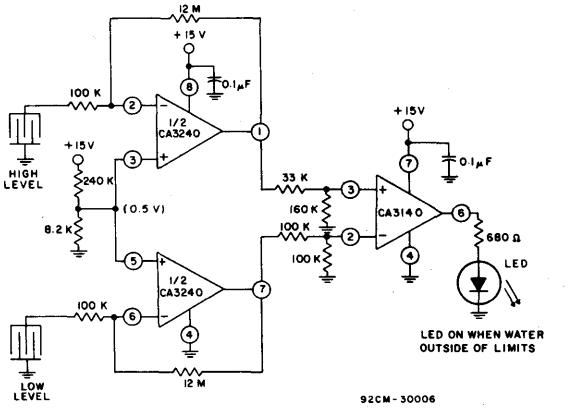


Fig. 27 – Dual level detector.

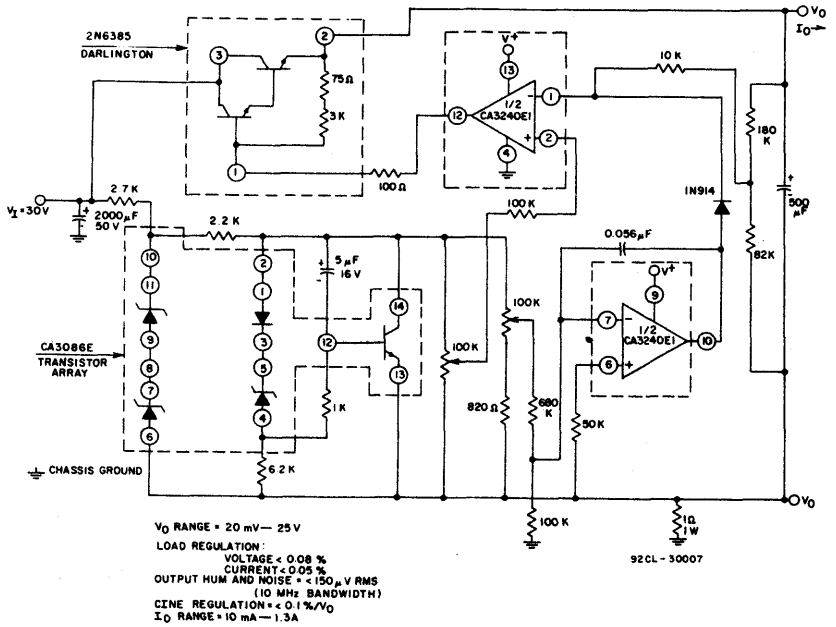


Fig. 28 — Constant-voltage/constant-current power supply.



TRANSIENT RESPONSE

TOP TRACE: OUTPUT VOLTAGE
(500 mV/cm AND 5 μs/cm)
BOTTOM TRACE: COLLECTOR OF LOAD
SWITCHING TRANSISTOR
LOAD = 100 mA TO 1A
(5 V/cm AND 5 μs/cm)

92CS-30034

Fig. 29 — Transient response.

Precision Differential Amplifier

Fig. 30 shows the CA3240E in the classical precision differential amplifier circuit. The CA3240E is ideally suited for biomedical applications because of its extremely high input impedance. To insure patient safety, an extremely high electrode series resistance is required to limit any current that might result in patient discomfort in the event of a fault condition. In this case, 10-MΩ resistors have been used to limit the current to less than 2 μA without affecting the performance of the circuit. Fig. 31 shows a typical electrocardiogram waveform obtained with this circuit.

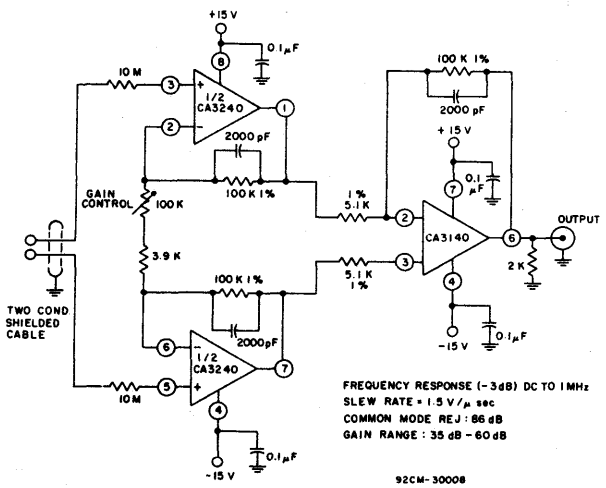
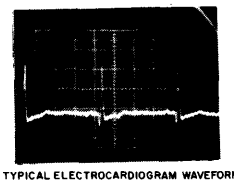


Fig. 30 — Precision differential amplifier.

Differential Light Detector

In the circuit shown in Fig. 32, the CA3240E converts the current from two photo diodes to voltage, and applies 1 V of reverse bias to the diodes. The voltages from the CA3240E outputs are subtracted in the second stage (CA3140) so that only the difference is amplified. In this manner, the circuit can be used over a wide range of ambient light conditions without circuit component adjustment. Also, when used with a light source, the circuit will not be sensitive to changes in light level as the source ages.



TYPICAL ELECTROCARDIOGRAM WAVEFORM

VERTICAL : 1.0 mV/DIV.
(AMPLIFIER GAIN = 100 X)
(SCOPE SENSITIVITY = 0.1V/DIV.
HORIZONTAL : > 0.2 SEC/DIV (UNCAL)

92CS-30033

Fig. 31 – Typical electrocardiogram waveform.

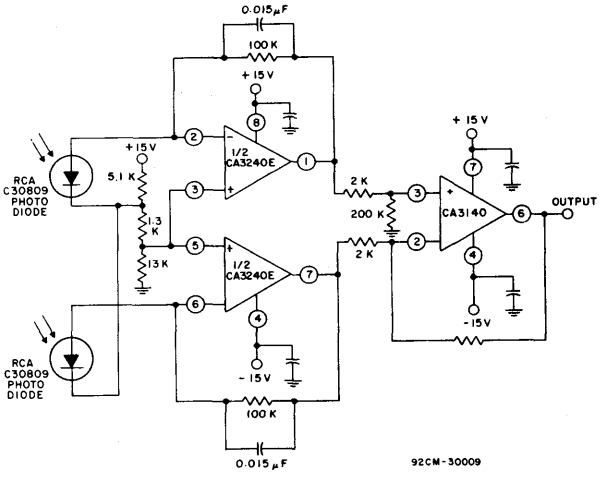
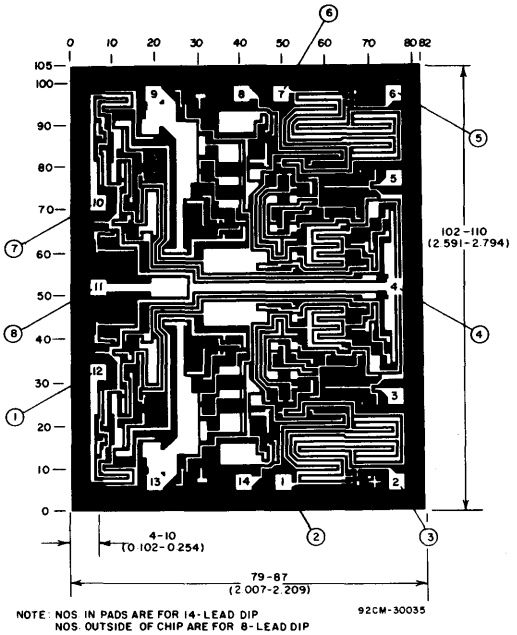


Fig. 32 – Differential light detector.

CA3240H Dimensions and Pad Layout



NOTE: NOS IN PADS ARE FOR 14-LEAD DIP
NOS. OUTSIDE OF CHIP ARE FOR 8-LEAD DIP

The photographs and dimensions represent a chip when it is part of the wafer. When the wafer is cut into chips, the cleavage angles are 57° instead of 90° with respect to the face of the chip. Therefore, the isolated chip is actually 7 mils (0.17 mm) larger in both dimensions.

Dimensions in parentheses are in millimeters and are derived from the basic inch dimensions as indicated. Grid graduations are in mils (10^{-3} inch).

BiMOS Dual Voltage Comparators

With MOS/FET Input, Bipolar Output

The RCA-CA3290B, CA3290A, and CA3290 types consist of a dual voltage-comparator on a single monolithic chip. The common-mode input voltage range includes ground when operated from a single supply. The low supply-current drain makes these comparators suitable for battery operation; their extremely low input currents allow their use in applications that employ sensors with extremely high source impedances. Package options are shown in the table below.

SELECTION CHART

Selection	Characteristic				Package & Suffix			
	Max. V_{IO} (mV)	Max. I_I (pA)	Min. A_{OL}	V^+ (V)	TO-5 Std.	TO-5 DIL-CAN	Plastic 8-Ld.	Plastic 14-Ld.
CA3290B	6	30	50K	44	T	S	–	–
CA3290A	10	40	25K	36	T	S	E	E1
CA3290	20	50	25K	36	T	S	E	E1

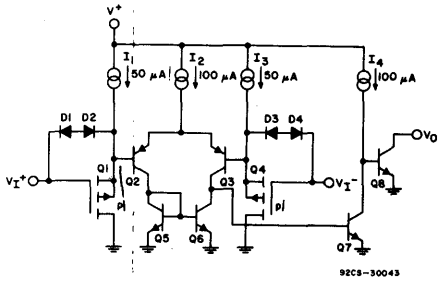


Fig. 1 – Basic CA3290 comparator.

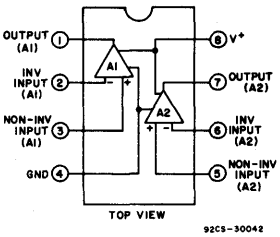
Features:

- **MOS/FET input stage:**
  - **(a) Very high input impedance (Z_{IN}) - 1.7 TΩ typ.**
  - **(b) Very low input current - 3.5 pA typ. at +5 V supply voltage**
  - **(c) Low input-offset voltage (V_{IO}) - to 6 mV max. (CA3290B)**
  - **(d) Wide common-mode input-voltage range (V_{ICR}) - can be swung 1.5 V (typ.) below negative supply-voltage rail**
  - **(e) No phase reversal of output signal for input signals down to 5 V below negative supply-voltage rail**
  - **(f) MOS/FET input stage - zener diode protected**
  - **(g) Virtually eliminates errors due to flow of input currents**
- **Wide supply-voltage range:**
  Single supply - 4 to 36 V dc
  Dual-supply - $^{+3.5}_{-0.5}$ to ±18 V dc
  (B-types up to 44 or ±22 V dc)
- **Very low supply-current drain - 0.8 mA at +5 V**
- **Differential input-voltage range - up to ±36 V**
- **Low output saturation voltage - 120 mV at 4 mA**
- **Output voltage compatible with TTL, DTL, ECL, MOS, and CMOS logic systems**
- **All types are rated for operation over the range of −55 to +125°C**
- **Stable V_{IO} vs. time due to source-follower inputs**

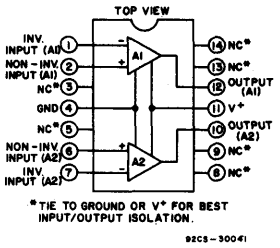
Applications:

- **High-source-impedance voltage comparators**
- **Long time delay circuits**
- **Square-wave generators**
- **A/D converters**
- **Window comparators**

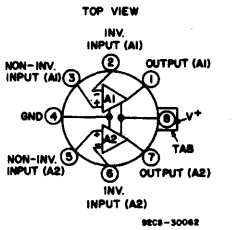
TERMINAL ASSIGNMENTS



E SUFFIX



EI SUFFIX



S and T SUFFIX

MAXIMUM RATINGS, *Absolute-Maximum Values:*

DC SUPPLY VOLTAGE:
- Single Supply:
  - CA3290B . . . . . +44 V
  - CA3290A, CA3290 . . . . . +36 V
- Dual Supply:
  - CA3290B . . . . . ±22 V
  - CA3290A, CA3290 . . . . . ±18 V

DIFFERENTIAL INPUT VOLTAGE . . . . . ±36 V or ±[(V^+−V^-)+5 V] (whichever is less)

COMMON-MODE INPUT VOLTAGE . . . . . V^++5 V to V^-−5 V

DEVICE DISSIPATION:
- Up to 55°C . . . . . 630 mW
- Above 55°C . . . . . Derate linearly at 6.67 mW/°C

OUTPUT-TO-V^- SHORT CIRCUIT DURATION* . . . . . CONTINUOUS

TEMPERATURE RANGE, ALL TYPES:
- Operating . . . . . −55 to +125°C
- Storage . . . . . −65 to +150°C

INPUT TERMINAL CURRENT . . . . . 1 mA

LEAD TEMPERATURE (DURING SOLDERING):
- AT DISTANCE 1/16 ± 1/32 INCH (1.59 ± 0.79 MM) FROM CASE FOR 10 SECONDS MAX. . . . . 265°C

*Short circuits from the output to V^+ can cause excessive heating and eventual destruction of the device.

CIRCUIT DESCRIPTION

The Basic Comparator

Fig. 1 shows the basic circuit diagram for one of the two comparators in the CA3290. It is generically similar to the industry-type "139" comparators, with PMOS transistors replacing p-n-p transistors as input stage elements. Transistors Q1 through Q4 comprise the differential input stage, with Q5 and Q6 serving as a mirror-connected active load and differential-to-single-ended converter. The differential input at Q1 and Q4 is amplified so as to toggle Q6 in accordance with the input-signal polarity. For example, if $+V_{IN}$ is greater than $-V_{IN}$, Q1, Q2, and current mirror transistors Q5 and Q6 will be turned off; transistors Q3, Q4, and Q7 will be turned on, causing Q8 to be turned off. The output is pulled positive when a load resistor is connected between the output and V^+.

ELECTRICAL CHARACTERISTICS at T_A = –55 to +125°C

CHARACTERISTIC	TEST CONDITIONS		VALUES						UNITS
			CA3290B		CA3290A		CA3290		
		V^+	Typ.	Max.	Typ.	Max.	Typ.	Max.	
Input Offset Voltage, V_{IO}	V_{IC}=1.4 V, V_O=1.4 V	5 V	3.5	–	4.5	–	8.5	–	mV
	V_{IC}=0 V, V_O=0 V	±15 V	3.5	–	8.5	–	8.5		
Temp. Coefficient of Input Offset Voltage, $\Delta V_{IO}/\Delta T$			8	–	8	–	8	–	μV/°C
Input Offset Current, I_{IO}	V_{IC}=1.4 V	5 V	2	22	2	28	2	32	nA
	V_{IC}=0 V	±15 V	7	22	7	28	7	32	
Input Current, I_I▲	V_{IC}=1.4 V	5 V	2.8	32	2.8	45	2.8	55	nA
	V_{IC}=0 V	±15 V	13	32	13	45	13	55	
Supply Current, I^+*	R_L = ∞	5 V	0.85	1.6	0.85	1	0.85	1.6	mA
		30 V	1.62	3.5	1.62	3	1.62	3.5	
Voltage Gain, A_{OL}	R_L=15 kΩ	±15 V	150	–	150	–	150	–	V/mV
			103	–	103	–	103	–	dB
Saturation Voltage	V^+=5 V, 4 mA, $+V_I$=0 V, $-V_I$=1 V	+125°C	0.22	0.7	0.22	0.7	0.22	0.7	V
		–55°C	0.1	–	0.1	–	0.1	–	
Output Leakage Current, I_{OL}		15 V	65	–	65	–	65	–	nA
		36 V	130	1k	130	1k	130	1k	

▲At T_A = +125°C
*At T_A = –55°C

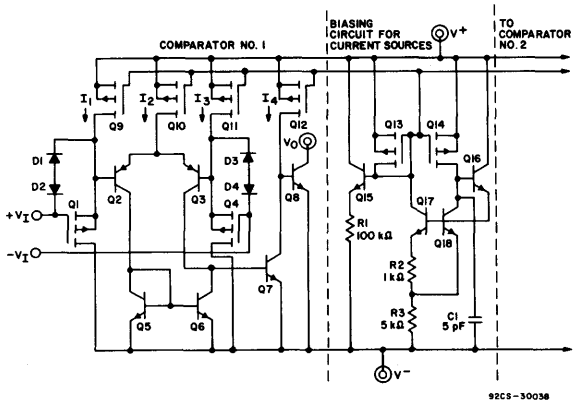


Fig. 2 – Schematic diagram of CA3290 (only one is shown).

In essence, Q1 and Q4 function as source-followers to drive Q2 and Q3, respectively, with zener diodes D1 through D4 providing gate-oxide protection against input voltage transients (e.g., static electricity). The current flow in Q1 and Q2 is established at approximately 50 microamperes by constant-current sources I_1 and I_3, respectively. Since Q1 and Q4 are operated with a constant current load, their gate-to-source voltage drops will be effectively constant as long as the input voltages are within the common-mode range. As a result, the input offset voltage ($V_{GS(Q1)} + V_{BE(Q2)} - V_{BE(Q3)} - V_{GS(Q4)}$) will not be degraded when a large differential dc voltage is applied to the device for extended periods of time at high temperatures.

Additional voltage gain following the first stage is provided by transistors Q7 and Q8. The collector of Q8 is open, offering the user a wide variety of options in applications. An additional discrete transistor can be added if it becomes necessary to boost the output sink-current capability.

The detailed schematic diagram for one comparator and the common current-source biasing is shown in Fig. 2. PMOS transistors Q9 through Q12 are the current-source elements identified in Fig. 1 as I_1 through I_4, respectively. Their gate-source potentials (V_{GS}) are supplied by a common bus from the biasing circuit shown in the right-hand portion of the Fig. 2. The currents supplied by Q10 and Q12 are twice those supplied by Q9 and Q11. The transistor geometries are appropriately scaled to provide the requisite currents with common V_{GS} applied to Q9 through Q12.

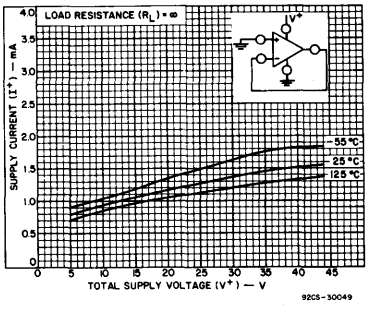


Fig. 3 – Supply current as a function of supply voltage (both amplifiers).

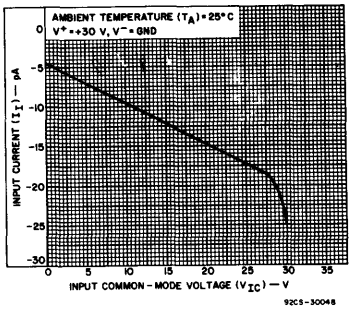


Fig. 4 – Input current as a function of input common-mode voltage.

ELECTRICAL CHARACTERISTICS AT T_A = 25°C

CHARACTERISTIC	TEST COND. V^+	LIMITS CA3290B Min.	CA3290B Typ.	CA3290B Max.	CA3290A Min.	CA3290A Typ.	CA3290A Max.	UNITS
Input Offset Voltage, V_{IO} V_{IC}=1.4 V, V_O=1.4 V	5 V	–	3	6	–	4	10	mV
V_{IC}=0 V, V_O=0 V	±15 V	–	3	6	–	4	10	mV
Input Current, I_I V_{IC}=1.4 V	5 V	–	3.5	30	–	3.5	40	pA
V_{IC}=0 V	±15 V	–	12	30	–	12	40	pA
Input Offset Current, I_{IO} V_{IC}=1.4 V	5 V	–	2	20	–	2	25	pA
V_{IC}=0 V	±15 V	–	7	20	–	7	25	pA
Common-Mode Input-Voltage Range, V_{ICR} V_O=1.4 V	5 V	V^+–3.5 V^-	V^+–3.1 V^-–1.5	–	V^+–3.5 V^-	V^+–3.1 V^-–1.5	–	V
V_O=0 V	±15 V	V^+–3.8 V^-	V^+–3.4 V^-–1.6	–	V^+–3.8 V^-	V^+–3.4 V^-–1.6	–	V
Supply Current, I^+ R_L = ∞	30 V	–	1.35	3	–	1.35	3	mA
	5 V	–	0.8	1.4	–	0.8	1.4	mA
Voltage Gain, A_{OL} R_L=15 kΩ	±15 V	50	800	–	25	800	–	V/mV
		94	118	–	88	118	–	dB
Output Sink Current V_O=1.4 V	5 V	6	30	–	6	30	–	mA
Saturation Voltage $+V_I$=0 V, $-V_I$=1 V, 4 mA	5 V	–	0.12	0.4	–	0.12	0.4	V
Output Leakage Current, I_{OL}	15 V	–	100	–	–	100	–	pA
	36 V	–	500	–	–	500	–	pA
Response Time R_L=5.1 kΩ Rising Edge	15 V	–	1.2	–	–	1.2	–	μs
Falling Edge		–	200	–	–	200	–	ns
Common-Mode Rejection Ratio, CMRR	±15 V	–	44	316	–	44	562	μV/V
	5 V	–	100	316	–	100	562	μV/V
Power-Supply Rejection Ratio, PSRR	±15 V	–	15	316	–	15	316	μV/V
Large-Signal Response Time R_L=5.1 kΩ	15 V	–	500	–	–	500	–	ns
	5 V	–	400	–	–	400	–	ns

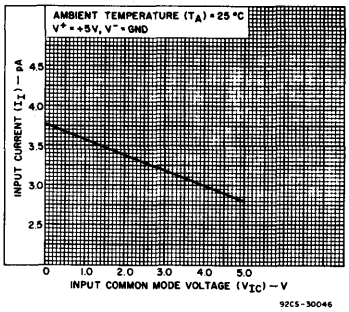


Fig. 5 – Input current as a function of input common-mode voltage.

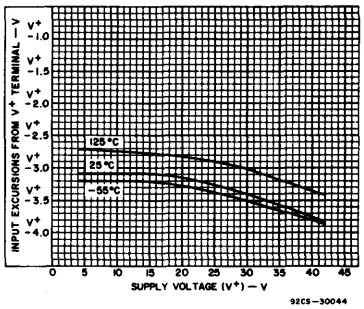


Fig. 6 – Positive common-mode input voltage range as a function of supply voltage.

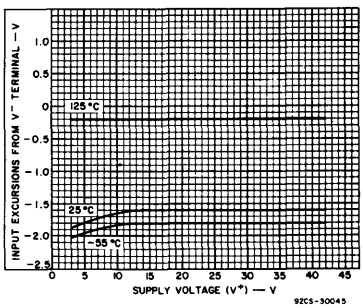


Fig. 7 – Negative common-mode input voltage range as a function of supply voltage.

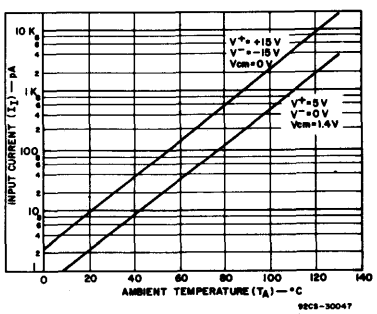


Fig. 8 – Input current as a function of ambient temperature.

ELECTRICAL CHARACTERISTICS AT $T_A = 25^oC$

CHARACTERISTIC	TEST COND. V^+	LIMITS CA3290 Min.	LIMITS CA3290 Typ.	LIMITS CA3290 Max.	UNITS
Input Offset Voltage, V_{IO} V_{IC}=1.4 V, V_O=1.4 V	5 V	–	7.5	20	mV
V_{IC}=0 V, V_O=0 V	±15 V	–	7.5	20	
Input Current, I_I V_{IC}=1.4 V	5 V	–	3.5	50	pA
V_{IC}=0 V	±15 V	–	12	50	
Input Offset Current, I_{IO} V_{IC}=1.4 V	5 V	–	2	30	pA
V_{IC}=0 V	±15 V	–	7	30	
Common-Mode Input-Voltage Range, V_{ICR} V_O=1.4 V	5 V	V^+–3.5 V^-	V^+–3.1 V^-–1.5	–	V
V_O=0 V	±15 V	V^+–3.8 V^-	V^+–3.4 V^-–1.6	–	
Supply Current, I^+ $R_L = \infty$	30 V	–	1.35	3	mA
	5 V	–	0.8	1.4	
Voltage Gain, A_{OL} R_L = 15 kΩ	±15 V	25	800	–	V/mV
		88	118	–	dB
Output Sink Current V_O=1.4 V	5 V	6	30	–	mA
Saturation Voltage $+V_I$=0 V, $-V_I$=1 V, 4 mA	5 V	–	0.12	0.4	V
Output Leakage Current, I_{OL}	15 V	–	100	–	pA
	36 V	–	500	–	
Response Time R_L=5.1 kΩ Rising Edge	15 V	–	1.2	–	μs
Falling Edge		–	200	–	ns
Common-Mode Rejection Ratio, CMRR	±15 V	–	44	562	μV/V
	5 V	–	100	562	
Power-Supply Rejection Ratio, PSRR	±15 V	–	15	316	μV/V
Large-Signal Response Time R_L=5.1 kΩ	15 V	–	500	–	ns
	5 V	–	400	–	

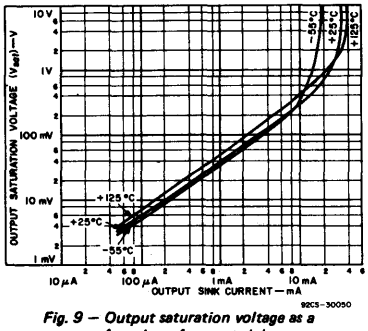
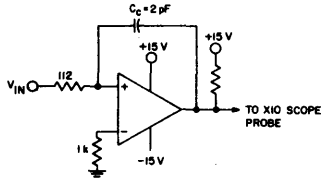
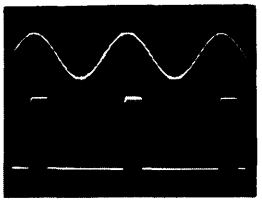


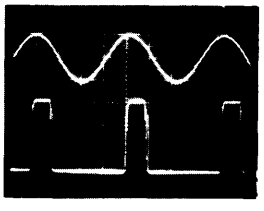
Fig. 9 – Output saturation voltage as a function of output sink current.





WITH C_c

TOP TRACE ≈ 4.5 mV/DIV = V_{IN}
BOTTOM TRACE = 10 V/DIV = V_{OUT}
H = 5 μs/DIV



WITHOUT C_c

TOP TRACE ≈ 4.5 mV/DIV
BOTTOM TRACE = 10 V/DIV
H = 5 μs/DIV

92CM-30059

Fig. 10 – Parasitic-oscillations test circuit and associated waveforms.

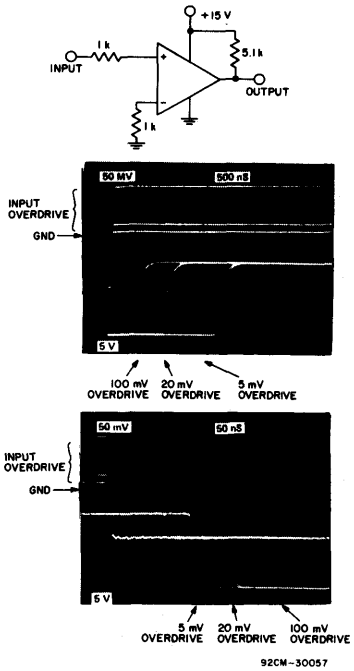


Fig. 11 – Non-inverting comparator response-time test circuit and waveforms.

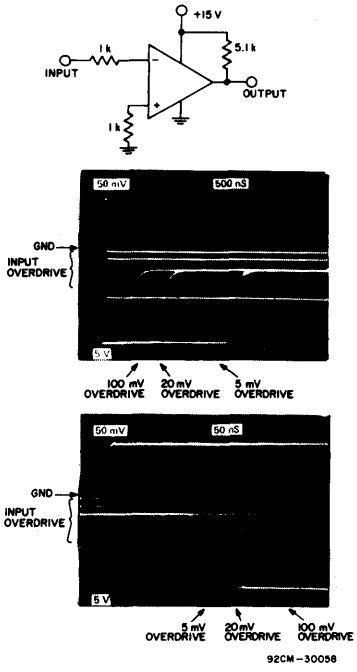


Fig. 12 – Inverting comparator response-time test circuit and waveforms.

OPERATING CONSIDERATIONS

Input Circuit

The use of MOS transistors in the input stage of the CA3290 series circuits provides the user with the following features for comparator applications:

1. Ultra-high input impedance (≅ 1.7 TΩ);
2. The availability of common-mode rejection for input signals at potentials below that of the negative power-supply rail;
3. Retention of the in-phase relationship of the input and output signals for input signals below the negative rail.

Although the CA3290 employs rugged bipolar (zener) diodes for protection of the input circuit, the input-terminal currents should not exceed 1 mA. Appropriate series-connected limiting resistors should be used in circuits where greater current flows might exist, allowing the signal input voltage to be greater than the supply voltage without damaging the circuit.

Output Circuit

The output of the CA3290 is the open collector of an n-p-n transistor, a feature providing flexibility in a broad range of comparator applications. An output ORing function can be implemented by parallel-connection of the open collectors. An output pull-up resistor can be connected to a power supply having a voltage range within the rating of the particular CA3290 in use; the magnitude of this voltage may be set at a value which is independent of that applied to the V^+ terminal of the CA3290.

Parasitic Oscillations

The ideal comparator has, among other features, ultra-high input impedance, high gain, and wide bandwidth. These desirable characteristics may, however, produce parasitic oscillations unless certain precautions are observed to minimize the stray capacitive coupling between the input and output terminals. Parasitic oscillations manifest themselves during the output voltage transition intervals as the comparator switches states. For high source impedances, stray capacitance can induce parasitic oscillations. The addition of a small amount (1 to 10 mV) of positive feedback (hysteresis) produces a faster transition, thereby reducing the likelihood of parasitic oscillations. Furthermore, if the input signal is a pulse waveform, with relatively rapid rise and fall times, parasitic tendencies are reduced.

When dual comparators, like the CA3290, are packaged in an 8-lead configuration, the output terminal of each comparator is adjacent to an input terminal. The lead-to-lead capacitance is approximately 1 pF, which may be sufficient to cause undesirable feedback effects in certain applications. Circuit factors such as impedance levels, supply voltage, toggling rate, etc., may increase the possibility of parasitic oscillations. To minimize this potential oscillatory condition, it is recommended that for source impedances greater than 1 kΩ a capacitor (≥ 1-2 pF) be connected between the appropriate input terminal and the output terminal. (See Fig. 10.)

The CA3290A and CA3290 are also supplied in a 14-lead dual-in-line plastic package. To minimize the possibility of parasitic oscillations the input and output terminals are positioned on opposite sides of the package. In addition, there are two leads between the output terminal of each comparator and its corresponding inverting input terminal, reducing the input/output coupling significantly. These leads (8, 9, 13, 14) should be tied to either the V^+ or V^- supply rail. If either comparator is unused, its input terminals should also be tied to either the V^+ or V^- supply rail.

TYPICAL APPLICATIONS

Light-Controlled One-Shot Timer

In Fig. 13 one comparator (A1) of the CA3290 is used to sense a change in photo diode current. The other comparator (A2) is configured as a one-shot timer and is triggered by the output of A1. The output of the circuit will switch to a low state for approximately 60 seconds after the light source to the photo diode has been interrupted. The circuit operates at normal room lighting levels. The sensitivity of the circuit may be adjusted by changing the values of R1 and R2. The ratio of R1 to R2 should be constant to insure constant reverse voltage bias on the photo diode.

Low-Frequency Multivibrator

In this application, one-half of the CA3290 is used as a conventional multivibrator circuit. Because of the extremely high input impedance of this device, large values of timing resistor (R1) may be used for long time delays with relatively small leakage timing capacitors. The second half of the CA3290 is used as an output buffer to insure that the multivibrator frequency will not be affected by output loading.

Window Comparator

Both halves of the CA3290 can be used in a high input-impedance window comparator as shown in Fig. 15. The LED will be turned "on" whenever the input signal is above the lower limit (V_L) but below the upper limit (V_U), as determined by the R1/R2/R3 resistor divider.

LED Bar Graph Driver

The circuit in Fig. 16 demonstrates the use of the CA3290 in a bar graph display. The non-inverting inputs of both comparators are tied to the voltage divider reference and the input signal is applied to both of the inverting inputs. The LED for a particular comparator will be turned "on" when the input voltage reaches the voltage on the resistor divider reference. The CA3290 is ideal for this application where input-signal loading is critical even though many comparator inputs are driven in parallel.

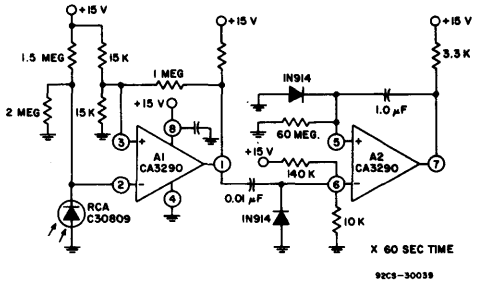


Fig. 13 – Light-controlled one-shot timer.

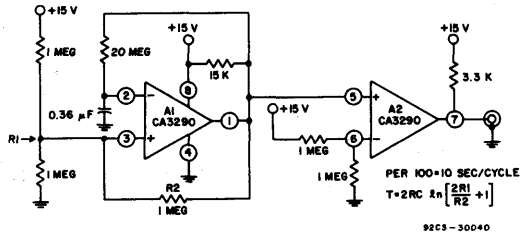


Fig. 14 – Low-frequency multivibrator.

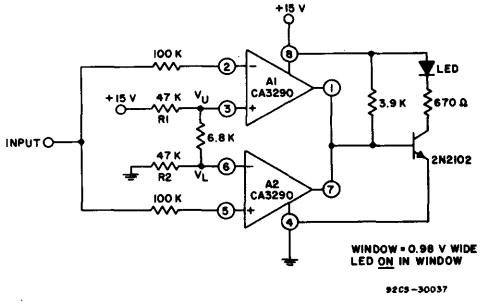


Fig. 15 – Window comparator.

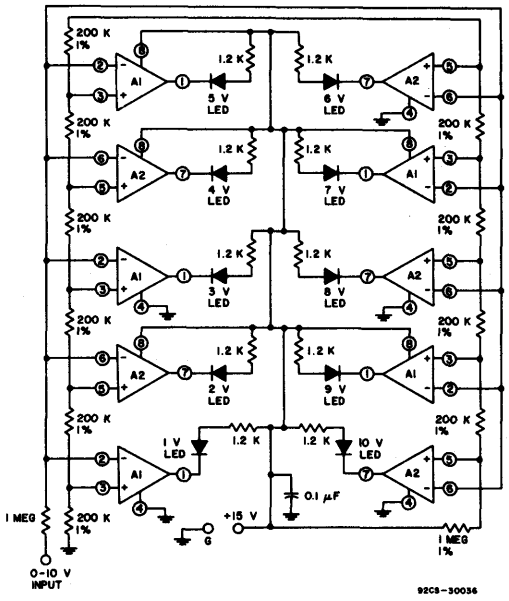
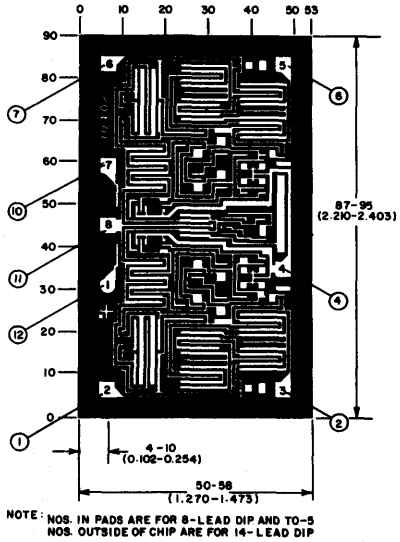


Fig. 16 – 1 LED bar-graph driver.



Dimensions and pad layout for the CA3290H.

The photographs and dimensions of each COS/MOS chip represent a chip when it is part of the wafer. When the wafer is cut into chips, the cleavage angles are 57° instead of 90° with respect to the face of the chip. Therefore, the isolated chip is actually 7 mils (0.17 mm) larger in both dimensions.

Dimensions in parentheses are in millimeters and are derived from the basic inch dimensions as indicated. Grid graduations are in mils (10^{-3} inch).

CA3401E, CA3401G

Quad Single-Supply Operational Amplifier

For Automotive Electronics and Industrial Control Systems

"G" Suffix Types – Hermetic Gold-CHIP in Dual-In-Line Plastic Package

"E" Suffix Types – Standard Dual-In-Line Plastic Package

The RCA-CA3401 is a high-gain monolithic quad operational amplifier designed specifically for applications using a single positive power supply. No external compensation is necessary. Closed-loop stability in each of the four independent amplifiers is maintained by a 3-pF on-chip capacitor. The CA3401 is ideally suited for applications in industrial control systems, automotive electronics, and general purpose amplifiers, e.g. oscillators, tachometers, active filters, and multichannel amplifiers.

The CA3401 is supplied in a 14-lead dual-in-line plastic package (E suffix), a hermetic gold-chip in 14-lead dual-in-line plastic package (G suffix), in chip form (H suffix), and as a hermetic gold-chip (HG suffix). It is a direct replacement for the Motorola MC3401P, and is pin-compatible with the Motorola MC3301P and the National Semiconductor LM3900N. The CA3401 can be operated over the temperature range of −55 to +125°C, although the limit values of certain specified electrical characteristics apply only over the range of 0 to +75°C.

Features:

- **Single-supply operation – +5 V to +18 Vdc**
- **Internally compensated**
- **Wide unity-gain bandwidth – 5 MHz typ.**
- **Low input bias current – 50 nA typ.**
- **High open-loop gain – 2000 V/V typ.**

Applications:

- **Automotive**
- **Constant-Current Sources**
- **Multivibrators**
- **Sample and Hold**
- **Square-Wave Generator**
- **Oscillators**
- **Tachometers**
- **Active Filters**
- **Multi-Channel Amplifiers**
- **Summing Amplifiers**

MAXIMUM RATINGS, *Absolute-Maximum Values at T_A = 25°C*

DC SUPPLY VOLTAGE	+18 V
INPUT SIGNAL CURRENT	5 mA
DEVICE DISSIPATION:	
Up to T_A = 25°C	625 mW
Above T_A = 25°C	Derate linearly 5 mW/°C
AMBIENT TEMPERATURE RANGE:	
Operating	−55 to + 125°C
Storage	−65 to + 150°C
LEAD TEMPERATURE (During soldering):	
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max.	300 °C

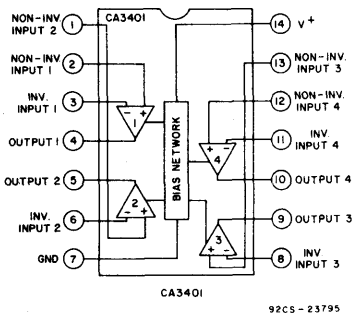


Fig.1 – Block diagram of CA3401.

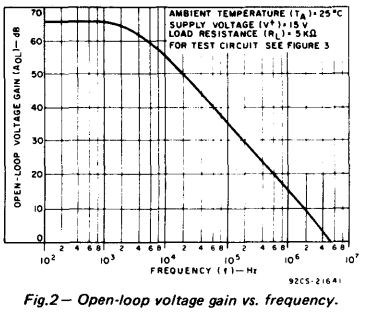


Fig.2 – Open-loop voltage gain vs. frequency.

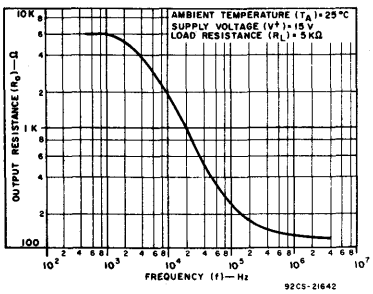


Fig.3 – Output resistance vs. frequency.

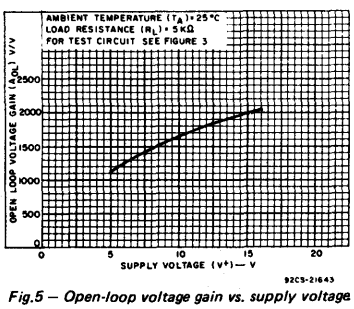


Fig.5 – Open-loop voltage gain vs. supply voltage

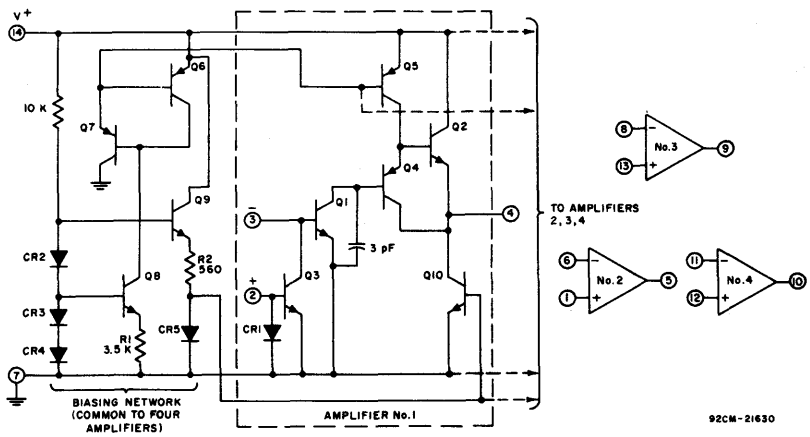


Fig.4 – Schematic diagram of CA3401.

ELECTRICAL CHARACTERISTICS AT T_A = 25°C, V^+ = 15 V (Unless Indicated Otherwise)

CHARACTERISTIC	TEST CONDITIONS	LIMITS			UNITS
		Min.	Typ.	Max.	
STATIC					
Output Voltage:					
High, V_{OH}		13.5	14.2	–	V
Low, V_{OL}		–	0.03	0.1	
Max. Undistorted Output Swing, V_{OP-P}	0°C<T_A<75°C	10	13.5	–	
Output Current:					
Source, I_{SOURCE}		5	10	–	mA
Sink, I_{SINK}		0.5	1	–	
Total Quiescent Current: I_Q					
Noninverting inputs open		–	6.9	10	mA
Noninverting inputs grounded		–	7.8	14	
Input Bias Current, I_{IB}	R_L = ∞ T_A = 25°C	–	50	300	nA
	R_L = ∞ 0°C ≤T_A≤75°C	–	–	500	
DYNAMIC					
Open-Loop Voltage Gain, A_{OL}	T_A = 25°C	1000	2000	–	V/V
	0°C≤T_A≤75°C	800	–	–	
Input Resistance, R_I		0.1	1	–	MΩ
Slew Rate, SR	C_L = 100 pF, R_L = 5 kΩ	–	0.6	–	V/μs
Unity Gain Gandwidth, BW		–	5	–	MHz
Phase Margin, ϕ		–	70	–	Degrees
Power Supply Rejection	f = 100 Hz	–	55	–	dB
Channel Separation, e_{01}/e_{02}	f = 1 kHz	–	65	–	dB

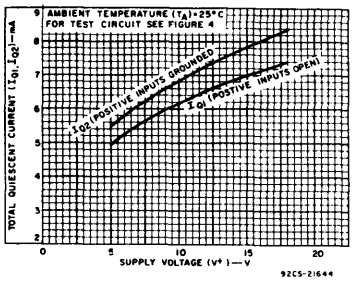


Fig.10 – Supply current vs. supply voltage.

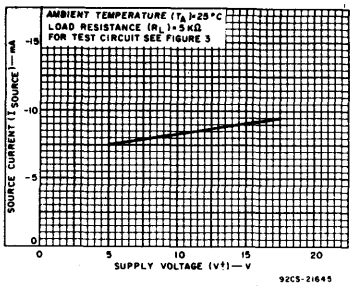


Fig.11 – Source current vs. supply voltage.

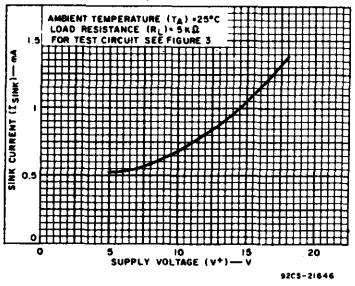


Fig.12 – Sink current vs. supply voltage.

TEST CIRCUITS

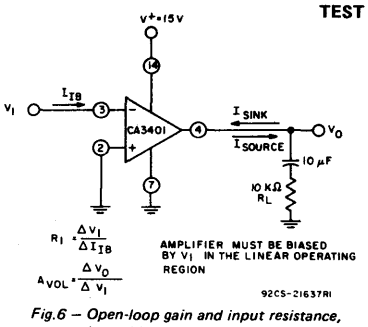


Fig.6 – Open-loop gain and input resistance, input bias current and output current test circuit.

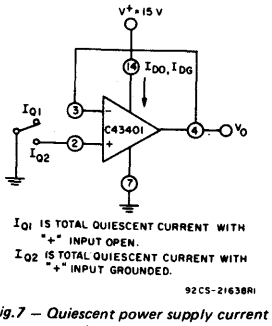


Fig.7 – Quiescent power supply current test circuit.

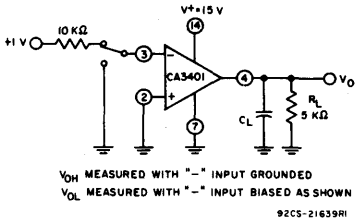


Fig.8 – Output voltage swing test circuit.

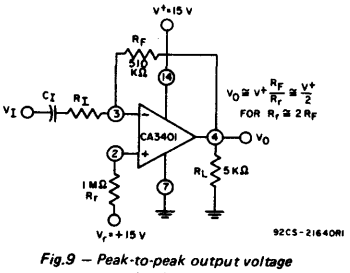


Fig.9 – Peak-to-peak output voltage test circuit.

CA3600E

COS/MOS Transistor Array

For Linear Circuit Applications

RCA-CA3600E is an array of COmplementary-Symmetry MOS Field-Effect Transistors* on a monolithic silicon substrate. It is comprised of three n-channel and three p-channel enhancement-type MOS transistors arrayed as shown in Fig. 1, and specified and tested for linear circuit operation. These transistors are uniquely suitable for service in complementary-symmetry circuits at supply voltages in the range of 3 to 15 volts and are useful at frequencies up to 5 MHz (untuned). Each transistor in the CA3600E can conduct currents up to 10 mA. This device is supplied in the 14-lead dual-in-line plastic package.

Formerly RCA Dev. No. TA6388.

* The theory and construction of COS/MOS transistors are described in the "RCA COS/MOS Integrated Circuits Manual," RCA Solid State Division Technical Series Publication No. CMS-271.

MAXIMUM RATINGS, *Absolute-Maximum Values at* $T_A = 25^oC$

DISSIPATION:

Any one transistor at T_A up to 55°C	150 mW
Total package at T_A up to 55°C	750 mW
Above T_A = 55°C	derate linearly 6.67 mW/°C

AMBIENT TEMPERATURE RANGE:

Operating	−55 to +125°C
Storage	−65 to +150°C

LEAD TEMPERATURE (During Soldering)

At distance not less than 1/16" ± 1/32" (1.59 ± 0.79 mm) from case for 10 s max.	265°C

The Following Ratings Apply for Each Transistor in the Device:

DRAIN-TO-SOURCE VOLTAGE, V_{DS}:	
n-channel	+15 V
p-channel	−15 V
DRAIN-TO-GATE VOLTAGE, V_{DG}:	
n-channel	+15 V
p-channel	−15 V
SOURCE-TO-SUBSTRATE VOLTAGE, V_{SB}:	
n-channel	+15 V
p-channel	−15 V
GATE-TO-SOURCE VOLTAGE, V_{GS}:	
p-channel transistors (p_1, p_2, p_3)	0 V(min.), $-V_D$(max.)
n-channel transistors (n_1, n_2, n_3)	0 V(min.), $+V_D$(max.)
COS/MOS transistor-pairs (p_1-n_1, p_2-n_2, p_3-n_3)	0 V(min.), $+V_{DD}$(max.)
DRAIN CURRENT, $\lvert I_D \rvert$	10 mA
GATE CURRENT, $\lvert I_G \rvert$	100 μA

The Following Rating Applies for Each COS/MOS Transistor-Pair in the Device:

DC SUPPLY VOLTAGE ($V_{DD} - V_{SS}$)	+15 V

Rules for Maintaining Electrical Isolation Between Transistors and Monolithic Substrate

Terminal No. 14 must be maintained at the most positive potential (or equally positive potential) with respect to any other terminal in the CA3600E.

Terminal No. 7 must be maintained at the most negative potential (or equally negative potential) with respect to any other terminal in the CA3600E.

Violation of these rules will result in improper transistor operation, circuit "latching," and/or possible permanent damage to the CA3600E.

Note: Users should observe the "Considerations in Handling CA3600E Devices"

Features:

- **High input resistance** . . . . . **100 GΩ (typ.)**
- **Low gate-terminal current** . . . . **10 pA (typ.)**
- **Matched p-channel pair:**
  Gate-voltage differential (I_D = -100 μA) ±20 mV (max.)
- **No "Popcorn" (burst) noise**
- **Stable transfer characteristics over an operating temperature range of −55°C to +125°C when operated in complementary circuit configuration at supply voltages in the 5 to 15 volt range (see Fig. 14)**
- **Integrated integral gate-protection system (see Fig. 34)**
- **High voltage gain (see Fig. 11)** . . . **up to 53 dB (typ.) per COS/MOS stage**
- **Individual MOS transistors have square-law characteristics, superior cross-modulation performance, and greater dynamic range than bipolar transistors**

Applications:

- **High input impedance, general-purpose amplifiers**
- **Pre-amplifiers**
- **Differential amplifiers**
- **Op-amps and comparators**
- **Constant-current sources and current mirrors**
- **Micropower amplifiers and oscillators**
- **Control of lamps, LED's, relays, and thyristors**
- **Timers**
- **Choppers**
- **Mixers**

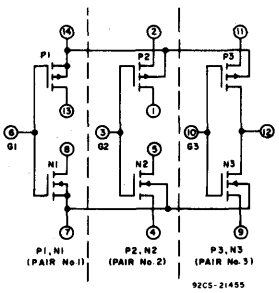


Fig. 1 – Schematic diagram for CA3600E COS/MOS transistor array.

1. Drain terminal, p-channel of pair no. 2
2. Source terminal, p-channel of pair no. 2
3. Common gate terminal of pair no. 2
4. Source terminal, n-channel of pair no. 2
5. Drain terminal, n-channel of pair no. 2
6. Common gate terminal of pair no. 1
7. Source terminal, n-channel of pair no. 1 and substrate connection for all n-channel transistors · · · V_{SS} terminal
8. Drain terminal, n-channel of pair no. 1
9. Source terminal, n-channel of pair no. 3
10. Common gate terminal of pair no. 3
11. Source terminal, p-channel of pair no. 3
12. Common drain terminal of pair no. 3
13. Drain terminal, p-channel of pair no. 1
14. Source terminal, p-channel of pair no. 1 and substrate connection for all p-channel transistors · · · V_{DD} terminal

Terminal Identification for Fig. 1.

TYPICAL CHARACTERISTICS CURVES

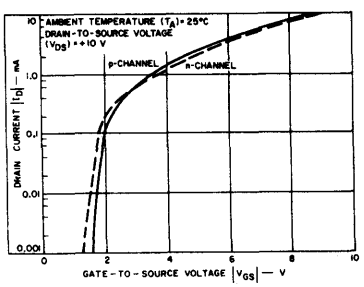


Fig. 2–Drain current vs. gate-to-source voltage.

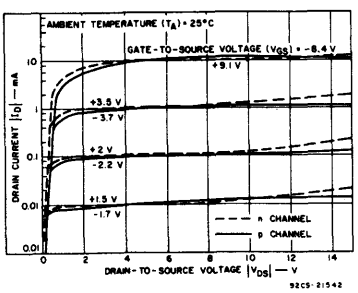


Fig. 3– Drain current vs. drain-to-source voltage.

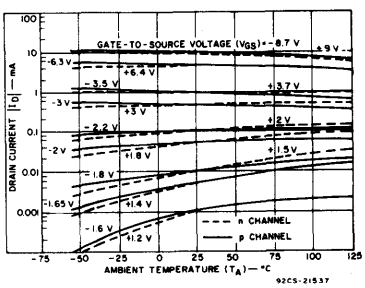


Fig. 4– Drain current vs. ambient temperature.

ELECTRICAL CHARACTERISTICS, *At T_A = 25°C*

CHARACTERISTIC	SYMBOL	TEST CONDITIONS	TYPICAL CURVE OR CIRCUIT FIG. NO.	LIMITS			UNIT
				Min.	Typ.	Max.	
For Each p-Channel MOS Transistor							
Drain Current	I_D	V_{DS}=−10 V,V_{GS}=−3.6 V	2,3,4	−0.5	−1.1	−2.0	mA
Gate-to-Source Threshold Voltage	$V_{GS(th)}$	I_D=−10 μA	–	–	−1.75	–	V
Gate-to-Source Voltage Differential (p_1 vs. p_2)	$\lvert V_{GS1}-V_{GS2}\rvert$	I_D=−100 μA,V_{DS}=−10 V	5	–	±4	±20	mV
Forward Transconductance	g_{fs}	I_D=−1 mA,f=1 kHz	6	–	920	–	μmho
Low-Frequency Noise Voltage	e_N	I_D=−1 mA,f=1 kHz,R_s=0 Ω	7	–	0,03	–	μV $\sqrt{Hz}$
Low-Frequency Noise Current	i_N	I_D=−1 mA,f=1 kHz,R_s=1 MΩ	7	–	0.2	–	pA $\sqrt{Hz}$
Current-Mirror Transfer Ratio (p_1/p_2)	I_{MTR}	I_1=−100 μA,V_{DS}=−10 V	30	0.7	1.1	1.5	–
Gate-Terminal Current	I_{GT}	V_{DS}=−10 V,V_{GS}=−3.5 V	–	–	±0.015	−40	nA
Input Capacitance	C_I	–	–	–	6.3	–	pF
Output Capacitance	C_O	–	–	–	3	–	pF
Input-to-Output Capacitance	$C_{I\text{-}O}$	–	–	–	0.75	–	pF
For Each n-Channel MOS Transistor							
Drain Current	I_D	V_{DS}=+10 V,V_{GS}=+3.6 V	2,3,4	0.4	0.9	1.6	mA
Gate-to-Source Threshold Voltage	$V_{GS(th)}$	I_D=10 μA	–	–	1.5	–	V
Gate-to-Source Voltage Differential (n_1 vs n_2)	$\lvert V_{GS1}-V_{GS2}\rvert$	I_D=100 μA,V_{DS}=+10 V	5	–	±30	–	mV
Forward Transconductance	g_{fs}	I_D=1 mA,f=1 kHz	6	–	860	–	μmho
Low-Frequency Noise Voltage	e_N	I_D=1 mA,f=1 kHz,R_s=0 Ω	7	–	0,2	–	μV $\sqrt{Hz}$
Low-Frequency Noise Current	i_N	I_D=1 mA,f=1 kHz,R_s=1 MΩ	7	–	0,3	–	pA $\sqrt{Hz}$
Current-Mirror Transfer Ratio (n_1/n_2)	I_{MTR}	I_1=100 μA,V_{DS}=+10 V	29	0.7	1.3	2.0	–
Gate-Terminal Current	I_{GT}	V_{DS}=+10 V,V_{GS}=+3.7 V	–	–	±0.01	+40	nA
Input Capacitance	C_I	–	–	–	5.5	–	pF
Output Capacitance	C_O	–	–	–	2.0	–	pF
Input-to-Output Capacitance	$C_{I\text{-}O}$	–	–	–	0.35	–	pF
For Each COS/MOS Transistor Pair							
Drain Current	I_{DD}	V_{DD}=+10 V	9,10	1.0	2.2	4.0	mA
Drain-to-Source Cutoff Current	$I_{DD(off)}$	V_{DD}=+10 V,V_{SS}=0 V Gate Voltage(V_G)=+10 V or 0 V	8	–	0.5	100	nA
DC Output Voltage	V_O	V_{DD}=+10 V	10	4.2	5.0	5.8	V
Forward Transconductance	g_{fs}	V_{DD}=+10 V, f = 1 kHz	6	–	2300	–	μmho
Slew Rate (Open-Loop)	SR	V_{DD}=+15 V	10	–	95	–	V/μs
Amplifier Voltage Gain	A_{OL}	V_{DD}=+10 V,f=1 kHz,R_b=22 MΩ R_s=50 Ω	10,11	–	32	–	dB
Gate-Terminal Current	I_{GT}	V_{DD}=+10 V	10	–	±0.005	±20	nA
Broadband Output Noise Voltage	E_{ON}	V_{DD}=+10 V,R_b=22 MΩ,R_s=10 kΩ	10,11	–	500		μV
Input Capacitance	C_I	–	–	–	11.8		pF
Output Capacitance	C_O	–	–	–	5.0		pF
Input-to-Output Capacitance	$C_{I\text{-}O}$	–	–	–	1.1		pF

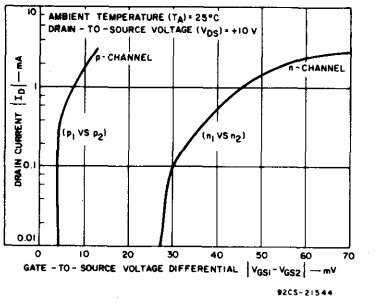


Fig. 5– Gate-to-source voltage differential vs. drain current.

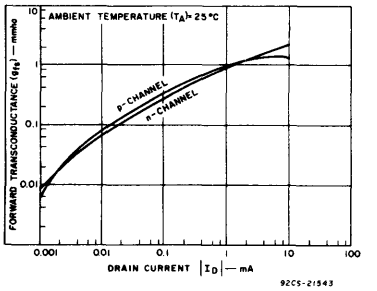


Fig. 6– Forward transconductance vs. drain current.

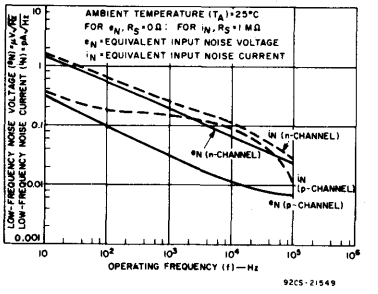


Fig. 7– Noise voltage and noise current vs. operating frequency.

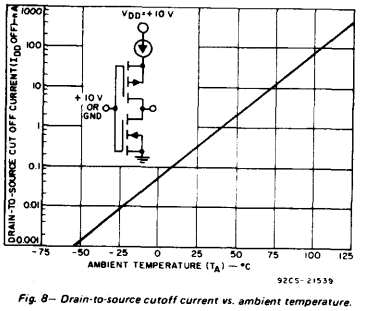


Fig. 8– Drain-to-source cutoff current vs. ambient temperature.

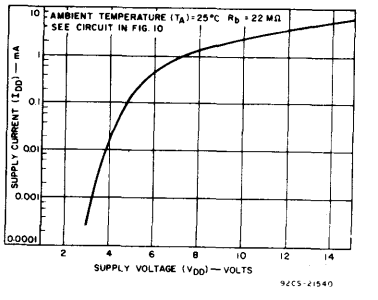


Fig.9 – Typical V_{DD} vs. I_{DD} characteristics for amplifier circuits of Fig.10 and Fig.15.

APPLICATIONS

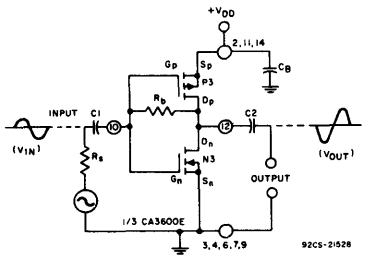


Fig. 10– COS/MOS transistor-pair biased for linear-mode operation.

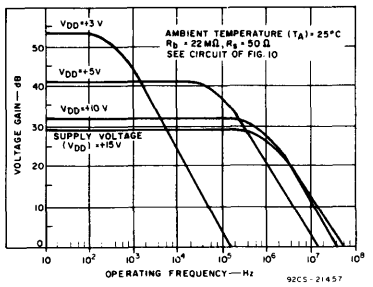


Fig. 11– Typical voltage gain vs. frequency characteristics for amplifier circuit of Fig. 10.

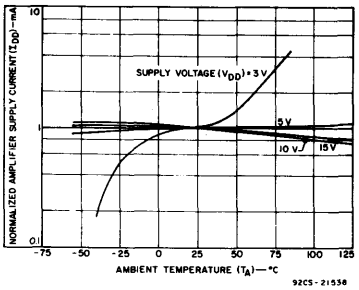


Fig. 12– Normalized amplifier supply current vs. ambient temperature characteristics for amplifier circuit of Fig. 10.

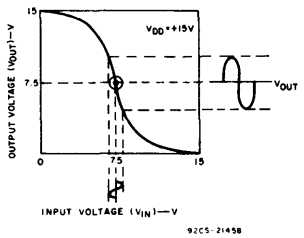


Fig. 13 – Representation of voltage-transfer characteristics for COS/MOS transistor pair.

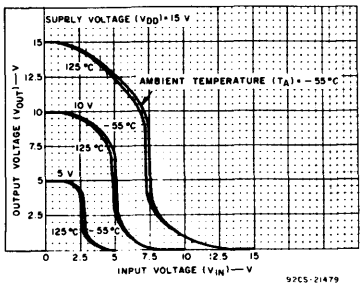


Fig. 14– Voltage transfer characteristics for COS/MOS transistor-pair amplifier in Fig. 10.

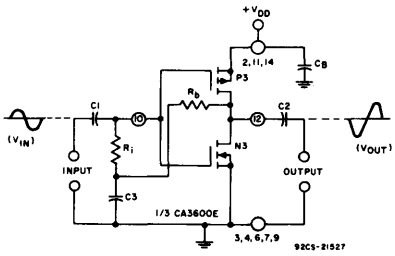


Fig. 15– Alternate method of biasing COS/MOS transistor-pair for linear-mode operation.

The Basic COS/MOS Linear Amplifier

P-n-p and n-p-n bipolar transistors have been used for many years in the design of so-called "true-complementary" linear amplifier circuits . Since mutually compatible p-channel and n-channel MOS/FET devices were not generally available, "true-complementary" amplifier circuits using MOS transistors were seldom used. Now, COS/MOS transistor technology[5] has made it possible to supply compatible p-channel/n-channel transistors in monolithic IC form such as the CA3600E COS/MOS transistor array shown in Fig. 1.

A "True-Complementary" Linear Amplifier Using COS/MOS Transistors

Fig. 10 shows the schematic diagram of a single-stage "true-complementary" linear amplifier using one pair of the complementary MOS transistors in the CA3600E, connected in a common-source circuit. Resistor R_b is used to bias the complementary pair for Class A operation, as described subsequently, and R_s represents the source resistance of the signal source. This generic amplifier is suitable for operation with a single or split voltage supply in the range of 3 to 15 volts. Fig. 11 shows voltage gain as a function of operating frequency at various supply voltages for the single-stage amplifier. This amplifier is capable of producing very high output-swing voltages (V_{out}); for example, its output voltages can be swung to within several millivolts of either supply-voltage "rail". Fig. 9 shows typical supply voltage (V_{DD}) vs. supply current (I_{DD}) characteristics for the single-stage amplifier. The curves in Fig. 12 show the normalized amplifier supply current as a function of ambient temperature at various supply voltages. When the amplifier is operating at V_{DD} = 3 V, the supply current changes rapidly as a function of temperature because the MOS transistors are operating in the proximity of their individual gate-source threshold voltages.

Voltage-Transfer Characteristics

Fig. 13 illustrates a voltage-transfer characteristic curve of a COS/MOS transistor pair connected in the amplifier circuit of Fig. 10, with a biasing resistor (R_b) connected between the drain and gate terminals (10,12). If the p- and n-channel transistors have identical characteristics, their channel resistances are equal, and the biasing method shown establishes a steady-state condition such that terminal 12 is at mid-potential between V_{DD} and ground. Thus, with negligibly small gate-source leakage resistances, under zero-signal conditions, the biasing resistor (R_b) establishes gate potential at the mid-point between V_{DD} and ground, i.e., $V_{in} = V_{out}$. Under these conditions the amplifier is biased for operation about the mid-point ("0") in the linear segment on the steep transition of the voltage-transfer characteristic as shown in Fig. 13. When the input signal (V_{in}) swings in the positive direction, there is a reduction in the instantaneous output voltage (V_{out}) with respect to ground. Negative-going input signals have inverse effects. Thus, phase-inversion occurs in the COS/MOS-pair amplifier. Power-supply current is constant during dynamic linear operation, i.e., Class A amplifier service. When the signal input-voltage level (V_{in}) becomes very large, the output signal (V_{out}) waveforms become distorted because the transistors are driven into the non-linear portions of their voltage-transfer characteristics. If the positive-going input-signal is sufficiently large, for example, the p-channel transistor can be driven to cutoff and the amplifier supply current (I_{DD}) is reduced to essentially zero.

Fig. 14 shows typical voltage-transfer characteristics of each COS/MOS pair in the CA3600E at several values of V_{DD}. The shape of these transfer characteristics is comparatively constant despite temperature changes from −55 to +125°C.

The biasing arrangement used in the circuit of Fig. 10 provides an easy method of establishing feedback for ac signals in accordance with the R_b/R_s ratio. When the feedback of ac signals is not desirable, the circuit of Fig. 15 may be used. The ac bypass capacitor (C_3) minimizes ac signal feedback.

Cascading Amplifier Stages of COS/MOS Transistor Pairs

Ultra-high-gain amplifiers can be designed by cascading stages of COS/MOS transistor pairs as shown in Fig. 16. The biasing system used is similar to that described above in connection with Fig. 10. The supply current for the three-stage amplifier shown in Fig. 16 is typically three times the values shown in Fig. 9. Gain and frequency-response characteristics of the amplifier are shown in Fig. 17.

Post-Amplifiers For Op-Amps

COS/MOS transistor-pairs can be advantageously applied as post-amplifiers for op-amps. Because the input impedance of the COS/MOS pair is comparatively high, the op-amp operates under essentially unloaded conditions. Each COS/MOS pair can sink and source output current up to about 10 mA. Additionally, the op-amp output can be directly coupled to bias the COS/MOS pair. A detailed description of the subject has been published previously.

The schematic diagram in Fig. 18 shows a COS/MOS transistor-pair serving as a post-amplifier to an RCA-CA3080 Operational Transconductance Amplifier. The approximate 30-dB gain in a single COS/MOS transistor-pair is an added increment to the 100-dB gain in the CA3080, yielding a total forward gain of about 130 dB. The open-loop slew rate of the circuit in Fig. 19 is approximately 65 V/μs. When compensated for the unity-gain voltage-follower mode shown in Fig. 19, the slew rate is about 1 V/μs. For greater current output, the two remaining transistor pairs of the CA3600E may be connected in parallel with the single stages shown in Figs. 18 and 19.

The use of the two-stage COS/MOS post-amplifier shown in Fig. 20 increases the total open-loop gain of the system to about 160 dB (100,000,000X). Open-loop slew rate remains at about 65 V/μs. A slew rate of about 1 V/μs is maintained with this circuit connected in the unity-gain voltage-follower mode, as shown in Fig. 21. These circuits operate in concert with stability.

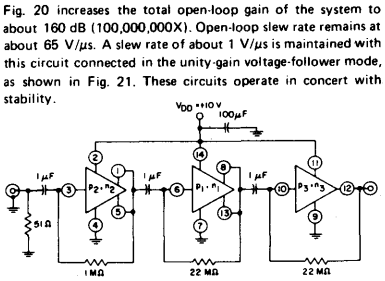


Fig. 16– High-gain amplifier uses cascaded COS/MOS transistor-pair in CA3600E.

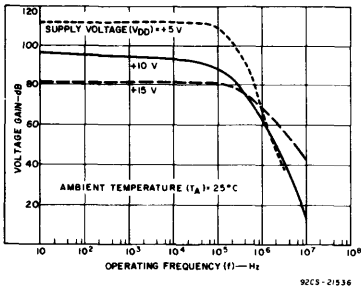


Fig. 17– Typical voltage gain vs. operating frequency characteristics for three-stage COS/MOS transistor-pair amplifier in Fig. 16.

APPLICATIONS – Post-Amplifiers for Op-Amps (Cont'd)

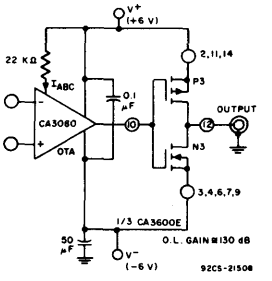


Fig. 18– COS/MOS transistor-pair used as post-amplifier to op-amp in open-loop circuit.

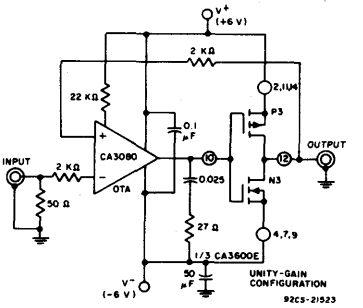


Fig. 19– COS/MOS transistor-pair used as post-amplifier to op-amp in unity-gain circuit.

Multivibrators, Threshold Detectors, and Comparators

Descriptions of several circuits using COS/MOS transistor-pairs in both monostable and astable multivibrators have been published. The characteristics of COS/MOS pairs are also ideal for mating with micropower op-amps in circuits such as the precision multistable circuits shown in Fig. 22. In these circuits precise timing and thresholds are assured by the stable characteristics of the input differential amplifier in the CA3080 Operational Transconductance Amplifier. Moreover, speed vs. power consumption tradeoffs can be made by adjustment of the Amplifier-Bias-Current (I_{ABC}) supplied to terminal 5 of the CA3080. The quiescent power consumption of the circuits shown in Fig. 22 is typically 6 mW, but can be made to operate in the micropower region by suitable modifications.

The schematic diagram of a programmable micropower comparator, shown in Fig. 23 employs the combination of an op-amp (CA3080A) and COS/MOS transistor-pairs in the CA3600E. Quiescent power consumption of the circuit is about 10 μW(typ.). When the comparator is strobed "ON", transistor P1 is driven into conduction and the OTA becomes active. Under these conditions, the circuit consumes 420 μW and responds to a differential-input signal in about 8 μs. By suitably biasing the CA3080A, the circuit response time can be decreased to about 150 ns but the power consumption is increased to 21 mW. The differential amplifier input common-mode range for this circuit is −1 V to +10.5 V. Voltage gain of this micropower comparator is typically 130 dB.

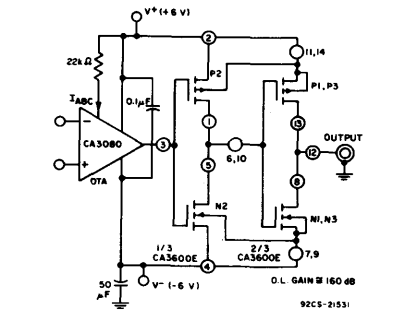


Fig. 20– COS/MOS transistor-pairs used as two-stage post-amplifier to op-amp in open-loop circuit.

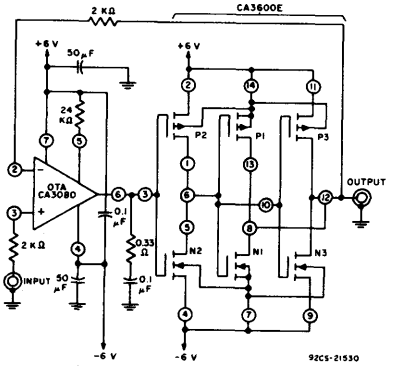


Fig. 21– Unity-gain amplifier uses COS/MOS transistor-pairs as two-stage post-amplifier to op-amp.

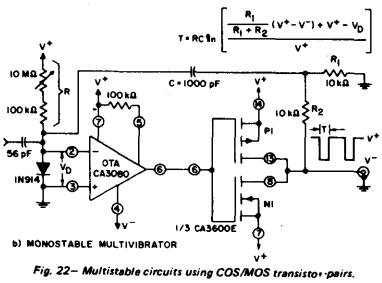


Fig. 22– Multistable circuits using COS/MOS transistor-pairs.

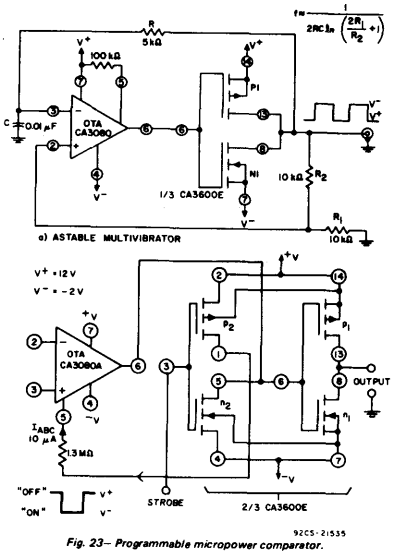
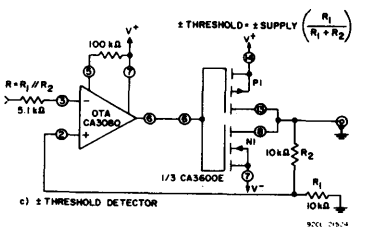


Fig. 23– Programmable micropower comparator.



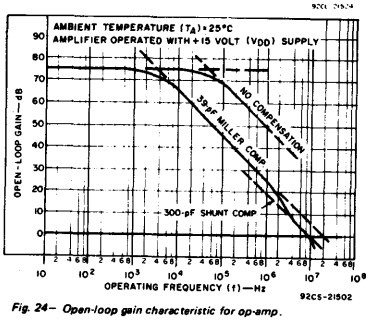


Fig. 24– Open-loop gain characteristic for op-amp.

Oscillator Circuits

Oscillator circuits using COS/MOS transistor-pairs have been widely used for several years in clock and watch circuits because of their low power consumption and good frequency stability. Details of their operating theory and characteristics have been published.

The design of COS/MOS oscillator circuits, like the design of any oscillator circuit, involves the provision of an amplifying section to operate compatibly with an appropriate feedback network. A single stage amplifier using a COS/MOS transistor-pair has already been described. A suitable feedback network to insure stable oscillator performance is easily added, as illustrated in connection with the crystal oscillator circuit shown in Fig. 25. The familiar pi-network has been connected between the input and output terminals, points "D" and "G", to provide the required 180° phase shift for stable oscillator performance. The frequency-determining crystal is an integral part of the pi-network feedback circuit. The resistors R_1 and R_2 decrease the total power consumption of the oscillator at a particular supply voltage and enhance the frequency stability. Variable frequency oscillators can be built by replacing the crystal with an appropriate inductance and tuning the pi-network by conventional means.

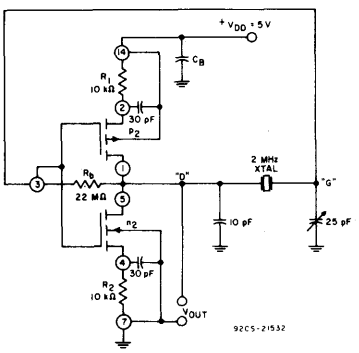


Fig. 25– Typical crystal-oscillator circuit using COS/MOS transistor-pair (1/3 CA3600E).

Current Mirrors Using MOS Transistors

Monolithic linear IC's using bipolar transistors frequently employ so-called "current-mirror" circuits. The theory and practical applications of current mirrors using bipolar transistors have been described in the literature. As shown in Fig. 26, a rudimentary form of "current-mirror" consists of a transistor Q_1 with a second transistor Q_2 connected as a diode. When both transistors have identical characteristics, a current I_1 forced to flow through Q_2 produces a current (I_2) of equal magnitude to flow in the collector of Q_1 (provided there is sufficient collector potential for Q_1). In a common form of application, a source of potential is used to force constant-current flow I_1, and thus to establish the flow of constant current I_2 through Q_1. Arrangements of this generic current-mirror type are frequently used when Q_1 acts as the common-emitter impedance in a differential-amplifier circuit. MOS transistors are also applicable as current mirrors, as shown in Fig. 29. The diode-connected MOS transistor N_2 functions as a transistor with 100 per-cent feedback. Therefore, the gate-to-source voltage (V_{GS}) in N_2 retains control of the drain current as in normal transistor action, i.e., $I_D \cong g_{fs}V_{GS}$, where g_{fs} is the forward transconductance of the device. If a current I_1 is forced into the diode-connected transistor (N_2), the gate-to-source voltage will rise until equilibrium is reached. Thus, a gate-to-source voltage is established in N_2 such that N_2 "sinks" the applied current I_1.

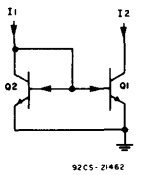


Fig. 26– Current mirror using n-p-n bipolar transistors.

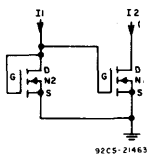


Fig. 27– Current mirror using n-channel MOS transistors.

If the gate and source terminals of another transistor (N_1) are connected in shunt with the gate and source terminals of N_2, as shown in Fig. 27, N_1 is also able to "sink" a mirror current approximately equal to that flowing in the drain lead of the diode-connected transistor N_2. It is assumed that both MOS transistors have identical characteristics, a prerequisite that is essentially established by the monolithic IC fabrication technology used in manufacturing the CA3600E COS/MOS transistor array.

Current mirrors can also be designed with p-channel MOS transistors as illustrated by the arrangement in Fig. 30 using transistors in the CA3600E. The characteristics of a current mirror using the p-channel transistors in the CA3600E are superior to those which can be achieved with a current mirror using the n-channel transistors because the characteristics of the p-channel transistors are more nearly matched. The data

Complementary Current Mirrors Using COS/MOS Transistor-Pairs

COS/MOS transistor-pairs can be applied advantageously in the design of Complementary Current-Mirrors, as shown in Fig. 30. Transistors P_1 and N_1 are series-connected and biased for linear operation as previously described, so that there is a current flow I_{D1} through P_1 and N_1. The potential developed between terminals 13 and 14 is applied as gate-source (2,3) voltage for P_2, forcing "mirror" operation of P_2 to produce a current source $I_{D2\text{-}P}$ equal to I_{D1}. Likewise, the potential developed between terminals 7 and 8 is applied as gate-source (3,4) voltage for N_2 forcing "mirror" operation of N_2 to produce a current-sink $I_{D2\text{-}N}$ equal to I_{D1}.

A variant of this complementary current mirror is used in the analog timer circuit shown in Fig. 28. Transistors P_2 and N_2 are series-connected together with a 60-megohm resistor to establish their drain current at 5 nA. The potential developed across terminals 1 and 2 also appears as the gate-source voltage for transistor P_1, thereby establishing a mirror-current source of 5 nA at terminal 13 to charge capacitor C_1 linearily. In this circuit, the "mirrored" current-sink available at terminal 8 (transistor N_1) is unused. This type of current-mirror configuration is exceptionally stable with temperature variations.

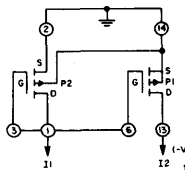


Fig. 28–Current mirror using p-channel MOS transistors in CA3600E.

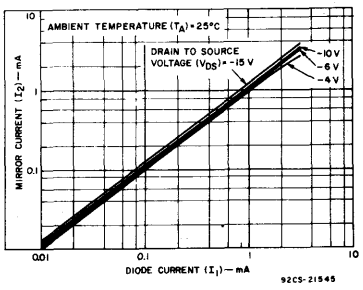


Fig. 29– Characteristics of current mirror circuit of Fig. 30 using p-channel transistors.

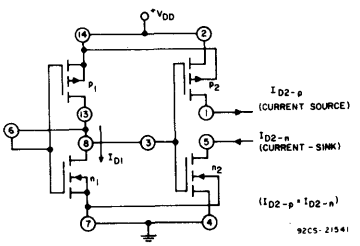


Fig. 30– Complementary current mirrors using COS/MOS transistor-pairs in CA3600E.

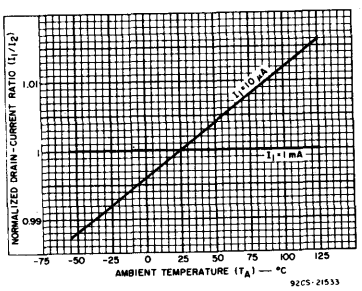


Fig. 31– Normalized drain current ratio vs. ambient temperature for typical current mirror using p-channel transistors (Fig. 28).

High-Current N-P-N Transistor Arrays

Four Individual Sealed-Junction High-Current N-P-N Transistors

The RCA-CA3724G and -CA3725G are high-current n-p-n transistor arrays each containing 4 individual sealed-junction high-current n-p-n transistors. They are intended for high-current, high-speed switching and driver applications.

These devices are alike except for breakdown voltage ratings.

The CA3724G and CA3725G are supplied in a 14-lead dual-in-line plastic package and operate over the full military temperature range of −55°C to +125°C. The transistor chips used in these packages are of the sealed-junction type to provide protection against the deteriorating effects of humidity and other surface contaminants without the need for a hermetic package enclosure.

The semiconductor junctions are sealed by utilizing a silicon nitride passivation layer. A multi-layered, highly corrosion-resistant, terminal-connection system of unique design is employed.

Applications:

- **Core-Memory Driver**
- **High-Speed Switching**
- **High-Current LED Driver**
- **High-Voltage Switching**
- **Relay and Solenoid Driver**
- **Lamp Driver**

Features:

- **High Current – 1 A**
- **High Breakdown Voltage:**
  CA3725G = 80 V dc min. $V_{(BR)CES}$ @ I_C = 10 μA
  CA3724G = 70 V dc min. $V_{(BR)CES}$ @ I_C = 10 μA
- **Fast Switching Speeds:**
  t_{on} = 30 ns typ. @ I_C = 500 mA
  t_{off} = 36 ns typ. @ I_C = 500 mA
- **"Hermetic Chip" Construction**
- **Silicon Nitride Passivated**
- **Platinum Silicide Ohmic Contacts**
- **Gold Chip-Metallization**
- **Electrically similar and pin compatible with industry types MPQ3724, MPQ3725; FPQ3724, FPQ3725; DH3724, DH3725; SP3724, SP3725 in similar packages**

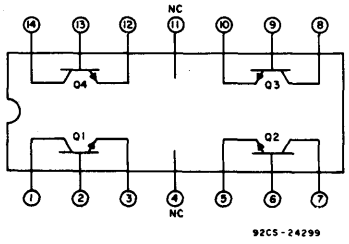


Fig. 1—Terminal diagram (top view).

MAXIMUM RATINGS, Absolute-Maximum Values at T_A = 25°C

		CA3724G	CA3725G	
COLLECTOR-TO-EMITTER VOLTAGE With Base Open	V_{CEO}	40	50	V
COLLECTOR-TO-BASE VOLTAGE With Emitter Open	V_{CBO}	70	80	V
EMITTER-TO-BASE VOLTAGE With Collector Open	V_{EBO}	6	6	V
COLLECTOR CURRENT	I_C	1.0	1.0	A
POWER DISSIPATION: At T_A up to 25°C:	P_D			
For Each Transistor		1.0	1.0	W
Total Package		2.0	2.0	W
At T_A above 25°C derate linearly		20		mW/°C
AMBIENT TEMPERATURE RANGE:				
Operating		−55 to +125	−55 to +125	°C
Storage		−65 to +150	−65 to +150	°C
LEAD TEMPERATURE (DURING SOLDERING): At distance 1/32" (3.17 mm) from seating plane for 10 s max.		300	300	°C

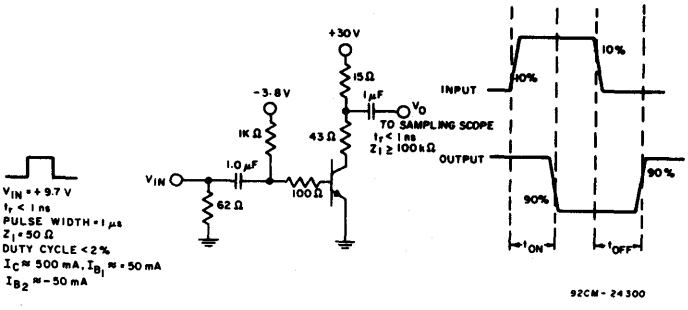


Fig. 2—Switching time test circuit.

ELECTRICAL CHARACTERISTICS AT T_A = 25°C

Characteristic	Test Conditions	Limits						Units
		CA3724G			CA3725G			
		Min.	Typ.	Max.	Min.	Typ.	Max.	
Collector-to-Emitter Sustaining Voltage, $V_{CEO(sus)}$*	I_C=10 mA, I_B=0	40	–	–	50	–	–	V
Collector-to-Emitter Breakdown Voltage, $V_{(BR)CES}$	I_C=10 μA, I_B=0	70	–	–	80	–	–	V
Collector-to-Base Breakdown Voltage, $V_{(BR)CBO}$	I_C=10 μA, I_E=0	70	–	–	80	–	–	V
Emitter-to-Base Breakdown Voltage, $V_{(BR)EBO}$	I_E=10 μA, I_C=0	6	–	–	6	–	–	V
Base-to-Emitter Saturation Voltage, $V_{BE(sat)}$*	I_C=500 mA, I_B=50 mA	0.75	–	1.0	0.75	–	1.0	V
Collector-to-Emitter Saturation Voltage, $V_{CE(sat)}$*	I_C=500 mA, I_B=50 mA	–	–	0.5	–	–	0.5	V
Collector-Cutoff Current, I_{CBO}	V_{CB}=40 V, I_E=0	–	–	1.7	–	–	1.7	μA
Static Forward-Current Transfer Ratio (Beta), h_{FE}	I_C=100 mA, V_{CE}=1.0 V	35	–	–	35	–	–	
	I_C=500 mA, V_{CE}=1.0 V	30	–	–	30	–	–	
	I_C=1 A, V_{CE}=1.0 V	20	–	–	20	–	–	
Small-Signal Forward-Current Transfer Ratio, h_{fe}	I_C=50 mA, V_{CE}=10 V, f=100 MHz	2.0	–	–	2.0	–	–	
Turn-On Time (See Test Ckt. Fig. 2), t_{on}	I_C=500 mA, I_{B1}=50 mA	–	–	40	–	–	40	ns
Turn-Off Time (See Test Ckt. Fig. 2), t_{off}	I_C=500 mA, I_{B1}= I_{B2}=50 mA	–	–	60	–	–	60	ns
Emitter-to-Base Capacitance, C_{eb}	I_C=0, V_{EB}=0.5 V	–	95	–	–	95	–	pF
Collector-to-Base Capacitance, C_{cb}	I_E=0, V_{CB}=10 V	–	12	–	–	12	–	pF

*Pulse Conditions: width = 300 μs; duty cycle = 1%.

Operational Amplifiers

CA6078AT – Micropower Type
CA6741T – General-Purpose Type

For Applications where Low Noise (Burst + 1/f) is a Prime Requirement

Virtually free from "popcorn" (burst) noise: device rejected if any noise burst exceeds 20 μV (peak), referred to input over a 30-second time period.

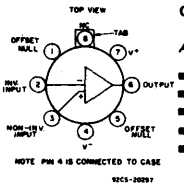
RCA-CA6078AT and CA6741T are low-noise linear IC operational amplifiers that are virtually free of "popcorn" (burst) noise.

These low-noise versions of the CA3078AT and CA3741T are a result of improved processing developments and rigid burst-noise inspection criteria. A highly selective test circuit (See Fig. 2) assures that each type meets the rigid low-noise standards shown in the data section. This low-burst-noise property also assures excellent performance throughout the 1/f noise spectrum.

In addition the CA6078AT and CA6741T offer the same features incorporated in the CA3078AT and CA3741T respectively, including output short-circuit protection, latch-free operation, wide common-mode and differential-mode signal ranges, and low-offset nulling capability.

For detailed data, characteristics curves, schematic diagram, dimensional outline, and test circuits, refer to the Operational Amplifier Data Bulletins File No. 531 and 535. In addition, for details of considerations in burst-noise measurements, refer to Application Note, ICAN-6732, "Measurement of Burst ("Popcorn") Noise in Linear IC's".

The CA6078AT and CA6741T utilize the hermetically sealed 8-lead TO-5 type package. The CA6078AT and the CA6741T can also be supplied on request with dual-in-line formed leads. These types are identified as the CA6078AS and CA6741S. This formed-lead configuration conforms to that of the 8-lead dual-in-line (Mini-Dip) package.



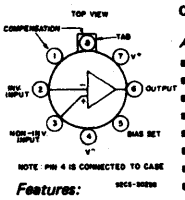
CA6741T

Applications:

- Low-noise AC amplifier
- Narrow-band or band-pass filter
- Integrator or differentiator
- DC amplifier
- Summing amplifier

Features:

- Internal phase compensation
- Input bias current: 500 nA max.
- Input offset current: 200 nA max.
- Open-loop voltage gain: 50,000 (94 dB) min.
- Input offset voltage: 5 mV max.



CA6078AT

Applications:

- Portable electronics
- Medical electronics
- DC amplifier
- Narrow-band or band-pass filter
- Integrator or differentiator
- Instrumentation
- Telemetry
- Summing amplifier

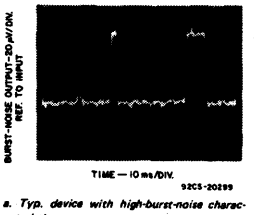
Features:

- Open-loop voltage gain: 40,000 (92 dB) min.
- Input offset voltage: 3.5 mV max.
- Operates with low total supply voltage: 1.5 V min. (± 0.75 V)
- Low quiescent operating current: adjustable for application optimization
- Input bias current: adjustable to below 1 nA

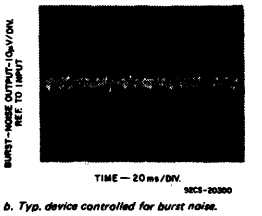
MAXIMUM RATINGS, *Absolute-Maximum Values at T_A = 25°C*

	CA6741T	CA6078AT
DC Supply Voltage (between V+ and V− terminals)	44 V	36 V
Differential-Mode Input Voltage	±30 V	±6 V
Common-Mode DC Input Voltage▲	±15 V	V+ to V−
Device Dissipation:		
Up to 75°C (CA6741T), Up to 125° (CA6078AT)	500 mW	250 mW
Above 75°C	Derate linearly 5 mW/°C	–
Temperature Range:		
Operating	−55 to +125 °C	−55 to +125 °C
Storage	−65 to +150 °C	−65 to +150 °C
Output Short-Circuit Duration●	No limitation	No limitation
Lead Temperature (During soldering):		
At distance 1/16 ±1/32 inch (1.59 ±0.79 mm) from case for 10 seconds max.	300 °C	300 °C

▲If Supply Voltage is less than ±15 volts, the Absolute Maximum Input Voltage is equal to the Supply Voltage.
●Short circuit may be applied to ground or to either supply.



a. Typ. device with high-burst-noise characteristic.



b. Typ. device controlled for burst noise.

Fig.1—Typ. waveforms of type with high burst noise and type controlled for burst noise.

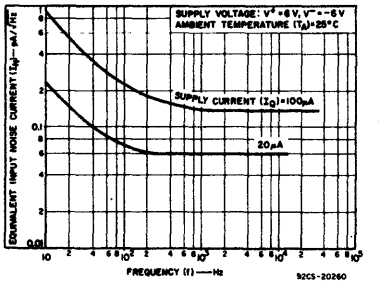


Fig.3—I_N vs. Frequency for CA6078AT.

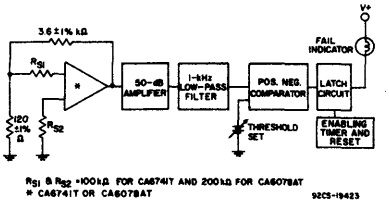


Fig.2—Block diagram of burst-noise "popcorn" test equipment.

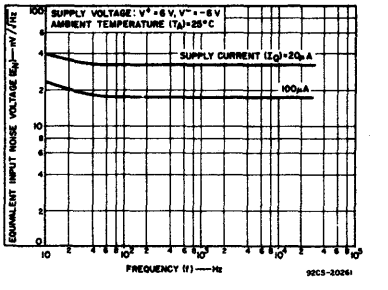


Fig.4—E_N vs. Frequency for CA6078AT.

ELECTRICAL CHARACTERISTICS – CA6078AT, *For Equipment Design.*

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	LIMITS			UNITS
		Supply Volts: $V^+ = 6$, $V^- = -6$ $T_A = 25°C$, $I_Q = 20\ \mu A$	MIN.	TYP.	MAX.	
Noise Characteristic						
"Popcorn" (Burst) Noise		Bandwidth = 1 kHz $R_{S1} = R_{S2} = 200\ k\Omega$	Device is rejected if the total noise voltage (burst + 1/f), referred to input, exceeds 20 μV peak, during a 30-sec. test period.			
Principal Characteristics (For detailed Electrical Characteristics refer to CA3078AT Data Bulletin, File No. 535.)						
Input Offset Voltage	V_{IO}	$R_S \leq 10\ k\Omega$	–	0.7	3.5	mV
Input Offset Current	I_{IO}		–	0.5	2.5	nA
Input Bias Current	I_{IB}		–	7	12	nA
Open-Loop Differential Voltage Gain	A_{OL}	$R_L \geq 10\ k\Omega$ $V_O = \pm 4V$	40,000 92	100,000 100	– –	dB
Common-Mode Input Voltage Range	V_{ICR}	$V^+ = V^- = 15$ V	±14	–	–	V
Common-Mode Rejection Ratio	CMRR	$R_S \leq 10\ k\Omega$	80	115	–	dB
Output Voltage Swing	$V_{O(P-P)}$	$R_L \geq 10\ \Omega$	±13.7	±14.1	–	V
		$R_L \geq 2\ k\Omega$	–	±14	–	
Supply Current	I_Q		–	20	25	μA

ELECTRICAL CHARACTERISTICS – CA6741T, *For Equipment Design.*

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	LIMITS			UNITS
		Supply Volts; $V^+ = 15$, $V^- = -15$ $T_A = 25°C$	MIN.	TYP.	MAX.	
Noise Characteristic						
"Popcorn" (Burst) Noise		Bandwidth = 1 kHz $R_{S1} = R_{S2} = 100\ k\Omega$	Device is rejected if the total noise voltage (burst + 1/f), referred to input, exceeds 20 μV peak, during a 30-sec. test period.			
Principal Characteristics (For detailed Electrical Characteristics refer to CA3741T Data Bulletin, File No. 531.)						
Input Offset Voltage	V_{IO}	$R_S \leq 10\ k\Omega$	–	1	5	mV
Input Offset Current	I_{IO}		–	20	200	nA
Input Bias Current	I_{IB}		–	80	500	nA
Open-Loop Differential Voltage Gain	A_{OL}	$R_L \geq 2\ k\Omega$ $V_O = \pm 10$ V	50,000 94	200,000 106	– –	dB
Common-Mode Input Voltage Range	V_{ICR}		±12	±13	–	V
Common-Mode Rejection Ratio	CMRR	$R_S \leq 10\ k\Omega$	70	90	–	dB
Output Voltage Swing	$V_{O(P-P)}$	$R_L \geq 10\ k\Omega$	±12	±14	–	V
		$R_L \geq 2\ k\Omega$	±10	±13		
Supply Current	I_Q		–	1.7	2.8	mA

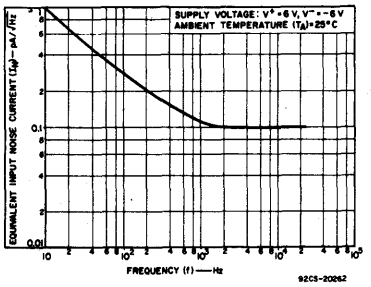


Fig.5–I_N vs. Frequency for CA6741T.

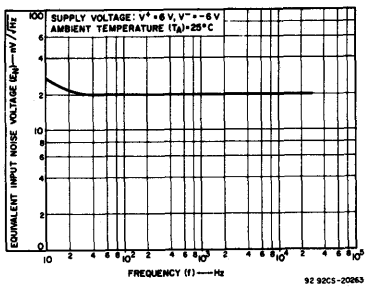


Fig.6–E_N vs. Frequency for CA6741T.

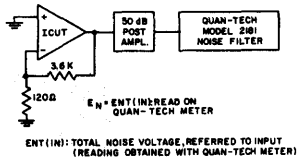


Fig.7–Test block diagram for E_N.

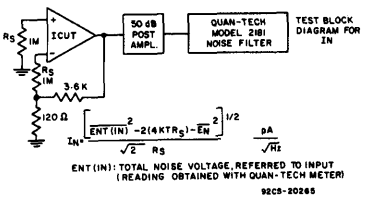


Fig.8–Test block diagram for I_N.

Linear Integrated Circuits for Consumer Applications

Technical Data

CA270 Types

TV Synchronous Demodulators

For Color and Black-and White TV Systems

The RCA-CA270AW, CA270BW, and CA-270CW are integrated circuits which perform the functions of synchronous detection of the TV if, video amplification and buffering, and noise inversion on dual-polarity waveforms. These devices also offer agc and afc facilities for use with n-p-n transistor if amplifiers and tuners. Both positive and negative polarities of video output are available. This feature provides great flexibility by permitting the designer to use either output for deriving the video and sound channels.

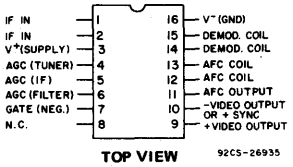
The RCA-CA270 series is pin-compatible and electrically similar to the industry series TCA270, but incorporates several improved features. In particular, improved white noise inversion and sync inversion systems force overshoots in the video waveform to be returned to accurately defined potentials. This design effectively removes dependence on both the degree of overshoot and temperature variations. In addition, reduced current consumption assures lower over-all power dissipation, thereby improving reliability.

The three types are electrically identical in most parameters. The CA270B has the most stringent limits on white level, video inversion, and afc dc offset. The CA270C has the least stringent limits on white level and video inversion, and no afc limits.

The CA270 series is supplied in a 16-lead staggered quad-in-line plastic package ("W" suffix).

Features:

- **Synchronous detector with single tuned coil**
- **Provides rf and if agc (forward)**
- **Tuner afc available with single quadrature coil**
- **Dual-polarity noise inverters**
- **Video amplifier**
- **Positive- and negative-polarity buffered video**
- **Differential if input**
- **Optional use of gating pulse**
- **Low-voltage, single-polarity power supply**



Terminal assignment.

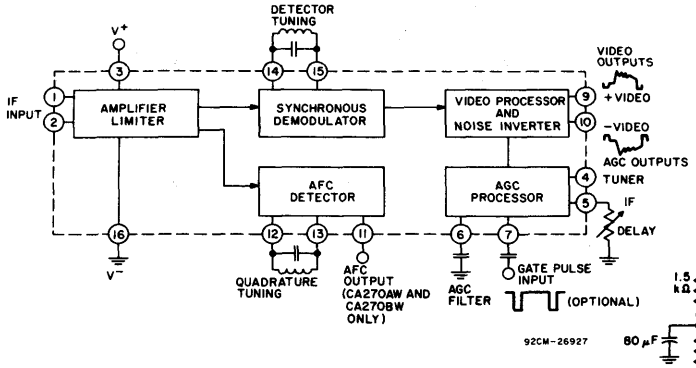


Fig. 1—Functional block diagram of CA270AW, CA270BW, and CA270CW TV synchronous demodulator.

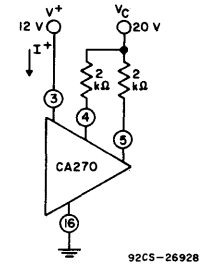


Fig. 2—Supply-current test circuit.

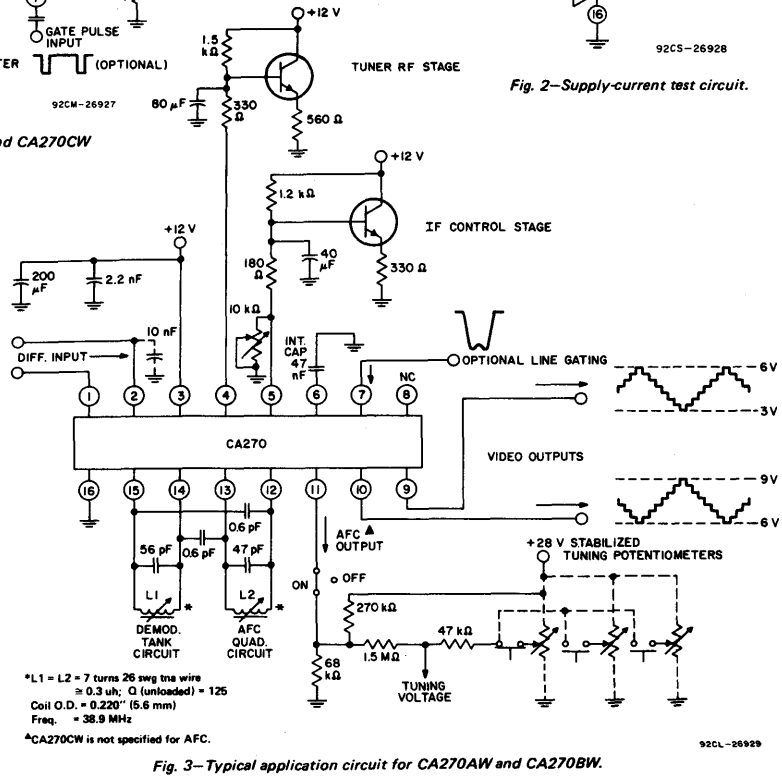


Fig. 3—Typical application circuit for CA270AW and CA270BW.

MAXIMUM RATINGS,
Absolute-Maximum Values at T_A=25°C:

DC SUPPLY VOLTAGE (Between Terminals 3 and 16 for 10 s max., with current limited to 100 mA) ..... 18 V

DEVICE DISSIPATION:
Up to T_A = 55°C ..... 750 mW
Above T_A = 55°C ... derate linearly 7.9 mW/°C

OPERATING TEMPERATURE RANGE
..... −40 to +55°C

STORAGE TEMPERATURE RANGE
..... −65 to +150°C

LEAD TEMPERATURE (During Soldering)
At distance 1/16" ± 1/32" (1.59 ± 0.79 mm) from case for 10 s max. ..... +265°C

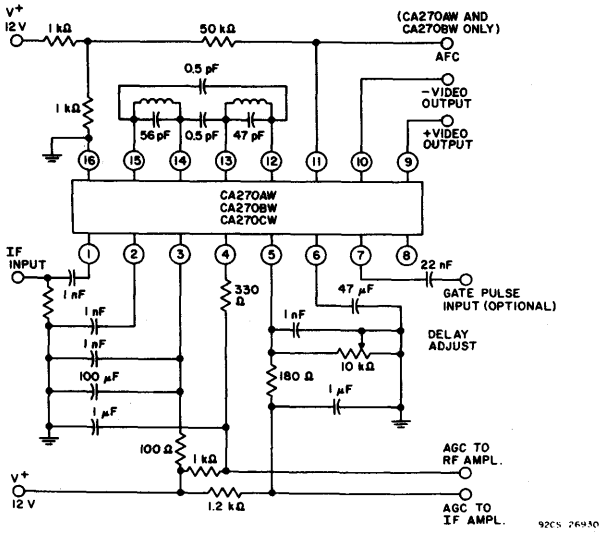


Fig. 4—Test circuit for CA270AW, CA270BW, and CA270CW.

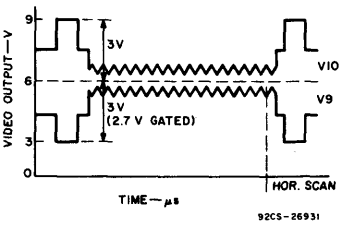


Fig. 5—Typical waveforms for video outputs.

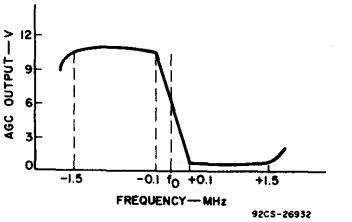


Fig. 6—Typical AFC characteristic.

ELECTRICAL CHARACTERISTICS at T_A = 25°C, Supply Voltage (V^+) = 12 V, and Referenced to Test Circuit (Fig. 4).

CHARACTERISTIC	TEST CONDITIONS		MIN.	TYP.	MAX.	UNITS
Supply Voltage, V^+	V^+=12 V		10.2	12	13.8	V
Supply Current, I^+ (See Fig. 2)	V^+=12 V		22	40	56	mA
Video Characteristics: DC Output Voltage, Term.9 (See Fig. 5)	Zero Signal	CA270AW	5.7	6	6.3	V
		CA270BW	5.8	6	6.2	
		CA270CW	5.5	6	6.5	
DC Output Voltage, Term.10 (See Fig. 5)	Zero Signal	CA270AW	5.6	6	6.4	V
		CA270BW	5.7	6	6.3	
		CA270CW	5.5	6	6.5	
Sync Tip Output Voltage, Term.9	Output=AGC threshold (non-gated)		–	3	–	V
AC Input Voltage, Terms.1,2	Input for output= AGC threshold		50	70	100	mV
Input Res.,Term.1			–	3.3	–	KΩ
Input Res., Term.2			–	3.3	–	KΩ
Video Bandwidth, Term.9	At output = –3 dB		–	5	–	MHz
Differential Gain	See Note 1		–	–	10	%
Differential Phase	See Note 1		–	–	10	deg
Intermod. Products: Beat Freq.,1.6 MHz	See Note 1 (95% sat. blue colour bar)		–	–	–60	dB
Beat Freq.,2.8 MHz			–	–	–67	dB
Rejection at Carrier Freq., Terms.9,10,11	F=Video Carrier;V_{IN} for Term.9(dc)=3.7V		–40	–	–	dB
Rejection,Twice Carrier Freq.,Terms.9,10,11	F=2X Video Carrier; V_{IN} for Term.9(dc) =3.7 V		–40	–	–	dB
AGC Characteristics: Sat.Voltage, Term.4	Zero Sig.; I_4 = 10 mA		–	–	0.3	V
Sat. Voltage, Term.5	Zero Sig.; I_5 = 10 mA		0.7	–	1.2	V

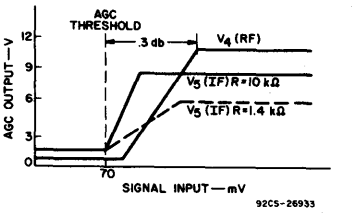


Fig. 7—Typical AGC characteristics.

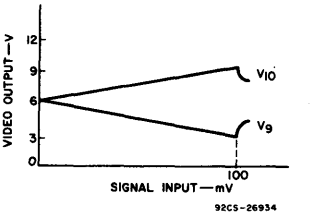


Fig. 8—Typical transfer characteristics.

ELECTRICAL CHARACTERISTICS at T_A = 25°C, Supply Voltage (V^+) = 12 V, (Cont'd) and Referenced to Test Circuit (Fig. 4).

CHARACTERISTIC	TEST CONDITIONS		MIN.	TYP.	MAX.	UNITS
Breakdown Voltage, Terms. 4,5	I_4 or I_5=1 mA (sink)		14	–	–	V
Control Current, Terms. 4,5			10	–	–	mA
Current Ratio I_4/I_5	I_5 = 1 mA		6	–	–	
Input Signal Increase with resp. to AGC Threshold (See Fig.7)	AGC from threshold to max.		–	–	0.5	dB
AGC Gating Pulse Input, Term. 7 (optional)	Pulse voltage=$V^+$to 0; See Note 2		2	–	V^+	V
Input Res.,Term.7			–	1.8	–	KΩ
AFC Characteristics: (See Fig. 6) Output Voltage, Term. 11	$f = f_o \pm 0.2$ MHz	CA270AW CA270BW CA270CW	10 10 –	– – –	– – –	V_{p-p}
Output Voltage, Term. 11	$f = f_o \pm 1.2$ MHz	CA270AW CA270BW CA270CW	10 10 –	– – –	– – –	V_{p-p}
DC Offset Voltage, Term. 11	Zero Sig.; measured across R_L = 50 KΩ to +6 V	CA270AW CA270BW CA270CW	–1.7 –1 –	– – –	1.7 1 –	V
Noise Inverter Characteristics: Inversion Threshold, Term. 9	Positive noise pulses		–	6.6	–	V
Inversion Threshold, Term. 9	Negative noise pulses		–	2.2	–	V
Noise Inversion Sensitivity,Term. 9	Signal inversion threshold for complete inversion		–	10	–	mV
Video Inversion Characteristics: Video Inversion, Term. 9 (at low carrier levels)	Carrier increase from 0 to 5 mV (appx.8% carrier)	CA270AW CA270BW CA270CW	– – –	– – –	0.2 0.1 0.3	V

Note 1: CCIR modulation system, peak white = 10% carrier.
Note 2: Maximum pulse amplitude must never exceed the supply voltage (V^+).

APPLICATIONS

The diagram shown in Fig. 3 is typical of the type of circuit used in a practical application of the CA270 series devices.

Video Detector

The if input signal may be applied push-pull to terminals 1 and 2, or single-ended to either terminal 1 as shown, or to terminal 2. These input terminals are internally biased.

The detector tank circuit can be tuned by applying a 50 mV cw signal of video if frequency to the input and adjusting the inductor L1 for maximum differential output between terminals 9 and 10. The input signal is then reduced to 25 mV and L1 is readjusted for maximum output.

AFC Detector

The afc quadrature tank circuit should be tuned only after the detector adjustment has been made. Using the same input signal, inductor L2 should be adjusted for 6 V dc output at terminal 11. The 0.5-pF quadrature phase-shift coupling capacitors can affect symmetry and actual values will depend on the layout used. When L1 and L2 are properly tuned, the output swing at terminal 11 will be 10 volts minimum for frequencies of ±0.2 MHz to ±1.2 MHz about the if carrier frequency.

AGC Detector

The agc threshold, corresponding to sync tip level, is approximately 3 volts at terminal 9. Full agc potential will be developed if the input signal increases by 0.5 dB maximum with respect to the threshold value. The agc control at terminal 4 is intended for tuner control. The agc control at terminal 5 is for forward agc control of n-p-n transistors in the if amplifier. When sinking 10 mA, the zero-signal agc voltage at terminal 4 is 0.3 volt maximum; at terminal 5, it is 1.2 volts maximum.

The design of the device is such that the sink current at terminal 4 is a minimum of 6 times that at terminal 5. The rf agc sink current begins to decrease when the if sink current is about one-sixth of that required to saturate the rf agc output at terminal 4. The rf agc delay may be adjusted by means of a variable resistor between terminal 5 and ground. This adjustment modifies the if system gain, thus affecting the rf delay threshold. At maximum gain the current into terminal 5 is large compared to the current in the variable resistor and adjustment is ineffective. As the signal increases and rf agc is applied, the terminal 5 sink current approaches zero and the if agc is determined by the value of the variable resistor.

A horizontal gating pulse may be applied to terminal 7 to gate the agc detector. The agc threshold (sync tip) decreases approximately 0.3 volt at terminal 9 when gating is used. The gating pulses must be negative-going with a recommended minimum amplitude of 3 volts. They may be ac or dc coupled, but the maximum peak value must not exceed the dc supply voltage at terminal 3. If dc coupling is used, the potential during flyback should be less than 0.5 volt and during scan, greater than 1.5 volts.

Noise Inverter

Noise pulses in excess of 6.6 volts at terminal 9, which would result in "white spots", are processed in the device by inverting and clamping them to near black level (approx. 3.6 V). Noise pulses at levels of less than 2.2 volts at terminal 9 which would result in sync noise interference, are inverted and returned to black level.

Complete inversion occurs for signals 10 mV above the inversion threshold.

RC Phase-Locked-Loop Stereo Decoder

For FM Multiplex Systems

RCA-CA758E is a monolithic silicon integrated circuit RC phase-lock loop stereo decoder intended for FM solid-state stereo multiplex systems.

The CA758E is pin compatible and electrically equivalent to industry types μA758, MC1311P, LM1800, and ULX2244.

The CA758E decodes the multiplexed stereo input signal into left and right channel audio output signals. The decoder also suppresses SCA (storecast) transmissions when present in the composite stereo signal.

The decoder uses a minimum of external components, and requires one adjustment (oscillator frequency) for complete alignment. In addition, the CA758E provides automatic mono-stereo mode switching and energizes a stereo indicator lamp.

The CA758E is supplied in a 16-lead dual-in-line plastic package and operates over an ambient temperature range of −40 to +85°C.

Features:

- **Low distortion (THD): 0.4% (typ.)**
- **Excellent SCA rejection: 70 dB typ.**
- **RC oscillator**
- **High-audio-channel separation: 45 dB**
- **Power supply range: 10 to 16 V dc**
- **Requires only one adjustment for complete alignment**
- **Low-impedance outputs**
- **Stereo indicator lamp drive: 150 mA typ.**

MAXIMUM RATINGS, *Absolute-Maximum Values at T_A = 25°C*

DC Supply Voltage	+18 V
DC Supply Voltage (for ≤ a 15-second period)	+22 V
DC Voltage at Term. 7 (Lamp Driver Circuit with Lamp "OFF")	+22 V
Device Dissipation:	
Up to T_A = 70°C	730 mW
Above T_A = 70°C derate linearly	9.1 mW/°C
Ambient Temperature Range:	
Operating	−40 to +85°C
Storage	−65 to +150°C
Lead Temperature (During soldering):	
At a distance not less than 1/32" (0.79 mm) from case for 10 s max.	+265°C

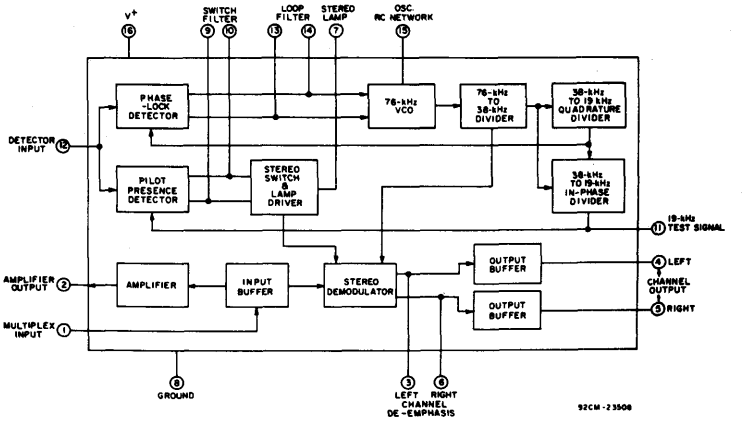
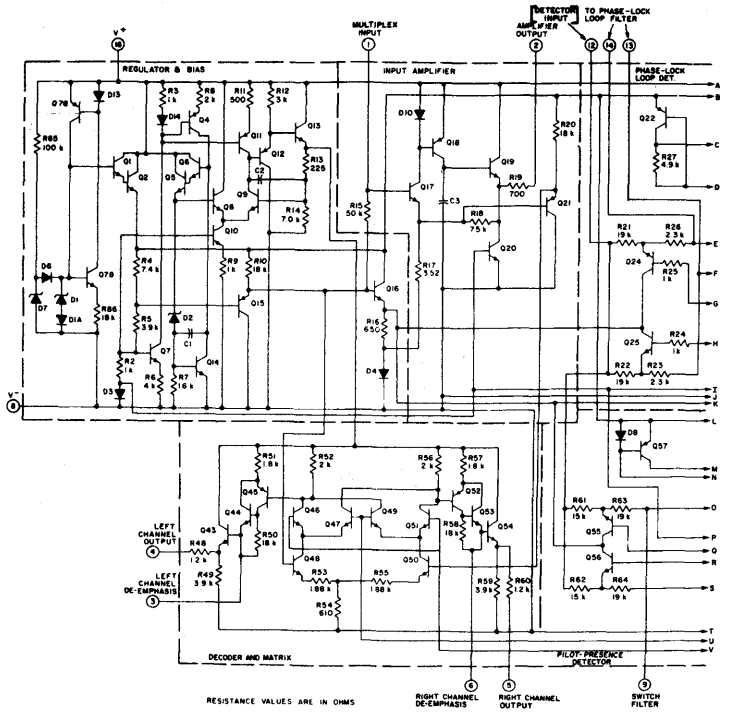


Fig. 1 — Functional block diagram of the CA758E.

ELECTRICAL CHARACTERISTICS

CHARACTERISTIC	TEST CONDITIONS (Referenced to Fig. 7 unless otherwise specified) V^+ = 12 V, T_A = 25°C, Multiplex Input Signal (L=R, pilot "OFF") = 300 mV RMS, 19-kHz Pilot Level = 30 mV RMS, f (modulation) = 400 Hz or 1 kHz	LIMITS Min.	LIMITS Typ.	LIMITS Max.	UNITS
Static Characteristics					
Total Current	Lamp "OFF"	–	26	35	mA
Maximum Available Lamp Current		75	150	–	mA
DC Voltage at Term. 7 (Lamp Driver)	I (Lamp) = 50 mA	–	1.3	1.8	V
DC Voltage Shift at either Term. 4 or 5 (Output)	Stereo-to-Mono Operation	–	30	150	mV
Dynamic Characteristics					
Power Supply Ripple Rejection	For a 200-Hz, 200-mV RMS Signal	35	45	–	dB
Input Resistance		20	35	–	kΩ
Output Resistance		0.9	1.3	2.0	kΩ
Channel Separation (Stereo)	At f = 100 Hz	–	40	–	dB
	f = 400 Hz	30	45	–	dB
	f = 10 kHz	–	45	–	dB
Channel Balance (Monaural)		–	0.3	1.5	dB
Voltage Gain	At f = 1 kHz	0.5	0.9	1.4	V/V
Pilot Input Level:					
19-kHz Input	Lamp "ON"	–	15	20	mV RMS
19-kHz Input	Lamp "OFF"	2.0	7.0	–	mV RMS
Hysteresis	Lamp "OFF"	3.0	7.0	–	dB
Capture Range (Deviation from 76-kHz Center Frequency)		±2.0	±4.0	±6.0	%
Total Harmonic Distortion	Multiplex Input Signal = 600 mV RMS (Pilot "OFF")	–	0.4	1.0	%
19-kHz Rejection		25	35	–	dB
38-kHz Rejection		25	45	–	dB
SCA (Storecast) Rejection	Measured Composite Signal: 80% Stereo, 10% Pilot, 10% SCA	–	70	–	dB
Voltage-Controlled Oscillator (VCO) Tuning Resistance	Total Resistance (Term. 15 to 8) required to set f_{REF} = 19 kHz ± 10 Hz (Term. 11)	21.0	23.3	25.5	kΩ
Voltage-Controlled Oscillator Frequency Drift	$0° \leq T_A \leq 25°C$	–	+0.1	±2	%
	$25° \leq T_A \leq 70°C$	–	−0.4	±2	%



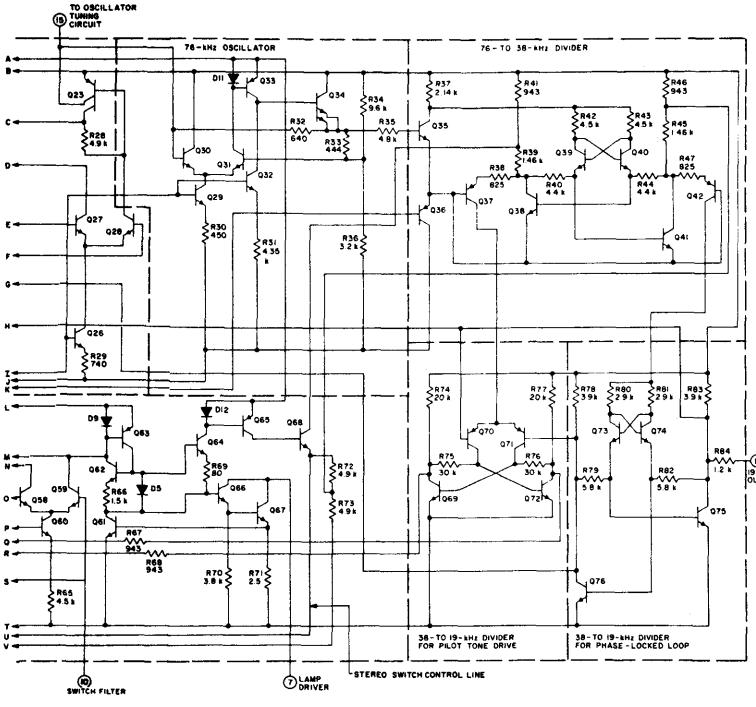
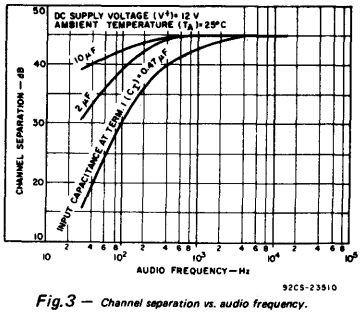


Fig. 2 – Schematic diagram of the CA758E.

92CL-23177

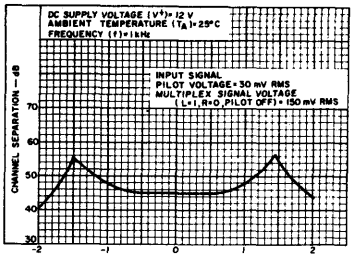
TYPICAL PERFORMANCE CHARACTERISTICS

(Referenced to Fig. 7)



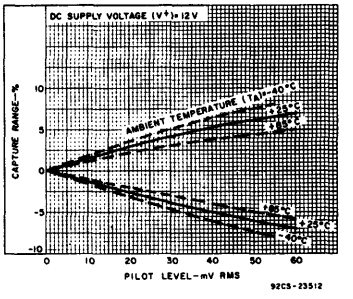
92CS-23510

Fig. 3 — Channel separation vs. audio frequency.



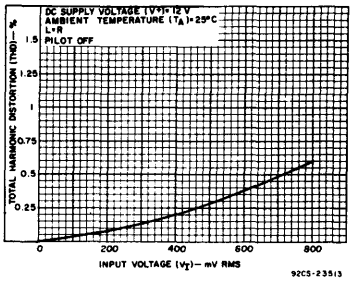
92CS-23511

Fig. 4 — Channel separation vs. oscillator free running frequency error.



92CS-23512

Fig. 5 — Capture range vs. pilot level.



92CS-23513

Fig. 6 — Total harmonic distortion vs. input level.

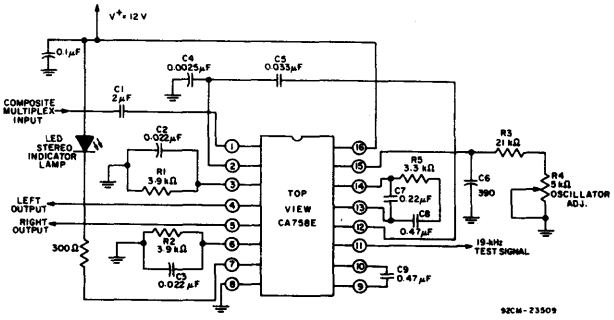


Fig. 7 — Test circuit for measurement of dynamic characteristics.

NOTES:

Tolerance on resistors is ±5% and tolerance on capacitors is ±20% unless otherwise specified.

C_1 = +100%, −20%

C_6 = ±1% in test circuit and ±5% in typical application.

R_3 = ±1%

R_4 = ±10%

R_1 and R_2 = ±1% in test circuit and ±5% in typical application.

TYPICAL PERFORMANCE CHARACTERISTICS (Referenced to Fig. 7)

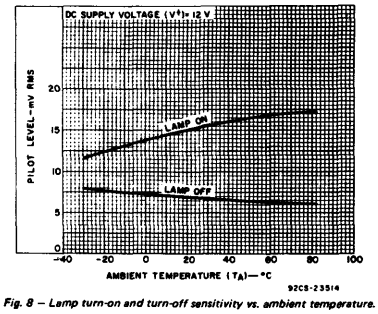


Fig. 8 — Lamp turn-on and turn-off sensitivity vs. ambient temperature.

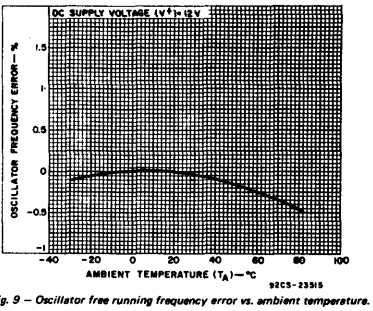


Fig. 9 — Oscillator free running frequency error vs. ambient temperature.

CA810, CA810A Types

Preliminary Data

7-Watt Audio Power Amplifier With Thermal Shut-Down

The RCA-CA810Q,CA810AQ, CA810QM and CA810AQM are monolithic audio amplifiers intended for class B operation. They are specifically designed for mobile equipment operating from 12-V battery supplies. They operate over a wide range of supply voltages (4 to 20 V) with very low harmonic and crossover distortion. The maximum repetitive peak output current is 2.5 A, and an integral thermal limiting circuit shuts the device down in case of output overload or excessive package temperature.

The CA8100, CA810AQ, CA810QM, and CA810AQM are supplied in modified 16-lead quad-in-line plastic packages ("Q" suffix) with integral wing-tab heat sinks. The tabs on the CA810Q and CA810AQ are bent down for p.c. board insertion, and on the CA810QM and CA810AQM they are flat and pierced for easy attachment to an external heat sink.

The CA810Q and CA810QM are electrically equivalent to types TBA810S and TBA810AS, respectively. It should be noted that pin-numbering conventions for these devices may differ from manufacturer to manufacturer, however the devices are pin compatible and interchangeability is not affected.

The CA810AQ and CA810AQM are electrically the same as the CA810Q and CA810QM, respectively, except for the inclusion of a load dump (overvoltage) voltage surge protection circuit. This feature makes the CA810AQ and CA810AQM ideally suitable for automotive applications.

Features:

- **Power output – 7 W with 4Ω load**
- **Supply voltage range – 4 to 20 V**
- **Peak output current – 2.5 A (max.)**
- **Very low harmonic and cross-over distortion**
- **Load dump voltage surge protection (CA810AQ and CA810AQM)**

MAXIMUM RATINGS, *Absolute-Maximum Values:*

PEAK SUPPLY VOLTAGE (50 ms) (CA810AQ, CA810AQM)			40 V
OPERATING SUPPLY VOLTAGE			20 V
OUTPUT PEAK CURRENT:			
REPETITIVE			2.5 A
NON-REPETITIVE			3.5 A
POWER DISSIPATION, P_D			
At T_A = 70°C			1 W
At T_{tab} = 100°C			5 W
THERMAL RESISTANCE, JUNCTION	CA810Q CA810AQ	CA810QM CA810AQM	
Junction to tab	12	10	°C/W
Junction to ambient	70*	80	°C/W
AMBIENT-TEMPERATURE RANGE:			
OPERATING	−40°C to (Refer to Fig.7 for typical high-temperature limit)		
STORAGE	−40 to +150°C		
LEAD TEMPERATURE (DURING SOLDERING):			
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10s max.			260°C

*Value obtained with tabs soldered to printed-circuit board.

ELECTRICAL CHARACTERISTICS, at T_A = 25°C

CHARACTERISTIC	SYMBOL	TEST CONDITIONS		LIMITS			UNITS
		Supply Voltage (V^+) = 14.4 V Unless Otherwise Specified		CA810Q,CA810AQ CA810QM,CA810AQM			
				MIN.	TYP.	MAX.	
Supply Voltage	V^+			4	–	20	V
Input Voltage	V_I			–	–	220	mV
Input Sensitivity	e_I	P_O = 6W, R_L = 4Ω, R1 = 56Ω, f = 1 kHz		–	80	–	mV
Quiescent Output Voltage	V_O			6.4	7.2	8	V
Quiescent Current Drain	I_O			–	12	20	mA
Input Noise Voltage	e_N	R_g = 0, BW (−3 dB) = 20 to 20,000 Hz		–	2	–	μV
Bias Current	I_{IB}			–	0.4	–	μA
Output Power	P_O	f = 1 kHz, R_L = 4Ω, THD = 10%	V^+ = 14.4 V	–	6	–	W
			V^+ = 6 V	–	1	–	
Input Resistance	R_I			–	5	–	MΩ
Total Harmonic Distortion	THD	P_O = 50 mW to 3W, R_L 4Ω, f = 1 kHz		–	0.3	–	%
Open-Loop Voltage Gain	A_{OL}	R_L = 4Ω, f = 1 kHz		–	80	–	dB
Closed-Loop Voltage Gain	A	R_L = 4Ω, f = 1 kHz, R1 = 56Ω		34	37	40	dB
Efficiency	η	P_O = 5W, R_L = 4Ω; f = 1 kHz		–	70	–	%

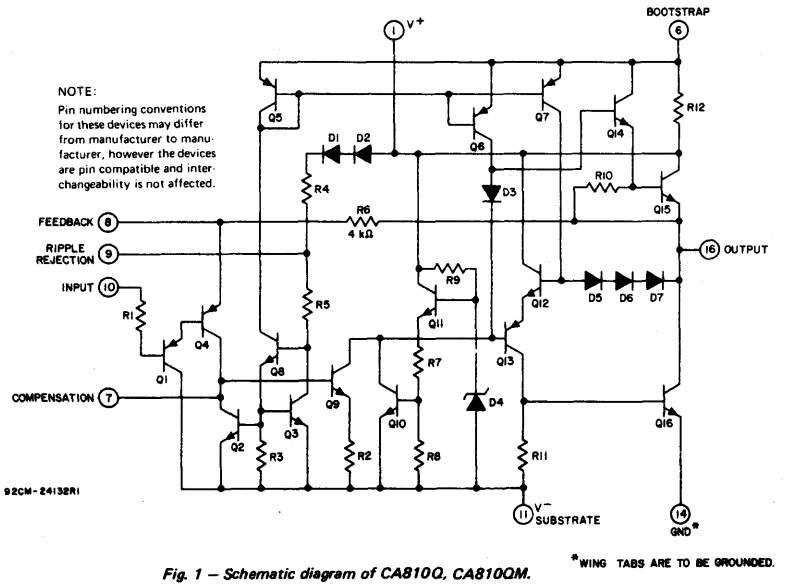


Fig. 1 – Schematic diagram of CA810Q, CA810QM.

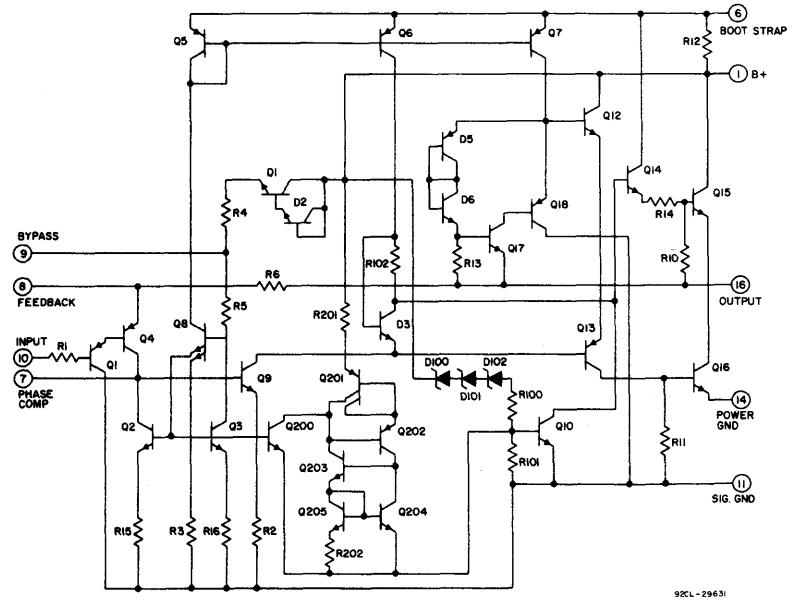


Fig. 3 – Schematic diagram of CA810AQ, CA810AQM.

Thermal Shut-Down

The thermal-limiting network incorporated in the CA810 Series circuits provides protection against damage due to excessive semiconductor temperatures that may result from high ambient temperatures and/or excessive dissipation, e.g., as encountered in sustained overloads. As indicated in Fig.2 the thermal-limiting feature automatically reduces the supply current (and output power) at the higher temperatures.

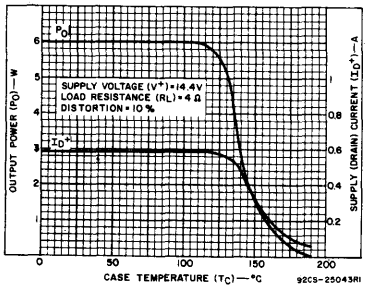


Fig. 2 – Typical output power and drain current as a function of case temperature for all types.

Load-Dump Voltage-Surge Protection

The maximum operating supply voltage of the CA810AQ and CA810AQM is 20 V, and internal protection is provided for peaks of up to 40 V, as shown in Fig. 4. Supply-voltage peaks of more than 40 V will require an LC network between the supply and terminal 5. An LC network, such as the one shown in Fig. 8, provides protection against supply-voltage surges of up to 120 V for 2 ms. This type of protection is ON when the supply voltage (pulsed or dc) exceeds 20 V.

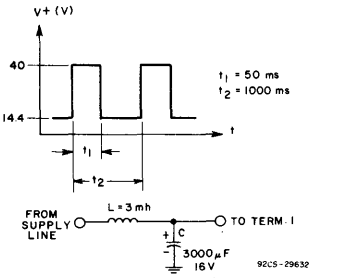


Fig. 4 – Load-dump (overvoltage) voltage surge protection network and timing diagram for CA810AQ and CA810AQM.

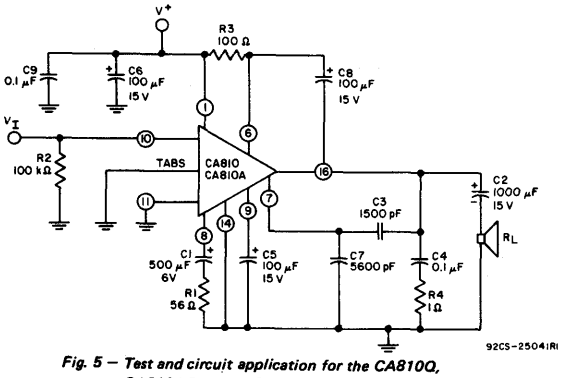
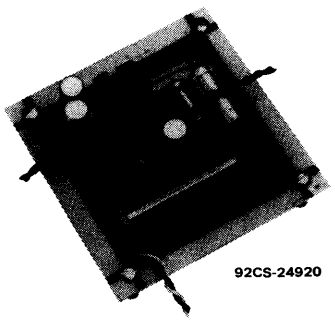


Fig. 5 – Test and circuit application for the CA810Q, CA810AQ and CA810M, CA810AQM.

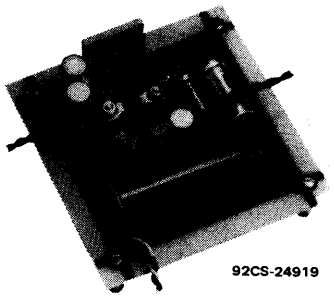


Fig. 6 – Bottom view of printed-circuit boards shown in Figs.7 and 8.



Circuit heat is dissipated by a combination of free air and printed-circuit board foil.

Fig. 7 – Component view of printed-circuit board for CA810Q and CA810AQ.



Circuit arrangement for use with chassis having a thermal resistance of $\leqslant 5^{\circ}C/W$. Vertical bracket should make good thermal contact to chassis.

Fig. 8 – Component view of printed-circuit board for CA810QM and CA810AQM.

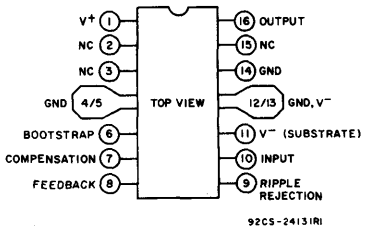


Fig. 9 – Terminal diagram of CA810Q, CA810AQ and CA810QM, CA810AQM. The wing tabs on the CA810Q and CA810AQ are bent down, and on the CA810QM and CA810AQM they are flat and pierced.

TV Horizontal Oscillator

For Color and Monochrome Receivers

The RCA-CA920AE* is a silicon monolithic integrated circuit intended for use in the horizontal stages of color and monochrome television receivers. This device performs the functions of a sync separator, noise gate, and horizontal oscillator with dual-time-constant switching in the fly-wheel loop. It also generates automatic phase control between horizontal flyback pulses and the horizontal oscillator frequency and provides fast edge switching drive for transistor or thyristor horizontal output stages.

The CA920AE is compatible with the industry type TBA920 in both lead arrangement and electrical operation, although the CA-920AE features reduced operating current.

The CA920AE is supplied in the 16-lead dual-in-line plastic package.

*Formerly Dev. Type No. TA6773.

Features:

- **Sync separator**
- **Noise gate input**
- **Internal precision timing ramp**
- **Dual-time-constant phase-locked loop**
- **Output suitable for transistor or thyristor deflection systems**
- **Reduced power dissipation**

MAXIMUM RATINGS, *Absolute Maximum Values:*

DC SUPPLY VOLTAGE	13.2 V
DEVICE DISSIPATION:	
Up to $T_A = 55^\circ C$	750 mW
Above $T_A = 55^\circ C$	Derate linearly at 7.9 mW/°C
AMBIENT TEMPERATURE RANGE:	
Operating	−40 to +85°C
Storage	−65 to +150°C
LEAD TEMPERATURE (During soldering):	
At a distance not less than 1/32" (0.79 mm) from case for 10 seconds max.	+265°C

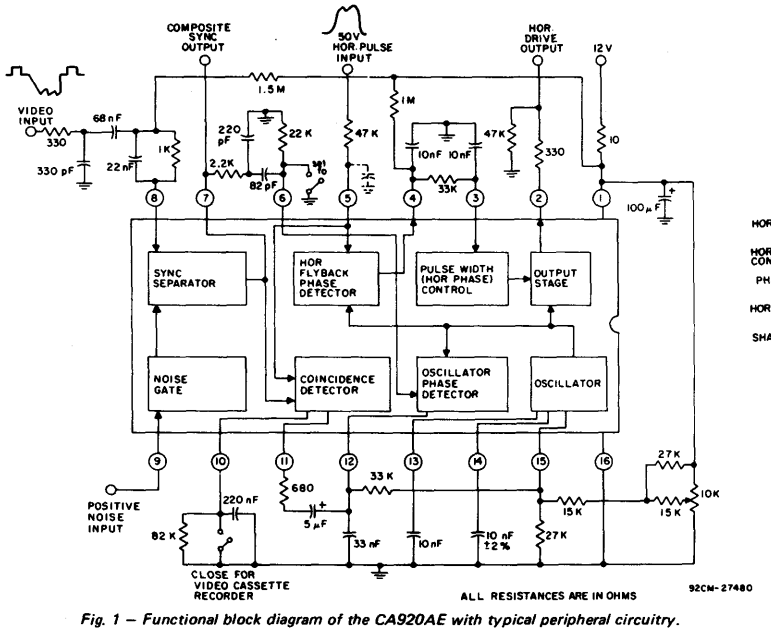
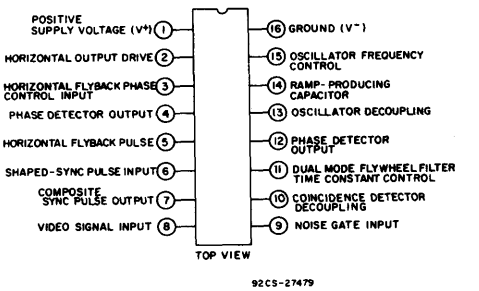


Fig. 1 – Functional block diagram of the CA920AE with typical peripheral circuitry.

TERMINAL ASSIGNMENT



ELECTRICAL CHARACTERISTICS at T_A = 25°C, and Supply Voltage (V^+) = 12 V, Unless otherwise specified. See Fig. 1.

CHARACTERISTIC		TEST CONDITIONS	LIMITS			UNITS
			Min.	Typ.	Max.	
Supply Current, Term. 1, I^+		Term. 2 open		22		mA
Video Characteristics (Term.8):						
Input Voltage	V_8	Peak to peak	1.5	3	6	V
Input Current	I_8	Peak			10	mA
Noise Gate Characteristics (Term.9):						
Input Current	I_9		0.03		10	mA
Reverse Input Current	I_9				−10	mA
Horizontal Flyback Positive Pulse Characteristics (Term.5):						
Input Voltage	V_5		1		3	V
Input Current	I_5		0.05	1	10	mA
Input Impedance	Z_5			0.4		kΩ
Positive Sync Characteristics (Term.7):						
Output Voltage	V_7	Peak to peak		10		V
Output Impedance	Z_7	Leading edge		50		Ω
Output Impedance	Z_7	Trailing edge		100		Ω
Horizontal Output Characteristics (Term.2):						
Output Current	I_2MAX	Peak			200	mA
Output Current	I_2AV	Average			20	mA
Output Pulse Width	t_W		12		32	μs
Output Impedance	Z_2	Leading edge		2.5		Ω
Output Impedance	Z_2	Trailing edge		15		Ω
Horizontal Oscillator Characteristics (Term.15):						
Free-Running Frequency	f_o	No sync input	14.84	15.625	16.41	kHz
Free-Running Frequency	f_o	V^+ = 4.5 V	14.06	(Note 1)	17.19	kHz
Oscillator Cut-out Voltage		V^+ varied		4.0		V
Oscillator Pull-in Range				±1.0		kHz
Phase Control (Note 2)					15	μs

Note 1: Free-running frequency at 12 V adjusted to 15.625 kHz.

Note 2: External delay between the leading edge of output pulse at Term. 3 and the start of the horizontal flyback pulse.

TV Sound IF and Audio Output Subsystems

"GQ" Suffix Type — Hermetic Gold-CHIP in Quad-In-Line Plastic Package

Features:

- **Nominal power output: 4 W at V^+=24 V, R_L=16Ω, dist. = 10%; 2 W at V^+=12 V, R_L = 8Ω, dist.=10%**
- **Wide power-supply range: 9 to 28 V**
- **Low quiescent current: 25 mA typ.**
- **5-kHz deviation sensitivity: 1 W output typ.**
- **3-dB limiting sensitivity: 50 μV typ.**
- **Excellent AM rejection: 50 dB typ.**
- **Differential peak detector — requires one tuned coil**
- **Electronic volume control with improved taper and single wire control**

The RCA-CA1190GQ combines the sound IF and audio output subsystems on a single monolithic integrated circuit to provide a television sound system. Each device includes a multistage IF amplifier-limiter, an FM detector, and an audio power amplifier that is designed to drive, primarily, an 8-, 16-, or 32-ohm speaker.

The CA1190GQ is electrically and mechanically equivalent to industry type TDA1190Z. The CA1190GQ differs from the TDA1190Z primarily in its provisions for external feedback components and a higher value volume control.

The CA1190GQ is supplied in the hermetic Gold-CHIP (G suffix) 16-lead quad-in-line plastic package with an integral bent-down wing-tab heat sink (Q suffix), intended for printed circuit board mounting.

The transistor chips used in the hermetic Gold-CHIP plastic package are of the sealed-junction type designed to provide protection against the deteriorating effects of humidity and other surface contaminants without the need for a hermetic package enclosure. The semiconductor junctions are sealed by utilizing a silicon nitride passivation layer. A multi-layered, highly corrosion-resistant, terminal-connection system of unique design is employed.

ELECTRICAL CHARACTERISTICS at T_A = 25°C, V^+ = 24 V, DC Volume Control Rx = 0 Ω, R_L = 16 Ω unless otherwise indicated. Refer to Fig. 1.

CHARACTERISTIC	TEST CONDITIONS	LIMITS Min.	LIMITS Typ.	LIMITS Max.	UNITS
Static Characteristics					
Current into Term. 14	P_o = 0	10	25	40	mA
Dynamic Characteristics					
IF Amplifier: Input Limiting Voltage, (At −3 dB point), V_1 (lim)	f_o = 4.5 MHz, f_m = 400 Hz, Δf = ± 25 kHz	–	50	100	μV
AM Rejection, AMR	f_o = 4.5 MHz, f_m = 400 Hz, Modulation Index = 0.3, V_{IN} = 1 mV	40	50	–	dB
Deviation Sensitivity	f_o = 4.5 MHz, f_m = 400 Hz, Δf = ± 25 kHz, V_I = 1 mV, Rx = 0, Deviation necessary to obtain 4 Vrms across 16 Ω (1 W)	–	5	–	kHz
Minimum Audio Output	f_o = 4.5 MHz, f_m = 400 Hz, Δf = ± 25 kHz, V_I = 1 mV, Rx = 15 kΩ	–	–	10	mVrms
Distortion at P_o = 1.5 W	f_o = 4.5 MHz, f_m = 400 Hz, Δf = ± 25 kHz, V_{IN} = 1 mV	–	–	3	%
Signal to Noise Ratio	V_{out} at Δf = 0 with Rx adjusted for V_{out} = 4 Vrms at Δf = ± 25 kHz	50	–	–	dB

MAXIMUM RATINGS, *Absolute-Maximum Valu*

DC SUPPLY VOLTAGE (Between Term. 14 V^+ and ground tabs)		+28	V
OUTPUT PEAK CURRENT:			
Repetitive		1.5	A
Non-repetitive		2	A
INPUT SIGNAL VOLTAGE (Between Terms. 1 and 2)		±3	V
DEVICE DISSIPATION:			
With Infinite Heat Sink —			
Up to T_A = 90°C	derate linearly	5	W
Above T_A = 90°C		83.3	mW/°C
With No Heat Sink — (free air) —			
Up to T_A = 25°C		1.75	W
Above T_A = 25°C	derate linearly	14	mW/°C
THERMAL RESISTANCE:			
Junction to ground tabs		12	°C/W
AMBIENT TEMPERATURE RANGE:			
Operating		−40 to +85	°C
Storage		−65 to +150	°C
LEAD TEMPERATURE (During Soldering): At a distance 1/16 in. ± 1/32 in. (1.59 ± 0.79 mm) from case for 10 seconds max.		+265	°C

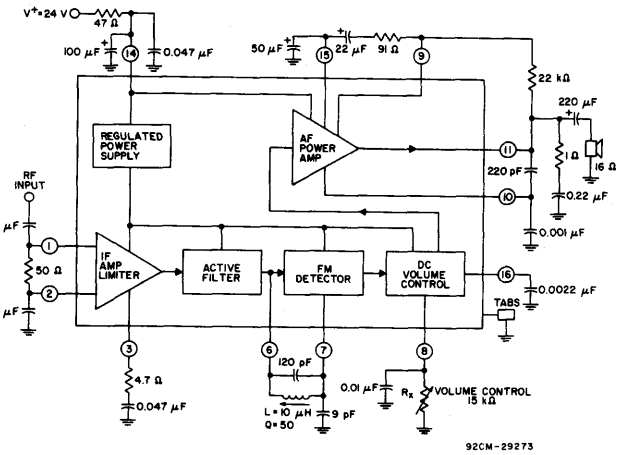


Fig.1 — Block diagram of the CA1190GQ in a typical application.

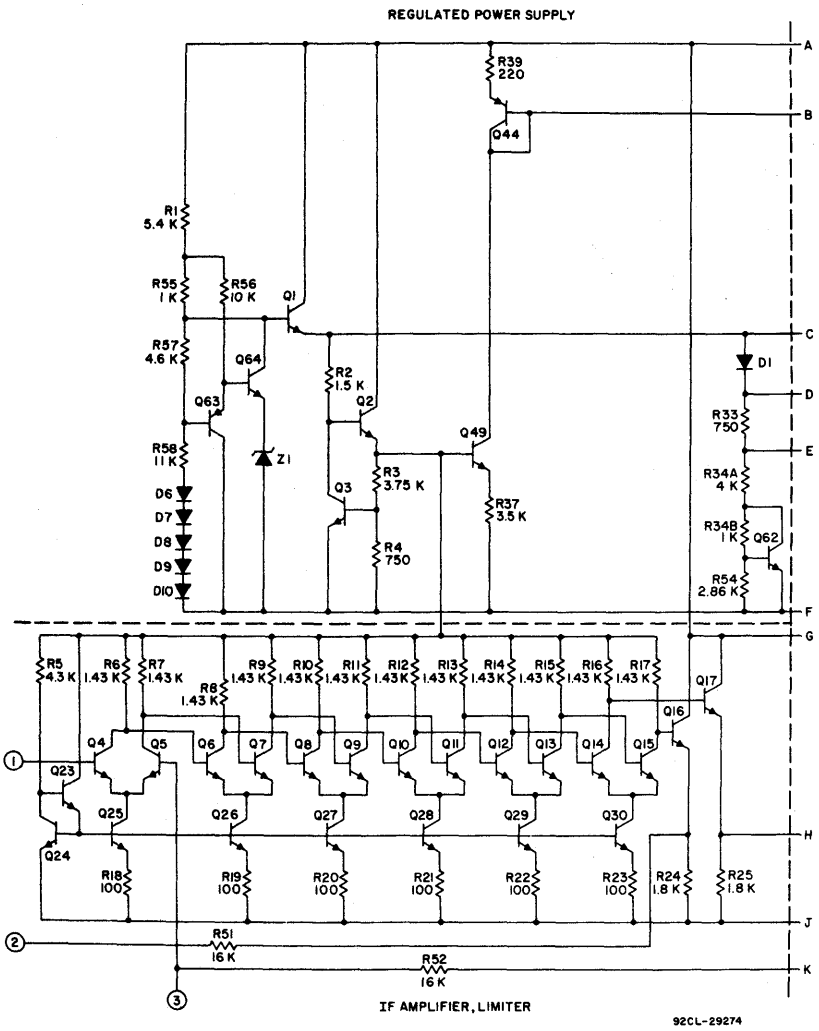


Fig.2 – Schematic diagram.

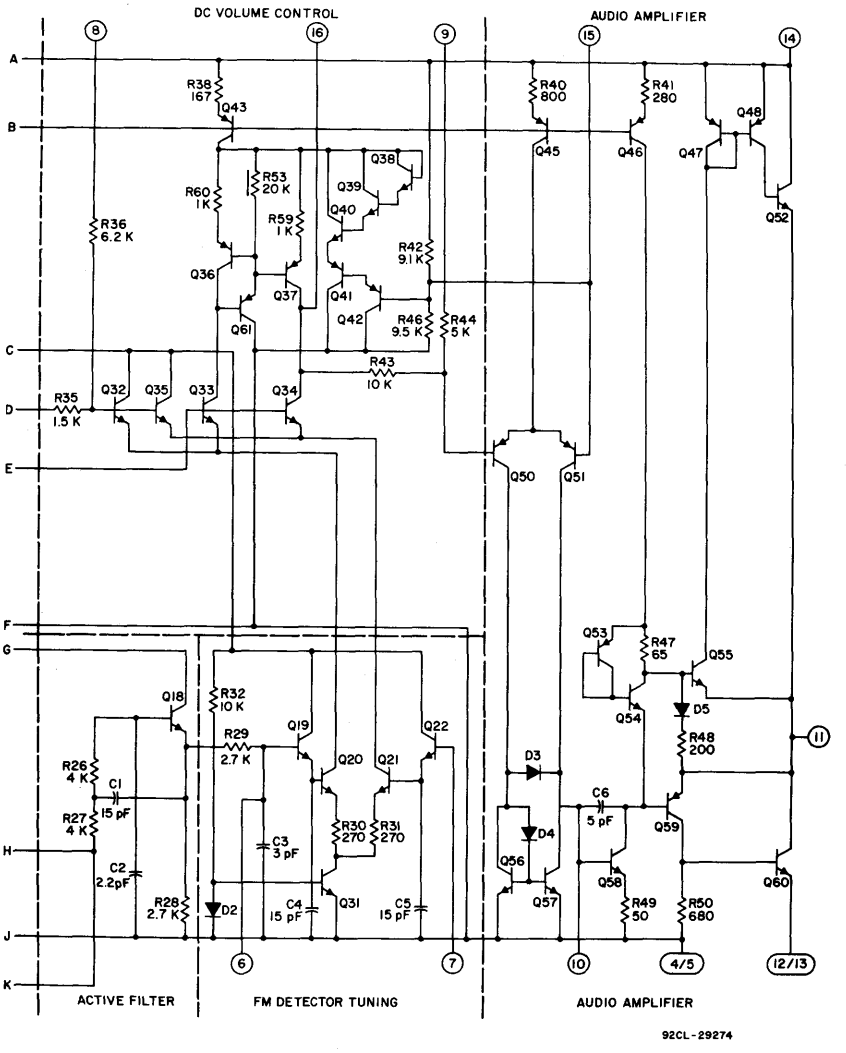


Fig.2 – Schematic diagram (cont'd).

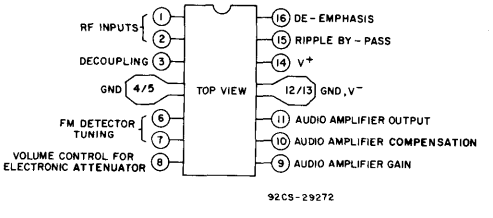


Fig.3 – Terminal diagram.

CA1310E

RC Phase-Locked-Loop Stereo Decoder

For FM Multiplex Systems

Features:

- **Low distortion (THD): 0.3% typ.**
- **Excellent SCA (storecast) rejection: 75 dB typ.**
- **RC oscillator**
- **High audio channel separation: 40 dB**
- **Operates from a wide range of power supplies: 8 to 14 V dc**
- **Requires only one adjustment for complete alignment**
- **Drives a stereo indicator lamp up to 75 mA — surge current limiting**

RCA-CA1310E is a monolithic silicon integrated circuit RC phase-lock-loop stereo decoder intended for FM solid-state stereo multiplex systems.

The CA1310E is a direct replacement for industry types MC1310P, LM1310, and SN76115N.

This decoder uses a minimum of external components. In addition the stereo decoder requires only one adjustment (*oscillator frequency*) for complete alignment.

The CA1310E is supplied in a 14-lead dual-in-line plastic package and operates over an ambient temperature range of −40 to +85°C.

MAXIMUM RATINGS, *Absolute-Maximum Values at T_A = 25°C*

DC Supply Voltage	14 V
Current (Lamp) at Term. 6	75 mA
Device Dissipation:	
Up to T_A = 25°C	625 mW
Above T_A = 25°C derate linearly	5 mW/°C
Ambient Temperature Range:	
Operating	−40 to +85°C
Storage	−65 to +150°C
Lead Temperature (During soldering):	
At distance not less than 1/32" (0.79 mm) from case for 10 s max.	+265°C

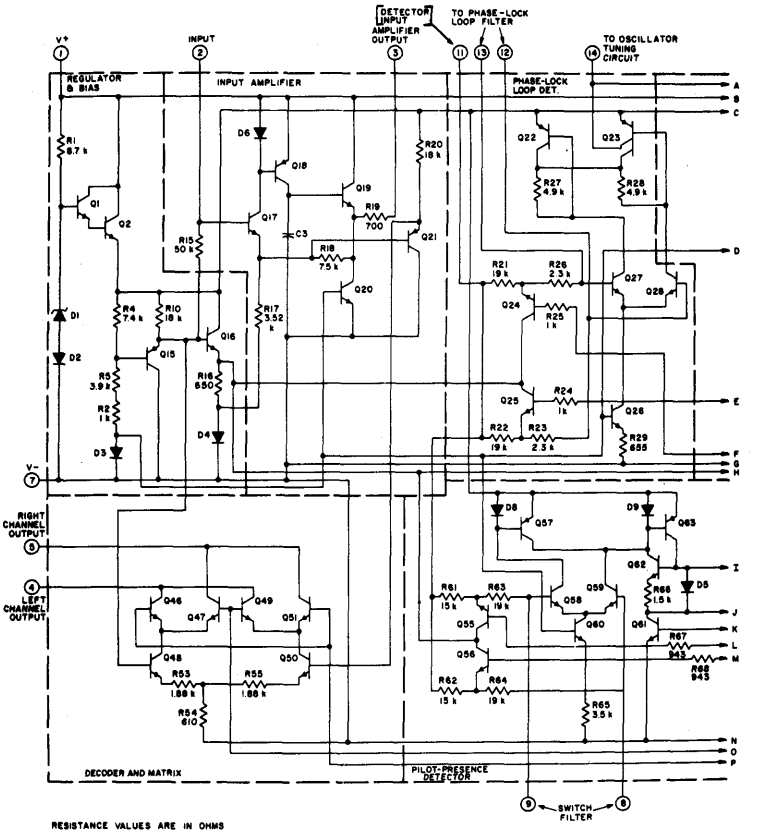


Fig. 2 – Schematic diagram of the CA1310E.

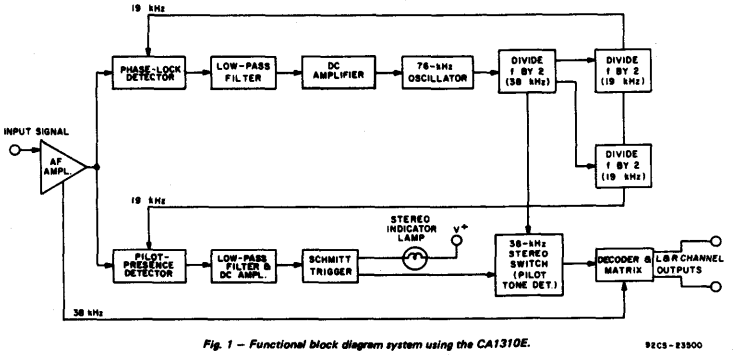


Fig. 1 – Functional block diagram system using the CA1310E. 92CS-23500

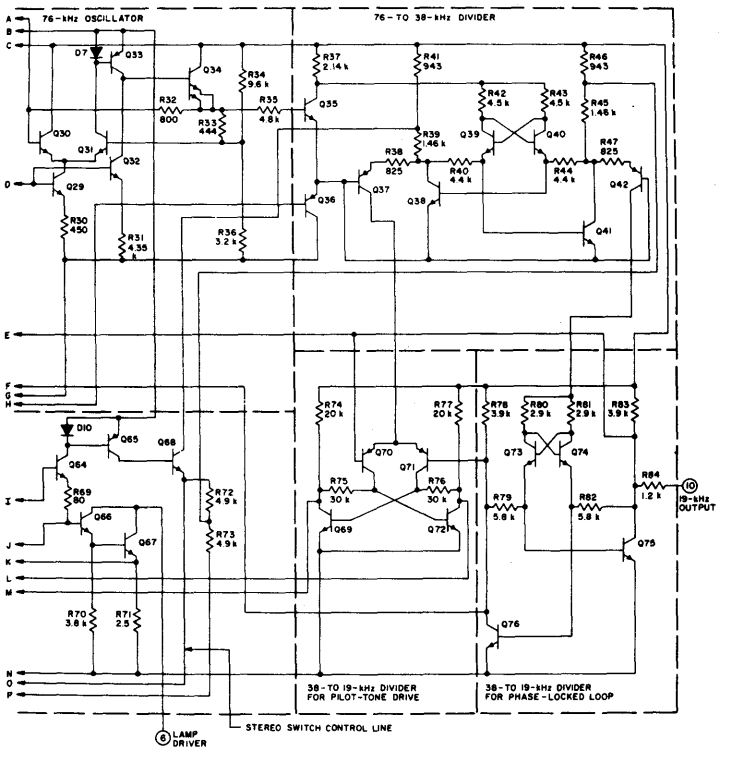
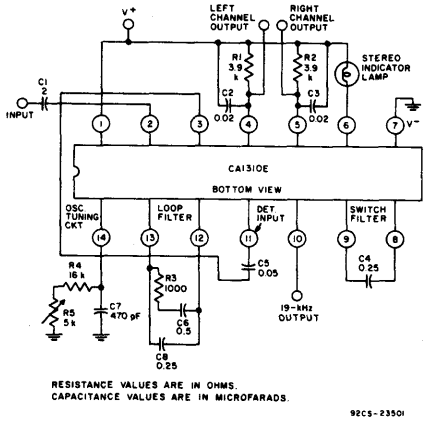


Fig. 2 – Schematic diagram of the CA1310E (Cont'd).



NOTES

A buffered 3-volt positive-going square wave is available at Term. 10. The alignment of the free-running oscillator frequency may be checked at this point with a frequency counter.

C1: A lower value input coupling capacitor may be used in place of the 2-μF value if reduced separation at low frequencies is acceptable.

C4: The time constant for the stereo switch level detector circuit is calculated by C4 x 53,000 ohms ±30% with a maximum dc voltage drop across C4 of 0.25 volt (Term. 8 positive) and a pilot level voltage of 100 mV RMS. Signal voltage across C4 is negligible.

C5: The recommended 0.05-μF capacitor provides a 1.75° phase lead at 19 kHz.

R1, R2: Load resistance values are related to supply voltage as follows:

Minimum Supply Voltage	8	10	12	V
Maximum Load Resistance	2.7	4.3	6.2	kΩ

R3, C6, C8: C8 may be omitted, R3 = 100 ohms and C6 = 0.25 μF, if relaxed circuit performance is acceptable.

R4, R5, C7: If a capture range greater than ±3% typ. is required, reduce value of C7 and increase values of R4, R5 proportionally. However, beat-note distortion is increased at high signal levels because of oscillator-phase jitter. R4, C7 = ±1% in test circuit and ±5% in typical application.

Fig. 3 – Test circuit for measurement of dynamic characteristics.

ELECTRICAL CHARACTERISTICS

CHARACTERISTIC	TEST CONDITIONS (Referenced to Fig. 3) $V^+ = 12$ V $T_A = 25^oC$ Composite Multiplex Input Signal = 560 mV RMS (2.8 V p-p) Only L or R Channel modulated; and with 100-mV RMS (10%) Pilot Level	LIMITS			UNITS
		Min.	Typ.	Max.	
Static Characteristics					
DC Supply Voltage	For 8-V operation, reduce load to 2.7 kΩ	8	–	14	V
Total Current	Lamp "OFF"	–	13	–	mA
Dynamic Characteristics					
Input Impedance		20	50	–	kΩ
Channel Separation (Stereo)	50 Hz – 15 kHz	30	40	–	dB
Audio Output Voltage (For any one channel)		–	485	–	mV RMS
Channel Balance (Monaural)	Pilot Tone "OFF"	–	–	1.5	dB
Capture Range (Permissible tuning error of internal oscillator)		–	±3.5	–	%
Total Harmonic Distortion		–	0.3	–	%
Ultrasonic Frequency Rejection: 19 kHz		–	34.4	–	dB
38 kHz		–	45	–	dB
SCA (Storecast) Rejection	f = 67 kHz, 9-kHz beat note measured with 1-kHz modulation "OFF"	–	75	–	dB
Stereo Switch Level: 19-kHz Input Level (For lamp on)		–	–	20	mV RMS
19-kHz Input Level (For lamp off)		5	–	–	mV RMS
Maximum Composite (Stereo) Input	0.5% THD	2.8	–	–	V p-p
Maximum Monaural Input	1% THD	2.8	–	–	V p-p

TV Video IF Amplifier

With AGC and Keyer Circuit

The RCA-CA1352E is a monolithic integrated circuit designed for use as an if amplifier in monochrome or color TV receivers. It features a high-gain gated AGC system with a 68-dB range (typ.). A delayed forward AGC output is adjustable by means of a potentiometer. Either positive- or negative-going sync may be used for this system.

The CA1352E is supplied in the 14-lead dual-in-line plastic package, and is directly interchangeable with the industry type 1352 in similar packages.

Features

- **High 45-MHz gain – 53 dB (typ.)**
- **High-gain gated AGC system – with either positive- or negative-going sync.**
- **Adjustable rf AGC delay to tuner**
- **AGC gain reduction – 68 dB (typ.)**

TYPICAL STATIC CHARACTERISTICS
at $T_A = 25°C$, $V^+ = 12$ V

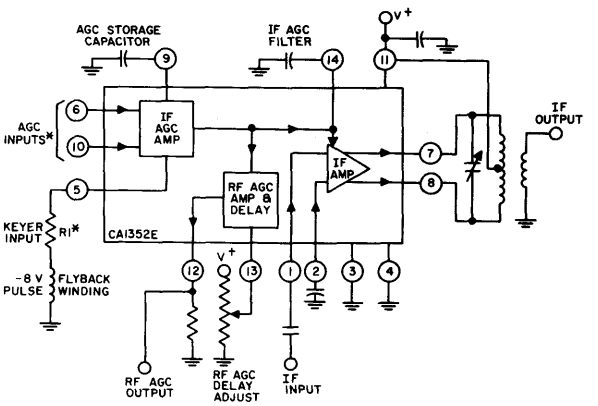
Total Current ($I_7 + I_8 + I_{11}$)	27 mA
Output Stage Current ($I_7 + I_8$)	5.7 mA

TYPICAL DYNAMIC CHARACTERISTICS
at $T_A = 25°C$, $V^+ = 12$ V

AGC Range	68 dB
Power Gain	53 dB
Minimum rf AGC Range (term. 12)	0.2 V
Maximum rf AGC Range (term. 12)	7 V

MAXIMUM RATINGS, *Absolute-Maximum Values*
At $T_A = 25$

SUPPLY VOLTAGE:	
Between terminals 4 and 11	18 V
Between terminals 7 or 8 and 4	18 V
INPUT VOLTAGE (terminal 1 or 2)	10 V p-p
AGC INPUT VOLTAGE (terminal 6 or 10)	6 V
DEVICE DISSIPATION:	
Up to $T_A = 55°C$	750 mW
Above $T_A = 55°C$ derate linearly at	7.9 mW/°C
AMBIENT TEMPERATURE RANGE:	
Operating	−40 to +85°C
Storage	−65 to +150°C
LEAD TEMPERATURE (During Soldering):	
At distance 1/16 ± 1/32 in. (1.59 ± 0.79 mm) from case for 10 seconds max.	+265°C



SYNC POLARITY	* VOLTAGE AT TERMINAL 6	* VOLTAGE AT TERMINAL 10	* VALUE OF RI – Ω
NEGATIVE	5.5 V, 2 V, 0 V	1 TO 4 V NOM = 2 V	0
POSITIVE	1 TO 8 V NOM = 4.5	4.5 V, 0 V	3.9k

92CS-24136R1

Fig. 1 – CA1352E block diagram and typical AGC test set-up.

CA1391E, CA1394E

TV Horizontal Processors

CA1391E – Positive Horizontal Sawtooth Input
CA1394E – Negative Horizontal Sawtooth Input

Features:

- **Internal shunt regulator**
- **Linear balanced phase detector**
- **Preset hold control capability**
- **±300-Hz pull-in (typ.)**
- **Low thermal frequency drift**
- **Small static phase error**
- **Variable output duty cycle**
- **Adjustable dc loop gain**

The RCA-CA1391E and CA1394E are monolithic integrated circuits designed for use in the low-level horizontal section of monochrome or color television receivers. Functions include a phase detector, an oscillator, a regulator, and a pre-driver.

The CA1391E and CA1394E are electrically equivalent and pin compatible with industry types 1391 and 1394 in similar packages.

These types are supplied in an 8-lead dual-in-line plastic (Mini-DIP) package, and operate over an ambient temperature range of 0 to +85°C.

MAXIMUM RATINGS, *Absolute-Maximum Values at T_A = 25°C*

DC SUPPLY CURRENT	40 mA
DC OUTPUT VOLTAGE	40 V
DC OUTPUT CURRENT	30 mA
SYNC INPUT VOLTAGE	5 V_{p-p}
SAWTOOTH INPUT VOLTAGE	5 V_{p-p}
DEVICE DISSIPATION:	
Up to T_A = 25°C	625 mW
Above T_A = 25°C derate linearly	5 mW/°C
AMBIENT TEMPERATURE RANGE:	
Operating	0 to +85°C
Storage	–65 to +150°C
LEAD TEMPERATURE (During Soldering):	
At distance 1/16 ± 1/32 in. (1.59 ± 0.79 mm) from case for 10 seconds max.	+260°C
THERMAL RESISTANCE	200°C/W

CIRCUIT OPERATION
(See schematic diagram, Fig.2)

The CA1391 and CA1394 contain the oscillator, phase detector, and predriver sections necessary for the television horizontal oscillator and AFC loop.

The oscillator is an RC type with terminal 7 used to control the timing. If it is assumed that Q7 is initially off, then an external capacitor connected from terminal 7 to ground charges through an external resistance connected between terminals 6 and 7. As soon as the voltage at terminal 7 exceeds the potential set at the base of Q8 by resistors R11 and R12, Q7 turns on, and Q6 supplies base current to Q5 and Q10. Transistor Q5 discharges the capacitor through R4 until the base bias of Q7 falls below that of Q8, at which time, Q7 turns off, and the cycle repeats.

The sawtooth generated at the base of Q4 appears across R3 and turn off Q3 whenever the sawtooth voltage rises to a value that exceeds the bias set at terminal 8. By adjusting the potential at terminal 8, the duty cycle at the pre-drive output (terminal 1) may be changed.

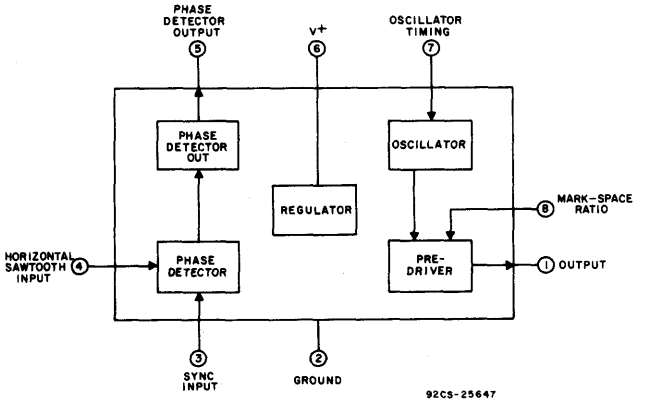


Fig.1 – Functional block diagram of the CA1391E, CA1394E.

ELECTRICAL CHARACTERISTICS at T_A = 25°C (See Fig.3)

CHARACTERISTIC	TEST CONDITIONS	LIMITS			UNITS
		Min.	Typ.	Max.	
Supply Voltage	S1, S5, S6 = 2 S2, S3, S4, S7, S8 = 1 Measure term. 6 to Gnd	8	–	9	V
Free-Running Frequency	S1, S5, S6 = 2 S2, S3, S4, S7, S8 = 1 Counter to term. 1	14734	–	16734	Hz
Output Leakage	S2, S3, S6, S8 = 1 S1, S4, S5, S7 = 2 Measure term. 1 to 25 V	–	10	–	mV
Output Saturation	S2, S3, S5, S6, S8 = 1 S1, S4, S7 = 2 Measure term. 1 to Gnd	–	60	–	mV
Phase Detector Bias	S2, S5, S6, S8 = 1 S1, S3, S4, S7 = 2 Measure term. 3 to Gnd	–	1.9	–	V
Phase Detector Leak	S5, S8 = 1 S1, S2, S3, S4, S6, S7 = 2 Measure term. 5 to +4 V	–2	–	+2	mV
Phase Detector Low	S1, S5, S8 = 1 S2, S3, S4, S6, S7 = 2 Measure term. 5 to +4 V	–0.55*	–	–	V
Phase Detector High	S1, S5, S6, S8 = 1 S2, S3, S4, S7 = 2 Measure term. 5 to +4 V	+0.55*	–	–	V
Phase Detector Balance	$V_{DET2} + V_{DET3}$	–100	–	+100	mV
Sync Diode	S1, S2, S3, S4, S6, S7 = 1 S5, S8 = 2	0.3	–	1.2	V
Static Phase Error	See Fig.4	–	0.5	–	μs
Oscillator Pull-in Range		–	±300	–	Hz
Oscillator Hold-in Range		–	±900	–	Hz

* Polarity reversed in the CA1391.

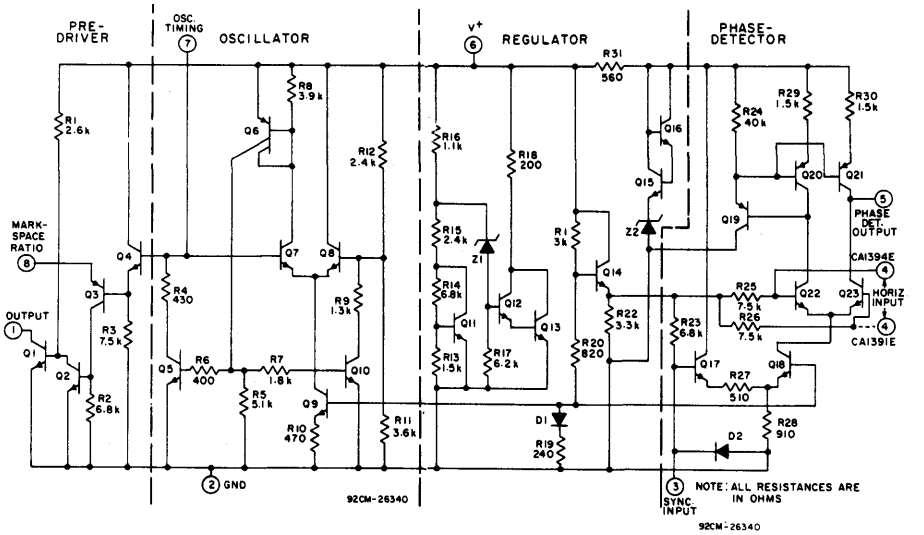


Fig.2 – Schematic diagram of CA1391E, CA1394E.

The phase detector is isolated from the remainder of the circuit by R31, Z2, Q15 and Q16. The phase detector consists of the comparator Q22 and Q23, and the gated current source Q18. Negative-going sync pulses at terminal 3 turn off Q17, and the current division between Q22 and Q23 is then determined by the phase relationship of the sync and the sawtooth waveform at terminal 4, which is derived from the horizontal flyback pulse. If there is no phase difference between the sync and sawtooth, equal currents flow in the collectors of Q22 and Q23 during each half of the sync pulse period. The current in Q22 is turned around by current mirror Q20 and Q21 so that there is no net output current at terminal 5 for balanced conditons. When a phase offset occurs, current flows either in or out of terminal 5. In circuit applications, this terminal is connected to terminal 7 through an external low-pass filter, thereby controlling the oscillator.

Shunt regulation for the circuit is obtained by using a V_{BE} and zener multiplier. Resistors R13 and R14 multiply the V_{BE} of Q11, and the ratio of R15 and R16 multiplies the voltage of the zener diode Z1.

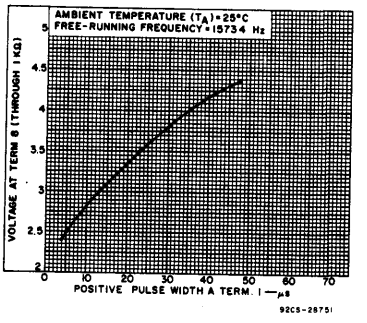


Fig.3 – Duty cycle at the pre-drive output (term.1) as it is affected by the input at term. 8.

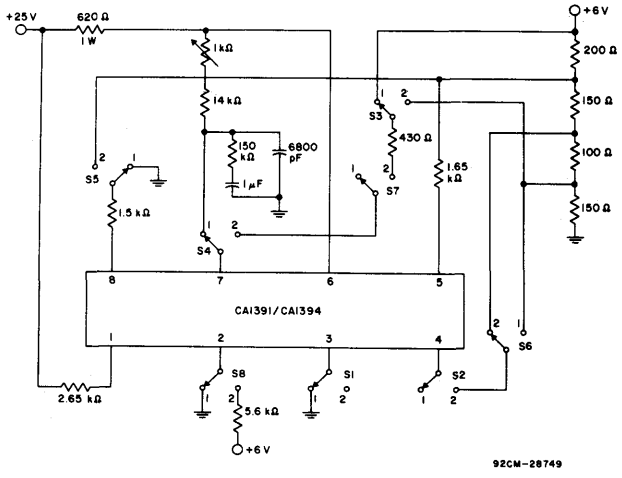


Fig.4 – DC test circuit.

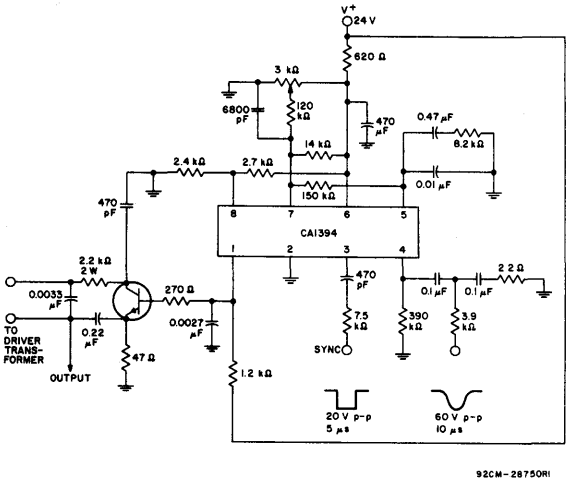


Fig.5 – Typical circuit application.

CA1398E

Television Chroma Processor

RCA-CA1398E is a monolithic silicon integrated-circuit chroma processor containing chroma-amplifier and gain-control, color-killer, color subcarrier oscillator, hue control, and ACC circuitry. It has been designed for interchangeability with other "1398"-type chroma-processor devices. It functions compatibly with the RCA-CA3125E Chroma Demodulator as well as other commercially available chroma demodulators in color-TV receivers. Fig. 2 shows a functional block diagram of a 2-package TV chroma system incorporating the CA1398E and CA3125E. The CA1398E is supplied in a 14-lead dual-in-line plastic package.

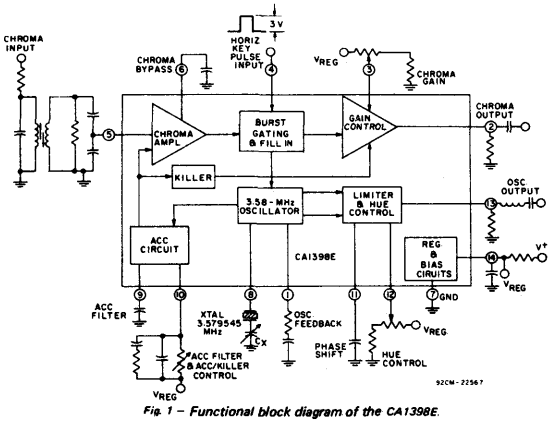


Fig. 1 – Functional block diagram of the CA1398E.

Features

- **Minimum number of external components required**
- **Injection-lock oscillator with internal feedback**
- **DC chroma gain control and hue control circuits**
- **Low-impedance internal voltage regulation**

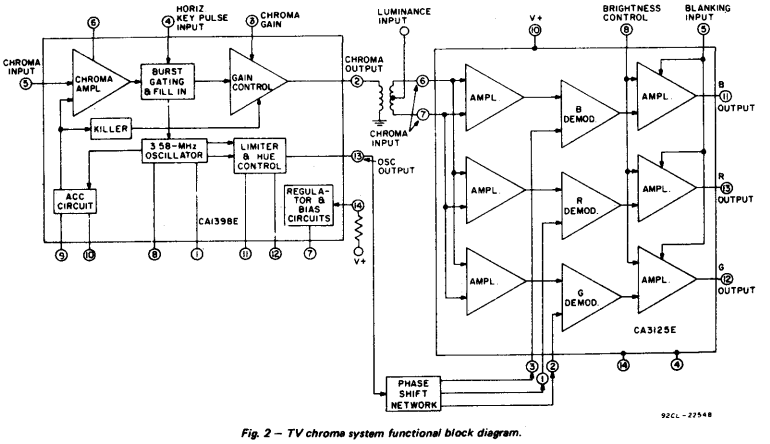


Fig. 2 – TV chroma system functional block diagram.

Maximum Ratings, *Absolute-Maximum Values at* $T_A = 25°C$

Peak Horizontal-Pulse Input Current	250 μA
Supply Current (Terminal 14)	35 mA
Ambient Temperature Range:	
Operating	−40 to +85°C
Storage	−65 to +150°C
Lead Temperature (During Soldering):	
At distance 1/16" ± 1/32" (1.59 ± 0.79 mm) from case for 10 s max.	265°C

ELECTRICAL CHARACTERISTICS at $T_A = 25°C$ and Referenced to Test Circuit (Fig. 4)

CHARACTERISTIC	TERMINAL MEASURED AND SYMBOL	TEST CONDITIONS: SWITCH POSITION (S1)	CONTROL SETTING: CHROMA	HUE	KILLER	V_{BURST} mV p-p	V_{CHROMA} mV p-p	LIMITS: MIN.	TYP.	MAX.	UNITS
Static Characteristics											
Regulated Supply Voltage	V_{14}	2	max.	max.	max.	0	0	8.9	9.5	11.5	V
Chroma Output Bias	V_{14} to V_2	2	max.	max.	max.	6	0	1.2	2.4	3.6	V
Regulator Impedance	See Note 1	2	max.	max.	max.	0	0	–	12	25	Ω
Dynamic Characteristics (Refer to Test Set-Up Procedure for Oscillator)											
Max. Chroma Gain	V_2	1	max.	max.	See Note 2	6	5	310	425	–	mV p-p
Min. Chroma Gain	V_2	1	min.	max.		6	5	–	–	7	mV p-p
ACC Action	V_2 (dB up from gain test)	1	max.	max.		50	50	2	7	11	dB
Killer Function: Kill	V_2	2	max.	max.		0	5	–	–	7	mV p-p
Unkill	V_2	1	max.	max.		15	5	100	–	–	mV p-p
Oscillator Lock-Up: Voltage	V_{13}	1	max.	max.		6	0	250	340	390	mV p-p
Phase (Referenced to burst)	ϕ_{13}	1	max.	max.		6	0	−20	0	+20	degrees
Hue Control Range: Voltage	V_{13}	1	max.	min.		6	0	250	340	390	mV p-p
Phase (Referenced to burst)	ϕ_{13}	1	max.	min.		6	0	95	110	140	degrees

Note 1 – Measure V_{14} at I_{SUPPLY} = 38 mA and 18 mA. Calculate the regulator impedance:
$Z_{reg.} = [V_{14}$ (at 38 mA) $- V_{14}$ (at 18 mA)$]/0.02$

Note 2 – Increase the killer potentiometer resistance from minimum until the circuit unkills. This condition is evidenced by a shift in bias voltage at Term. 2. Maintain this potentiometer setting for all the dynamic tests.

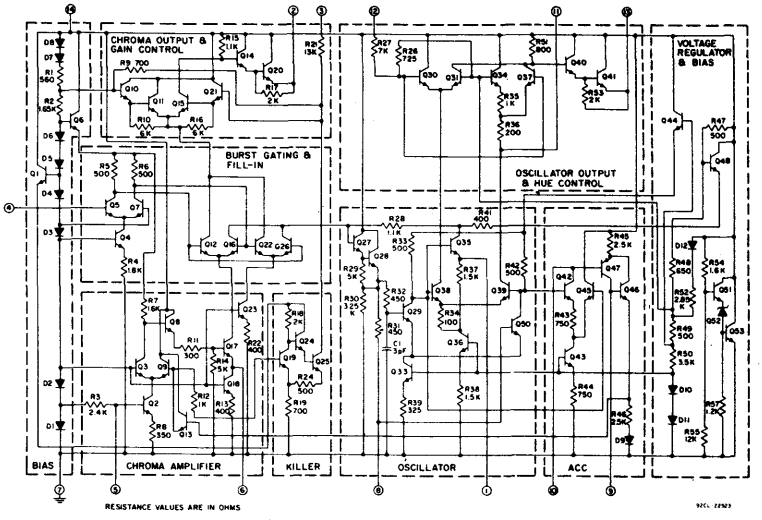


Fig. 3 – Schematic diagram of the CA1398E.

TEST SET-UP PROCEDURE FOR OSCILLATOR

Remove the horizontal keying and chroma inputs and adjust C_X to obtain a free-running oscillator frequency of 3.579545 MHz ±10 Hz. Under the same Test Conditions described in the Electrical Characteristics Chart for Oscillator Lock-Up, vary L1 (approx. 20 μH) and/or C1 (approx. 1000 pF) to obtain the initial conditions for amplitude and phase oscillator lock-up.

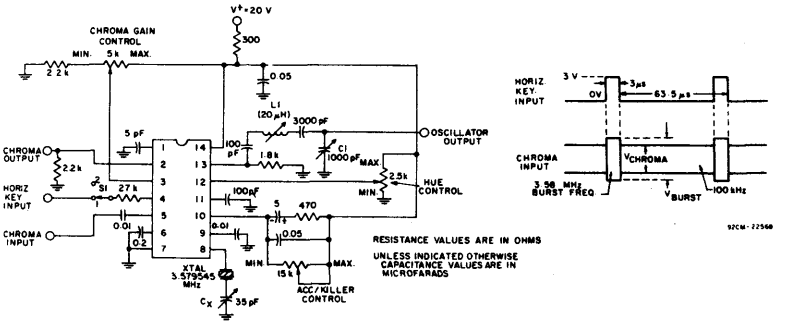
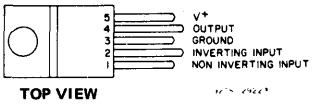


Fig. 4 – Typical static and dynamic characteristics test circuit for the CA1398E.

CA2002, CA2002M — Preliminary Data

8-Watt Audio Power Amplifier

For Automobile Radios



TOP VIEW

TERMINAL ASSIGNMENT

Features:

- **Hermetic Gold-CHIP encapsulated in a 5-lead plastic TO-220-style package (Versa-V)**
- **Output short-circuit and thermal overload protection**
- **Drives load impedance as low as 1.6 Ω**
- **Load dump voltage surge protection**
- **Output current capability of up to 3.5A**
- **Few external components**
- **Versa-V power transistor package-requires no electrical insulation**

The RCA-CA2002 is a monolithic silicon class B audio power amplifier designed for driving loads as low as 1.6 Ω. It provides a high output current capability (up to 3.5A), very low harmonic and cross-over distortion, and load-dump voltage-surge protection.

The CA2002 is supplied in a hermetic tri-metal Gold-CHIP encapsulated in the 5-lead plastic TO-220-style Versa-V package. All leads (except term. 3) are electrically insulated from the mounting flange, eliminating the need for insulating hardware.

The VERSA-V package is available with two lead configurations. The CA2002 has a vertical-mount lead form, and the CA2002M has a horizontal-mount lead form.

MAXIMUM RATINGS, *Absolute-Maximum Values:*

PEAK SUPPLY VOLTAGE (50 ms)	40 V
DC SUPPLY VOLTAGE	28 V
OPERATING SUPPLY VOLTAGE	18 V
OUTPUT PEAK CURRENT:	
REPETITIVE	3.5 A
NON-REPETITIVE	4.5 A
POWER DISSIPATION, P_D at $T_A = 90^\circ C$	15 W
THERMAL RESISTANCE, JUNCTION TO CASE	$4^\circ C/W$
AMBIENT-TEMPERATURE RANGE:	
OPERATING	See Figure 16
STORAGE	-40 to $+150^\circ C$
LEAD TEMPERATURE (DURING SOLDERING):	
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 12 s max.	$260^\circ C$

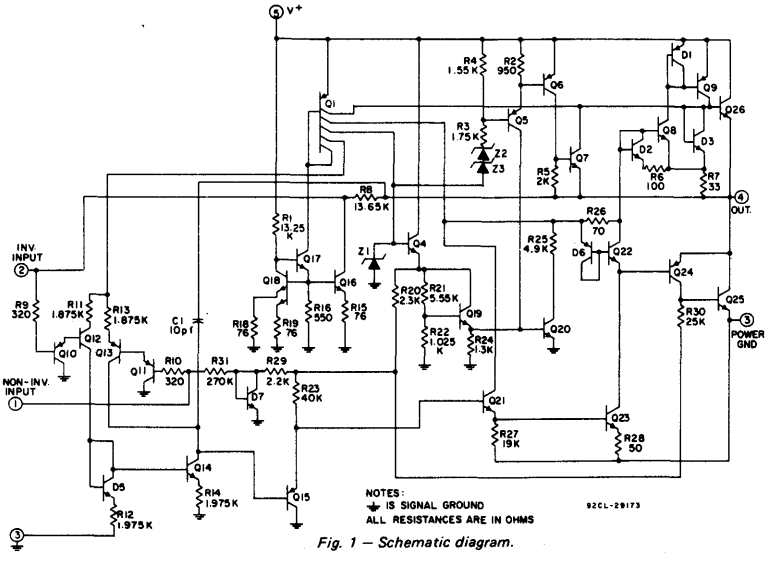


Fig. 1 – Schematic diagram.

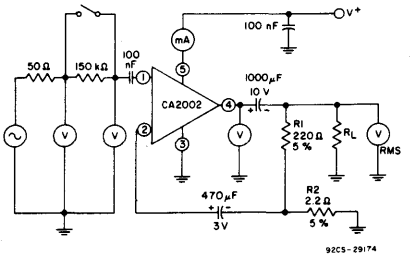


Fig. 2 – Test circuit.

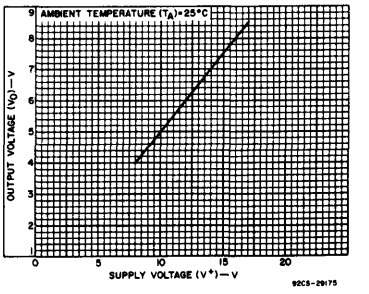


Fig. 3 – Typical quiescent output voltage as a function of supply voltage.

ELECTRICAL CHARACTERISTICS at T_A = 25°C, V^+ = 14.4 V
Unless otherwise specified (See Figure 2)

CHARACTERISTIC	TEST CONDITIONS	LIMITS Min.	Typ.	Max.	UNITS
Supply Voltage, V^+		8	–	18	V
Quiescent Output Voltage, V_O	Measure at Term. 4	6.4	7.2	8	V
Quiescent Drain Current, I_D	Measure at Term. 5	–	45	80	mA
Output Power, P_O	THD = 10%, A = 40 dB, f = 1 KHz; V^+ = 14.4V, R_L = 4 Ω	4.8	5.2	–	W
	V^+ = 14.4V, R_L = 2 Ω	7	8	–	W
	V^+ = 16 V, R_L = 4 Ω	–	6.5	–	W
	V^+ = 16 V, R_L = 2 Ω	–	10	–	W
Input Saturation Voltage, $V_{I(RMS)}$		400	–	–	mV
Input Sensitivity, e_I	A = 40 dB, f = 1 KHz; P_O = 0.5 W, R_L = 4 Ω	–	15	–	mV
	P_O = 0.5 W, R_L = 2 Ω	–	11	–	mV
	P_O = 5.2 W, R_L = 4 Ω	–	55	–	mV
	P_O = 8 W, R_L = 2 Ω	–	50	–	mV
Frequency Response (–3 dB)	R_L = 4 Ω, C_X = 39 nF, R_X = 39 Ω (See Figs.15,20)	40 to 15000			Hz
Input Resistance, R_I (Term. 1)	f = 1 KHz	70	150	–	KΩ
Open-Loop Voltage Gain, A_{OL}	R_L = 4 Ω, f = 1 KHz	–	80	–	dB
Closed-Loop Voltage Gain, A	R_L = 4Ω, f = 1 KHz	39.5	40	40.5	dB
Input Noise Voltage, e_N	Freq. Resp. = 40 to 15,000 Hz (–3 dB)	–	4	–	μV
Input Noise Current, i_N	Freq. Resp. = 40 to 15,000 Hz (–3 dB)	–	60	–	pA
Efficiency, η	A = 40 dB, f = 1 KHz; P_O = 5.2 W, R_L = 4 Ω	–	68	–	%
	P_O = 8 W, R_L = 2 Ω	–	58	–	%
Power Supply Rejection Ratio, PSRR	R_L = 4 Ω, A = 40 dB, Rg = 10 KΩ, f_{ripple} = 100 Hz, V_{ripple} = 0.5 V	30	35	–	dB

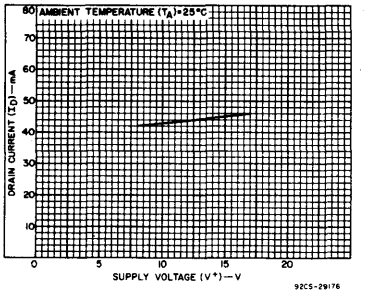


Fig. 4 – Typical quiescent drain current as a function of supply voltage.

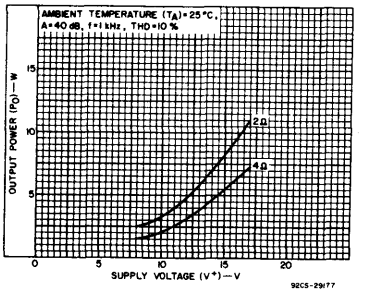


Fig. 5 – Typical output power as a function of supply voltage.

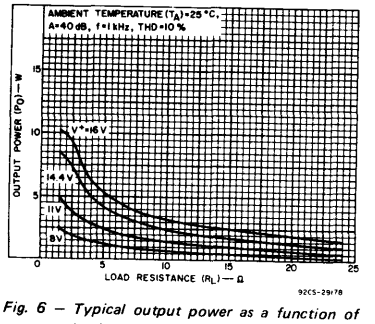


Fig. 6 – Typical output power as a function of load resistance.

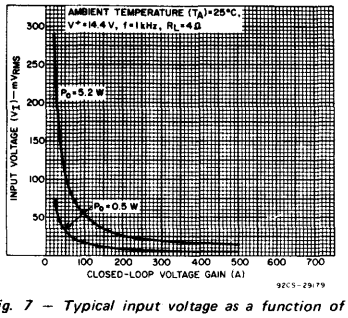


Fig. 7 – Typical input voltage as a function of closed-loop voltage gain.

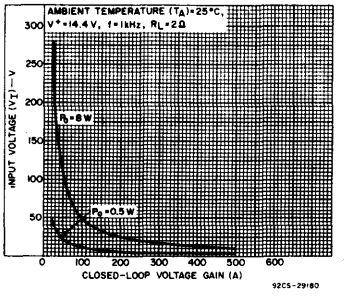


Fig. 8 – Typical input voltage as a function of closed-loop voltage gain.

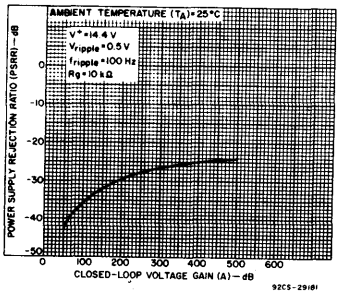


Fig. 9 – Typical power supply rejection ratio as a function of closed-loop voltage gain.

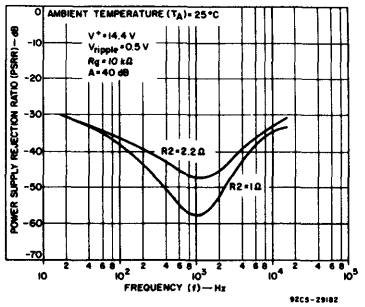


Fig. 10 – Typical power supply rejection ratio as a function of frequency.

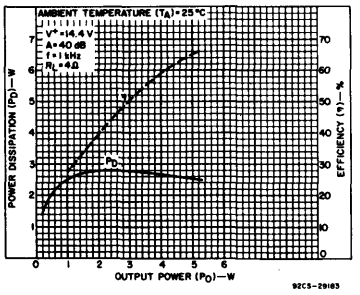


Fig. 11 – Typical power dissipation and efficiency as a function of output power.

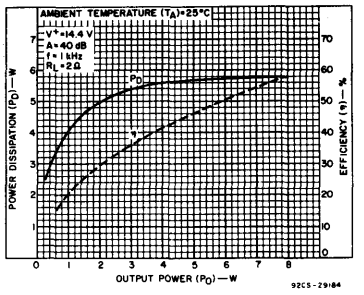


Fig. 12 – Typical power dissipation and efficiency as a function of output power.

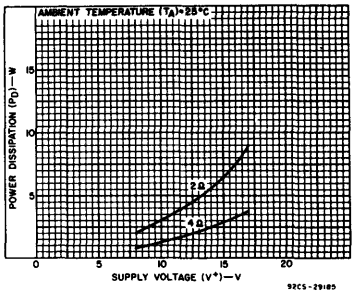


Fig. 13 – Maximum power dissipation as a function of supply voltage (sine-wave operation).

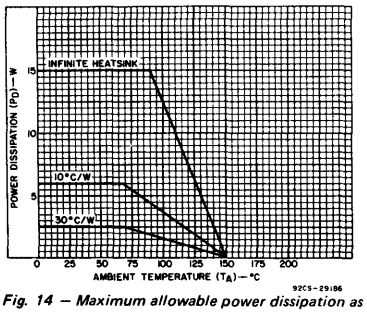


Fig. 14 – Maximum allowable power dissipation as a function of ambient temperature.

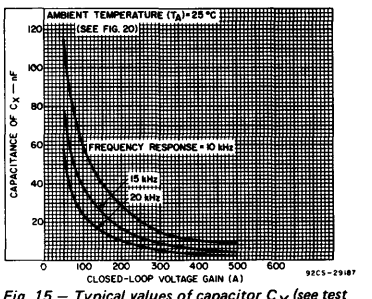


Fig. 15 – Typical values of capacitor C_X (see test circuit, figure 20) for different values of frequency response..

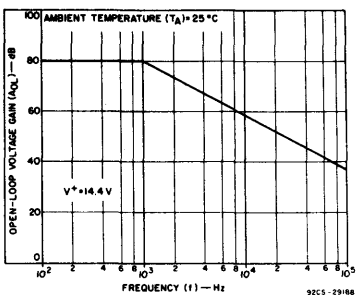


Fig. 16 – Open-loop voltage gain as a function of frequency.

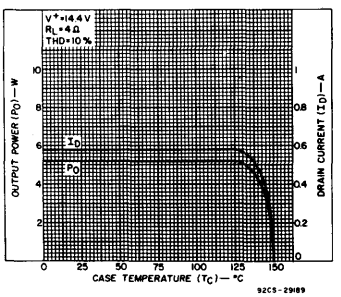


Fig. 17 – Output power and drain current as a function of case temperature.

12-Watt Audio Power Amplifier

The RCA-CA2004 is a monolithic silicon class B audio power amplifier designed for driving loads as low as 3.2 Ω. It provides a high output current capability (up to 3.5 A), and very low harmonic and cross-over distortion.

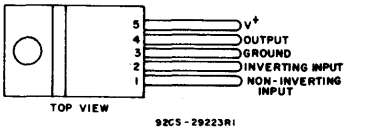
The CA2004 is supplied in a hermetic tri-metal Gold-CHIP encapsulated in the 5-lead plastic TO-220-style VERSA-V package. All leads (except term. 3) are electrically insulated from the mounting flange, eliminating the need for insulating hardware. The VERSA-V package is available with two lead configurations. The CA2004 has a vertical-mount lead form, and the CA2004M has a horizontal-mount lead form.

Features:

- **Hermetic Gold-CHIP encapsulated in a 5-lead plastic TO-220-style package (VERSA-V)**
- **Thermal overload protection**
- **Drives load impedance as low as 3.2 Ω**
- **Deflection amplifier capability**
- **Output current capability of up to 3.5 A**
- **Few external components**
- **VERSA-V power transistor package-requires no electrical insulation**

MAXIMUM RATINGS, *Absolute-Maximum Values:*

DC SUPPLY VOLTAGE	28 V
OPERATING SUPPLY VOLTAGE	26 V
OUTPUT PEAK CURRENT:	
REPETITIVE	3.5 A
NON-REPETITIVE	4.5 A
POWER DISSIPATION, P_D at T_A = 90°C	15 W
THERMAL RESISTANCE, JUNCTION TO CASE	4°C/W
AMBIENT-TERMPERATURE RANGE:	
OPERATING	0 to +125°C
STORAGE	−40 to +150°C
LEAD TEMPERATURE (DURING SOLDERING):	
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 12 s max.	260°C



TERMINAL ASSIGNMENT

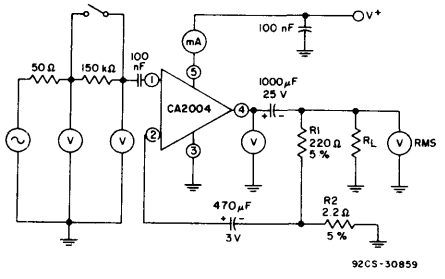


Fig. 1 – Test circuit.

Thermal Shut-Down

Thermal shut-down occurs if the output overloads (temporary or permanent), the ambient temperature is excessive, or the junction temperature is excessive. None of these conditions results in device damage. They merely cause a temporary automatic reduction of output power and drain current.

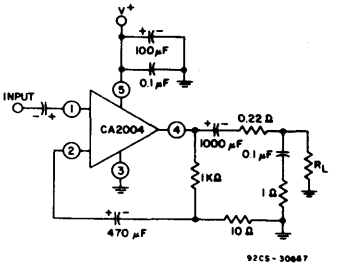


Fig. 2 – Typical application.

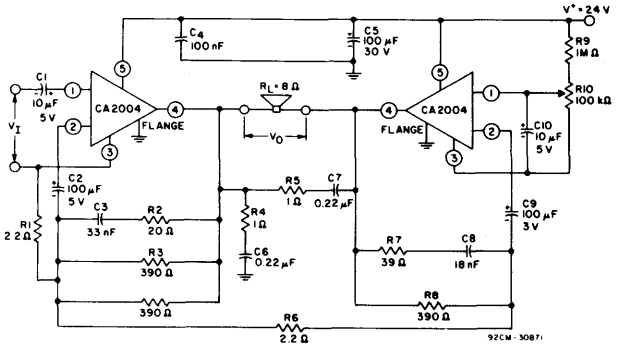


Fig. 3 – 25 W circuit-bridge application.

ELECTRICAL CHARACTERISTICS at T_A = 25°C, V^+ = 24 V
Unless otherwise specified (See Figure 1)

CHARACTERISTIC	TEST CONDITIONS	LIMITS			UNITS
		Min.	Typ.	Max.	
Supply Voltage, V^+		8	–	26	V
Quiescent Output Voltage, V_O	Measure at Term. 4	11	12	13	V
Quiescent Drain Current, I_D	Measure at Term. 5	–	40	100	mA
Output Power, P_O	THD = 10%, A = 40 dB, f = 1 KHz, R_L = 4 Ω	10	12	–	W
	THD = 10%, A = 40 dB, f = 1 KHz, R_L = 8 Ω	–	8	–	W
Input Saturation Voltage, $V_{I(RMS)}$		400	–	–	mV
Input Resistance, R_I (Term.1)	f = 1 KHz	70	150	–	KΩ
Open-Loop Voltage Gain, A_{OL}	R_L = 8 Ω, f = 1 KHz	–	80	–	dB
Closed-Loop Voltage Gain, A	R_L = 8 Ω, f = 1 KHz	39.5	40	40.5	dB
Input Noise Voltage, e_N	Freq. Resp. = 40 to 15,000 Hz (–3 dB)	–	4	–	μV
Power Supply Rejection Ratio, PSRR	R_L = 4 Ω, A = 40 dB, Rg = 10 KΩ, f_{ripple} = 100 Hz, V_{ripple} = 0.5 V	30	35	–	dB

Ultra-High-Gain Wide-Band Amplifier Array

- Three Individual General-Purpose Amplifiers
- Ideal for service in Remote-Control Amplifiers — — e.g., TV Receivers
- Available in two electrically identical versions: CA3035 with straight leads; CA3035V1 with formed leads

HIGHLIGHTS

- Three separate amplifiers — gain and bandwidth for each amplifier can be adjusted with suitable external circuitry
- Amplifiers operable independently or in cascade
- Exceptionally high cascade voltage gain — 129 dB typ. at 40 kHz
- Low noise performance
- Wide-band response
- All amplifiers single-ended — only one power supply required
- Wide operating temperature range — -55°C to +125°C
- Built-in temperature compensation
- Hermetically sealed, all-welded 10-lead TO-5-style metal package with straight or formed leads

ABSOLUTE-MAXIMUM RATINGS:

Operating Temperature Range ..... -55°C to +125°C
Storage Temperature Range ..... -65°C to +150°C
Device Dissipation ..... 300 mW
Input Voltage ..... 1 V p-p
Supply Voltage ..... +15V
Lead Temperature (During Soldering):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max. ..... +265°C

ELECTRICAL CHARACTERISTICS AT T_A = 25°C

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS	TEST CIRCUITS AND CHARACTERISTICS CURVES	LIMITS CA3035, CA3035V1 Min.	Typ.	Max.	UNITS
STATIC CHARACTERISTICS							
Quiescent Operating Voltage	V3 V5 V7	V_{CC} = +9V	Fig.3	– – –	2 1.9 4.9	– – –	V V V
Total Current Drain	I_d	V_{CC} = +9V, R_{L3} = 5KΩ	Fig.3	3.5	5	7.5	mA
DYNAMIC CHARACTERISTICS							
Voltage Gain: Amplifier No.1 Amplifier No.2 Amplifier No.3	A_1 A_2 A_3	f = 40 kHz, V_{CC} = +9V		40 40 38	44 46 42	– – –	dB dB dB
Output Voltage Swing	Vout V_{1out} V_{2out} V_{3out}	R_{L1} = 10KΩ R_{L2} = 10KΩ R_{L3} = 5KΩ Sinusoidal Output, V_{CC} = +9V		– – –	2 2.6 8	– – –	Vp-p Vp-p Vp-p
Input Resistance: Amplifier No.1 Amplifier No.2 Amplifier No.3	R_1in R_2in R_3in	f = 40 kHz		– – –	50K 2K 670	– – –	Ω Ω Ω
Output Resistance	R_1out R_2out R_3out	f = 40 kHz		– – –	270 170 100K	– – –	Ω Ω Ω
Bandwidth at -3dB point: Amplifier No.1 Amplifier No.2 Amplifier No.3	BW_1 BW_2 BW_3	V_{CC} = +9V	Fig.5 Fig.6 Fig.7	– – –	500 2.5 2.5	– – –	kHz MHz MHz
Noise Figure Amplifier No.1	NF_1	f = 1 kHz, R_S = 1 KΩ	Fig.4	–	6	7	dB
Sensitivity		V_{CC} = +13 V Relay (K_1) Current = 7.5 mA	Fig.2	–	100	150	μV

SCHEMATIC DIAGRAM FOR CA3035 AND CA3035V1

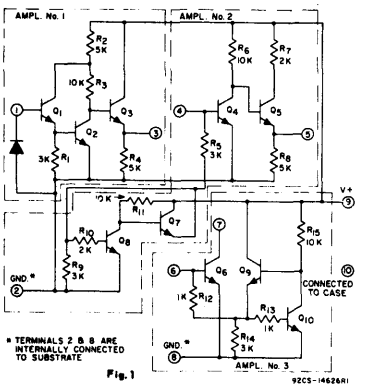


Fig. 1

TYPICAL REMOTE CONTROL SYSTEM

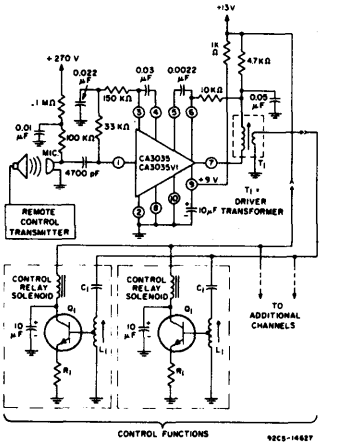


Fig.2

STATIC CHARACTERISTICS TEST CIRCUIT

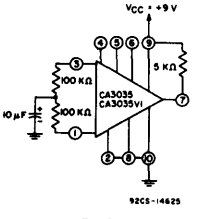
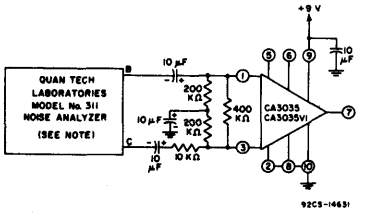


Fig.3

NOISE FIGURE TEST CIRCUIT



NOTE: SET ALL INTERNAL POWER SUPPLIES ON QUAN TECH NOISE ANALYZER TO ZERO VOLTS.

Fig.4

CA3041

Wideband Amplifier, FM Detector, AF Preamplifier/Driver

For Sound Sections of TV Receivers Using Tube-Type AF Output Amplifiers

FEATURES

- high-sensitivity – input limiting voltage (knee) = 150 μV typ. at 4.5 MHz
- large audio drive voltage capability
- excellent AM rejection – 58 dB typ. at 4.5 MHz
- inherent high stability – internally shielded
- internal Zener-diode-regulated voltage supply
- low harmonic radiation
- wide frequency capability – <100 kHz to > 20 MHz
- low harmonic distortion

RCA Integrated Circuit Type CA3041 provides, in a single monolithic silicon chip, a major subsystem for the sound sections of TV receivers. As shown in the Schematic Diagram (Fig.1) and the TV Receiver Block Diagrams (Fig.12) the CA3041 contains a multistage wide-band if-amplifier/limiter section, an FM-detector stage, a Zener-diode-regulated power-supply section, and an af-amplifier section specifically designed to drive directly a 6AQ5 beam power tube or other audio output tube of similar characteristics.

In FM receivers, the CA3041 can be used to provide if amplification and limiting, FM detection, and af preamplification.

The CA3041 provides exceptional versatility of circuit design because the if-amplifier/limiter section, FM detector section, and af-preamplifier/driver section can be used independently of each other.

The CA3041 utilizes a 14-lead dual-in-line plastic package with leads specially formed to facilitate automatic insertion of the device in suitably punched printed-circuit boards.

MAXIMUM RATINGS, *Absolute Maximum Values:*

OPERATING-TEMPERATURE RANGE ..... –40° to +85°C
STORAGE-TEMPERATURE RANGE ..... –65° to +150°C
LEAD TEMPERATURE (During Soldering):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max. ..... +265°C
MAXIMUM INPUT-SIGNAL VOLTAGE:
Between Terminals 1 and 3 ..... ±3 V
MAXIMUM DEVICE DISSIPATION:
At Ambient Temperatures up to +25°C ..... 950 mW
At Ambient Temperatures above +25°C ..... Derate at 10.8 mW/°C

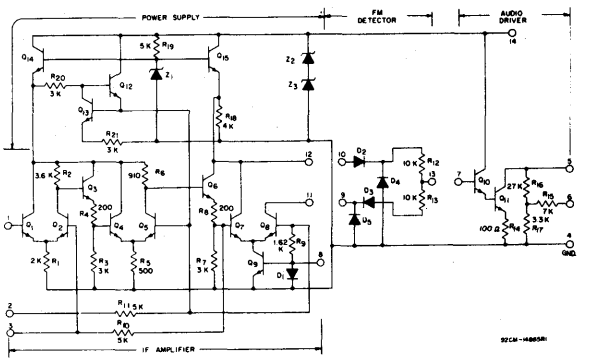


Fig.1 - Schematic diagram.

ABSOLUTE-MAXIMUM VOLTAGE AND CURRENT LIMITS AT T_A = 25°C

Indicated voltage or current limits for each terminal may be applied under the specified voltage conditions for other terminals. All voltages are with respect to ground (Terminal 4).

TERMINAL	VOLTAGE OR CURRENT LIMITS		VOLTAGE CONDITIONS AT OTHER TERMINALS													
			1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	-3 V	+3 V	–	AT SAME DC VOLTAGE AS TERMINAL 1	EXTERNAL DC VOLTAGE IS NOT NORMALLY APPLIED TO THIS TERMINAL	GROUND (VOLTAGE REFERENCE TERMINAL)	CONNECTED TO +140 V THROUGH 47 kΩ RESISTOR*	CONNECTED TO TERMINAL 7 THROUGH 100 kΩ RESISTOR*	AF-INPUT TERMINAL (EXTERNAL DC VOLTAGE IS NOT NORMALLY APPLIED TO THIS TERMINAL)	DC VOLTAGE IS NOT NORMALLY APPLIED TO THIS TERMINAL	AT SAME DC VOLTAGE AS TERMINAL 12 (EXCEPT TERMINAL 9)	AT SAME DC VOLTAGE AS TERMINAL 12 (EXCEPT TERMINAL 10)	AT SAME DC VOLTAGE AS TERMINAL 12 (EXCEPT TERMINAL 11)	EXTERNAL DC VOLTAGE IS NOT NORMALLY APPLIED TO THIS TERMINAL	AT SAME DC VOLTAGE AT TERMINAL 12 (EXCEPT TERMINAL 13)	CONNECTED TO +140 V DC THROUGH 6.2-kΩ RESISTOR*
2	-3 V	+3 V	-3 to +3													
3	-3 V	+3 V	-3 to +3													
4	GROUND (VOLTAGE REFERENCE TERMINAL)		-3 to +3													
5	20 mA		-3 to +3													
6	0 V	+10 V	-3 to +3													
7	10 mA		-3 to +3													
8	10 mA		-3 to +3													
9	10 mA		-3 to +3													
10	10 mA		-3 to +3													
11	+2.5 V	+5 V	-3 to +3													
12	+2.5 V	+5 V	-3 to +3													
13	+2.5 V	+5 V	-3 to +3													
14	50 mA		-3 to +3													

* Any other combination of DC Supply Voltage and Series Resistance which will not cause the Maximum Device Dissipation Limit or any of the Maximum Voltage or Current Limits for the CA3041 to be exceeded may be used.

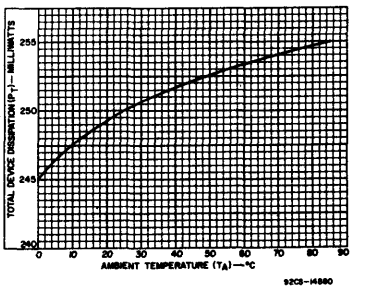


Fig.2 - Typical dissipation characteristic for CA3041.

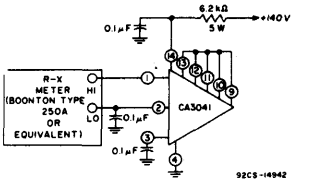


Fig.3 - Test setup for measurement of input-impedance components.

ELECTRICAL CHARACTERISTICS, *at an Ambient Temperature, T_A, of 25°C, and a DC Supply Voltage, V_{CC}, of +140 Volts applied to Terminal 14 through a resistance of 6.2 kΩ, unless otherwise indicated. Any other combination of DC Supply Voltage and Series Resistance which will not cause the Maximum Dissipation Limit or any of the Maximum Voltage or Current Limits for the CA3041 to be exceeded may be used.*

CHARACTERISTICS (See Page 7 for Definitions of Terms)	SYMBOLS	TEST CONDITIONS: SETUP AND PROCEDURE Fig.	TEST CONDITIONS: SPECIAL CONDITIONS	LIMITS: TYPE CA3041 Min.	Typ.	Max.	Units	TYPICAL CHARACTERISTICS CURVES Fig.
Total Device Dissipation	P_T	11	T_A = 0°C	220	245	270	mW	2
			T_A = +25°C	225	250	275	mW	
			T_A = +85°C	230	255	280	mW	
Zener Regulating Voltage (DC Supply Voltage at Terminal 14)	V_{14}	-		10.5	11.2	12.3	V	-
Quiescent Operating Current (into Terminal 11)	I_{11}	11		0.25	0.63	1	mA	-
9-Volt Current Drain (Quiescent Operating Current into Terminal 14)	I_{14}	11	V_{CC} = +9 V applied directly to Terminal 14	7	11	16	mA	-
Input-Impedance Components: Parallel Input Resistance	R_i	3	f = 4.5 MHz	-	11	-	kΩ	-
Parallel Input Capacitance	C_i	3		-	5	-	pF	-
Output-Impedance Components: Parallel Output Resistance	R_o	-		-	100	-	kΩ	-
Parallel Output Capacitance	C_o	-		-	4	-	pF	-
Input Limiting Voltage (Knee)	$V_{i(lim)}$	7		-	150	200	μV (rms)	4
Amplitude-Modulation Rejection	AMR	10		45	58	-	dB	9
IF-Amplifier Voltage Gain	$A_{(IF)}$	5		-	67	-	dB	4
Recovered AF Voltage: 1. At FM-Detector Output	$V_{o(af)}$	-	R_L = 50 kΩ, Δf = ±25 kHz THD = 0.7% (typ.)	-	250	-	mV (rms)	-
2. At AF-Driver Output in Test Setup		-	THD < 5%	8	9	-	V (rms)	-
Total Harmonic Distortion	THD	7	$V_{o(af)}$ = 8 V(rms)	-	1.5	5	%	-
Discriminator Output Resistance	$R_{o(dis)}$	-	f = 1 kHz	-	10	-	kΩ	-
AF-Amplifier Input Resistance	$R_{i(af)}$	-		-	100	-	kΩ	-
AF-Amplifier Output Resistance	$R_{o(af)}$	-		-	30	-	kΩ	-
AF-Driver Voltage Gain	A_{af}	6		-	41	-	dB	8

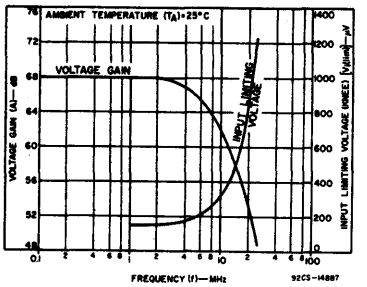
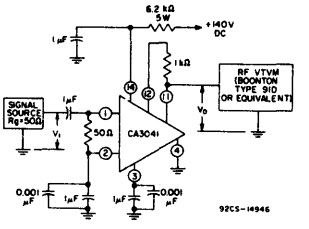


Fig.4 - Typical IF-amplifier voltage gain and input-limiting voltage (knee) characteristics.



PROCEDURE:

A - Voltage Gain:

1) Set input frequency at desired value, v_i = 100 μV rms.
2) Record v_o.
3) Calculate Voltage Gain A from A = 20 $\log_{10} v_o/v_i$
4) Repeat Steps 1, 2, and 3 for each frequency and/or for temperature desired.

Fig.5 - Test setup for measurement of IF-amplifier voltage gain.

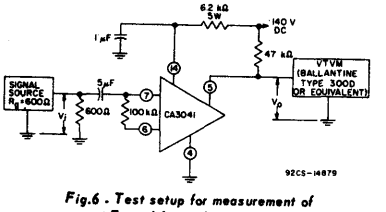
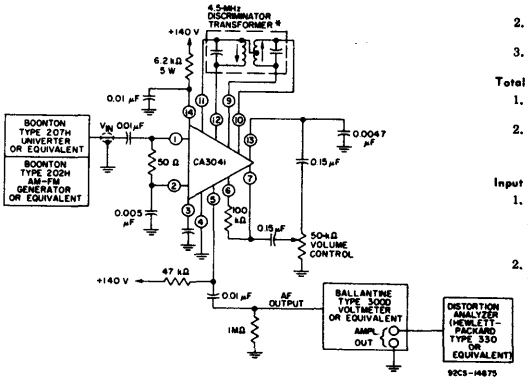


Fig.6 - Test setup for measurement of AF-amplifier voltage gain.



* TRW Electronics, Des Plaines, Illinois. Part No. EO23874, or equivalent.

Fig.7 - Test setup for measurement of input limiting voltage (Knee), recovered AF voltage, and total harmonic distortion.

PROCEDURES:

Recovered AF Voltage:

1. Set Input Signal Generator as follows:
   Output frequency = 4.5 MHz
   Modulating frequency = 1 kHz
   Deviation = ± 25 kHz
   Output level for V_{in} = 100 mV rms
2. Set volume control for maximum af output.
3. Measure af output voltage and record as Recovered AF Voltage.

Total Harmonic Distortion:

1. Adjust volume control for an af output voltage of 300 mV rms.
2. Measure Total Harmonic Distortion of the output signal in accordance with the Operating Instructions for the Distortion Analyzer.

Input Limiting Voltage (Knee):

1. Decrease V_{in} until the af output voltage is 3 dB less than the value set in Step 1 of the procedure for measurement of Total Harmonic Distortion (300 mV - 3 dB = 210 mV)
2. Measure resulting value of V_{in} and record as Input Limiting Voltage (Knee).

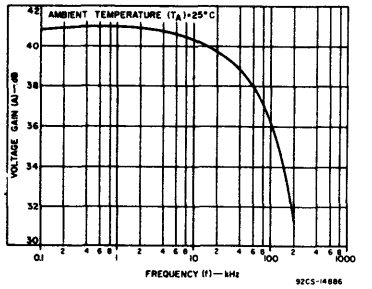


Fig.8 - Typical AF-driver voltage-gain characteristic

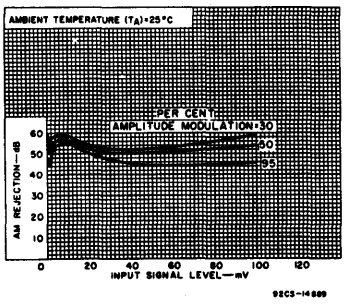
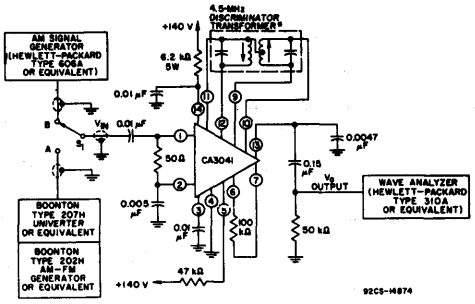


Fig.9 - Typical AM rejection characteristics for CA3041.

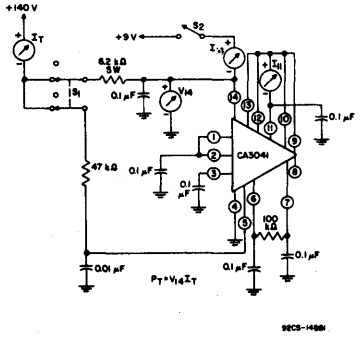


* TRW Electronics, Des Plaines, Illinois. Part No. EO23874, or equivalent.

PROCEDURES:

1. Set FM Signal Generator as follows:
   Output frequency = 4.5 MHz
   Modulating frequency = 1000 Hz
   Deviation = ± 25 kHz
   Output level for V_{in} = 100 mV rms
2. Set AM Signal Generator as follows:
   Output frequency = 4.5 MHz
   Modulating frequency = 1000 Hz
   Per cent modulation = 30
   Output level for V_{in} = 10 mV rms
3. With S_1 in Position A measure AF Output Voltage and record as $V_{o(FM)}$.
4. With S_1 in Position B measure AF Output Voltage and record as $V_{o(AM)}$.
5. Determine AM Rejection from $AMR = V_{o(FM)}/V_{o(AM)}$

Fig.10 - Test setup for measurement of AM rejection.



PROCEDURES:

Total Device Dissipation:

1. Close S_1, open S_2.
2. Measure and record V_{14} and I_T.
3. Determine Total Device Dissipation from $P_T = V_{14}I_T$.

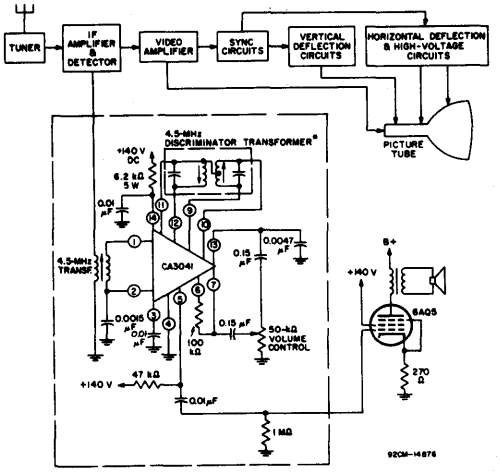
Quiescent Operating Current into Terminal 11:

1. Close S_1, open S_2.
2. Measure I_{11} and record as Quiescent Operating Current into Terminal 11.

9-Volt Current Drain:

1. Open S_1, close S_2.
2. Measure I_{14} and record as 9-Volt Current Drain.

Fig.11 - Test setup for total dissipation, quiescent operating current into terminal No.11, and 9-volt current drain.



* TRW Electronics, Des Plaines, Illinois. Part No. EO23874, or equivalent.

Fig.12 - Block diagram of typical TV receiver using CA3041.

Wideband Amplifier, FM Detector, AF Preamplifier/Driver

CA3042

For Sound Sections of TV Receivers Using Transistor-Type AF Output Amplifiers

FEATURES

- high sensitivity – input limiting voltage (knee) = 150 μV typ. at 4.5 MHz
- 6-mA audio drive capability
- excellent AM rejection – 58 dB typ. at 4.5 MHz
- inherent high stability – internally shielded
- internally Zener-diode-regulated voltage supply
- low harmonic radiation
- wide frequency capability – <100 kHz to >20 MHz
- low harmonic distortion

RCA Integrated Circuit Type CA3042 provides, in a single monolithic silicon chip, a major sub-system for the sound sections of TV receivers. As shown in the Schematic Diagram (Fig.1) and the TV Receiver Block Diagrams (Figs.2A and 2B) the CA3042 contains a multistage wide-band if-amplifier section, an FM-detector stage, a Zener-diode-regulated power-supply section, and an af-amplifier section specifically designed to drive directly an n-p-n audio output transistor or a high-gain audio output pentode tube.

In FM receivers, the CA3042 can be used to provide if amplification and limiting, FM detection, and af preamplification.

The CA3042 provides exceptional versatility of circuit design because the if-amplifier/limiter section, FM detector section, and af-preamplifier/driver section can be used independently of each other.

The CA3042 utilizes a 14-lead dual-in-line plastic package with leads specially formed to facilitate automatic insertion of the device in suitably punched printed-circuit boards.

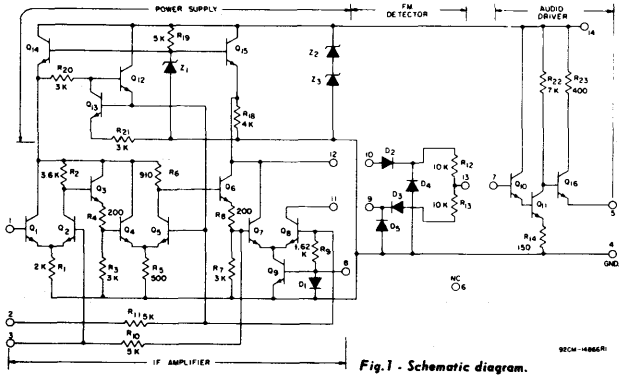


Fig.1 - Schematic diagram.

MAXIMUM RATINGS, *Absolute-Maximum Values:*

OPERATING-TEMPERATURE RANGE	−40° to +85°C
STORAGE-TEMPERATURE RANGE	−65° to +150°C
LEAD TEMPERATURE (During Soldering):	
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max.	+265°C
MAXIMUM INPUT-SIGNAL VOLTAGE:	
Between Terminals 1 and 3	±3 V
MAXIMUM DEVICE DISSIPATION:	
At Ambient up to +25°C	950 mW
Temperatures above +25°C	Derate at 10.8 mW/°C

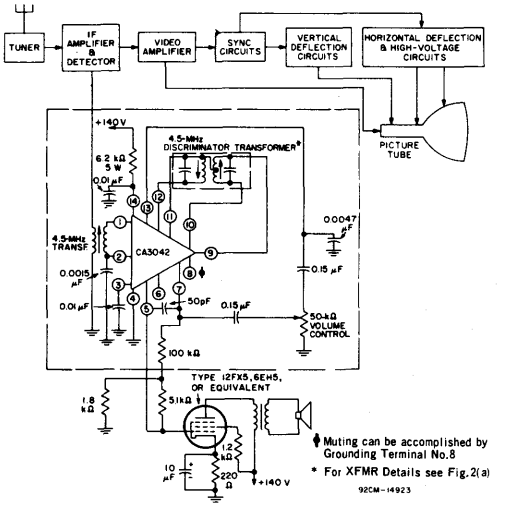


Fig.2(b) - Block diagram of typical TV receiver utilizing the CA3042 and a 12FX5, 6EH5, or equivalent.

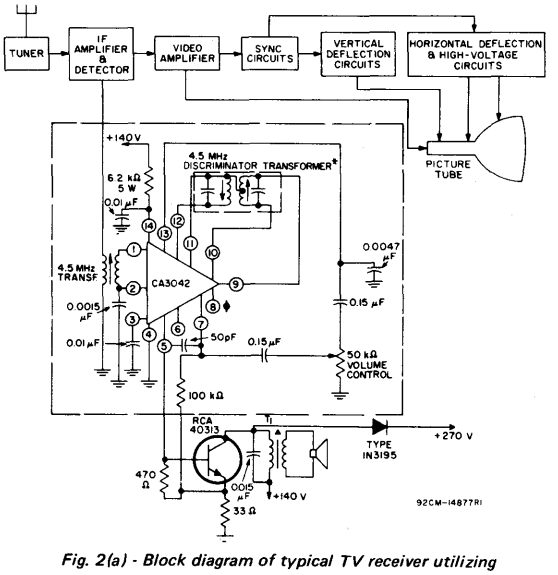
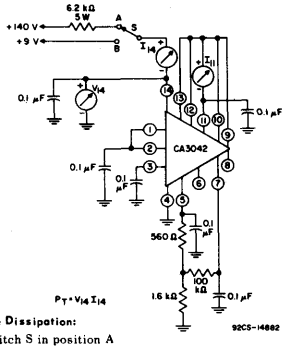


Fig. 2(a) - Block diagram of typical TV receiver utilizing transistor RCA-40313.



PROCEDURES: $P_T = V_{14}I_{14}$

Total Device Dissipation:

1. Set switch S in position A
2. Measure and record V_{14} and I_{14}.
3. Determine Total Device Dissipation from $P_T = V_{14}I_{14}$

Quiescent Operating Current into Terminal 11:

1. Turn switch S to position B
2. Measure I_{11} and record as Quiescent Operating Current into Terminal 11.

9-Volt Current Drain:

1. Set switch S in position B
2. Measure I_{14} and record as 9-Volt Current Drain.

Fig.3 - Test setup for measurement of total device dissipation, quiescent current into terminal No.11, and 9-volt current drain.

CA3042

ABSOLUTE-MAXIMUM VOLTAGE AND CURRENT LIMITS AT T_A = 25°C

Indicated voltage or current limits for each terminal may be applied under the specified voltage conditions for other terminals. All voltages are with respect to ground (Terminal 4).

TERMINAL	VOLTAGE OR CURRENT LIMITS		VOLTAGE CONDITIONS AT OTHER TERMINALS													
			1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	-3 V	+3 V	–	AT SAME DC VOLTAGE AS TERMINAL 1	EXTERNAL DC VOLTAGE IS NOT NORMALLY APPLIED TO THIS TERMINAL	GROUND (VOLTAGE REFERENCE TERMINAL)	AF-DRIVER OUTPUT TERMINAL (EXTERNAL DC VOLTAGE IS NOT NORMALLY APPLIED TO THIS TERMINAL)	NO CONNECTION	AF-INPUT TERMINAL (EXTERNAL DC VOLTAGE IS NOT NORMALLY APPLIED TO THIS TERMINAL)	MUTING TERMINAL (EXTERNAL DC VOLTAGE IS NOT NORMALLY APPLIED TO THIS TERMINAL EXCEPT THAT TERMINAL MAY BE GROUNDED TO OBTAIN MUTING ACTION)	AT SAME DC VOLTAGE AS TERMINAL 12 (EXCEPT TERMINAL 9)	AT SAME DC VOLTAGE AS TERMINAL 12 (EXCEPT TERMINAL 10)	AT SAME DC VOLTAGE AS TERMINAL 12 (EXCEPT TERMINAL 11)	EXTERNAL DC VOLTAGE IS NOT NORMALLY APPLIED TO THIS TERMINAL	AT SAME DC VOLTAGE AT TERMINAL 12 (EXCEPT TERMINAL 13)	CONNECTED TO +140 V DC THROUGH 6.2-kΩ RESISTOR*
2	-3 V	+3 V	-3 to +3													
3	-3 V	+3 V	-3 to +3													
4	GROUND (VOLTAGE REFERENCE TERMINAL)		-3 to +3													
5	20 mA		-3 to +3													
6	NO CONNECTION		-3 to +3													
7	10 mA		-3 to +3													
8	10 mA		-3 to +3													
9	10 mA		-3 to +3													
10	10 mA		-3 to +3													
11	+2 V	+10 V	-3 to +3													
12	+2.5 V	+10 V	-3 to +3													
13	0 V	+10 V	-3 to +3													
14	50 mA		-3 to +3													

* Any other combination of DC Supply Voltage and Series Resistance which will not cause the Maximum Device Dissipation Limit or any of the Maximum Voltage or Current Limits for the CA3042 to be exceeded may be used.

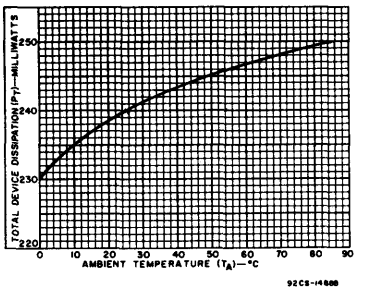


Fig.4 - Typical dissipation characteristic.

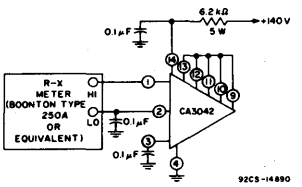
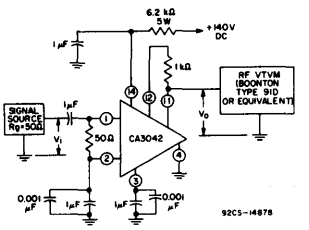


Fig.5 - Test setup for measurement of input-impedance components.



PROCEDURE Voltage Gain:

1. Set input frequency at desired value, v_i = 100 μV rms.
2. Record v_o.
3. Calculate Voltage Gain A from A = 20 $\log_{10} v_o/v_i$.
4. Repeat Steps 1, 2, and 3 for each frequency and/or for temperature desired.

Fig.6 - Test setup for measurement of IF amplifier voltage gain.

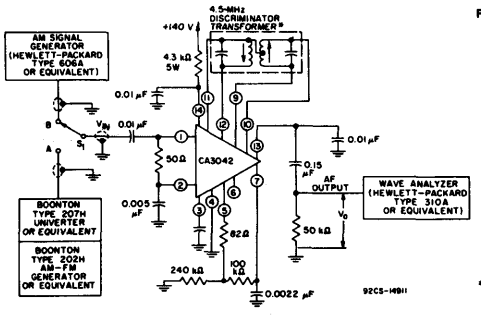


Fig.7 - Test setup for measurement of AM rejection.

PROCEDURES:

1. Set FM Signal Generator as follows:
   Output Frequency = 4.5 MHz
   Modulating frequency = 1000 Hz
   Deviation = ±25 kHz
   Output level for V_{in} = 100 mV rms
2. Set AM Signal Generator as follows:
   Output frequency = 4.5 MHz
   Modulating frequency = 1000 Hz
   Per cent modulation = 30
   Output level for V_{in} = 10 mV rms
3. With S_1 in Position A measure AF Output Voltage and record as $V_{o(FM)}$.
4. With S_1 in Position B measure AF Output Voltage and record as $V_{o(AM)}$.
5. Determine AM Rejection from AMR = $\frac{V_{o(FM)}}{V_{o(AM)}}$

* TRW Electronics, Des Plaines, Illinois. Part No. EO23874, or equivalent.

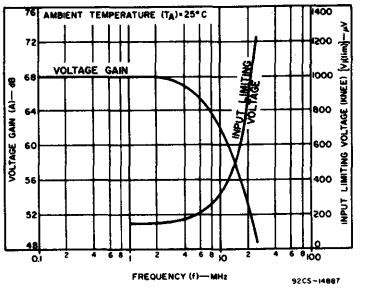


Fig.8 - Typical IF amplifier voltage gain and input limiting voltage (knee) characteristics.

ELECTRICAL CHARACTERISTICS, *at an Ambient Temperature, T_A, of 25°C, and a DC Supply Voltage, V_{CC}, of +140 Volts applied to Terminal 14 through a resistance of 6.2 kΩ, unless otherwise indicated. Any other combination of DC Supply Voltage and Series Resistance which will not cause the Maximum Dissipation Limit or any of the Maximum Voltage or Current Limits for the CA3042 to be exceeded may be used.*

CHARACTERISTICS (See Page 7 for Definitions of Terms)	SYMBOLS	TEST CONDITIONS: SETUP AND PROCEDURE Fig.	TEST CONDITIONS: SPECIAL CONDITIONS	LIMITS: TYPE CA3042 Min.	Typ.	Max.	Units	TYPICAL CHARACTERISTICS CURVES Fig.
Total Device Dissipation	P_T	3	T_A = 0°C	200	230	260	mW	4
			T_A = +25°C	210	240	270	mW	
			T_A = +85°C	220	250	280	mW	
Zener Regulating Voltage (DC Supply Voltage at Terminal 14)	V_{14}	–		10.5	11.2	12.3	V	–
Quiescent Operating Current (into Terminal 11)	I_{11}	3		0.25	0.63	1	mA	–
9-Volt Current Drain (Quiescent Operating Current into Terminal 14)	I_{14}	3	V_{CC} = +9 V applied directly to Terminal 14	8	12	18	mA	–
Input-Impedance Components: Parallel Input Resistance	R_i	5	f = 4.5 MHz	–	11	–	kΩ	–
Parallel Input Capacitance	C_i	5	f = 4.5 MHz	–	5	–	pF	–
Output-Impedance Components: Parallel Output Resistance	R_o	–	f = 4.5 MHz	–	100	–	kΩ	–
Parallel Output Capacitance	C_o	–	f = 4.5 MHz	–	4	–	pF	–
Input Limiting Voltage (Knee)	$V_{i(lim)}$	11	f = 4.5 MHz	–	150	200	μV (rms)	8
Amplitude-Modulation Rejection	AMR	7	f = 4.5 MHz	45	58	–	dB	
IF-Amplifier Voltage Gain	A(IF)	6	f = 4.5 MHz	–	67	–	dB	8
Recovered AF Voltage: 1. At FM-Detector Output	V_o(af)	11	f = 4.5 MHz, Δf = ±25 kHz, R_L = 50 kΩ, THD = 0.7% (typ.)	–	250	–	mV (rms)	–
2. At AF-Driver Output in Test Setup		11	f = 4.5 MHz, Δf = ±25 kHz, R_L = 322 Ω, THD < 5%	500	800	–	mV (rms)	–
3. At AF-Driver Output in TV-Receiver Sound System		2A or 2B	f = 4.5 MHz, Δf = ±25 kHz, R_L = 150 kΩ, THD = 1.5% (typ.)	–	3	–	V (rms)	–
Total Harmonic Distortion: 1. In Test Setup	THD	11	f = 4.5 MHz, Δf = ±25 kHz, V_o(af) = 500 mV (rms)	–	1.5	5	%	–
2. In TV Receiver Sound System		2A or 2B	f = 4.5 MHz, Δf = ±25 kHz, V_o(af) = 1.3 V (rms)	–	1	–	%	–
FM-Detector Output Resistance	R_o(det)	–	f = 1 kHz	–	10	–	kΩ	–
AF-Driver Input Resistance	R_i(af)	–	f = 1 kHz	–	100	–	kΩ	–
AF-Driver Output Resistance	R_o(af)	–	f = 1 kHz	–	250	–	Ω	–
AF-Driver Voltage Gain	A_{af}	9	f = 1 kHz, R_S = 50 Ω, C_1 = 0	–	30	–	dB	10

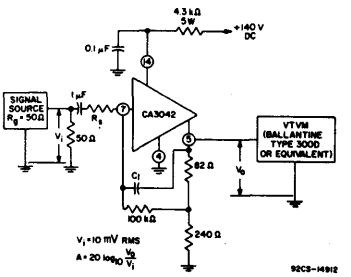


Fig.9 - Test setup for measurement of AF amplifier voltage gain.

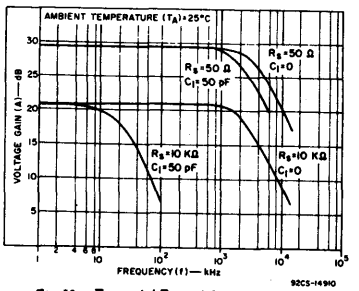
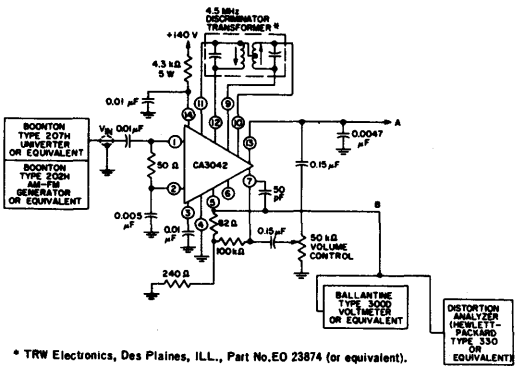


Fig.10 - Typical AF amplifier voltage gain characteristics.



* TRW Electronics, Des Plaines, ILL., Part No.EO 23874 (or equivalent).

PROCEDURES:

Recovered AF Voltage:

1. Set Input Signal Generator as follows:
   - Output frequency = 4.5 MHz
   - Modulating frequency = 1 kHz
   - Deviation = ± 25 kHz
   - Output level for V_{in} = 100 mV rms
2. Set volume control for maximum af output
3. Measure af output voltage and record as Recovered AF Voltage.

Total Harmonic Distortion:

1. Adjust volume control for an af output voltage of 500 mV rms.
2. Measure Total Harmonic Distortion of the output signal in accordance with the Operating Instructions for the Distortion Analyzer.

Input Limiting Voltage (Knee):

1. Decrease V_{in} until the af output voltage is 3 dB less than the value set in Step 1 of the procedure for measurement of Total Harmonic Distortion (500 mV - 3 dB = 350 mV)
2. Measure resulting value of V_{in} and record as Input Limiting Voltage (Knee).

Fig.11 - Test setup for measurement of input limiting voltage (knee), recovered AF voltage, and total harmonic distortion.

CA3043
Special-Function Subsystem

HIGH-GAIN IF AMPLIFIER, LIMITER, FM DETECTOR, AND AF PREAMPLIFIER/DRIVER

For FM IF Amplifier Applications in Communications Receivers and High-Fidelity FM Receivers up to 20 MHz

FEATURES

- high sensitivity - - input limiting voltage (knee) 50 μV typ. at 10.7 MHz
- excellent AM rejection - - 58 dB typ. at 10.7 MHz
- inherent high stability - - internally shielded
- internal Zener-diode regulated voltage supply
- low harmonic radiation
- wide frequency capability - - <100 kHz to >20 MHz
- low harmonic distortion
- hermetic 12-lead TO-5 style package

RCA Integrated Circuit Type CA3043 provides in a single monolithic silicon chip, a major sub-system for the IF sections of Communications and high-fidelity FM receivers. As shown in the Schematic Diagram (Fig.2) and the FM Receiver Block Diagram (Fig.1), the CA3043 contains a multistage if-amplifier/limiter section, an FM-detector stage, a Zener-diode regulated power-supply section, and an af-amplifier section. In FM receivers, the CA3043 can be used to provide if amplification and limiting, FM detection, and af preamplification. The CA3043 provides exceptional versatility of circuit design because the if-amplifier/limiter section, FM detector section, and af-preamplifier/driver section can be used independently of each other.

The four stage emitter-follower-coupled if amplifier section provides 80-dB voltage gain at 10.7 MHz, and features an output stage with exceptionally good limiting characteristics because of its transistor constant-current sink.

The FM detector section is distinguished by circuitry which provides forward bias to the detector diodes, D2 and D3, and also provides a reference voltage for AFC.

The audio amplifier provides a low-impedance drive for subsequent audio amplifiers.

The power supply section provides zener-regulated, decoupled voltages for the IF amplifier, detector, and audio amplifier sections.

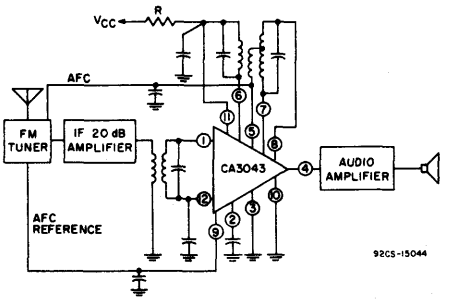


Fig.1 - Typical application of the CA3043 as a high-gain limiter, amplifier-detector in an FM receiver.

ABSOLUTE-MAXIMUM RATINGS at T_A = 25°C

DISSIPATION:
At T_A = 25°C to T_A = 85°C . . . . . 450 mW
Above T_A = 85°C . . . . . Derate linearly 5 mW/°C

LEAD TEMPERATURE (During Soldering):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 min) from case for 10 seconds max . . . . . + 265°C

TEMPERATURE RANGE:
Operating . . . . . -55°C to + 125°C
Storage . . . . . -65°C to + 150°C

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS	TEST CIRCUIT AND PROCEDURE	LIMITS TYPE CA3043			UNITS
			Fig.	Min.	Typ.	Max.	
STATIC CHARACTERISTICS							
Current Drain at 6V into Pin No.11	I_{11}	V_{CC} = +6V	3	10	16	20	mA
Regulator Voltage Pin No.11	V_{11}	V_{CC} = +30V, R_S = 750 Ω	3	6.9	7.4	8	V
Total Device Dissipation	P_T		3	200	225	260	mW
Quiescent Operating Current into Pin No.6	I_6		3	-	0.65	-	mA
DYNAMIC CHARACTERISTICS at V_{CC} = +30V, R_S = 750 Ω, f = 10.7 MHz							
Voltage Gain	A_V		4	72	80	-	dB
Input Limiting Voltage (knee)	v_i(lim)	v_o(af) at -3 dB point	6	-	50	-	μV (RMS)
Limiting Current from Pin No.6	I_6(lim)		4	-	0.42	-	mA (RMS)
Recovered AF Voltage	v_o(af)	v_i = 1 mV (RMS) f (modulating) = 1 kHz Deviation = ± 75 kHz	6	75	110	150	mV (RMS)
Amplitude-Modulation Rejection	AMR	v_i = 10 mV f (modulating) = 1 kHz % modulation = 50%	8	-	58	-	dB
Total Harmonic Distortion	THD	v_i = 1 mV (RMS)	6	-	0.3	-	%
Input Impedance Components:							
Parallel Input Resistance	R_I		-	-	7	-	kΩ
Parallel Input Capacitance	C_I		-	-	5	-	pF

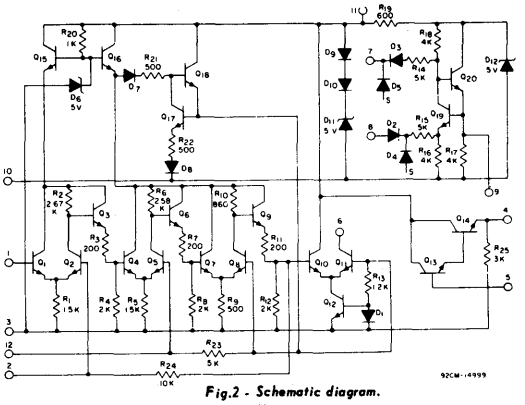
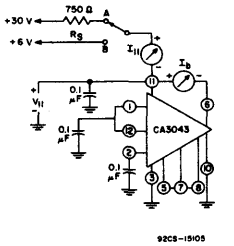


Fig.2 - Schematic diagram.

Notes:
S = Substrate
Terminal No.3 wire-connected to the case.
Terminal No.10 connected to the case through the substrate
Terminals No.3 and 10 which are connected to the substrate should be connected to the most negative point in the circuit.
Diodes D_4 and D_5, act as capacitors and are used to balance the detector substrate capacitances.



Switch in Position A for:
Regulator-Voltage, Quiescent-Operating-Current, and Device Dissipation Test
Switch in Position B for Current into Pin No.11

Fig.3 - Regulator voltage, device dissipation, quiescent operating current, and current at 6 volts into Pin No. 11.

MAXIMUM VOLTAGE RATINGS

The following chart gives the range of voltages which can be applied to the terminals listed horizontally with respect to the terminals listed vertically. For example, the voltage range between horizontal terminal 5 and vertical terminal 3 is +6 to 0 volts.

TERMINAL No.	1	2	3	4	5	6	7	8	9	10	11	12
1		+4 -4	0 -5	*	*	*	*	*	*	0 -5	*	Note(1)
2			0 -3	*	*	*	*	*	*	0 -3	*	*
3				+6 0	+6 0	+15 +2	+6 0	+6 0	+6 0	0	Note(2)	+3 0
4					+2 -4	*	*	*	*	0 -6	*	*
5						*	*	*	*	0 -6	+6 0	*
6							*	*	*	-2 -15	*	*
7								Note(1)	*	0 -6	*	*
8									*	0 -6	*	*
9										0 -6	*	*
10											Note(2) 0	+3 0
11												*
12												

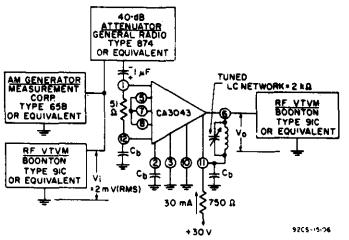
MAXIMUM CURRENT RATINGS

TERMINAL No.	I_{IN} mA	I_{OUT} mA
1	-	-
2	-	-
3	0.1	40
4	-	20
5	-	-
6	-	-
7	-	-
8	-	-
9	-	20
10	0.1	40
11	40	0.1
12	-	-

Note 1: These terminals should be connected through a dc resistance to any terminal which does not exceed 100 ohms.

Note 2: Pin 11 may be connected to any positive voltage source through a suitable resistor provided its current rating is not exceeded.

* Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if the specified limits between all other terminals are not exceeded.



Voltage Gain = $20 \log_{10} 100 \frac{V_o}{V_i}$

C_b - Bypass Capacitor, 0.1 μF electrolytic in parallel with 0.01 μF

$I_{6(lim)} = \frac{V_o}{2K\Omega}$, v_i = 100 mV(RMS)

* Output circuit should be completely shielded from the input circuit at the socket.

Fig.4 - Voltage gain test circuit.

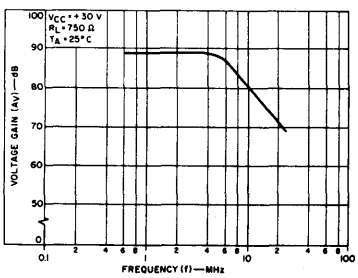
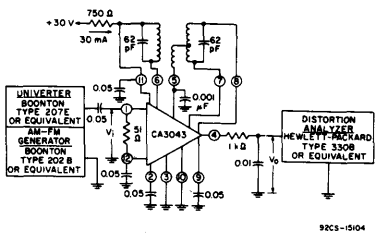


Fig.5 - Voltage gain vs frequency.



PROCEDURE:

1. Recovered Audio Voltage $v_{o(af)}$ –
   Set input frequency to 10.7 MHz,
   v_i = 1 mV(RMS), modulating frequency = 1 kHz
   Deviation = ±75 kHz
   Record v_o as measured on the Distortion Analyzer meter scale.
   This is the recovered Audio Voltage $v_{o(af)}$
2. 3 dB Limiting Sensitivity $v_{i(lim)}$ –
   Reduce v_i until $v_{o(af)}$ drops 3 dB.
   Record this value of v_i as $v_{i(lim)}$
3. Total Harmonic Distortion THD –
   Reset v_i to 1 mV(RMS) and operate Distortion Analyzer per manufacturer's instructions to measure THD.

* See Fig.9 for details on Discriminator Transformer.

Fig.6 - Input limiting voltage (knee), recovered AF voltage, and total harmonic distortion test circuit.

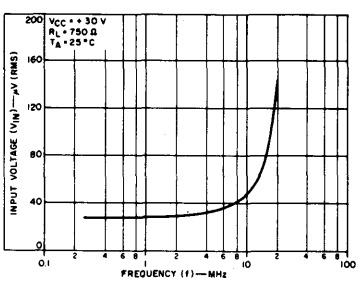
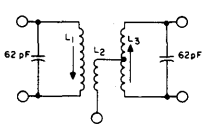
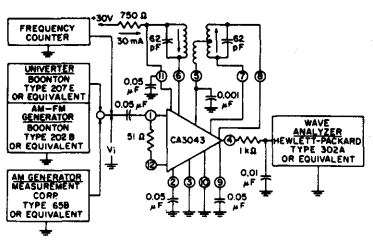


Fig.7 - Input limiting voltage (knee) at -3 dB point vs frequency.



Coil Form, Outside Diameter = 7/32"
Can = 1/2" square X 1-1/8" long
Slugs - Radio Industries Type MP34/MP100 Material
L_1 & L_3 = 20 Turns 5-44 litz wire universal wound
L_2 = 10 Turns 5-44 litz wire wound bifilar with L_1
L_1 & L_3 coupling adjusted to 520 kHz peak to peak separation on S curve when operated in circuit shown in Fig.6.

Fig.9 - 10.7-MHz discriminator transformer for CA3043.



PROCEDURE:

A. Connect FM Generator to CA3043 input.
   Set frequency to 10.7 MHz, v_i = 10 mV, modulating frequency = 1 kHz
   Deviation = ±75 kHz.
   Tune Wave Analyzer to peak reading at 1 kHz and record recovered Audio Voltage $v_{o(af)FM}$.

B. Disconnect FM Generator and Connect AM Generator to CA3043 input.
   Set frequency to 10.7 MHz, v_i = 10 mV, modulating frequency = 1 kHz, percent modulation = 50%.
   Tune Wave Analyzer to peak reading and record recovered audio voltage $v_{o(af)AM}$
   Amplitude Modulation Rejection Ratio = $20 \log_{10} \frac{v_{o(af)FM}}{v_{o(af)AM}}$

Fig.8 - Amplitude modulation rejection test circuit.

CA3044, CA3044V1

Special-Function Subsystem

WIDE-BAND AMPLIFIER/PHASE DETECTOR WITH ZENER DIODE VOLTAGE REGULATOR

For AFC (Automatic Frequency Control) Applications

The RCA CA3044 and CA3044V1 represent a second generation of integrated circuits designed primarily for AFC (Automatic-Frequency-Control) applications.

The CA3044V1 is electrically identical to the CA3044 but is supplied with formed leads for easier PC board design and construction.

FEATURES

- Primarily intended for AFC (automatic frequency control) Applications
- Internal Zener Diode Voltage Regulator
- Differential Input Amplifier/Limiter
- Full-Wave Diode Bridge Detector
- Differential Output Voltage Amplifier
- Available in Two Electrically Identical Versions of the 10-lead TO-5 style package,
  - CA3044 *With Straight Leads;*
  - CA3044V1 With Formed Leads
- Wide Operating Temperature Range; -55 to +125°C

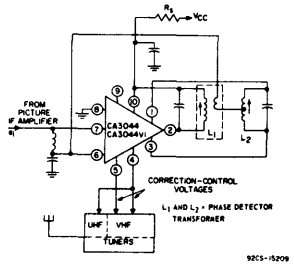


Fig.1 - Block diagram of Typical Automatic Fine Tuning (AFT) Application using CA3044 or CA3044V1 in Color-TV Receiver.

ABSOLUTE-MAXIMUM RATINGS

DISSIPATION:
At T_A = 25°C ..... 830 mW
Above T_A = 25°C ..... Derate linearly 5.6 mW/°C

TEMPERATURE RANGE:
Operating ..... -55°C to +125°C
Storage ..... -65°C to +150°C

LEAD TEMPERATURE (During Soldering):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max. ..... + 265C

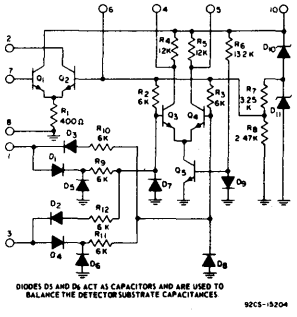


Fig.2 - Schematic diagram CA3044, CA3044V1

MAXIMUM VOLTAGE RATINGS at T_A = 25°C

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range between vertical terminal 2 and horizontal terminal 6 is +20 to 0 volts.

TERMINAL No.	9	10	1	2	3	4	5	6	7	8
9	NO INTERNAL CONNECTION									-
10			+20 0	+20 -10	+20 0	+20 0	+20 0	+20 0	+20 0	▲
1				*	+12 -12	*	*	+6 -6	*	+6 0
2					*	*	*	+20 0	*	+20 0
3						*	*	+6 -6	*	+6 0
4							*	*	*	+12 0
5								*	*	+12 0
6									+5 -5	+5 0
7										+8 -5
8										REF. SUB-STRATE

MAXIMUM CURRENT RATINGS

TERMINAL No.	I_{IN} mA	I_{OUT} mA
9	-	-
10	50	50
1	5	5
2	20	20
3	5	5
4	5	5
5	5	5
6	5	5
7	5	5
8	50	50

▲ Terminal No. 10 may be connected to any positive voltage source through a suitable dropping resistor—provided the dissipation rating is not exceeded.

* Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if the specified limits between all other terminals are not exceeded.

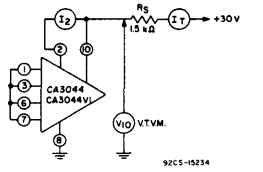


Fig.3 - Test setup: Measurement of total device dissipation, Zener regulating voltage, quiescent operating current (terminal 2).

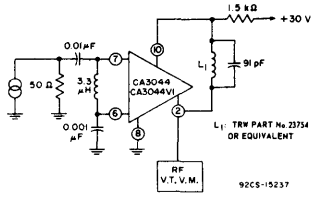


Fig.4 - Input limiting sensitivity test circuit.

ELECTRICAL CHARACTERISTICS at T_A = 25°C, Unless Otherwise Specified

CHARACTERISTICS	SYMBOLS	TEST CIRCUITS FIG.	TEST CONDITIONS	LIMITS CA3044 and CA3044V1 MIN.	TYP.	MAX.	UNITS	CHARACTERISTIC CURVES FIG.
STATIC CHARACTERISTICS								
Device Dissipation	P_T	3	V_{CC} = 30 V, R_S = 1.5 kΩ, T_A = -55°C	90	120	150	mW	-
Device Dissipation	P_T	3	V_{CC} = 30 V, R_S = 1.5 kΩ, T_A = 25°C	110	140	170	mW	-
Device Dissipation	P_T	3	V_{CC} = 30 V, R_S = 1.5 kΩ, T_A = +125°C	130	160	190	mW	-
9-Volt Current Drain	I_T	3	V_{10} = 9 V	2.5	4	5.5	mA	-
Zener Regulating Voltage - DC Supply Voltage at Terminal 10	V_{10}	3	V_{CC} = 30 V, R_S = 1.5 kΩ	10.5	11.2	11.9	V	-
Quiescent Operating Current into Terminal 2	I_2	3		1	2	4	mA	-
Quiescent Operating Voltage at Terminal 4	V_4	-		5.0	6.5	8.0	V	-
Quiescent Operating Voltage at Terminal 5	V_5	-		5.0	6.5	8.0	V	-
Output Offset Voltage between Terminals 4 and 5	V_{4-5}	-		-1.5	0	1.5	V	-
DYNAMIC CHARACTERISTICS (AS RF AMPLIFIER)								
Input Limiting Voltage (Knee)	V_i Limiting	4	f=45.75 MHz	-	75	-	mV	-
Input Admittance	y_{11}	-	f = 45.75 MHz, V_{CC} = 30 V, R_S = 1.5 kΩ	-	0.5+j1.1	-	mmho	-
Reverse Transfer Admittance	y_{12}	-		-	3.8+j3.4	-	μmho	-
Forward Transfer Admittance	y_{21}	-		-	-11.7 +10.1	-	mmho	-
Output Admittance	y_{22}	-		-	0.077+j0.9	-	mmho	-
OUTPUT vs FREQUENCY DEVIATION - AFC								
Correction-Control Voltage at Terminal 4	V corr. (4)	5	V_{CC} = +30 V, V_{in} = 200 mV RMS, f_o = MHz as indicated	% of V_{10}		% of V_{10}		
			45.750 - 0.025	85	-	-	V	6,7
			45.750 + 0.025	-	-	33	V	
			45.750 - 0.900	75	-	-	V	7
			45.750 + 0.900	-	-	43	V	
			45.750 - 1.500	-	-	85	V	
			45.750 + 1.500	33	-	-	V	
Correction-Control Voltage at Terminal 5	V corr. (5)	5	45.750 - 0.025	-	-	33	V	6,7
			45.750 + 0.025	85	-	-	V	
			45.750 - 0.900	-	-	43	V	7
			45.750 + 0.900	75	-	-	V	
			45.750 - 1.500	33	-	-	V	
			45.750 + 1.500	-	-	85	V	

DYNAMIC CONTROL VOLTAGE CHARACTERISTICS

The CA3044 and CA3044VI are specifically intended for use in the AFT system of color television receivers. Each device is tested so that the control voltages generated by the circuit meet the critical requirements of the system. Figure 5 is the schematic diagram of the test circuit.

Figure 6 and 7 show the control voltages generated at terminals 4 and 5 of the Integrated Circuit as a function of the frequency deviation from the nominal center frequency. Figure 6 shows the region within 25 KHz of the center frequency while Figure 7 covers the entire bandwidth of the system. The horizontal reference lines on the figures are generated by a voltage divider connected between the power supply voltage on Terminal 10 and ground. The dynamic control voltages are compared with these references according to the Output vs Frequency Deviation Table. For example: when the frequency deviation is -25 KHz the control voltage at Terminal 4 is greater than the reference A voltage; the control voltage at Terminal 5 is less than the reference B voltage.

The shape of the correction voltage characteristics is dependent to a large degree upon transformer characteristics and the parts layout. In order to closely duplicate the curves shown, the printed circuit board shown in Figure 8 and the parts layout shown in Figure 9 should be followed as closely as possible.

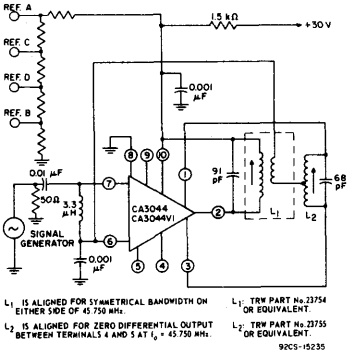


Fig.5 - Correction voltage test circuit for CA3044 and CA3044VI.

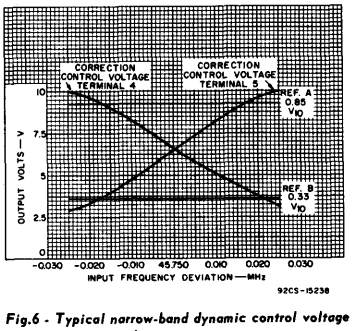


Fig.6 - Typical narrow-band dynamic control voltage characteristics.

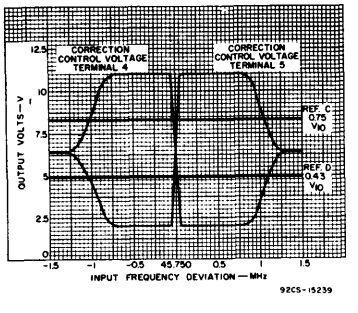


Fig.7 - Typical wide-band dynamic control voltage characteristics.

DEFINITIONS OF TERMS

Input Limiting Voltage (Knee) [v_i(lim)]

The input signal voltage which will cause the output signal to decrease 3 dB from its maximum level.

Total Device Dissipation (P_T)

The total power drain of the device with no signal applied and no external load current.

Quiescent Operating Voltage

The dc voltage at the output terminal, with respect to ground, with no signal applied.

Quiescent Operating Current

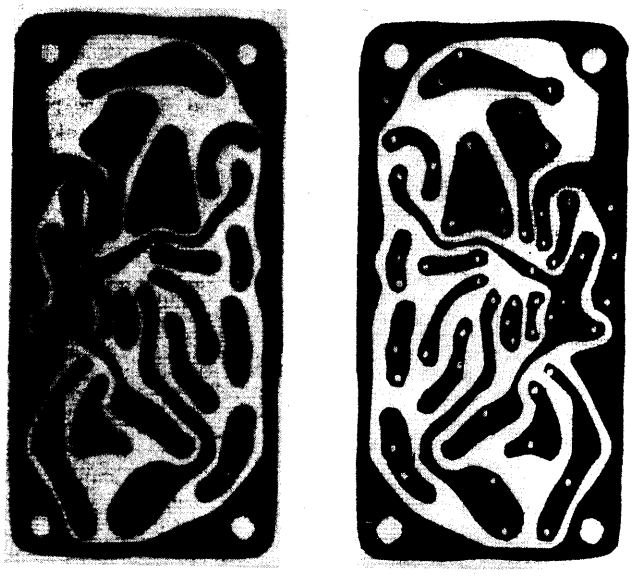
The average (dc) value of the current in either output, terminal, with no signal applied.

Output Offset Voltage

The dc voltage between output terminals with no signal applied.

Control Voltage

The dc voltage at either output terminal with respect to ground with an RF signal of specified frequency applied.



a) Top view b) Bottom view

Fig.8 - Printed Circuit Board for Test Circuit -- Full Size

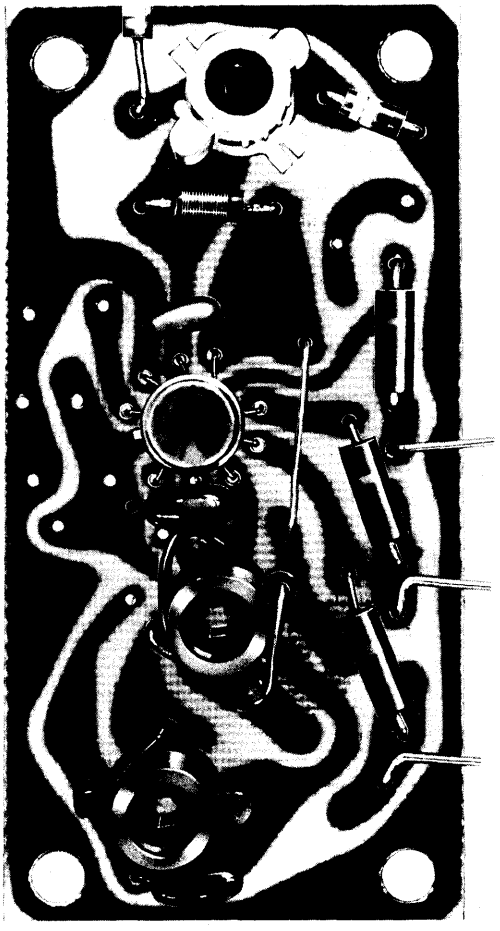


Fig.9 - Top view of wired test board.

Four Independent AC Amplifiers

Special-Function Sub-System for Stereo Preamplifiers, Magnetic Pickups, Tape Heads, etc.

APPLICATIONS

- **Full-function stereo preamplifiers**
- **Tape recorder and playback preamplifiers**
- **Tone Generators**

FEATURES

- **Four AC amplifiers on a common substrate**
- **Independently accessible inputs and outputs**
- **Operates from single-ended supply**

EACH AMPLIFIER

- **High voltage gain . . . . 53 dB min.**
- **High input resistance . . 90 k Ω typ.**
- **Undistorted output voltage 2 V rms min.**
- **Output Impedance . . . 1 k Ω typ.**
- **Open-loop bandwidth . . 300 kHz typ.**

The RCA-CA3052 is a silicon monolithic integrated circuit designed specifically for stereo preamplifier service. The circuit consists of four independent ac amplifiers which can operate from a single-ended supply.

The CA3052 can operate as an equalizer amplifier in tape recorders, magnetic cartridge phonograph applications, and tone control amplifiers. It can provide all of the amplification necessary for a full-function stereo preamplifier.

The CA3052 is supplied in a 16-lead dual-in-line plastic package.

RCA-CA3052 is schematically identical with the CA3048 Amplifier Array (File No. 377). Each amplifier of the CA3048 is tightly specified for equivalent output noise under a variety of test methods. The CA3052 is specified using RIAA test methods for equivalent input noise using one test method for amplifiers 1 and 4, and an appropriately different method for amplifiers 2 and 3.

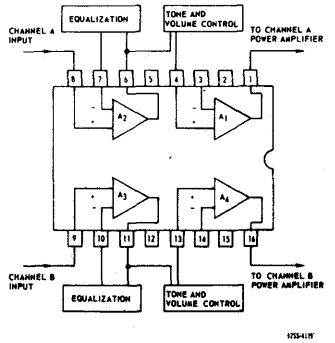


Fig. 1 – Block diagram of stereo preamplifier using CA3052.

ABSOLUTE-MAXIMUM RATING at T_A = 25°C:

POWER SUPPLY VOLTAGE . . . . . +16 V

AC INPUT VOLTAGE . . . . . 0.5 V rms

DISSIPATION:

Up to T_A = 55°C . . . . . 750 mW

Above T_A = 55°C . . . . . Derate linearly at 7.7 mW/°C

TEMPERATURE RANGE:

Operating . . . . . –40°C to +85°C

Storage . . . . . –65°C to +150°C

LEAD TEMPERATURE (During Soldering):

At distance 1/16 ±1/32 inch (1.59 ±0.79 mm) from case for 10 seconds max. . . . . +265°C

MAXIMUM VOLTAGE RATINGS

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range between vertical terminal 2 and horizontal terminal 4 is +2 to -3.6 volts.

TERMINAL No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1		+16 0	*	*	*	*	*	*	*	*	*	*	*	*	0 -16	*
2			*	+2 -3.6	0	*	*	+2 -3.6	+2 -3.6	*	*	+16 0	+2 -3.6	*	+16 0	0 -16
3				+5 -5	*	*	*	*	*	*	*	*	*	*	*	*
4					+3.6 -2	*	*	*	*	*	*	*	*	*	*	*
5						0 -16	*	+2 -3.6	+2 -3.6	*	0 -16	+16 0	+2 -3.6	*	+16 0	*
6							*	*	*	*	*	*	0 -16	*	*	*
7								+5 -5	*	*	*	*	*	*	*	*
8									*	*	*	*	*	*	*	*
9										+5 -5	*	*	*	*	*	*
10											*	*	*	*	*	*
11												*	*	*	*	*
12													0 -16	*	*	*
13														+5 -5	*	*
14															*	*
15																+16 0
16																

* Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if the specified limits between all other terminals are not exceeded.

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	LIMITS CA3052			UNITS
			MIN.	TYP.	MAX.	
STATIC						
Current drain per amplifier pair	I_{12} or I_{15}	V_{CC} = +12 V	9.5	13.5	17.5	mA
DC Voltage at Output Terminals	V_1, V_6, V_{11}, V_{16}	V_{CC} = +12 V	6.1	6.9	8.1	V
DC Voltage at Feedback Terminals	V_3, V_7, V_{10}, V_{14}	V_{CC} = +12 V	1.7	2.0	2.3	V
DC Voltage at Input Terminals	V_4, V_8, V_9, V_{13}	V_{CC} = +12 V	2.2	2.5	2.8	V
DYNAMIC each amplifier with no AC feedback unless otherwise noted–terminals 3, 7, 10, & 14 bypassed to ground						
Open-Loop Gain	A_{OL}	V_{CC} = +12 V E_{IN} = 2 mV f = 10 kHz	53	58	–	dB
Open-Loop Output Voltage Swing	$V_{O(rms)}$	V_{CC} = +12 V f = 1 kHz THD = 5%	2.0	2.4	–	V
Open-Loop -3 dB Bandwidth	BW	V_{CC} = +12 V E_{IN} = 2 mV	–	300	–	kHz
Open-Loop Total Harmonic Distortion	THD	V_{CC} = +12 V, f = 1 kHz E_{OUT} = 2 V rms	–	0.65	–	%
Input Resistance	R_I	V_{CC} = +12 V, f = 1 kHz	–	90	–	kΩ
Input Capacitance	C_I	V_{CC} = +12 V, f = 1 MHz	–	9	–	pF
Output Resistance	R_O	V_{CC} = +12 V, f = 1 kHz	–	1	–	kΩ
Feedback Capacitance (Output to non-inverting Input)	C_{FB}	V_{CC} = +12 V f = 1 MHz	–	<0.1	–	pF
Equivalent Input Noise Voltage (Amplifiers 1 & 4), "C" Filter at Output*	E_{NI}‡	V_{CC} = +10 V R_S = 5 kΩ A = 45 dB	–	1.7	6.4	μV
Equivalent Input Noise Voltage (Amplifiers 2 & 3) RIAA Compensated*	E_{N2}‡	V_{CC} = +10 V R_S = 5 kΩ A = 64 dB (1 kHz)	–	4	15.0	μV
Inter-Amplifier Audio Separation "Cross Talk"[11]		V_{CC} = +12 V f = 1 kHz 0 dB = 0.78 V	–	<-45	–	dB
Inter-Amplifier Capacitance (Any amplifier output to any *other* amplifier input)	C	V_{CC} = +12 V f = 1 MHz	–	< 0.02	–	pF

*Per IHF Standard Methods of Measurement for Audio Amplifiers IHF–A–201, 1966

‡ ac feedback included in test circuit

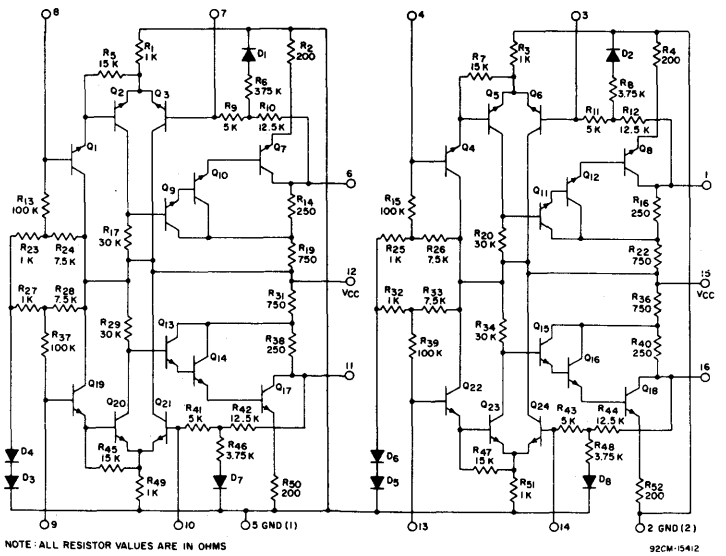


Fig. 2 – Schematic diagram for CA3052.

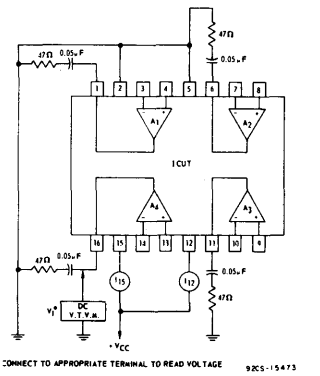
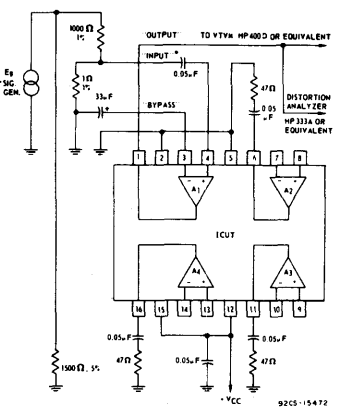


Fig. 3 – Test circuit for measurement of collector supply voltage and currents.



* Sig Gen should be a low distortion type (0.2% THD or less) HP206A or equivalent.

● Adjustment of E_g to 2 volts will make E_S = 2 mV.

Test Circuit shows Amplifier #1 under test, to test Amplifiers 2, 3, or 4; Connect terminals as shown in Table.

AMPLIFIER	TERMINALS		
	OUTPUT	INPUT	BYPASS
1	1	4	3
2	6	8	7
3	11	9	10
4	16	13	14

Fig. 6 – Test circuit for measurement of distortion, open-loop gain, and bandwidth characteristics.

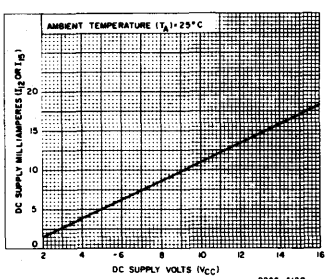


Fig. 4 – Typical DC supply current vs supply voltage.

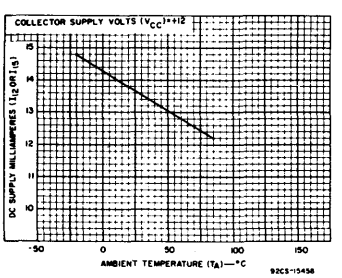


Fig. 5 – Typical DC supply current vs ambient temperature.

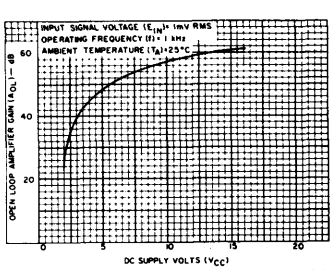


Fig. 7 – Typical amplifier gain vs DC supply voltage.

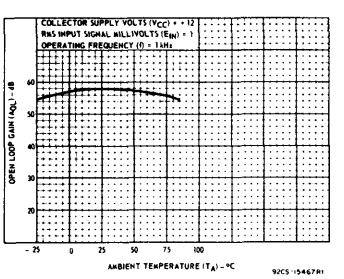


Fig. 8 – Typical open-loop gain vs ambient temperature.

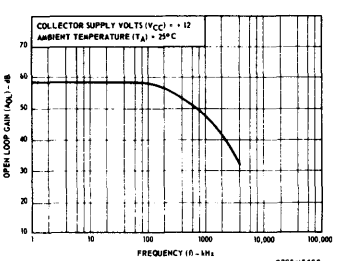


Fig. 9 – Typical open-loop gain vs frequency.

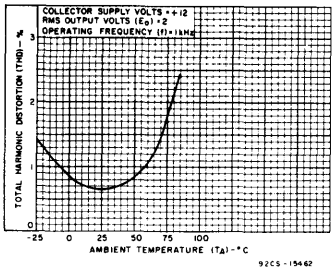
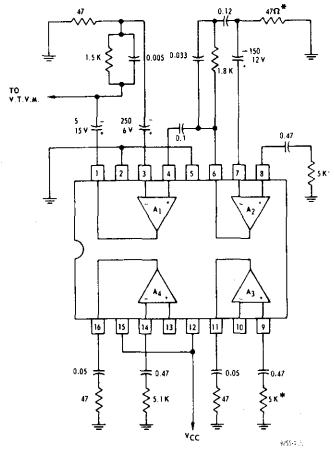
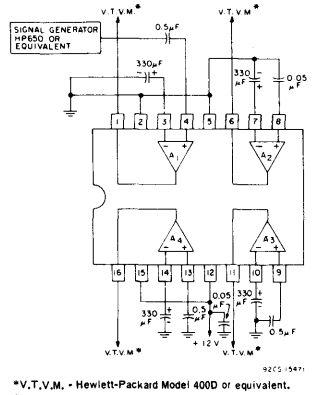


Fig. 10 – Typical total harmonic distortion vs ambient temperature.



*Resistors are low noise precision (1%) Metal Film type.

Fig. 13 – Test circuit for equivalent input noise voltage measurement, RIAA compensated.

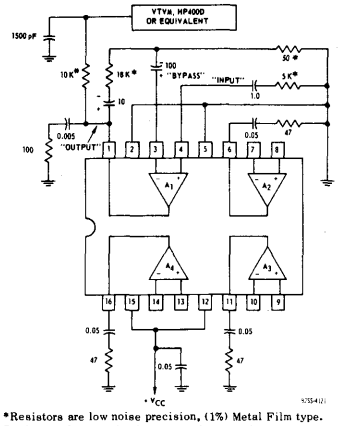


*V.T.V.M. - Hewlett-Packard Model 400D or equivalent.

Procedure:

1. Adjust Signal Generator for 0 dB output at reference terminal.
2. Read voltage at other output terminals (Figure shows terminal #1 used as reference).

Fig. 14 – Test circuit for measurement of inter-amplifier audio separation "cross talk" characteristic.



*Resistors are low noise precision, (1%) Metal Film type. Resistor values are in ohms; capacitance values are in microfarads, unless otherwise specified.

Fig. 11 – Test circuit for measurement of equivalent input noise voltage of amplifiers 1 and 4.

OPERATING CONSIDERATIONS

Economical Gain Control

The CA3052 is designed to permit flexibility in the methods by which amplifier gain can be controlled. Fig. 15 shows a curve of the gain of an amplifier when the internal resistive feedback of the device is used in conjunction with an external resistor. Although measured gain of various amplifiers will not be uniform, because of tolerances of internal resistances, this method is very economical and easy to apply.

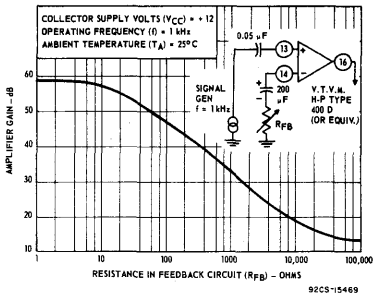
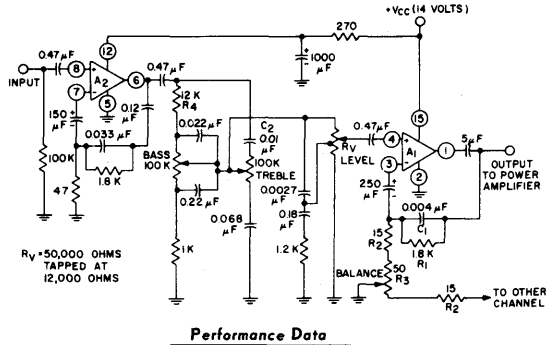


Fig. 12– Typical amplifier gain vs feedback resistance.

Stability

The CA3052, as in other devices having high gain-band-width product, requires some attention to circuit layout, design, and construction to achieve stability.

Should the CA3052 be left unterminated, socket capacitance alone will provide sufficient feedback to cause high frequency oscillations; therefore, all test circuits in this data bulletin include loading networks that provide stability under all conditions.



Performance Data

Gain at 1-kHz reference	47 dB
Boost at 100 Hz	11.5 dB
Boost at 10 kHz	11.5 dB
Cut at 100 Hz	10 dB
Cut at 10 kHz	9 dB

Noise:

At maximum volume (input shorted)	> 70 dB below 1 volt
At minimum volume	> 80 dB below 1 volt
Total harmonic distortion (at 1-kHz reference and an output of 1 volt)	< 0.3 per cent

92CM-29305

Fig. 15 – Schematic of one channel of a complete stereo preamplifier.

TV Automatic Fine Tuning Circuit

RCA-CA3064 and CA3064E represent the third generation of integrated circuits designed primarily for AFC (Automatic-Frequency-Control) applications. They provide all of the signal-processing components needed (with the exception of the tuned-phase-detector transformer) to derive the AFT correction signals from the output of the video-if amplifier. The CA3064 is supplied in the 10-formed-lead TO-5 style package, and the CA3064E in the 14-lead dual-in-line plastic package. Both types operate over the temperature range of −55 to +125°C.

The CA3064 and CA3064E are functionally similar to the CA3044 and CA3044V1 but embody a higher-gain input amplifier which provides a 20-dB improvement in sensitivity. The increased sensitivity extends the application of a proven AFT system to the low-level if-amplifier stages in TV receivers.

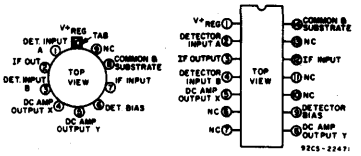
Because the CA3064 and CA3064E are functionally similar to the CA3044 and CA3044V1, refer to Application Note ICAN-5831, "Application of the RCA CA3044 and CA3044V1 Integrated Circuits in Automatic Fine-Tuning Systems" for general application information.

Features:

- **Cascode type high-gain amplifier (18 mV input for rated output)**
- **Internal voltage regulator**
- **Differential detector**
- **For use with either color or monochrome**
- **Differential amplifier**
- **Bipolar outputs**
- **Wide operating-temperature range; −55 to +125°C**

MAXIMUM RATINGS, *Absolute-Maximum Values:*

DEVICE DISSIPATION:
Up to T_A = 25°C . . . . . 700 mW
Above T_A = 25°C . . . . . derate linearly 5.6 mW/°C
AMBIENT TEMPERATURE RANGE:
Operating . . . . . −55 to +125°C
Storage . . . . . −65 to +150°C
LEAD TEMPERATURE (During Soldering):
At distance 1/16" ± 1/32" (1.59 mm ± 0.79 mm) from case for 10 s max.. . . . . 265°C



(a) CA3064 (b) CA30S4E

Fig.2 – Terminal assignment diagrams.

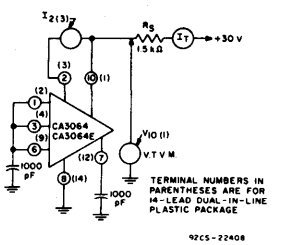


Fig. 3 – Test setup: Measurement of total device dissipation, zener regulating voltage, quiescent operating current at terminal 2 (3).

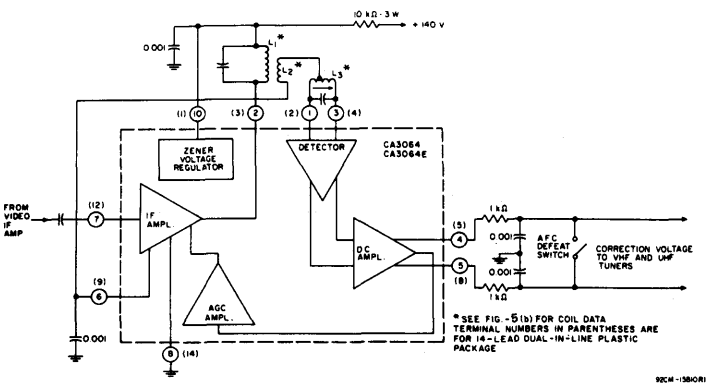


Fig.1 – Block diagram of typical operating circuit utilizing the CA3064 and CA3064E.

ELECTRICAL CHARACTERISTICS at T_A = 25°C, Unless Otherwise Specified

CHARACTERISTICS	SYMBOLS	TEST CIRCUITS FIG.	TEST CONDITIONS		LIMITS CA3064, CA3064E MIN.	TYP.	MAX.	UNITS	CHARACTERISTIC CURVES FIG.
STATIC CHARACTERISTICS									
Device Dissipation	P_D	3	V^+ = 30V, R_S = 1.5kΩ	T_A −25°C	-	135	150	mW	-
				+25°C	130	140	150		-
				+85°C	-	145	150		-
Current Drain at 10.5 Volts	I_T	3	$V_{10(1)}$ = 10.5 V		4	6.5	9.5	mA	-
Zener Regulated Voltage – DC Supply Voltage at terminal 10(1)*	$V_{10(1)}$	3			10.9	11.8	12.8	V	-
Quiescent Operating Current into Terminal 2(3)	$I_{2(3)}$	3			1	2	4	mA	-
Quiescent Operating Voltage at Terminal 4(5)	$V_{4(5)}$	-	V^+ = 30 V, R_S = 1.5 kΩ		5	6.9	8	V	-
Quiescent Operating Voltage at Terminal 5(8)	$V_{5(8)}$	-			5	6.9	8	V	-
Output Offset Voltage between Terminals 4 and 5(5 and 8)	V_{4-5} (5-8)	-			-1	0	1	V	-
DYNAMIC CHARACTERISTICS (AS RF AMPLIFIER IN TO-5 STYLE PACKAGE)									
Input Voltage Sensitivity	V_I sensitivity	5	V^+ = +30 V, V_I = 18 mV		Correction Voltage Output as shown in table below.				
Input Admittance	y_{11}	-	f = 45.75 MHz, V^+ = 30 V, R_S = 1.5 kΩ		-	0.41 + j1.0	-	mmho	-
Reverse Transfer Admittance	y_{12}	-			-	0 +j3.4	-	μmho	-
Forward Transfer Admittance	y_{21}	-			-	24.5 - j29	-	mmho	-
Output Admittance	y_{22}	-			-	0.04 + j0.9	-	mmho	-
OUTPUT vs FREQUENCY DEVIATION - AFC									
Correction-Control Voltage at Terminal 4(5)	V corr. 4(5)	5	V^+ = +30 V, V_I = 18 mV RMS, f_0 = MHz as indicated		% of V_{10} (1)		% of V_{10} (1)		
			45.750 - 0.030		85	-	-	V	6,7
			45.750 + 0.030		-	-	25	V	
			45.750 - 0.900		80	-	-	V	7
			45.750 + 0.900		-	-	35	V	
			45.750 - 1.500			-	80	V	
			45.750 + 1.500		35	-		V	
Correction-Control Voltage at Terminal 5(8)	V corr. 5(8)	5	45.750 - 0.030		-	-	25	V	6,7
			45.750 + 0.030		85	-	-	V	
			45.750 - 0.900		-	-	35	V	7
			45.750 + 0.900		80	-	-	V	
			45.750 - 1.500		35	-	-	V	
			45.750 + 1.500		-	-	80	V	

* Terminal numbers in parentheses are for 14-lead dual-in-line plastic package.

CA3064, CA3064E

MAXIMUM VOLTAGE RATINGS at T_A = 25°C

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range between vertical terminal 2 (3) and horizontal terminal 6 (9) is +20 to 0 volts. Terminal nos. in parentheses are for the 14-lead dual-in-line plastic package.

TERMINAL No.	9(6,7, 10,11, 13)	10 (1)	1 (2)	2 (3)	3 (4)	4 (5)	5 (8)	6 (9)	7 (12)	8 (14)
9(6,7,♦ 10,11, 13)	◄ NO INTERNAL CONNECTION ►									
10 (1)			+12 0	+10 -10	+12 0	+12 0	+12 0	+10 0	+20 0	▲
1 (2)				*	+10 - 10	*	*	+5 - 5	*	+5 - 6
2 (3)					*	*	*	+20 0	*	+20 0
3 (4)						*	*	+5 - 6	*	+5 - 6
4 (5)							*	*	*	+12 0
5 (8)								*	*	+12 0
6 (9)									+5 - 2	+2 0
7 (12)										+2 - 10
8 (14)										REF.SUBSTRATE & CASE■

MAXIMUM CURRENT RATINGS

TERMINAL No.	I_{IN} mA	I_{OUT} mA
9(6,7, 10,11, 13)	-	-
10 (1)	50	50
1 (2)	1	0.1
2 (3)	20	20
3 (4)	1	0.1
4 (5)	5	5
5 (8)	5	5
6 (9)	5	5
7 (12)	1	1
8 (14)	50	50

▲ Terminal number 10 (1) may be connected to any positive voltage source greater than the internal zener regulating voltage through a suitable dropping resistor – provided the dissipation rating is not exceeded.

■ This terminal should be connected to the most negative potential of the complete circuit.

* Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if the specified limits between all other terminals are not exceeded.

♦ It is recommended that unused terminals 6,7,10,11, and 13 on the 14-lead dual-in-line-plastic package and terminal 9 on the TO-5 package be grounded to act as shields.

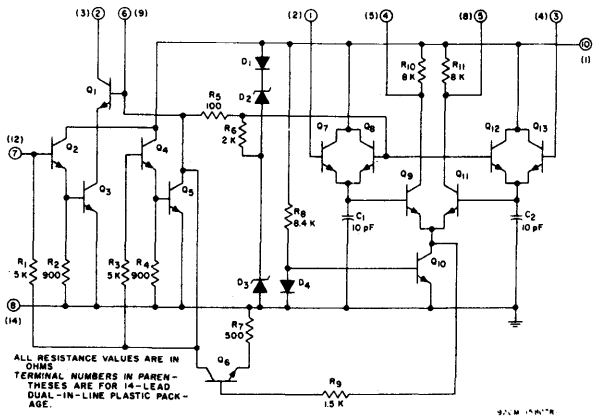
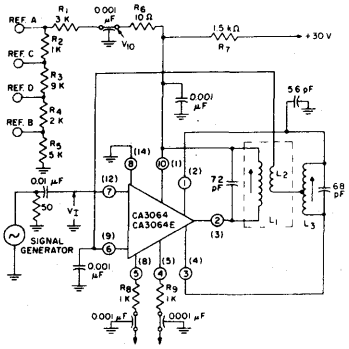


Fig.4 – Schematic diagram for CA3064 and CA3064E.



L_1 IS ALIGNED FOR SYMMETRICAL BANDWIDTH ON EITHER SIDE OF 45.750 MHz
L_2 TERTIARY WINDING WOUND ON L_1 COIL FORM
L_3 IS ALIGNED FOR ZERO DIFFERENTIAL OUTPUT BETWEEN TERMINALS 4 AND 5 AT f_o = 45.750 MHz
* FOR COIL CONSTRUCTION DATA, SEE FIG. 4(b).

REFERENCE VOLTAGE PERCENTAGES	
Ref. A	85% of $V_{10(1)}$
Ref. B	25% of $V_{10(1)}$
Ref. C	80% of $V_{10(1)}$
Ref. D	35% of $V_{10(1)}$

Coil	RCA Distributor Part No.
(L_1, L_2)	122 213
L_3	122 203

Fig.5 (a) – Correction voltage test circuit for CA3064 and CA3064E.

COIL DATA FOR DISCRIMINATOR WINDINGS

L_1 – Discriminator Primary: 3-1/6 turns; #20 Enamel-covered wire – close-wound, at bottom of coil form. Inductance of L_1 = 0.165 μH; Q_o = 120 at f_o = 45.75 MHz. Start winding at terminal #6; finish at Terminal #1. See Notes below.

L_2 – Tertiary Windings: 2-1/6 turns; #20 Enamel-covered wire – close wound over bottom end of L_1. Start winding at Terminal #3; finish at Terminal #4. See Notes below.

L_3 – Discriminator Secondary: 3-1/2 turns; center-tapped, space wound at bottom of coil form. Inductance of L_3 = 0.180 μH; Q_o = 150 at f_o = 45.75 MHz. Start winding at Terminal #2; finish at Terminal #5; connect center tap to Terminal #7. See Notes below.

Notes: 1. Coil Forms; Cylindrical; –0.30" Dia. max.
2. Tuning Core: 0.250" Dia. x 0.37" Length.
: Material: Carbinal J or equivalent
3. Coil Form Base: See drawing below.
4. End of coil nearest terminal board to be designated the winding start end.

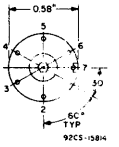


Fig.5 (b) Coil form base terminal diagram.

IF Amplifier-Limiter, FM Detector, Electronic Attenuator, Audio Driver

For Television Sound-System Applications

The RCA CA3065 Television Sound System is a monolithic integrated circuit which combines a multistage IF amplifier limiter, an FM detector, an electronic attenuator, a zener diode regulated power supply, and an audio amplifier-driver that is designed to directly drive an npn power transistor or high-transconductance tube. Because the circuit is so inclusive, a minimum number of external components is required. A block diagram of the integrated circuit television sound system is shown in Fig. 1.

The CA3065 with its advanced circuit design provides a high-performance multistage subsystem for the sound system of a television receiver. A particular feature of the CA3065 is the electronic attenuator which performs the conventional volume control function. Volume control is accomplished when the bias levels in the attenuator are changed by means of a variable resistor connected between Terminal 6 and ground (attenuation in excess of 60 dB is attained). Because no audio signal is present in this control, hum or noise pickup can be bypassed. In most cases, only a single unshielded wire is required between the IF board and the variable resistor (volume control).

The CA3065 utilizes a 14-lead dual-in-line plastic package with leads specially formed to facilitate automatic insertion of the device into suitably punched printed-circuit boards.

FEATURES:

- Electronic attenuator - replaces conventional volume control
- Differential peak detector - requires one single tuned coil
- Internal Zener diode regulated supply
- Inherent high stability
- Excellent AM rejection - 50 dB typ. at 4.5 MHz
- Low harmonic distortion
- High sensitivity - 200 μV limiting (knee) at 4.5 MHz
- Audio drive capability - 6 mA p-p
- Undistorted audio output voltage - 7 V p-p

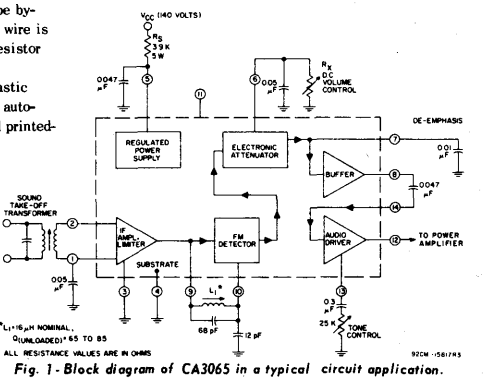


Fig. 1 - Block diagram of CA3065 in a typical circuit application.

MAXIMUM RATINGS, *Absolute Maximum Values*, at T_A = 25°C

Input Signal Voltage (between Terminals 1 and 2)	±3	V
Power Supply Current (Terminal 5)	50	mA
Power Dissipation:		
Up to T_A = 25°C	850	mW
Above T_A = 25°C	Derate linearly 6.67	mW/°C
Ambient Temperature Range:		
Operating	- 40 to + 85	°C
Storage	- 65 to + 150	°C
Lead Temperature (During Soldering):		
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max.	+265	°C

MAXIMUM VOLTAGE RATINGS at T_A = 25°C

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range of the vertical terminal 9 with respect to terminal 3 is 0 to +4 volts.

TERMINAL No.	4	5	6	7	8	9	10	11	12	13	14	1	2	3
4		SUBSTRATE CONNECTION – ALWAYS CONNECT TO TERMINAL 3												
5			+13 0	+13 0	+13 0	*	*	INTERNAL CONNECTION DO NOT USE	+13 0	+13 0	*	*	*	NOTE 1
6				*	*	*	*		*	*	*	*	*	+13 −5
7					+1 −4	*	*		*	*	*	*	*	+13 0
8						*	*		*	*	*	*	*	*
9							*		*	*	*	*	*	+4 0
10									*	*	*	*	*	+4 −5
11									INTERNAL CONNECTION DO NOT USE					
12										+4 −1	*	*	*	*
13											*	*	*	*
14												*	*	+3 −5
1													+5 −5	+5 −5
2														+4 −5
3														

Note 1: Terminal No. 5 may be connected to any positive voltage through a suitable resistor provided that the current and dissipation ratings of the CA3065 are not exceeded.

*Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if specified limits between all other terminals are not exceeded.

MAXIMUM CURRENT RATINGS

TERMINAL No.	I_{IN} mA	I_{OUT} mA
4	SUBSTRATE: CONNECT TO TERMINAL 3	
5	50	1
6	1	1
7	1	1
8	0.5	6
9	1	1
10	1	0.1
11	INT. CONN. DO NOT USE	
12	0.5	6
13	1	2
14	1	0.1
1	1	0.1
2	1	0.1
3	0.1	50

CA3065

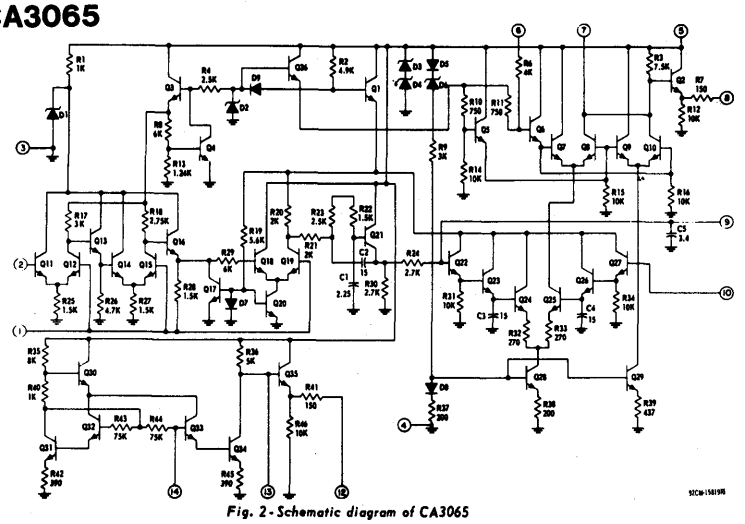


Fig. 2 - Schematic diagram of CA3065

The resistance values included on the schematic diagram have been supplied as a convenience to assist Equipment Manufacturers in optimizing the selection of "outboard" components of equipment designs. The values shown may vary as much as ±30%.

RCA reserves the right to make any changes in the Resistance Values provided such changes do not adversely affect the published performance characteristics of the device.

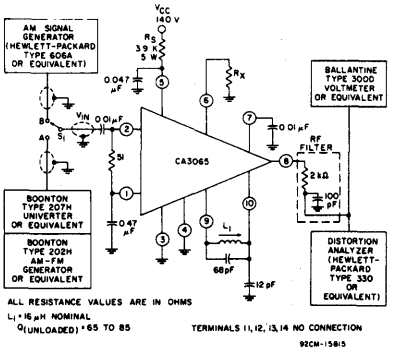


Fig. 3 - Input limiting voltage, AM rejection, recovered audio, total harmonic distortion, maximum attenuation, maximum "play-through" test circuit.

ELECTRICAL CHARACTERISTICS at T_A = 25° C, V_{CC} = +140 V applied to Terminal 5 through R_S = 3.9 kΩ, and DC Volume Control (R_X) = 0 unless otherwise indicated.

CHARACTERISTIC	SYMBOL	SPECIAL TEST CONDITIONS	LIMITS Min.	LIMITS Typ.	LIMITS Max.	UNITS
Static Characteristics						
Zener Regulating Voltage Terminal No. 5	V_5		10.3	11.2	12.2	V
Current into Terminal 5	I_5	Connect Terminal 5 to +9 V	10	16	24	mA
Total Device Dissipation	P_T		343	370	400	mW
Terminal Voltages: 1	V_1		-	2	-	V
6	V_6		-	4.8	-	
7	V_7		-	6.1	-	
9	V_9		-	3.7	-	
12	V_{12}		4	5.1	5.8	
Dynamic Characteristics						
IF AMPLIFIER Input Limiting Voltage (at −3 dB point)	$V_{i(lim)}$	f_0 = 4.5 MHz, f_m = 400 Hz, Deviation = ±25 kHz,	-	200	400	μV
AM Rejection	AMR	Amplitude Modulation = 30% f = 4.5 MHz	40	50	-	dB
Transconductance Magnitude	$\|G_m\|(IF)$	f = 4.5 MHz IF Input Terminals: 2, 1	-	500	-	mmho
Phase Angle	$\theta(IF)$	IF Output Terminals: 9, 3	-	46	-	degrees
Feedback Capacitance	C_{fb}	f = 1 MHz; Terminals 2 and 9	-	0.02	-	pF
Input Impedance Components: Parallel Input Resistance	$R_i(IF)$	Measured between Terminal Nos. 1 and 2 f = 4.5 MHz	-	17	-	kΩ
Parallel Input Capacitance	$C_i(IF)$		-	4	-	pF
Output Impedance Components: Parallel Output Resistance	$R_o(IF)$	Measured between Terminal No. 9 and gnd f = 4.5 MHz	-	3.25	-	kΩ
Parallel Output Capacitance	$C_o(IF)$		-	7.5	-	pF
DETECTOR Recovered AF Voltage	$V_o(af)$	f = 4.5 MHz; V_i = 100 mV ∴ f = ±25 kHz fm = 400 Hz	0.5	0.75	-	V(rms)
Total Harmonic Distortion	THD		-	0.9	2	%
Output Resistance: Terminal 7	R_o		-	7.5	-	kΩ
Terminal 8			-	300	-	Ω
ATTENUATOR		See Fig. 7				
Max. Attenuation	-	R_X = ∞	60	80	-	dB
Max. "Play-through" Voltage*	-	R_X = 0Ω	-	0.075	1	mV
AUDIO AMPLIFIER Voltage Gain	A(af)	V_i = 0.1 V(rms), f = 400 Hz	17.5	20	-	dB
Total Harmonic Distortion	THD	V_O = 2 V(rms), f = 400 Hz	-	1.5	-	%
Undistorted Output Voltage	-	THD = 5%, f = 400 Hz	2	2.5	-	V(rms)
Input Resistance	$R_i(af)$	f = 400 Hz	-	70	-	kΩ
Output Resistance	$R_o(af)$	f = 400 Hz	-	270	-	Ω

*"Playthrough" voltage is the unwanted signal, measured at Terminal 8, when the volume control is set for minimum output.

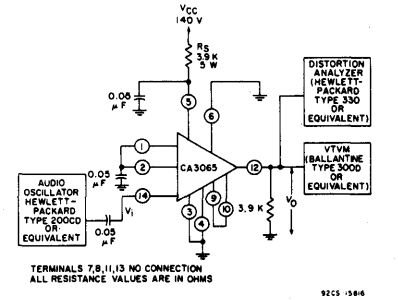
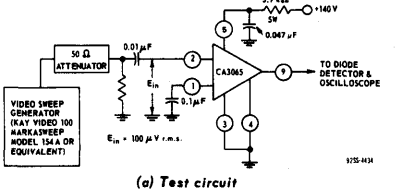
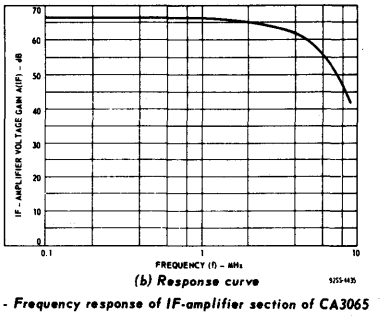


Fig. 4 - Audio voltage gain (undistorted output) test circuit.

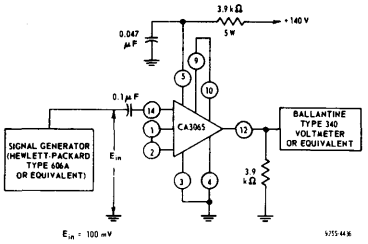


(a) Test circuit

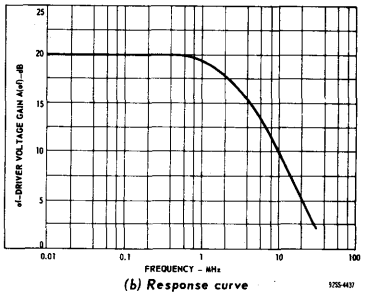


(b) Response curve

Fig. 5 - Frequency response of IF-amplifier section of CA3065



(a) Test circuit



(b) Response curve

Fig. 6 - Frequency response of af-amplifier section of CA3065

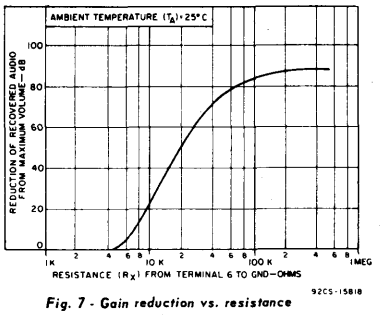


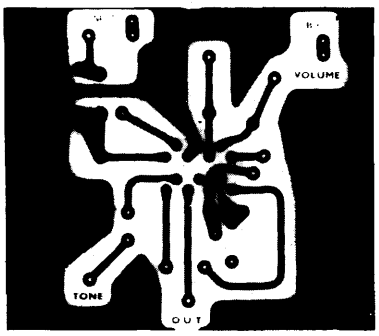
Fig. 7 - Gain reduction vs. resistance (terminal 6 to gnd)

OPERATING CONSIDERATIONS

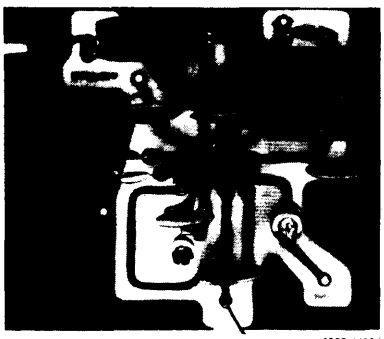
The CA3065 may be used to drive a video output transistor or a high-transconductance output tube.

As in all TV receivers, precaution should be taken to prevent destruction of the CA3065 in the event of cascade arcs originating in the picture tube or in the output tube. In the case of arcing in the output tube a resistor of 150 k in series with terminal No. 12 and the grid of the tube is usually sufficient protection.

To prevent damage from picture tube arcs, a careful analysis of board layout and coupling modes (electrostatic or magnetic) may be necessary to suggest alternate layouts or appropriate locations for the placement of spark gaps to absorb the high energy discharge.



*(a) Printed circuit board – bottom view**



*(b) Parts layout – top view**

Fig. 8 - Recommended parts layout for TV receiver sound strip using CA3065.

* A 200 mil square grid was used in the layout of passive components on the printed circuit board. The Quad-in-line formed leads conform to a standard grid spacing of 100 mil centers.

CA3066, CA3067

Television Chroma System

The RCA CA3066 and CA3067 are monolithic silicon integrated circuits that constitute a complete chroma system for color television receivers. The CA3066 provides subcarrier regeneration and total chroma signal processing prior to demodulation; the CA3067 performs the demodulation and tint control functions. Each device utilizes a 16-lead quad-in-line plastic package.

System Features

CA3066
CHROMA SIGNAL PROCESSOR

- Complete Color Sync Circuit
- Blanked Chroma Amplifier
- Chroma Band-Pass Amplifier
- Low Output Impedance Chroma Driver
- ACC Detector-Amplifier
- Killer Detector-Amplifier
- DC Chroma Gain Control
- Zener Diode for Regulated Voltage Reference
- Short-Circuit Protection on All Terminals

CA3067
CHROMA DEMODULATOR

- Balanced Chroma Demodulators
- Color Difference Matrix
- DC Tint Control
- Three Low Output Impedance Drivers for Direct Coupling
- Reference Subcarrier Limiter
- Zener Diode for Regulated Voltage Reference
- Internal RF Filtering

CA3066 CHROMA SIGNAL PROCESSOR

The CA3066 contains substantially all the color processing circuitry exclusive of the tint control and demodulating circuits. The chroma amplifier sections of the CA3066 consist of the chroma and bandpass amplifiers. The chroma amplifier receives the chroma input signal at terminal No. 1. This amplifier is gain controlled by the automatic chroma control (ACC) detector-amplifier. The chroma signal is internally coupled from the output of the chroma amplifier to the input of the chroma bandpass amplifier and burst separator amplifier. The horizontal keying pulse (+8V) is used to gate the burst portion of the chroma signal from the input of the bandpass amplifier to the input of the burst separator amplifier. The bandpass amplifier is gain controlled by the dc chroma gain control and can also be controlled by the killer detector-amplifier. The bandpass amplifier output is internally coupled to the chroma output amplifier stage of the CA3066. The coils of the chroma amplifier and the bandpass amplifier are stagger-tuned to provide a combined typical bandpass of 3.08 to 4.08 MHz. The burst separator amplifier injects the burst signal into the 3.58 MHz oscillator. The oscillator amplitude is dependent on the terminal No. 9 impedance to ground and is also responsive to the burst signal amplitude at terminal No. 11. The ACC detector and killer detector sense the burst level or absence of burst, respectively, by monitoring the oscillators response to the burst injection level. The thresholds for the ACC and killer are independently adjusted by resistors R2 and R1 at terminals No. 9 and No. 4, respectively. The chroma output is at terminal No. 14 and the oscillator output is at terminal No. 8. Terminal No. 6 is a zener diode for use as a regulated voltage reference at 11.9 volts. When the zener reference element is not used, the power supply voltage should be maintained at 11.2 ± 0.5 volts.

CA3066

ELECTRICAL CHARACTERISTICS at T_A = 25°C and V^+ = 11.2 V

CHARACTERISTICS	SYMBOL	TEST CONDITIONS	LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNITS	TEST FIG. AND CURVES
Static Characteristics							
Voltages:							
ACC Reference	V_2		–	0.5	–		
Burst-Chroma Ampl. Bias Current Term.	V_3		–	2.9	–		
Killer Reference	V_4		–	1.0	–		
Zener Reg. Reference	V_6		10.6	11.9	12.6	V	2
Oscillator Input	V_7		–	1.4	–		
Oscillator Output	V_8		–	2.35	–		
Balance (ACC Control)	V_9		–	1.65	–		
Chroma Output	V_{14}		–	4.6	–		
Currents:							
Total Supply	I_5		14	24	33		
Burst Separator Output	I_{11}	S_1 Closed	–	6.5	–	mA	
Band-Pass Ampl. Output	I_{13}		–	4.8	–		
Chroma Ampl. Output	I_{16}		–	1.27	–		
Dynamic Characteristics							
Oscillator Output	v_8	v_1 = 0 $v_{p\text{-}p}$ v_1 = 1.25 $v_{p\text{-}p}$	0.8 –	1.2 2.5	– 3.5	$v_{p\text{-}p}$	4
Chroma Output:							
100%	v_{14}	v_1 = 1.25 $v_{p\text{-}p}$	0.5	1.0	–	$v_{p\text{-}p}$	3, 4
Killed		v_1 = 0.025 $v_{p\text{-}p}$	–	–	12		
ACC Detector Output	v_2	v_1 = 1.25 $v_{p\text{-}p}$	–	0.9	–	V	4
Small-Signal Input Resistance (Term. No.1)	r_i		–	50	–	kΩ	–
Small-Signal Input Capacitance (Term. No.1)	c_i		–	2.4	–	pF	
Small-Signal Output Impedance (Term. No.14)	r_o		–	250	–	Ω	

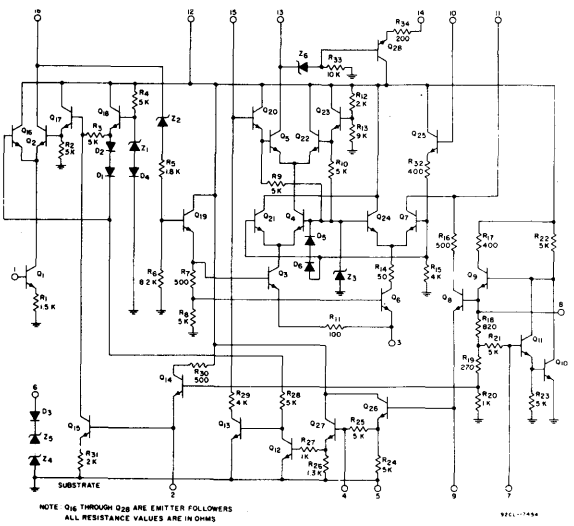


Fig. 1 - CA3066 schematic diagram.

CA3066

MAXIMUM RATINGS, *Absolute-Maximum Values at T_A = 25°C*

Supply Voltages and Currents (see charts below)

Device Dissipation:
Up to T_A = 70°C . . . . . 600 mW
Above T_A = 70°C . . . . *derate linearly 7.7 mW/°C*

Ambient Temperature Range:
Operating . . . . . −40 to +85 °C
Storage . . . . . −65 to +150 °C

Lead Temperature (During soldering for 10s max. at not less than 1/32" from package) . . . +265 °C

Voltage with respect to Terminal No. 5.

Terminal No.	$V_{min.}$ (volts)	$V_{max.}$ (volts)
6	See Note N1	
7	–	–
8	–	–
9	–	–
10	−5.0	N2
11	0.0	18.0
12	0.0	12.0
13	0.0	15.0
14	–	–
15	0.0	N2
16	0.0	15.0
1	−5.0	5.0
2	–	–
3	–	–
4	–	–

Current

Terminal No.	I_I mA	I_O mA
6	20	0.1
7	5	0.1
8	1	2
9	0.1	2
10	1	0.1
11	10	1
12	50	1
13	10	1
14	0.1	6
15	3	1
16	6	1
1	1	0.1
2	0.1	2
3	0.1	20
4	1	1

N1 Terminal No. 6 is connected to a zener reference element, that, if used, should be biased by a positive voltage through a resistor that limits the current to a value which is less than the maximum current rating of terminal No. 6.

N2 The upper voltage limit cannot exceed the power supply input voltage at terminal 12.

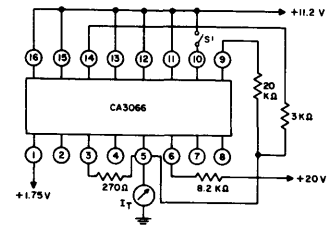


Fig. 2 - Static characteristics test circuit for CA3066.

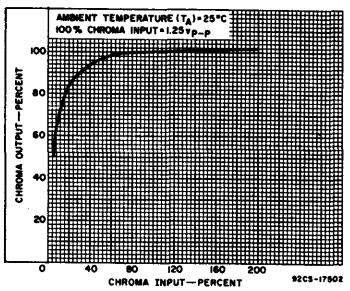


Fig. 3 - Typical ACC characteristic of chroma output vs chroma input for CA3066.

CA3067 CHROMA DEMODULATOR

The CA3067 contains the separate functional systems of a dc tint control and a demodulator. The phase shift of the tint amplifier system is accomplished by functional control of the fixed phase signal from the CA3066 oscillator output. This regenerated reference subcarrier is applied to terminal No. 3 and driven differentially into phase shift circuits. The tint adjustment controls the vector addition of phase shifted signals after which a limiting amplifier removes any remaining amplitude modulation. The output of the tint amplifier at terminal No. 1 is phase separated for the required reference subcarrier phase at terminal No. 6 and No. 12 (terminal No. 12 lags terminal No. 6 by approximately 76°). These terminals are inputs to the demodulator drive amplifiers. The demodulators consist of two sets of balanced detectors which receive their reference subcarrier from the demodulator drive amplifiers. The chroma signal input from the CA3066 is applied to terminal No. 14. The chroma signal differentially drives the demodulators. The demodulation components are matrixed and dc-shifted in voltage to give R–Y, G–Y, and B–Y color difference components with close dc balance and proper amplitude ratios. The output amplifiers of the CA3067 are specially designed to meet the low-impedance driving source requirements of the high-level color output amplifiers. A special feature of the CA3067 is R–C filtering of high frequency demodulation components. Terminal No. 4 is a zener diode for use as a regulated voltage reference at 11.9V. When the zener reference element is not used, the power supply should be maintained at +11.2 ±0.5 volts.

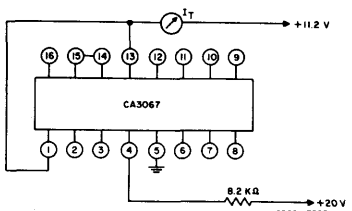


Fig. 5 - Static characteristics test circuit for CA3067.

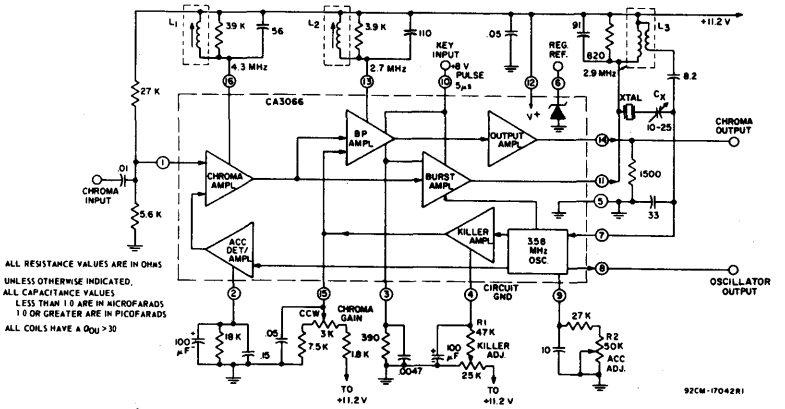


Fig. 4 - Dynamic characteristics test circuit for CA3066.

DYNAMIC CHARACTERISTICS TEST PROCEDURE

Steps 1, 2, and 3 are performed with no Chroma input ($v_1 = 0$)

1. Adjust ACC potentiometer for $V_2 = +0.65V$.
2. Adjust Killer potentiometer for $V_4 = +1.2V$.
3. Adjust capacitor C_X (crystal trimmer) so that frequency of oscillator is 3.579545 MHz.
4. Unless otherwise noted, the chroma gain control is at maximum gain (fully clockwise).
5. The chroma input test signal is a 52.5 μs "line" at subcarrier frequency, and 10 cycles of burst at 46.5% of the "line" amplitude. The chroma input (v_1) is in peak-to-peak volts of "line" amplitude.
6. The chroma output (v_{14}) is the same as the chroma input (v_1) except that the burst is removed and keying overshoot occurs in the retrace period. The chroma output is in peak-to-peak volts of "line" amplitude.
7. The oscillator output (vg) is the CW output at terminal No. 8 and is in peak-to-peak volts. Some modulation of oscillation dampening between burst injection is visible.

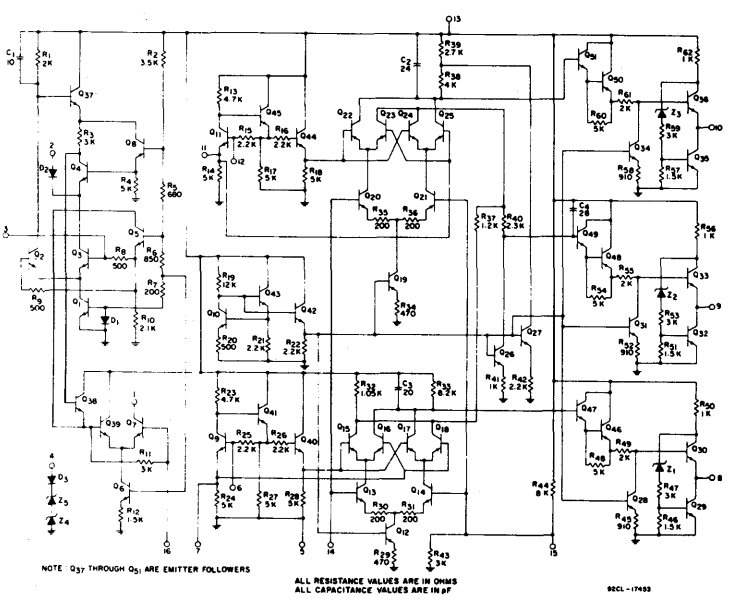


Fig. 6 - CA3067 schematic diagram.

CA3066, CA3067

ELECTRICAL CHARACTERISTICS at T_A = 25°C and V^+ = 11.2 V

CHARACTERISTIC	SYMBOL	TEST CONDITIONS	LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNITS	TEST FIG. AND CURVES
Static Characteristics							
Voltages:							
Tint Control Input	V_2	I_2 = 0.25 mA	–	3.5	–		
Reference Subcarrier	V_3		–	2.1	–		
Zener Regulator Ref.	V_4		10.6	11.9	12·6	V	9
B–Y, R–Y Oscillator Ref. Inputs	V_6, V_{12}		–	5.7	–		
Balance (B–Y, R–Y)	V_7, V_{11}		–	5.0	–		
B–Y, G–Y, R–Y Outputs	V_8, 9, 10		4.2	5.0	5.8		9, 11, 12
Difference Outputs*	$\Delta V_8, \Delta V_9, \Delta V_{10}$		–0.3	–	0.3		
Chroma Inputs	V_{14}, V_{15}		–	3.0	–		9
Tint Ampl. Balance	V_{16}		–	4.7	–		
Currents:							
Tint Ampl. Output (min.)	I_1(min.)	V_{16} = 8 V	0.16	0.37	–	mA	
Total Supply	$I_1 + I_{13}$		15	24	33		
Dynamic Characteristics							
Tint Amplifier Output							
Sensitivity	V_1	V_3 = 7 mV (RMS)	160	250	–	mV (RMS)	
Limiting Knee		V_3 = 35mV (RMS)	–	300	–		
Limiting		V_3 = 350mV (RMS)	–	–	380		
Tint Ampl. Phase Ref.▲	ϕ_6	V_3 = 70mV (RMS)	185	220	235	deg.	
Tint Ampl. Phase Shift‡	$\Delta\phi_6$	V_3 = 70mV (RMS)	90	105	–	deg.	
Demodulated Chroma Output:							
R–Y	V_{10}	V_3 = 70mV (RMS) V_{14} = 35mV (RMS)	150	250	–	$V_{(RMS)}$	10
Ratio of G–Y to R–Y	V_9/V_{10}		0.28	0.36	0.44		
Ratio of B–Y to R–Y	V_8/V_{10}		1.0	1.2	1.4		
Color Difference Output BW at 3.3 dB	$BW_{Diff.}$		450	550	–	kHz	
Color Difference Outputs (max. input signals):							
R–Y	v_{10}	V_3 = 70mV (RMS) V_{14} = 212mV (RMS)	–	3.0	–	v_{p-p}	
G–Y	v_9		–	1.1	–		
B–Y	v_8		–	3.6	–		
Small Signal Input Resistance							
Terminal No. 3	r_i		–	550	–	Ω	
Terminal Nos. 6 & 12			–	22	–		
Small Signal Output Resistance							
Terminal Nos. 8, 9, & 10	r_o		–	5	–		

$$*\Delta V_8 = V_8 - \left(\frac{V_8 + V_9 + V_{10}}{3}\right), \Delta V_9 = V_9 - \left(\frac{V_8 + V_9 + V_{10}}{3}\right), \Delta V_{10} = V_{10} - \left(\frac{V_8 + V_9 + V_{10}}{3}\right)$$

▲ Terminal No. 3 is phase reference

‡ read phase shift as tint control is varied

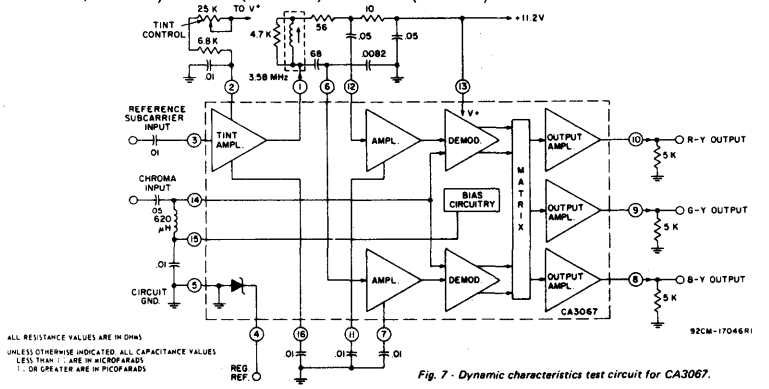


Fig. 7 - Dynamic characteristics test circuit for CA3067.

DYNAMIC CHARACTERISTICS TEST PROCEDURE

1. The reference subcarrier input (v_3) is a 3.58 MHz CW signal from a 50Ω source.
2. The chroma input (v_{14}) is a 3.53 MHz CW signal from a 50Ω source.
3. Phase and amplitude at terminal Nos. 1, 3, 6 and 12 are measured with a vector voltmeter (HP8405A or equivalent).
4. Signals at terminal Nos. 8, 9, and 10 are measured with an ac voltmeter (HP400E or equivalent) or an oscilloscope.
5. Unless otherwise noted the Tint control is at maximum resistance.

CA3067

MAXIMUM RATINGS, *Absolute-Maximum Values at T_A = 25°C*

Supply Voltages and Currents (see charts below)

Device Dissipation:

Up to T_A = 70°C . . . . . 600 mW

Above T_A = 70°C . . . . derate linearly 7.7 mW/°C

Ambient Temperature Range:

Operating . . . . . −40 to +85 °C

Storage . . . . . −65 to +150 °C

Lead Temperature (During soldering for 10s max. at not less than 1/32" from package) . . . +265 °C

Voltage with respect to Terminal No. 5

Terminal No.	$V_{min.}$ (volts)	$V_{max.}$ (volts)
6	0	N2
7	0	N2
8	0	N2
9	0	N2
10	0	N2
11	0	N2
12	0	N2
13	0	12
14	−3	N2
15	0	N2
16	N3	N3
1	0	15
2	0	N2
3	0	5
4	N1	

Current

Terminal No.	I_i (mA)	I_o (mA)
6	3	3
7	3	3
8	20	20
9	20	20
10	20	20
11	3	3
12	3	3
13	50	1
14	1	0.1
15	6	2
16	N3	N3
1	3	3
2	3	0.1
3	3	3
4	20	0.1

N1 Terminal No. 4 is connected to a zener reference element, that, if used, should be biased by a positive voltage through a resistor that limits the current to a value which is less than the maximum current rating of terminal No. 4.

N2 The upper voltage limit cannot exceed the power supply input voltage at terminal 13.

N3 Terminal No. 16 should be bypassed for normal operation.

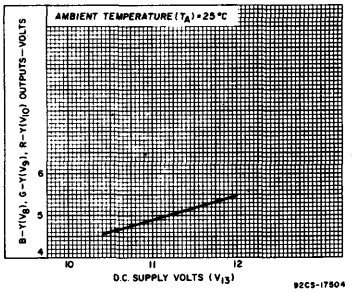


Fig. 8 - DC voltage at color-difference outputs vs supply voltage for CA3067.

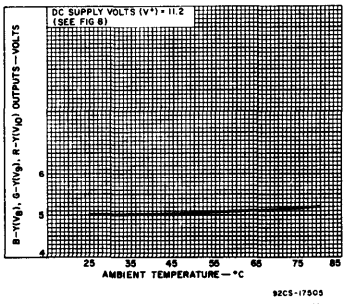


Fig. 9 - Temperature drift of DC voltage at color-difference outputs for CA3067.

Television Video IF System

RCA-CA3068 is a monolithic integrated circuit that incorporates an entire video TV-IF subsystem on a single chip. Innovations in integrated circuit design, in addition to the many active devices and closely matched components utilized in the circuit, make the CA3068 ideally suited for use in color and black-and-white TV receivers.

The primary functions performed by the IF subsystem are video IF amplification, linear detection, video output amplification, AGC from a keyed supply, AGC delay for tuner, sound carrier detection, sound carrier amplification, and a buffered AFT output. The advanced circuit design of the CA3068 also includes secondary functions for improved noise immunity and minimal airplane flutter. An isolated zener reference diode, incorporated in the IC, provides a convenient and economical means for controlling the regulated voltage supply. The inherent wide bandwidth capability (10-70 MHz) and high overall gain (87 dB) make the CA3068 suitable for other AM IF applications whose frequencies range within this bandwidth.

The CA3068 utilizes a unique 20-lead quad-in-line plastic package. This package also includes a wrap-around shield that serves to minimize interlead capacitances.

FEATURES:

- High-gain wide-band IF amplifier: 75 dB typ. at 45 MHz
- Gain reduction with excellent stability: 50 dB typ. at 45 MHz
- Video detector with linear characteristics
- Video amplifier: 12 dB gain
- Impulse noise limiter
- Keyed AGC with noise immunity circuits
- Delayed AGC for tuner
- Buffered AFT output
- Separate sound IF intercarrier amplification
- Sound carrier detector
- 4.5 MHz sound carrier amplifier
- Isolated zener reference diode for regulated voltage supply
- See ICAN-6303, "A Single IC for the Complete PIX-IF System in TV Receivers" for Schematic Diagram

MAXIMUM RATINGS, *Absolute Maximum Values, at $T_A = 25°C$*

DC Supply Voltage:		
Between Terminals 15 and 5*	11.3	V
Terminal 7 (Collector to ground)	20	V
Terminal 9 (Collector to ground)	20	V
DC Current (into Terminal 18)	2	mA
Device Dissipation:		
Up to $T_A = 60°C$	600	mW
Above $T_A = 60°C$	derate linearly 6.7 mW/°C	
Ambient Temperature Range:		
Operating	−40 to +85	°C
Storage	− 65 to +150	°C
Lead Temperature (During soldering):		
At distance not less than 1/32" (0.79 mm) from case for 10 seconds max.	+265	°C

* This rating does not apply when using the internal zener reference in conjunction with the pass transistor.

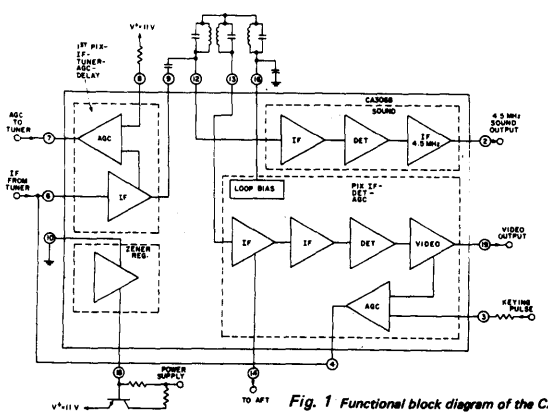
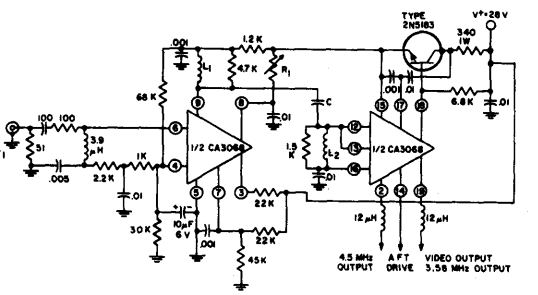
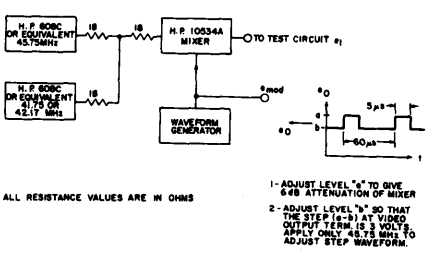


Fig. 1 Functional block diagram of the CA3068.



(a) Test setup for measurement of video sensitivity, sync. tip level, delay bias, AFT drive voltage.



(b) Test setup for measurement of sound and chroma outputs.

Fig. 3 – Typical dynamic test circuit diagrams.

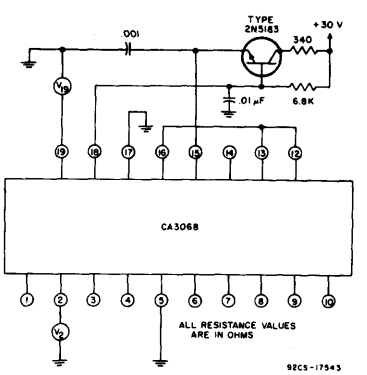


Fig. 2 – Test circuit for measurement of white level (V_{1g}) and terminal 2 voltage (V_2).

CA3068

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTIC	SYMBOL	TEST CONDITIONS	LIMITS			UNITS
			Min.	Typ.	Max.	
Static (DC) Characteristics						
Quiescent Circuit Current	I_{15}	–	15	–	45	mA
DC Voltages:						
Terminal 2 (Sound)	V_2	–	–	6	–	V
Terminal 3 (Keying Input)	V_3	–	6.4	–	10	V
Terminal 7 (1) (AGC)	V_7	–	16	–	21	V
Terminal 7 (2) (AGC)	V_7	–	–	1	–	V
Terminal 8 (AGC Delay)	V_8	–	–	4	–	V
Terminal 9 (Cascode Collector)	V_9	–	–	8.5	–	V
Terminal 16 (Bias)	V_{16}	–	1.1	–	2.3	V
Terminal 18 (Zener)	V_{18}	$V_5 = V_{17} = 0$ V, I_{18} = 1 mA	10.6	11.9	13.2	V
Terminal 19 (White Level)	V_{19}	–	6	–	10	V
Dynamic Characteristics						
Video Sensitivity	e_I	f_O = 45.75 MHz, Mod. (AM) = 85% at 400 Hz; Adjust e_I for 4 $V_{p\text{-}p}$ at Term. 19	40	100	200	μV
Sync. Tip Level Voltage	V_{19}	f_O = 45.75 MHz, e_I(CW) = 10 mV	0.4	0.8	1.6	V
Automatic Fine Tuning (AFT) Drive Level Voltage	V_{14}		–	15	–	mV
Delay Bias Voltage:	V_7					
At e_I = 10 mV		f_O = 45.75 MHz, e_I(CW) = 20 mV;	16	–	–	V
At e_I = 30 mV		Adjust R_1 for V_7 = 14 V	0.5	–	2	V
3.58 MHz Chroma Output Voltage	V_{19}	f_O = 45.75 MHz, e_I(step mod.) = 10 mV; f_1 = 42.17 MHz, e_I(step mod.) = 3.33 mV	0.5	0.8	–	V
4.5 MHz Sound Output Voltage	V_2	f_O = 45.75 MHz, e_I(step mod.) = 10 mV; f_2 = 41.25 MHz, e_I(step mod.) = 2.5 mV	50	200	–	mV
Parallel Input Impedance:						
Resistance at Term. 6	R_{I-6}	f_O = 45.75 MHz Impedance and Admittance measured at bias conditions as developed by circuit shown in Fig. 7	4	–	–	kΩ
Capacitance at Term. 6	C_{I-6}		–	2	–	pF
Resistance at Term. 12	R_{I-12}		–	4.5	–	kΩ
Capacitance at Term. 12	C_{I-12}		–	4	–	pF
Resistance at Term. 13	R_{I-13}		–	5	–	kΩ
Capacitance at Term. 13	C_{I-13}		–	4	–	pF
Parallel Output Impedance:						
Resistance at Term. 9	R_{O-9}		30	–	–	kΩ
Capacitance at Term. 9	C_{O-9}		–	3	–	pF
Cascode Transfer Characteristics: Magnitude of Forward Transadmittance	$\lvert Y_f \rvert$		–	50	–	mmho
Reverse Transfer Capacitance	C_r		–	0.001	–	pF

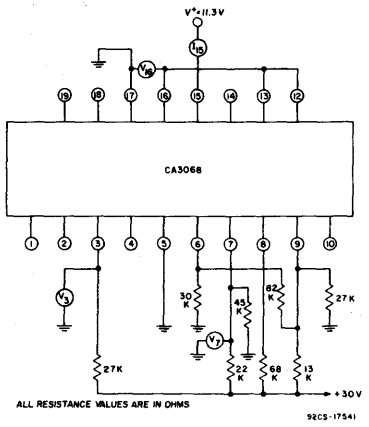


Fig. 4 – Test circuit for measurement of quiescent current (I_{15}), keying terminal voltage (V_3), bias voltage (V_{16}), AGC terminal voltage 1 (V_7), and cascode collector voltage (V_9).

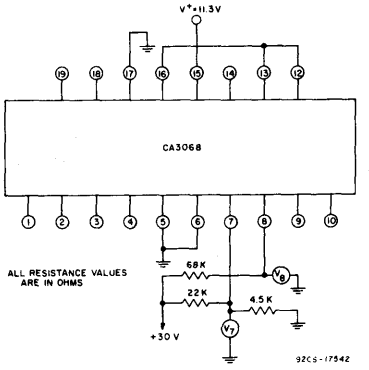


Fig. 5 – Test circuit for measurement of AGC terminal voltage 2 (V_7) and terminal 8 voltage (V_8).

Television Chroma System

The RCA CA3070, CA3071, and CA3072 are monolithic silicon integrated circuits that constitute a complete chroma system for color television receivers. The CA3070 is a complete subcarrier regeneration system featuring a new concept of phase control applied to the oscillator circuit. The CA3071 is a chroma amplifier system and the CA3072 performs the demodulation function.

The CA3070 utilizes the 16-lead plastic dual-in-line package; the CA3071 and CA3072 are supplied 14-lead plastic dual-in-line packages.

SYSTEM FEATURES

CA3070

- Voltage Controlled Oscillator
- Keyed APC & ACC Detectors
- DC Hue Control
- Shunt Regulator

CA3071

- ACC Controlled Chroma Amplifier
- DC Chroma Gain Control
- Color Killer
- Amplifier Short-Circuit Protection

CA3072

- Synchronous Detector with Color Difference Matrix
- Emitter-Follower Output Amplifiers with Short-Circuit Protection

CA3070 Chroma Signal Processor

The CA3070 is a complete subcarrier regeneration system with automatic phase control applied to the oscillator. An amplified chroma signal from the CA3071 is applied to terminals No. 13 and No. 14, which are the automatic phase control (APC) and the automatic chroma control (ACC) inputs. APC and ACC detection is keyed by the horizontal pulse which also inhibits the oscillator output amplifier during the burst interval.

The ACC system uses a synchronous detector to develop a correction voltage at the differential output terminal Nos. 15 & 16. This control signal is applied to the input terminal Nos. 1 & 14 of the CA3071. The APC system also uses a synchronous detector. The APC error voltage is internally coupled to the 3.58 MHz oscillator at balance; the phase of the signal at terminal No. 13 is in quadrature with the oscillator.

To accomplish phasing requirements, an RC phase shift network is used between the chroma input and terminal Nos. 13 and 14. The feedback loop of the oscillator is from terminal Nos. 7 and 8 back to No. 6. The same oscillator signal is available at terminal Nos. 7 and 8, but the dc output of the APC detector controls the relative signal levels at terminal Nos. 7 or 8. Because the output at terminal No. 8 is shifted in phase compared to the output at terminal No. 7, which is applied directly to the crystal circuit, control of the relative amplitudes at terminal Nos. 7 and 8 alters the phase in the feedback loop, thereby changing the frequency of the crystal oscillator. Balance adjustments of dc offsets are provided to establish an initial no-signal offset control in the ACC output, and a no-signal, on-frequency adjustment through the APC detector-amplifier circuit which controls the oscillator frequency. The oscillator output stage is differentially controlled at terminal Nos. 2 and 3 by the hue control input to terminal No. 1. The hue phase shift is accomplished by the external R, L, and C components that couple the oscillator output to the demodulator input terminals. The CA3070 includes a shunt regulator to establish a 12-volt dc supply.

Maximum Voltage and Current Ratings at T_A = +25°C

Terminal No.	Voltage▲ Min. Volts	Voltage▲ Max. Volts	Current Terminal No.	Current I_I mA	Current I_O mA
1	0	*	1	20	1
2	0	+16	2	–	–
3	0	+16	3	–	–
4	–5	N2	4	20	1
6	–	–	10	N3	1
7	–	–	11	–	–
8	–	–	12	–	–
10	0	N3	13	20	1
11	0	N1	14	20	1
12	0	N1			
13	0	N1			
14	0	N1			
15	0	+16			
16	0	+16			

▲ With respect to terminal No.5 and with terminal No. 10 connected through 470Ω to +24 V.

N1 Regulated voltage at terminal No. 10.

N2 Controlled by max. input current.

N3 Limited by dissipation.

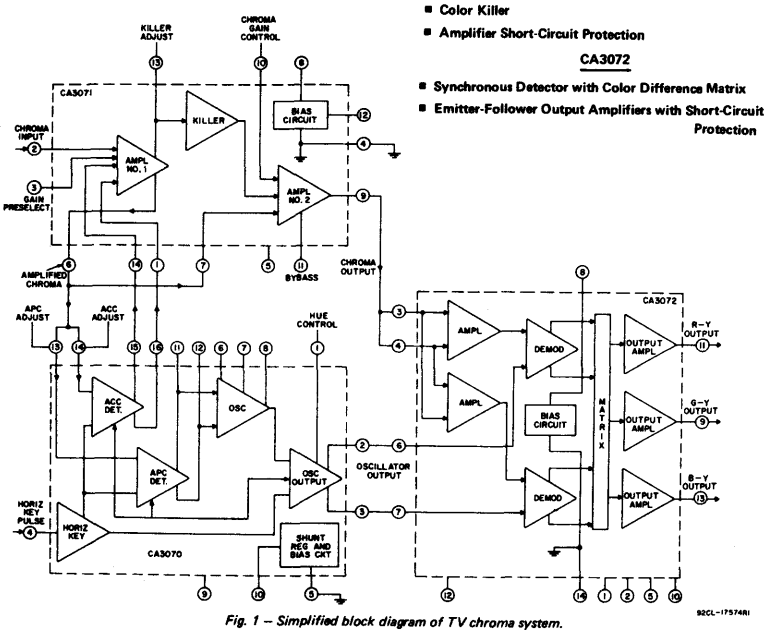


Fig. 1 – Simplified block diagram of TV chroma system.

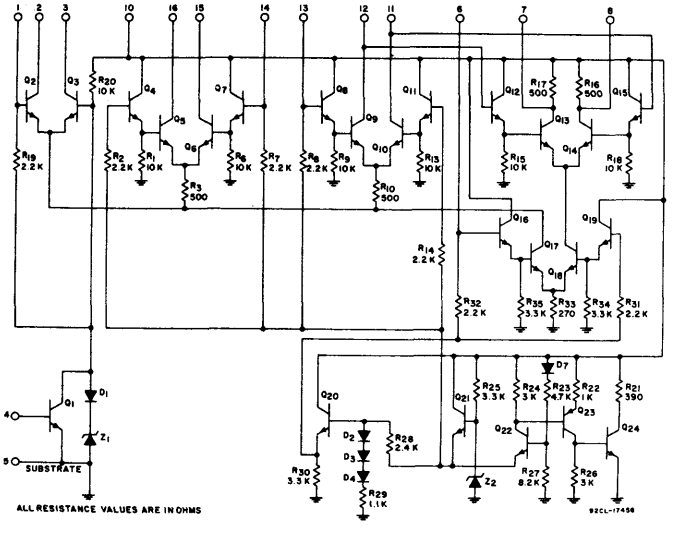


Fig. 2 – Schematic diagram CA3070.

CA3070, CA3071, CA3072 Types

MAXIMUM RATINGS, *Absolute Maximum-Values at T_A = 25°C*

Device Dissipation:
- Up to T_A = +70°C ..... 530 mW
- Above T_A = +70°C ... Derate Linearly at 6.7 mW/°C

Ambient Temperature Range:
- Operating ..... −40 to +85 °C
- Storage ..... −65 to +150 °C

Lead Temperature (During Soldering):
- At distance 1/32 in. (3.17 mm) from seating plane for 10 s max. ..... +265 °C

ELECTRICAL CHARACTERISTICS, at T_A = 25°C and V^+ = +24 V unless otherwise specified

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS	LIMITS CA3070 MIN.	LIMITS CA3070 TYP.	LIMITS CA3070 MAX.	UNITS	TEST CIRCUITS FIG.
Static Characteristics							
Voltage:							
Hue Control	V_1	Switch in position 2	6.9	7.7	8.6	V	3c
Oscillator Input	V_6		–	2.8	–	V	3a
APC Input	V_{13}		–	6.5	–	V	3a
Regulator	V_{10}	V^+ = 21 V	11	12.3	13.5	V	3a
Regulator Change	V_{10}	V^+ = 27 V	−0.2	–	+0.2	V	3a
Horizontal Key Input	V_4	I_4 = −10 μA	5	–	–	V	3a
Currents:							
Oscillator Output	I_2		–	5.8	–	mA	3c
APC Output	I_{11}, I_{12}		–	1.45	–	mA	3b
ACC Output	I_{15}, I_{16}		–	1.45	–	mA	3b
Dynamic Characteristics							
Oscillator Outputs:							
Terminal No. 2	V_2	S_1 in position 1	0.75	1.0	–	V_{p-p}	4
Terminal No. 3	V_3	S_1 in position 2	0.75	1.0	–	V_{p-p}	4
ACC Detected Output	$V_{16}-V_{15}$	S_1 in position 1	115	150	–	mV	4
Oscillator Pull-In Range	–		–	±400	–	Hz	4

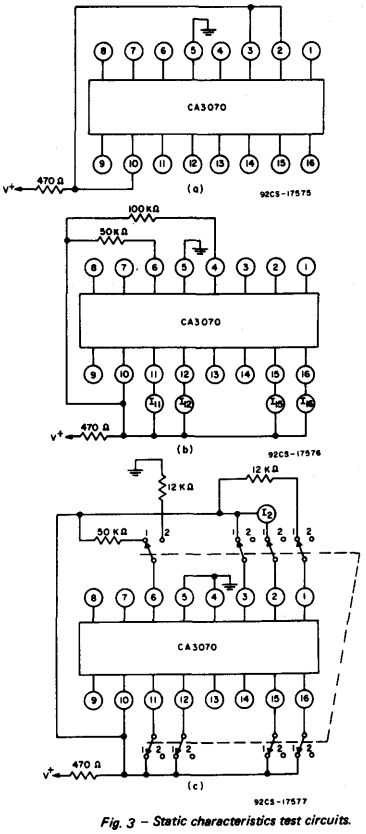


Fig. 3 – Static characteristics test circuits.

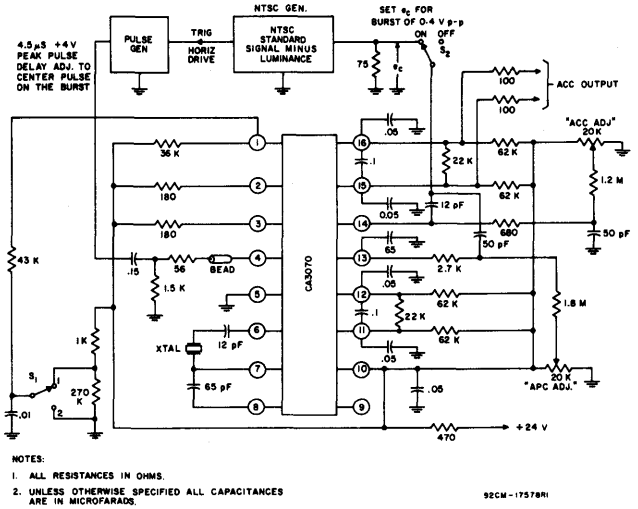


Fig. 4 – CA3070 Dynamic test circuit.

Dynamic Test Initial Adjustments

1. APC ADJUST: With S2 in "OFF" position adjust the "APC ADJ" potentiometer to set oscillator frequency at 3.579545 MHz ±25 Hz. With S1 in position 1 measure frequency at terminal No. 2 output, using crystal probe shown in Fig. 6.
2. ACC ADJUST: With S2 in "OFF" position adjust "ACC ADJ" potentiometer to give an ACC output reading of 0 ±2 mV.

Procedure to Pull-in Range Measurement

1. Set S1 in position 1 and connect the crystal probe to terminal No. 2.
2. Turn S2 to "OFF" and set "APC ADJ." arm to ground.
3. Turn S2 to "ON" and gradually adjust "APC ADJ" until oscillator "locks" as witnessed by a sharp increase in ACC output voltage between terminal Nos. 15 and 16.
4. Turn S2 to "OFF" and adjust capacitor Cp of crystal probe for maximum deflection on Ballantine Meter.
5. Switch Ballantine meter to "Amplifier" position and read oscillator frequency on counter.
6. Repeat steps 2 – 5 with "APC ADJ" arm set to terminal No. 10 instead of to ground.

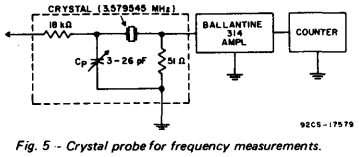


Fig. 5 - Crystal probe for frequency measurements.

CA3071 Chroma Amplifier

The CA3071 is a combined two-stage chroma amplifier and functional control circuit. The input signal is received from the video amplifier and applied to terminal No. 2 of the input amplifier stage. The first amplifier stage is part of the ACC system and is controlled by differential adjustment from the ACC input terminal Nos. 1 and 14. The output of the 1st amplifier is directed to terminal No. 6 from where the signal may be applied to the ACC detection system of the CA3070 or an equivalent circuit. The output at terminal No. 6 is also applied to terminal No. 7 which is the input to the 2nd amplifier stage. Another output of the 1st amplifier at terminal No. 13 is directed to the killer adjustment circuit.

The dc voltage level at terminal No. 13 rises as the ACC differential voltage decreases with a reduction in the burst amplitude. At a pre-set condition determined by the killer adjustment resistor the killer circuit is activated and causes the 2nd chroma amplifier stage to be cut off. The 2nd chroma amplifier stage is also gain controlled by the adjustment of dc voltage at terminal No. 10. The output of the 2nd chroma amplifier stage is available at terminal No. 9. The typical output termination circuit that is shown, provides differential chroma drive signal to the demodulator circuit. Both amplifier outputs utilize emitter-followers with short-circuit protection.

MAXIMUM RATINGS, *Absolute Maximum-Values at* $T_A = 25°C$

DC Supply Voltage (Terminal 8 to Terminal 4)	30	VDC
Device Dissipation:		
Up to $T_A = +70°C$	530	mW
Above $T_A = +70°C$	Derate Linearly at 6.7	mW/°C
Ambient Temperature Range:		
Operating	−40 to +85	°C
Storage	−65 to +150	°C
Lead Temperature (During Soldering):		
At distance 1/32 in (3.17 mm) from seating plane for 10 s max.	+265	°C

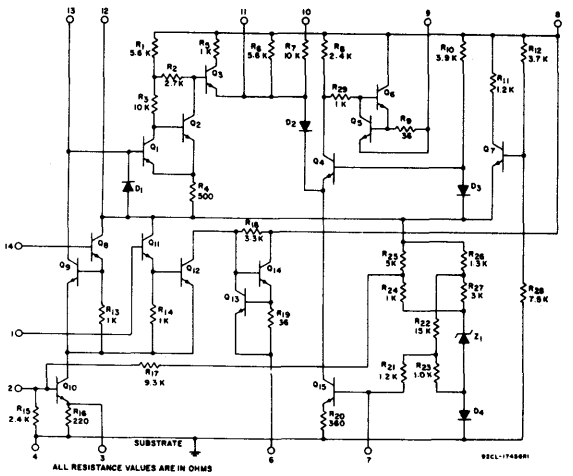


Fig. 6 -Schematic diagram for CA3071.

Maximum Voltage and Current Ratings @ $T_A = +25°C$

Terminal No.	Current I_I mA	Current I_O mA	Terminal No.	Voltage* MIN VOLTS	Voltage* MAX VOLTS
1	5	1.0	1	−5	+15
2	5	1.0	2	−5	+5
3	10	10	3	0	+2
6	1.0	20	6	0	+24
7	5	1.0	7	−5	+5
9	1.0	20	8	0	+30
12	1.0	5	9	0	+24
14	5	1.0	10	0	+24
			11	0	+24
			12	0	+20
			13	0	+20
			14	−5	+15

* With reference to terminal No. 4 and with +24 V on terminal No. 8 except for the rating given for terminal No. 8.

ELECTRICAL CHARACTERISTICS, at $T_A = 25°C$

CHARACTERISTICS	SYMBOLS (Measure)	SPECIAL TEST CONDITIONS	LIMITS CA3071 MIN.	LIMITS CA3071 TYP.	LIMITS CA3071 MAX.	UNITS	CURVES & TEST CIRCUITS FIG.
Static Characteristics							
Voltages							
Bias Reference Terminal	V_{12}	S_1 Open, S_2 Open	-	17.3	-	V	7
Ampl. No. 1 Chroma Input	V_2	S_1 Open, S_2 Open	-	1.75	-		
Ampl. No. 1 Chroma Output Balanced	V_6	S_1 Open, S_2 Open	-	20	-		
Unbalanced	V_6	S_1 Open, S_2 Closed	-	13.5	-		
Ampl. No. 2 Chroma Input	V_7	S_1 Open, S_2 Open	-	1.5	-		
Ampl. No. 2 Chroma Output	V_9	S_1 Closed, S_2 Open	-	20.6	-		
Supply Current	I_T	S_1 Open, S_2 Open	17	24.5	31	mA	
Dynamic Characteristics							
Amplifier No. 1 Voltage Gain	A_{V1}	E_g 30 mV$_{RMS}$ Measure v_6	14	-	-	dB	8
Amplifier No. 2 Voltage Gain	A_{V2}	V_g 1.0 V (RMS) Measure v_7	-	14	-	dB	
Max. Chroma Output Voltage	v_9		-	2	-	V_{RMS}	11
10% Chroma Gain Control Reference Voltage	$V_8 - V_{10}$	E_g - 50 mV$_{RMS}$, adjust Chroma Gain Control to Change v_9 to 10% of Maximum Chroma Output	2.1	3.8	6.8	V	8
Output Voltage, Killer Off	v_9	S_1 in Position 2 E_g - 50 mV$_{RMS}$, adjust "Killer Adjust" for an abrupt decrease in V_9	-	-	12	mV RMS	
Output Voltage, Chroma Off	v_9	E_g - 50 mV$_{RMS}$, adjust Chroma control to min. Chroma Output	-	-	12	mV RMS	
Bandwidth: Amplifier No. 1	BW		-	12	-	MHz	9, 10
Amplifier No. 2			-	30	-		
Ampl. No. 1 Input Impedance	r_{i1}		-	2	-	kΩ	8
	c_{i1}		-	4	-	pF	
Ampl. No. 1 Output Impedance	r_{o1}		-	85	-	Ω	
Ampl. No. 2 Input Impedance	r_{i2}		-	2.1	-	kΩ	
	c_{i2}		-	3.5	-	pF	
Ampl. No. 2 Output Impedance	r_{o2}		-	85	-	Ω	

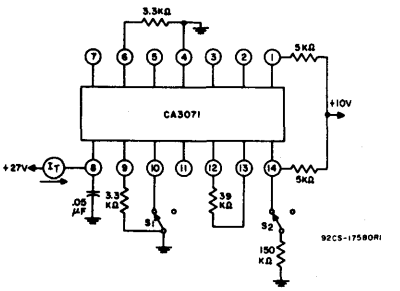


Fig. 7 - Static characteristics test circuit—CA3071.

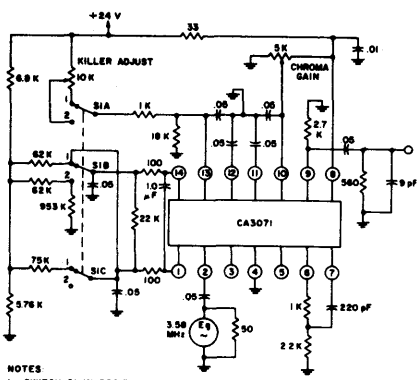


Fig. 8 - Dynamic characteristics circuit-CA3071.

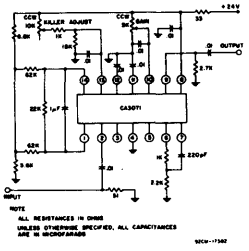


Fig. 9 – CA3071 Wideband amplifier circuit.

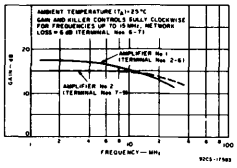


Fig.10— Frequency response for wideband amplifier CA3071.

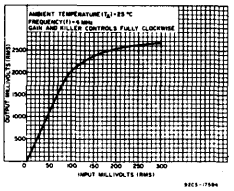


Fig. 11 – Typical CA3071 wideband amplifier linearity

CA3072 Chroma Demodulator

The CA3072 has two sets of synchronous detectors with matrix circuits to achieve the R-Y, G-Y, and B-Y color difference output signals. The chroma input signal is applied to terminal Nos. 3 and 4 while the oscillator injection signal is applied to terminal Nos. 6 and 7. The color difference signals, after matrix, have a fixed relationship of amplitude and phase nominally equal dc voltage levels. The outputs of the CA3072 are suitable for driving high level color difference or R, G, B output amplifiers. Emitter-follower output stages used to drive the high level color amplifiers have short-circuit protection.

MAXIMUM RATINGS, *Absolute Maximum-Values at* T_A = 25° C

DC Supply Voltage (Terminal 8 to Terminal 14)	27 V
Reference Input Voltage	5 V_{p-p}
Chroma Input Voltage	5 V_{p-p}
Device Dissipation:	
Up to T_A = +70°C	530 mW
Above T_A = +70°C	Derate Linearly at 6.7 mW/°C
Ambient Temperature Range:	
Operating	-40 to +85°C
Storage	-65 to +150°C
Lead Temperature (During Soldering):	
At distance 1/32 in (3.17 mm) from seating plane for 10 s max	+265°C

Maximum Voltage and Current Ratings

at T_A = +25°C

Voltage*			Current		
Terminal No.	MIN VOLTS	MAX VOLTS	Terminal No.	I_I mA	I_O mA
3	0	+5	3	–	–
4	0	+5	4	–	–
6	0	+12	6	–	–
7	0	+12	7	–	–
8	0	+27	8	–	–
9	0	+20	9	1.0	20
11	0	+20	11	1.0	20
13	0	+20	13	1.0	20

*With reference to terminal No. 14 and with the voltage between terminal No. 8 and terminal No. 14 at +24 V except as given in rating for terminal No. 8.

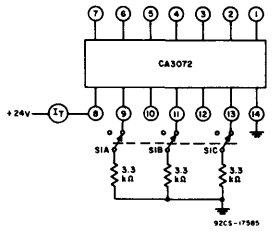


Fig. 13 — Static characteristics test circuit—CA3072.

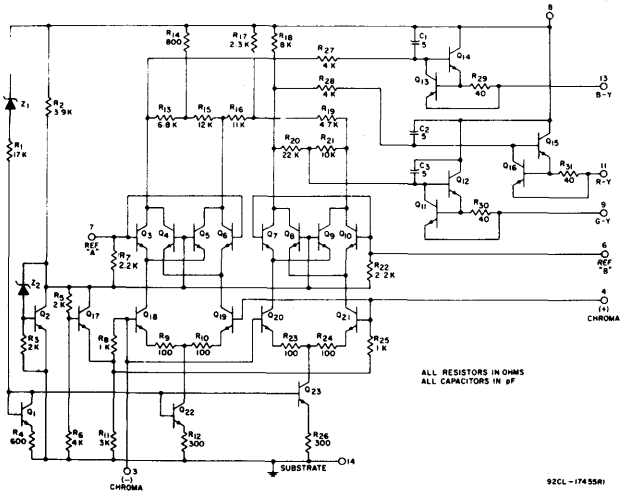


Fig. 12 — Schematic diagram for CA3072.

ELECTRICAL CHARACTERISTICS, at T_A = 25° C and V^+ = +24 V unless otherwise specified

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS	LIMITS CA3072 MIN.	TYP.	MAX.	UNITS	TEST CIRCUITS FIG.
Static Characteristics							
Supply Current With Output Loads	I_T	S_1 Closed	16.5	–	26.5	mA	13
With No Output Loads		S_1 Open	–	9			
G-Y, R-Y, B-Y Outputs	V_9, V_{11}, V_{13}	S_1 Closed	13.2	14.7	15.8	V	
Chroma Inputs	V_3, V_4	S_1 Open	–	3.3	–		
Reference Subcarrier	V_6, V_7	S_1 Open	–	6.2	–		
Dynamic Characteristics							
Demodulator Unbalance	v_9,v_{11},v_{13}	$V_3 = V_4$ = 0	–	–	0.8	V_{p-p}	14
Maximum Color Difference Output Voltage	v_{13}	$V_3 = V_4$ = 0.6 Vp-p	8.0	–	–	V_{p-p}	
	v_{11}		5.5	–	–		
	v_9		1.2	–	–		
Chroma Input Sensitivity	v_3	Adjust e_c for 5.0 v_{p-p} @ term No. 13 (B-Y)	–	0.2	0.35		
Relative R-Y Output	v_{11}		3.5	–	4.2		
Relative G-Y Output	v_9		0.75	–	1.25		
V_{DC} Difference Between any two Output Terminals	$\lvert V_9 \rvert - \lvert V_{11} \rvert$ $\lvert V_9 \rvert - \lvert V_{13} \rvert$ $\lvert V_{11} \rvert - \lvert V_{13} \rvert$	e_c = 0	–	–	0.6	V	
Input Impedance Reference Subcarrier Inputs	r_i6, 7		–	1.7	–	kΩ	
	c_i6, 7		–	6	–	pF	
Input Impedance at Chroma Inputs	r_i3, 4		–	0.95	–	kΩ	
	c_i3, 4		–	6	–	pF	
Output Resistance	r_o9, r_o11, r_o13		–	180	–	Ω	

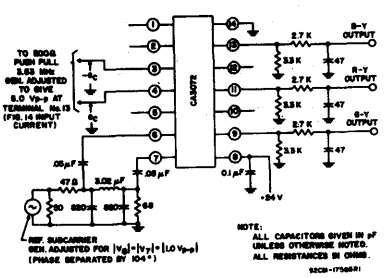


Fig. 14 - Dynamic characteristics test circuit for CA3072.

Application Information

TYPICAL APPLICATION CIRCUIT FOR THE CHROMA SYSTEM

The circuit of *Fig.15*: is a complete signal processing system for color TV. The RCA types CA3070, CA3071 and CA3072 monolithic integrated circuits are respectively used as the subcarrier regenerator, chroma amplifier, and chroma demodulator.

The input to the system is the chroma signal which may be taken from the first or second video stage and is coupled into the CA3071 chroma amplifier through a bandpass filter. The outputs from the system are the color difference signals which are intended to drive high level amplifiers. Luminance mixing may be external to the picture tube or, the difference signals may be amplified and applied to the picture tube grid or cathode, where they are internally mixed with the luminance signal.

Other input requirements to the system are the power supply voltage of +24 volts and the horizontal keying pulse. The power supply voltage should be maintained within ±3 volts of the recommended value of +24 volts. The total current for the system is approximately 70 milliamperes. The horizontal keying pulse input to the subcarrier regenerator is approximately +4 volts peak and centered on the burst as seen at terminal Nos. 13 and 14 of the CA3070. The pulse width should be maintained as close as possible to the recommended value of 4.5 microseconds.

CA3070 Circuit Operation

The CA3070 circuit as shown in Fig. 2, consists of an oscillator, automatic phase control (APC) detector, automatic chroma control (ACC) detector, gated oscillator output amplifier and a shunt regulator. The shunt regulator provides the necessary bias stability for the 3.579545 MHz oscillator, as well as the bias to all functions of the CA3070 circuit. The regulation voltage is nominally +12 volts as measured at terminal No. 10.

The APC and ACC detectors are synchronous detectors which are keyed by the horizontal input pulse. This form of detection eliminates the need for a burst separator as an individual amplifier stage. When a positive pulse is present at terminal No. 4, the oscillator output is cutoff and the oscillator drive signal is diverted to the APC and ACC detectors. Referring to Fig. 2, the APC detector (Q_9 & Q_{10}) and the ACC detector (Q_5 & Q_6) are emitter driven from the

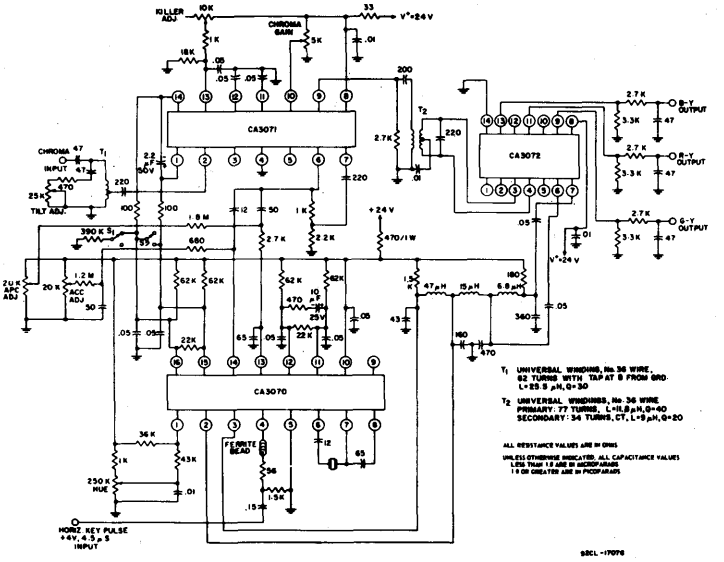


Fig. 15 - Typical chroma system for color-TV receivers utilizing RCA-CA3070, CA3071, and CA3072.

oscillator transistor (Q_{17}), when the oscillator output amplifier transistors (Q_2 & Q_3) are cutoff. The chroma signal is applied to terminal Nos. 13 and 14. There is oscillator current drive to the APC and ACC detectors during the keying interval; burst separation is effectively accomplished by the gating action of the detectors. A further advantage of the keying action is the high gain made possible as a result of the low average current flow of the APC and ACC detectors. High resistor values of 62 kilohms at the detector output terminals provide proper detector bias consistent with the duty factor of the keying pulse. For a wider keying pulse, it is necessary that smaller values of detector load resistors be used.

In the absence of the keying pulse (line period), the resistor, R_{20}, biases the oscillator's output amplifier transistors (Q_2 & Q_3) on by keeping their emitters at a higher potential than the base bias voltages of Q_5, Q_6, Q_9, and Q_{10}. The 3.58 MHz signal is now present at terminal Nos. 2 & 3. Photographs of oscilloscope traces for one line period at the terminal Nos. 1, 2, and 3 are shown in Fig. 16. The effect of the keying pulse is shown in Fig. 16a, and the cutoff of the oscillator output amplifier is shown in Fig. 16(b) and 16c.

The oscillator section of the CA3070 consists of the loop formed by Q_{18} and the emitter driven differential pair, Q_{13} & Q_{14}. The signal output from terminal Nos. 7 & 8 is coupled through the series tuned crystal circuit back through terminal No. 6 to Q_{16} & Q_{17}. The collector of Q_{17} drives the oscillator output amplifier and the APC & ACC detectors. Q_{17} is emitter coupled to transistor Q_{18}. The oscillator frequency and phase control is accomplished by the differential drive from the APC detector to transistors Q_{12} & Q_{15} which control the balance of Q_{13} & Q_{14}. The resulting phase of the feedback loop is determined by the relative amplitudes of the oscillator output signal at terminal Nos. 7 and 8. The 65 pF capacitor between terminal No. 7 and 8 provides the phase shifting component as the balance of Q_{13} and Q_{14} is varied. In this way the APC detector controls the crystal frequency at which the phase shift is cancelled in the feedback loop.

The controls for the CA3070 subcarrier regenerator circuit are the APC balance, the ACC balance, and the hue control. The hue control is a dc balance adjustment of the oscillator output amplifier transistors Q_2 & Q_3. A phase delay network between the output terminals Nos. 2 & 3 determines the range of the hue control, which for the value shown in Fig. 15, is approximately 90°.

The ACC adjustment sets the initial balance of the ACC drive to the input of the CA3071 in Fig. 15 (terminal Nos. 1 and 14 of the CA3071). The APC is a frequency adjustment of the oscillator through the balance control of the APC detector.

As a setup adjustment, for both the ACC and APC, switch S1 is opened and S2 is closed. The chroma input to the system is removed and the dc voltage at terminal No. 6 of the CA3071 is noted. The switch S2 is then opened and the ACC adjusted to set the voltage at terminal No. 6 to that previously noted. Alternatively, the differential dc voltage at terminal Nos. 15 & 16 of the CA3070 may be set to 0 mV (±2 mV) when S1 and S2 are open, and the CA3071 is removed from the circuit.



Fig. 16(a) - CA3070 terminal No. 1 7.5 V oscillator "gate off" pulse.



Fig. 16(b) - CA3070 terminal No. 2, 3.5 $V_{p\text{-}p}$ oscillator output; one horizontal line, (gated off during burst).



Fig. 16(c) - CA3070 terminal No. 3, 2.0 $V_{p\text{-}p}$ oscillator output; one horizontal line, (gated off during burst).

With the chroma signal still removed, the APC adjustment sets the frequency of the oscillator to 3.579545 MHz. Due to the gated off interval, a counter will not accurately record the frequency at the oscillator output amplifier terminals. Two simple and accurate methods are as follows: (1) a buffered crystal filter circuit, connected to the oscillator output amplifier terminals will continue to ring and fill the gated off window providing the proper interface to a counter; (2) the other method involves monitoring the demodulated output at the color difference output terminals of the CA3072. A zero beat signal, at the color difference outputs may be seen on an oscilloscope.

When these adjustments are made, similar oscilloscope traces should be seen as shown in Fig. 17.

CA3071 CIRCUIT OPERATION

The CA3071 is the basic amplifier and control circuit of the chroma system. It contains the gain control functions of the ACC loop, the color killer, and the dc chroma gain control. The CA3071 is a wide band amplifier having two stages of voltage gain. Curves of frequency-response and linearity are shown in Figs. 10 & 11 for the wideband circuits shown in Fig. 9. This is the same basic amplifier as the one in the system shown in Fig. 15 except for the omission of the tuned circuits and the ACC loop connection. The amplifiers have bandwidths of greater than 10 MHz, and are usable well beyond 30 MHz. The signal swing of the wide band amplifier is in excess of 5 $V_{p\text{-}p}$, even with the typical load coupling as shown in Fig. 15. Fig. 18 (a, b and c) show the oscilloscope traces for an NTSC signal at the chroma input. The overall frequency-response curves are shown in Fig. 19.

CA3071 operation is as follows (Refer to Figs. 6 &15). The input chroma signal is applied to terminal No. 2. This signal is amplified in a cascode differential circuit from Q_{10} to Q_{12} and the output is an emitter follower, Q_{14} (Terminal No. 6.) The signal is divided in the Q_9 & Q_{12} differential amplifier, depending on the applied ACC error signal amplitude at terminal Nos. 1 & 14. The ACC error signal is derived from terminal Nos. 15 & 16 of the CA3070 and after filtering, is applied to terminal Nos. 1 & 14 of the CA3071.

At low signal drive, the 390 kilohm resistor at switch S1 (normally closed) unbalances the differential amplifier for high signal gain through Q_{12}. As the burst level at the chroma input increases, the ACC drive changes differentially in a positive direction at terminal No. 14 and a negative direction at terminal No. 1. At strong signal levels the gain is reduced by diverting the balance of ac current in the differential amplifier from Q_{12} to Q_9, which is shunted to ac ground at terminal Nos. 12 and 13. The ACC loop is completed through the chroma signal at terminal No. 6 of the CA3071 to terminal No. 14 (input) of the CA3070. A typical ACC characteristic is shown in Fig. 23.

The chroma signal is buffer connected from terminal No. 6 to terminal No. 7 of the CA3071 and is amplified in the 2nd stage of voltage gain. Both the color killer adjustment and the dc chroma gain control are applied to the 2nd stage to control the chroma output at terminal No. 9. The color killer section of the CA3071 is a Schmitt trigger & amplifier circuit consisting of transistors Q_1, Q_2 and Q_3. Under maximum chroma output conditions, the diode D_2 is reversed biased, and the signal path is through Q_{15}, Q_4 and Q_5 to terminal No. 9. When the color killer circuit is actuated, or the chroma gain control is adjusted to a higher positive voltage at terminal No. 10, the anode voltage of diode D_2 is increased to draw current from the signal path at the emitter of Q_4. This decreases the chroma gain as the potential at terminal No. 10 is increased. When the potential at terminal No. 10 is the same as terminal No. 8, the chroma output at terminal 9 is cutoff.

The color killer circuit provides an abrupt voltage swing at the anode of D_2 to cutoff the chroma output when the Schmitt trigger circuit is forward biased at terminal No. 13. In the circuit of Fig. 18, the color killer adjustment is a resistance divider circuit which establishes the threshold of burst level at which the killer operates the chroma amplifier.

CA3072 CIRCUIT OPERATION

The CA3072 is a chroma demodulator having full color difference signal demodulation capability. The chroma signal is applied to terminal Nos. 3 & 4 and the reference subcarrier signal is applied to terminals Nos. 6 & 7 of the CA3072. The output color difference signals are B-Y at terminal No. 13, R-Y at terminal No. 11, and G-Y at terminal No. 9. The typical level of differential chroma drive required at terminal Nos. 3 & 4 is 400 $mV_{p\text{-}p}$. The amplitude of chroma at terminal No. 6 & 7 is approximately 1.0 volt at 104° relative phase difference which results in a B-Y output amplitude of $5V_{p\text{-}p}$. The voltages of the R-Y & G-Y outputs are at 3.8 and 1.0 $V_{p\text{-}p}$ respectively, when there is $5V_{p\text{-}p}$ output at B-Y. These comparative signals are based upon a complete phase rotation of the chroma relative to the subcarrier signal reference. The relative demodulation phase and amplitude ratios of the Fig. 15 circuit are shown in the oscilloscope trace photographs of Fig. 21. Using the hue control setting for B-Y phase at the B-Y output, the G-Y color-difference signal is approximately −104° and the R-Y color-difference signal is approximately +106°. Since the amplitude ratios are a function of the applied signal phase relationship, the NTSC color difference output signals are shown here primarily for phase reference conditions.

CHROMA SYSTEM CONSTRUCTION

Fig. 25 shows the complete CA3070, CA3071 and CA3072 chroma system in the Fig. 18 circuit. Table I lists the dc terminal voltages for the system. The chroma gain and hue controls, as well as the switches S1 and S2 are removed. The template circuit board layout is also shown for duplication purposes. It should be noted that a few component values are modified in Fig. 18 from the dynamic circuit values of the data sheet. These are necessary for system matching and overall filter requirements.



Fig. 17(a) - CA3070 terminal No. 6, oscillator waveform 1.1 $V_{p\text{-}p}$ 3.58 MHz.

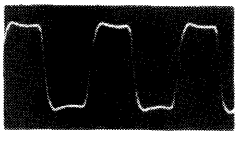


Fig. 17(b) - CA3070 terminal No. 7, oscillator waveform 1.4 $V_{p\text{-}p}$ 3.58 MHz.



Fig. 17(c) - CA3070 terminal No. 8, oscillator waveform 1.6 $V_{p\text{-}p}$ 3.58 MHz.



Fig. 18(a) - CA3071 chroma input 1.25 $V_{p\text{-}p}$; one horizontal line of NTSC input signal.

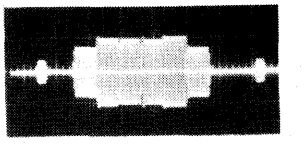


Fig. 18(b) - CA3071 terminal No. 6, amplifier No. 1 chroma output 2.3 $V_{p\text{-}p}$; one horizontal line for 1.25 $V_{p\text{-}p}$ chroma input

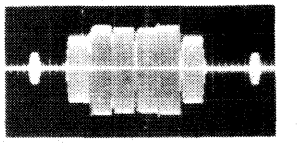


Fig. 18(c) - CA3071 terminal No. 9, amplifier No. 2 chroma output 5.5 $V_{p\text{-}p}$; one horizontal line for 1.25 $V_{p\text{-}p}$ chroma input

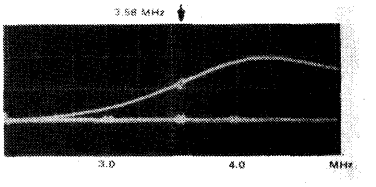


Fig. 19(a) - Frequency response sweep curve between terminal Nos. 2 & 6 for CA3071. f = 250 KHz/div.

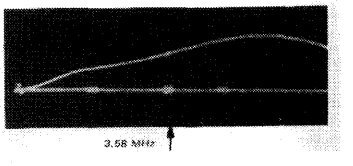


Fig. 19(b) - Frequency response sweep curve between terminal No. 2 of CA3071 and terminal No. 3 of CA3072. f = 250 KHz/div.

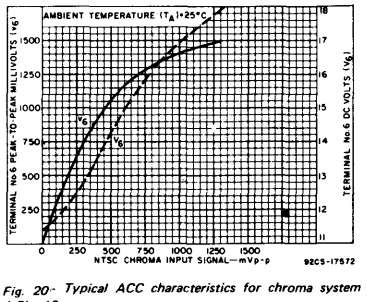


Fig. 20 - Typical ACC characteristics for chroma system of Fig. 18



Fig. 21(a) - CA3072 - terminal No. 3 or 4, chroma input signal, 220 mV_{p-p}, one horizontal line



Fig. 21(b) - CA3072 - terminal No. 6 or 7, reference subcarrier 1.2 V_{p-p}, one horizontal line



Fig. 21(c) - CA3072 terminal No. 13, 4.8 v_{p-p} B-Y output, one horizontal line

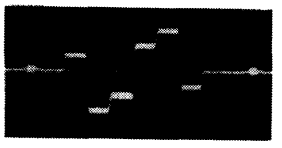


Fig. 21(d) - CA3072 - terminal No. 11, 5.2 v_{p-p} R-Y output, one horizontal line



Fig. 21(e) - CA3072 - terminal No. 9, 1.2 v_{p-p} G-Y output, one horizontal line

CA3088E

AM Receiver Subsystem

Includes: AM Converter, IF Amplifiers, Detector and Audio Preamplifier

For Applications in a Variety of AM Broadcast and Communications Receivers and Applications Requiring an Array of Amplifiers

Features:

- **Excellent overload characteristics**
- **AGC for IF amplifier**
- **Buffered output signal for tuning meter**
- **Internal Zener diode provides voltage regulation**
- **Two IF amplifier stages**
- **Low-noise converter and first IF amplifier**
- **Low harmonic distortion (THD)**
- **Delayed AGC for RF amplifier**
- **Terminals for optional inclusion of tone control**
- **Operates from wide range of power supplies: V^+ = 6 to 16 volts**
- **Optional AC and/or DC feedback on wide-band amplifier**
- **Array of amplifiers for general-purpose applications**
- **Suitable for use with optional external RF stage, either MOS or bipolar**

RCA-CA3088E*, a monolithic integrated circuit, is an AM subsystem that provides the converter, IF amplifier, detector, and audio preamplifier stages for an AM receiver.

The CA3088E also provides internal AGC for the first IF amplifier stage, delayed AGC for an optional external RF amplifier, a buffer stage to drive a tuning meter, and terminals facilitating the optional use of a tone control.

Fig. 2 is a functional diagram of the CA3088E. The signal from the low-noise converter is applied to the first IF amplifier and is then coupled to the second IF amplifier. This IF signal is then detected and externally filtered. The resultant audio signal is applied to an audio preamplifier. Optionally, a tone control circuit may be connected at the junction of the detector circuit and the audio preamplifier. The gain of the first IF amplifier stage is controlled by an internal AGC circuit. The CA3088E supplies a delayed AGC signal output for use with an external RF amplifier. A buffered output signal is also available for driving a tuning meter. A DC voltage, internally regulated by a Zener diode, supplies the second IF amplifier, the AGC and tuning meter circuits and may also be used with any other stage.

The CA3088E features four independent transistor amplifiers, each incorporating internal biasing for temperature tracking. These amplifiers are particularly useful in general-purpose amplifier, oscillator, and detector applications in a wide variety of equipment designs.

*Formerly Developmental Type TA5842.

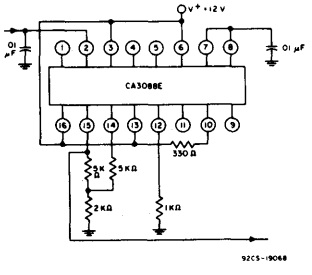


Fig.1–Test circuit for DC characteristics.

MAXIMUM RATINGS, *Absolute Maximum Values, at T_A = 25°C*

DC SUPPLY VOLTAGE:		
Across Term. 5 and Terms. 3, 6, 13, 16, respectively	16	V
DC CURRENT:		
At Terms. 3, 6, 13, 16, respectively	10	mA
At Term. 10	30	mA
DEVICE DISSIPATION:		
Up to T_A = 50°C	760	mW
Above T_A = 50°C	derate linearly 7.6	mW/°C
AMBIENT TEMPERATURE RANGE:		
Operating	−55 to +125	°C
Storage	−65 to +150	°C
LEAD TEMPERATURE (During soldering):		
At distance not less than 1/32" (0.79 mm) from case for 10 seconds max.	+265	°C

The CA3088E is supplied in the 16-lead dual- in-line plastic package.

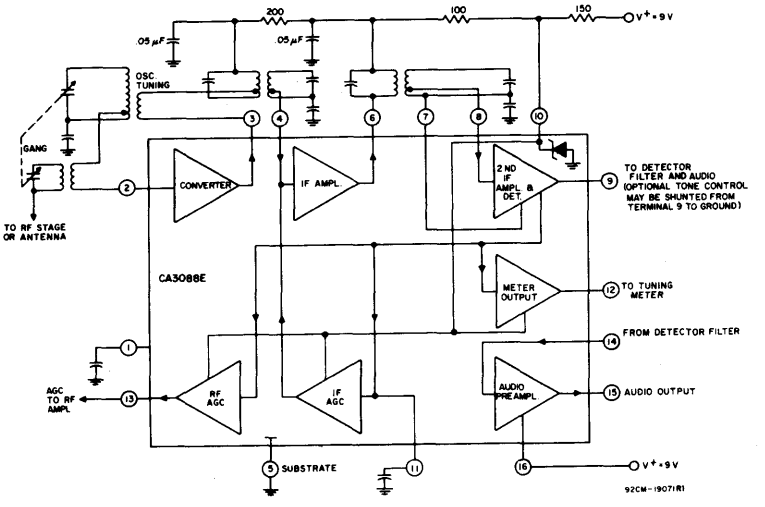


Fig.2–Functional block diagram of the CA3088E.

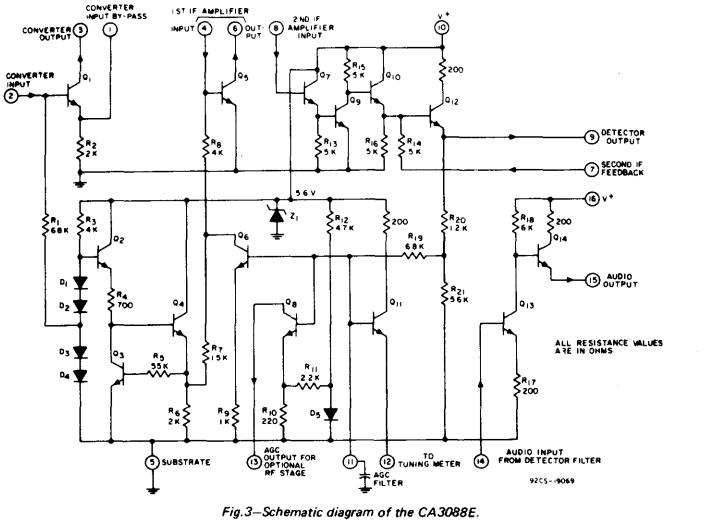


Fig.3–Schematic diagram of the CA3088E.

TYPICAL ELECTRICAL CHARACTERISTICS

CHARACTERISTIC	SYMBOL	TEST CONDITIONS T_A = 25°C, V+ = 12 V	TEST CIRCUIT FIG. NO.	TYPICAL VALUES	UNITS
Static (DC) Characteristics					
DC Voltages:					
Terms. 1, 4, 9, 11	$V_{1, 4, 9, 11}$		1	0.7	V
Terms. 2, 7, 8	$V_{2, 7, 8}$			1.4	V
Term. 10	V_{10}			5.6	V
Term. 12	V_{12}			0	V
Term. 15	V_{15}			3.5	V
DC Current:					
Term. 3	I_3		1	0.35	mA
Term. 6	I_6			1.0	mA
Term. 10	I_{10}			20	mA
Term. 13	I_{13}			0	mA
Term. 16	I_{16}			1.2	mA
Dynamic Characteristics					
Detector Output		30% Modulation	4	75	mV RMS
Audio Amplifier Gain	A_{AF}	f = 1 kHz	4	30	dB
Audio Distortion		V_{OUT} = 100 mV	4	0.2	%
Sensitivity:		f_{IN} = 1 MHz			
At Converter Stage Input		Signal-to-Noise Ratio (S/N) = 20 dB	2	200	μV/m
At RF Stage Input			4	100	μV/m
Total Harmonic Distortion	THD	30% Modulation	4	1.0	%
Input Resistance:	R_I	No AGC, Input signal frequency (f_{IN}) = 1 MHz			
At Transistor Q1				3500	Ω
At Transistor Q5				2000	Ω
Input Capacitance:	C_I				
At Transistor Q1				12	pF
At Transistor Q5				17	pF
Feedback Capacitance:	C_{FB}				
At Transistor Q1				1.5	pF
At Transistor Q5				1.5	pF

The typical characteristics for the CA3088E are intended for guidance purposes in evaluating this device for equipment design.

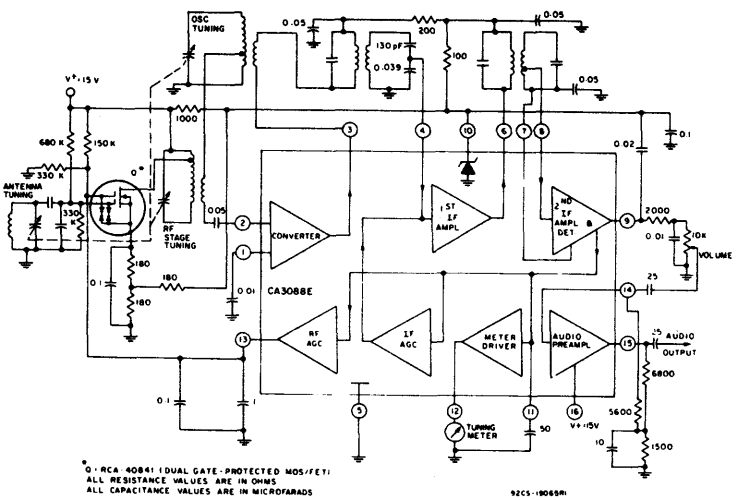


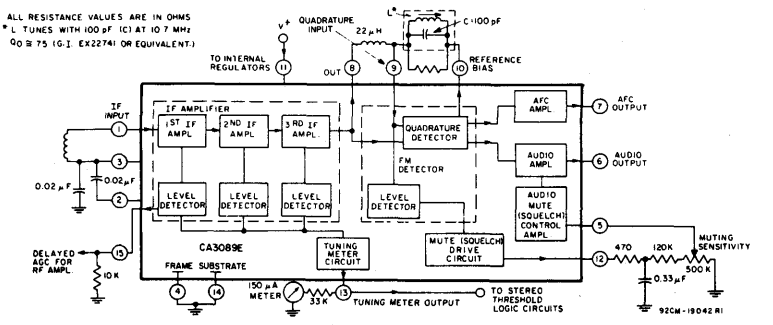
Fig.4—Typical AM broadcast receiver using the CA3088E with optional RF amplifier stage.

CA3089E

FM IF System

Includes---IF Amplifier, Quadrature Detector, AF Preamplifier, and Specific Circuits for AGC, AFC, Muting (Squelch), and Tuning Meter

For FM IF Amplifier Applications in High-Fidelity Automotive, and Communications Receivers



Block diagram of the CA3089E.

The CA3089E is supplied in the 16-lead dual-in-line plastic package.

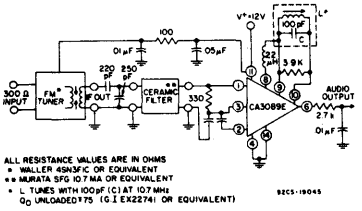
Features:

- **Exceptional limiting sensitivity: 12 μV typ. at −3 dB point**
- **Low distortion: 0.1% typ. (with double-tuned coil)**
- **Single-coil tuning capability**
- **High recovered audio: 400 mV typ.**
- **Provides specific signal for *control of interchannel muting* (squelch)**
- **Provides specific signal for direct drive of a tuning meter**
- **Provides delayed AGC voltage for RF amplifier**
- ***Provides a specific circuit for flexible AFC***
- **Internal supply-voltage regulators**

RCA-CA3089E is a monolithic integrated circuit that provides all the functions of a comprehensive FM-IF system. Fig. 1 is a block diagram showing the CA3089E features, which include a three-stage FM-IF amplifier/limiter configuration with level detectors for each stage, a doubly-balanced quadrature FM detector and an audio amplifier that features the optional use of a muting (squelch) circuit.

The advanced circuit design of the IF system includes desirable deluxe features such as delayed AGC for the RF tuner, an AFC drive circuit, and an output signal to drive a tuning meter and/or provide stereo switching logic. In addition, internal power supply regulators maintain a nearly constant current drain over the voltage supply range of +8.5 to +16 volts.

The CA3089E is ideal for high-fidelity operation. Distortion in a CA3089E FM-IF System is primarily a function of the phase linearity characteristic of the outboard detector coil.



Performance data at f_0 = 98 MHz, f_{MOD} = 400 Hz, Deviation = ±75 kHz:

-3dB Limiting Sensitivity	**2μV (Antenna Level)**
20dB Quieting Sensitivity	**1μV (Antenna Level)**
30dB Quieting Sensitivity	**1.5μV (Antenna Level)**

Fig. 1- Typical FM tuner using the CA3089E with a single-tuned detector coil.

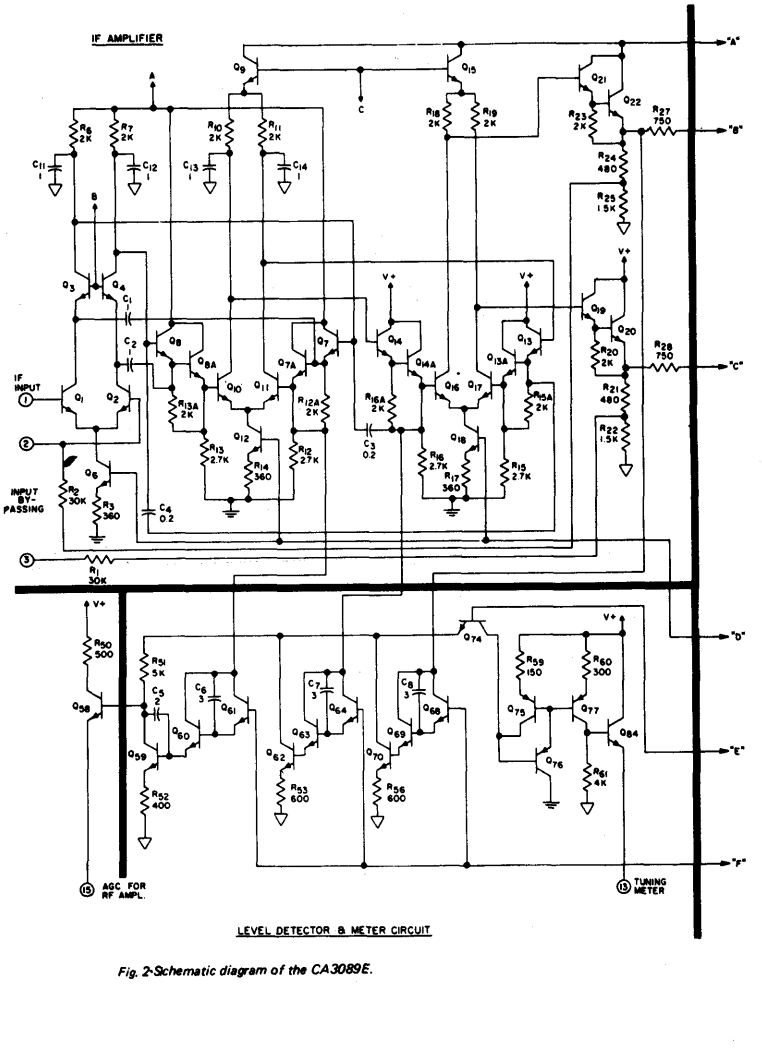
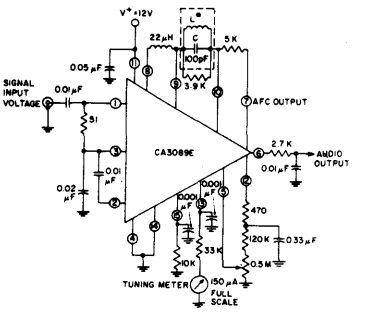


Fig. 2-Schematic diagram of the CA3089E.

MAXIMUM RATINGS, *Absolute Maximum Values, at* $T_A = 25^oC$

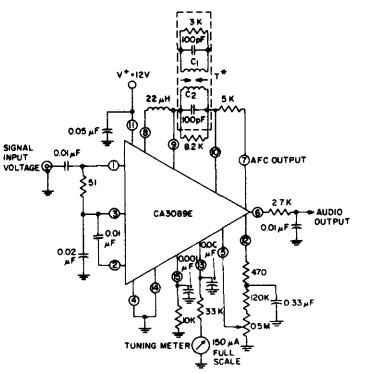
DC Supply Voltage:		
Between Terminals 11 and 4	16	V
Between Terminals 11 and 14	16	V
DC Current (out of Terminal 15)	2	mA
Device Dissipation:		
Up to $T_A = 60^oC$	600	mW
Above $T_A = 60^oC$	derate linearly 6.7 mW/^oC	
Ambient Temperature Range:		
Operating	−55 to +125	^oC
Storage	-65 to +150	^oC
Lead Temperature (During Soldering):		
At distance not less than 1/32" (0.79mm) from case for 10 seconds max.	+265	^oC



ALL RESISTANCE VALUES ARE IN OHMS
*L TUNES WITH 100 pF (C) AT 10.7 MHz
Q_o(UNLOADED) ≥ 75 (G I AUTOMATIC MFG DIV. EX22741 OR EQUIVALENT)

92CM-19040R

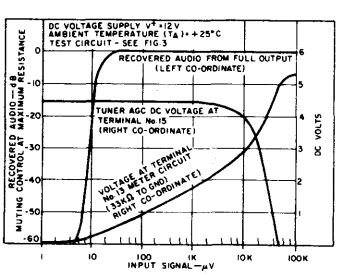
Fig.3-Test circuit for CA3089E using a single-tuned detector coil.



ALL RESISTANCE VALUES ARE IN OHMS
*T: PRI. - Q_o(UNLOADED) ≥ 75 (TUNES WITH 100 pF (C1) 20 t OF 34 e ON 7/32" DIA FORM
SEC - Q_o(UNLOADED) ≥ 75 (TUNES WITH 100 pF (C2) 20 t OF 34 e ON 7/32" DIA FORM
k Q (PER CENT OF CRITICAL COUPLING) ≅ 70 %
(ADJUSTED FOR COIL VOLTAGE V_C = 150 mV
ABOVE VALUES PERMIT PROPER OPERATION OF MUTE (SQUELCH) CIRCUIT
"E" TYPE SLUGS, SPACING 4 mm

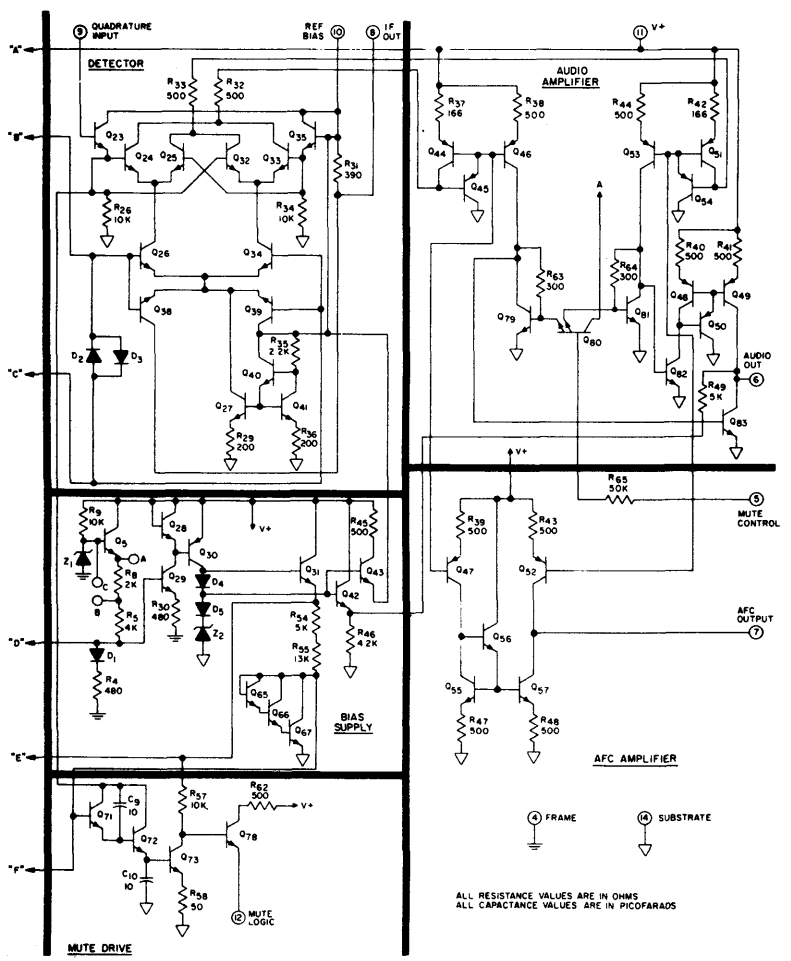
92CM-19041R1

Fig.4-Test circuit for CA3089E using a double-tuned detector coil.



92CS-19990

Fig.5-Muting action, tuner AGC, and tuning meter output as a function of input signal voltage.



ALL RESISTANCE VALUES ARE IN OHMS
ALL CAPACTANCE VALUES ARE IN PICOFARADS

Fig.2-Schematic diagram of the CA3089E.

ELECTRICAL CHARACTERISTICS, at T_A = 25°C, V^+ = 12 Volts

CHARACTERISTIC	SYMBOL	TEST CONDITIONS		Circuit Fig. No.	LIMITS Min.	Typ.	Max.	UNITS
Static (DC) Characteristics								
Quiescent Circuit Current	I_{11}	No signal input, Non muted		3, 4	16	23	30	mA
DC Voltages:								
Terminal 1 (IF Input)	V_1				1.2	1.9	2.4	V
Terminal 2 (AC Return to Input)	V_2				1.2	1.9	2.4	V
Terminal 3 (DC Bias to Input)	V_3				1.2	1.9	2.4	V
Terminal 6 (Audio Output)	V_6				5.0	5.6	6.0	V
Terminal 10 (DC Reference)	V_{10}				5.0	5.6	6.0	V
Dynamic Characteristics								
Input Limiting Voltage (-3 dB point)	V_I(lim)	–		3, 4	–	12	25	μV
AM Rejection (Term. 6)	AMR	V_{IN} = 0.1 V, AM Mod. = 30%	f_0 = 10.7 MHz,		45	55	–	dB
Recovered AF Voltage (Term. 6)	V_O(AF)				300	400	500	mV
Total Harmonic Distortion: *		V_{IN} = 0.1 V	$f_{mod.}$ = 400 Hz, Deviation = ±75 kHz					
Single Tuned (Term. 6)	THD			3	–	0.5	1.0	%
Double Tuned (Term. 6)	THD			4	–	0.1	–	%
Signal plus Noise to Noise Ratio (Term. 6)	S + N/N			3, 4	60	67	–	dB

* THD characteristics are essentially a function of the phase characteristics of the network connected between terminals 8, 9, and 10.

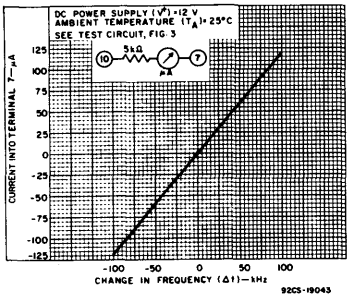
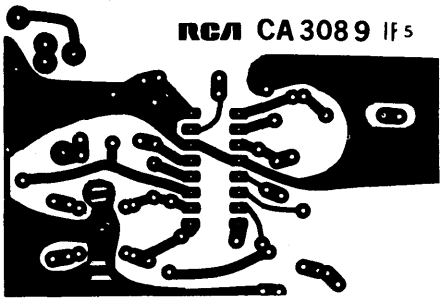
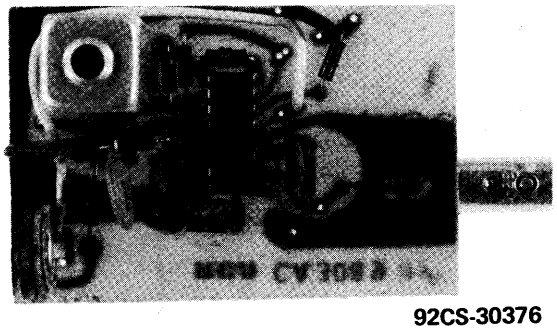


Fig.6-AFC characteristics (current at Term. 7 as a function of change in frequency).



a) Bottom view of printed-circuit board.



92CS-30376

b) Component side – top view.

CA3089E and outboard components mounted on a printed-circuit board.

Stereo Multiplex Decoder

For FM Stereo Multiplex Systems

RCA-CA3090AQ , a monolithic silicon integrated circuit, is a stereo multiplex decoder intended for FM multiplex systems.

The CA3090AQ is the successor to the CA3090Q; it offers three major advantages over the CA3090Q as follows:

1. Can directly drive a stereo indicator lamp with a current drain of up to 100 mA.
2. Stereo Defeat/Enable control-voltage specifications.
3. Capable of operation with lower distortion.

This stereo multiplex decoder requires only one low-inductance tuning coil (requires only one adjustment for complete alignment), provides automatic stereo switching, energizes a stereo indicator lamp, and operates from a wide range of voltage supplies.

Figure 1 shows the block diagram for the CA3090AQ. The input signal from the detector is amplified by a low-distortion preamplifier and simultaneously applied to both the 19-kHz and 38-kHz synchronous detectors. A 76-kHz signal, generated by a local voltage-controlled oscillator (VCO), is counted down by two frequency dividers to a 38-kHz signal and to two 19-kHz signals in phase quadrature.

The 19-kHz pilot-tone supplied by the FM detector is compared to the locally generated 19-kHz signal in a synchronous detector. The resultant signal controls the voltage controlled oscillator (VCO) so that it produces an output signal to phase-lock the stereo decoder with the pilot tone. A second synchronous detector compares the locally generated 19-kHz signal with the 19-kHz pilot tone. If the pilot tone exceeds an externally adjustable threshold voltage, a Schmitt trigger circuit is energized. The signal from the Schmitt trigger lights the stereo indicator, enables the 38-kHz synchronous detector, and automatically switches the CA3090AQ from monaural to stereo operation. The output signal from the 38-kHz detector and the composite signal from the preamplifier are applied to a matrixing circuit from which emerge the resultant left and right channel audio signals. These signals are applied to their respective left and right post amplifiers for amplification to a level sufficient to drive most audio amplifiers.

The CA3090AQ may be used without the stereo defeat/enable function (see Fig. 6) if a control voltage for this function is not readily available. In this case, Terminal 4 should be grounded.

The CA3090AQ utilizes the 16-lead quad-in-line plastic package and operates over the ambient temperature range of −55°C to +125°C.

Features:

- Requires the use of only one low-inductance tuning coil
- Automatic stereo switching
- Directly drives a stereo indicator lamp up to 100 mA
- Includes driver for stereo-lamp indicator
- Operates from a wide range of power supplies: 10 to 16 volts
- Requires only one adjustment for alignment
- Switching from monaural to stereo and stereo to monaural produces no audible thumps
- Low distortion: under 0.22% (typ.)
- Separate dc input permits stereo defeat or enable
- High signal output: directly drives audio amplifiers
- Excellent SCA (storecast) rejection: 55 dB typ.
- High audio channel separation: 40 dB typ.

MAXIMUM RATINGS, *Absolute-Maximum Values at* T_A = 25°C

DC SUPPLY VOLTAGE	16 V
CURRENT AT TERM. 12	100 mA
INPUT SIGNAL VOLTAGE (COMPOSITE)■	400 mV
AMBIENT TEMPERATURE RANGE:	
Operating	−55 to +125°C
Storage	−65 to +150°C
LEAD TEMPERATURE (DURING SOLDERING):	
At distance not less than 1/32" (0.79 mm) from case for 10 s max.	+265°C

■ For stereo operation, a minimum input signal voltage (composite) of 40 mV is required

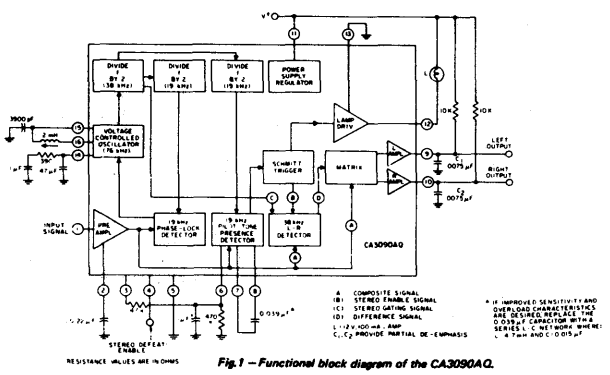


Fig. 1 – Functional block diagram of the CA3090AQ.

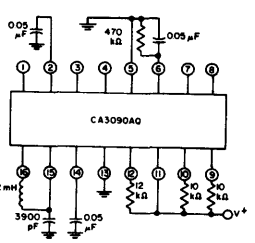


Fig. 2 – Test circuit for DC characteristics.

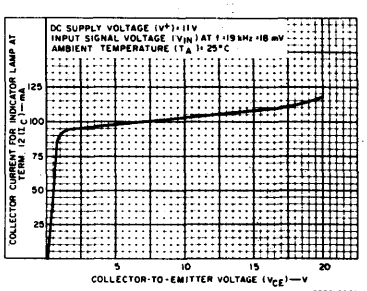


Fig. 3 – Indicator lamp characteristics (I_C vs. V_{CE}).

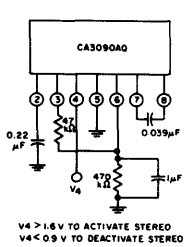


Fig. 4 – Test circuit for use with stereo defeat/enable.

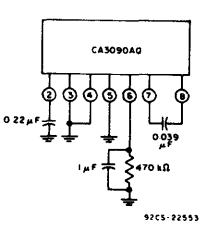


Fig. 5 – Test circuit for use without stereo defeat/enable.

CA3090AQ

ELECTRICAL CHARACTERISTICS

CHARACTERISTIC	TERMINAL MEASURED AND SYMBOL	TEST CONDITIONS (T_A = 25°C, V+ = 12 V (unless specified otherwise))	LIMITS Min.	Typ.	Max.	UNITS
Static Characteristics						
Total Current (Terms. 9, 10, 11)	I_{total}	Lamp OFF	–	22	27	mA
DC Voltage: Term. 1	V_1		1.6	2.3	3.1	V
Term. 6 (Indicator Lamp OFF)	V_6		–	2.1	3.6	V
Terms. 9 and 10	$V_{9\ \&\ 10}$		4.7	6.4	8.4	V
Term. 12 (Indicator Lamp OFF)	V_{12}	V^+ = 16 V	12.7	–	–	V
Voltage Differential (Term. 2–Term. 1)	$V_2 - V_1$		–	0	0.1	V
Current at Term. 12 (In actual use external circuit resistance (e.g. lamp should limit Term. 12 to the maximum rated value of 100 mA.)		V_{IN} (at f = 19 kHz) = 18 mV	75	100	–	mA
Dynamic Characteristics						
Input Impedance	Z_{IN}		–	50k	–	Ω
Channel Separation (L + R Reference)*		V_{IN} = 180 mV	25	40	–	dB
Channel Balance (Monaural)			–	0.3	3	dB
Monaural Gain			3	6	9	dB
Stereo/Monaural Gain Ratio*			–	±0.3	±3	dB
Indicator Lamp – Turn-ON Voltage		19-kHz pilot-tone @ Term. 1	–	4	–	mV
Capture Range (Deviation from 76-kHz center frequency)		19-kHz pilot-tone voltage = 18 mV	±6.6	±10	–	%
Distortion (75-μs de-emphasis): 2nd Harmonic		V_{IN} = 240 mV	–	0.2	–	%
3rd, 4th, and 5th Harmonic			–	<0.1	–	%
19-kHz Rejection			–	35	–	dB
38-kHz Rejection			–	48	–	dB
SCA (storecast) Rejection			–	70	–	dB
Stereo Defeat Voltage (V_4)			–	1.2	<0.9	V
Stereo Enable Voltage (V_4)			>1.6	1.2	–	V

* For stereo operation, test conditions require a composite stereo input signal (modulated at 1 kHz) including a 19-kHz (18 mV) pilot-tone signal.

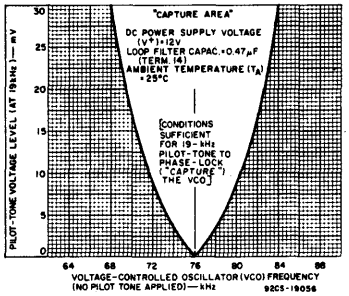


Fig. 7 – Pilot-tone voltage level vs. VCO frequency with no pilot-tone applied.

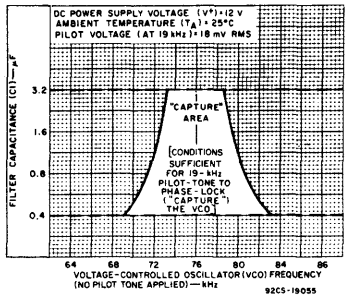


Fig. 8 – Filter capacitance vs. VCO frequency with no pilot-tone applied.

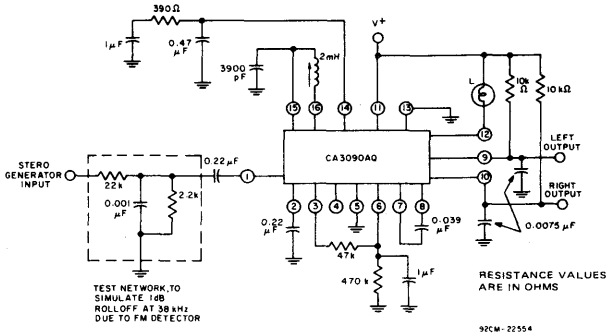
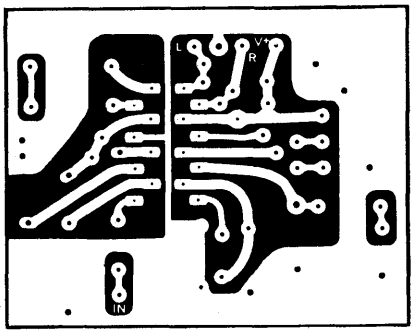
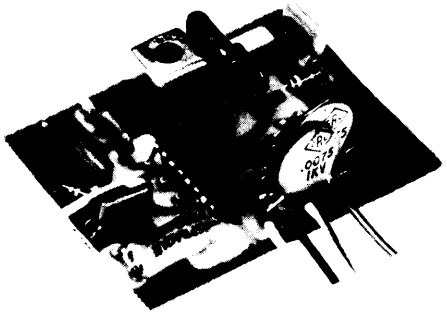


Fig. 6 – Test circuit for measurement of dynamic characteristics.



A–Foil side.



B–Component side.

Fig. 9 – Photographs of the CA3090AQ and outboard components mounted on a 2 X 2¼-inch printed-circuit board to constitute a complete stereo multiplex decoder.

TV Signal Processors ("Jungle" Circuits)

For Color and Monochrome Receivers

The RCA-CA3120E and CA3142E are monolithic silicon integrated circuit TV signal processors for use in color or monochrome receivers. These circuits provide low-impedance video output signals, stripped synchronization signals in both polarities, and AGC output signals for IF (reverse) and tuner (forward and/or reverse).

The circuit designs of the CA3120E and CA3142E feature impulse noise inversion, delay techniques to reduce the deleterious effects of impulse noise in the receiver AGC and sync circuits. In addition, they incorporate standard AGC strobing techniques. The AGC noise lockout circuit is deleted in the CA3142E.

These devices are supplied in the 16-lead dual-in-line plastic package.

Features:

- **Internal impulse noise processing**
- **Sync separator – low impedance, dual polarity**
- **Strobed AGC system**
- **IF AGC output**
- **Delayed outputs for forward or reverse AGC tuners**
- **Automatic noise threshold and AGC detector level control**
- **High-impedance video input**
- **Low-impedance video output**
- **Choice of external time constants for sync separator**
- **Negative power supply not required**
- **RF AGC delay externally controlled**

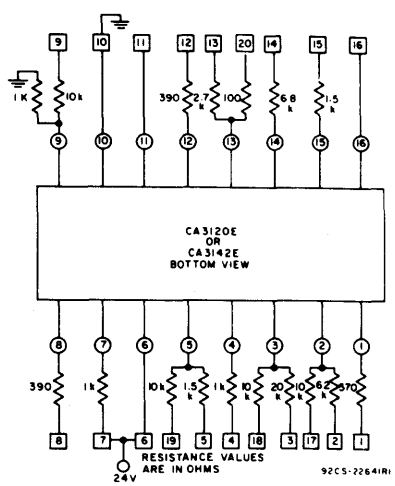


Fig. 2 – Test circuit for measuring electrical characteristics of the CA3120E and CA3142E. Refer to Figs. 7 and 8 for switch selector positions.

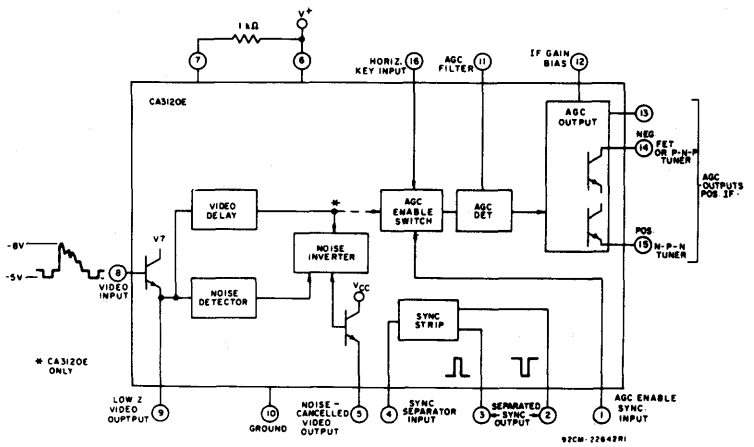


Fig. 1 – Simplified block diagram of the CA3120E and CA3142E.

MAXIMUM RATINGS, Absolute-Maximum Values at $T_A = 25^oC$

DC SUPPLY VOLTAGE	30 V
DEVICE DISSIPATION:	
Up to $T_A = 55^oC$	750 mW
Above $T_A = 55^oC$ . . . . . Derate linearly at 7.9	mW/°C
AMBIENT TEMPERATURE RANGE:	
Operating	–40 to +85 °C
Storage	–65 to +150 °C
LEAD TEMPERATURE (During soldering):	
At a distance not less than 1/32" (0.79 mm) from case for 10 seconds max.	+265 °C

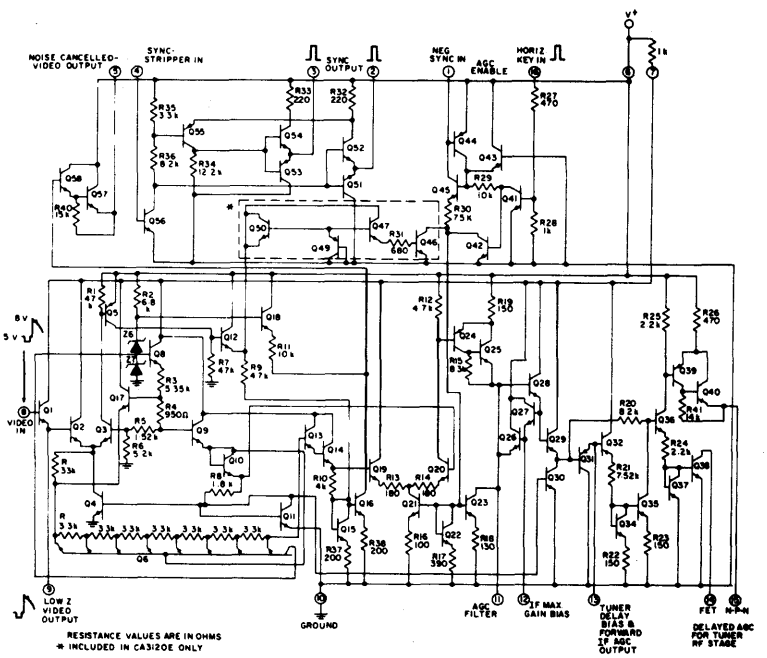


Fig. 3 – Schematic diagram of the CA3120E and CA3142E.

CIRCUIT DESCRIPTION*

An AGC sample-and-hold system generates control voltages proportional to the video level. The sync-tip voltage is compared to an internal reference voltage during the horizontal synchronization (retrace) interval. The control voltages (AGC outputs) are supplied to the tuner's RF stage and the IF amplifier to maintain the video level at a constant value.

The composite positive and negative output sync signals are developed across a low impedance source (totem-pole circuit) at an amplitude of approximately 20 volts peak-to-peak.

Video Chain and Impulse Noise Inverter – The input video signal applied at Terminal 8 is white "positive" with a required amplitude in the range of 2 to 4 volts. The DC level of the sync peaks, AGC threshold voltage (V_{TH}) is approximately 5 volts. The level is maintained at 5 volts by the AGC loop in the circuit, comprised of the CA3120E or CA3142E and the TV receiver RF and IF amplifiers. A low source impedance video signal is available from the emitter of Q1 (Terminal 9 in Fig. 3). The external resistor (R_{X1} in Fig.9) reduces the dissipation of Q1. The emitter-follower output of Q1 is directly coupled to a differential comparator stage (Q2, Q3). Unless a negative-going pulse is present, Q2 functions as an emitter follower and also cuts off transistors Q3, Q5, and Q12.

The output of Q2 is applied through a signal delay network, consisting of transistor Q60 and associated resistors, to the Darlington followers (Q13 and Q14). The delayed video signal at Q14 is fed via its emitter to an AGC comparator Q19 and to the junction of a noise-cancelling amplifier stage (Q16). The noise-cancelled video signal is inverted and amplified by Q16 and then connected to a Darlington emitter-follower output stage (Q57, Q58).

If impulse noise is present on the video signal, Q3 conducts and turns on transistors Q5 and Q12. Q5 inverts and "stretches" the noise pulse width. The output of Q5 is applied to an emitter follower stage (Q12). The signal from Q12, in turn, is applied to the summing junction to the noise-cancelling amplifier Q16. The noise pulse, which has now been amplified, inverted and stretched, is added to the delayed video signal from the emitter of Q14.

Because the video signal has been delayed approximately 300 nanoseconds and the noise pulse has been widened ("stretched") approximately 500 nanoseconds, the output of the combined signal no longer contains impulse noise signals. The derived noise-gating pulse "surrounds" and effectively eliminates the effects of the impulse noise.

The noise-cancelled video signal, amplified and buffered, is available at Terminal 5 for use in the sync-separator stage. The peak-to-peak amplitude of the noise-cancelled output signal is approximately twice the amplitude of the input video signal at Terminal 8.

Sync Separator (See Figure 4) – The sync separator stage (Q56) clamps the detected sync tips to a fixed reference voltage (≅ 0.7 V) across its base-emitter junction, and amplifies a portion of the sync signal to provide dual polarity sync-signal outputs at Terminals 2 (negative) and 3 (positive). The output signals are derived from low-impedance complementary emitter-follower stages; a base current of 100 microamperes into Terminal 4 is sufficient to generate full-amplitude sync signals.

The choice of coupling the noise-cancelled video-signal from the emitter-follower (Terminal 5) to the sync separator (Terminal 4) is a user option. Fig. 5 shows three typical coupling networks.

Fig. 6 illustrates the operation of the AGC circuits. An input ramp signal, simulating the potential to which the AGC filter capacitor may be charged, is applied to Terminal 11. The forward IF AGC output voltage appears at Terminal 13. Under low-signal level conditions (represented by A to B in Fig. 6) the output level is approximately 1.4 volts less than the voltage applied to Terminal 12.

The circuit designer should select the voltage at Terminal 12 to provide the maximum IF gain required for the system. At intermediate signal level conditions (represented by B to C in Fig. 6), the IF AGC signal follows the AGC filter potential. The tuner(s) will operate at maximum gain for good signal-to-noise ratios at these equivalent input signal levels. Point C is a turnover point determined by the open-circuit potential of the tuner-delay bias potentiometer. At this potential, further change in the IF AGC output is inhibited (for good dynamic range) and the tuner AGC potentials are activated (represented by C to D).

The output at Terminal 14 with suitable level shifting is used for tuners requiring reverse AGC, such as MOSFET or electron-tube types. The output at Terminal 15 is used for tuners requiring forward AGC, such as tuners utilizing n-p-n bipolar transistors.

* For additional information refer to the IEEE Transactions on Broadcast and TV Receivers," August 1970, pp. 185–195, Vol. BTR No.3.

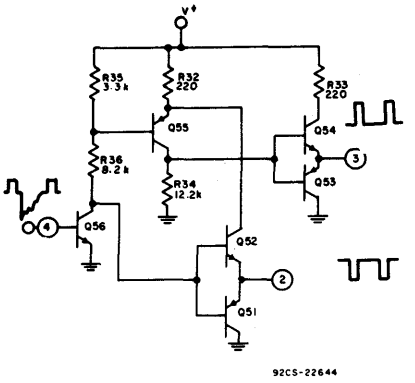


Fig. 4 – Sync separator stage.

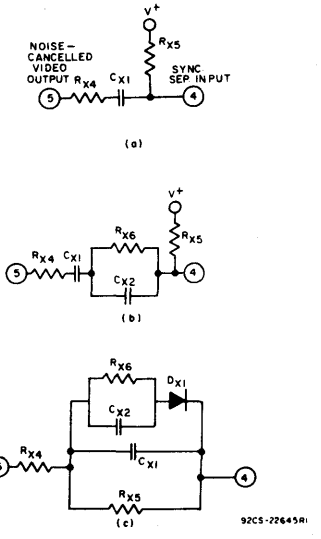


Fig. 5 – Typical coupling networks (Term. 5 to Term. 4).

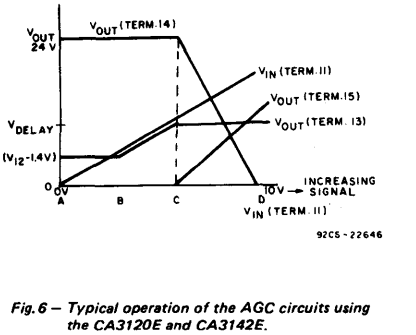


Fig. 6 – Typical operation of the AGC circuits using the CA3120E and CA3142E.

CHARACTERISTIC	TEST CONDITIONS — SWITCH NUMBERS / SWITCH POSITION																TERMINAL MEASURED
	1	*2*	*3*	*4*	*5*	*8*	*9*	*11*	*12*	*13*	*14*	*15*	*16*	*17*	*18*	*19*	*20*	
I_{T24}	2	3	1	2	1	2	3	1	1	3	2	1	2	2	2	1	5	*2 6 7 9 14*
V_{TH}	2	1	2	1	1	4	3	4	4	3	1	2	2	2	2	1	3	*8*
V_5	2	1	2	1	1	4	3	4	4	3	1	2	2	2	2	2	3	*19*
$V_{TH(SEP)}$	3	1	2	1	1	*	3	3	4	1	1	2	1	2	2	2	1	*
$I_{4(OFF)}$	3	1	2	4	2	1	1	1	1	1	1	2	1	2	2	1	1	I_4
V_{2L}	1	2	2	3	2	1	1	1	1	1	1	2	1	1	2	1	1	V_{17}
V_{2H}	3	3	1	1	2	1	1	1	1	1	1	2	1	1	2	1	1	V_{17}
V_{3L}	3	3	1	1	2	1	1	1	1	1	1	2	1	2	1	1	1	V_{18}
V_{3H}	3	3	1	3	2	1	1	1	1	1	1	2	1	2	1	1	1	V_{18}
$I_{11(CH)}$	2	1	2	5	2	1	1	5	4	3	1	2	2	2	2	1	5	I_{11}
$I_{11(DISCH)}$	2	1	2	5	1	2	3	6	4	3	1	2	2	2	2	1	5	I_{11}
$I_{11(LEAK)}$	2	1	2	5	2	1	1	6	4	3	2	2	1	2	2	1	5	I_{11}
V_{11}	2	1	2	5	1	2	3	2	3	3	1	2	2	2	2	1	5	V_{11}
V_{12}	3	1	2	5	2	1	1	3	4	3	1	2	1	2	2	1	5	V_{12}
$V_{13(LOW)}$	3	1	2	5	2	2	3	1	1	2	1	2	1	2	2	1	2	V_{13}
$V_{13(HIGH)}$	3	1	2	5	2	2	3	7	4	3	2	1	1	2	2	1	4	V_{20}
$I_{14(OFF)}$	3	1	2	5	2	2	3	3	4	3	3	1	1	2	2	1	5	I_{14}
$I_{14(ON)}$	3	1	2	5	2	2	3	8	4	3	3	1	1	2	2	1	5	I_{14}
$I_{15(OFF)}$	3	1	2	5	2	2	3	3	4	3	2	3	1	2	2	1	5	I_{15}
$I_{15(ON)}$	3	1	2	5	2	2	3	8	4	3	2	3	1	2	2	1	5	I_{15}

CAUTION: Remove power before selecting or adjusting switches.

* Reduce voltage at Terminal 8 until V_{19} decreases. $V_{TH(SEP)} = V_{TH} - V_8$.

NOTE: Switch numbers in italics correspond to numbers in square boxes in Figs. 2 and 8.

Fig.7 – Test condition values for associated switches 1 through 20 (switches 6, 7, and 10 are omitted). Refer to Figs. 2 and 8 for test circuit and test-condition selector-switch arrangements.

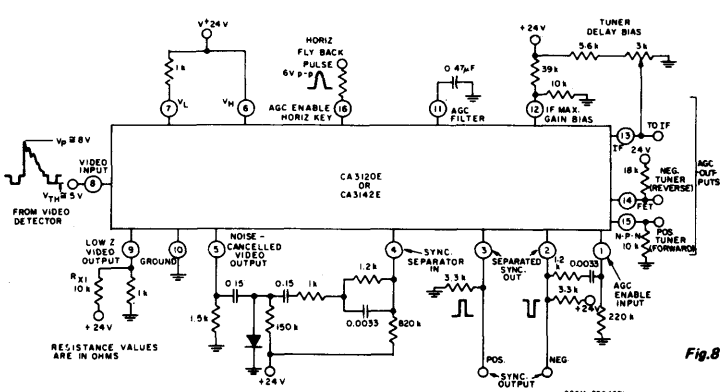
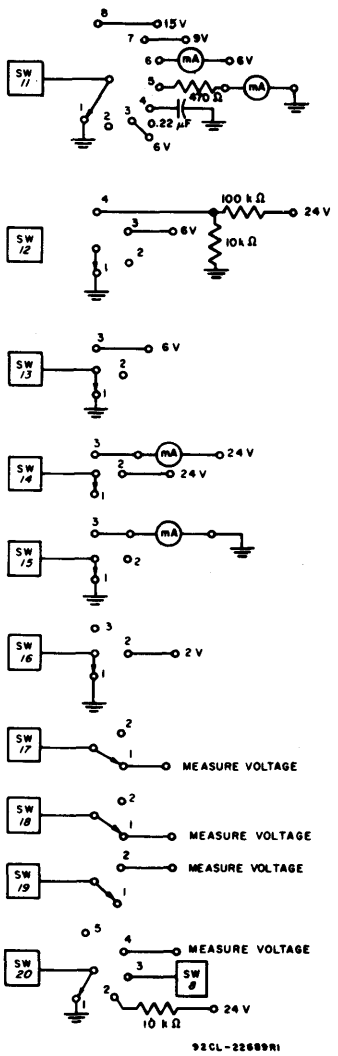


Fig.9 – Typical application using the CA3120E and CA3142E.



NOTE: The italicized numbers in the square boxes refer to the 17 switches (switches 6, 7, and 10 are omitted) of the test circuit and correspond to those given in Figs. 2 and 7.

CAUTION: Remove power before selecting or adjusting switches

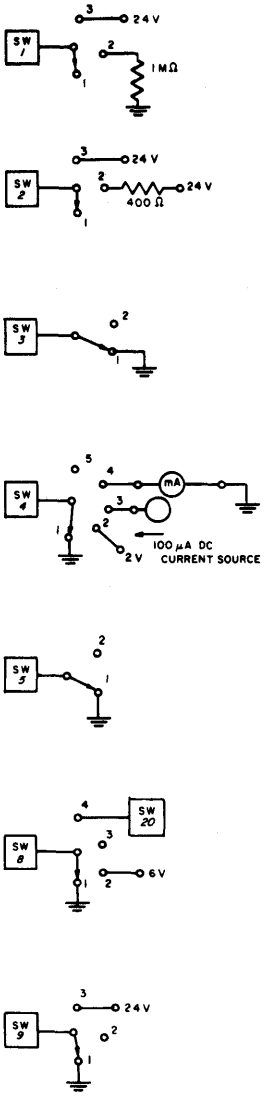
Fig.8 – Test condition selector switch arrangement for measuring the electrical characteristics of the CA3120E and CA3142E.

(Figure 8 continued on the next page)

CA3120E, CA3142E

ELECTRICAL CHARACTERISTICS at $T_A = 25^oC$, Supply Voltage (V^+) = 24 V and Referenced to Test Circuits and Test Conditions (Figs. 2, 7, and 8).

CHARACTERISTICS	TERMINAL MEASURED AND SYMBOL	CA3120E CA3142E LIMITS			UNITS
		Min.	Typ.	Max.	
Supply Current (Pulse Test)	I_{T24}	20	–	40	mA
AGC Threshold (Sync Tip Level at Video Input)	V_{TH}	4.5	–	5.5	V
Video Input Amplitude (White Positive)	V_8	–	3	–	Vp-p
Video Output Amplitude (Low Impedance)	V_9	–	3	–	Vp-p
Noise Cancelled Video Output at V_{TH} (Black Positive, Gain ≅ 2)	V_5	3.6	–	9.2	V
AGC to Noise Separation	$V_{TH\ (SEP)}$	1.1	–	2.2	V
Sync Input Current for Full Amplitude Outputs	$I_{4\ (ON)}$	–	–	100	μA
Maximum Leakage Current at Terminal 4	$I_{4\ (OFF)}$	–	–	±6	μA
Sync Outputs:					
Negative Sync Low	$V_{2(L)}$	0	–	2.6	V
Negative Sync High	$V_{2(H)}$	23.8	–	24	V
Positive Sync Low	$V_{3(L)}$	0	–	0.2	V
Positive Sync High	$V_{3(H)}$	20.1	–	24	V
AGC Filter:					
Charge Current (Pulse Test)	$I_{11(CH)}$	12	–	36	mA
Discharge Current	$I_{11(DISCH)}$	1.1	–	2.6	mA
Leakage Current	$I_{11(LEAK)}$	–	–	±6	μA
AGC Enable:					
Horizontal Keying	$V_{16\ (ON)}$	3	–	6	V
Negative Sync Input Current	$I_{1\ (ON)}$	–	1	–	mA
Maximum IF Gain-Clamp Voltage	V_{11}	4.8	–	5.7	V
Maximum IF Gain Bias	V_{12}	4.2	–	5.2	V
IF AGC Voltage:					
Low	$V_{13\ (LOW)}$	0	–	3.3	V
High	$V_{13(HIGH)}$	5.7	–	6	V
Tuner Currents:					
Reverse AGC (FET) OFF Current	$I_{14\ (OFF)}$	–	–	±6	μA
Reverse AGC (FET) ON Current	$I_{14\ (ON)}$	1.8	–	5.5	mA
Forward AGC (n-p-n) OFF Current	$I_{15\ (OFF)}$	–	–	±6	μA
Reverse AGC (n-p-n) ON Current	$I_{15\ (ON)}$	4.5	–	15	mA
Internal Noise-Lockout Time (CA3120E only)	T	1	–	63	μs



NOTE: The italicized numbers in the square boxes refer to the 17 switches (switches 6, 7, and 10 are omitted) of the test circuit and correspond to those given in Figs. 2 and 7.

CAUTION: Remove power before selecting or adjusting switches

Fig.8 – Test condition selector switch arrangement for measuring the electrical characteristics of the CA3120E and CA3142E.

TV Chroma Amplifier/ Demodulator

Provides Complete System for Processing Chroma When Used with RCA-CA3070 or CA3170 "G" Suffix Type-Hermetic Gold-CHIP in Dual-in-Line Plastic Package

Features:

- **Excellent linearity in dc chroma gain-control circuit**
- **Improved filtering resulting in reduced 7.2 MHz output from the color demodulators**
- **Current limiting for short-circuit protection**
- **Good tolerance to B+ supply variations**
- **Good temperature coefficient stability**
- **Gold-CHIP for increased reliability**

RCA-CA3121G is a monolithic silicon integrated circuit chroma amplifier/demodulator with ACC and killer control for color-TV receivers. It is designed to function compatibly with the CA3070 or CA3170 in a two-package chroma system. Figs. 5 and 6 show a functional block diagram and the outboard circuitry of a typical two-package chroma system incorporating the CA3121G and CA3170, respectively.

The CA3121G is supplied in a 16-lead dual-in-line plastic package with hermetic Gold-CHIP (G suffix).

The transistor chips used in the hermetic Gold-CHIP plastic packages are of the sealed-junction type designed to provide protection against the deteriorating effects of humidity and other surface contaminents without the need for a hermetic package enclosure. The semiconductor junctions are sealed by utilizing a silicon nitride passivation layer. A multilayered, highly corrosion-resistant, terminal-connection system of unique design is employed.

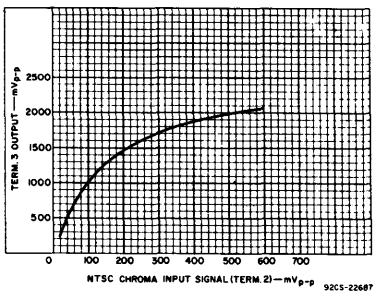


Fig. 2 – Typical ACC plot for the CA3121G when used with the CA3070.

MAXIMUM RATINGS, Absolute-Maximum Values at T_A = 25°C

Supply Voltage	30 V
Device Dissipation:	
Up to T_A = 15°C	1 W
Above T_A = 55°C	derate linearly 10.5 mW/°C
Operating Temperature Range	−40 to +85°C
Storage Temperature Range	−65 to +150°C
Lead Temperature (During Soldering)	
At distance 1/16" ±1/32" (1.59 ±0.79 mm) from case for 10 s max.	+265°C

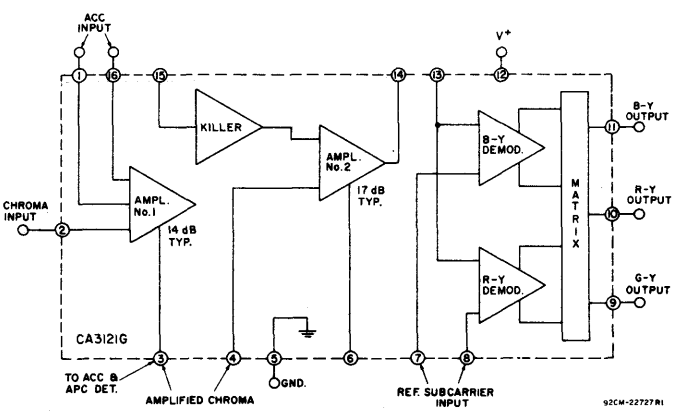


Fig. 1 – Functional block diagram of the CA3121G.

CIRCUIT OPERATION

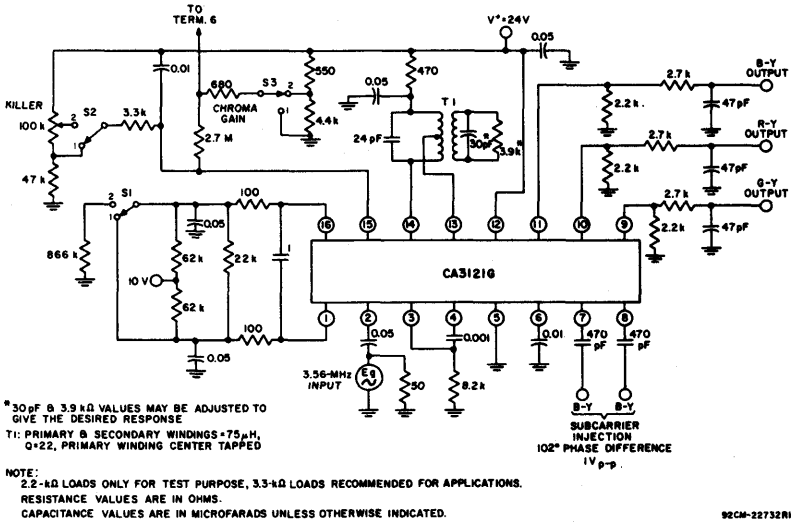
The CA3121G consists of three basic circuit sections: (1) amplifier No.1, (2) amplifier No.2, and (3) demodulator. Amplifier No.1 contains the circuitry for automatic chroma control (ACC) and color-killer sensing. The output of amplifier No.1 (Terminal 3) is coupled to the Chroma Signal Processor (CA3070, CA3170 or equivalent) for ACC and automatic phase control (APC) operation and to the input of amplifier No.2 (Terminal 4) containing the chroma gain control circuitry. The signal from the color-killer circuit in amplifier No.1 acts upon amplifier No.2 to greatly reduce its gain.

The output from amplifier No.2 (Terminal 14) is applied, through a filtering network, to the demodulator input (Terminal 13).

The demodulator also receives the R-Y and B-Y demodulation subcarrier signals (Terminals 7 and 8) from the oscillator output of the Chroma Signal Processor. The R-Y and B-Y demodulators and the matrix network contained in the demodulator section of the CA3121G reconstruct the G-Y signal to achieve the R-Y, G-Y, and B-Y color difference signals. These high-level outputs signals with low impedance outputs are suitable for driving high-level R, G, and B output amplifiers. Internal capacitors are included on each output to filter out unwanted harmonics. For additional operating information and signal waveforms, refer to Television Chroma System (utilizing RCA-CA3070, CA3071, CA3072), File No. 468.

ELECTRICAL CHARACTERISTICS at T_A = 25°C and Reference to Test Circuit (Fig. 3)

CHARACTERISTIC, TERMINAL MEASURED, AND SYMBOL	TEST CONDITIONS	LIMITS			UNITS
		Min.	Typ.	Max.	
Supply Current I_T	–	–	40	50	mA
Input Sensitivity V_2	Vary Eg; set V_4 for 55 mV RMS	6	10	15	mV RMS
Second-Stage Sensitivity V_4	Vary Eg; set V_{11} for 2 V RMS	25	55	100	mV RMS
Output Voltage (Killer off) V_{11}	Switch Position: S1=2, S2=2, S3=2 Adjust killer potentiometer until output drops	–	–	70	mV RMS
Demodulator Characteristics:					
Output Voltages V_9, V_{10}, V_{11}	–	13	14.3	15.6	V
DC Output Balance (Between any 2 outputs)	–	–0.6	–	+0.6	V
Unbalance V_9, V_{10}, V_{11}	Eg=0; Swith Position: S1=1, S2=1, S3=1	–	–	0.8	Vp-p
Relative Outputs– R-Y V_{10}	Vary Eg; set V_{11} for 2 V RMS	1.4	1.52	1.68	V RMS
G-Y V9		0.3	0.4	0.5	V RMS
Relative Phase– R-Y V_{10}	Vary Eg; set V_{11} for 2 V RMS; read phase of V_{10} and V_9	–101	–106	–111	degrees
G-Y V_9	with V_{11} as reference	112	104	96	degrees
Max. Output Voltage V_{11}	Eg = 750 mV	2.8	–	–	V RMS



*30 pF & 3.9 kΩ VALUES MAY BE ADJUSTED TO GIVE THE DESIRED RESPONSE

T1: PRIMARY & SECONDARY WINDINGS=75 μH, Q=22, PRIMARY WINDING CENTER TAPPED

NOTE:
2.2-kΩ LOADS ONLY FOR TEST PURPOSE, 3.3-kΩ LOADS RECOMMENDED FOR APPLICATIONS.
RESISTANCE VALUES ARE IN OHMS.
CAPACITANCE VALUES ARE IN MICROFARADS UNLESS OTHERWISE INDICATED.

92CM-22732R1

Fig. 3 – Typical characteristics test circuit for the CA3121G.

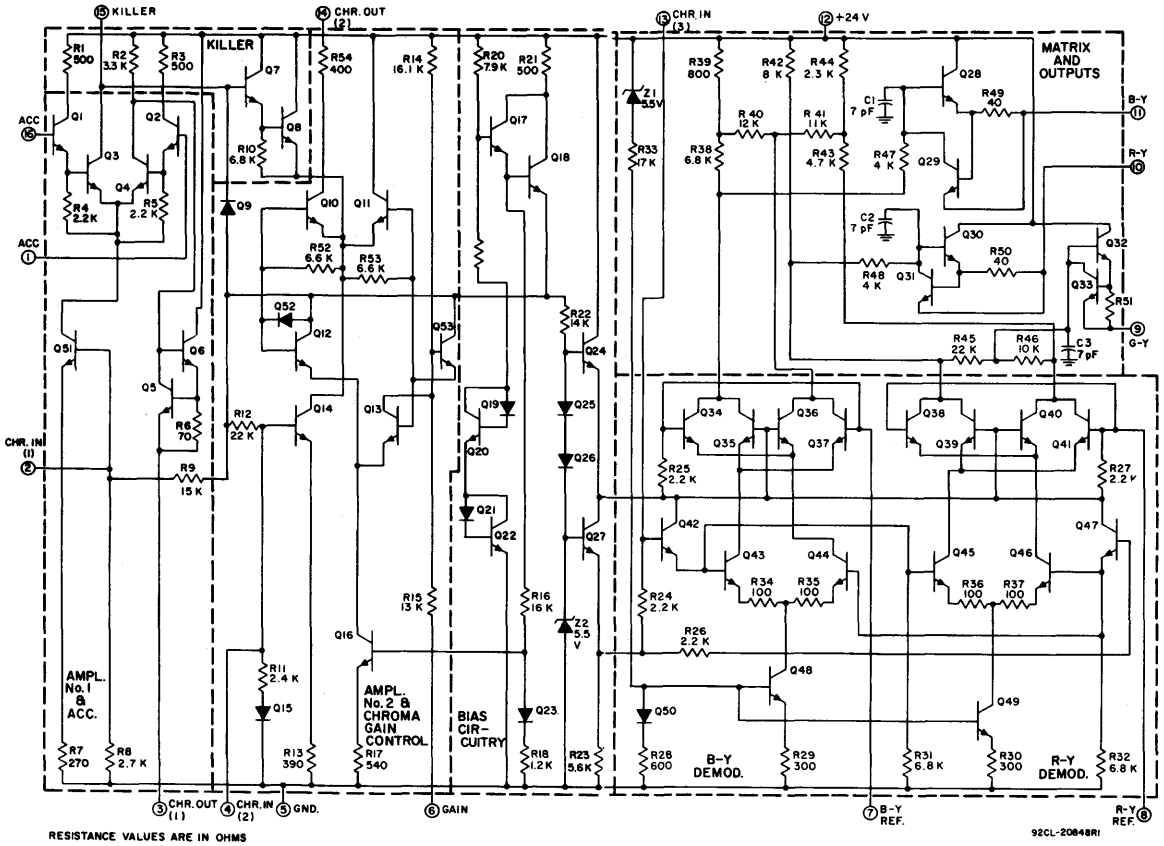


Fig. 4 – Schematic diagram of the CA3121G .

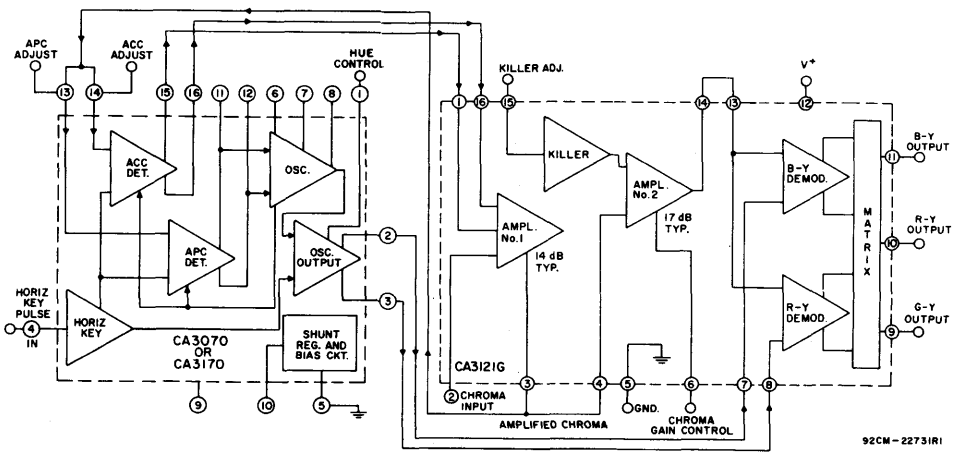


Fig. 5 – Simplified functional diagram of a two-package TV chroma system utilizing the CA3121G and CA3070 or CA3170 .

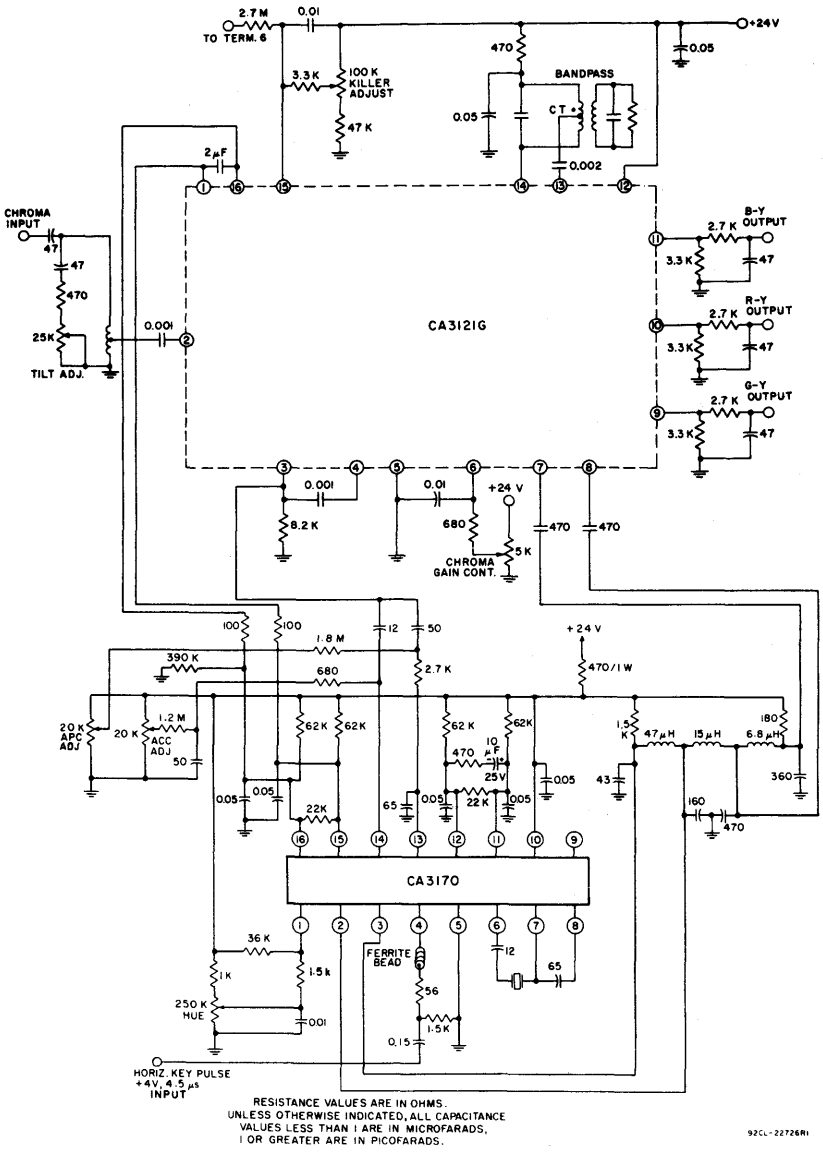


Fig. 6 – Outboard circuitry of a typical two-package chroma system for color-TV receivers utilizing the CA3121G and CA3170.

AM Radio Receiver Subsystem

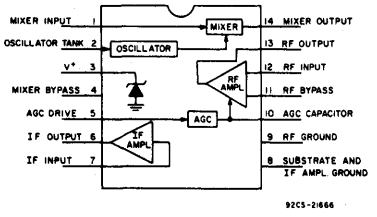
Includes RF Amplifier, IF Amplifier, Mixer, Oscillator, AGC Detector, and Voltage Regulator

The CA3123E* is a monolithic silicon integrated circuit that provides an rf amplifier, if amplifier, mixer, oscillator, AGC detector, and voltage regulator on a single chip. It is intended for use in super-heterodyne AM radio receiver applications particularly in automobiles. The CA3123E is supplied in a 14-lead dual-in-line plastic package and operates over the temperature range of −55° to 125°C.

* Formerly RCA Dev. No. TA6155

Features:

- Low-noise, low-R_b' rf stage in cascode connection — eliminates Miller-Effect regeneration and allows controlled power rise by the choice of external components
- Mixer-oscillator stage with internal feedback — eliminates need for tapped or multi-winding oscillator coils
- Cascode if amplifier with controlled output impedance and negligible Miller Effect — eliminates regeneration and selectivity skewing
- Frequency-counter AGC circuit — allows control of AGC response by selection of the coupling capacitor
- Integral regulation with built-in surge protection
- Separately accessible amplifiers



Terminal assignment diagram.

MAXIMUM RATINGS, *Absolute-Maximum Values:*

DC SUPPLY VOLTAGE:
- At Terminal No. 3 (V^+) ..... 9 V
- At Terminal No. 6 (IF Output) ..... 40 V
- At Terminal No. 13 (RF Output) ..... 20 V
- At Terminal No. 14 (Mixer Output) ... 20 V

DC CURRENT:
- Into Terminal No. 3 (V^+) ..... 35 mA

DEVICE DISSIPATION:
- Up to T_A = 55°C ..... 750 mW
- Above T_A = 55°C ..... derate linearly 6.67 mW/°C

AMBIENT TEMPERATURE RANGE:
- Operating ..... −55 to +125°C
- Storage ..... −65 to +150°C

LEAD TEMPERATURE (During Soldering):
- At distance 1/16" ± 1/3" (1.59 mm ± 0.79 mm) from case for 10 s max. ..... 265°C

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTIC	SYMBOL	TEST CONDITIONS	LIMITS Min.	LIMITS Typ.	LIMITS Max.	UNITS
Static Characteristics In Circuit of Fig. 3						
DC Voltage:						
At Terminals 1, 4	V_1,V_4			4.7		V
At Terminals 2, 3, 14	V_2,V_3,V_{14}			6.8		V
At Terminal 5	V_5			0.25		V
At Terminal 6	V_6			12		V
At Terminal 7	V_7			0.76		V
At Terminals 8, 9	V_8,V_9			0		V
At Terminals 10, 11	V_{10},V_{11}			0.71		V
At Terminal 12	V_{12}			0.71		V
At Terminal 13	V_{13}			4.0		V
DC Current:						
Into Terminals 1, 4, 5, 7, 8, 9, 10, 11, 12	I_1,I_4,I_5,I_7, I_8,I_9,I_{10},I_{11},I_{12}			0		mA
Into Terminal 2	I_2			1.2		mA
Into Terminal 3	I_3			15		mA
Into Terminal 6	I_6			4.3		mA
Into Terminal 13	I_{13}			4.5		mA
Into Terminal 14	I_{14}			0.170		mA
Performance Characteristics In Circuit of Fig. 3						
Sensitivity		Input Signal to Dummy Antenna at f_{IN}=1 MHz, 30% AM Modulation at f_{MOD}=400 Hz, for 11 mV output at V_O	–	2.3	5	μV
Signal-to-Noise Ratio	S/N	Ratio of Output at V_O with Modulation ON and then OFF, Input Signal=100 μV, 30% AM Modulation at f_{MOD}=400 Hz	34	43	–	dB
Overload Distortion		Input Signal set at 1 MHz, 90% AM Modulation, Distortion at V_O must be ≤ 10%	160000	400000	–	μV

Dynamic Characteristics For Indicated Stages In Circuit of Fig. 3

Stage	Parallel Capacitance Input pF	Parallel Capacitance Output pF	Parallel Resistance Input Ω	Parallel Resistance Output Ω	Transconductance μmhos
RF Amplifier	80	6	750	2 x 10^6 min.	140000
IF Amplifier	35	3.5	950	10^4	80000
Mixer	6	2	2000	2 x 10^6 min.	2500 (Mixer) 3000 (Amplifier)

TYPICAL CHARACTERISTICS

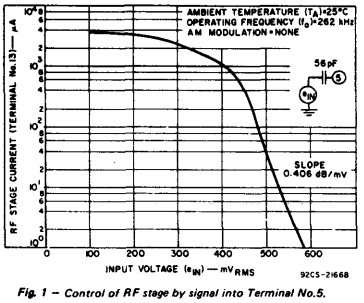


Fig. 1 – Control of RF stage by signal into Terminal No.5.

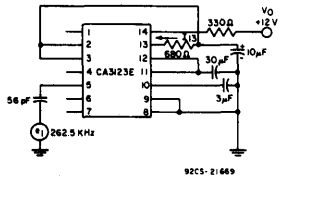
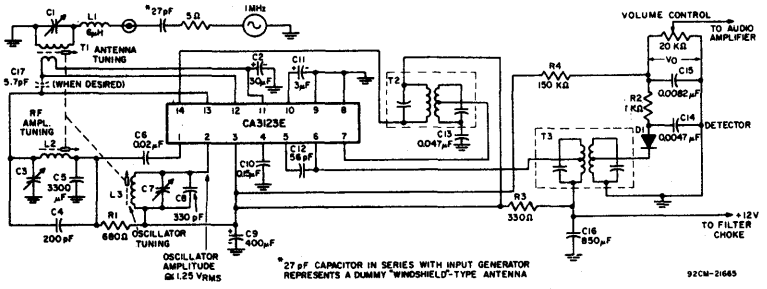


Fig. 2–Test circuit for Fig. 1.

CA3123E



Transformer	Symbol	Frequency	Inductance μh (≈)	Capacitance pF (≈)	Q (≈)	Total Turns To Tap Turns Ratio	Coupling
First IF: Primary	T_2	262 kHz	2840	130	60	none	critical ≈0.017 ≈1/Q
Secondary			2840	130	60	or 30:1 31:1	
Second IF: Primary	T_3	262 kHz	2840	130	60	8.5:1	– critical ≈0.017 ≈1/Q
Secondary			2840	130	60	8.5:1	
Antenna: Primary	T_1	1 MHz	195	(C_1)–130	65		
Secondary		Adjusted to an impedance of 75 Ω with primary resonant at 1 MHz. Coupling should be as tight as practical. Wire should be wound around end of coil away from tuning core.					
Coils	L_1	7.9 MHz	6		50		
	L_2	1 MHz	55		50		
	L_3	1.262 MHz	41		40		

Fig. 3– Schematic diagram of AM radio receiver using CA3123E.

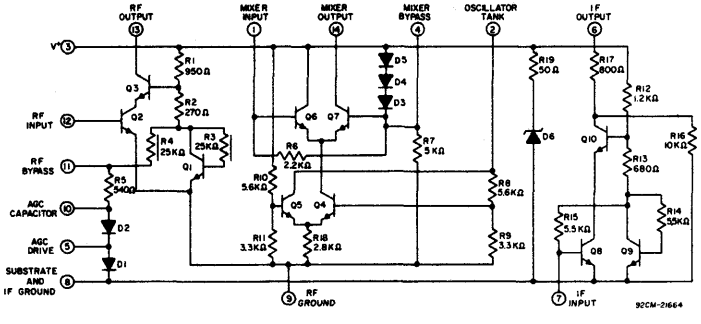
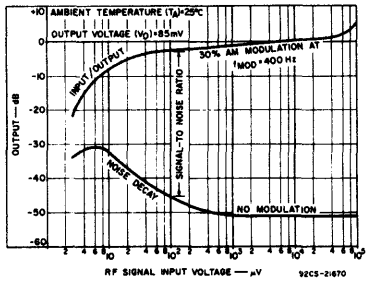


Fig. 4 – Schematic diagram of CA3123E.

PERFORMANCE CHARACTERISTICS IN CIRCUIT OF FIG. 3



Signal-to-noise performance.

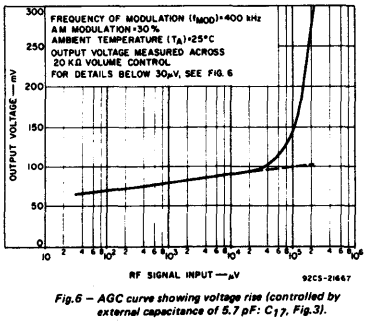


Fig.6 – AGC curve showing voltage rise (controlled by external capacitance of 5.7 pF: C_{17}, Fig.3).

Change in slope in the vicinity of 40000 μV signal input voltage is the result of the use of C_{17} (5.7 pF) in Fig.3. The dotted curve indicates expected performance if C_{17} = 0.

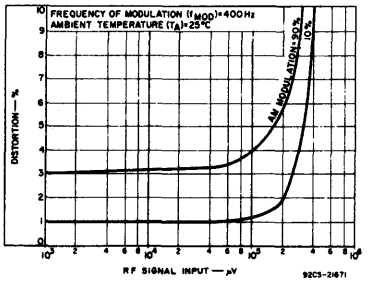


Fig.7 – Overload response.

Television Chroma Demodulator

RCA-CA3125E is a monolithic silicon integrated-circuit chroma demodulator having three separate demodulators with independent phase control. It is designed to function compatibly with the CA1398E IC Chroma Processor as well as other commercially available Chroma Processors in R-G-B Systems of color-TV receivers. The CA3125E is supplied in a 14-lead dual-in-line plastic package.

MAXIMUM RATINGS, *Absolute-Maximum Values at* $T_A = 25°C$

SUPPLY VOLTAGE	25 V
SUPPLY CURRENT	20 mA
AMBIENT-TEMPERATURE RANGE:	
Operating	−40°C to +85°C
Storage	−65°C to +150°C
LEAD TEMPERATURE (DURING SOLDERING):	
At distance 1/16" ± 1/32" (1.59 ± 0.79 mm) from case for 10 s max.	265°C

Features:

- **Luminance input**
- **Blanking control input**
- **Three separate demodulators with independent phase control**
- **Low output offset voltage** ..... **0.4 V**

TYPICAL STATIC CHARACTERISTICS AT $T_A = 25°C$, $V^+ = +20$ VOLTS

SUPPLY CURRENT	9.6 mA
BRIGHTNESS CONTROL VOLTAGE:	
Measured with 8 volts at Terminals 11, 12, and 13	1.4 V
MAX. OUTPUT DIFFERENCE VOLTAGE:	
Measured between any two of Terminals 11, 12, and 13	±0.4 V
MAXIMUM DC DETECTOR UNBALANCE VOLTAGE:	
DC voltage shift on Terminals 11, 12, and 13 when Terminals 1, 2, and 3 are alternately biased 0.5 volt positive, then negative with reference to Terminal 14	+150 mV

TYPICAL DYNAMIC CHARACTERISTICS AT $T_A = 25°C$, $V^+ = +20$ volts

BLUE CHROMA GAIN:	
Peak-to-peak voltage at Terminal 11 with 1.0 volt peak-to-peak applied differentially between Terminals 6 and 7, and with a subcarrier injection voltage of 1 volt peak-to-peak	7.36 Vp-p
RED GAIN RATIO:	
$\frac{\text{Peak-to-peak voltage at Terminal 13}}{\text{Peak-to-peak voltage at Terminal 11}} \times 100$	100%
GREEN GAIN RATIO:	
$\frac{\text{Peak-to-peak voltage at Terminal 12}}{\text{Peak-to-peak voltage at Terminal 11}} \times 100$	30%
LUMINANCE GAIN:	
Peak-to-peak voltage measured at Terminals 11, 12, and 13, with a peak-to-peak voltage of 0.1 volt applied to Terminals 6 and 7 (common mode), and with no subcarrier injection	0.7 Vp-p

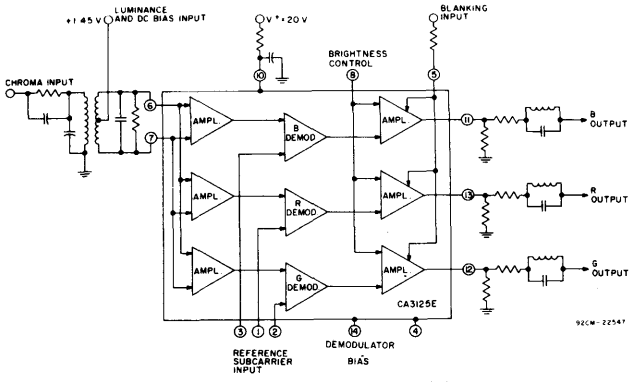


Fig. 1 – Functional block diagram of the CA3125E.

CA3126Q

TV Chroma Processor

RCA-CA3126Q is a monolithic silicon integrated circuit designed for chroma processing applications in color TV receivers. It is compatible with the CA3067 chroma demodulator as well as other chroma demodulators.

MAXIMUM RATINGS, *Absolute-Maximum Values at* $T_A = 25^oC$

DEVICE DISSIPATION:
- Up to $T_A = 55^oC$ ..... 750 mW
- Above $T_A = 55^oC$ ..... derate linearly 7.9 mW/°C

DC SUPPLY VOLTAGE (Across Terms. 5 and 12)[A] ..... 13.2 V

DC CURRENT:
- Into Term. 12 ..... 38 mA
- Into Term. 14 ..... 20 mA

DC VOLTAGE (Terminal 9):
- Negative Rating ..... −5 V
- Positive Rating ..... 3 V

AMBIENT TEMPERATURE RANGE:
- Operating ..... −40 to +85°C
- Storage ..... −65 to +150°C

LEAD TEMPERATURE (During Soldering):
- At a distance not less than 1/32 in. (0.79 mm) from case for 10 seconds max. ..... +265°C

[A]This rating does not apply when using the internal zener reference in conjunction with an external pass transistor.

The CA3126Q is supplied in the 16-lead quad-in-line plastic package.

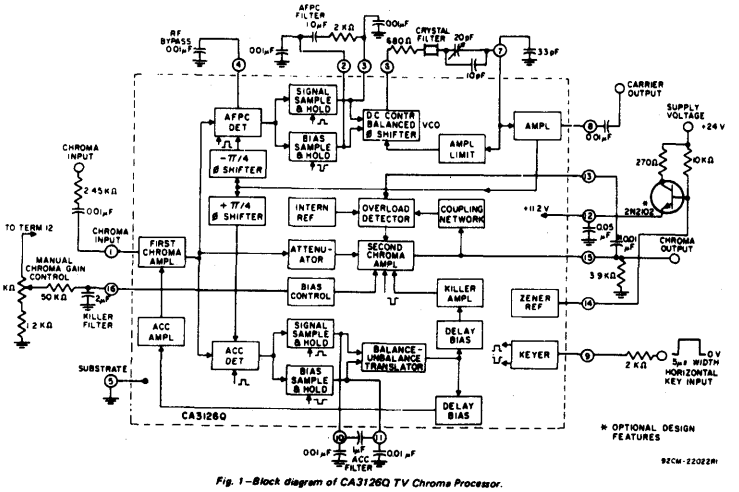


Fig. 1–Block diagram of CA3126Q TV Chroma Processor.

ELECTRICAL CHARACTERISTICS

Test Conditions: $T_A = 25^oC$, chroma control at maximum position for all characteristics tests except for chroma output test. For this test, control should be set at minimum position. Electrical characteristics referenced to test circuit, Fig.2.

CHARACTERISTIC	TERMINAL, MEASUREMENT, AND SYMBOL	SWITCH POS.		CHROMA INPUT	LIMITS			UNITS
		S1	S2	TP1	Min.	Typ.	Max.	
Static Characteristics								
Voltage Regulator	V_{12}	2	2	0	10.1	11.2	12.1	V
Supply Current	I_{12}	2	2	0	16	25	38	mA
Dynamic Characteristics (See Note 1)								
Pull-in Range*	V_8	*	2	0.5 $V_{p\text{-}p}$	±250	–	–	Hz
Oscillator Output	V_8	2	2	0	0.6	1.0	–	$V_{p\text{-}p}$
100% Chroma Output	V_{15}	1	2	0.5 $V_{p\text{-}p}$	1.4	2.7	–	$V_{p\text{-}p}$
Overload Detector	V_{15}	1	1	0.5 $V_{p\text{-}p}$	0.4	–	0.7	$V_{p\text{-}p}$
Minimum Chroma Output	V_{15}	1	2	0.5 $V_{p\text{-}p}$	–	–	20	$mV_{p\text{-}p}$
200% Chroma Output	V_{15}	1	2	1 $V_{p\text{-}p}$	70	100	140	% of 100% reading
20% Chroma Output	V_{15}	1	2	0.1 $V_{p\text{-}p}$	40	–	105	
Kill Level	V_{TP1}	1	2	vary	5	–	60	$mV_{p\text{-}p}$

Note 1: Except for pull-in range testing, tune oscillator trimmer capacitor for free-running frequency of 3.579545 MHz ± 10 Hz.
*Set Switch 1 to Position 2, detune oscillator ± 250 Hz, set Switch 1 to Position 1, and check for oscillator pull-in.

Features:

- Phase-locked subcarrier regeneration utilizes sample-and-hold techniques
- Automatic chrominance control (ACC)/killer detector employs sample-and-hold techniques
- Supplementary ACC with an overload detector to prevent oversaturation of the picture tube
- Sinusoidal subcarrier output
- Keyed chroma output
- Emitter-follower buffered outputs for low output impedance
- Linear dc saturation control
- Internal zener-regulated reference potentials
- Only the initial crystal filter tuning is required. . . no killer or ACC adjustments required at any time
- Few external components required
- Compensation for temperature and supply variations
- All terminals protected against short circuits

CIRCUIT DESCRIPTION

The following paragraphs briefly describe the circuit operation of the CA3126Q (shown in Fig. 1). A detailed description of the operation of various portions of the CA3126Q is given in ICAN-6247, "Application of the CA3126Q Chroma-Processing IC Using Sample-and-Hold Techniques".

The chroma input is applied to Terminal 1 through the desired band-shaping network. A 2,450-ohm resistor should be placed in series with Terminal 1 to minimize oscillator pickup in the first chroma amplifier. This amplifier supplies signals to the second chroma amplifier and to the ACC and AFPC detectors. The first chroma amplifier is gain-controlled by the ACC amplifier.

A horizontal keying pulse is applied to Terminal 9. This pulse must be present to ensure proper operation of the oscillator circuit. The subcarrier burst is sampled during the keying interval in the AFPC detector. The error voltage, produced at Terminal 2 and proportional to the burst phase, is compared to the quiescent bias voltage at Terminal 3 by the sample-and-hold circuitry. This "compared" voltage controls the phase-shifting network in the phase-locked loop. The operation of the AFPC loop is independent of any external adjustments or voltages except for an initial capacitor adjustment to set the free-running frequency.

The regenerated oscillator signal at Terminal 8 is applied internally to the AFPC and ACC detectors through +45- and −45-degree phase-shifter networks to establish the proper phase relationship for these detectors. The ACC detector, which also samples the burst during the keying interval, produces a correction voltage proportional to the burst amplitude. The correction voltage is compared to the quiescent bias level using sample-and-hold circuitry similar to that used in the AFPC portion of the circuit. The "compared" voltage is applied internally to the ACC amplifier and killer amplifier. Because the amplifier gains and killer threshold are determined by the ratios of the internal resistors, these functions are independent of external voltages or controls.

The attenuated chroma signal is fed to the second chroma amplifier, where the burst is removed by keyer action. The killer amplifier, the chroma gain control, and the overload detector control the action of the second chroma amplifier, whose gain is proportional to the dc voltage at Terminal 16. The overload detector (Terminal 13) receives a sample of the chroma output (Terminal 15) and detects the peak of the signal. The detected voltage is stored in an external capacitor connected to Terminal 16. This stored voltage on Terminal 16 affects the gain of the second chroma in the same manner as the chroma gain control.

APPLICATIONS INFORMATION

General Considerations

The block diagram shown in Fig. 1 is typical of the type of circuit used in the practical application of the CA3126Q. Several items are critical for proper operation of the circuit.

1. A series resistor of approximately 2,450 ohms (or high source impedance) must be used at the chroma input, Terminal 1. This high impedance minimizes pickup of unbalanced currents, particularly of the subcarrier oscillator signal.

2. When the overload detector is used, a large resistor (nominally 47,000 ohms) must be placed in series with Terminal 16 to set the required RC time constant. The same RC network series serves to set the killer time constant.
3. The setting of the free-running oscillator frequency requires the presence of the keying pulse. The free-running frequency will be erroneous if Terminal 1 is dc shorted during the setting operation because of the dc offset voltage introduced to the AFPC detector.
4. Care must be taken in PC board designs to provide reasonable isolation between the oscillator portion of the circuit (Terminals 6, 7, and 8) and the chroma input (Terminal 1).

Overload Detector

The overload detector accomplishes two purposes:

1. It prevents oversaturation due to low burst-to-chroma ratios.
2. It prevents overload conditions due to noise.

Both of these conditions are discussed in more detail in ICAN-6247. The extent to which the overload detector is used depends upon the individual receiver design goals. If greater than 0.5-volt peak-to-peak output is desired, the chroma output at Terminal 15 can be tapped to yield any desired degree of overload detector action.

Chroma Gain Control

The chroma gain control operates by varying the base bias on current source transistor Q25. To ensure proper temperature tracking of the chroma gain control, it is essential that the control be operated from a supply source derived from the reference voltage at Terminal 12. Because the control operates from a current source, chroma gain is much more predictable and far less temperature sensitive than controls that steer current by means of a differential amplifier. The typical chroma gain characteristic for the CA3126Q is shown in Fig. 3.

Subcarrier Regenerator Oscillator

The oscillator filter consists of a 3.579545-MHz crystal, a 680-ohm resistor, and a 10-pF capacitor connected in series across Terminals 6 and 7. A 33-pF capacitor, shunt connected from Terminal 7 to ground, rolls off higher-order harmonics, thereby preventing oscillation at the crystal third-harmonic frequency. A curve of the typical static phase error as a function of the free-running oscillator frequency is shown in Fig. 4. It should be noted that the slope of the curve determines the dc gain of the phase-locked loop, i.e., 40 Hz per degree.

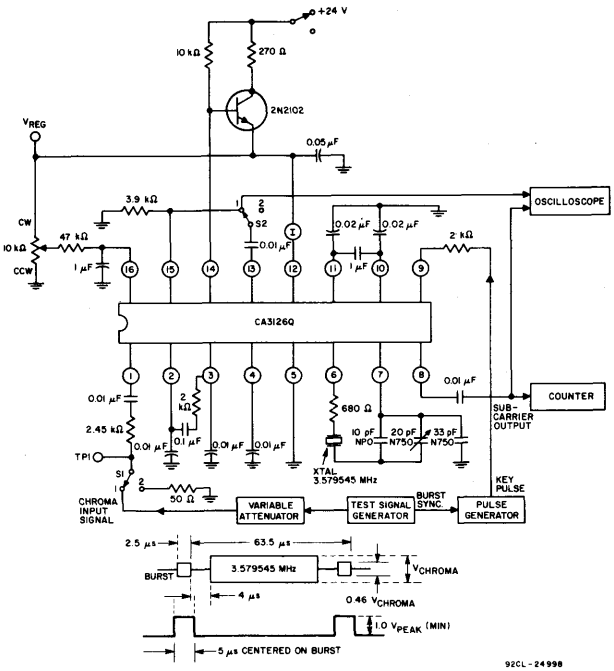


Fig. 2 – Test circuit for CA3126Q.

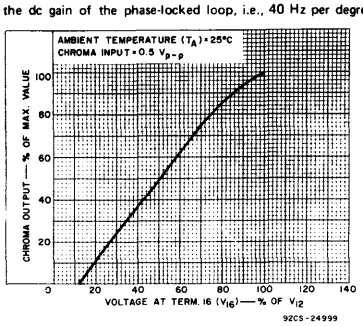


Fig. 3 – Chroma gain control.

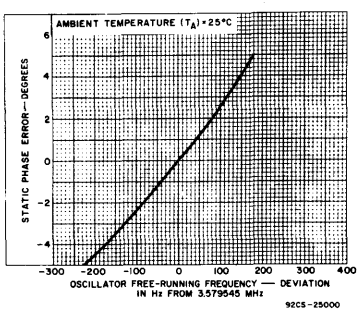


Fig. 4 – Static phase error.

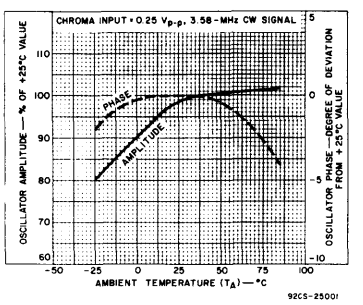


Fig. 5 – Amplitude and phase variations of oscillator output vs. temperature.

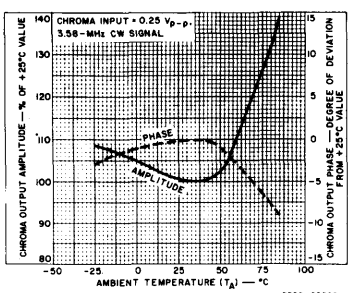


Fig. 6 – Amplitude and phase variations of chroma output vs. temperature.

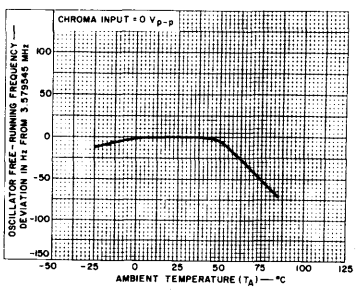


Fig. 7 – Variation of oscillator free-running frequency vs. temperature.

Thermal Considerations

The circuit of the CA3126Q is thermally compensated to achieve the optimal operating characteristics over the normal operating temperature range of TV receivers. Figs. 5 and 6 show the oscillator- and chroma-output amplitudes and phases as a function of temperature (Terminals 8 and 15), respectively. Both the oscillator- and chroma-output amplitudes and phases are measured relative to the chroma-input phase. The performance of the oscillator free-running frequency as a function of temperature is shown in Fig. 7. All the temperature plots are characteristic of the test circuit with the indicated component types and values given in Fig. 2.

CA3128Q

Preliminary Data

TV Chroma Processor for PAL Systems

The RCA-CA3128Q is a monolithic silicon integrated circuit designed primarily for PAL chroma processing applications in color TV receivers. For a circuit description of the CA3128Q and an explanation of this device in PAL systems, refer to "A New Chroma Processing IC Using Sample -and-Hold Techniques" by L. A. Harwood (ST6144).

Features:

- Phase-locked subcarrier regeneration utilizes sample-and-hold techniques in the automatic frequency phase control (AFPC) servo loop
- Automatic chrominance control (ACC)/ killer detector employs sample-and-hold techniques
- Supplementary ACC with an overload detector to prevent oversaturation of the picture tube
- Sinusoidal subcarrier output
- Keyed chroma output
- Emitter-follower buffered outputs for low output impedance
- Linear dc saturation control
- PAL identification output
- Only the initial crystal filter tuning is required . . . no killer and ACC adjustments required at any time
- Few external components required
- Compensation for temperature and supply variations
- All terminals protected against short circuits

MAXIMUM RATINGS, *Absolute-Maximum Values at* $T_A = 25°C$

DC SUPPLY VOLTAGE (Between Terms. 12 and 5)	13.2	V
DC VOLTAGE (Term. 9):		
Positive Value	+3	V
Negative Value	−5	V
DEVICE DISSIPATION:		
Up to $T_A = 55°C$	750	mW
Above $T_A = 55°C$	derate linearly at 7.9 mW/°C	
AMBIENT TEMPERATURE RANGE:		
Operating	−40 to +85	°C
Storage	−65 to +150	°C
LEAD TEMPERATURE (DURING SOLDERING):		
At a distance not less than 1/32" (0.79 mm) from case for 10 seconds max.	+265	°C

TYPICAL STATIC CHARACTERISTICS at $T_A = 25°C$:

DC Supply Current (I_{12}) with $V_{12} = 11.2$ V dc	25	mA

TYPICAL DYNAMIC CHARACTERISTICS at $T_A = 25°C$ with a Burst-to-Chroma Ratio of 46.5%:

100% Chroma Output Voltage at $V_{I(p-p)} = 0.5$ V	3.5	Vp-p
Oscillator-Level Output Voltage	1	Vp-p
Killer Threshold Input Voltage	0.018	Vp-p
Pull-in Frequency	500	Hz
PAL Identification Output Voltage	1	Vp-p

The CA3128Q is supplied in the 16-lead quad-in-line plastic package.

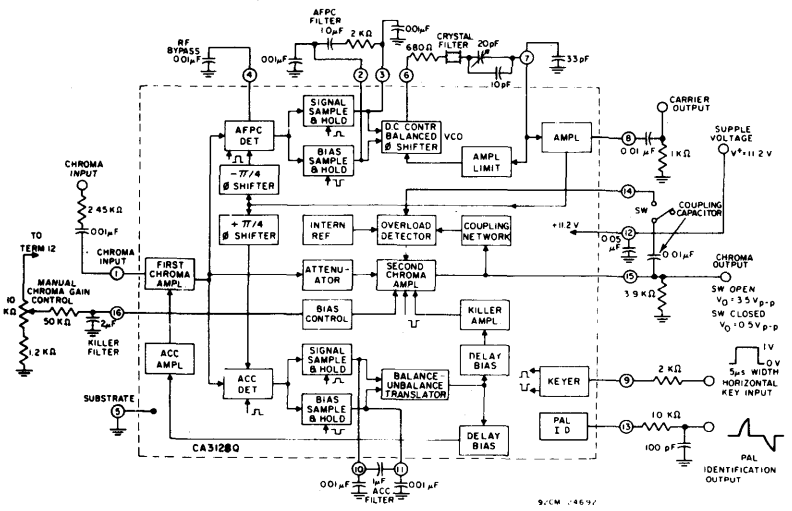


Fig. 1 – Block diagram of CA3128Q TV Chroma Processor.

5-Watt Audio Amplifiers

With Integral Heat Sink

RCA-CA3131EM and CA3132EM are audio amplifiers with integral preamplifier stages on single integrated-circuit monolithic chips.

Utilizing a uniquely designed package with an integral heat sink, these devices can provide a power-output signal in excess of five watts at an ambient temperature of 25°C.

The CA3131EM employs an internal feedback network that sets the over-all gain of the amplifier to typically 48 dB.

The CA3132EM omits the internal feedback network. This arrangement offers the circuit designer a wide latitude in the choice of an external feedback network more suitable to a specific application.

Both types are encapsulated in a 16-lead dual-in-line plastic package with 4 center leads removed.

The CA3131EM and CA3132EM are electrically equivalent to and pin compatible with types SN76013 and SN76023, respectively.

MAXIMUM RATINGS, *Absolute-Maximum Values:*

SUPPLY VOLTAGE, V^+	28 V
CONTINUOUS OUTPUT POWER, P_O (with R_L = 8 Ω and V^+ = 24 V)	8 W RMS
MINIMUM RECOMMENDED LOAD IMPEDANCE, R_L	8 Ω
AMBIENT OPERATING TEMPERATURE, T_A (at 6 W RMS Output Power)	70 °C
STORAGE TEMPERATURE RANGE	−55 to +150 °C

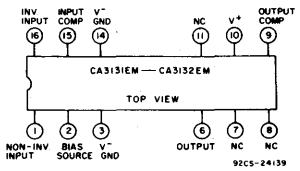


Fig. 1—Terminal assignment of the CA3131EM and CA3132EM.

Determining External Component Values (Refer to Figs. 2 & 3)

The dc quiescent output voltage is set by the voltage at Terminal 1. This voltage, in turn, is set by the internal voltage at Terminal 2 less I_I (input current, fixed by $R_A + R_B$, for Q4). The voltage at Terminal 2 is set slightly above half the supply voltage to allow for the voltage drop across $R_A + R_B$. Filter R_BC3 attenuates any ac ripple injected from the supply line and prevents positive feedback to Terminal 1. The rejection of supply voltage is a direct function of the filter attenuation.

The input impedance of the audio amplifiers is a function of the closed-loop gain and the magnitude of the Q8 current. In practice the input impedance is well above 1 megohm. The input signal, applied through C2, sees an impedance equivalent to the resistance of R_A connected in parallel with the amplifier input impedance. Hence, the value of R_A in most cases is dominant in establishing the input signal impedance.

The value of C1 depends on the regulation of the power supply. It is possible for the amplifier to work with a value of C1 as low as 0.1 μF to attenuate high-frequency signals in the supply line. Ideally, C1 should be placed as near Terminal 10 as possible. An electrolytic capacitor should be used for C1 if the power supply is poorly regulated to avoid ripple at the output.

Capacitor C6 at Terminal 15 provides over-all compensation. If a 1000-pF capacitor is used for C6, then the first breakpoint for a 46-dB closed-loop gain occurs at 200 kHz. Higher capacitance values will cause the constant current from Q10 to charge C6 on the positive voltage swing and thus limit the slew rate at high-signal levels. Because p-n-p transistor Q19 has a lower gain-bandwidth product (f_T) than the n-p-n transistors, C7 is connected to Terminal 9 to compensate for gain losses occurring in the negative voltage swings.

The use of the filter networks C8 and R_D at the output Terminal 6 is a standard requirement for class B audio outputs driving reactive speaker loads. Capacitor C8 compensates for the speaker inductance and R_D limits the current surges through C8.

The value of the coupling capacitor C9 to the load determines the low-frequency response of the amplifier.

Closed-Loop Gain

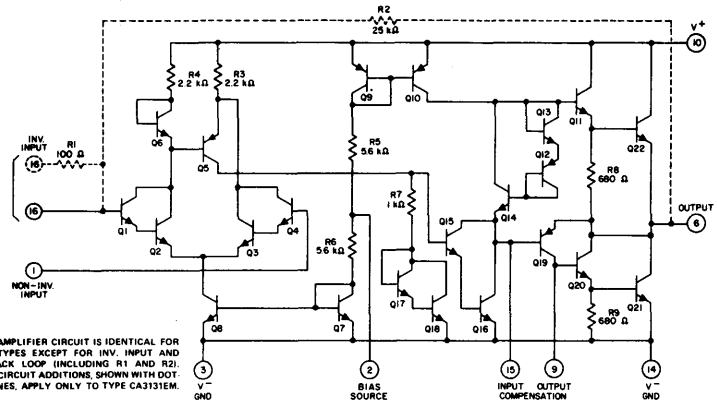
The closed-loop gain for either type is set by the ratio (R1 + R2)/R1. These resistors are included in the CA3131EM circuit and are external when used with the CA3132EM. In either type, the low-frequency value (−3 dB point) is reached when the impedance of C5 equals the value of R1.

Features:

- **Power Output: 4 W min., 5 W typ.**
- **Complete amplifier including: preamplifier stages, power-output amplifier, and integral heat sink**
- **High power-supply rejection ratio**
- **Operating voltage: V^+ = 24 V typ.**
- **Available with internal feedback (CA3131EM) or without feedback (CA3132EM)**

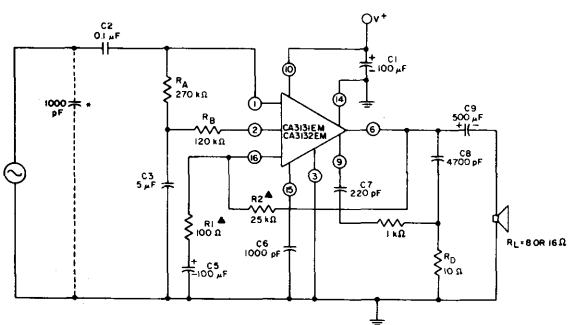
ELECTRICAL CHARACTERISTICS at T_A = 25°C, V^+ = 24 V

Characteristic	Symbol	Conditions	Values Min.	Values Typ.	Unit
Input Impedance	Z_I		200k	–	Ω
Power Output	P_O	At clipping onset			
		R_L = 8 Ω	4	–	W
		R_L = 16 Ω	3	–	W
Closed-Loop Gain — CA3131EM	A	f = 1 kHz	46	48	dB
Supply Current	I^+	Zero signal	–	10	mA
Total Harmonic Distortion	THD	P_O = 50 mW–4 W, R_L = 8 Ω	–	1	%
		P_O = 50 mW–3 W, R_L = 15 Ω	–	1	%
Noise Voltage	V_n	f=20 Hz–20 kHz	–	1.5	mV RMS



NOTE: AMPLIFIER CIRCUIT IS IDENTICAL FOR BOTH TYPES EXCEPT FOR INV. INPUT AND FEEDBACK LOOP (INCLUDING R1 AND R2). THESE CIRCUIT ADDITIONS, SHOWN WITH DOTTED LINES, APPLY ONLY TO TYPE CA3131EM.

Fig. 2—Schematic diagram of types CA3131EM and CA3132EM.



* A 1000-pF capacitor is required if input has an open circuit.

▲ External resistors R1 and R2 are used only with the CA3132EM. When testing the CA3131EM, omit R1 and R2 and connect the (+) termination of C5 to Terminal 16.

Fig. 3—Test circuit for types CA3131EM and CA3132EM.

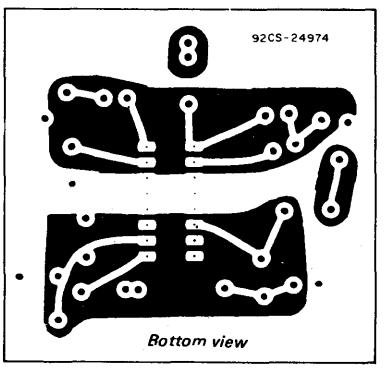


Fig. 4—Printed-circuit board (actual size) containing the test circuit, shown in Fig. 3, for the CA3131EM.

CA3134G, CA3134GM, CA3134GQM

TV Sound IF and Audio Output Subsystems

The RCA-CA3134 combines the sound if and audio output subsystems on a single monolithic integrated circuit to provide a television sound system for color or black and white applications. Each device includes a multistage if amplifier-limiter, and fm detector, and an audio power amplifier that is designed to drive an 8-, 16-, or 32-ohm speaker.

The CA3134 is supplied in the hermetic Gold-CHIP, which is of the sealed-junction type designed to provide protection against the deteriorating effects of humidity and other surface contaminants without the need for a hermetic package enclosure. This hermetic chip is encapsulated in a 16-lead plastic "power stud" dual-in-line package, which has an inherently low junction-to-case (stud) thermal resistance. This package lends itself to a wide variety of heat-sink methods, depending on the application requirements.

The CA3134G is supplied in the 16-lead plastic "power stud" dual-in-line package. The CA3134GM and CA3134GQM are similar to the CA3134G except that they incorporate a tin-plated copper-strap heat sink. The CA3134GQM also has quad-formed leads.

MAXIMUM RATINGS, *Absolute-Maximum Values:*

	CA3134G	CA3134GM, CA3134GQM	
DC SUPPLY VOLTAGE (Between Term. 1, V^+ and Terms.4, audio-output ground and 13, substrate)	33	33	V
INPUT SIGNAL VOLTAGE (Between Terms. 14 and 15)	±3	±3	
DEVICE DISSIPATION:			
With Infinite Heat Sink—			
Up to T_A = 70°C	6.5	–	
Above T_A = 70°C, derate linearly	83.3	–	mW/°C
With no Heat Sink—			
Up to T_A = 25°C	1.4	–	W
Above T_A = 25°C, derate linearly	11.1	–	mW/°C
With Copper-Strap Heat Sink—			
Soldered to PC Board			
Up to T_A = 25°C	–	3.9	W
Above T_A = 25°C, derate linearly	–	31.2	mW/°C
Unsoldered			
Up to T_A = 25°C	–	2.5	W
Above T_A = 25°C, derate linearly	–	20	mW/°C
THERMAL RESISTANCE			
Junction to Stud	12	12	°C/W
AMBIENT TEMPERATURE RANGE:			
Operating	−40 to +85		°C
Storage	−65 to +150		°C
LEAD TEMPERATURE (During Soldering):			
At a distance 1/16 in. ±1/32 in. (1.59 ±0.79 mm) from case for 10 seconds max.	+265		°C

Features:

- **Output power 3W (typ.) at V^+ = 24V, R_L = 16Ω**
- **Power amplifier with current limiting and thermal shutdown**
- **Wide power-supply range: 12V to 33V**
- **Low quiescent current: 30 mA typ.**
- **5-kHz deviation sensitivity: 1W output typ.**
- **3-dB limiting sensitivity: 200 μV typ.**
- **Excellent AM rejection: 50 dB typ.**
- **Differential peak detector—requires one tuned coil**
- **Electronic volume control with improved taper**
- **Optional unattenuated audio output**
- **Optional power-supply ripple by-pass**
- **Hermetic Gold-CHIP**

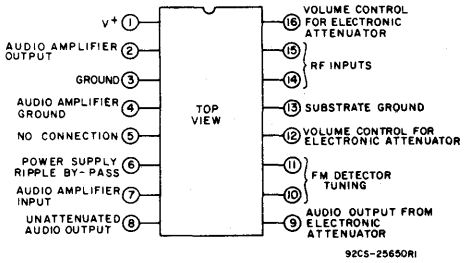


Fig. 1 – Terminal diagram of the CA3134G, CA3134GM, and CA3134GQM.

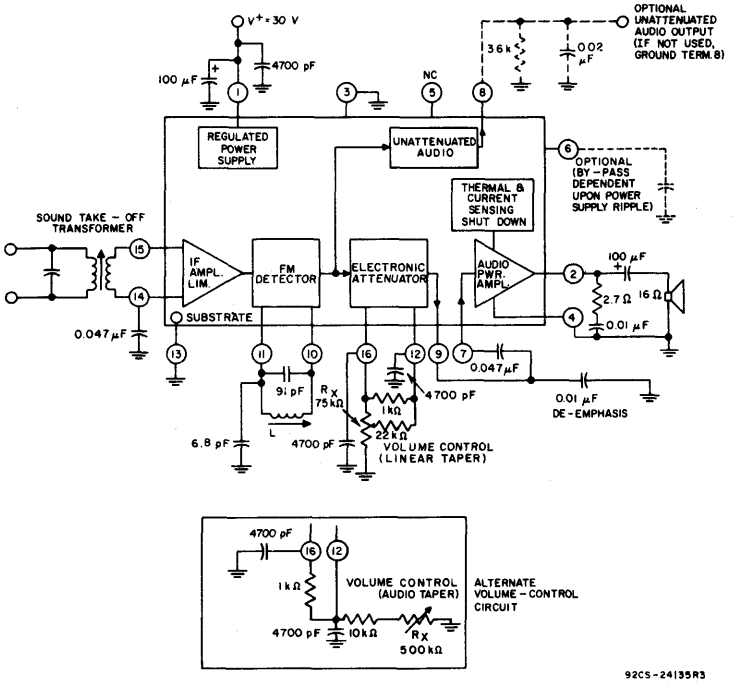


Fig. 2 – Block diagram of the CA3134 in a typical circuit application.

ELECTRICAL CHARACTERISTICS
Test Conditions: T_A = 25°C, V^+ = +30 V (applied to Term. 1), DC Volume Control, R_X = 75 kΩ, R_L = 16 Ω, unless otherwise indicated. Refer to Fig.2.

CHARACTERISTIC	SPECIAL TEST CONDITIONS	LIMITS			UNITS
		Min.	Typ.	Max.	
Static Characteristics					
Current into Term. 1, I_1	P_O = 0	15	30	45	mA
Dynamic Characteristics					
IF AMPLIFIER: Input Limiting Voltage, V_{15}(lim) (at −3 dB point)	f_O = 45 MHz, f_m = 400 Hz, Δf = ±25 kHz	–	200	400	μV
AM Rejection, AMR	f_O = 4.5 MHz, f_m = 400 Hz, Modulation Index = 0.3, V_{15} = 20 mV	40	50	–	dB
Input Resistance, R_I	V_{15} = 35 mV	–	25	–	kΩ
Input Capacitance, C_I	V_{15} = 35 mV	–	3	–	pF
DETECTOR: Recovered af Voltage (Term. 9), V_O(af)	f_O = 4.5 MHz, f_m = 400 Hz, Δf = ±25 kHz, V_{15} = 100 mV	–	700	–	mV
Total Harmonic Distortion, (THD)		–	0.8	3	%
Output Resistance, R_O	At Term. 9	–	7.5	–	kΩ
ATTENUATOR: Maximum Attenuation	R_X = 0	–	10	15	mV
UNATTENUATED AUDIO: Recovered af Voltage (Term. 8), V_O(af)	f_O = 4.5 MHz, f_m = 400 Hz, Δf = ±25 kHz, V_{15} = 100 mV	–	600	–	mV
Total Harmonic Distortion (THD)		–	0.8	–	%
AUDIO POWER AMPLIFIER: Voltage Gain, A(af)	f = 1 kHz	–	35	–	dB
System Total Harmonic Distortion THD (System)	P_O = 1 W (I_T = 140 mA typ.)	–	1.5	–	%
	P_O = 2 W (I_T = 180 mA typ.)	–	1.6	3	%
Power Output, P_O	THD (System) = 10% (I_T = 210 mA typ.)	–	5*	–	W
Input Resistance, (R_I(af)	f = 1 kHz	–	100	–	kΩ

* With suitable heat sink for the CA3134G.

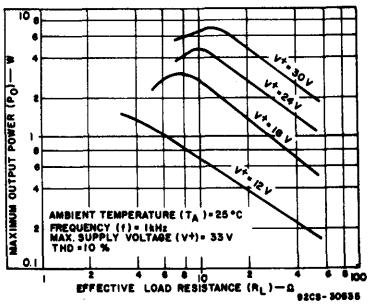


Fig. 3 – Maximum output power as a function of effective load resistance.

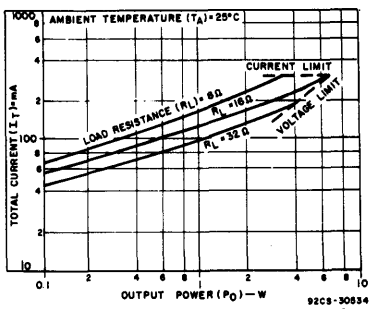


Fig. 4 – Total supply current as a function of output power.

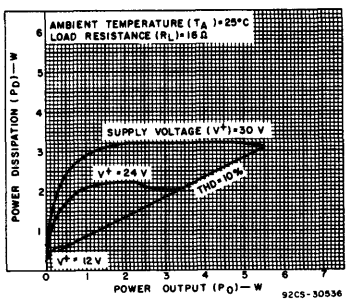


Fig. 5 – Power dissipation as a function of output power.

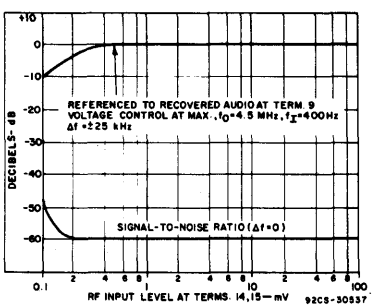


Fig. 6 – Recovered audio, and signal-to-noise ratio as a function of rf input level.

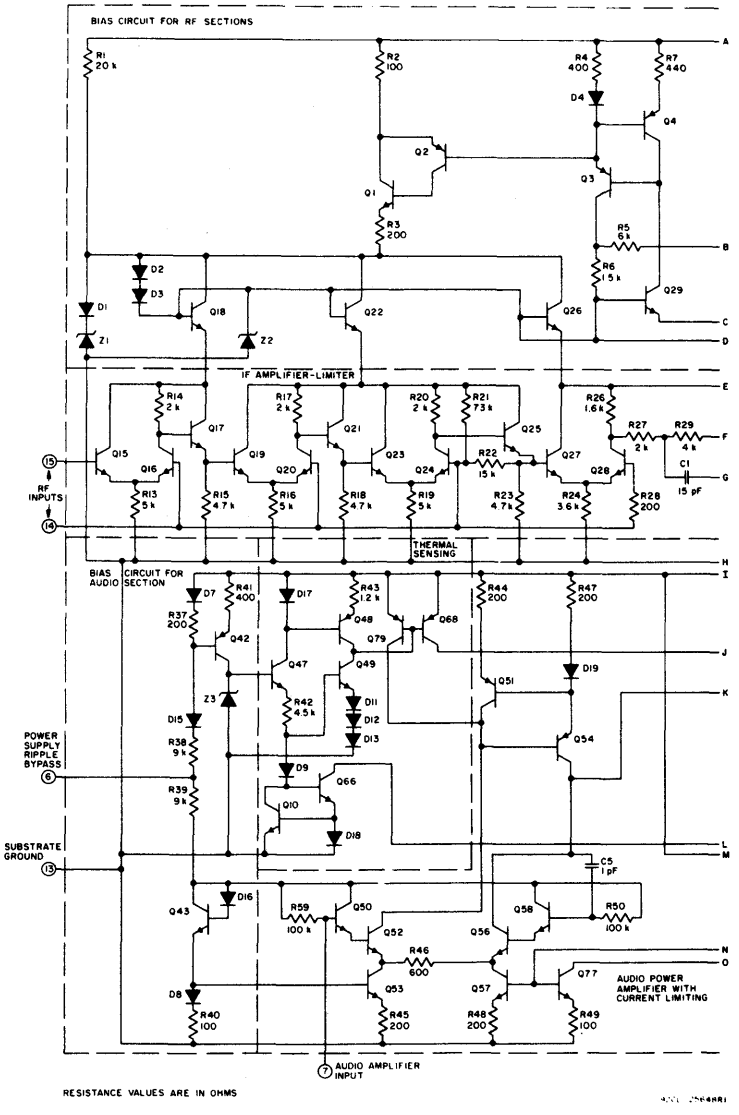


Fig. 7 – Schematic diagram of the CA3134

OPERATING CONSIDERATIONS

The CA3134GM, which incorporates the tin-plated copper-strap heat sink, was used to obtain the power measurement values given in this data bulletin.

A heat sink, similar to the type attached to the CA3134GM, may also be used with the stud-type CA3134G. A recommended procedure for attaching the heat sink to the CA3134G is described as follows:

Apply a non-conductive epoxy (Uniset structural adhesive or equivalent) to the top side of the plastic package. Then apply a conductive epoxy (DuPont 5504A or equivalent) in the hole of the heat sink and around the stud projecting from the plastic package. To assure good thermal conduction, use sufficient conductive epoxy to allow the excess to be forced through the hole when the heat sink is fitted over the stud. Stress applied to the stud should be limited to less than 0.34 newton-meters (3 in-lb) of torque 66.7 newtons (15 lb) of tension, and 444.8 newtons (100 lb) of compression.

The assembly when soldered to a 7.6 cm (3 in.) x 10.2 cm (4 in.) PC board has an overall thermal resistance ($\theta_{Junction\text{-}to\text{-}Ambient}$) of 32°C/W.

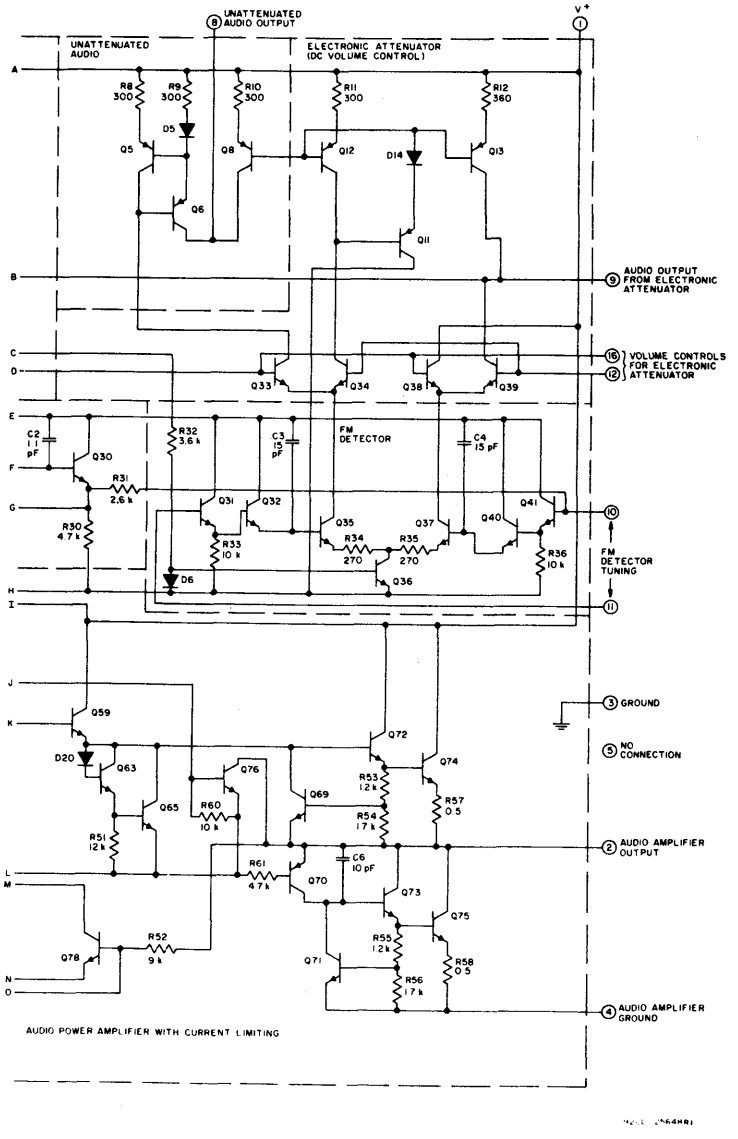


Fig.7 – Schematic diagram of the CA3134 .

CA3135G

TV Luminance Processor

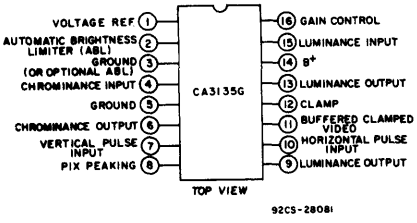
"G" Suffix Type – Hermetic Gold-CHIP in Dual-In-Line Plastic Package

The RCA-CA3135G monolithic silicon integrated circuit operates from a 12-V supply and is used as a low-level luminance processor in TV applications. It performs the function of video and chroma amplification and allows the gain of both channels to be adjusted with a single control voltage. The dc level of "black" is maintained by clamping the level of the "back porch" (black-level reference) of the blanking interval. This clamping feature provides for 100% dc restoration. Vertical blanking is applied to the luminance as well as to the chrominance channel so that vertical interval test signals (VITS) interference is eliminated. Automatic brightness limiting (ABL) is accomplished by gain reduction in the luminance and chrominance channels while maintaining black level.

The CA3135G is supplied in the hermetic Gold-CHIP 16-lead dual-in-line plastic package (G suffix). The chips used in the hermetic Gold- Chip plastic packages are of the sealed-junction type designed to provide protection against the deteriorating effects of humidity and other surface contaminants without the need for a hermetic package enclosure.

System Features:

- **Single gain control for luminance and chrominance channels**
- **100% dc restoration with "back porch" clamp**
- **Vertical blanking of both luminance and chrominance channels**
- **Automatic brightness limiting**
- **Operates from a 12-V supply**
- **Hermetic Gold-CHIP construction**
- **Gold-CHIP metallization**
- **Silicon-nitride passivated**
- **Platinum-silicide ohmic contacts**



Terminal Assignment

MAXIMUM RATINGS, *Absolute-Maximum Values:*

DC SUPPLY VOLTAGE:
- At terminal 9 . . . . . 28 V
- At terminal 14 . . . . . 15 V

DC SUPPLY CURRENT:
- At terminal 9 . . . . . 30 mA
- At terminal 14 . . . . . 50 mA

DEVICE DISSIPATION:
- Up to $T_A = 55^\circ C$ . . . . . 750 mW
- Above $T_A = 55^\circ C$ . . . . . Derate linearly at 7.9 mW/°C

AMBIENT TEMPERATURE RANGE:
- Operating . . . . . −40 to +85°C
- Storage . . . . . −65 to +150°C

LEAD TEMPERATURE (During Soldering):
- At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max. . . . . +265°C

STATIC ELECTRICAL CHARACTERISTICS at $T_A = 25^\circ C$ (See Fig. 3)

CHARACTERISTIC	TEST CONDITIONS	LIMITS			UNITS
		Min.	Typ.	Max.	
Supply-Voltage Drop	S2 = closed S3 = open S1,S4,S5,S6,S7,S8 = 1 Measure across 10 Ω resistor	130	215	300	mV
First-Stage Bias	S2,S3 = closed S1,S6 = 2 S4,S5,S7,S8 = 1 Measure term. 13 to ground	1.7	2.7	3.7	V
Chroma Bias	S2,S3 = closed S1,S4,S5,S7 = 1 S6,S8 = 2 Measure term. 6 to ground	7.3	8	9.1	V
Clamp Video Level	S2 = open Ref.= +12 V S3 = closed S1,S4,S5,S6,S7,S8 = 1 Measure across 82 kΩ	—	−8.7	—	V
Video Bias Level	S2 = open S3 = closed S1 = 1 S4,S5,S6,S7,S8 = 2 Measure across 1 kΩ	–	9	–	V
Luminance Blanking	S2 = open S3 = closed S1,S8 = 1 S4,S5,S6,S7 = 2 Measure across 1 kΩ	–	−50	–	mV
Chroma Blank	Setup same as above, measure term. 6	10.38	11.2	11.58	V
Chroma Input Impedance	Max. horizontal input = 10 V_{p-p} Max. vertical input = 2 V_{p-p}	–	2.5	–	kΩ
Chroma Output Impedance		–	150	–	Ω
Luminance Input Impedance		–	50	–	Ω
Luminance Output Impedance		–	10	–	kΩ

DYNAMIC ELECTICAL CHARACTERISTICS at T_A = 25°C (See Fig. 4)

CHARACTERISTIC	TEST CONDITIONS	LIMITS			UNITS
		Min.	Typ.	Max.	
Min. Video Gain	S1, S2 = 1; S3, S4 = 2 V_{IN} = 70 mV_{RMS}, f = 100 kHz, V_{16} = 12 V	0.2	0.35	0.5	V_{RMS}
Max. Video Gain	S2 = 1; S1, S3, S4 = 2 V_{IN} = 70 mV_{RMS}, f = 100 kHz, V_{16} = 0 V	1.6	2.1	2.6	V_{RMS}
Limited Video Gain	S2, S4 = 1, S1, S3 = 2 V_{IN} = 70 mV_{RMS}, f = 100 kHz, V_{16} = 0 V	–	0.3	–	V_{RMS}
Min. Chroma Gain	S1, S3 = 1; S2, S4 = 2; V_{16} = 12 V; chroma in = 530 mV_{RMS}, f = 3.58 MHz	–	0.095	–	V_{RMS}
Max. Chroma Gain	S3 = 1; S2 = 2, V_{16} = 0 V; chroma in; S1 = 2, S4 = 2 530 mV_{RMS}, f = 3.58 MHz	0.5	0.65	0.8	V_{RMS}
Video Freq. Response	S2 = 1, S1, S3, S4 = 2 V_{IN} = 70 mV_{RMS}; V_{16} = 0 V; f = 3.58 MHz	1	1.9	2.8	V_{RMS}
Chroma Phase Angle	S3 = 1; S2 = 2; V_{16} = 0 V; chroma in; S1 = 2, S4 = 2 530 mV_{RMS}, f = 3.58 MHz	12	19.5	27	Degrees
Chroma Gain with V^+ Variation	Vary V^+ from 10.8 V (REF.) to 13.2 V V_{16} = 50% of V^+; S1, S3 = 1 S2, S4 = 2	–	1.5	–	dB
Video Gain with V^+ Variation	Vary V^+ from 10.8 V (REF.) to 13.2 V V_{16} = 50% of V^+; S1, S2 = 1 S3, S4 = 2	–	1.5	–	dB

Typical max. luminance input before clipping (f = 100 kHz):

V_{16}	INPUT
+12 V	2.5 V_{p-p}
+6 V	0.75 V_{p-p}
0 V	0.45 V_{p-p}

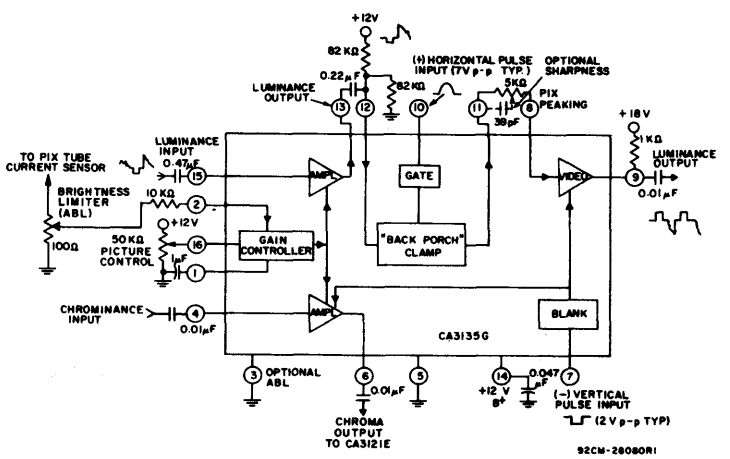


Fig. 1 – Block diagram.

CIRCUIT DESCRIPTION

(See fig. 2 for schematic diagram).

A video (luminance) signal from the receiver's "second detector" is coupled through a capacitor to term. 15 with sync-negative polarity. For purposes of the following amplifier, the level is clamped at the most negative point (sync tips) at the input (this is not the point at which the final "black"-level clamping, or dc restoration, is performed). The capacitor at term. 15 is charged on the most negative excursions of the signal by conduction of Q4. Positive signal excursions lift the emitter of Q4 into cutoff. The signal voltage on R3 develops a signal current in Q6. The current passes through Q7 and Q8, the division of current depends on the condition of the gain-adjusted signal voltage on the load resistors (discussed below). The gain-adjusted signal voltage on the load resistors is converted to current by the emitter-follower Q14 into R9, and fed into the current mirror, Q15, Q16, and Q17. The output of the current mirror develops a voltage across R13. The dc level is shifted by withdrawing some current from the input to the mirror. The fixed dc-level shifting current is developed in R6 and its diode string and is mirrored in Q13. Because the dc level is altered by adjustment of the gain, compensating dc currents that depend on these adjustments are fed into the mirrors through R13 and R29. The compensations are arranged so that, as gain is varied, the dc level of "black" is approximately constant at the output term. 13. The output is driven by emitter follower Q18, which has a short-circuit pulldown protection circuit, R14 and Q19. A constant-current source Q20 loads the emitter-follower to prevent distortion in the emitter-follower that may result from using a resistive load. The constant current is derived by mirroring the current in the diode D23. The resistor R15 prevents serious interaction with another current source mirrored from this point in case Q20 saturates.

The video output signal at term. 13 is coupled by a capacitor to term. 12. The polarity has not been inverted by the first amplifier, and sync is in the negative direction at this point. Black-level clamping is accomplished by application of a flyback pulse to term. 10. Between pulse peaks, Q29 is not conducting, and the base of Q24 goes up to the supply voltage so that term. 12 can be at any voltage between ground and the supply. While the flyback pulse is positive, that is during the blanking interval, the base of Q24 is held at about 2.8 volts. The most positive signal excursion during that time will cause Q24 to conduct with the result that the capacitor feeding term. 12 is charged until the most positive point of the signal is just at the conduction point, about 3.5 volts. The most positive part of the signal *during blanking* is the "back porch" or black-level reference. During trace time, the signal swings more positive, but the dc level of black is preserved regardless of the levels

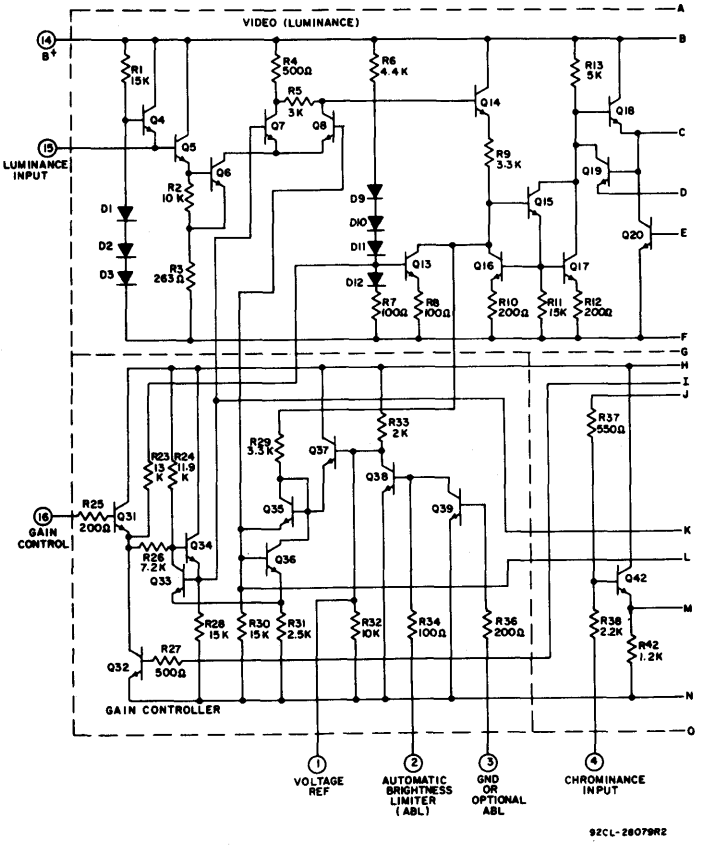


Fig. 2 – Schematic diagram

of sync or video signals. Term. 12 is a high-impedance point, and the emitter-follower Q26 is used to bring the signal out to term. 11.

The signal voltage at term. 11 is directly coupled through a resistor to term. 8, generating a current in term. 8. This current is amplified 10 times by the current mirror Q51, Q52, and Q53. Blanking during the vertical retrace interval is accomplished at Q50 via term. 7. Term. 7 is normally high enough to keep Q49 in saturation. A negative pulse from the vertical circuit cuts Q49 off, allowing some of the current through R51 to saturate Q50. When Q50 sinks the term. 8 input current, there is no output from term. 9 – as if the signal were blacker-than-black. The output current from term. 9 is used to drive the receiver's RGB matrix and the amplifiers that drive the picture tube.

The chrominance signal is taken from the first chroma amplifier following the automatic chroma control (ACC) and coupled through a capacitor to term. 4. The signal is attenuated by R38 and R37 and applied to an emitter-follower amplifier which drives the emitter of Q43. The current is steered through Q40 and Q41 depending on the gain-control conditions to the load resistors. An emitter-follower Q46 feeds term. 6, and R46 and Q45 provide short-circuit protection. The chroma amplifier is also blanked via the input at term. 7. The negative pulse at term. 7 allows the current through R51 to feed the base of Q44 (as well as the base of the video blanker, Q50). When Q44 saturates, the current is cut off in Q43 to disable the amplifier.

The combined gain control for the video and chroma sections is operated by varying the voltage on term. 16 between ground and the positive supply. Term. 16 has an emitter-follower Q31 loaded by a current source Q32. The voltage on term. 16 then determines whether the flow of current in R31 goes through Q36 or through Q33 to the resistors R24 and R26. The current on the Q33 side, a portion of the total current, is varied linearly by the control voltage. The gain-control amplifiers are slaves which follow the linear current control. The transistors Q34 and Q35 are driven as Darlington stages to reduce base-current effects in the control circuit. The normal gain-control function causes a change in the voltage on the base of Q34 with respect to the reference voltage at the base of Q35. The gain can also be changed by altering this reference voltage. This change in reference voltage is also used for "brightness limiting". The picture-tube current is sensed, and, when it exceeds some predetermined level, a voltage applied to term. 2 turns Q38 ON to reduce the reference voltage, thereby reducing the gain. Under these conditions, there is a closed feedback loop; the gain is set at a point such that the picture-tube current is just sufficient to cause a little conduction in Q38.

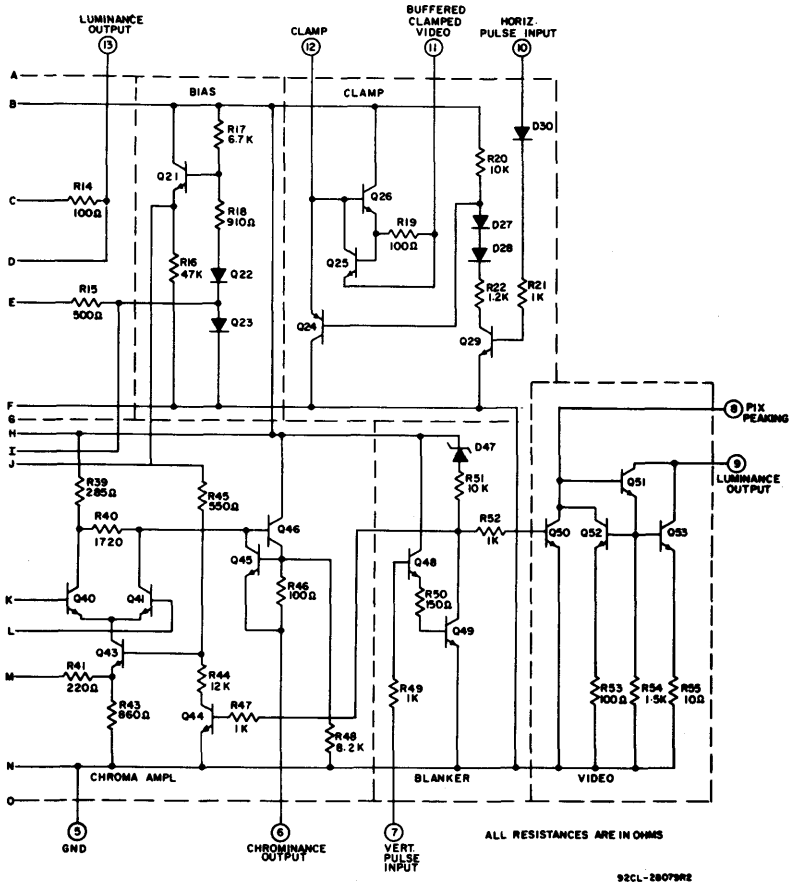


Fig. 2 – Schematic diagram

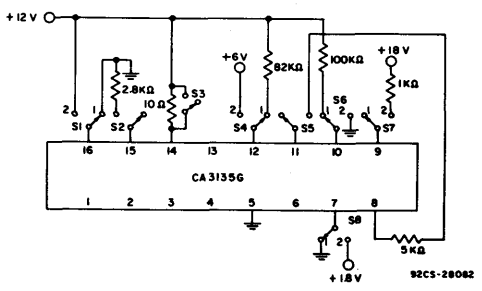


Fig. 3 – Static characteristics test circuit.

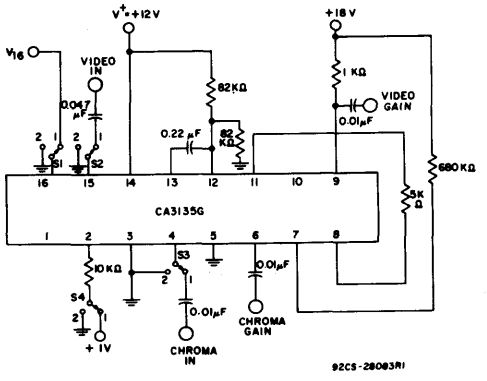


Fig. 4 – Dynamic characteristics test circuit.

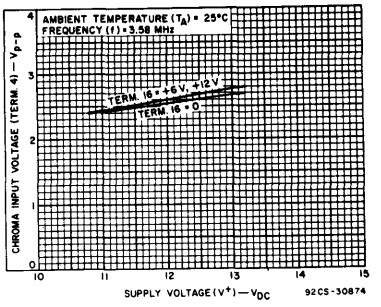


Fig. 5 – Typical chroma amplifier maximum linear voltage as a function of supply voltage.

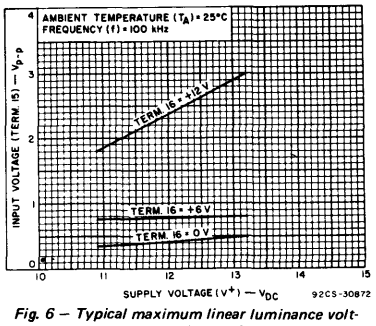


Fig. 6 – Typical maximum linear luminance voltage at terminal 15 as a function of supply voltage.

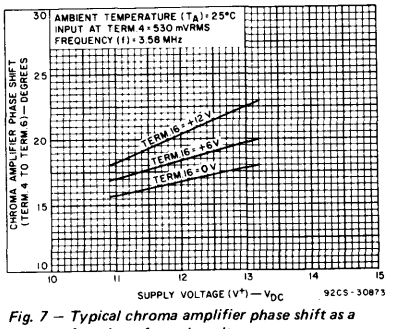


Fig. 7 – Typical chroma amplifier phase shift as a function of supply voltage.

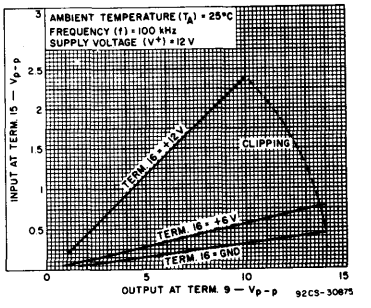


Fig. 8 – Input voltage as a function of output voltage.

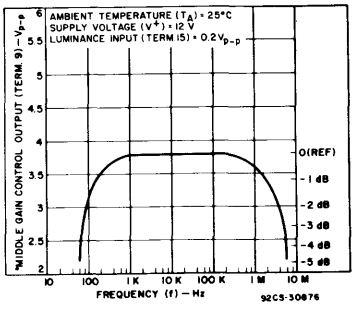


Fig. 9 – Typical gain-bandwidth response.

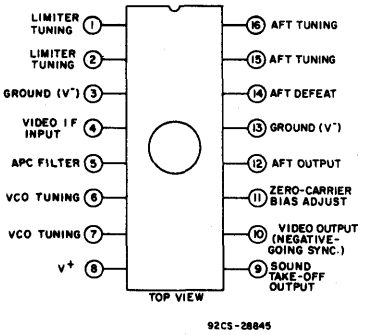
TV Video IF Phase-Locked-Loop Synchronous Detector for Color TV Receivers

Features:

- PLL carrier oscillator with wide pull-in and hold-in range
- Excellent low-level detector linearity
- Noise inversion at video output
- Wide range, variable zero-carrier level adjustment
- Automatic Fine Tuning (AFT) Detector
- Separate output for sound take-off
- 12-volt power supply

The RCA-CA3136E is a linear IC synchronous detector employing a phase-locked oscillator to demodulate the 45.75-MHz video if signals in color-TV receivers. The CA3136E features AFT voltage for dc control of the tuner; an adjustment for the zero-carrier dc level at the video output terminal; an amplifier arrangement for inverting noise impulses toward the black level; and a separate output terminal (non-inverting) for the sound if.

The CA3136E is supplied in a 16-lead plastic "power-stud" dual-in-line package.



TERMINAL DIAGRAM

MAXIMUM RATINGS, *Absolute-Maximum Values:*

Power Supply Voltage	15 V
Power Supply Current	100 mA
Input Signal Voltage	1 Vrms
Device Dissipation:	
With no Heat Sink:	
Up to T_A = 25°C	1.4 W
Above T_A = 25°C	derate linearly at 11.1 mW/°C
With Infinite Heat Sink:	
Up to T_A = 70°C	6.5 W
Above T_A = 70°C	derate linearly at 83.3 mW/°C
Thermal Resistance:	
$R_{\theta JS}$ (Junction to Stud)	12 °C/W
Ambient Temperature Range:	
Operating	−40 to +85°C
Storage	−65 to +150°C
Lead Temperature (During Soldering):	
At a distance 1/16 in. ± 1/32 in. (1.59 ± 0.79 mm) from case for 10 seconds max.	265°C

SUGGESTED GENERAL ALIGNMENT PROCEDURE

Fig. 1 shows a block diagram of the CA3136 in a typical circuit indicating the internal functions as well as the external circuitry and signals. A 45.75-MHz, 100-mVrms (50-ohm) signal is applied to the VIDEO IF INPUT (Terminal 4). While monitoring the VIDEO OUTPUT (Terminal 10), make the following adjustments in the indicated sequence; (1) adjust the VCO TUNING coil for a dc signal (lock). (2) Adjust the LIMITER TUNING coil for a minimum dc voltage on Terminal 10. (3) Adjust the VCO TUNING coil for 5.2 Vdc on Terminal 5 (with 12 volt supply on Terminal 8). (4) Close the AFT DEFEAT switch and note the dc voltage at the AFT OUTPUT (Terminal 12). (5) Return the AFT DEFEAT switch to its open position, and adjust the AFT TUNING coil for the same dc voltage noted when the AFT DEFEAT switch was closed. (6) Remove the rf input and adjust the ZERO CARRIER BIAS potentiometer for 7 volts dc on the VIDEO OUTPUT (Terminal 10). This final adjustment completes the alignment procedure.

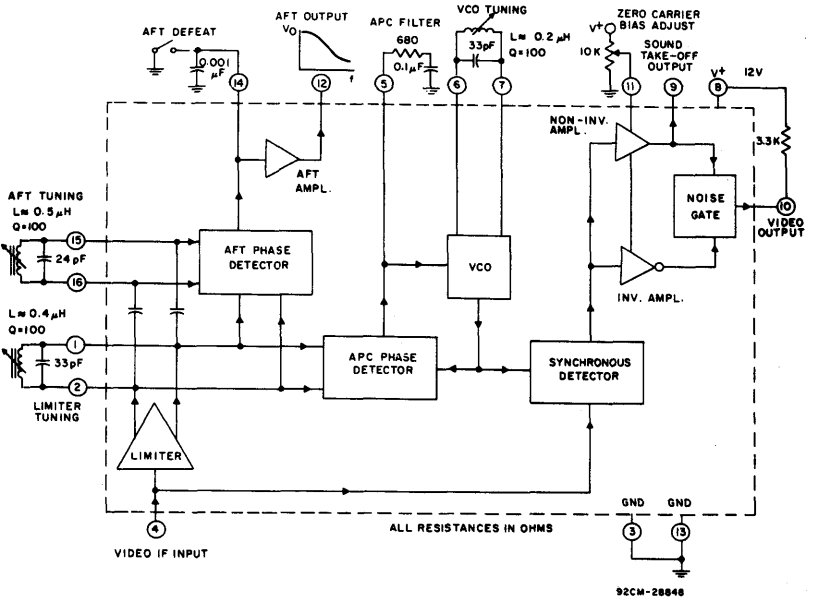


Fig. 1 – Block diagram of the CA3136 in a typical circuit application.

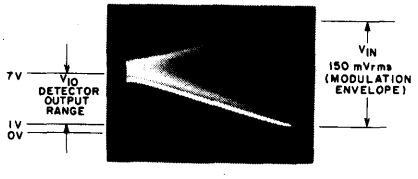


Fig. 2 – Typical detector output linearity.

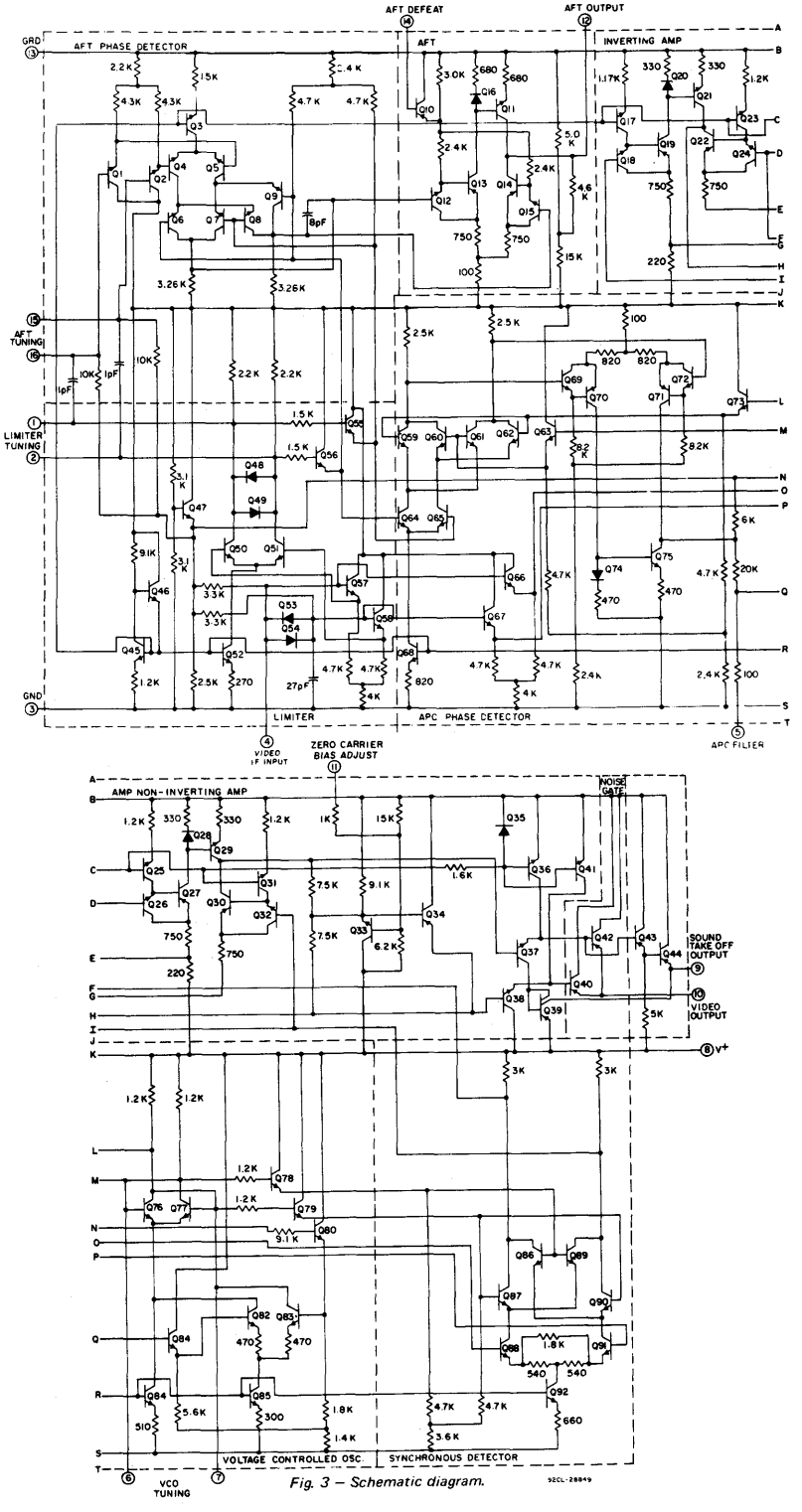


Fig. 3 – Schematic diagram.

92CL-28849

TYPICAL ELECTRICAL CHARACTERISTICS

At V^+ = 12 VDC, f_c = 45 MHz, T_A = 25°C

CHARACTERISTIC	SYMBOL	TEST CONDITIONS	VALUE	UNITS
Supply Current	$I_8 + I_{10}$		60	mA
Video-Output Voltage	V_{10}	Zero Carrier Bias Adjust	7	V_{DC}
Noise-Inversion Offset Voltage	V_{10}	Referenced to Zero-Carrier Level	0.3	V_{DC}
Sound IF-Take-Off Output Voltage	V_9	V_{10} = 7 V_{DC}	7.7	V_{DC}
AFT Output Voltage	V_{12}	AFT Defeat Switch Closed	3	V_{DC}
Oscillator Pull-In Range			3	MHz
Oscillator Hold-In Range			6	MHz
Detector Conversion Gain			30	dB
Video Bandwidth			9	MHz
Carrier Rejection at Video Output: fc = 45 MHz			30	dB
2 fc = 90 MHz			40	dB
Video IF Parallel Input Impedance: Resistance at Term. 4	R_p		4	kΩ
Capacitance at Term. 4	C_p		5	pF
Sound Take-Off Output Resistance at Term. 9	R_o	1 MHz	50	Ω
Video Output Resistance at Term. 10	R_o	1 MHz	50	Ω

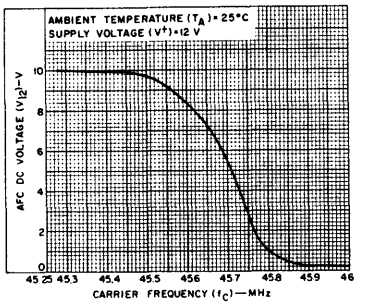


Fig. 4 – Typical AFT output of CA3136.

CA3137E

TV Chroma Demodulator

Features:

- **Balanced chroma demodulators**
- **Color difference matrix (6500°K)**
- **DC tint control**
- **Three low-output-impedance drivers for direct coupling**
- **Reference subcarrier limiter**
- **Internal RF filtering**
- **DC chroma gain control**
- **Dynamic "flesh correction" — corrects purple and green flesh colors without affecting primary red, green, and blue colors**
- **Requires few external components**
- **No tuning adjustments are necessary**

The RCA-CA3137E is a monolithic silicon integrated circuit that performs the demodulation, dynamic "flesh correction", tint control, and chroma gain-control functions. It is designed to function compatibly with the CA3126Q Chroma Processor, and is supplied in the 16-lead dual-in-line plastic package.

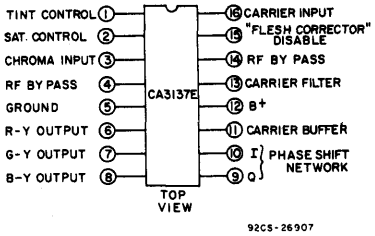


Fig.1 – CA3137E terminal assignment.

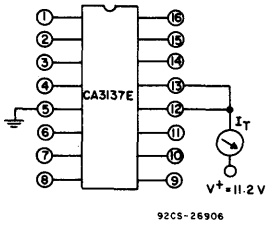


Fig.2 – DC test circuit.

MAXIMUM RATINGS, Absolute-Maximum Values at T_A = 25°C

DC SUPPLY VOLTAGE (between Terms. 5 and 12)	13.2 V
DEVICE DISSIPATION:	
Up to T_A = 55°C	750 mW
Above T_A = 55°C	derate linearly 7.9 mW/°C
AMBIENT-TEMPERATURE RANGE:	
Operating	−40 to +85°C
Storage	−65 to +150°C
LEAD TEMPERATURE (During soldering):	
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 s max.	+265°C

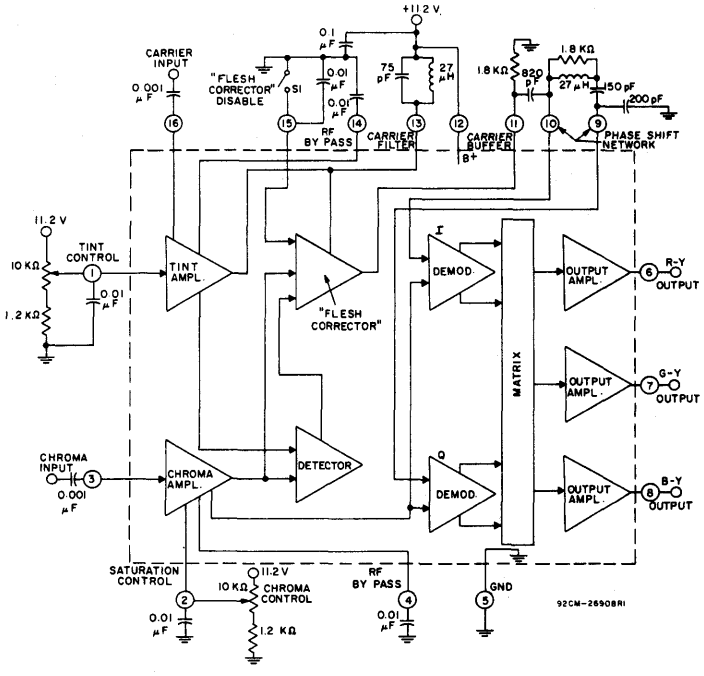


Fig.3 – Functional diagram and typical dynamic test circuit.

ELECTRICAL CHARACTERISTICS AT T_A = 25°C, V^+ = 11.2 V

CHARACTERISTIC	SYMBOL	TEST CONDITIONS	LIMITS			UNITS
			Min.	Typ.	Max.	
STATIC (See Fig.2)						
Supply Current	I_T		–	35	47	mA
Reference Subcarrier Input	V_{16}		–	6.7	–	VDC
Oscillator Reference Inputs	V_9, V_{10}		–	3.8	–	VDC
R-Y, G-Y, B-Y Outputs	V_6, V_7, V_8		–	5	–	VDC
Chroma Input	V_3		–	1.2	–	VDC
DYNAMIC (See Fig.3)						
Tint and Sensitivity Limiting	V_{11}	V_{16} = 200 mV p-p @ 3.58 MHz	200	300	–	mVp-p
Tint Limiting	V_{11}	V_{16} = 800 mV p-p @ 3.58 MHz	–	425	600	mVp-p
Tint Amplifier* Phase Reference	ϕV_{11}	V_{16} = 400 mV p-p, Term.1 = 11.2 VDC	–35	–25	–15	Degrees
Tint Control▲ Range	$\Delta\phi_{11}$	V_{16} = 800 mV p-p, Term.1 = 1.2 VDC	–130	–110	–80	Degrees
Ratio G-Y to R-Y	V_7/V_6	V_{16} = 400 mV p-p,	28	33	38	%
Ratio B-Y to R-Y	V_8/V_6	V_3 = 40 mV p-p	108	120	132	%
Demodulated Chroma Output R-Y	V_6	V_{16} = 400 mV p-p, V_3 = 40 mV p-p	350	550	–	mV p-p
Color Difference Output (Bandwidth at 3 dB)		V_3 = 40 mV p-p	–	900	–	kHz
Maximum Color Difference Outputs: R-Y	V_6	V_{16} = 400 mV p-p, V_3 = 300 mVp-p	1.5	2.2	–	V_{p-p}
G-Y	V_7		0.42	0.7	–	
B-Y	V_8		1.6	2.65	–	
"Flesh Detector" Reference:		Set-Up: Term.2 = 1.6 V; Term.1 = 11.2 V; Term.16 = 400 mV p-p @ 0° Reference Angle; Term.3 = 40 mV p-p @ 10° Reference Angle; S_1 Closed (Term.15 at GND)	Reference Set-Up			
"Flesh Detector": Phase	ϕ_{11}	Same Set-up except S_1 open	–	0	–	Degrees
Amplitude	V_{11}		–	275	–	%
"Flesh Detector": Phase	ϕ_{11}	Same Set-up except Term.3 at 190° angle	–	0	–	Degrees
Amplitude	V_{11}		–	100	–	%
Small-Signal Output Resistance (Terms.6,7,8)	r_o		–	50	–	Ω
Small-Signal Input Resistance: Term.3	r_i		–	3	–	kΩ
Terms.9&10			–	2.5	–	

* Phase angle of term. 11 referenced to term. 16 phase angle.

▲ Phase angle of term. 11 with term. 1 = 1.2 V minus phase angle of term. 11 with term. 1 = 11.2 V.

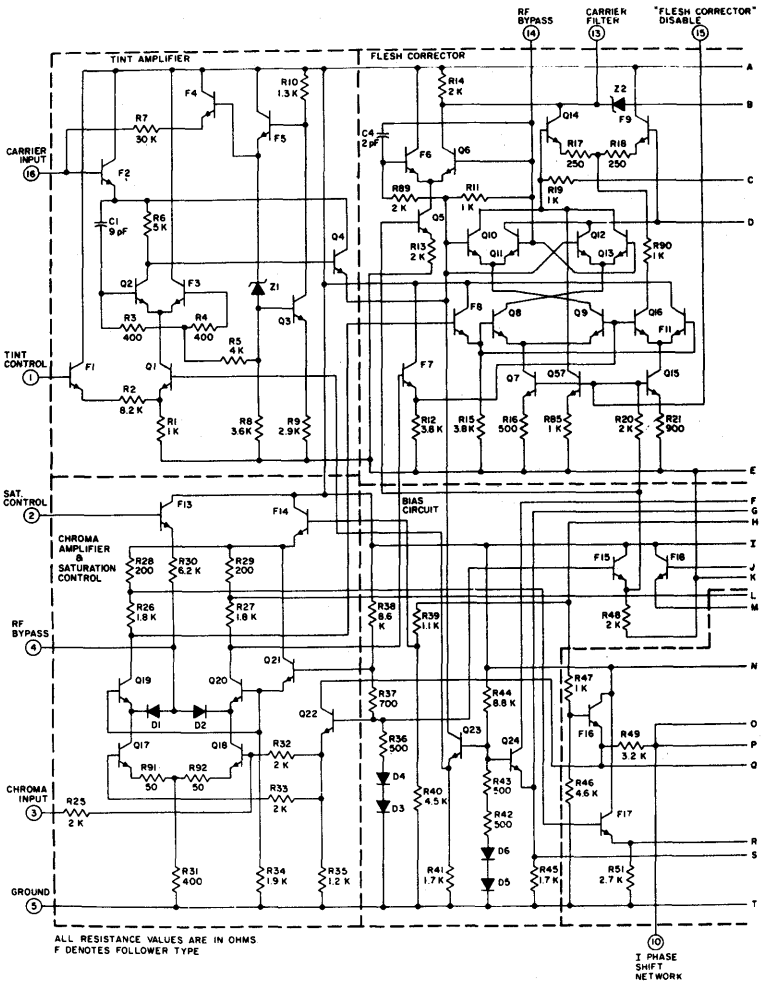


Fig.4 – CA3137E Schematic diagram.

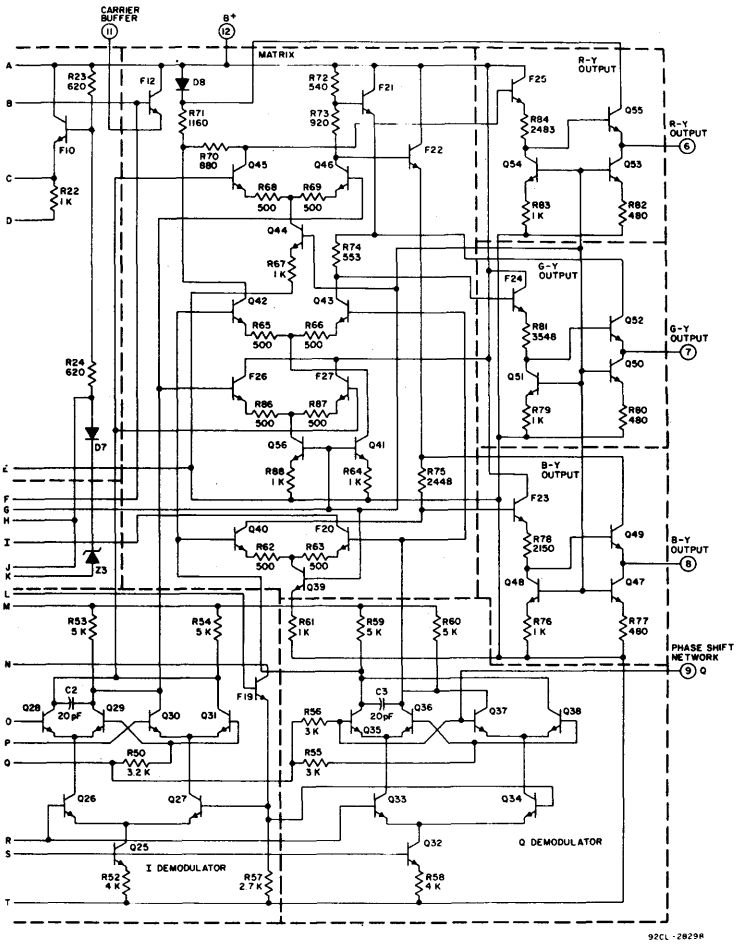


Fig.4 – CA3137E Schematic diagram.

CA3139E, CA3139Q

TV Automatic Fine Tuning Circuit

With Intercarrier Mixer/Amplifier
For Color and Monochrome Receivers

Features:

- **Cascode-type high-gain amplifier (15-mV input for rated output)**
- **AFT differential peak detector**
- **Differential amplifier**
- **Bipolar outputs**
- **Five-stage intercarrier mixer/amplifier**
- **Internal voltage regulator**
- **For use in either color or monochrome receivers**

The RCA-CA3139 is a monolithic TV Automatic Fine Tuning (AFT) circuit that provides an AFT voltage and an amplified 4.5-MHz intercarrier sound signal. When connected to an output of an IF amplifier the CA3139 provides the signal processing (amplification and detection) necessary to generate the AFT correction signals required by the TV tuner. It also mixes the video and sound IF carriers and amplifies the resultant 4.5-MHz intercarrier sound signal. This sound output may then be connected to an FM detector such as the RCA-CA3134 "TV Sound IF and Audio Output Subsystem", or the RCA-CA3065 "FM Detector and Audio Driver".

The AFT portion of the CA3139 is similar to the RCA-CA3064 AFT circuit with the following exceptions: (a) the AFT filter capacitors are external and user selectable, allowing the detector to operate as a peak detector and resulting in a higher effective gain for the TV signal; (b) the detector bias resistor is external and user selectable, allowing the gain of the AFT and intercarrier signals to be adjusted; (c) the dynamic resistance of the shunt regulator has been decreased.

The CA3139 is supplied in a 14-lead dual-in-line plastic package (CA3139E) or a 14-lead plastic package with quad-formed leads (CA3139Q).

MAXIMUM RATINGS,
Absolute-Maximum Values:

DEVICE DISSIPATION:
Up to T_A = 25°C ..... 630 mW
Above T_A = 25°C. . . . .derate linearly 6.7 mW/°C
AMBIENT TEMPERATURE:
Operating ..... −40 to +85°C
Storage ..... −65 to +150°C
LEAD TEMPERATURE (During Soldering):
At distance 1/16" ± 1/32"
(1.59 mm ± 0.79 mm)
from case for 10 s max. ..... 265°C

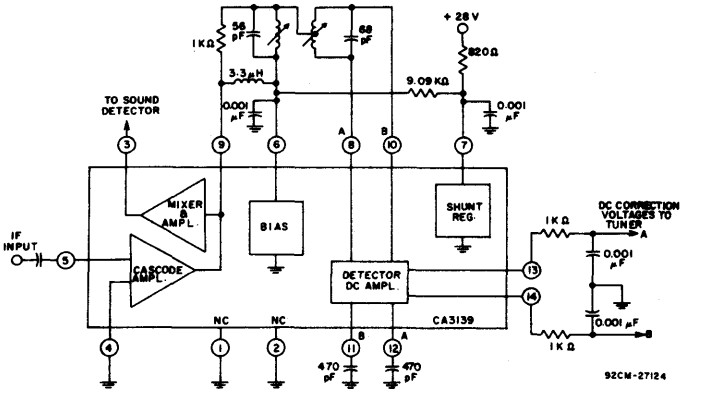


Fig. 1 – Block diagram and typical application of CA3139.

MAXIMUM VOLTAGE RATINGS at T_A = 25°C

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range between vertical terminal 3 and horizontal terminal 12 is +8 to −1.5 volts.

Terminal No.	1,2♦	3	4■	5	6	7▲	8	9	10	11	12	13	14	MAXIMUM CURRENT RATINGS I_{IN}, I_{OUT} mA
1, 2♦	← NO INTERNAL CONNECTION →													
3			+10 −0	+9 −1.5	+8 −1.5	+0 −10	+8 −1.5	+8 −1.5	+8 −1.5	+8 −1.5	+8 −1.5	+8 −1.5	+8 −1.5	10
4■				+0 −2	+0 −3	+0 −11	+0 −3	+0 −3	+0 −3	+0 −3	+0 −3	+0 −11	+0 −11	50
5					+0 −5	+0 −14	+2 −5	+1 −5	+2 −5	+2 −5	+2 −5	+1 −8	+1 −8	1
6						+0 −14	+2 −2	+0 −2	+2 −2	+1 −3	+1 −3	+0 −10	+0 −10	2
7▲							+15 −0	+13 −0	+15 −0	+13 −0	+13 −0	+10 −0	+10 −0	50
8								+1 −5	+5 −5	+5 −5	+1 −5	+0 −14	+0 −14	2
9									+10 −2	+8 −2	+8 −2	+0 −10	+0 −10	10
10										+1 −5	+5 −5	+1 −10	+1 −10	2
11											*	*	*	2
12												*	*	2
13													+14 −14	2
14														2

▲ Terminal number 7 may be connected to any positive voltage source greater than the internal zener regulating voltage through a suitable dropping resistor – provided the dissipation rating is not exceeded.

■ This terminal should be connected to the most negative potential of the complete circuit.

* Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if the specified limits between all other terminals are not exceeded.

♦ It is recommended that unused terminals 1 and 2 be grounded to act as shields.

ELECTRICAL CHARACTERISTICS at T_A = 25°C, V^+ = 28 V (Unless Otherwise Specified)

See Test Circuit, Fig. 2

CHARACTERISTIC	TEST CONDITIONS	LIMITS Min.	LIMITS Max.	UNITS
NO SIGNAL INPUT				
Supply Current, I^+		15	20	mA
Low Voltage at Term. 7[1]	V^+ = 20.8 V	11	14.5	V
Shunt Reg. Voltage		12	14.5	V
Quiescent Voltage at Term. 3		4.5	10	V
Quiescent Voltage[2] at Terms. 13 and 14	Term. 13 connected to Term. 14	6	8.5	V
Quiescent Difference Voltage, Terms. 13 to 14		−0.8	+0.8	V
Quiescent Voltage at Term. 6		1.4	2.6	V
SIGNAL INPUT = 15 mV_{RMS} (Unless Otherwise Specified), Note 3				
Correction Voltage at Term. 13	f = 44.65 MHz	2.2	4.7	V
	f = 45.69 MHz	1.2	4.4	
	f = 45.81 MHz	9.6	13.8	
	f = 46.85 MHz	9.1	12.1	
Correction Voltage at Term. 14	f = 44.65 MHz	9.1	12.1	V
	f = 45.69 MHz	9.6	13.8	
	f = 45.81 MHz	1.2	4.4	
	f = 46.85 MHz	2.2	4.7	
4.5 MHz Output	Two-Tone Input f1 = 45.75 MHz at 15 mV f2 = 41.25 MHz at 5 mV	50	200	mV_{RMS}

NOTES: 1. I_7 = 12 mA maximum at V_7 = 11 V.

2. $V_{13} = 0.55\ V_Z \pm 0.7$ V

3. Resistor from term. 6 to term. 7 = 9.09 KΩ. Crossover steepens and "bow tie" width increases when resistor is decreased in value. Total peak swing decreases slightly.

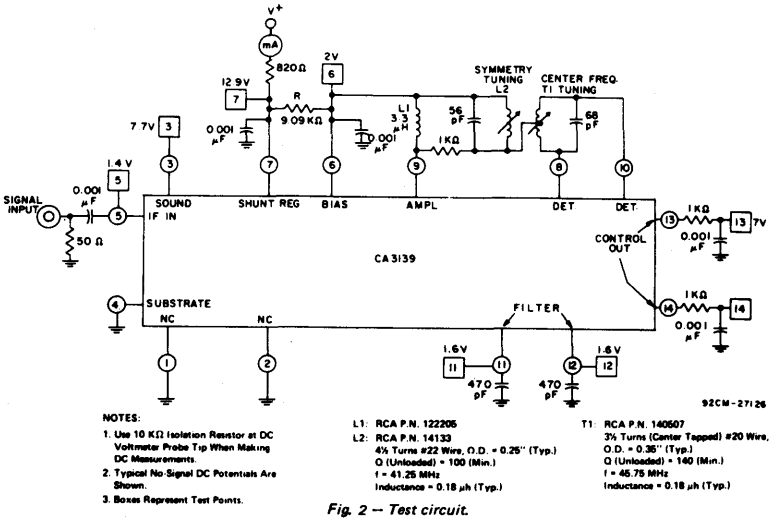


Fig. 2 -- Test circuit.

CIRCUIT DESCRIPTION

The CA3139 consists of five functional circuits as shown in the block diagram, Fig. 1 (see Fig. 5 for schematic diagram).

1) **Cascode Amplifier** – Consists of emitter-follower Q1, common-emitter amplifier Q2, and common-base amplifier Q3.

2) **Bias Circuit** – Consists of Q4 and resistors R1, R4, R5, and an external resistor (user selectable) connected to the voltage regulator, terminal 7. The nominal value of the external resistor is 9.1 kΩ. Reduced values will raise the gain of the cascode amplifier chain, and higher values will reduce the gain. If the gain is increased, the AFT "Bow Tie" width will increase and the crossover slope will increase (become steeper). The input transistor Q1 is internally biased, so AC coupling is normally used to the input terminal 5.

3) **Intercarrier Mixer/Amplifier** – The output of the cascode amplifier at terminal 9 is also internally connected to the intercarrier mixer/amplifier chain consisting of transistors Q13 through Q17 and associated components. The video IF carrier at 45.75-MHz and the FM sound IF carrier at 41.25-MHz are down-converted to a 4.5-MHz FM signal by Q14. A low-pass filter removes the carriers and upper conversion signal components. The 4.5-MHz FM signal is further amplified and filtered by Q16 and C3. The FM sound output signal is at terminal 3. The gain with respect to a 5-mV sound carrier (tested with a 15-mV video carrier) input signal at terminal 5 is 10 to 40 when the resistor is connected between terminals 6 and 7 is 9.09 kΩ.

4) **AFT Detector and DC Amplifier** – Consists of Q6 through Q12 and related components. The detector inputs at terminals 8 and 10 are connected to the external discriminator transformer and biased through the transformer at terminal-6 potential. The total current through transistors Q7 and Q8 is held constant by the current-mirror transistors Q10, Q11, and Q12. External filter capacitors connected to terminals 11 and 12 assure that peak detection is accomplished. The AFT output voltages are shown in the Electrical Characteristics chart, and a graphical representation is shown in Fig. 4.

5) **Voltage Regulator** – An active shunt regulator, consisting of D1, D2, Z1, Z2, and Q5, is included to reduce the dynamic resistance.

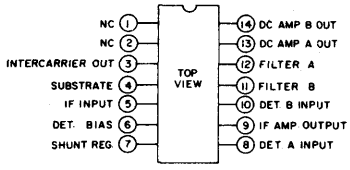


Fig. 3 – Terminal assignment.

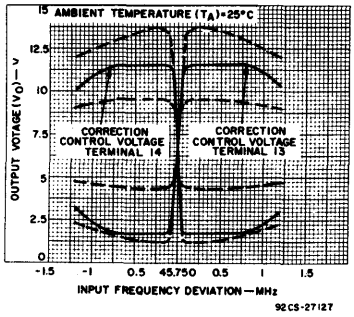


Fig. 4 – Dynamic control-voltage characteristics.

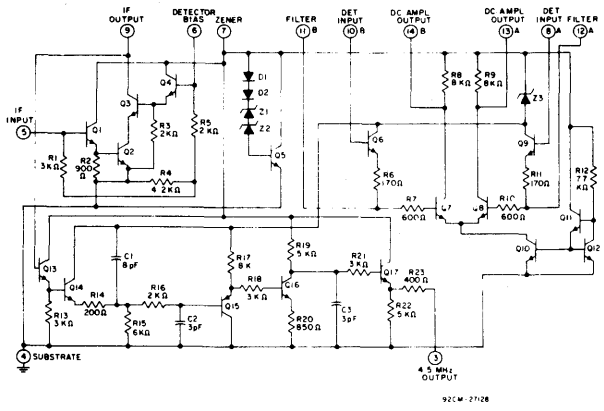


Fig. 5 – Schematic diagram of CA3139.

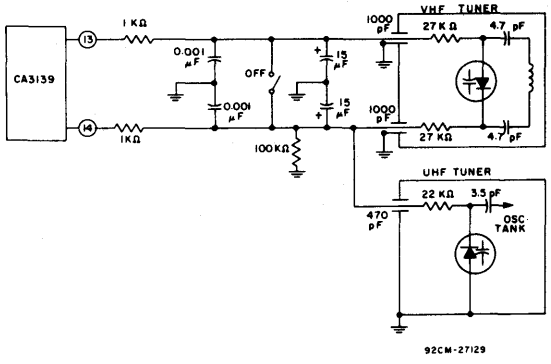


Fig. 6 – Typical tuner connection.



Fig. 7 – Template of CA3139Q circuit board (actual size, bottom view).

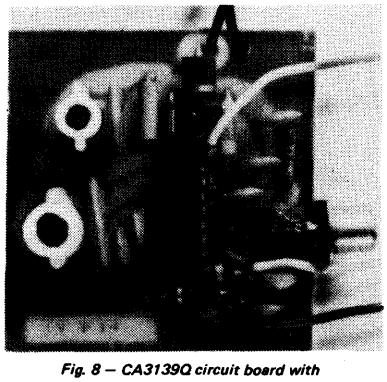


Fig. 8 – CA3139Q circuit board with components.

TV Luminance Processor

The CA3143E is a monolithic silicon integrated circuit that performs the luminance processing functions of amplification; contrast, brightness and peaking control; blanking; and black-level clamping.

This device, when used in conjunction with the CA3126Q chroma processor and the CA3137E chroma demodulator, will provide a luminance/chrominance system having excellent tracking of controls. The CA3143E is supplied in a 14-lead dual-in-line plastic package.

Features:

- **Black-level clamping**
- **Linear dc controls for brightness, contrast, and peaking**
- **Horizontal and vertical blanking**
- **Operates with standard or tapped delay line**

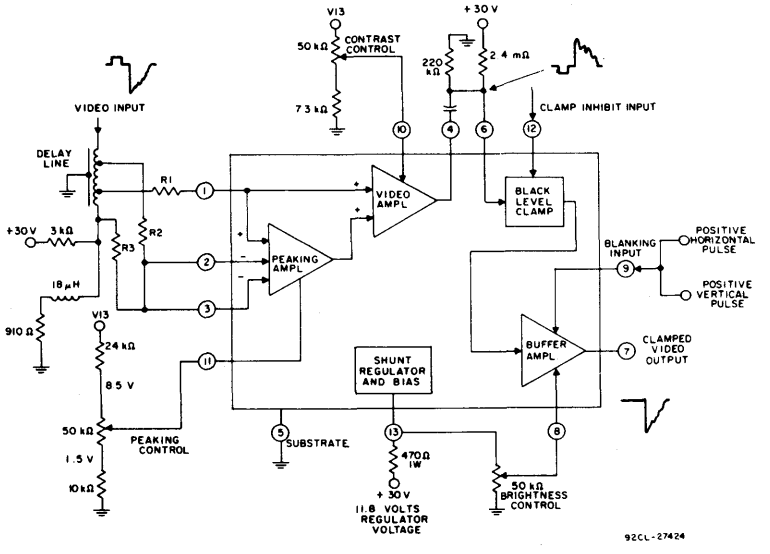


Fig. 1 – Functional block diagram.

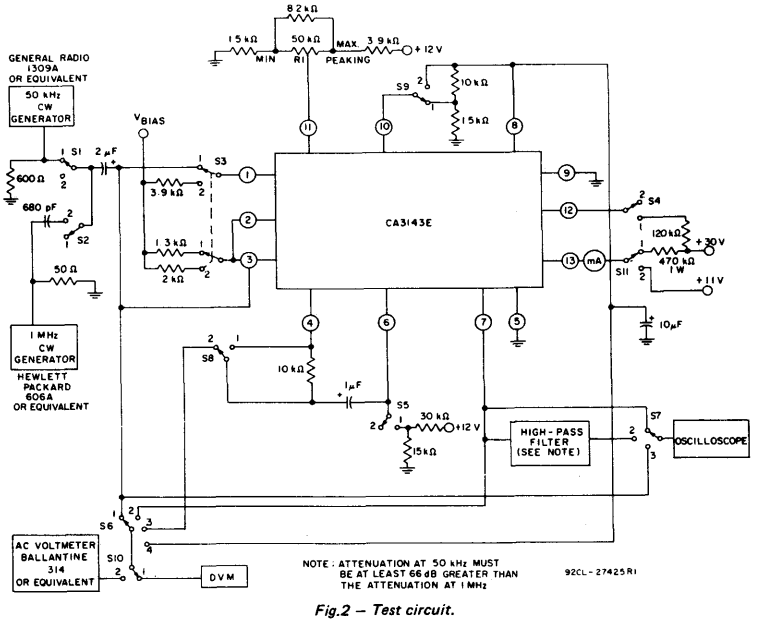


Fig. 2 – Test circuit.

CIRCUIT DESCRIPTION

Fig. 1 is a block diagram of the CA3143E indicating the internal functions as well as external circuitry and signals. The video input signal with positive-going sync is applied to the input of the tapped delay line. Signals from fixed taps of the delay line are applied to terminals 1, 2 and 3 of the CA3143E. In referring to Fig.4, the signal from the delay line tap A is applied to the video input at terminal 1. The signals from taps B and C are summed where $V_A + V_B = V_{sum}$. The signal (V_{sum}) is then applied to the parallel connection of the peaking input terminals, 2 and 3. The video input signal is applied to a non-inverting input of the peaking amplifier while the peaking input signal (V_{sum}) is applied to an inverting input of the peaking amplifier.

Low-frequency video components are unattenuated, while high-frequency components are attenuated as a function of the delay-line tap points. The peaking amplifier is a differential amplifier, so that the output is proportional to V_1 minus V_{sum}. At low frequencies, the signal at terminals 2 and 3 is unattenuated, and the peaking amplifier produces no output at these frequencies. However, at high frequencies the signal at terminals 2 and 3 is attenuated thus, the peaking amplifier output consists of high-frequency video. The peaking control setting determines the amplitude of the peaking signal which is then fed to the video amplifier, where it is added to the video input signal and amplified. The setting of the peaking control does not substantially affect the dc quiescent voltage at terminal 4.

The low-impedance video amplifier output is at terminal 4. The signal is fed through an external coupling capacitor to terminal 6, the black-level clamp input. The action of the black-level clamp is such that it clamps to the black level rather than to the sync level. Refer to the circuit diagram in Fig.3. Consider the situation where no signal is applied to terminal 12. Terminal 6 is biased through diode D2. The signal at terminal 6 will clamp its most negative excursion (sync pulse) to the anode voltage of D2. However, if a positive pulse is applied to terminal 12 during the sync interval, the anode of D2 is forced to ground due to saturation of Q17. The clamp is thus disabled, and terminal 6 will clamp to the next lower signal level, the black level.

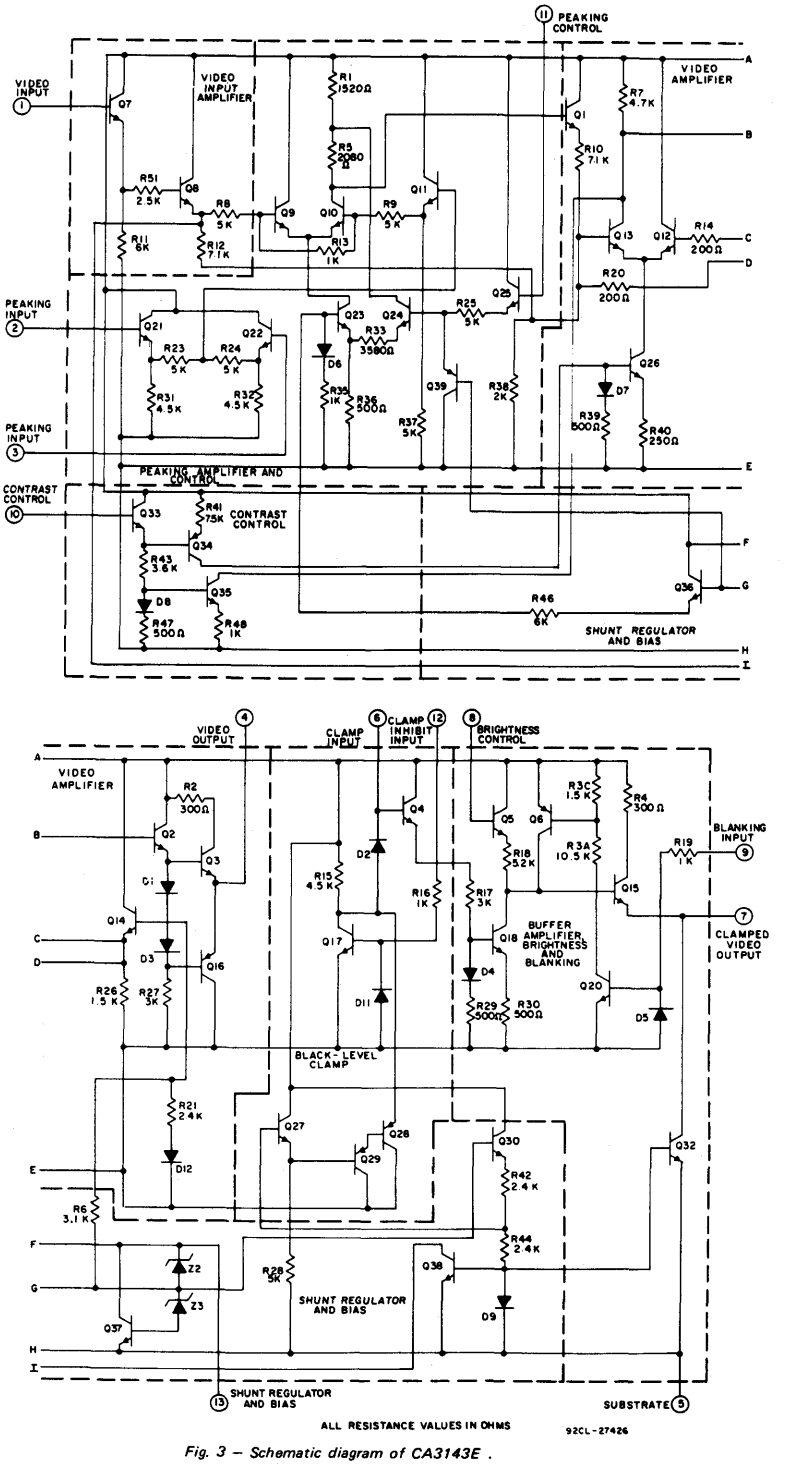


Fig. 3 – Schematic diagram of CA3143E .

MAXIMUM RATINGS, *Absolute-Maximum Values*:

DC SUPPLY CURRENT (Into Terminal 13)* . . . . . 59.5 mA

DEVICE DISSIPATION:*

Up to T_A = 55°C . . . . . 750 mW

Above T_A = 55°C . . . . . derate linearly 7.9 mW/°C

AMBIENT–TEMPERATURE RANGE:

Operating . . . . . –40 to +85°C

Storage . . . . . –65 to +150°C

LEAD TEMPERATURE (During soldering):

At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 s max. . . . . +265°C

* Although the CA3143E is rated for maximum dissipation of 750 mW, it is recommended that the current into terminal 13 be limited by external circuit resistance to 39 mA for a typical voltage at terminal 13 of 11.8 volts.

ELECTRICAL CHARACTERISTICS at T_A = 25°C

Characteristic	Bias Volts (V)	Test Conditions: Switch Numbers S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	LIMITS Min.	Typ.	Max.	UNITS
		Switch Positions For Characteristics Measurements														
STATIC																
Voltage: At Term. 13 (V13)	6.1	2	1	1	2	2	4	1	2	2	1	1	11	11.8	13.2	V
Quiescent Voltage At Term. 4 (V4)	6.1	2	1	1	2	2	3	1	2	2	1	1	3.3	4	5.7	V
Quiescent Voltage At Term. 7 (V7)	6.1	2	1	1	2	2	2	1	2	2	1	1	7.1	7.7	8.3	V
Current into Term.13 (Term.13 Connected to +11 V) (I13)	6.1	2	1	1	2	2	3	1	2	2	1	2	10	19	30	mA
DYNAMIC																
Wide-Band Gain (Note 1)	5.8	1	1	1	2	1	2	1	1	1	2	1	6	8.3	11	dB
Contrast Gain Reduction (Note 2)	5.8	1	1	1	2	1	2	1	1	2	2	1	27	30	–	dB
Peaking Gain (Note 1)	5.8	1	1	2	2	1	2	1	1	1	2	1	15	18.4	22	dB
Peaking Gain Reduction (Note 3)	5.8	1	1	2	2	1	2	1	1	1	2	1	16	18	–	dB
Max. Intermodulation Distortion: 2V (Note 4)	5.8	1	–	1	1	1	2	–	2	1	2	1	–	20	–	%
3V (Note 5)	5.8	1	–	1	1	1	2	–	2	1	2	1	–	40	–	%

Note 1: Set 50-kHz generator for 100 mVp-p. Adjust R1 Peaking Control (See Fig.2) for minimum setting. Measure wide-band gain at terminal 7.

Note 2: Set 50-kHz generator for 100 mVp-p. Adjust R1 for minimum setting. Measure contrast gain reduction at terminal 7.

Note 3: Set 50-kHz generator for 100 mV-p-p. Adjust R1 for maximum setting. Measure peaking gain reduction at terminal 7.

Note 4: Adjust R1 for minimum setting. With S2 at switch position1 and S7 at switch position 3, set 50-kHz generator for 2 Vp-p. Then with S2 at switch position 2, set 1 MHz generator for 100 mVp-p. Then with S7 at switch position 2, measure downward modulation of the 1-MHz signal due to the 50-kHz signal.

Note 5: Repeat step 4 except that the 50-kHz generator must be set at 3 Vp-p.

The clamped video signal at terminal 6 is amplified and inverted at terminal 7. Blanking is accomplished by applying horizontal and vertical sync pulses to terminal 9. The pulses turn ON p-n-p transistor Q6 which shorts the base of transistor Q15 to the terminal 13 supply voltage. The brightness control function is accomplished by varying the voltage on terminal 8. The gain of the inverter stage remains constant, but the dc reference voltage follows the terminal 8 voltage. The contrast control function is accomplished by varying the voltage of terminal 10. Increasing the voltage on terminal 10 lowers the gain of the video amplifier. This reduction in gain does not substantially affect the dc quiescent voltage at terminal 4.

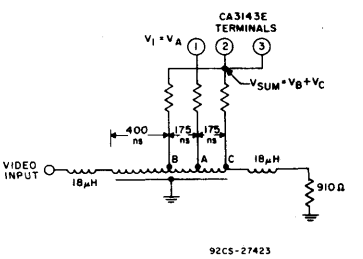
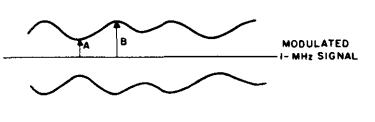


Fig.4 – Tapped delay line.



A = Amplitude of 50 kHz signal at deepest trough

B = Peak amplitude of 50 kHz signal

Downward Modulation = $\frac{B-A}{B}$

CA3144G

TV Luminance Processor

The CA3144G is a monolithic silicon integrated circuit that performs the luminance processing functions of amplification; contrast, brightness and peaking control; blanking; and black-level clamping.

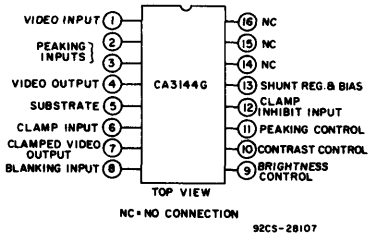
This device, when used in conjunction with the CA3126Q chroma processor and the CA3137E chroma demodulator, will provide a luminance/chrominance system having excellent tracking of controls. The CA3144G is supplied in a 16-lead hermetic Gold-CHIP dual-in-line plastic package ("G" suffix).

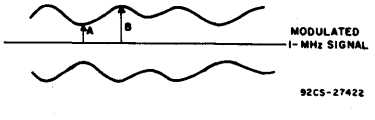
The semiconductor junctions in this device are sealed by utilizing a silicon nitride passivation layer. A multi-layered, highly corrosion-resistant, terminal-connection system of unique design is employed.

Features:

- **Black-level clamping**
- **Linear dc controls for brightness, contrast, and peaking**
- **Horizontal and vertical blanking**
- **"Hermetic Chip" construction**
- **Silicon nitride passivated**
- **Platinum silicide ohmic contacts**
- **Gold-CHIP metallization**
- **Operates with standard or tapped delay line**

TERMINAL ASSIGNMENT





A = Amplitude of 50-kHz signal at deepest trough
B = Peak amplitude of 50-kHz signal

Downward Modulation = $\frac{B-A}{B}$

MAXIMUM RATINGS, *Absolute-Maximum Values:*

DC SUPPLY CURRENT (Into Terminal 13)* . . . . . 57 mA

DEVICE DISSIPATION: *

Up to $T_A = 55^\circ C$ . . . . . 750 mW

Above $T_A = 55^\circ C$ . . . . . derate linearly 7.9 mW/°C

AMBIENT-TEMPERATURE RANGE:

Operating . . . . . −40 to +85°C

Storage . . . . . −65 to +150°C

LEAD TEMPERATURE (During soldering):

At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 s max. . . . . +265°C

* Although the CA3144G is rated for maximum dissipation of 750 mW, it is recommended that the current into terminal 13 be limited by external circuit resistance to 39 mA for a typical voltage at terminal 13 of 12.3 volts.

ELECTRICAL CHARACTERISTICS at $T_A = 25^\circ C$

Characteristic	Bias Volts (V)	Test Conditions: Switch Numbers S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	LIMITS Min.	Typ.	Max.	UNITS
		Switch Positions For Characteristics Measurements														
STATIC																
Voltage: At Term. 13 (V13)	6.5	2	1	1	2	2	4	1	2	2	1	1	11	12.3	13.2	V
Quiescent Voltage At Term. 4 (V4)	6.5	2	1	1	2	2	3	1	2	2	1	1	3.3	4	5.7	V
Quiescent Voltage At Term. 7 (V7)	6.5	2	1	1	2	2	2	1	2	2	1	1	7.1	7.7	8.3	V
Current into Term.13 (Term.13 Connected to +11 V) (I13)	6.5	2	1	1	2	2	3	1	2	2	1	2	10	18	30	mA
DYNAMIC																
Wide-Band Gain (Note 1)	7.3	1	1	1	2	1	2	1	1	1	2	1	1	3	5	dB
Contrast Gain Reduction (Note 2)	7.3	1	1	1	2	1	2	1	1	2	2	1	27	30	–	dB
Peaking Gain (Note 1)	7.3	1	1	2	2	1	2	1	1	1	2	1	9	13	17	dB
Peaking Gain Reduction (Note 3)	7.3	1	1	2	2	1	2	1	1	1	2	1	16	18	–	dB
Max. Intermodulation Distortion: 3.8 V (Note 4)	7.3	1	–	1	1	1	2	–	2	1	2	1	–	20	–	%
5 V (Note 5)	7.3	1	–	1	1	1	2	–	2	1	2	1	–	40	–	%

Note 1:
Set 50-kHz generator for 200 mV_{rms}. Adjust R1 peaking control for minimum setting (see Fig. 2). Measure wide-band gain at terminal 7.

Note 2:
Set 50-kHz generator for 200 mV_{rms}. Adjust R1 for minimum setting. Measure contrast gain reduction at terminal 7.

Note 3:
Set 50-kHz generator for 200 mV_{rms}. Adjust R1 for maximum setting. Measure peaking gain reduction at terminal 7.

Note 4:
Adjust R1 for minimum setting. With S2 at switch position 1 and S7 at switch position 3, set 50-kHz generator for 3.8 V_{p-p}. Then with S2 at switch position 2, set 1-MHz generator for 200 mV_{rms}. Then with S7 at switch position 2, measure downward modulation of the 1-MHz signal due to the 50-kHz signal.

Note 5:
Repeat step 4 except that the 50-kHz generator must be set at 5 V_{p-p}.

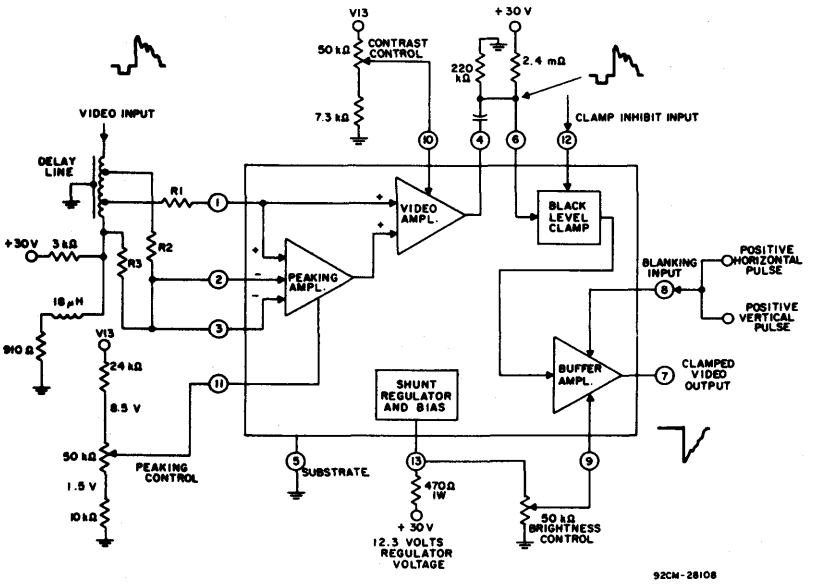


Fig. 1 – Functional block diagram.

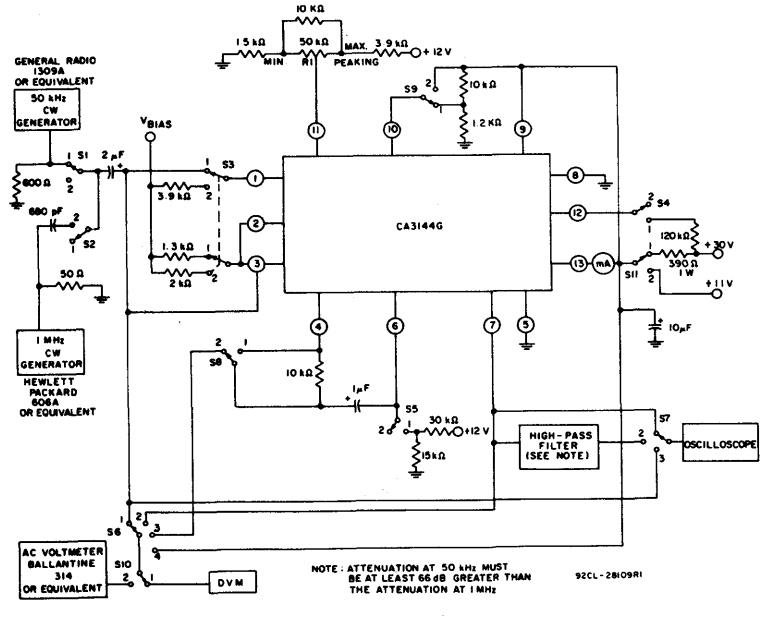


Fig. 2– Test circuit.

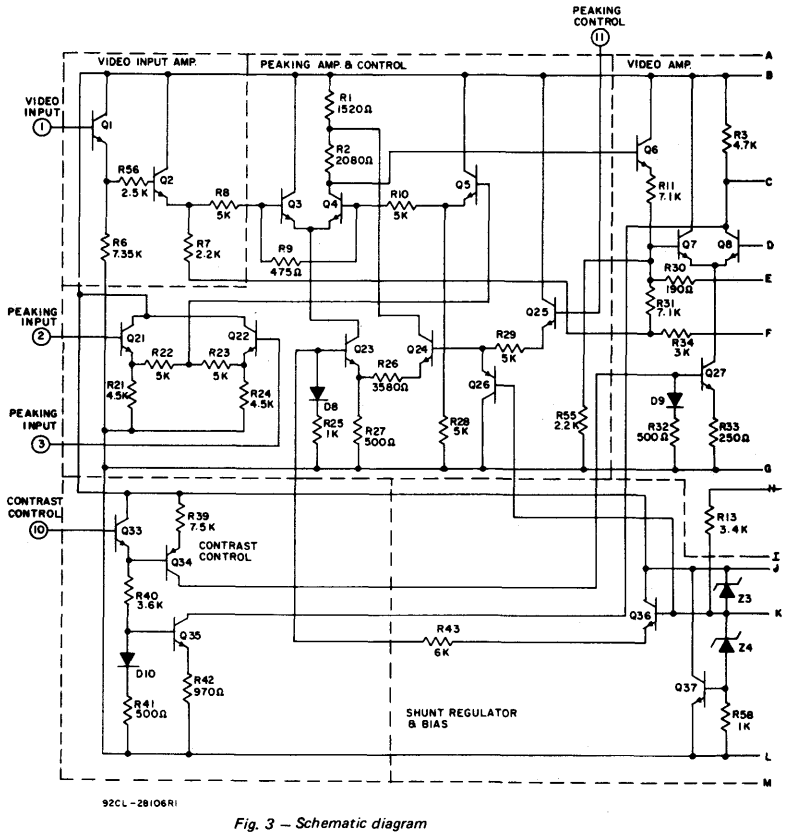


Fig. 3 – Schematic diagram

CIRCUIT DESCRIPTION

Fig. 1 is a block diagram of the CA3144G indicating the internal functions as well as external circuitry and signals. The video input signal with negative-going sync is applied to the input of the tapped delay line. Signals from fixed taps of the delay line are applied to terminals 1, 2, and 3 of the CA-3144G. In referring to Fig. 4, the signal from the delay line tap A is applied to the video input at terminal 1. The signals from taps B and C are summed where $V_A + V_B = V_{sum}$. The signal (V_{sum}) is then applied to the parallel connection of the peaking input terminals, 2 and 3. The video input signal is applied to a non-inverting input of the peaking amplifier while the peaking input signal (V_{sum}) is applied to an inverting input of the peaking amplifier.

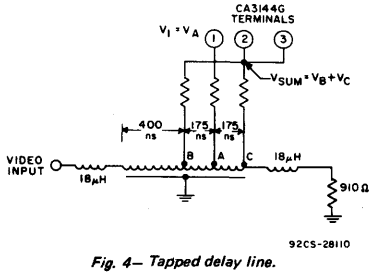


Fig. 4– Tapped delay line.

Low-frequency video components are unattenuated, while high-frequency components are attenuated as a function of the delay-line tap points. The peaking amplifier is a differential amplifier, so that the output is proportional to V_1 minus V_{sum}. At low frequencies, the signal at terminals 2 and 3 is unattenuated, and the peaking amplifier produces no output at these frequencies. However, at high frequencies the signal at ter-

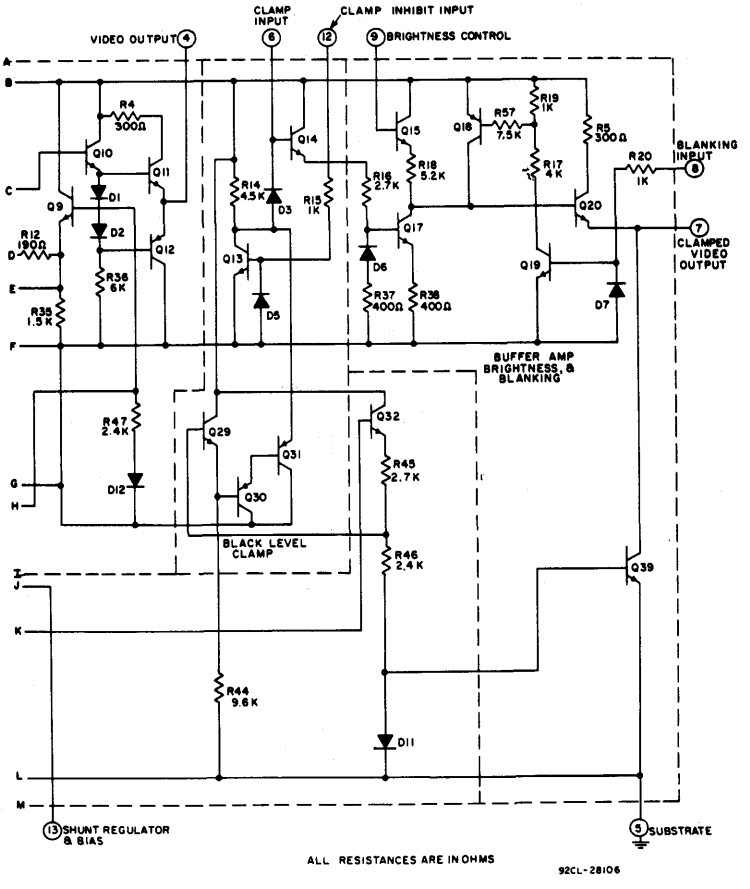


Fig. 3– Schematic diagram

minals 2 and 3 is attenuated thus, the peaking amplifier output consists of high-frequency video. The peaking control setting determines the amplitude of the peaking signal which is then fed to the video amplifier, where it is added to the video input signal and amplified. The setting of the peaking control does not substantially affect the dc quiescent voltage at terminal 4.

The low-impedance video amplifier output is at terminal 4. The signal is fed through an external coupling capacitor to terminal 6, the black-level clamp input. The action of the black-level clamp is such that it clamps to the black level rather than to the sync level. Refer to the circuit diagram in Fig. 1. Consider the situation where no signal is applied to terminal 12. Terminal 6 is biased through diode D3. The signal at terminal 6 will clamp its most negative excursion (sync pulse) to the anode voltage of D3. However, if a positive pulse is applied to terminal 12 during the sync interval, the anode of D3 is forced to ground due to saturation of Q13. The clamp is thus disabled, and terminal 6 will clamp to the next lower signal level, the black level.

The clamped video signal at terminal 6 is amplified and inverted at terminal 7. Blanking is accomplished by applying horizontal and vertical sync pulses to terminal 8. The pulses turn ON p-n-p transistor Q18 which shorts the base of transistor Q20 to the terminal 13 supply voltage. The brightness control function is accomplished by varying the voltage on terminal 9. The gain of the inverter stage remains constant, but the dc reference voltage follows the terminal 8 voltage. The contrast control function is accomplished by varying the voltage of terminal 10. Increasing the voltage on terminal 10 lowers the gain of the video amplifier. This reduction in gain does not substantially affect the dc quiescent voltage at terminal 4.

CA3151G

Single Chip TV Chroma Processor/Demodulator

"G" Suffix Type – Hermetic Gold-CHIP in Dual-In-Line Plastic Package

System Features:

- **All chroma processing and demodulating circuitry on a single chip in a 24-lead plastic package**
- **Phase-locked subcarrier regeneration utilizing sample-and-hold techniques**
- **Supplementary ACC with overload detector to prevent over saturation of the picture tube**
- **Linear dc controls for chroma gain and tint**
- **Dynamic "flesh correction" – corrects purple and green flesh colors without affecting primary colors**
- **Balanced chroma demodulators with low output impedance for direct coupling**
- **Internal rf filtering**
- **Requires few external components**
- **Low system dissipation—nominal 0.5 W**

The RCA-CA3151G is a monolithic silicon integrated circuit that performs the complete chroma processor and demodulating functions for color TV. The single chip contains all the features of the CA3126 chroma processor and the CA3137 chroma demodulator.

The CA3151G is supplied in the hermetic Gold-CHIP 24-lead dual-in-line plastic package (G suffix). The transistor chips used in the hermetic Gold-CHIP plastic packages are of the sealed-junction type designed to provide protection against the deteriorating effects of humidity and other surface contaminants without the need for a hermetic package enclosure. The semiconductor junctions are sealed by utilizing a silicon nitride passivation layer. A multi-layered, highly corrosion-resistant, terminal-connection system of unique design is employed.

MAXIMUM RATINGS, *Absolute-Maximum Values:*

DC SUPPLY VOLTAGE:
Between Terms. 18 and 7 . . . . . 13.2 V
DEVICE DISSIPATION:
Up to T_A = 55°C . . . . . 825 mW
Above T_A = 55°C . . . Derate linearly at 8.7 mW/°C
AMBIENT TEMPERATURE RANGE:
Operating . . . . . −40 to +85°C
Storage . . . . . −65 to +150°C
LEAD TEMPERATURE (During Soldering):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max. . . . . +265°C

Preliminary Data

ELECTRICAL CHARACTERISTICS at T_A = 25°C, V^+ = 11.6 V

CHARACTERISTIC	TEST CONDITIONS							TYPICAL VALUE	UNITS
	S_1	S_2	S_3	Chroma In	Burst In	V_4	V_{17}		
STATIC (See Fig. 1)									
Supply Current, I_T								42	mA
R-Y, G-Y, B-Y, Outputs, V_8,V_9,V_{10}								5.3	Vdc
Oscillator Reference Inputs, V_{11},V_{12}								3.7	Vdc
Chroma Demodulator Input, V_{13}								2.9	Vdc
Chroma Processor Input, V_1								2.2	Vdc
DYNAMIC (See Fig. 2)									
Minimum Oscillator Pull-In Range*, V_{12}	2	1	1	273 mV_{p-p}		1.5 V	7 V	±300	Hz
Oscillator Level, V_{12}	2	1	1					0.6	V_{p-p}
100 Percent ACC, V_{13}	1	1	1					1	V_{p-p}
Minimum Gain Control, V_{13}	1	1	1			11.6 V		20	mV_{p-p}
50 Percent Gain Control, V_{13}	1	1	1			6 V		50	% of 100% ACC Value
200 Percent ACC, V_{13}	1	1	1	546 mV_{p-p}		1.5 V		100	% of 100% ACC Value
20 Percent ACC, V_{13}	1	1	1	54.6 V_{p-p}				100	% of 100% ACC Value
Maximum Kill Output, V_{13}	1	1	1	54.6 mV_{p-p}	4 mV_{p-p}			20	mV_{p-p}
Minimum Unkill Output, V_{13}	1	1	1		30 mV_{p-p}			400	mV_{p-p}
Overload Detector (OLD), V_{13}	1	1	2	546 mV_{p-p}	273 mV_{p-p}			1	V_{p-p}
R-Y Sensitivity, V_{10} E_g = 282 mV_{p-p}, 3.53 MHz	1	2	1	0				0.8	V_{p-p}
R-Y Ratio B-Y/R-Y, V_8**	1	2	1					120	%
G-Y Ratio G-Y/R-Y, V_9**	1	2	1					33	%
Max. R-Y Output, V_{10} E_g = 2 V_{p-p}, 3.53 MHz	1	2	1					3	V_{p-p}
Minimum Tint Control Range, ϕ13	1	1	1	273 mV_{p-p}			0 V to 11.6 V	80	Degrees

* Tune C_2 to 3,579,845 Hz with S_1 in position 2. Put S_1 in position 1, and check for pull in. Repeat for frequency tuned to 3,579,245 Hz. For other tests, frequency tuned to 3,579,545 ± 10 Hz.

** All input levels up to 2 V_{p-p}.

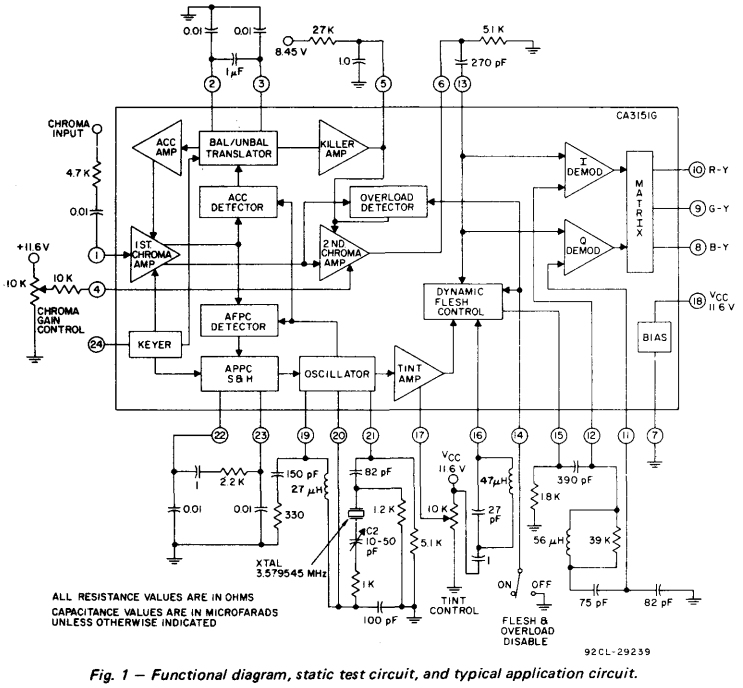


Fig. 1 – Functional diagram, static test circuit, and typical application circuit.

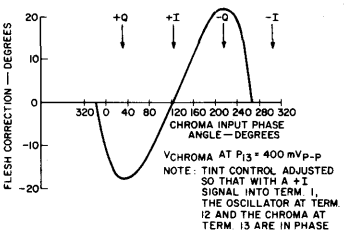
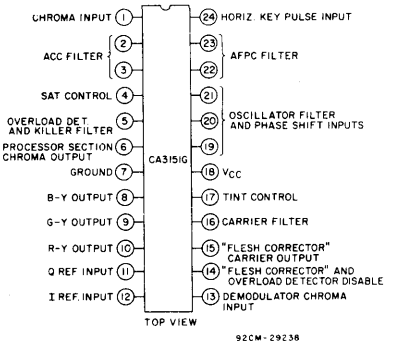


Fig. 2 – "Flesh" correction of oscillator phase angle as a function of chroma input phase angle.

TERMINAL DIAGRAM



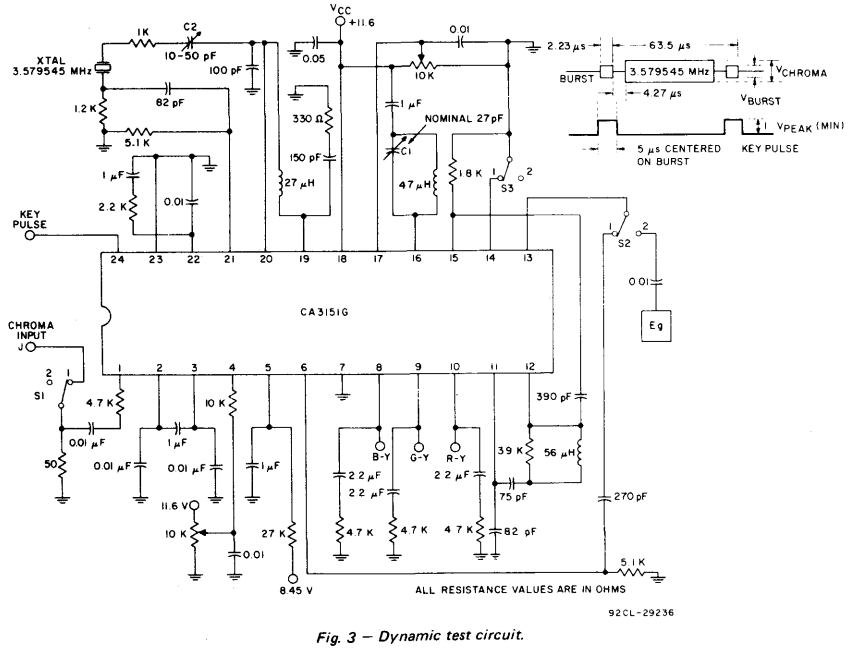


Fig. 3 – Dynamic test circuit.

CA3153G

Television Video IF System

"G" Suffix Type -- Hermetic Gold-CHIP in Dual-In-Line Plastic Package

The RCA-CA3153G is a monolithic silion integrated circuit designed to perform if amplification, video detection, and video-amplifier functions in color and monochrome TV receivers. The signal-to-noise performance has been improved compared to the RCA-CA3068*. The AGC performance has also been improved through the use of a sample and hold keyed system. The RCA-CA3153G is designed to interface with the RCA-CA3139# Automatic Fine Tuning (aft) circuit, and intercarrier amplifier.

* The CA3068 is described in RCA data bulletin File No. 467.

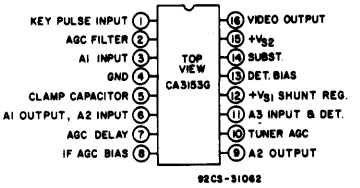
The CA3153G is supplied in the hermetic Gold-CHIP 16-lead dual-in-line plastic package (G suffix). The transistor chips used in the hermetic Gold-CHIP plastic packages are of the sealed-junction type designed to provide protection against the deteriorating effects of humidity and other surface contaminants without the need for a hermetic package enclosure. The semiconductor junctions are sealed by utilizing a silicon nitride passivation layer. A multi-layered, highly corrosion-resistant, terminal-connection system of unique design is employed.

\# The CA3139 is described in RCA data bulletin File No. 905.

System Features:

- **Improved agc**
  Fast response
  Sample and hold keyed
- **High gain wideband IF amplifiers**
- **Delayed agc output for tuner**
- **Gain reduction with excellent stability**
- **Linear video detector**
- **Video amplifier**
- **Low noise**
- **Internal shunt regulator**
- **For color or monochrome**
- **Gold-chip metalization**

TERMINAL DIAGRAM



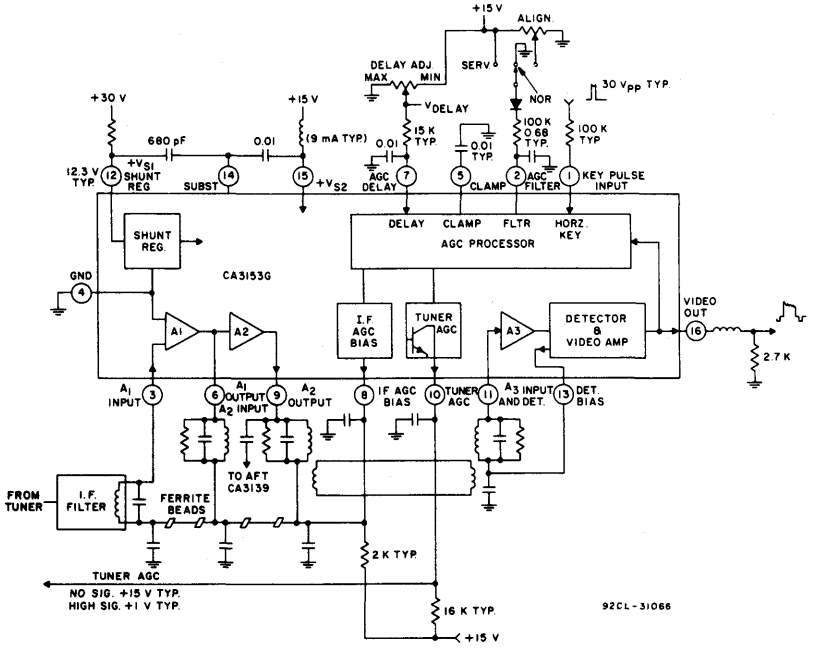


Fig. 1 – Functional block diagram of the IF amplifier-system of CA3153G with typical peripheral circuitry.

MAXIMUM RATINGS, *Absolute-Maximum Values:*

DC SUPPLY VOLTAGE:
- Between Terms. 15 and 4 . . . . . 16 V
- Between 470 Ω connected to Term. 12 and 4 . . . . . 35 V

DC SUPPLY CURRENT:
- At Term. 15 . . . . . 20 mA
- At Term. 12 . . . . . 30 mA

DEVICE DISSIPATION:
- Up to T_A = +55°C . . . . . 750 mW
- Above T_A = +55°C . . . . . Derate linearly at 7.9 mW/°C

AMBIENT TEMPERATURE RANGE:
- Operating . . . . . −40 to +85°C
- Storage . . . . . −65 to +150°C

LEAD TEMPERATURE (During Soldering):
- At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max. . . . . +265°C

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTIC	TEST CONDITIONS	LIMITS		UNITS
		Min.	Max.	
Operating Supply Voltage, V_{15}	See Note 1	12	14.2	V
Supply Current, I_{15}		3	15	mA
Shunt Regulator Voltage, V_{12}		10.9	13	V
Shunt Regulator Current, I_{12}	V_{12} = 10.5 V	6	20	mA
Tuner AGC High Voltage, V_{10}		18.5	21	V
Tuner AGC Low Voltage, V_{10}		0.3	1.3	V
AGC Current, I_2	Non-Keyed	80	500	μA
AGC Current (Peak), I_2	Keyed Source Current	0.7	3	mA
AGC Current (Peak), I_2	Keyed Sink Current	150	680	μA
Horizontal Key Input	Through 100 kΩ connected to Term. 1	25	35	V
Video Output High Voltage, V_{16}	At Zero Carrier	7	10	V
Video Output Low Voltage, V_{16}	At 30 mV Input	0.9	2	V
Sensitivity Voltage, V_{16}	At 400 μV Input	0.9	5	V
Noise		–	12	$mV_{(RMS)}$
Chroma	45.75 MHz, 10 mV; 42.17 MHz, 3 mV	0.7	1.6	$V_{(RMS)}$
AFT Drive		35	85	$mV_{(RMS)}$
Distortion	50 kHz, 80% Modulated, Sync TIP Equiv. 30 $mV_{(RMS)}$	–	10	%
Delay Voltage	Through 15kΩ connected to Term. 7. See note 2	0	V_{15}	V

Note 1: V_{15} MIN. should be at least 0.6 V above Terminal 12 potential. Lower voltage may cause some "white" compression.

Note 2: Zero voltage corresponds to maximum delay at signal input = 30 $mV_{(RMS)}$.

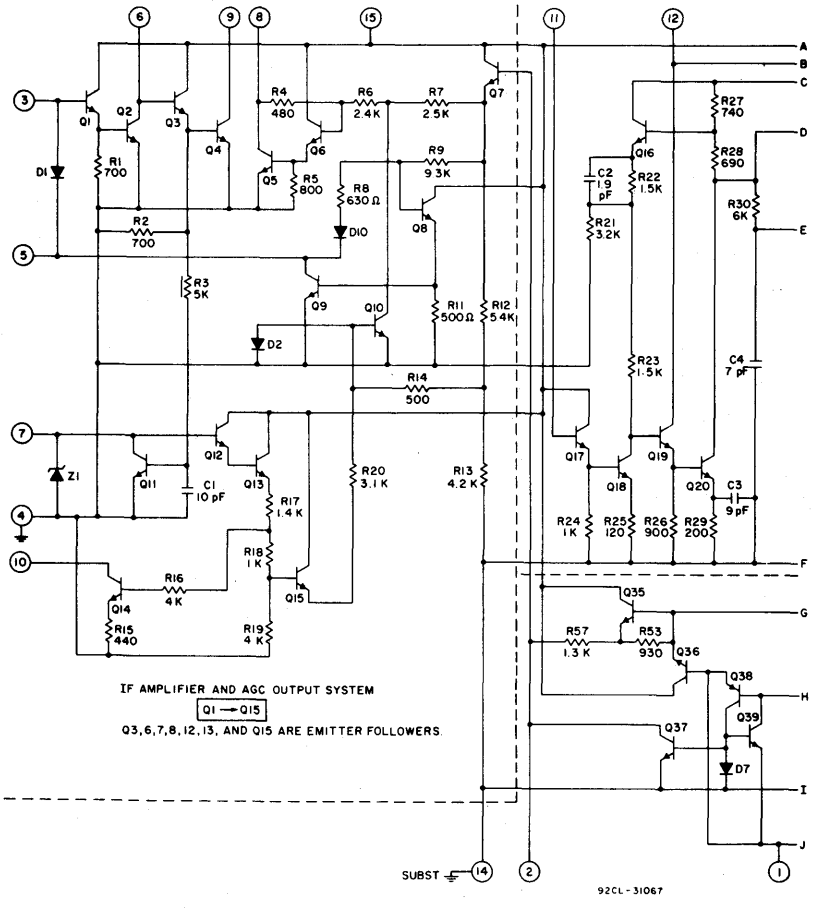


Fig. 2 – Schematic diagram for the CA31536.

AGC System (See Fig. 3)

The AGC system employs a sample-and-hold system to allow a fast-acting agc and reduce the effect of the vertical synchronizing signal on the video output stage. An override path is provided to allow a lower-gain agc system when the key pulse is not locked to the sync signal (for example, during channel selection).

The negative-going sync signal at the video output, Terminal 16, is applied to transistor Q41 through resistors R51 and R52 which act as current-limiting and filtering components. The sync signal is inverted and amplified by transistor Q41. The video portion of the signal is cutoff by the saturation voltage of Q41. When the TV system is in synchronization, the positive sync pulse at the collector of Q41 is coincident with the key input at Terminal 1, Transistor Q42 is turned off by the key pulse. Capacitor C13 is charged by the positive sync pulse through diode D9. The amplitude of the potential at C13 is proportional to the video-signal amplitude. The voltage is transfered through transistors Q40,

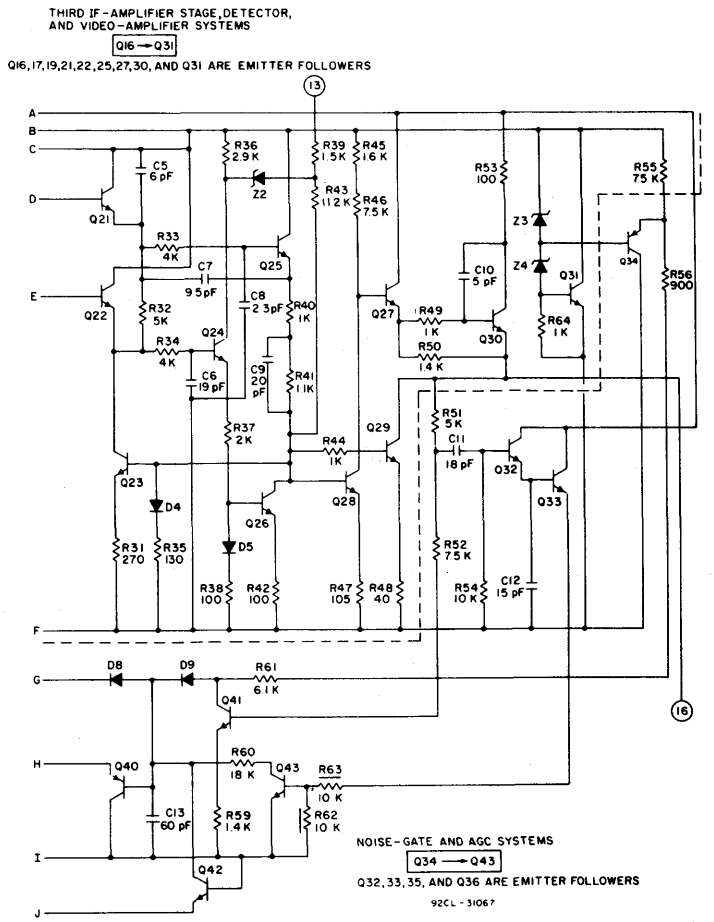


Fig. 2 – Schematic diagram for the CA3153G.

Q38, Q36, and Q35 to resistor R57 to form the charge current for the external agc filter capacitor at Terminal 2.

A constant-current discharge path for the capacitor at Terminal 2 is provided by current mirror components D7 and Q37 during the key-pulse duration. Thus the external agc filter capacitor is charged or discharged during the key-pulse interval only by the difference in current between the charge- and discharge currents. At the end of the key-pulse duration, C13 is discharged, and the charge and discharge current paths at Terminal 2 are turned off. Diode D8 provides a lower-gain agc path for turn-on during channel acquisition.

Noise-Gate System (See Fig. 3)

The circuit components, C11, R54, Q32, Q33, and Q43 perform the function of a statistical system to reduce agc gain during "spike" noise. The noise gate turns on for large amplitude fast signals and reduces the agc loop gain.

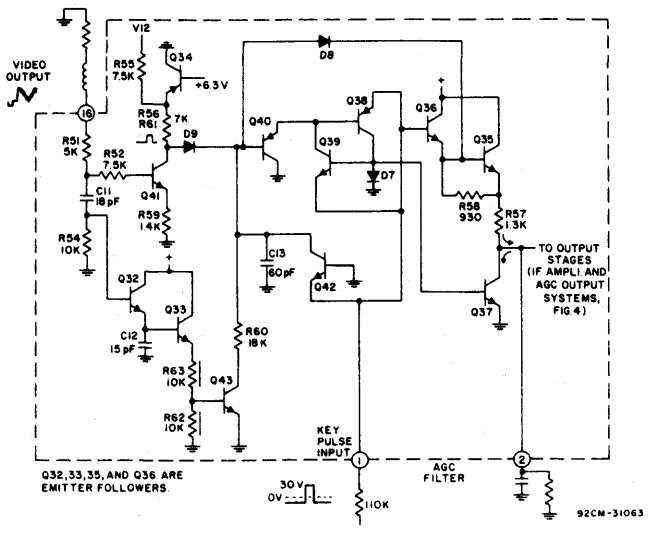


Fig. 3 – Noise-gate and AGC system of CA3153G (Q34 → Q43).

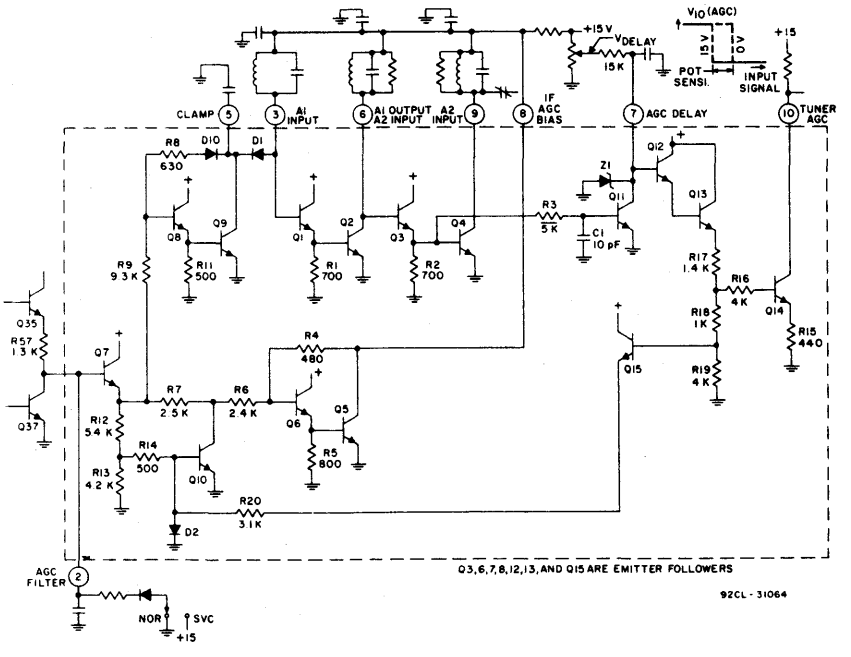


Fig. 4 – IF amplifier and AGC output system of CA3153G (Q1 → Q15).

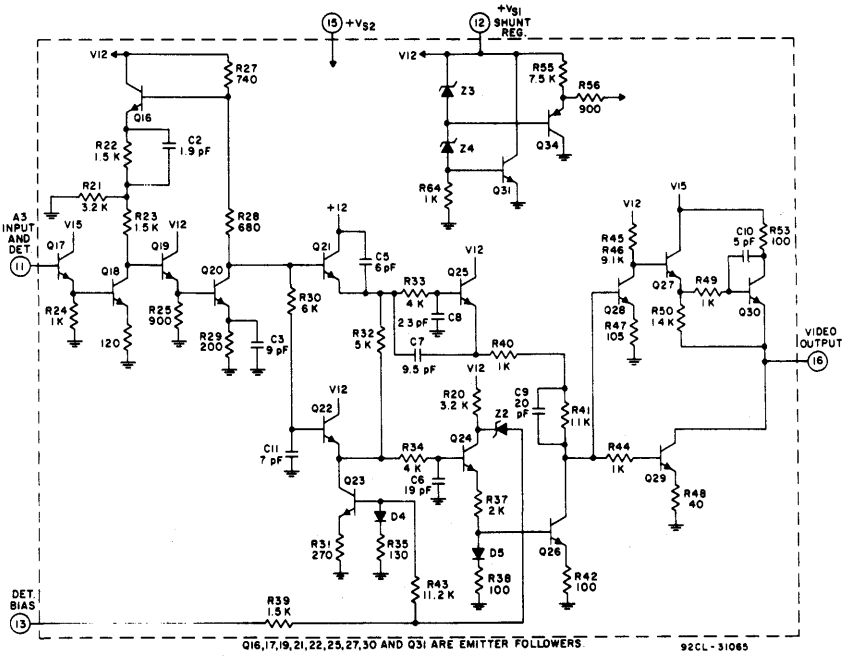


Fig. 5 – Third IF-amplifier stage, detector, and video-amplifier systems of CA3153G (Q16 → Q31).

CA3159G

Preliminary Data

Horizontal Processor and AGC Detector

The CA3159G is a monolithic integrated circuit designed for use as a horizontal processor and AGC detector in color or black-and-white TV receivers. It performs the functions of AGC, sync separation, and noise immunity, and a 31.5 kHz oscillator is provided for use with vertical-countdown circuits.

The CA3159G is supplied in a 16-lead dual-in-line plastic package with a hermetic "Gold-CHIP" (G suffix). These chips are of the sealed-junction type designed to provide protection against the deteriorating effects of humidity and other surface contaminants without the need for a hermetic package enclosure.

Features:

- **AGC voltage**
- **Separated sync**
- **31.5 kHz oscillator**
- **Gates AGC and sync for noise immunity**

MAXIMUM RATINGS, *Absolute-Maximum Values:*

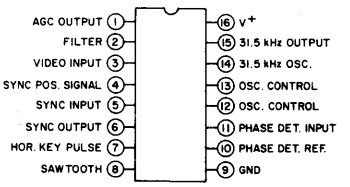
DC SUPPLY VOLTAGE . . . . . +30 V
DC SUPPLY CURRENT . . . . . 30 mA
DEVICE DISSIPATION:
Up to $T_A = 55^\circ C$ . . . . . 750 mW
Above $T_A = 55^\circ C$ . . . . . Derate linearly at 7.9 mW/°C
AMBIENT TEMPERATURE RANGE:
Operating . . . . . −40 to +85°C
Storage . . . . . −65 to +150°C
LEAD TEMPERATURE (DURING SOLDERING):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm)
from case for 10s max. . . . . +265°C

ELECTRICAL CHARACTERISTICS at $T_A = 25^\circ C$, $V^+ = 28$ V, all switches open unless otherwise specified. (See Fig. 2)

CHARACTERISTIC	TEST CONDITIONS	TERM. MEAS.	TYPICAL VALUES	UNITS
AGC Voltage	S1, S9 closed	1	1.85	V
Noise Inverter[1]	S2, S9 closed	1	0.7	V
Shift Threshold[1]	S2, S3 closed	1	20	V
Sync Level	S1, S9 closed	4	18	V
Positive Pulse[2]	S5 closed	7	25	V
Positive Sawtooth[3]	S5, S6 closed	8	3	V
Sync Low	S3, S4 closed	6	1.5	V
Supply Current		16	20	mA
Free-Running Freq.[4]	S7,S8,S9 closed	15	31.5	kHz
Duty Cycle	S7,S8,S9 closed	15	48	%

1. A = 3 V, B = 1.2 V, − 1 mA to term. I
2. C = 0.2 mA
3. C = 0.2 mA, D = 5 mA
4. Adjust LI, $V^+ = 20$ V

CA3159G
TERMINAL ASSIGNMENT



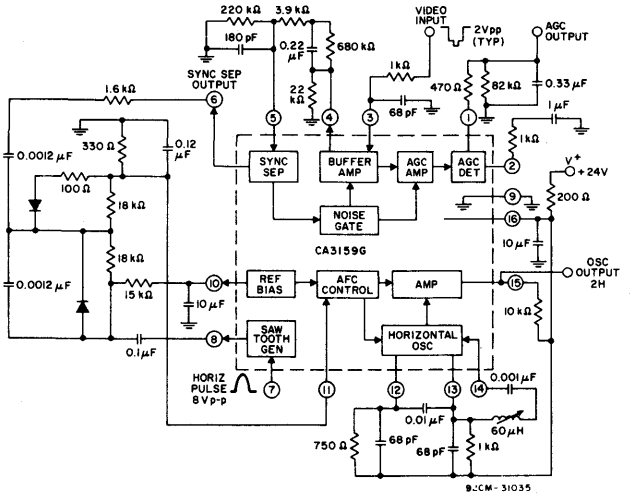


Fig. 1 – Functional block diagram of CA3159G.

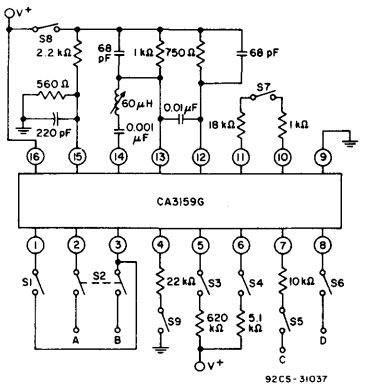


Fig. 2 – DC test circuit.

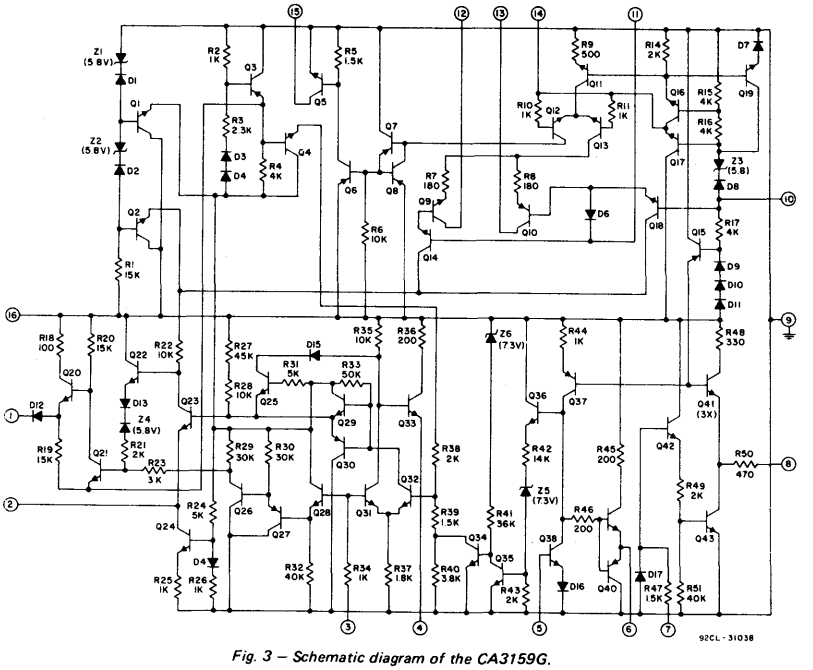


Fig. 3 – Schematic diagram of the CA3159G.

Circuit Description

The negative sync video input at terminal 3 is the detected video if. This video signal is buffered and V_{be} compensated by emitter-followers Q28, Q27, and Q26. The buffered video signal is applied between the base of Q21 and a temperature-stable 2-V reference. Q21 is normally in saturation, and the negative sync pulse imparts a positive swing to the base of Q20. Q20 is used as a peak rectifier driving a capacitor at terminal 1. The voltage at terminal 1 is the AGC control voltage that sets the if gain such that the sync pulses drop to just below the 2 V level, driving Q21 out of saturation.

The above description is for a normal video signal; the presence of noise pulses more negative than the sync tip level would lower the gain to that level, thus disturbing the picture. A gated noise-inversion threshold is provide at the base of Q32 to compensate for these noise pulses. The threshold is about 1.5 V during trace time, but is reduced to about 1 V during coincidence of the sync and flyback pulses. When the video signal is more negative than the noise threshold, Q32 conducts and pulls the base and emitter of Q30 low. Without noise, Q23 conducts 0.5 mA with its collector at 7 V, which holds Q22 in cutoff. Q29 has an emitter load provided by an external 1 kΩ resistor and a series capacitor: when its base is switched low, its collector switches high. The resulting flow of current in Q23 overrides the normal negative-going pulse in the direct signal path and holds Q21 in saturation.

The video input to terminal 3 also operates the sync channel, beginning with Q31. Because Q32 is normally cut off, Q31 acts as an amplifer with a moderate gain to its collector, and a positive sync signal appears at terminal 4. If the noise pulse is more negative than the noise threshold at the base of Q32, the base of Q30 is pulled down as discussed above. In addition to operating the AGC noise inverter, the Q30 current passes through Q25 to the amplifier load resistor, R35, and cancels the potentially positive pulse at that point.

The positive sync signal at terminal 4 is coupled through an RC network to terminal 5 for sync separation. In essence, the network permits Q38 to clamp the positive peaks, so the most positive part of the signal is amplified by Q38 while the rest is beyond cutoff. The separated sync, a negative pulse at the collector of Q38, follows two paths. First, the sync operates an output driver to terminal 6, which drives the outboard diode phase detector. Second, the negative pulse cuts off the current through Q36, which otherwise holds Q35 in saturation, thus enabling a current in R41 to turn Q34 on and thereby shift the noise threshold voltage.

Terminal 7 receives a positive flyback pulse that supplies R41 with the signal to complete the coincidence gate that alters the noise threshold when sync and flyback pulses are in phase. The buffered and clipped flyback pulse also turns Q43 on, which, in conjunction with an external integrating capacitor, forms a sawtooth waveform. This sawtooth (at flyback rate) is phase compared with the sync pulse that was separated from the video input.

The phase detector works against an internal bias point brought out to terminal 10, and the phase detector output applied to terminal 11 is slightly positive or negative relative to terminal 10. This voltage differential with terminal 10 determines the division of current between Q9 and Q10, which are part of the voltage controlled oscillator. The oscillator consists of the current source Q11, differential amplifier Q12 and Q13, and differential amplifier Q9 and Q10. The frequency is determined primarily by a series LC circuit connected between terminals 13 and 14 (terminals 12 and 13 have resistor loads to the positive supply). If the entire oscillator current passes through Q10 to terminal 13, the oscillator operates at the frequency at which the phase shift in the LC circuit is zero. If the current is sent through Q9 to terminal 12, however, it must go through an external capacitor between terminals 12 and 13 and then through the original LC circuit and the circuit is tuned differently. Intermediate proportions of current division will produce intermediate oscillator frequencies. The oscillator current output from Q12 provides base drive for the 31.5 kHz output at terminal 15.

CA3163G

Preliminary Data

VHF/UHF Prescaler

Features:

- **Broadband operation – 90 to 1000 MHz**
- **High sensitivity**
- **Standard 5 V power supply**
- **Dual mode operation – VHF/UHF**
- **Complementary ECL outputs**
- **Independent VHF & UHF input terminals**
- **Hermetic Gold-CHIP construction**

The RCA-CA3163G* is an integrated-circuit prescaler intended for use in TV frequency synthesis tuning systems over an input frequency range of 90 to 1000 MHz. It performs division by 256 in the uhf mode and division by 64 in the vhf mode.

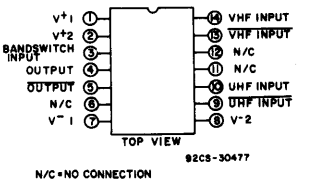
The mode of operation can be selected by means of the bandswitch and the separate uhf and vhf input terminals provided. The output is a complementary emitter-coupled stage with controlled slew rate for harmonic suppression.

All input terminals should be ac coupled to the appropriate input signal source. Because of high sensitivity, unbuffered coupling from the local oscillator is possible in most cases. In the uhf mode, which is activated by applying a high level to the bandswitch input terminal, all eight divider stages are operative, resulting in division by 256. In the vhf mode, activated by a low level at the vhf input terminal, two divider stages are bypassed, resulting in division by 64. As a result, approximately the same range of output frequencies are generated for both the uhf and vhf TV bands. An internal amplifier/multiplexer provides this control while isolating both inputs and amplifying the vhf signal. In addition, harmonic output is reduced above 40 MHz by limiting output signal rise and fall times and maintaining a balanced load.

The CA3163G is supplied in the 14-lead dual-in-line hermetic Gold-CHIP package. The chips used are of the sealed-junction type designed to provide protection against the deteriorating effects of humidity and other surface contaminants without the need for a hermetic package enclosure.

* Formerly RCA Developmental No.TA10535.

TERMINAL DIAGRAM



N/C = NO CONNECTION

MAXIMUM RATINGS, *Absolute-Maximum Values:*

DC SUPPLY VOLTAGE	5.5	V
DC BANDSWITCH VOLTAGE	20	V
RMS INPUT VOLTAGE	0.5	V
DEVICE DISSIPATION:		
UP TO T_A = 70°C	600	mW
ABOVE T_A = 70°C, derate linearly at	7.5	mW/°C
AMBIENT TEMPERATURE RANGE:		
OPERATING	0 to 70	°C
STORAGE	−55 to +150	°C
LEAD TEMPERATURE (DURING SOLDERING):		
AT DISTANCE 1/16 ± 1/32 INCH (1.59 ± 0.79 MM) FROM CASE FOR 10 SECONDS MAX.	+265	°C

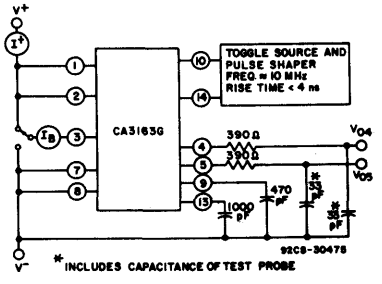


Fig. 1 – DC characteristics test circuit.

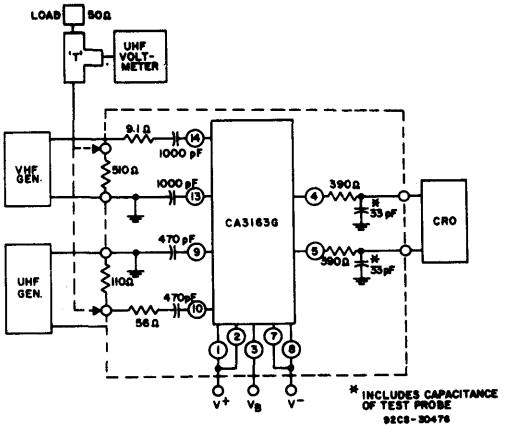


Fig. 2 – AC characteristics test circuit.

ELECTRICAL CHARACTERISTICS At T_A = 25°C, V^+ = 5 VDC, V^- = 0 VDC; see Figs. 1 & 2

CHARACTERISTIC	TEST CONDITIONS	LIMITS			UNITS
		Min.	Typ.	Max.	
Supply Current, I^+	Terms. (1+2), Fig. 1	30	60	90	mA
UHF Bandswitch Input Voltage, V_{BH}	High level	2.4	–		V
VHF Bandswitch Input Voltage, V_{BL}	Low level	–	–	0.8	V
UHF Bandswitch Input Current, I_{BH}	V_{BH} = 20 VDC, Fig. 1	–	–	0.5	mA
VHF Bandswitch Input Current, I_{BL}	V_{BL} = 0 VDC, Fig. 1	–	–	–1	mA
UHF Sensitivity Level Input Voltage, $V_{IN}(U)$	f_{IN} = 450 to 950 MHz, $f_{OUT} = f_{IN}/256$, Fig. 2	–	–	80	mVRMS
VHF Sensitivity Level Input Voltage, $V_{IN}(V)$	f_{IN} = 90 to 275 MHz, $f_{OUT} = f_{IN}/64$, Fig. 2	–	–	40	mVRMS
Output Voltage, V_O	Terms, 4 or 5, Fig. 2	0.65	1	–	V_{p-p}
Output Voltage Rise of Fall Time, t_r.t_f		–	70	–	ns

CA3166E

Preliminary Data

Operational Amplifier Bandswitch

BiMOS input operational amplifier, frequency band-select switch, and AFT mode switch

For Frequency-Synthesizer Television Tuning Systems

The RCA-CA3166E incorporates bipolar, PMOS, and CMOS processes on a single monolithic chip to provide three functional blocks for use in frequency-synthesizer type TV tuners. Included are an input operational amplifier, a band-select switch, and an AFT mode switch.

The operational amplifier features internal bias and phase compensation, high-impedance PMOS input transistors, diode clipper input limiting, output short-circuit protection, and static charge protection. The operational amplifier is used to amplify an error signal that is proportional to the detected phase difference between the desired channel frequency and an internally generated reference signal.

The band-select switch has two logic inputs, a control voltage input, and three outputs (UHF, VHF Low, VHF High) with a drive capability of 90 mA each, for controlling the tuner varactor diodes.

The AFT mode switch is a CMOS transmission gate with static charge protection and an enable logic input for selecting the "AFT ON" or "AFT DEFEAT" mode.

The CA3166E is supplied in the 14-lead dual-in-line plastic package.

Features:

- **Three independent functions – input operational amplifier, AFT mode switch, and band-select switch**
- **Input operational amplifier has internal biasing circuitry and high-impedance PMOS input transistors**
- **Internal diode clipper limiting at operational amplifier inputs and short-circuit protection at the outputs**
- **Static charge protection for both PMOS and CMOS circuit components**
  - **AFT mode switch utilizes CMOS transmission gate and enable logic inputs control AFT mode**
  - **Logic-controlled band-select switch**
  - **Three band-select-switch outputs, each with 90-mA drive capability (typ.) at input voltage up to 28 V dc**
  - **High voltage-rating for wide dynamic control range of error signals and switch functions**

MAXIMUM RATINGS, *Absolute-Maximum Values:*

DC SUPPLY-VOLTAGE RANGE, V^+	+29.5 to +35 V
DIFFERENTIAL INPUT VOLTAGE	±1.5 V
DC SUPPLY CURRENT I^+	20 mA
BAND-SELECT SWITCH INPUT VOLTAGE RANGE, V_{BS}	0 to +28 V
DEVICE DISSIPATION:	
Up to +70°C	600 mW
Above +70°C	Derate linearly at 11.1 mW/°C
AMBIENT TEMPERATURE RANGE:	
Operating	0 to +70°C
Storage	−55 to +150°C
LEAD TEMPERATURE (DURING SOLDERING):	
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max.	+265°C

Bandswitch Truth Table

LOGIC INPUTS		OUTPUTS		
A	B	UHF	VHF LOW	VHF HIGH
0	0	0	0	0
0	1	0	1	0
1	0	0	0	1
1	1	1	0	0

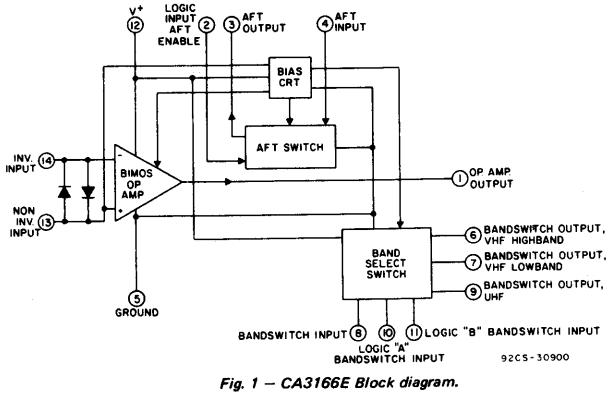
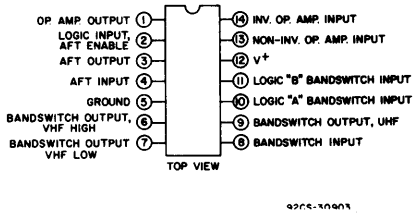


Fig. 1 – CA3166E Block diagram.

CA3166E TERMINAL ASSIGNMENT



Operational Amplifier (See Fig. 2)
Electrical Characteristics at T_A = 25°C, $V^{\pm}$ = 32.5 V, V_{BS} = 18 V, Terms 4 & 5 grounded

CHARACTERISTIC	TEST CONDITIONS	TYPICAL VALUES	UNITS
Input Bias Voltage, V_{13}	I_{13} = 4 mA, Feedback = 1 MΩ	2.5	V_{DC}
Input Bias Voltage, V_{13}	I_{13} = 6 mA, Feedback = 1 MΩ	2.6	V_{DC}
Input Bias Voltage, V_{14}	I_{14} = 4 mA, Feedback = 1 MΩ	3.3	V_{DC}
Diode Voltage (term. 14 to term. 13)	I_{14} = 4 mA, Term. 13 = Reference	0.8	V_{DC}
Diode Voltage (term. 13 to term. 14)	I_{13} = 4 mA, Term. 14 = Reference	0.8	V_{DC}
Output Voltage Low, V_{OL}	I_{14} = 4 mA, Resistance between Terms. 1 and 12 = 10 kΩ	0.2	V_{DC}
Output Voltage High, V_{OH}	V_{14} = 0 V, I_{13} = 4 mA, Resistance between Terms. 1 and 12 = 10 kΩ	28	V_{DC}
Input Offset Voltage, V_{IO}	V_{13} = 0 V, Term. 1 connected to Term. 14	10	mV
Supply Current, I^+	V_4 = 1 V, Feedback (Terms. 1 to 14) = 1 MΩ	14	mA
Output Sink Current, I_{OL}	I_{14} = 4 mA, V_1 = 32.5 V	25	mA
Output Source Current, I_{OH}	I_{13} = 4 mA, V_1 = V_{14} = 0V	−15	mA
Input Bias Current, I_{IB} (term. 14)	V_{13} = 0 V, Term. 1 connected to Term. 14	0.5	nA
Common-Mode Rejection Ration, CMRR		65	dB
Power Supply Rejection Ratio, PSRR		75	dB
Open-Loop Voltage Gain, A_{OL}		80	dB

Band-Select Switch (See Fig. 3)
Electrical Characteristics at T_A = 25°C, V^+ = 32.5 V, V_{BS} = 18 V, Terms. 4 & 5 grounded
Terms. 6, 7, 9 = 100 kΩ to ground

CHARACTERISTIC	TEST CONDITIONS	TYPICAL VALUES	UNITS
Logic Inputs "A" & "B" Sink Current		100	μA
Logic Inputs "A" & "B" Source Current	I_9 = −90 mA, V_{10} = V_{11} = 2.4 V	−5	μA
Output Leakage Current, Terms. 6, 7, 9		2	μA
Output Saturation Voltage: Term. 9	I_9 = −90 mA, V_{10} = V_{11} = 2.4 V	0.6	V
Term. 9	I_9 = −60 mA, V_{10} = V_{11} = 24 V	0.3	V
Term. 7	I_7 = −90 mA, V_{10} = 0 V, V_{11} = 24 V	0.6	V
Term. 7	I_7 = −60 mA, V_{10} = 0 V, V_{11} = 2.4 V	0.3	V
Term. 6	I_6 = −90 mA, V_{10} = 2.4 V, V_{11} = 0 V	0.6	V
Term. 6	I_6 = −60 mA, V_{10} = 24 V, V_{11} = 0 V	0.3	V

AFT Mode Switch

Electrical Characteristics at T_A = 25°C, V^+ = 32.5 V, V_{BS} = 18 V, Terms. 5, 10, 11 grounded

CHARACTERISTIC	TEST CONDITIONS	TYPICAL VALUES	UNITS
Logic Input Current Low	V_2 = 0 V, $R_{TERM.\ 3}$ = 10 MΩ, V_4 = 13.5 V	−100	μA
Logic Input Current High	V_2 = 2.4 V, $R_{TERMS.\ 3}$ = 1 kΩ, V_4 = 1 V	2	μA
Input Current	V_2 = 0 V, $R_{TERM.\ 3}$ = 10 MΩ, V_4 = 13.5 V	2	μA
Output Leakage Current	V_2 = 0.6 V, V_3 = 8 V, V_4 = 0 V	1	nA
Output Sink Current	V_2 = 2.4 V, V_3 = 1.8, V_4 = 0 V	2	mA
Output Offset Voltage	V_2 = 2.4 V, V_4 = 3 V	0.1	V
Output Voltage, "ON"	V_2 = 2.4 V, $R_{TERM.\ 3}$ = 1 kΩ, V_4 = 1 V	0.8	V
Output Voltage, "ON"	V_2 = 2.4 V, $R_{TERM.\ 3}$ = 1 kΩ, V_4 = 13.5 V	10	V

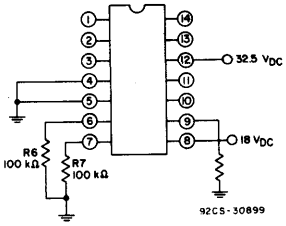


Fig. 3 – Bandswitch test circuit.

TEST	CONDITIONS (V)			
	V^+	V_{CM}	V_{BS}	V_{REF}
V_{01}	35	0	18	−15
V_{02}	33	−2	16	−17
V_{03}	29.5	0	18	−15
V_{04}	35	0	18	−5

$$CMRR = 20\log\frac{(V_{01}-V_{02})}{200}$$

$$PSRR = 20\log\frac{(V_{01}-V_{03})}{550}$$

$$A_{OL} = -20\log\frac{(V_{01}-V_{04})}{1000}$$

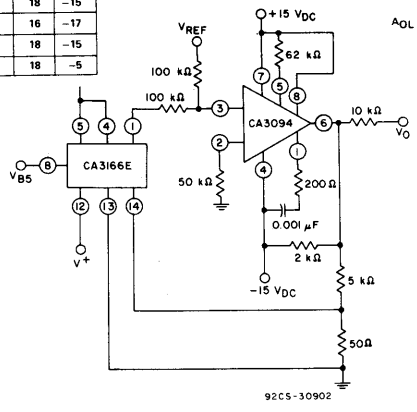


Fig. 2 – Operational amplifier test circuit for CMRR, PSRR, and A_{OL}.

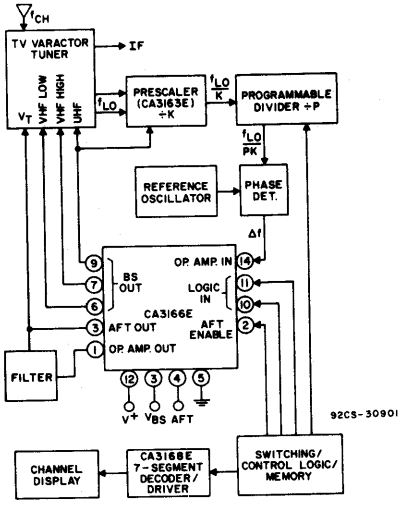


Fig. 4 – Block diagram of a typical digital tuning system.

Turner Operation

Fig. 4 shows a typical digital TV tuning system employing the CA3166E. This system consists of a phase-locked loop (PLL) and a programmable divider to generate a tuner local-oscillator frequency that is an integral multiple of a reference-oscillator frequency. The output of the local oscillator is connected to a prescaler (CA3163) which divides the frequency to values that can be processed by a programmable divider. The amount of division is established by the control logic and depends on the desired channel to be viewed. This signal and a reference signal are combined in a phase detector to produce an error signal proportional to the frequency separation. The error signal is then amplified by the CA3166E and filtered to provide a dc voltage to the varactors of the tuner voltage-controlled oscillator (VCO). The VCO frequency is thus corrected to reduce its difference with the reference.

Logic-control signals are applied to terminals 10 and 11 (band-select switch) of the CA3166E, and the proper varactor circuits for UHF, low-band VHF, or high-band VHF are selected. The truth table for the selection logic is shown on page 3.

An analog switch for AFT operation is included in the CA3166E for automatic correction of frequency transmission errors.

Preliminary Data

CA3168E

2-Digit BCD-to-7-Segment Decoder/Driver

For Common-Anode LED Displays

The RCA-CA3168E is a monolithic integrated circuit intended for 2-digit display such as "numbers" for TV and "CB" channel selection, and other 0-99 numerical or counting for consumer or industrial indicator applications. It consists of two independent BCD-to-7-segment decoder/drivers. Two sets of BCD inputs are buffered with p-n-p differential amplifier stages internally referenced to 1.7 V. Each of the eight input terminals draws less than 15 μA and is provided with an internal protection circuit.

Decoding is accomplished with I^2L ROM's. The fourteen output terminals are buffered with Darlington pairs driving common-emitter output transistors. Each output is capable of sinking 25 mA for an LED common-anode display device. The supply-voltage range (V_{CC}) is intended to be 4.5 V to 6 V. The output voltage (V_O) must not exceed 12 V, which provides for a wide range of common-anode voltage sources.

The CA3168E is supplied in the 24-lead dual-in-line plastic package.

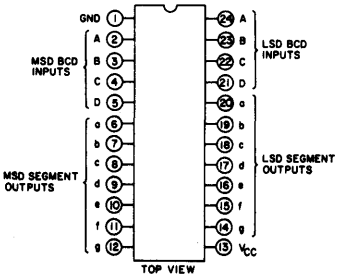
Features:

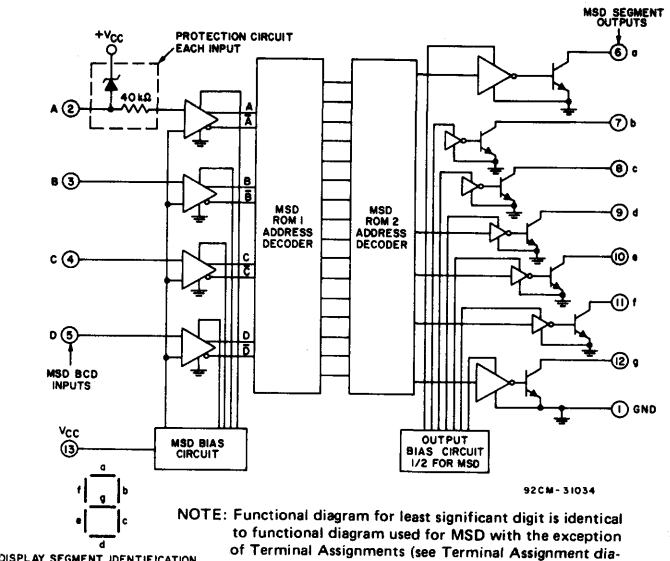
- **Separate BCD inputs and segment outputs for each digit**
- **Input loading less than 15 μA**
- **I^2L logic with buffered inputs and outputs**
- **Internal input overrange protection circuit**
- **5-V supply operation**
- **Internal biasing circuits**
- **Output drive capability of 25 mA per segment**
- **Open collector outputs drive indicators directly**

MAXIMUM RATINGS, *Absolute-Maximum Values:*

SUPPLY-VOLTAGE, V_{CC}	6 V
INPUT-VOLTAGE (MIN./MAX.)	$-0.3/V_{CC}$ V
INPUT CURRENT (PROTECTION CIRCUIT)	±10 mA
OUTPUT VOLTAGE, V_O	12 V
OUTPUT SEGMENT CURRENT, $I_{DISPLAY}$	25 mA
AMBIENT TEMPERATURE RANGE:	
Operating	0 to +70°C
Storage	−55 to +150°C
POWER DISSIPATION:	
Up to +70°C	400 mW
Above +70°C	derate linearly at 8.7 mW/°C
LEAD TEMPERATURE (DURING SOLDERING):	
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max.	+265°C

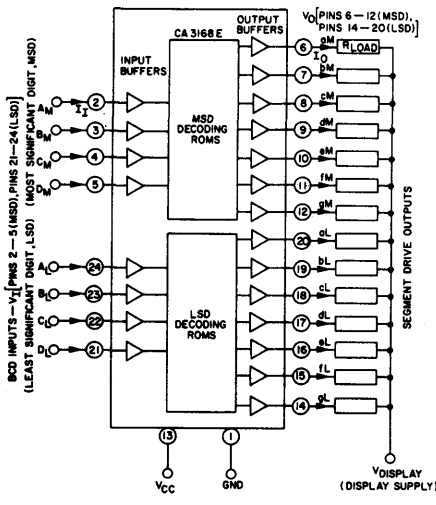
CA3168E
TERMINAL ASSIGNMENT





NOTE: Functional diagram for least significant digit is identical to functional diagram used for MSD with the exception of Terminal Assignments (see Terminal Assignment diagram). A separate LSD Bias circuit, and ½ of the Output Bias Circuit is used for LSD.

Fig. 1 – Functional diagram for Most Significant Digit (MSD).



NOTE: See truth table for test sequence of input/output logic tests and Minimum $R_{LOAD} = \frac{V_{DISPLAY} - V_{OL}}{\text{Max. } I_{DISPLAY}}$ For each of the 14 segment drive output terminals. (LED is not used in test circuit)

Fig. 2 – Test circuit.

TYPICAL ELECTRICAL CHARACTERISTICS at V_{CC} = 5 V, V_1 = GND, V_{DISP} = 12 V, and T_A = 25°C, See Fig. 2
Unless Otherwise Specified

CHARACTERISTIC	TEST CONDITIONS	LIMITS			UNITS
		Min.	Typ.	Max.	
Input Voltage High, V_{IH}		2.4	5	V_{CC}	V
Input Voltage Low, V_{IL}		0	–	0.6	V
Input Current High, I_{IH}	All BCD Inputs = 5 V	–	–	15	μA
Input Current Low, I_{IL}	All BCD inputs = 0 V	–10	–	–	μA
On-State Output Voltage, V_{OL}	$I_{O(Sink)}$ = 25 mA	–	–	1	V
Off-State Output Current, I_{OH}		–	5	50	μA
Power Supply Drain Current, I_{CC}	V_{CC} = 6 V	–	17	25	mA
Input Capacitance, C_I		–	5	–	pF

TRUTH TABLES

Most Significant Digit (MSD)

INPUTS				OUTPUTS							DISPLAY
A	B	C	D	a	b	c	d	e	f	g	
0	0	0	0	0	0	0	0	0	0	1	0
0	0	0	1	1	0	0	1	1	1	1	1
0	0	1	0	0	0	1	0	0	1	0	2
0	0	1	1	0	0	0	0	1	1	0	3
0	1	0	0	1	0	0	1	1	0	0	4
0	1	0	1	0	1	0	0	1	0	0	5
0	1	1	0	0	1	0	0	0	0	0	6
0	1	1	1	0	0	0	1	1	1	1	7
1	0	0	0	0	0	0	0	0	0	0	8
1	0	0	1	0	0	0	0	1	0	0	9
1	0	1	0	0	1	1	0	0	0	1	C
1	0	1	1	0	0	0	1	0	0	0	A
1	1	0	0	0	0	1	1	0	0	0	P
1	1	0	1	0	1	1	0	0	0	0	E
1	1	1	0	1	1	1	1	1	1	0	–
1	1	1	1	1	1	1	1	1	1	1	BLANK

Least Significant Digit (LSD)

INPUTS				OUTPUTS							DISPLAY
A	B	C	D	a	b	c	d	e	f	g	
0	0	0	0	0	0	0	0	0	0	1	0
0	0	0	1	1	0	0	1	1	1	1	1
0	0	1	0	0	0	1	0	0	1	0	2
0	0	1	1	0	0	0	0	1	1	0	3
0	1	0	0	1	0	0	1	1	0	0	4
0	1	0	1	0	1	0	0	1	0	0	5
0	1	1	0	0	1	0	0	0	0	0	6
0	1	1	1	0	0	0	1	1	1	1	7
1	0	0	0	0	0	0	0	0	0	0	8
1	0	0	1	0	0	0	0	1	0	0	9
1	0	1	0	1	0	0	1	0	0	0	H
1	0	1	1	1	0	0	0	0	1	1	J
1	1	0	0	1	1	1	0	0	0	1	L
1	1	0	1	0	1	1	1	0	0	0	F
1	1	1	0	1	1	1	1	1	1	0	–
1	1	1	1	1	1	1	1	1	1	1	BLANK

TV Chroma System

"G" Suffix Type-Hermetic Gold-CHIP in Dual-In-Line Plastic Package

The RCA-CA3170G is a monolithic silicon integrated circuit that performs the functions of subcarrier regeneration, ACC and APC detection, and tint control in color television receivers. It is designed to function compatibly with the CA3121E TV Chroma Amplifier/Demodulator in a 2-package chroma system.

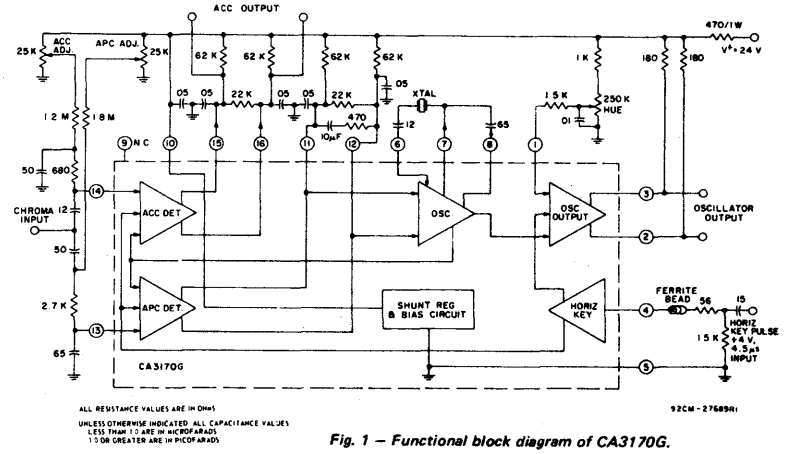
The CA3170G is a TV Chroma System of advanced design that incorporates all the features of the CA3070E but with the added advantage of the modified Hue Control Characteristic. With the CA3170G, the designer can provide a front panel hue control that functions linearly over its entire range, a particularly desirable consumer feature.

MAXIMUM RATINGS, *Absolute-Maximum:*

DEVICE DISSIPATION:*
- Up to T_A = 55°C . . . . . 750 mW
- Above T_A = 55°C . . . . . derate linearly 7.9 mW/°C

AMBIENT-TEMPERATURE RANGE:
- Operating . . . . . −40 to +85°C
- Storage . . . . . −65 to +150°C

LEAD TEMPERATURE (During soldering):
- At distance 1/16 ±1/32 inch (1.59 ±0.79 mm) from case for 10 s max. . . . . +265°C



ALL RESISTANCE VALUES ARE IN OHMS
UNLESS OTHERWISE INDICATED ALL CAPACITANCE VALUES LESS THAN 1.0 ARE IN MICROFARADS 1.0 OR GREATER ARE IN PICOFARADS

Fig. 1 – Functional block diagram of CA3170G.

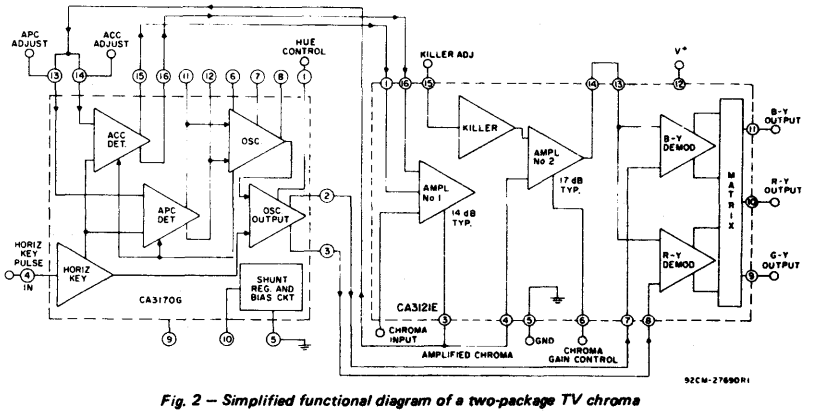


Fig. 2 – Simplified functional diagram of a two-package TV chroma system utilizing the CA3170G and CA3121E.

Features:

- **Voltage-controlled oscillator**
- **Keyed APC and ACC detectors**
- **DC hue control**
- **Shunt regulator**

The CA3170G is supplied in the 16-lead dual-in-line plastic package with a hermetic Gold-CHIP (G suffix). The chips used in the hermetic Gold-CHIP plastic packages are of the sealed-junction type designed to provide protection against the deteriorating effects of humidity and other surface contaminants without the need for a hermetic package enclosure. The semiconductor junctions are sealed by utilizing a silicon nitride passivation layer. A multilayered, highly corrosion-resistant, terminal-connection system of unique design is employed.

CIRCUIT DESCRIPTION

The CA3170G is a complete subcarrier regeneration system with automatic phase control applied to the oscillator. An amplified chroma signal from the CA3121E is applied to terminals No. 13 and No. 14, which are the automatic phase control (APC) and the automatic chroma control (ACC) inputs. APC and ACC detection is keyed by the horizontal pulse which also inhibits the oscillator output amplifier during the burst interval. The ACC system uses a synchronous detector to develop a correction voltage at the differential output terminal Nos. 15 & 16. This control signal is applied to the input terminal Nos. 1 & 16 of the CA3121E. The APC system also uses a synchronous detector. The APC error voltage is internally coupled to the 3.58 MHz oscillator at balance; the phase of the signal at terminal No. 13 is in quadrature with the oscillator. To accomplish phasing requirements, an RC phase shift network is used between the chroma input and terminal Nos. 13 and 14. The feedback loop of the oscillator is from terminal Nos. 7 and 8 back to No. 6. The same oscillator signal is available at terminal Nos. 7 and 8, but the dc output of the APC detector controls the relative signal levels at terminal Nos. 7 or 8. Because the output at terminal No. 8 is shifted in phase compared to the output at terminal No. 7, which is applied directly to the crystal circuit, control of the relative amplitudes at terminal Nos. 7 and 8 alters the phase in the feedback loop, thereby changing the frequency of the crystal oscillator. Balance adjustments of dc offsets are provided to establish an initial no-signal offset control in the ACC output, and a no-signal, on-frequency adjustment through the APC detector-amplifier circuit which controls the oscillator frequency. The oscillator output stage is differentially controlled at terminal Nos. 2 and 3 by the hue control input to terminal No. 1. The hue

ELECTRICAL CHARACTERISTCS, at T_A = 25°C and V^+ = +24 V unless otherwise specified

CHARACTERISTICS	SPECIAL TEST CONDITIONS	LIMITS CA3170G MIN.	TYP.	MAX.	UNITS
Static Characteristics					
Voltage: Hue Control, V_1	See Fig.7				V
Oscillator Input, V_6	S_1 CLOSED, S_3 OFF; S_2, S_4, S_5 OPEN, See Fig. 8	–	2.6	–	
APC Input, V_{13}		–	5.4	–	
Regulator, V_{10}	V^+ = 21 V	11	12.3	13.5	
Regulator Change, V_{10}	V^+ = 27 V	-0.2	–	+0.2	
Horizontal Key Input, V_4	I_4 = -10 μA	5	–	–	
Currents: Oscillator Output, I_2	S_1, S_2, S_4, S_5 CLOSED, S_3 in position 2, See Fig. 8	–	5.8	–	mA
APC Output, I_{11}, I_{12}	S_1, S_5 OPEN, S_2, S_4 CLOSED,	–	1.45	–	
ACC Output, I_{15}, I_{16}	S_3 in position 1, See Fig. 8	–	1.45	–	
Dynamic Characteristics (See Figure 6)					
Oscillator Outputs: Terminal No. 2, V_2	S_1 in position 1	0.75	1.0	–	V_{p-p}
Terminal No. 3, V_3	S_1 in position 2	0.75	1.0	–	
ACC Detected Output $V_{16} - V_{15}$	S_1 in position 1	115	150	–	mV
Oscillator Pull-In Range	S_1 in position 1	–	±400	–	Hz

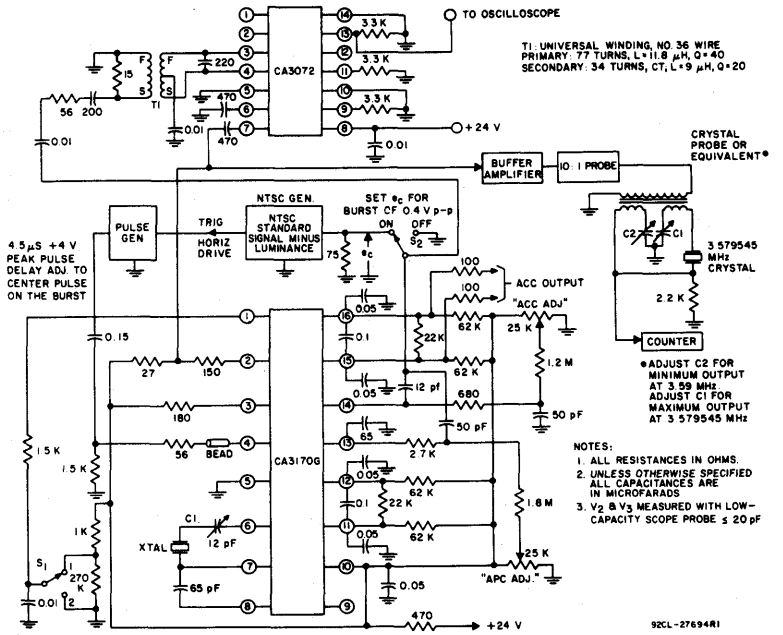


Fig. 5 – Dynamic characteristics test circuit.

phase shift is accomplished by the external R, L, and C components that couple the oscillator output to the demodulator input terminals. The CA3170G includes a shunt regulator to establish a 12-volt dc supply.

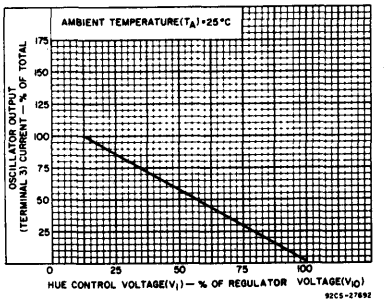


Fig. 3 – Typical hue control characteristic.

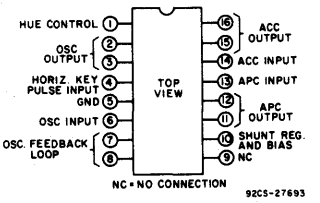


Fig. 4 – Terminal diagram of the CA3170G.

DYNAMIC TEST PROCEDURE

1. With S2 in "OFF" position, short terminals 11 and 12. Then with S1 in 1 position, adjust CX for a frequency of 3.579545 MHz ± 5 Hz. Measure the frequency using the frequency counter or by zero beat indication on the oscilloscope.
2. Remove short from terminals 11 and 12, and adjust "APC" control for zero beat on the oscilloscope. With S2 in "ON" position, pattern on oscilloscope must lock.
3. With S2 in "OFF" position adjust "ACC" control to give output reading of 0 ± 2 mV between terminals 15 and 16. Then with S2 in "ON" position, read "ACC" output.
4. Example of pull-in testing to ± 200 Hz: With S2 in "OFF" position, adjust CX for frequency of 3.579545 + 200 Hz. Then with S1 in position 1 and S2 in "ON" position, pattern on oscilloscope must lock.
5. Repeat Step 4 with CX adjusted to - 200 Hz.

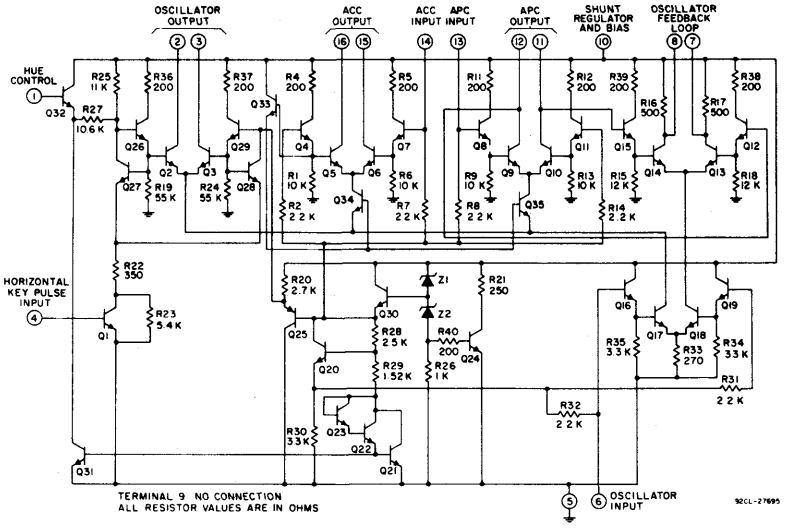


Fig. 6 – Schematic diagram of the CA3170G.

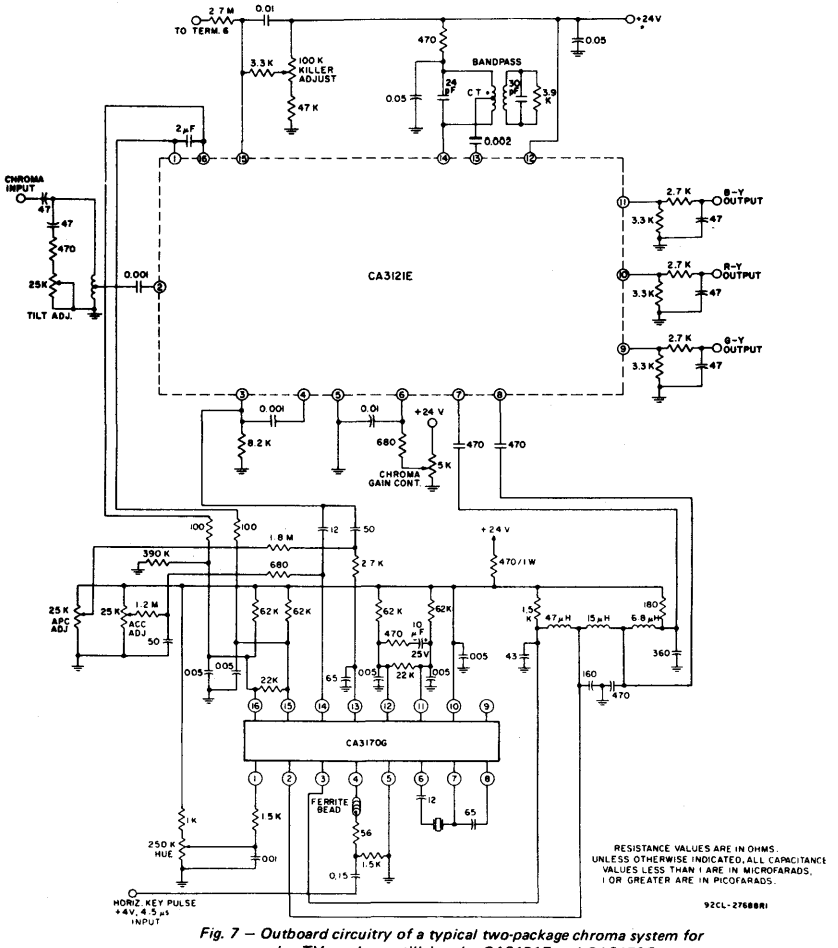


Fig. 7 – Outboard circuitry of a typical two-package chroma system for color-TV receivers utilizing the CA3121E and CA3170G.

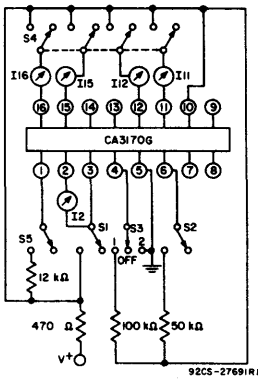


Fig. 8 – Static characteristics test circuit

CA3172G

TV Chroma Demodulator

The RCA-CA3172G is a monolithic silicon integrated circuit intended for use as a chroma demodulator in TV applications. It is operated from a 24-volt supply.

The device has synchronous detectors with matrix circuits to achieve the R-Y, G-Y, and B-Y color-difference output signals. The chroma input signal is applied to terminal Nos. 3 and 4, while the oscillator injection signal is applied to terminal Nos. 6 and 7. The color-difference signals, after matrix, have a fixed relationship of amplitude and phase.

The outputs of the CA3172G are suitable for driving high-level color-difference or R, G, and B output amplifiers. The emitter-follower stages used to drive the high-level color amplifiers have short-circuit protection.

The CA3172G is supplied in a 14-lead dual-in-line plastic package.

System Features:

- **Synchronous detector with color-difference matrix**
- **Emitter-follower output amplifier with short-circuit protection**
- **Typical R-Y output ratio of 0.95 and 89°, G-Y output ratio of 0.33 and 244°, and B-Y output ratio of 1.0 and 0°**

MAXIMUM RATINGS, *Absolute Maximum-Values at T_A = 25°C*

DC SUPPLY VOLTAGE (Terminal 8 to Terminal 14)	27	V
REFERENCE INPUT VOLTAGE	5	V_{p-p}
CHROMA INPUT VOLTAGE	5	V_{p-p}
DEVICE DISSIPATION:		
Up to T_A = +70°C	530 mW	
Above T_A = +70°C	Derate Linearly at 6.7 mW/°C	
AMBIENT TEMPERATURE RANGE:		
Operating	-40 to +85°C	
Storage	-65 to +150°C	
LEAD TEMPERATURE (During Soldering):		
At distance 1/32 in. (3.17 mm) from seating plane for 10 s max.	+265°C	

ELECTRICAL CHARACTERISTICS, *at T_A = 25°C and V^+ = +24 V unless otherwise specified*

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS	LIMITS CA3172G			UNITS
			MIN.	TYP.	MAX.	
Static Characteristics[a]						
Supply Current With Output Loads	I_T	S_1 Closed	16.5	–	28.5	mA
With No Output Loads		S_1 Open	–	9	–	
G-Y, R-Y, B-Y Outputs	V_9 V_{11}, V_{13}	S_1 Closed	13	14.5	15.5	V
Chroma Inputs	V_3, V_4	S_1 Open	–	3.6	–	
Reference Subcarrier	V_6, V_7	S_1 Open	–	6.4	–	
Dynamic Characteristics[b]						
Demodulator Unbalance	V_9, V_{11}, V_{13}	$V_3 = V_4 = 0$	–	–	0.6	V_{p-p}
Maximum Color Difference Output Voltage	V_{13}	$V_3 = V_4 = 0.35\ V_{p-p}$	5	–	–	V_{p-p}
Chroma Input Sensitivity	V_3	Adjust e_c for 5.0 V_{p-p} @ term No. 13 (B-Y)	–	0.2	0.35	
R-Y Output Ratio	V_{11}		–	0.95	–	
G-Y Output Ratio	V_9		–	0.32	–	
V_{DC} Difference Between any two Output Terminals	$\lvert V_9\rvert - \lvert V_{11}\rvert$ $\lvert V_9\rvert - \lvert V_{13}\rvert$ $\lvert V_{11}\rvert - \lvert V_{13}\rvert$	e_c = 0	–	–	0.6	V
Input Impedance	R_i6, 7		–	1.7	–	kΩ
Reference Subcarrier	C_i6, 7		–	6	–	pF
Input Impedance at	R_i3, 4		–	0.95	–	kΩ
Chroma Inputs	C_i3, 4		–	6	–	pF
Output Resistance	R_o9, R_o11, R_o13		–	180	–	Ω

[a] Test circuit Fig. 3 [b] Test circuit Fig. 4

Maximum Voltage and Current Ratings at T_A = +25°C

Voltage*			Current		
Terminal No.	MIN VOLTS	MAX VOLTS	Terminal No.	I_I mA	I_O mA
3	0	+5	3	–	–
4	0	+5	4	–	–
6	0	+12	6	–	–
7	0	+12	7	–	–
8	0	+27	8	–	–
9	0	+20	9	1.0	20
11	0	+20	11	1.0	20
13	0	+20	13	1.0	20

* With reference to terminal No. 14 and with the voltage between terminal No. 8 and terminal No. 14 at +24 V except as given in rating for terminal No. 8

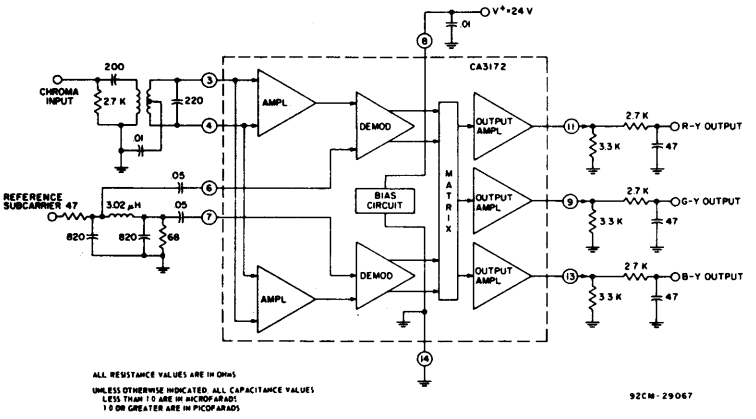


Fig. 1 – Functional diagram of RCA-CA3172.

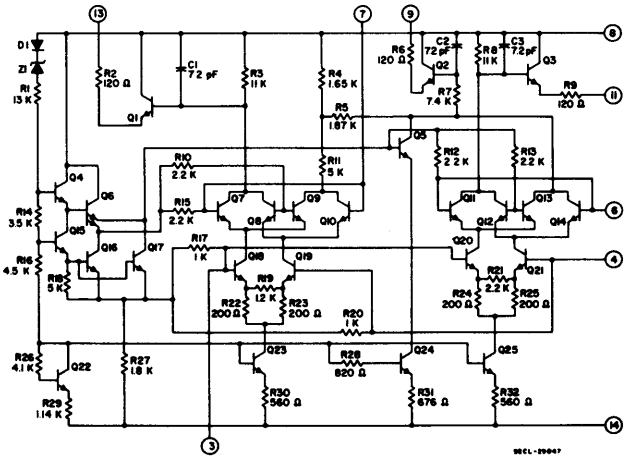


Fig. 2 – Schematic diagram for CA3172.

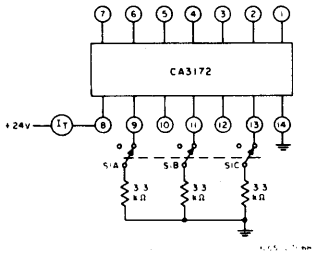


Fig. 3 – Static characteristics test circuit.

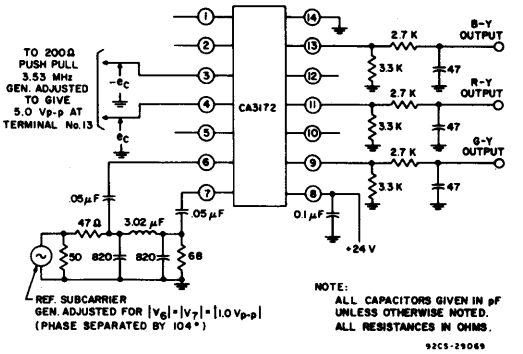


Fig. 4 – Dynamic characteristics test circuit.

CA3189E

FM IF System

Includes IF Amplifier, Quadrature Detector, AF Preamplifier, and Specific Circuits for AGC, AFC, Tuning Meter, Deviation-Noise Muting, and ON Channel Detector

For FM IF Amplifier Applications in High-Fidelity, Automotive, and Communications Receivers

Features:

- **Exceptional limiting sensitivity: 12 μV typ. at −3 dB point**
- **Low distortion: 0.1% typ. (with double-tuned coil)**
- **Single-coil tuning capability**
- **Improved S + N/N Ratio**
- **Externally programmable recovered audio level**
- **Provides specific signal for control of inter-channel muting (squelch)**
- **Provides specific signal for direct drive of a tuning meter**
- **On channel step for search control**
- **Provides programmable AGC voltage for RF amplifier**
- **Provides a specific circuit for flexible audio output**
- **Internal supply-voltage regulators**
- **Externally programmable "on" channel step width, and deviation at which muting occurs**

The RCA-CA3189E* is a monolithic integrated circuit that provides all the functions of a comprehensive FM-IF system. Fig. 1 shows a block diagram of the CA3189E, which includes a three-stage FM-IF amplifier/limiter configuration with level detectors for each stage, a doubly-balanced quadrature FM detector and an audio amplifier that features the optional use of a muting (squelch) circuit.

The advanced circuit design of the IF system includes desirable deluxe features such as programmable delayed AGC for the RF tuner, an AFC drive circuit, and an output signal to drive a tuning meter and/or provide stereo switching logic. In addition, internal power-supply regulators maintain a nearly constant current drain over the voltage supply range of +8.5 to +16 volts.

The CA3189E is ideal for high-fidelity operation. Distortion in a CA3189E FM-IF System is primarily a function of the phase linearity characteristic of the outboard detector coil.

The CA3189E has all the features of the CA3089E plus additions. See CA3189E features compared to the CA3089E in Table I.

The CA3189E utilizes the 16-lead dual-in-line plastic package and can operate over the ambient temperature range of −40°C to +85°C.

*Formerly Developmental Type No. TA10038.

MAXIMUM RATINGS, *Absolute-Maximum Values at T_A = 25°C:*

DC SUPPLY VOLTAGE (between Terms. 11 and 4)	16 V
(between Terms. 11 and 14)	16 V
DC CURRENT (Out of Term. 15)	2 mA
DEVICE DISSIPATION:	
Up to T_A = 85°C	640 mW
Above T_A = 85°C	derate linearly at 9.9 mW/°C
AMBIENT-TEMPERATURE RANGE:	
Operating	−40 to +85°C
Storage	−65 to +150°C
LEAD TEMPERATURE (During soldering):	
At distance not less than 1/32 inch (0.79 mm) from case for 10s max.	+265°C

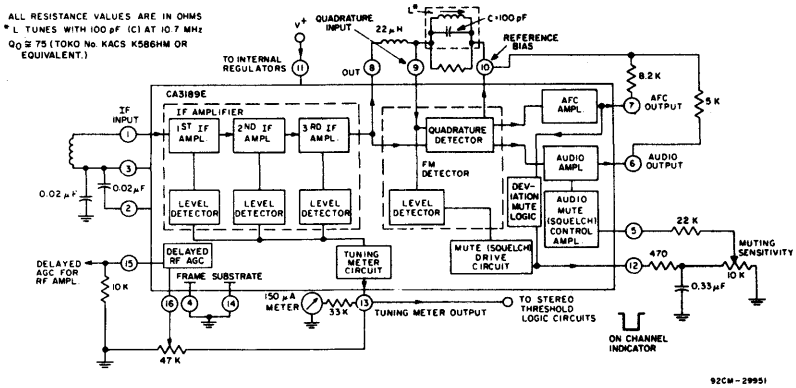
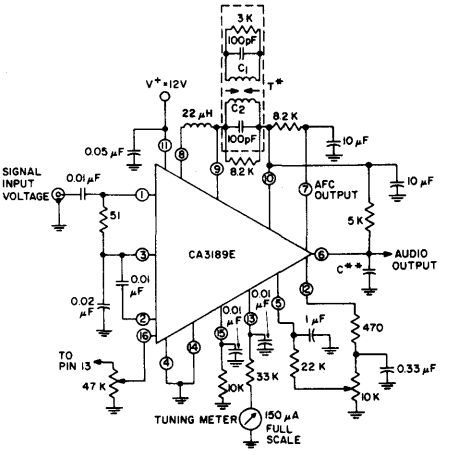


Fig. 1 – Block diagram of the CA3189E.

ELECTRICAL CHARACTERISTICS, at T_A = 25°C, V^+ = 12 Volts

CHARACTERISTIC	SYMBOL	TEST CONDITIONS			Circuit or Fig. No.	LIMITS Min.	Typ.	Max.	UNITS
Static (DC) Characteristics									
Quiescent Circuit Current	I_{11}					20	31	40	mA
DC Voltages: Terminal 1 (IF Input)	V_1					1.2	1.9	2.4	V
Terminal 2 (AC Return to Input)	V_2	No signal input, Non muted			2,6	1.2	1.9	2.4	V
Terminal 3 (DC Bias to Input)	V_3					1.2	1.9	2.4	V
Terminal 15 (RF AGC)	V_{15}					7.5	9.5	11	V
Terminal 10 (DC Reference)	V_{10}					5	5.6	6	V
Dynamic Characteristics									
Input Limiting Voltage (−3 dB point)	V_I(lim)					–	12	25	μV
AM Rejection (Term. 6)	AMR	V_{IN} = 0.1 V, AM Mod. = 30%	f_O = 10.7 MHz,		2,6	45	55	–	dB
Recovered AF Voltage (Term. 6)	V_O(AF)					325	500	650	mV
Total Harmonic Distortion:* Single Tuned (Term. 6)	THD	V_{IN} = 0.1 V	$f_{mod.}$ = 400 Hz,		6	–	0.5	1	%
Double Tuned (Term. 6)	THD		Deviation ±75 kHz		2	–	0.1	–	%
Signal plus Noise to Noise Ratio (Term. 6)	S + N/N				2,6	65	72	–	dB
Deviation Mute Frequency	$f_{DEV.}$		$f_{mod.}$ = 0		4,6,7	–	±40	–	kHz
RF AGC Threshold	V_{16}				2,6	–	1.25	–	V
On Channel Step	V_{12}	V_{IN} = 0.1 V	$f_{DEV.}$ < ±40 kHz		6	–	0	–	V
			$f_{DEV.}$ > ±40 kHz			–	5.6	–	

*THD characteristics are essentially a function of the phase characteristics of the network connected between terminals 8, 9, and 10.



ALL RESISTANCE VALUES ARE IN OHMS

*T: PRI. $-Q_0$(UNLOADED) ≅ 75 (TUNES WITH 100 pF (C1) 20 t OF 34 e ON 7/32" DIA. FORM

SEC. $-Q_0$(UNLOADED) ≅ 75 (TUNES WITH 100 pF (C2) 20 t OF 34 e ON 7/32" DIA. FORM

kQ (PER CENT OF CRITICAL COUPLING) ≅ 70 %

(ADJUSTED FOR COIL VOLTAGE V_C) = 150 mV

ABOVE VALUES PERMIT PROPER OPERATION OF MUTE (SQUELCH) CIRCUIT

"E" TYPE SLUGS, SPACING 4 mm

**C = 0.01 μF FOR 50 μs DEEMPHASIS (EUROPE)

= 0.015 μF FOR 75 μs DEEMPHASIS (USA)

92CM-29954

Fig. 2 – Test circuit for CA3189E using a double-tuned detector coil.

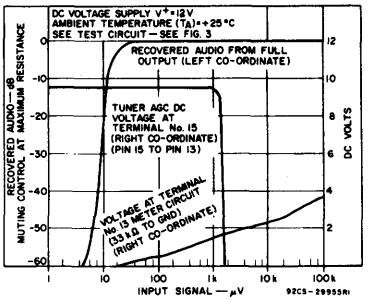


Fig. 3 – Muting action, tuner AGC, and tuning meter output as a function of input signal voltage.

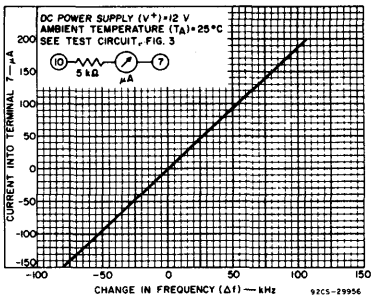


Fig. 4 – AFC characteristics (current at Term. 7 as a function of change in frequency).

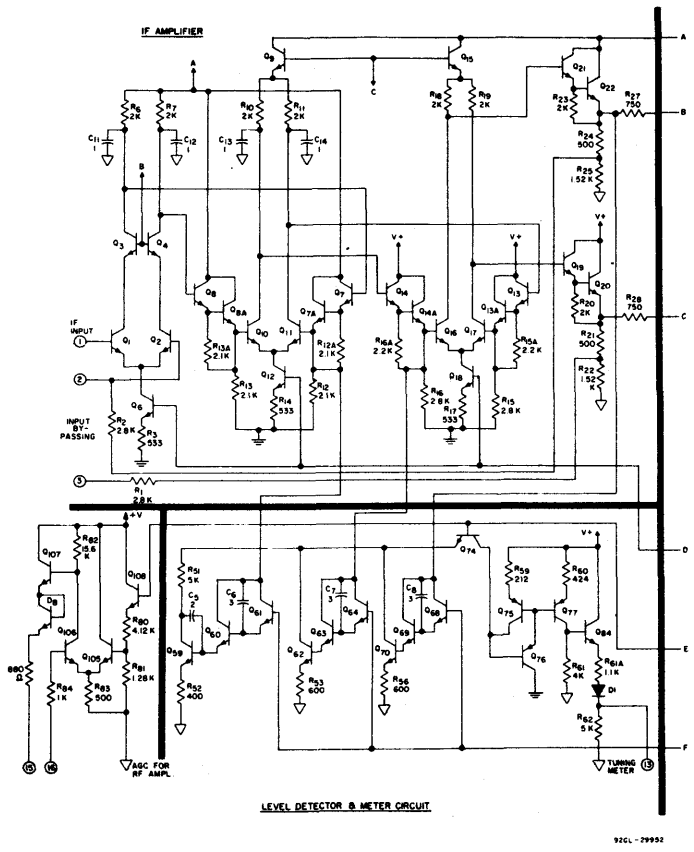


Fig. 5 – Schematic diagram of the CA3189E

TABLE I – CA3189E Features Compared to CA3089E

FEATURES	CA3189E	CA3089E
Low Limiting Sensitivity (12 μV typ.)	Yes	Yes
Low Distortion	Yes	Yes
Single-coil Tuning Capability	Yes	Yes
Programmable Audio Level	Yes	No
S/N Mute	Yes	Yes
Deviation Mute	Yes	No
Flexible AFC	Yes	Yes
Programmable AGC Threshold and Voltage	Yes	No
Typical S + N/N > 70 dB	Yes	No
Meter Drive Voltage Depressed at Very-Low Signal Levels	Yes	No
On-Channel Step Control Voltage	Yes	No

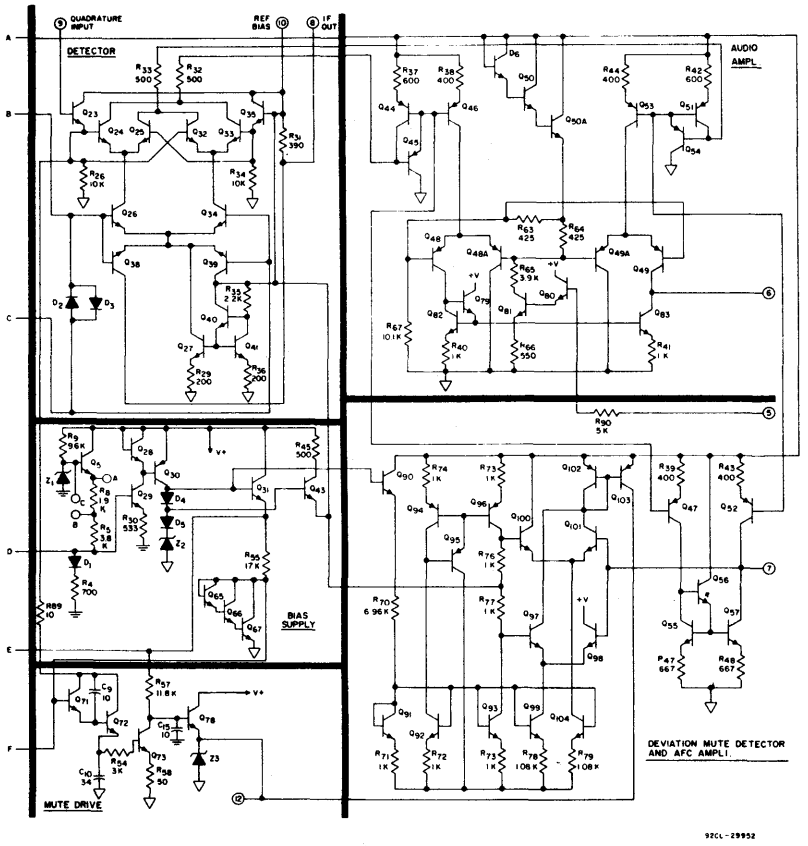


Fig. 5 – Schematic diagram of the CA3189E

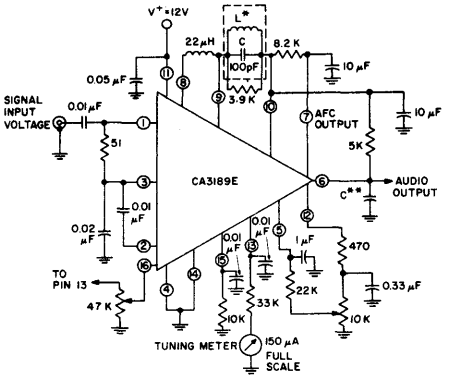


Fig. 6 – Test circuit for CA3189E using a single-tuned detector coil.

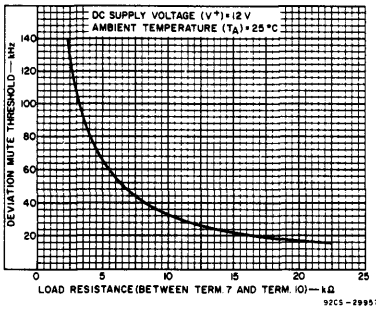


Fig. 7 – Deviation mute threshold as a function of load resistance (between Term. 7 and Term. 10).

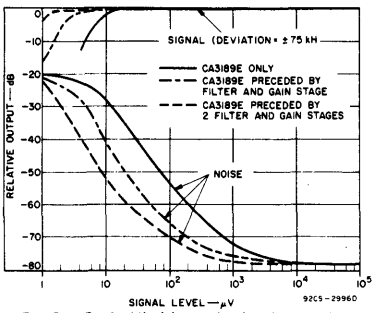


Fig. 8 – Typical limiting and noise characteristics.

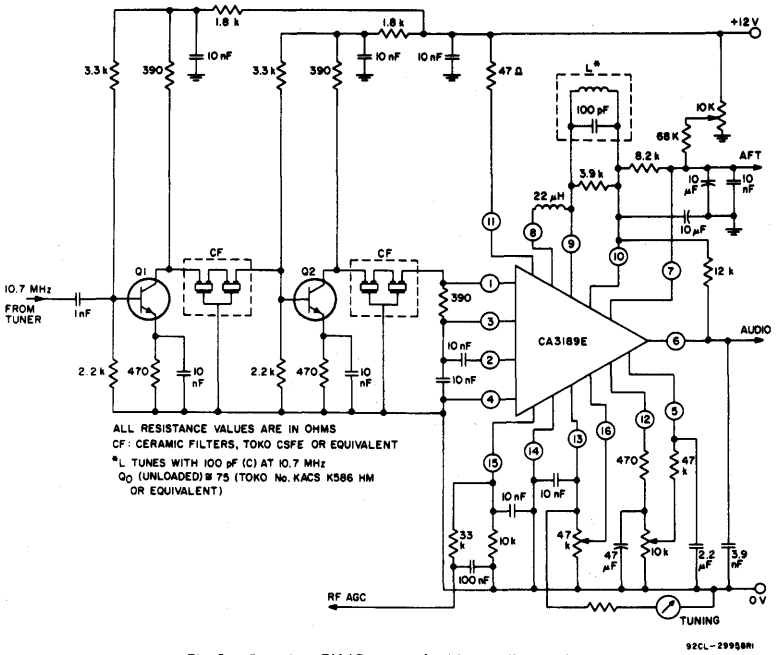


Fig. 9 – Complete FM IF system for high-quality receivers.

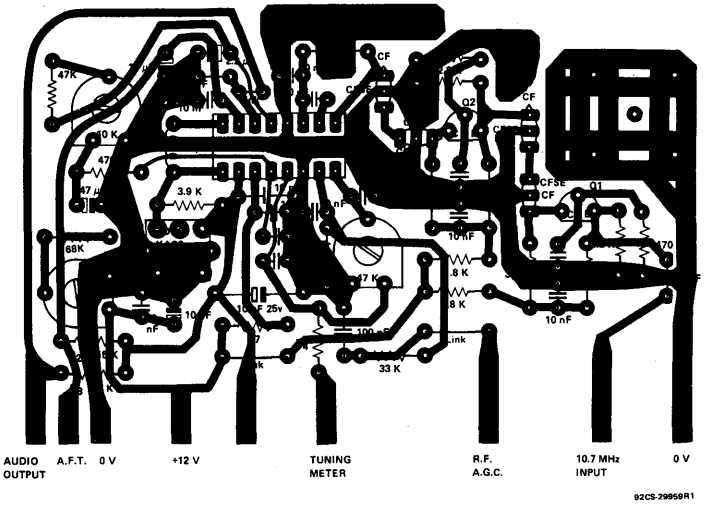


Fig. 10 – Printed circuit-board and component layout for circuit shown in Fig. 9.

COMPLETE FM IF SYSTEM FOR HIGH-QUALITY TUNERS

The circuit, Fig. 9, provides a complete FM IF system for a high-quality receiver. Either one or two stages of amplification and bandpass filtering may be desired, depending on the receiver requirements. Figure 8 shows typical limiting and noise characteristics for each circuit configuration which can be compared to the CA3189E alone.

TV Chroma Amplifier/ Demodulator

Provides Complete System for Processing Chroma When Used with RCA-CA3070 or CA3170 "G" Suffix Type-Hermetic Gold-CHIP in Dual-In-Line Plastic Package

Features:

- Excellent linearity in dc chroma gain-controlled circuit
- Improved filtering resulting in reduced 7.2-MHz output from the color demodulators
- Current limiting for short-circuit protection
- Good tolerance to B+ supply variations
- Good temperature coefficient stability
- Gold-CHIP for increased reliability

The RCA-CA3221G is a monolithic silicon integrated circuit chroma amplifier/demodulator with ACC, saturation control, and killer control for use in NTSC color TV receivers. It is designed to function compatibly with the CA3070 or CA3170 in a 2-package chroma system. The CA3221G is functionally identical to the industry standard CA3121, but has a modified saturation control as well as a modified color difference matrix.

The CA3221G is supplied in the 16-lead dual-in-line plastic package with a hermetic Gold-CHIP (G suffix). The transistor chips used in the hermetic Gold-CHIP plastic packages are of the sealed-junction type designed to provide protection against the deteriorating effects of humidity and other surface contaminants without the need for a hermetic package enclosure. The semiconductor junctions are sealed by utilizing a silicon nitride passivation layer. A multi-layered, highly corrosion-resistant, terminal-connection system of unique design is employed.

MAXIMUM RATINGS at T_A = 25°C

Supply Voltage	30 V
Device Dissipation:	
Up to T_A = 55°C	1 W
Above T_A = 55°C	derate linearly 10.5 mW/°C
Operating Temperature Range	−40 to +85°C
Storage Temperature Range	−65 to +150°C
Lead Temperature (During Soldering)	
At distance 1/16" ±1/32" (1.59 ±0.79 mm) from case for 10 s max.	+265°C

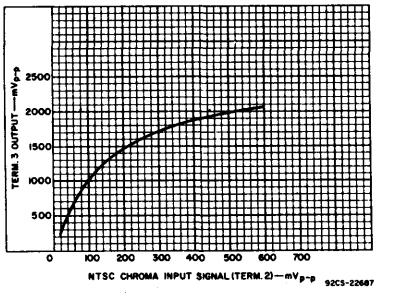


Fig. 2 – Typical ACC plot for the CA3221G when used with the CA3070.

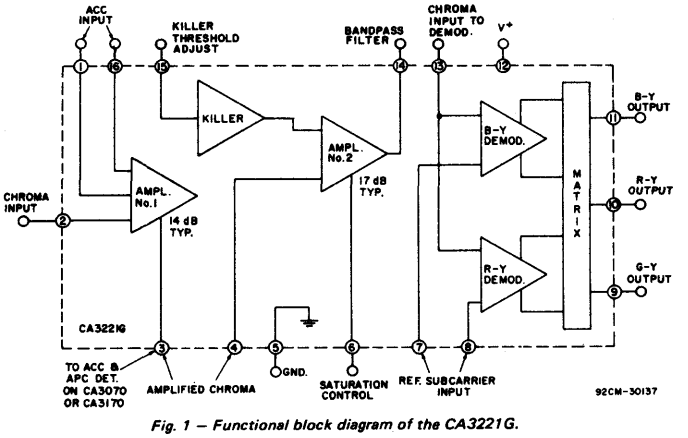


Fig. 1 – Functional block diagram of the CA3221G.

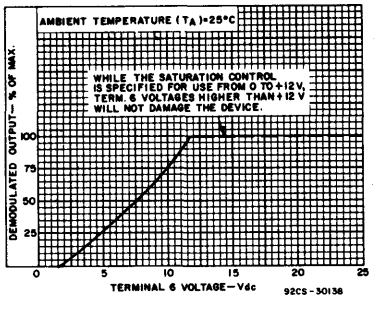


Fig. 3 – Saturation control characteristic.

ELECTRICAL CHARACTERISTICS at T_A = 25°C and Referenced to Test Circuit (Fig.7)

CHARACTERISTIC, TERMINAL MEASURED, AND SYMBOL	TEST CONDITIONS	LIMITS			UNITS
		Min.	Typ.	Max.	
Supply Current, I_T	–	–	40	50	mA
Input Sensitivity, V_2	Vary Eg; set V_{11} for 2 V RMS S3 = 1	4	12	20	mV RMS
Second-Stage Sensitivity, V_4	Vary Eg; set V_{11} for 2 V RMS S3 = 1	30	53	75	mV RMS
Output Voltage (Killer off)	Switch Positions: S1=2, S2=2, S3=1 Adjust killer potentiometer until output drops	–	–	70	mV RMS
Saturation Control Characteristics: * V_{11} 50% Gain	Vary Eg; set V_{11} for 2 V RMS with S3 = 1. Set S3 = 2 measure V_{11}	0.71	0.95	1.16	V RMS
0% Gain	Same as above, S3 = 3	–	–	20	mV RMS
Demodulator Characteristics: Output Voltages, V_9, V_{10}, V_{11}	–	13.5	14.5	15.5	V
DC Output Balance (Between any 2 outputs)	–	–0.6	–	+0.6	V
Unbalance, V_9, V_{10}, V_{11}	Eg=0; Switch Position: S1=1, S2=1, S3= 1	–	–	0.8	Vp-p
Relative Outputs– R-Y, V_{10}	Vary Eg; set V_{11} for 2 V RMS, S3 = 1	1.75	1.85	1.95	V RMS
G-Y, V_9		0.6	0.7	0.8	V RMS
Relative Phase – R-Y, V_{10}	Vary Eg; set V_{11} for 2 V RMS; read phase of V_{10} and V_9	–	90	–	degrees
G-Y, V_9	with V_{11} as reference	–	244	–	degrees
Max. Output Voltage, V_{11}	Eg = 750 mV	2.8	–	–	V RMS

*See Fig. 3 for saturation control characteristic.

CIRCUIT OPERATION

The CA3221G consists of three basic circuit sections: (1) amplifier No. 1, (2) amplifier No. 2, and (3) demodulator. Amplifier No. 1 contains the circuitry for automatic chroma control (ACC) and color-killer sensing. The output of amplifier No. 1 (Terminal 3) is coupled to the Chroma Signal Processor (CA3070 or equivalent) for ACC and automatic phase control (APC) operation and to the input of amplifier No. 2 (Terminal 4) containing the chroma gain control circuitry. The signal from the color-killer circuit in amplifier No. 1 acts upon amplifier No. 2 to greatly reduce its gain under weak signal conditions.

The output from amplifier No. 2 (Terminal 14) is applied, through a Bandpass Filter, to the demodulator input (Terminal 13). The demodulator also receives the R-Y and B-Y demodulator subcarrier signals (Terminals 7 and 8) from the oscillator output of the chroma signal processor. The R-Y and B-Y demodulators and the matrix network contained in the demodulator section of the CA3221G reconstruct the G-Y signal to achieve the R-Y, G-Y, and B-Y color difference signals. These high-level outputs signals with low impedance outputs are suitable for driving high-level R, G, B output amplifiers. Internal capacitors are included on each output to filter out unwanted harmonics. For additional operating information and signal waveforms, refer to Television Chroma System (utilizing RCA-CA3070, CA3071, CA3072), File No. 468.

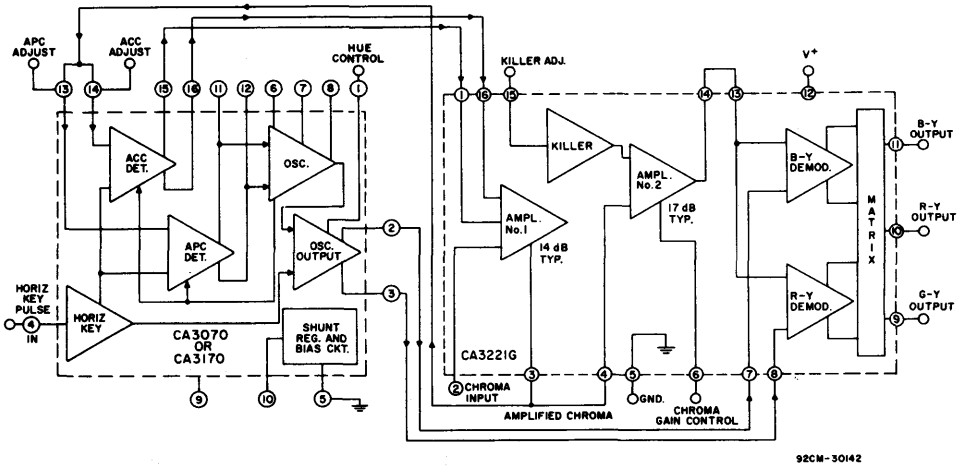


Fig. 4 – Simplified functional diagram of a two-package TV chroma system utilizing the CA3221G and CA3070 or CA3170.

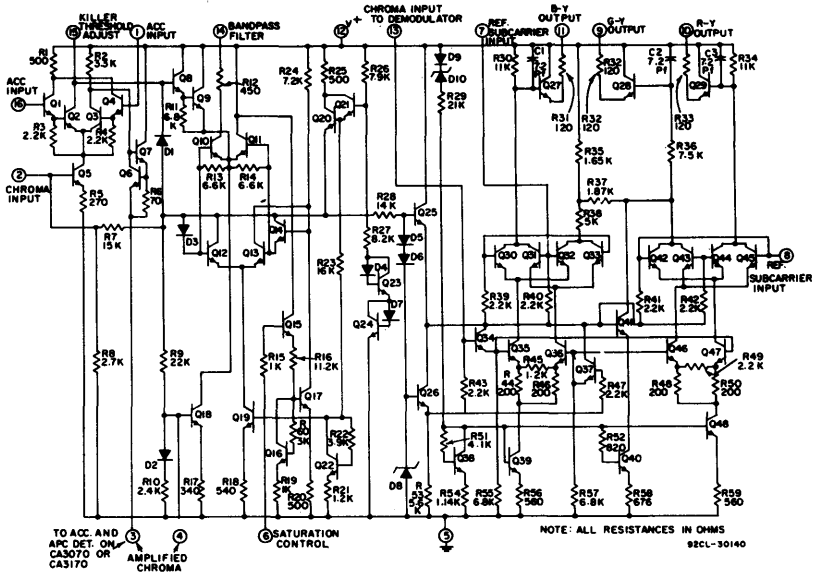


Fig. 5 – Schematic diagram of CA3221G.

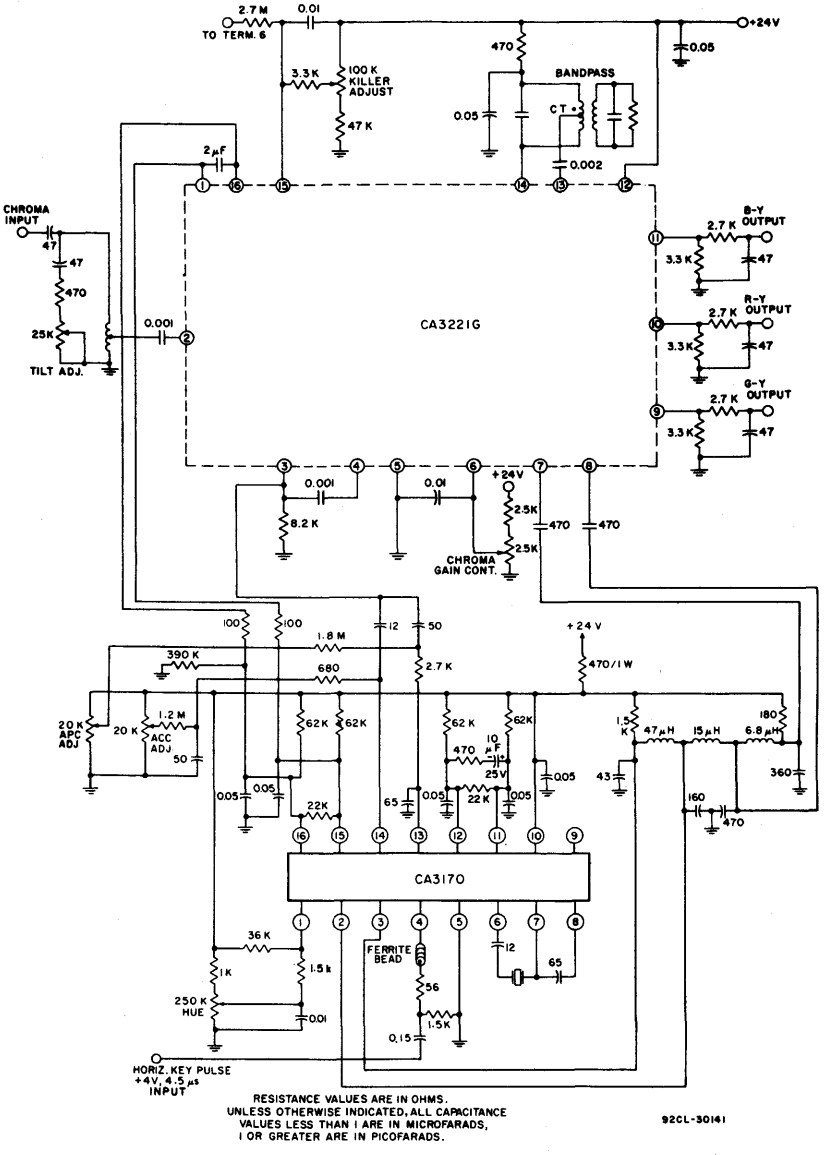
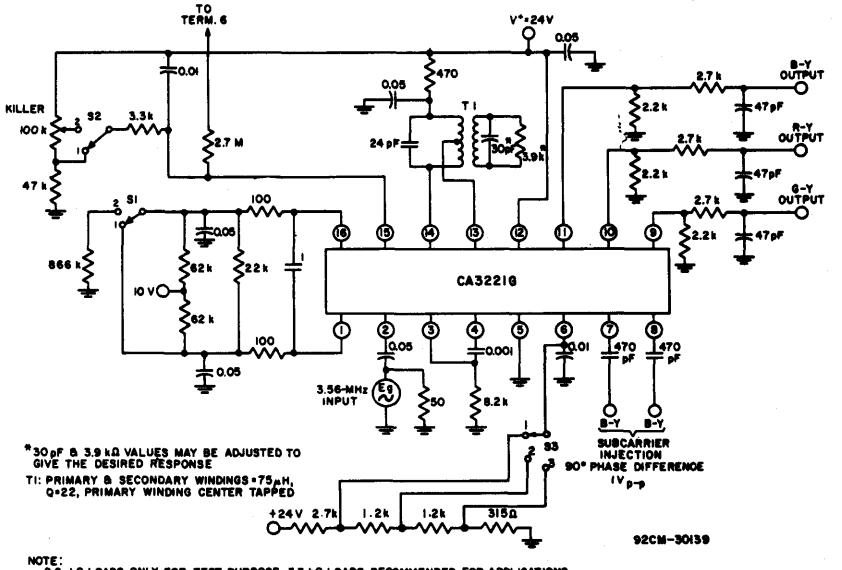


Fig. 6 — Outboard circuitry of a typical two-package chroma system for color-TV receivers utilizing the CA3221G and CA3170.



*30 pF & 3.9 kΩ VALUES MAY BE ADJUSTED TO GIVE THE DESIRED RESPONSE

T1: PRIMARY & SECONDARY WINDINGS=75 μH, Q=22, PRIMARY WINDING CENTER TAPPED

NOTE:
2.2-kΩ LOADS ONLY FOR TEST PURPOSE, 3.3-kΩ LOADS RECOMMENDED FOR APPLICATIONS.
RESISTANCE VALUES ARE IN OHMS.
CAPACITANCE VALUES ARE IN MICROFARADS UNLESS OTHERWISE INDICATED.

Fig. 7 – Typical characteristics test circuit for the CA3221G.

MOS Field-Effect Transistors
Technical Data

3N128, 3N143

Silicon MOS Transistors

N-Channel Depletion Types

For Amplifier, Mixer, & Oscillator Applications in Military & Industrial VHF Communications Equipment Operating up to 250 MHz

RCA-3N128 and 3N143 are N-channel depletion-type silicon insulated-gate field-effect transistors utilizing the MOS* construction. The 3N128 is intended primarily for VHF amplifier service in military and industrial applications. It also is extremely well suited for use in dc and low-frequency amplifier applications requiring a transistor having high power gain, very high input impedance, and low gate leakage.

The 3N143 is designed for use as a VHF mixer and oscillator. Because of their improved transfer characteristic and increased dynamic range the 3N128 and 3N143 provide substantially better cross-modulation performance in linear amplifier applications than conventional (bipolar) transistors and are free from diode-current loading common to junction type FET's. These transistors are hermetically sealed in JEDEC TO-72 metal packages.

Maximum Ratings, *Absolute-Maximum Values at T_A = 25° C:*

*DRAIN-TO-SOURCE VOLTAGE, V_{DS} . . . . . +20 V
*DRAIN-TO-GATE VOLTAGE, V_{DG} . . . . . +20 V
*GATE-TO-SOURCE VOLTAGE, V_{GS}:
Continuous dc . . . . . +1, -8 V
Peak ac . . . . . ±15 V
*DRAIN CURRENT, I_D . . . . . 50 mA
*TRANSISTOR DISSIPATION, P_T:
At Ambient up to 25°C . . . . . 330 mW
Temperatures above 25° . . . . . Derate 2.2 mW/°C
*AMBIENT TEMPERATURE RANGE:
Storage and Operating . . . . . -65 to +175°C
*LEAD TEMPERATURE (During soldering):
At distances not closer than 1/32 inch to seating surface for 10 seconds maximum . . . . . 265 °C

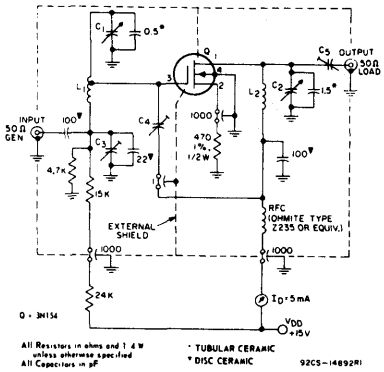
*In accordance with Jedec Registration Data Format JS9-RDF11B.

ELECTRICAL CHARACTERISTICS: *(At T_A = 25°C)*

Measured with Substrate Connected to Source Unless Otherwise Specified.

	CHARACTERISTIC	SYMBOL	CONDITIONS	LIMITS 3N128 MIN.	3N128 TYP.	3N128 MAX.	3N143 MIN.	3N143 TYP.	3N143 MAX.	UNITS
*	Gate Leakage Current	I_{GSS}	V_{DS} = 0, V_{GS} = -8 V T_A = 25°C	-	0.1	50	-	0.1	1000	pA
			V_{DS} = 0, V_{GS} = -8 V T_A = 125°C	-	-	5	-	-	200	nA
*	Zero-Bias Drain Current	I_{DSS}	V_{DS} = 15 V, V_{GS} = 0	5	15	25	5	15	30	mA
*	Drain-to-Source Cutoff Current	$I_{D(off)}$	V_{DS} = 20 V, V_{GS} = -8 V	-	-	50	-	-	50	μA
*	Gate-to-Source Cutoff Voltage	$V_{GS(off)}$	V_{DS} = 15 V, I_D = 50 μA	-0.5	-3	-8	-0.5	-3	-8	V
*	Forward Transconductance	g_{fs}	V_{DS} = 15 V, I_D = 5 mA, f = 1 kHz	5,000	7,500	12,000	5,000	7,500	12,000	μmho
*	Drain-to-Source Channel Resistance	$r_{DS(on)}$	V_{DS} = 0, V_{GS} = 0, f = 1 kHz	-	200	-	-	200	-	Ω
*	Small-Signal Short-Circuit Reverse Transfer Capacitance▲	C_{rss}	V_{DS} = 15 V, I_D = 5 mA, f = 0.1 to 1 MHz	0.15	0.25	0.35	0.12	0.25	0.38	pF
*	Small-Signal Short-Circuit Input Capacitance	C_{iss}	V_{DS} = 15 V, I_D = 5 mA, f = 0.1 to 1 MHz	-	5.5	7	-	5.5	7	pF
			Common-Source Configuration							
*	Input Admittance	Y_{is}	f = 200 MHz	-	0.4 + J7.3	-	-	-	-	mmho
*	Forward Transfer Admittance	Y_{ss}	V_{os} = 15 Volts	-	7 - J2	-	-	-	-	mmho
*	Output Admittance	Y_{os}	I_D = 5 mA	-	0.28 + J1.8	-	-	-	-	mmho
*	Maximum Available Power Gain	MAG	V_{DS} = 15 V, I_D = 5 mA, f = 200 MHz		21	-	-	-	-	dB
*	Insertion Power Gain (Fixed Neutralization) See Fig. 1	G_{PS}		13.5	16	-	-	-	-	dB
	Power Gain (Conversion (See Fig. 3)	$G_{P(C)}$	V_{DS} = 15 V, I_D = 1 mA, f_{in} = 200 MHz, f_{out} = 30 MHz	-	-	-	10	13.5	-	dB
	Noise Figure (See Fig. 1 & 2)	NF	V_{DS} = 15 V, I_D = 5 mA, f = 200 MHz	-	3.5	5	-	-	-	-

*Inaccordance with **JEDEC** Registration Data Format JS9-RDF-11B.
▲Three-Terminal Measurement: Source Returned to Guard Terminal.



C_1, C_2: 1.5-5 pF variable air capacitor: E. F. Johnson Type 160-102 or equivalent

C_3: 1-10 pF piston-type variable air capacitor: JFD Type VAM-010, Johanson Type 4335, or equivalent

C_4, C_5: 0.3-3 pF piston-type variable air capacitor: Roanwell Type MH-13 or equivalent

L_1: 5 turns silver-plated 0.02" thick, 0.07"-0.08" wide copper ribbon. Internal diameter of winding = 0.25"; winding length approx. 0.65". Tapped at 1-1/2 turns from C_1 end of winding

L_2: Same as L_1 except winding length approx. 0.7"; no tap.

Fig. 1 - Test circuit used to measure 200-MHz maximum usable power gain and noise figure for 3N128

Performance Features

- **Large dynamic range**
- **Greatly reduces spurious responses in receiver front ends**
- **Permits use of vacuum-tube biasing techniques**
- **Excellent thermal stability**
- **Superior crossmodulation capability**

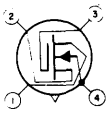
Device Features

- **Low noise figure (3N128) – 3.5 dB typ. at 200 MHz**
- **High VHF amplifier gain (3N128) – 16 dB typ. at 200 MHz**
- **Low input capacitance – 5.5 pF typ.**
- **High transconductance – 7500 μmho typ.**
- **High input resistance – 10^{14} Ω typ.**
- **High conversion gain (3N143, mixer) – 13.5 dB typ. at 200 MHz**

Applications

- **VHF amplifiers, mixers, converters and if-amplifiers in communication receivers.**
- **High-impedance timing circuits**
- **Detectors, oscillators, frequency multipliers, phase splitters, pulse stretchers and current limiters**
- **Electrometer amplifiers**
- **Voltage-controlled attenuators**
- **High impedance differential amplifiers**

TERMINAL DIAGRAM



1 - Drain
2 - Source
3 - Insulated Gate
4 - Bulk (Substrate) and Case

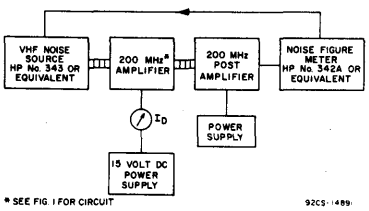
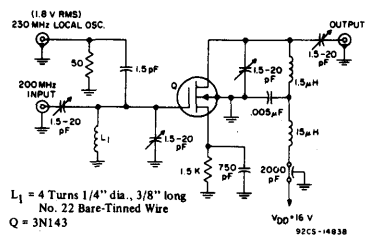


Fig. 2 - Noise figure measurement setup for 3N128



L_1 = 4 Turns 1/4" dia., 3/8" long No. 22 Bare-Tinned Wire
Q = 3N143

Fig. 3 - Conversion power gain test circuit for 3N143

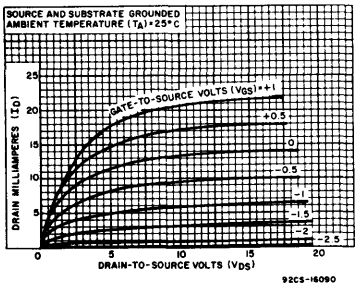


Fig. 4 - Drain current vs. drain-to-source voltage

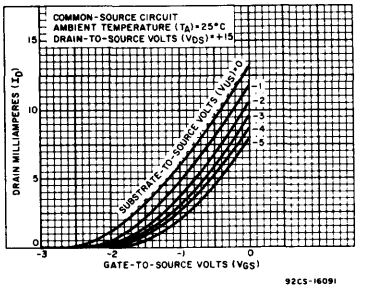


Fig. 5 - Drain current vs. gate-to-source voltage (V_{GS})

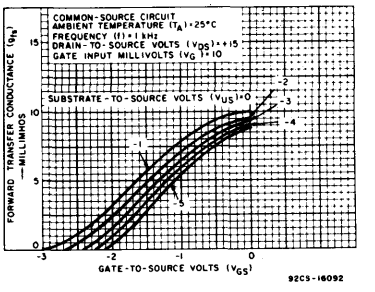


Fig. 6 - Forward transconductance vs. gate bias voltage

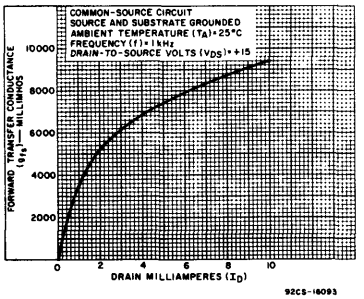


Fig. 7 - Forward transconductance vs. drain current

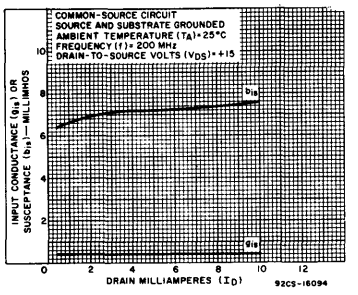


Fig. 8 - Input admittance vs. drain current

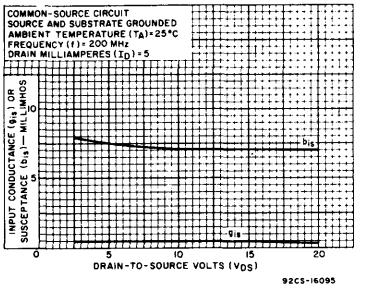


Fig. 9 - Input admittance vs. drain-to-source voltage

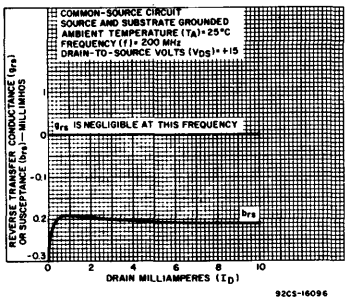


Fig. 10 - Reverse transadmittance vs. drain current

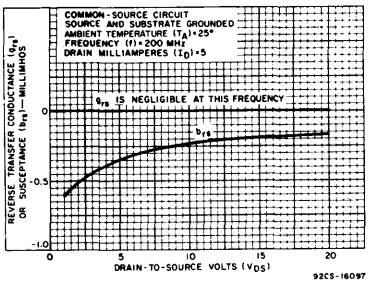


Fig. 11 - Reverse transadmittance vs. drain-to-source voltage

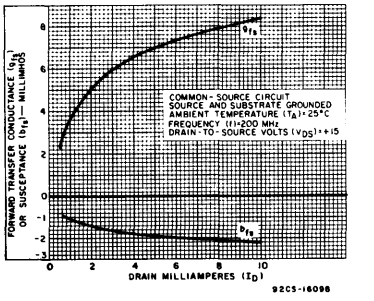


Fig. 12 - Forward transadmittance vs. drain current

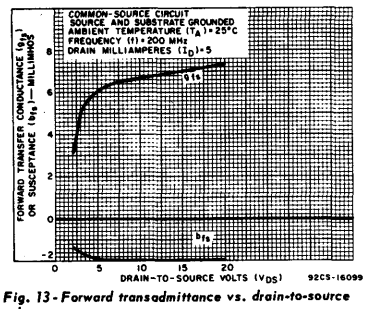


Fig. 13 - Forward transadmittance vs. drain-to-source voltage

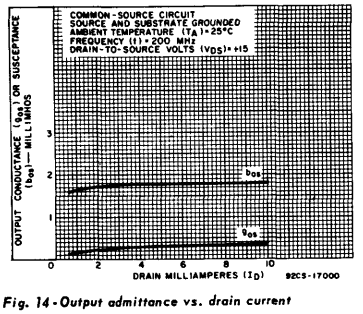


Fig. 14 - Output admittance vs. drain current

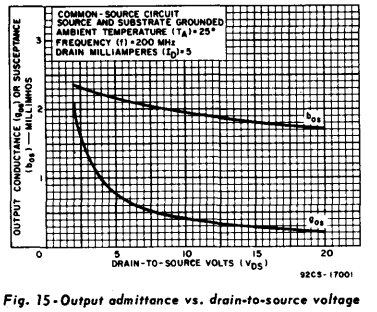


Fig. 15 - Output admittance vs. drain-to-source voltage

3N138

SILICON INSULATED-GATE FIELD-EFFECT TRANSISTOR

N-Channel Depletion Type

For Critical Chopper Applications and Multiplex Service in Instrumentation and Control Circuits

RCA-3N138 is a silicon, insulated-gate field-effect transistor of the N-channel depletion type, utilizing the MOS* construction. It is intended primarily for critical chopper and multiplex applications up to 60MHz.

The insulated gate provides a very high value of input resistance (10^{14} ohms typ.) which is relatively insensitive to temperature and is independent of gate-bias conditions (positive, negative, or zero bias). The 3N138 also features extremely low feed-through capacitance (0.18pF typ.) and zero inherent offset voltage.

The 3N138 is hermetically sealed in the JEDEC TO-72 package and features a gate metallization that covers the entire source-to-drain channel.

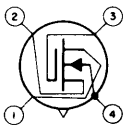
* Metal-Oxide-Semiconductor.

Maximum Ratings, Absolute-Maximum Values:

(Substrate connected to source unless otherwise specified)

DRAIN-TO-SOURCE VOLTAGE, V_{DS}	+35 max.	V
DRAIN-TO-SUBSTRATE VOLTAGE, V_{DB}	+35, −0.3 max.	V
SOURCE-TO-SUBSTRATE VOLTAGE, V_{SB}	+35, −0.3 max.	V
DC GATE-TO-SOURCE VOLTAGE, V_{GS}	±10 max.	V
PEAK GATE-TO-SOURCE VOLTAGE, V_{GS}	±14 max.	V
PEAK VOLTAGE, GATE-TO-ALL OTHER TERMINALS: V_{GS}, V_{GD}, V_{GB}, non-repetitive	±45 max.	V
DRAIN CURRENT, I_D (Pulse duration 20 ms, duty factor ≤ 0.10)	50 max.	mA
TRANSISTOR DISSIPATION, P_T: At ambient temperatures up to 25°C	330 max.	mW
above 25°C	Derate linearly at 2.2 mW/°C	
AMBIENT TEMPERATURE RANGE: Storage	−65 to +150	°C
Operating	−65 to +125	°C
LEAD TEMPERATURE (During Soldering): At distances ≥ 1/32" to seating surface for 10 seconds max.	265 max.	°C

TERMINAL DIAGRAM



1 - Drain
2 - Source
3 - Insulated Gate
4 - Bulk (Substrate) and Case

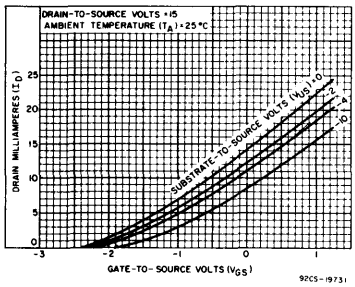


Fig. 1 – Drain Current vs Gate-to-Source Voltage

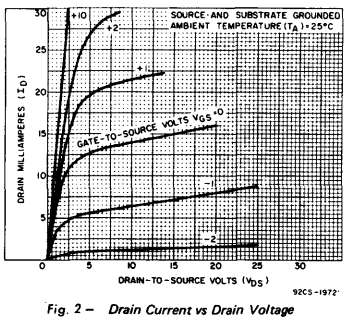


Fig. 2 – Drain Current vs Drain Voltage

Features

- excellent thermal stability
- zero inherent offset voltage
- low leakage current: 10 pA max.
- low "on" resistance — $r_{DS}(on)$ = 240Ω typ. (V_{GS} = 0V)
- high "off" resistance — $R_{DS}(off)$ = 10^{11}Ω typ.
- low feedback capacitance — C_{rss} = 0.18pF typ.
- low input capacitance — C_{iss} = 3pF typ.

Applications

- Servo Amplifiers
- Telemetry Amplifiers
- Computer Operational Amplifiers
- Sampling Circuits
- Electrometer Amplifiers

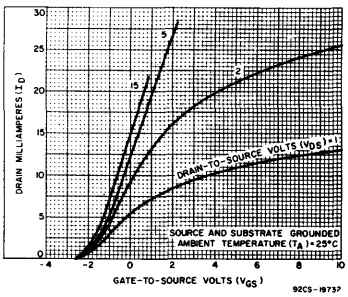


Fig. 3 – Drain Current vs Gate-to-Source Voltage

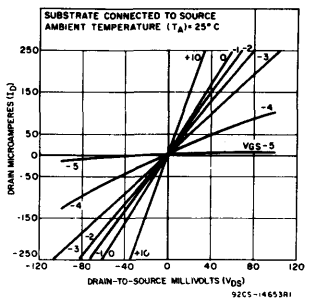


Fig. 4 – Low-Level Drain Current vs Drain-to-Source Voltage

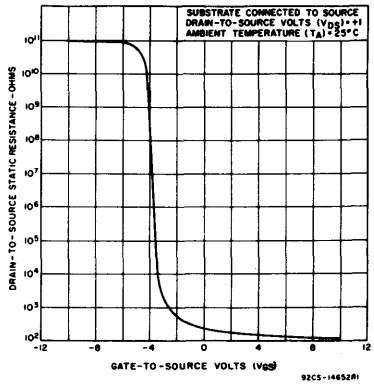


Fig. 5 – Drain-to-Source Static Resistance vs Gate-to-Source Voltage

ELECTRICAL CHARACTERISTICS, at T_A = 25° C, Unless Otherwise Specified. Substrate Connected to Source.

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	LIMITS Type 3N138 Min.	Typ.	Max.	UNITS
Gate-Leakage Current	I_{GSS}	V_{GS} = ±10, V_{DS} = 0, T_A = 25°C V_{GS} = ±10, V_{DS} = 0, T_A = 125°C	— —	0.1 20	10 200	pA pA
Drain-to-Source "ON" Resistance	$r_{DS}(on)$	V_{GS} = 0, V_{DS} = 0, f = 1 KHz, T_A = 25°C V_{GS} = +10, V_{DS} = 0, f = 1 KHz, T_A = 25°C V_{GS} = 0, V_{DS} = 0, f = 1 KHz, T_A = 125°C	— — —	240 135 350	350 — —	Ω Ω Ω
Drain-to-Source "OFF" Resistance	$R_{DS}(off)$	V_{GS} = −10, V_{DS} = +1	2×10^8	10^{11}	—	Ω
Drain-to-Source Cutoff Current	$I_{D}(off)$	V_{GS} = −10, V_{DS} = +1, T_A = 25°C V_{GS} = −10, V_{DS} = +1, T_A = 125°C	— —	0.01 0.01	5 0.5	nA μA
Small-Signal, Short-Circuit, Reverse Transfer Capacitance	C_{rss}	V_{GS} = −10, V_{DS} = 0, f = 1 MHz	—	0.25	0.4	pF
Small-Signal, Short-Circuit, Input Capacitance	C_{iss}	V_{GS} = −10, V_{DS} = 0, f = 1 MHz	—	3	5	pF
Zero-Gate-Bias Forward Transconductance	g_{fs}	V_{DS} = 12, I_D = 5 mA	—	6000	—	μmho
Offset Voltage	V_{os}	V_{GS} = ±10, V_{DS} = 0	—	0*	—	V

* In measurements of Offset Voltage, thermocouple effects and contact potentials in the measurement setup may cause erroneous readings of 1 microvolt or more. There errors may be minimized by the use of solder having a low thermal e.m.f. such as Leeds & Northrup No.107-1.0.1, or equivalent.

SILICON MOS TRANSISTOR

N-Channel Depletion Type

For Audio, Video, and RF Amplifier Applications in Communications, Instrumentation and Control Circuits

RCA 3N139 is a silicon, insulated-gate field-effect transistor of the N-channel depletion type, utilizing the MOS* construction. It is a general purpose transistor especially suited for audio, video, and rf applications, and for wide-band amplifier designs. The insulated gate provides a very high input resistance (10^{14} Ω typ.) which is relatively insensitive to temperature and is independent of gate-bias conditions (positive, negative, or zero bias). The 3N139 also has a high transconductance, a low value of input capacitance (3 pF typ.), and a very low feedback capacitance (0.19 pF typ.).

The 3N139 is hermetically sealed in the standard 4-lead JEDEC TO-72 package.

Maximum Ratings, *Absolute-Maximum Values:*

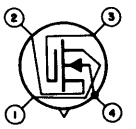
DRAIN-TO-SOURCE VOLTAGE, V_{DS}	+35 max.	V
DRAIN-TO-SUBSTRATE VOLTAGE, V_{DB}	+35, -0.3 max.	V
SOURCE-TO-SUBSTRATE VOLTAGE, V_{SB}	+35, -0.3 max.	V
DC GATE-TO-SOURCE VOLTAGE, V_{GS}	±10 max.	V
PEAK GATE-TO-SOURCE VOLTAGE, V_{GS}	±14 max.	V
PEAK VOLTAGE, GATE-TO-ALL OTHER TERMINALS; V_{GS}, V_{GD}, V_{GB}, non-repetitive	±42 max.	V
DRAIN CURRENT, I_D	50 max.	mA
TRANSISTOR DISSIPATION, P_T:		
At ambient temperatures up to 25°C	330	mW
above 25°C	Derate linearly at 2.2	mW/°C
AMBIENT TEMPERATURE RANGE:		
Storage	-65 to +175	°C
Operating	-65 to +175	°C
LEAD TEMPERATURE (During Soldering):		
At distance not closer than 1/32 inch to seating surface for 10 seconds max.	265 max.	°C

* Metal-Oxide-Semiconductor

FEATURES

- high input resistance $R_{GS} = 10^{14}$ Ω typ.
- low input capacitance $C_{iss} = 3$ pF typ.
- low feed back capacitance $C_{rss} = 0.2$ pF typ.
- low gate leakage current $I_{GSS} = 0.1$ nA typ.
- high drain-to-source voltage: +35 max. V

TERMINAL ARRANGEMENT



1 - Drain
2 - Source
3 - Insulated Gate
4 - Bulk (Substrate) and Case

ELECTRICAL CHARACTERISTICS, at T_A = 25° C Unless Otherwise Specified. Bulk (Substrate) Connected to Source

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS				LIMITS			UNITS
		FREQUENCY f	DC DRAIN-TO-SOURCE VOLTAGE V_{DS}	DC GATE-TO-SOURCE VOLTAGE V_{GS}	DC DRAIN CURRENT I_D				
		MHz	V	V	mA	Min.	Typ.	Max.	
Drain-to-Source Cutoff Current	$I_{D(OFF)}$		15	-8		—	—	50	μA
Zero-Bias Drain Current*	I_{DSS}		15	0		5	15	25	mA
Gate Reverse Current	I_{GSS}	T_A = 25°C	0	±10		—	—	1	nA
		T_A = 100°C	0	±10		—	—	100	nA
Gate-to-Source Cutoff Voltage	$V_{GS(OFF)}$		15		0.05	-2	-4	-6	V
Small-Signal, Short-Circuit Reverse-Transfer Capacitance (Drain-to-Gate)	C_{rss}	1	15		5	0.05	0.2	0.4	pF
Input Resistance	r_{is}	100	15		5		12	—	kΩ
Input Capacitance	C_{iss}	100	15		5	—	3	10	pF
Output Resistance	r_{os}	100	15		5		6	—	kΩ
Output Capacitance	C_{oss}	100	15		5	—	1.4	—	pF
Forward Transconductance	g_{fs}	1 kHz	15		5		5	—	mmho

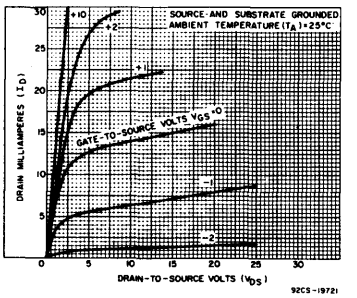


Fig. 1 – Drain Current vs Drain Voltage

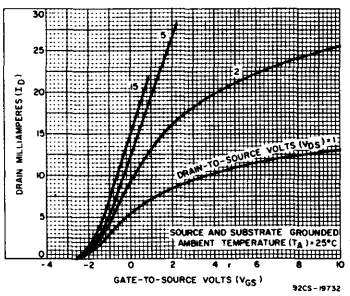


Fig. 2 – Drain Current vs Gate-to-Source Voltage

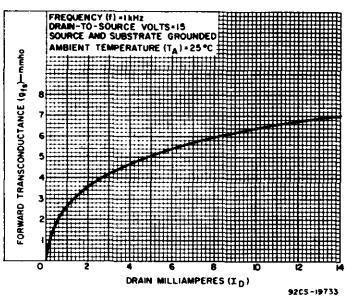


Fig. 3 – 1 kHz forward transconductance vs drain current

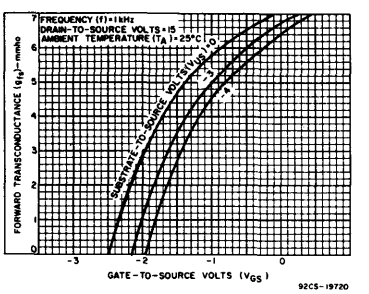


Fig. 4 – 1 kHz forward transconductance vs gate-to-source voltage

3N140, 3N141

SILICON DUAL INSULATED-GATE FIELD-EFFECT TRANSISTORS

N-Channel Depletion Types

For Amplifier and Mixer Applications Up to 300 MHz

RCA-3N140 and 3N141* are n-channel silicon, depletion type, dual insulated-gate, field-effect transistors utilizing the MOS** construction. They have exceptional characteristics for rf-amplifier and mixer applications at frequencies up to 300 MHz. These transistors feature a series arrangement of two separate channels, each channel having an independent control gate.

The 3N140, used in a common-source configuration in which gate No.2 is ac grounded, reduces oscillator feedthrough to the antenna thereby minimizing oscillator radiation. The 3N141 provides excellent isolation between the oscillator and rf signals because each of the two signal frequencies being mixed has its own control element.

The mixing function performed by the 3N141 is unique in that the signal applied to gate No.2 is used to modulate the input-gate (gate No.1) transfer characteristic. This technique is superior to conventional "square law" mixing, which can only be accomplished in the nonlinear region of the device transfer characteristic.

The use of the 3N141 as described provides high useful conversion gains at all vhf frequencies, and the reduction in spurious responses is substantial and easily obtainable in simple circuits.

Maximum Ratings, *Absolute-Maximum Values, at* $T_A = 25°C$

DRAIN-TO-SOURCE VOLTAGE, V_{DS}	0 to +20 V
GATE No.1-TO-SOURCE VOLTAGE, V_{G1S}:	
Continuous (dc)	-8 to +1 V
Peak ac	-8 to +20 V
GATE No.2-TO-SOURCE VOLTAGE, V_{G2S}:	
Continuous (dc)	-8 to 40% of V_{DS} V
Peak ac	-8 to +20 V
DRAIN-TO-GATE VOLTAGE, V_{DG1} OR V_{DG2}	+20 V
DRAIN CURRENT, I_D (Pulsed): Pulse duration ≤ 20 ms, duty factor ≤ 0.15	50 mA
TRANSISTOR DISSIPATION, P_T:	
At ambient temperatures up to 25°C	400 mW
At ambient temperatures above 25°C	derate linearly at 2.67 mW/°C
AMBIENT TEMPERATURE RANGE:	
Storage and Operating	-65 to +175 °C
LEAD TEMPERATURE (During soldering):	
At distances ≥ 1/32 inch from seating surface for 10 seconds max.	265 °C

The 3N140 and 3N141 are hermetically sealed in metal JEDEC TO-72 packages.

* Formerly Dev. Nos. TA2644 and TA7274, respectively.

** Metal-Oxide-Semiconductor.

APPLICATIONS

- RF amplifier and mixer in military and industrial communications equipment
- aircraft and marine vehicular receivers
- CATV and MATV equipment
- telemetry and multiplex equipment

PERFORMANCE FEATURES

- wide dynamic range permits large-signal handling before overload
- dual-gate permits simplified agc circuitry
- virtually no agc power required
- greatly reduces spurious responses in fm receivers
- permits use of vacuum-tube biasing techniques
- excellent thermal stability
- superior cross-modulation performance and greater dynamic range than bipolar or single-gate FET's

DEVICE FEATURES

- low gate leakage currents -- I_{G1SS} & I_{G2SS} = 1 nA max. at $T_A = 25°C$
- high forward transconductance -- g_{fs} = 6000 μmho min.
- high unneutralized RF power gain -- G_{ps} = 16 dB min. at 200 MHz
- low VHF noise figure -- 4.5 dB max. at 200 MHz

ELECTRICAL CHARACTERISTICS, at $T_A = 25°C$ Unless Otherwise Specified. Common-Source Circuit.

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	LIMITS						UNITS
			TYPE 3N140 RF AMPLIFIER			TYPE 3N141 MIXER			
			MIN.	TYP.	MAX.	MIN.	TYP.	MAX.	
Gate No.1-to-Source Cutoff Voltage	$V_{G1S}(off)$	V_{DS} = +16V, I_D = 200 μA, V_{G2S} = +4V	-	-2	-4	-	-2	-4	V
Gate No.2-to-Source Cutoff Voltage	$V_{G2S}(off)$	V_{DS} = +16V, I_D = 200 μA, V_{G1S} = 0	-	-2	-4	-	-2	-4	V
Gate No.1 Leakage Current	I_{G1SS}	V_{G1S} = -20V, V_{G2S} = 0, V_{DS} = 0, T_A = 25°C	-	-	1	-	-	1	nA
		V_{G1S} = +1V, V_{G2S} = 0, V_{DS} = 0, T_A = 25°C	-	-	1	-	-	1	nA
		V_{G1S} = -20V, V_{G2S} = 0, V_{DS} = 0, T_A = 125°C	-	-	0.2	-	-	0.2	μA
Gate No.2 Leakage Current	I_{G2SS}	V_{G2S} = -20V, V_{G1S} = 0, V_{DS} = 0, T_A = 25°C	-	-	1	-	-	1	nA
		V_{G2S} = +1V, V_{DS} = 0, V_{G1S} = 0, T_A = 25°C	-	-	1	-	-	1	nA
		V_{G2S} = -20V, V_{G1S} = 0, V_{DS} = 0, T_A = 125°C	-	-	0.2	-	-	0.2	μA
Zero-Bias Drain Current	I_{DSS}*	V_{DD} = +14V, V_{G1S} = 0, V_{G2S} = +4	5	18	30	5	18	30	mA
Forward Transconductance (Gate No.1 to Drain)	g_{fs}	V_{DD} = +14V, I_D = 10 mA, V_{G2S} = +4V, f = 1 kHz	6000	10000	18000	6000	10000	18000	μmho
Cutoff Forward Transconductance (Gate No.1 to Drain)	$g_{fs}(off)$	V_{DD} = +14V, V_{G1S} = -0.5V, V_{G2S} = -2V, f = 1 kHz	-	-	100	-	-	-	μmho
Small-Signal, Short-Circuit Input Capacitance▲	C_{iss}	V_{DS} = +13V, I_D = 10 mA, V_{G2S} = +4V, f = 1 MHz	3	5.5	7	3	5.5	7	pF
Small-Signal, Short-Circuit Reverse Transfer Capacitance (Drain to Gate No.1)▲	C_{rss}	V_{DS} = +13V, I_D = 10 mA, V_{G2S} = +4V, f = 1 MHz	0.01	0.02	0.03	0.01	0.02	0.03	pF
Small-Signal Short-Circuit Output Capacitance	C_{oss}	V_{DS} = +13V, I_D = 10 mA, V_{G2S} = +4V, f = 1 MHz	-	2.2	-	-	2.2	-	pF
Power Gain (See Fig.1 for Measurement Circuit)	G_{ps}	V_{DD} = +15V, R_S = 270Ω, f = 200 MHz, R_G = 50Ω	16	18	-	-	-	-	dB
Conversion Power Gain (See Fig.2 for Measurement Circuit)	G_{psc}	V_{DD} = +15V, R_S = 120Ω, f_{IN} = 200 MHz, f_{OUT} = 30 MHz, Oscillator injection voltage● = 2.5 V (rms)	-	-	-	13	17	-	dB
Measured Noise Figure (See Fig.1 for Measurement Circuit)	NF	V_{DD} = +15V, R_S = 270Ω, f = 200 MHz, R_G = 50Ω	-	3.5	4.5	-	-	-	dB

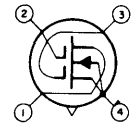
* Pusle test: Pulse duration ≤ 20 ms, duty factor ≤ 0.15.

▲ Capacitance between Gate No.1 and all other terminals.

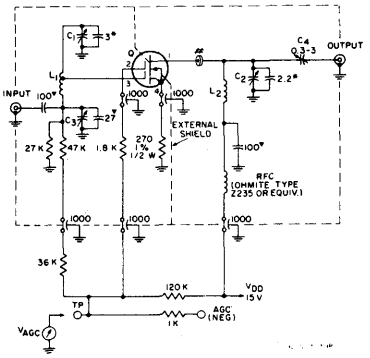
▲ Three-Terminal Measurement with Gate No.2 and Source Returned to Guard Terminal.

● Measured from gate No.2 to source.

TERMINAL DIAGRAM



LEAD 1 - DRAIN
LEAD 2 - GATE No.2
LEAD 3 - GATE No.1
LEAD 4 - SOURCE, SUBSTRATE AND CASE



Q = 3N140.

▼ Disc ceramic. All resistors in ohms

* Tubular ceramic. All capacitors in pF

Ferrite bead (1/2 used); Indiana General No.H1742C-(A-147), F-1157-1-H

C_1, C_2: 1.5-5 pF variable air capacitor: E.F. Johnson Type 160-102 or equivalent.

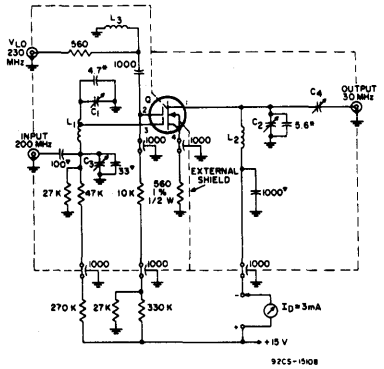
C_3: 1-10 pF piston-type variable air capacitor: JFD Type VAM-010, Johanson Type 4335, or equivalent.

C_4: 0.3-3 pF piston-type variable air capacitor: Roanwell Type MH-13 or equivalent.

L_1: 5 turns silver-plated 0.02" thick, 0.07 ±0.08" wide copper ribbon. Internal diameter of winding = 0.25", winding length approx. 0.65". Tapped at 1-1/2 turns from C_1 end of winding.

L_2: Same as L_1 except winding length approx. 0.7"; no tap.

Fig.1 - 200 MHz power gain and noise figure test circuit for type 3N140.



Q = 3N141.

▼ Disc ceramic.

* Tubular ceramic.

All resistors in ohms

All capacitors in pF

C_1, C_2: 1.5-5 pF variable air capacitor: E.F. Johnson Type 160-102 or equivalent.

C_3: 1-10 pF piston-type variable air capacitor: JFD Type VAM-010, Johanson Type 4335, or equivalent.

C_4: 0.3-3 pF piston-type variable air capacitor: Roanwell Type MH-13 or equivalent.

L_1: 5 turns silver-plated 0.02" thick, 0.07 ±0.08" wide copper ribbon. Internal diameter of winding = 0.25", winding length approx. 0.65". Tapped at 1-1/2 turns from C_1 end of winding.

L_2: Ohmite Z-144 RF choke or equivalent.

L_3: J.W. Miller Co. #4580 0.1 μH RF choke or equivalent.

Note: If 50Ω meter is used in place of sweep detector, a low pass filter must be provided to eliminate local oscillator voltage from load.

Fig.2 - Conversion power gain test circuit for type 3N141.

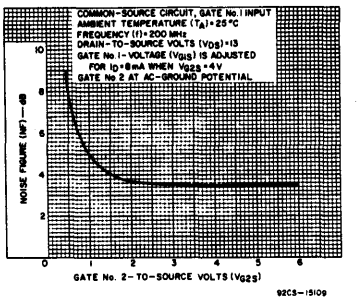


Fig.3 - NF vs V_{G2S}.

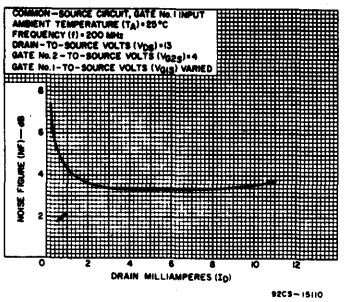


Fig.4 - NF vs I_D.

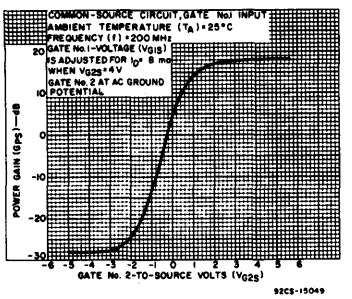


Fig.5 - G_{PS} vs V_{G2S} (For 3N140).

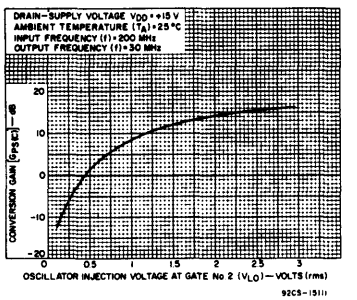


Fig.6 - $G_{PS(C)}$ vs V_{LO} (For 3N141).

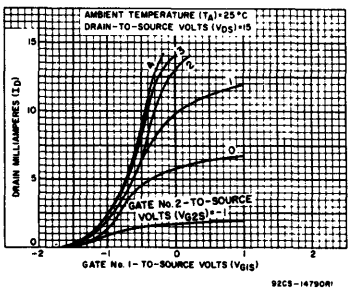


Fig.7 - I_D vs V_{G1S}.

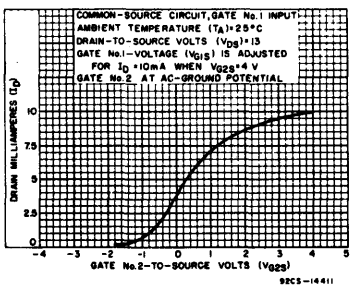


Fig.8 - I_D vs V_{G2S}.

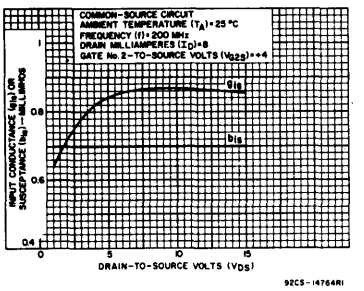


Fig.9 - y_{is} vs V_{DS}.

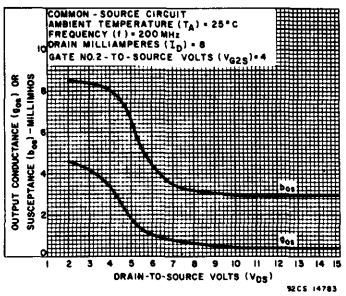


Fig.10 - y_{os} vs V_{DS}.

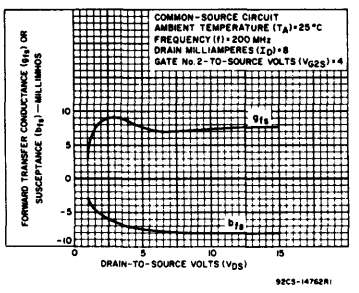


Fig.11 - y_{fs} vs V_{DS}.

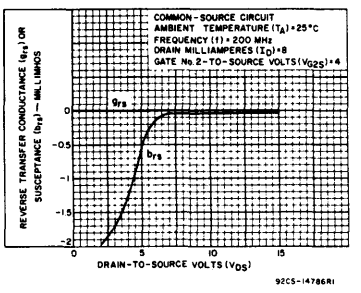


Fig.12 - y_{rs} vs V_{DS}.

3N140, 3N141

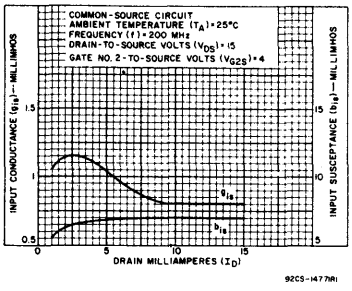


Fig.13 - y_{is} vs I_D.

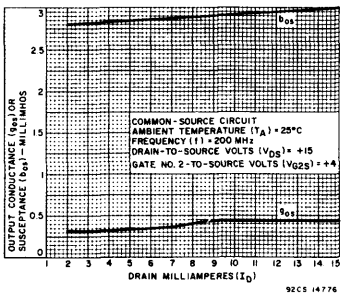


Fig.14 - y_{os} vs I_D.

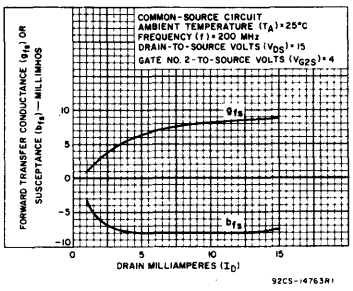


Fig.15 - y_{fs} vs I_D.

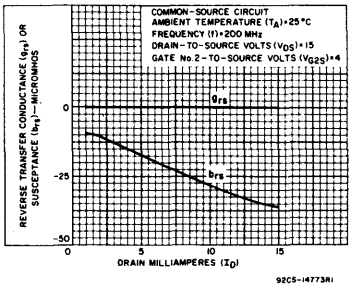


Fig.16 - y_{rs} vs I_D.

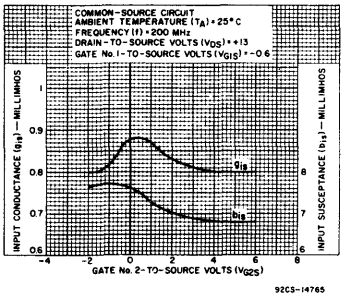


Fig.17 - y_{is} vs V_{G2S}.

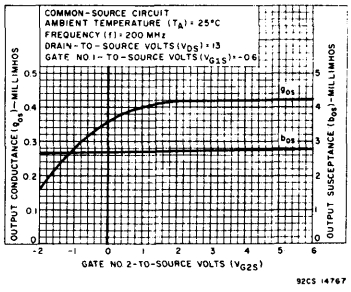


Fig.18 - y_{os} vs V_{G2S}.

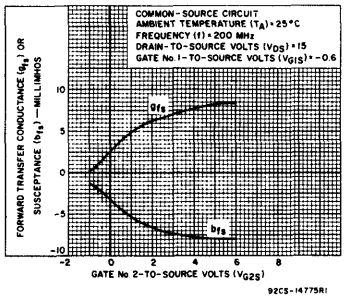


Fig.19 - y_{fs} vs V_{G2S}.

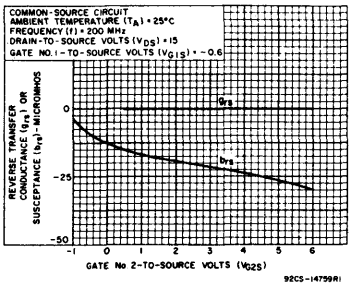


Fig.20 - y_{rs} vs V_{G2S}.

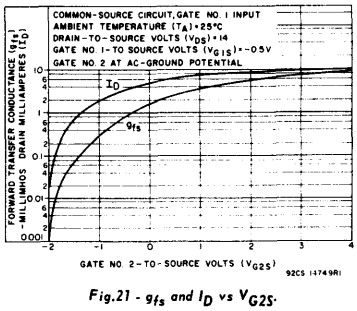


Fig.21 - g_{fs} and I_D vs V_{G2S}.

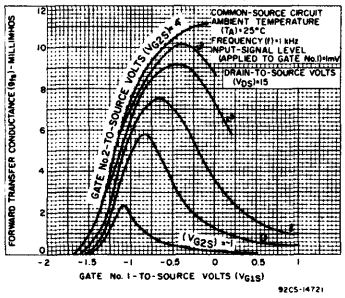


Fig.22 - g_{fs} vs V_{G1S}.

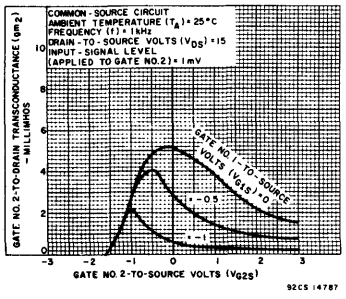


Fig.23 - g_{fs2} vs V_{G2S}.

Silicon MOS Transistor N-Channel Depletion Type

For Industrial and Military Applications to 175 MHz

The-3N142 is a silicon, insulated-gate field-effect transistor of the N-channel depletion type utilizing the MOS■ construction.

The-3N142 is intended primarily for use as the rf amplifier in FM receivers and general amplifier applications at frequencies up to 175 MHz.

The wide dynamic range of the 3N142 reduces cross-modulation effects in AM receivers and minimizes the generation of spurious responses in FM receivers.

■ Metal-Oxide-Semiconductor

Applications

- **RF amplifier, Mixer, and Oscillator in:**
  CB and Mobile Communication Receivers
  Aircraft and Marine Receivers
  CATV and MATV Equipment
- **Industrial Control Circuits**
- **Variable Attenuators**
- **Current Limiters**
- **Instrumentation Equipment**
- **High-Impedance Timing Circuits**

Maximum Ratings, Absolute-Maximum Values at T_A = 25° C

*DRAIN-TO-SOURCE VOLTAGE, V_{DS}	±20	V
*DRAIN-TO-GATE VOLTAGE, V_{DG}	±20	V
*GATE-TO-SOURCE VOLTAGE, V_{GS}:		
Continuous	+1 to −8	V
Peak ac	±15	V
*DRAIN CURRENT, I_D	50	mA
*TRANSISTOR DISSIPATION, P_T:		
At ambient temperatures up to 25°C	330	mW
At ambient temperatures above 25°C	Derate at 2.2 mW/°C	
*AMBIENT TEMPERATURE RANGE:		
Storage	−65 to +175	°C
Operating	−65 to +175	°C
*LEAD TEMPERATURE (During Soldering):		
At distances ≥ 1/32" from seating surface for 10 seconds max.	265	°C

* In accordance with JEDEC Registration Data Format JS-9 RDF11-B

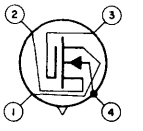
Performance Features

- **Large dynamic range**
- **Enhanced signal-handling capability for low cross-modulation**
- **Dual-polarity gate permits positive and negative swing without degradation of input impedance**
- **Reduced spurious responses in FM receivers**
- **Permits use of vacuum-tube biasing techniques**
- **Excellent thermal stability for critical oscillator designs**

Device Features

- **High input resistance – 1000 megohms**
- **Low feedback capacitance – 0.35 pF max.**
- **Low noise figure – 2.5 dB typ.**
- **High useful power gain – neutralized – 16 dB min. at 100 MHz**
- **Hermetically sealed TO – 72 metal package**

TERMINAL DIAGRAM



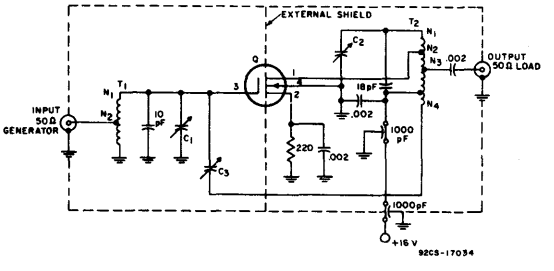
LEAD 1 - DRAIN
LEAD 2 - SOURCE
LEAD 3 - INSULATED GATE
LEAD 4 - BULK (SUBSTRATE) AND CASE

ELECTRICAL CHARACTERISTICS: (At T_A = 25° C)

Measured with Substrate Connected to Source Unless Otherwise Specified.

	CHARACTERISTICS	SYMBOLS	CONDITIONS	LIMITS Min.	Typ.	Max.	UNITS
*	Gate Leakage Current	I_{GSS}	V_{DS} = 0, V_{GS} = -8 V, T_A = 25° C	-	0.0001	1	nA
			V_{DS} = 0, V_{GS} = -8 V, T_A = 125° C	-	-	200	nA
			V_{DS} = 0, V_{GS} = +1, T_A = 25° C	-	0.0001	1	nA
			V_{DS} = 0, V_{GS} = +1, T_A = 125° C	-	-	200	nA
*	Zero-Bias Drain Current**	I_{DSS}	V_{DS} = 15 V, V_{GS} = 0	5	15	25	mA
*	Drain-to-Source Cutoff Current	$I_{D(off)}$	V_{DS} = 20 V, V_{GS} = -8 V	-	-	50	μA
*	Gate-to-Source Cutoff Voltage	$V_{GS(off)}$	V_{DS} = 15 V, I_D = 50 μA	-0.5	-3	-8	V
*	Forward Transconductance	g_{fs}	V_{DS} = 15 V, I_D = 5 mA, f = 1 kHz	5000	7500	12,000	μmho
*	Drain-to-Source Channel Resistance	$r_{DS(on)}$	V_{DS} = 0, V_{GS} = 0, f = 1 kHz	-	200	-	Ω
*	Small-Signal Short-Circuit Reverse Transfer Capacitance‡	C_{rss}	V_{DS} = 15 V, I_D = 5 mA, f = 0.1 to 1 MHz	0.10	0.22	0.35	pF
*	Small-Signal Short-Circuit Input Capacitance	C_{iss}	V_{DS} = 15 V, I_D = 5 mA, f = 0.1 to 1 MHz	-	5.5	7	pF
*	Input Admittance	Y_{is}	Common Source Configuration f = 100 MHz V_{DS} = 15V I_D = 5 mA	-	0.155+J3.45	-	mmho
*	Forward Transfer Admittance	Y_{fs}		-	7.5-J0.9	-	mmho
*	Output Admittance	Y_{os}		-	0.21+J0.9	-	mmho
*	Maximum Available Power Gain	MAG	V_{DS} = 15 V, I_D = 5 mA, f = 100 MHz	-	26	-	dB
	Maximum Usable Power Gain (Fixed Neutralization)	MUG		-	17	-	
*	Insertion Power Gain** (Fixed Neutralization)	G_{ps}		16	-	-	dB
*	Noise Figure**	NF	V_{DS} = 15 V, I_D = 5 mA, f = 100 MHz	-	2.5	4	dB

* In accordance with JEDEC Registration Data Format JS-9 RDF-11B ‡ Three-Terminal Measurement: Source Returned to Guard Terminal
**See Fig. 1



T_1 N_1 = 6 Turns #20 Tinned Copper Wire; ¼" I.D. ½" Long
Q_o = 205, N_1/N_2 = 4.85

T_2 $N_1 + N_4$ = 6¾ Turns #20 Tinned Copper Wire ¼" I.D. $^9/_{16}$" Long
Q_o = 190 N_1/N_2 = 1.9 N_1/N_3 = 12.3 N_1/N_4 = 8

C_1 = 10 pF Variable Air Capacitor (Hammarlund Mac-10 or Equivalent)
C_2 = 5 pF Variable Air Capacitor (Hammarlund Mac-5 or Equivalent)
C_3 = 0.7-3 pF Piston-Type Variable Air Capacitor (Erie 535C or Equivalent)
Q = 3N142

Fig. 1 - Test Set Up for 100 MHz Insertion Power Gain and Noise Figure

For characteristics curves, refer to types 3N128 and 3N143.

3N152

Silicon MOS Transistor N-Channel Depletion Type

For Low-Noise RF Applications in Military & Industrial VHF Communications Equipment Operating up to 250 MHz

RCA-3N152 is an N-channel depletion-type silicon insulated gate field-effect transistor utilizing the MOS■ construction. It is intended primarily for VHF amplifier applications up to 250 MHz in military and industrial equipment.

Because of its improved transfer characteristic and exceptionally wide dynamic range, the 3N152 with the substrate in the reversed bias mode can provide substantially better cross-modulation performance in linear amplifier applications than conventional bipolar transistors. The insulated gate with its extremely low reverse (leakage) current eliminates the problem of diode-current loading of the input circuit under strong input conditions, which is common to junction-type FET's. These features in addition to low feedback capacitance permit the design of circuits providing superior high-frequency operation and high gain without neutralization. The 3N152 utilizes full-gate construction and is hermetically sealed in a JEDEC TO-72 metal package.

■ Metal-Oxide-Semiconductor.

Maximum Ratings, *Absolute-Maximum Values at T_A = 25°C:*

* DRAIN-TO-SOURCE VOLTAGE, V_{DS}	+20 max.	V
* DRAIN-TO-GATE VOLTAGE, V_{DG}	+20	V
* GATE-TO-SOURCE VOLTAGE, V_{GS}:		
* CONTINUOUS (dc)	+1, -8 max.	V
* PEAK ac	±15 max.	V
* DRAIN CURRENT, I_D	50 max.	mA
TRANSISTOR DISSIPATION:		
At ambient temperatures up to 25°C	330 max.	mW
above 25°C	derate at 2.2 mW/°C	
* AMBIENT TEMPERATURE RANGE:		
Storage	-65 to +175	°C
Operating	-65 to +175	°C
* LEAD TEMPERATURE (During Soldering):		
At distances not closer than 1/32 inch to seating surface for 10 seconds maximum	265 max.	°C

*In accordance with Jedec Registration Data Format JS-9 RDF 11-B.

Features

- Low gate leakage current — I_{GSS} = 0.1 pA typ.
- Low feedback capacitance — C_{rss} = 0.25 pF typ.
- High forward *transconductance* — g_{fs} = 7500 μmho typ.
- High vhf power gain — G_{PS} = 16 dB typ. at 200 MHz
- Low vhf noise figure — NF = 2.5 dB typ. at 200 MHz
- Exceptionally good cross-modulation characteristics

Performance

- Large dynamic range
- Greatly reduced spurious responses
- Permits use of vacuum-tube biasing techniques
- Excellent thermal stability
- Superior cross-modulation performance and greater dynamic range than bipolar transistors

ELECTRICAL CHARACTERISTICS AT T_A = 25°C

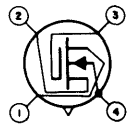
Measured with Substrate Connected to Source Unless Otherwise Specified

	CHARACTERISTICS	SYMBOLS	CONDITIONS	LIMITS 3N152 Min.	Typ.	Max.	UNITS
*	Gate Leakage Current	I_{GSS}	V_{DS} = 0, V_{GS} = -8V, T_A = 25°C	-	0.0001	1	nA
			V_{DS} = 0, V_{GS} = -8 V, T_A = 125°C	-	-	200	nA
*	Zero-Bias Drain Current	I_{DSS}	V_{DS} = 15 V, V_{GS} = 0	5	15	30	mA
	Drain-to-Source Cutoff Current	$I_{D(off)}$	V_{DS} = 20 V, V_{GS} = -8V	-		50	μA
*	Gate-to-Source-Cutoff Voltage	$V_{GS(off)}$	V_{DS} = 15 V, I_D = 50 μA	-0.5	-3	-8	V
*	Forward Transconductance	g_{fs}	V_{DS} 15 V, I_D = 5 mA, f = 1 kHz	5000	7500	12,000	μmho
	Drain-to-Source Channel Resistance	$r_{DS(on)}$	V_{DS} = 0, V_{GS} = 0, f = 1 kHz	-	200	-	Ω
*	Small-Signal Short-Circuit Reverse Transfer Capacitance▲	C_{rss}	V_{DS} = 15 V, I_D 5 mA, f 0.1 to 1 MHz	0.15	0.25	0.35	pF
	Small-Signal Short-Circuit Input Capacitance	C_{iss}	V_{DS} 15 V, I_D = 5 mA, f = 0.1 to 1 MHz	-	5.5	7	pF
	Input Admittance	Y_{is}	Common Source Configuration f 200 MHz	-	0.4 + J7.3	-	mmho
	Forward Transfer Admittance	Y_{fs}	V_{DS} 15 V,	-	7-J2	-	mmho
	Output Admittance	Y_{os}	I_D 5 mA	-	0.28 + J1.8	-	mmho
	Power Gain Maximum Available Gain	MAG		-	21	-	dB
	Insertion Power Gain (Fixed ...ization) See Fig.1	G_{PS}	V_{DS} 15 V, I_D 5 mA, f 200 MHz	14.5	16	-	dB
	Noise Figure (See Figs. 1 & 2)	NF	V_{DS} 15 V, I_D 5 mA, f 200 MHz	-	2.5	3.5	dB

▲ Three-Terminal Measurement. Source Returned to Guard Terminal.

* In accordance with JEDEC Registration Data Format JS-9 RDF-11B.

TERMINAL ARRANGEMENT



1 - Drain
2 - Source
3 - Insulated Gate
4 - Bulk (Substrate) and Case

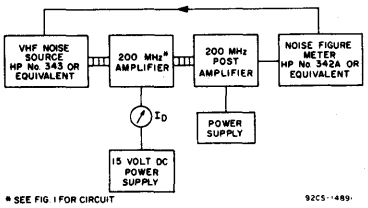
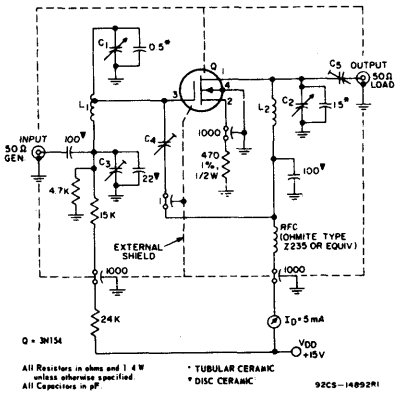


Fig. 2 - Noise figure measurement setup.



C_1, C_2: 1.5-5 pF variable air capacitor: E. F. Johnson Type 160-102 or equivalent

C_3: 1-10 pF piston-type variable air capacitor: JFD Type VAM-010, Johanson Type 4335, or equivalent

C_4, C_5: 0.3-3 pF piston-type variable air capacitor: Roanwell Type MH-13 or equivalent

L_1: 5 turns silver-plated 0.02" thick, 0.07"-0.08" wide copper ribbon. Internal diameter of winding = 0.25"; winding length approx. 0.65". Tapped at 1-1/2 turns from C_1 end of winding

L_2: Same as L_1 except winding length approx. 0.7"; no tap

Fig. 1 - Test circuit used to measure 200-MHz maximum usable power gain and noise Figure.

For characteristics curves, refer to types 3N128 and 3N143.

SILICON INSULATED GATE FIELD-EFFECT TRANSISTOR

N-Channel Depletion Type

RCA 3N153 is a silicon, insulated-gate field-effect transistor of the N-channel depletion type, utilizing the MOS* construction. It is intended primarily for critical chopper and multiplex applications up to 60 MHz.

The insulated gate provides a very high value of input resistance (10^{10} ohms typ) which is relatively insensitive to temperature and is independent of gate-bias conditions (positive, negative, or zero bias). The 3N153 also features extremely low feedback capacitance (0.34 pF typ) and virtually zero inherent offset voltage.

This transistor features a Terminal Arrangement in which the gate and source connections are interchanged to provide maximum isolation between the output (drain) and the input (gate) terminals. Although this new basing configuration does not appreciably change the measured device feedback capacitance, it permits the use of external inter-terminal shields to reduce the feedback due to external capacitances, particularly on printed circuit boards. This feature makes it possible to minimize feedthrough capacitance.

The 3N153 is hermetically sealed in the JEDEC TO-72 package and features a gate metallization that covers the entire source-to-drain channel.

Maximum Ratings, *Absolute-Maximum Values:*

(Substrate connected to source unless otherwise specified)

DRAIN-TO-SOURCE VOLTAGE, V_{DS}	+20	max.	V
DRAIN-TO-SUBSTRATE VOLTAGE, V_{DB}	+20, -0.3	max.	V
SOURCE-TO-SUBSTRATE VOLTAGE, V_{SB}	+20, -0.3	max.	V
DC GATE-TO-SOURCE VOLTAGE, V_{GS}	+6, -8	max.	V
PEAK GATE-TO-SOURCE VOLTAGE, v_{GS}	±14	max.	V
DRAIN CURRENT, I_D (Pulse duration 20 ms, duty factor ≤ 0.10)	50	max.	mA
TRANSISTOR DISSIPATION, P_T:			
At ambient temperatures from -65 to +25°C	400	max.	mW
above 25°C	derate linearly at 2.67		mW/°C
AMBIENT TEMPERATURE RANGE:			
Storage	-65 to +175		°C
Operating	-65 to +175		°C
LEAD TEMPERATURE (During soldering):			
At distance ≥ 1/32" to seating surface for 10 seconds max.	265	max.	°C

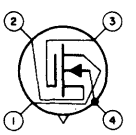
FEATURES

- excellent thermal stability
- virtually zero inherent offset voltage
- low leakage current: 50 pA max.
- low "on" resistance — $r_{DS(on)}$ = 200 Ω typ.
- high "off" resistance — $R_{DS(off)}$ = 10^{10} Ω typ.
- low feedback capacitance — C_{rss} = 0.34 pF typ.
- low input capacitance — C_{iss} = 6 pF typ.

APPLICATIONS

- Choppers
- Multiplexers
- Servo Amplifiers
- Computer Operational Amplifiers
- Sampling Circuits
- Electrometer Amplifiers

TERMINAL DIAGRAM



1 - Drain
2 - Source
3 - Insulated Gate
4 - Bulk (Substrate) and Case

* Metal-Oxide-Semiconductor

ELECTRICAL CHARACTERISTICS, at T_A = 25° C, Unless Otherwise Specified. Substrate Connected to Source.

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	LIMITS Type 3N153 Min.	Typ.	Max.	UNITS
Gate-Leakage Current	I_{GSS}	V_{GS} = +6,-8V; V_{DS} = 0V; T_A = 25°C V_{GS} = +6,-8V; V_{DS} = 0V; T_A = 125°C	- -	0.1 -	50 1	pA nA
Static Drain-to-Source "ON" Resistance	$r_{DS(on)}$	V_{GS} = 0V, V_{DS} = 0V	-	200	300	Ω
Drain-to-Source "OFF" Resistance	$R_{DS(off)}$	V_{GS} = -8V, V_{DS} = +1V	10^9	10^{10}	-	Ω
Drain-to-Source Cutoff Current	$I_{D(off)}$	V_{GS} = -8V, V_{DS} = +1V, T_A = 25°C V_{GS} = -8V, V_{DS} = +1V, T_A = 125°C	- -	0.1 0.1	1 1	nA μA
Small-Signal, Short-Circuit, Reverse Transfer Capacitance	C_{rss}	V_{GS} = -8V, V_{DS} = 0V, f = 1 MHz V_{DS} = 15V, I_D = 5 mA, f = 1 MHz	- -	0.34 0.25	0.5 0.38	pF pF
Small-Signal, Short-Circuit, Input Capacitance	C_{iss}	V_{GS} = -8V, V_{DS} = 0V, f = 1 MHz	-	6	8	pF
Small-Signal, Drain-to-Source Capacitance	C_{ds}	V_{DS} = 0V, V_{GS} = -8V, f = 1 MHz	-	-	3	pF
Zero-Gate-Bias Forward Transconductance	g_{fs}	V_{GS} = 0V, V_{DS} = +15V	-	10,000	-	μmho
Offset Voltage	V_o	V_{GS} = +6,-8V; V_{DS} = 0V	-	0*	-	V

* In measurements of Offset Voltage, thermocouple effects and contact potentials in the measurement setup may cause erroneous readings of 1 microvolt or more. These errors may be minimized by the use of solder having a low thermal e.m.f., such as Leeds & Northrup No.107-1.0.1, or equivalent.

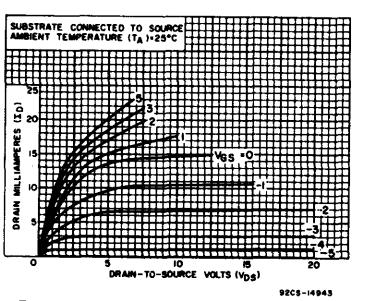


Fig.1 - Drain current vs. drain-to-source voltage.

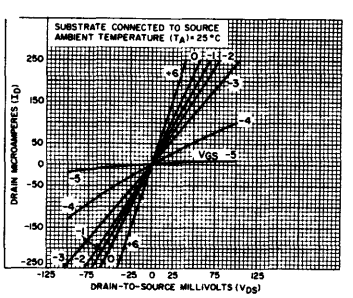


Fig.2 - Low-level drain current vs. drain-to-source voltage.

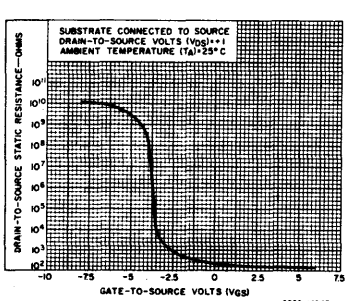


Fig.3 - Drain-to-source static resistance vs. gate-to-source voltage.

3N154

Silicon MOS Transistor N-Channel Depletion Type

For Critical Amplifier Applications in Military & Industrial VHF Communications Equipment Operating up to 250 MHz

RCA 3N154 is an n-channel depletion-type silicon insulated-gate field-effect transistor utilizing the MOS■ construction. It is intended primarily for vhf amplifier applications up to 250 MHz in military and industrial equipment.

Because of its improved transfer characteristic and exceptionally wide dynamic range, the 3N154 can provide substantially better crossmodulation performance in linear amplifier applications than conventional bipolar transistors. The extremely low gate leakage current eliminates diode-current loading of the input circuit under strong signal conditions, a problem which is common to junction-type FET's. These features, in addition to low feedback capacitance, permit the design of circuits providing superior high-frequency operation and high gain without neutralization. The 3N154 utilizes full-gate construction and is hermetically sealed in a JEDEC TO-72 metal package.

■ Metal-Oxide-Semiconductor

Maximum Ratings, *Absolute-Maximum Values at* $T_A = 25^oC$:

*DRAIN-TO-SOURCE VOLTAGE, V_{DS}	+20	V
*DRAIN-TO-GATE VOLTAGE, V_{DG}	+20	V
*GATE-TO-SOURCE VOLTAGE, V_{GS}:		
* CONTINUOUS (dc)	+1, -8	V
* PEAK ac	±15	V
*DRAIN CURRENT, I_D▲	50	mA
*TRANSISTOR DISSIPATION:		
At ambient temperatures up to 25°C	330	mW
above 25°C, derate at	2.2	mW/°C
*AMBIENT TEMPERATURE RANGE:		
Storage	-65 to +175	°C
Operating	-65 to +175	°C
LEAD TEMPERATURE (During Soldering):		
At distances not closer than 1/32 inch to seating surface for 10 seconds maximum	265	°C

In accordance with JEDEC Registration Data Format JS9-RDF-11B

▲ Pulsed:
Pulse duration ≤ 20 ms
Duty factor ≤ 0.15

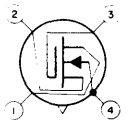
Device Feature:

- Closely controlled IDSS – 10 to 25 mA
- Low gate leakage current – I_{GSS} = 0.1 pA typ.
- Low feedback capacitance – C_{rss} = 0.25 pF typ.
- High forward transconductance – g_{fs} = 7500 μmho typ.
- High vhf power gain – G_{PS} = 16 dB typ. at 200 MHz
- Low vhf noise figure – NF = 3.5 dB typ. at 200 MHz
- Exceptionally good cross-modulation characteristics

Performance Features

- Large dynamic range
- Greatly reduced spurious responses
- Permits use of vacuum-tube biasing techniques
- Excellent thermal stability
- Superior cross-modulation performance and greater dynamic range than bipolar transistors

TERMINAL DIAGRAM



1 - Drain
2 - Source
3 - Insulated Gate
4 - Bulk (Substrate) and Case

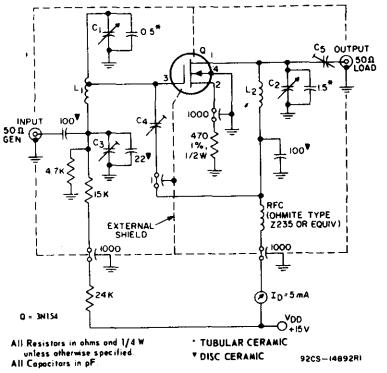
ELECTRICAL CHARACTERISTICS: (At T_A = 25° C)

Measured with Substrate Connected to Source Unless Otherwise Specified.

	CHARACTERISTICS	SYMBOLS	CONDITIONS	LIMITS 3N154 Min.	Typ.	Max.	UNITS
*	Gate Leakage Current	I_{GSS}	V_{DS} = 0, V_{GS} = -8 V, T_A = 25° C	-	0.0001	0.05	nA
			V_{DS} = 0, V_{GS} = -8 V, T_A = 125° C	-	-	5	nA
			V_{DS} = 0, V_{GS} = +1, T_A = 25° C	-	0.0001	0.05	nA
			V_{DS} = 0, V_{GS} = +1, T_A = 125° C	-	-	5	nA
*	Zero-Bias Drain Current	I_{DSS}	V_{DS} = 15 V, V_{GS} = 0	10	15	25	mA
	Drain-to-Source Cutoff Current	$I_{D(off)}$	V_{DS} = 20 V, V_{GS} = -8 V	-	-	50	μA
*	Gate-to-Source Cutoff Voltage	$V_{GS(off)}$	V_{DS} = 15 V, I_D = 50 μA	-0.5	-3	-8	V
	Forward Transconductance	g_{fs}	V_{DS} = 15 V, I_D = 5 mA, f = 1 kHz	5000	7500	12,000	μmho
	Drain-to-Source Channel Resistance	$r_{DS(on)}$	V_{DS} = 0, V_{GS} = 0, f = 1 kHz	-	200	-	Ω
*	Small-Signal Short-Circuit Reverse Transfer Capacitance	C_{rss}	V_{DS} = 15 V, I_D = 5 mA, f = 0.1 to 1 MHz	0.15	0.25	0.35	pF
	Small-Signal Short-Circuit Input Capacitance▲	C_{iss}	V_{DS} = 15 V, I_D = 5 mA, f = 0.1 to 1 MHz	-	5.5	7	pF
	Input Admittance	Y_{is}	Common Source Configuration f = 200 MHz, V_{DS} = 15 V, I_D = 5 mA	-	0.4 + J7.3	-	mmho
	Forward Transfer Admittance	Y_{fs}		-	7 - J2	-	mmho
	Output Admittance	Y_{os}		-	0.28 + J1.8	-	mmho
	Maximum Available Power Gain	MAG	V_{DS} = 15 V, I_D = 5 mA, f = 200 MHz	-	21	-	dB
*	Insertion Power Gain (Fixed Neutralization) (see Fig. 1)	G_{PS}		13.5	16	-	dB
*	Noise Figure (see Figs. 1 & 2)	NF	V_{DS} = 15 V, I_D = 5 mA, f = 200 MHz	-	3.5	5	dB

* In Accordance with JEDEC Registration Data Format JS-9 RDF-11B

▲ Three-Terminal Measurement: Source Returned to Guard Terminal



C_1, C_2: 1.5-5 pF variable air capacitor: E. F. Johnson Type 160-102 or equivalent

C_3: 1-10 pF piston-type variable air capacitor: JFD Type VAM-010, Johanson Type 4335, or equivalent

C_4, C_5: 0.3-3 pF piston-type variable air capacitor: Roanwell Type MH-13 or equivalent

Q = 3N154

L_1: 5 turns silver-plated 0.02" thick, 0.07"-0.08" wide copper ribbon. Internal diameter of winding = 0.25"; winding length approx. 0.65". Tapped at 1-1/2 turns from C_1 end of winding

L_2: Same as L_1 except winding length approx. 0.7"; no tap.

Fig. 1 - Test circuit used to measure 200-MHz maximum usable power gain and noise figure

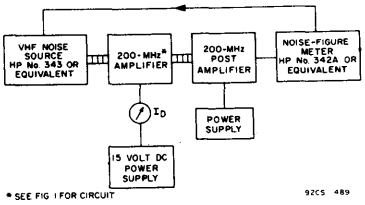


Fig. 2 - Noise figure measurement setup

For characteristics curves, refer to types 3N128 and 3N143.

SILICON DUAL INSULATED-GATE FIELD-EFFECT TRANSISTOR

N-Channel Depletion Type

For Military and Industrial Low-Noise RF-Amplifier Applications Up to 300 MHz

The 3N159 is an n-channel silicon, depletion type, dual insulated-gate, field-effect transistor utilizing the MOS** construction. It has exceptional characteristics for rf-amplifier applications at frequencies up to 300 MHz. This transistor features a series arrangement of two separate channels, each channel having an independent control gate.

Type 3N159 has an exceptionally low-noise figure, which makes this type particularly suitable for critical vhf applications. When used in a common-source configuration in which gate No.2 is ac grounded, this device reduces oscillator feedthrough to the antenna thereby minimizing oscillator radiation.

The 3N159 is hermetically sealed in the metal JEDEC TO-72 package.

** Metal-Oxide-Semiconductor.

APPLICATIONS

- RF amplifier in military and industrial communications equipment
- aircraft, marine and vehicular receivers
- CATV and MATV equipment
- telemetry and multiplex equipment

Maximum Ratings, *Absolute-Maximum Values:* at T_A = 25°C

DRAIN-TO-SOURCE VOLTAGE, V_{DS} . . . . . 0 to +20 V

GATE-No.1-TO-SOURCE VOLTAGE, V_{G1S}:
- Continuous (dc) . . . . . -8 to +1 V
- Peak ac . . . . . -8 to +20 V

GATE No.2-TO-SOURCE VOLTAGE, V_{G2S}:
- Continuous (dc) . . . . . -8 to 40% of V_{DS} V
- Peak ac . . . . . -8 to +20 V

DRAIN-TO-GATE VOLTAGE:
- V_{DG1} or V_{DG2} . . . . . +20 V

DRAIN CURRENT, I_D
- Pulsed: Pulse duration ≤ 20 ms, duty factor ≤ 0.15 . . . . . 50 mA

TRANSISTOR DISSIPATION, P_T:
- At ambient temperatures: up to 25°C . . . . . 400 mW; above 25°C . . . . . derate linearly at 2.67 mW/°C

AMBIENT TEMPERATURE RANGE:
- Storage and Operating . . . . . -65 to +175 °C

LEAD TEMPERATURE (During soldering):
- At distances ≥ 1/32 inch from seating surface for 10 seconds max. . . . . 265 °C

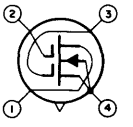
PERFORMANCE FEATURES

- wide dynamic range permits large-signal handling before overload
- dual-gate permits simplified agc circuitry
- virtually no agc power required
- greatly reduces spurious responses in FM receivers
- permits use of vacuum-tube biasing techniques
- excellent thermal stability
- superior cross-modulation performance and greater dynamic range than bipolar or single-gate field-effect transistors

DEVICE FEATURES

- low gate leakage currents — — I_{G1SS} & I_{G2SS} = 1 nA max.
- high forward transconductance — — g_{fs} = 7000 μmho min.
- high unneutralized RF power gain — — G_{ps} = 16 dB min. at 200 MHz
- low vhf noise figure — — NF = 3.5 dB max. at 200 MHz

TERMINAL DIAGRAM



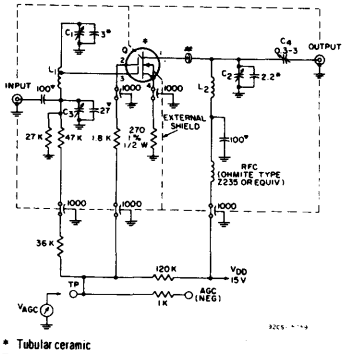
LEAD 1 - DRAIN
LEAD 2 - GATE No.2
LEAD 3 - GATE No.1
LEAD 4 - SOURCE, SUBSTRATE AND CASE

ELECTRICAL CHARACTERISTICS, at T_A = 25°C unless otherwise specified

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	LIMITS 3N159 Min.	Typ.	Max.	UNITS
Gate-No.1-to-Source Cutoff Voltage	$V_{G1S(off)}$	V_{DS} = +16V, I_D = 200 μA, V_{G2S} = +4V	-	-2	-4	V
Gate-No.2-to-Source Cutoff Voltage	$V_{G2S(off)}$	V_{DS} = +16V, I_D = 200 μA, V_{G1S} = 0	-	-2	-4	V
Gate-No.1-Leakage Current	I_{G1SS}	V_{G1S} = -20V, V_{G2S} = 0, V_{DS} = 0, T_A = 25°C	-	-	1	nA
		V_{G1S} = +1V, V_{G2S} = 0, V_{DS} = 0, T_A = 25°C	-	-	1	nA
		V_{G1S} = -20V, V_{G2S} = 0, V_{DS} = 0, T_A = 125°C	-	-	0.2	μA
Gate-No.2-Leakage Current	I_{G2SS}	V_{G2S} = -20V, V_{G1S} = 0, V_{DS} = 0, T_A = 25°C	-	-	1	nA
		V_{G2S} = +1, V_{DS} = 0, V_{G1S} = 0, T_A = 25°C	-	-	1	nA
		V_{G2S} = -20V, V_{G1S} = 0, V_{DS} = 0, T_A = 125°C	-	-	0.2	μA
Zero-Bias Drain Current	I_{DSS}*	V_{DD} = +14V, V_{G1S} = 0, V_{G2S} = +4V	5	18	30	mA
Forward Transconductance (Gate-No.1-to-Drain)	g_{fs}	V_{DD} = +14V, I_D = 10 mA, V_{G2S} = +4V, f = 1 kHz	7000	10,000	18,000	μmho
Cutoff Forward Transconductance (Gate-No.1-to-Drain)	$g_{fs(off)}$	V_{DD} = +14V, V_{G1S} = -0.5V, V_{G2S} = -2V, f = 1 kHz	-	-	100	μmho
Small-Signal, Short-Circuit Input Capacitance▲	C_{iss}	V_{DS} = +13V, I_D = 10 mA, V_{G2S} = +4V, f = 1 MHz	3	5.5	7	pF
Small-Signal, Short-Circuit, Reverse Transfer Capacitance (Drain-to-Gate No.1)♦	C_{rss}	V_{DS} = +13V, I_D = 10 mA, V_{G2S} = +4V, f = 1 MHz	0.01	0.02	0.03	pF
Small-Signal, Short-Circuit Output Capacitance	C_{oss}	V_{DS} = +13V, I_D = 10 mA, V_{G2S} = +4V, f = 1 MHz	-	2.2	-	pF
Maximum Usable Power Gain (See Fig.1 for Measurement Circuit)	MUG	V_{DD} = +15V, R_S = 270Ω, R_G = 50Ω, f = 200 MHz	16	18	22	dB
Measured Noise Figure (See Fig.1 for Measurement Circuit)	NF	V_{DD} = +15V, R_S = 270Ω, f = 200 MHz, R_G = 50Ω	-	2.5	3.5	dB

* Pulse Test: Pulse duration ≤ 20 ms, duty factor ≤ 0.15.
▲ Capacitance between Gate No.1 and all other terminals.
♦ Three-Terminal Measurement with Gate No.2 and Source Returned to Guard Terminal.

For characteristics curves refer to types 3N140, 3N141.



* Tubular ceramic
▼ Disc ceramic
Ferrite bead (1/2 used); Indiana General No. H 1742C-(A-147) or F1157-1-H or equivalent.
‡ VHF plug in socket Jettron CD72-148 and CD72149 (part No.7977-1) or equivalent.

C_1, C_2: 1.5-5 pF variable air capacitor: E. F. Johnson Type 160-102 or equivalent.
C_3: 1-10 pF piston-type variable air capacitor: JFD Type VAM-010, Johanson Type 4335, or equivalent.
C_4: 0.3-3 pF piston-type variable air capacitor: Roanwell Type MH-13 or equivalent.

L_1: 5 turns silver-plated 0.02" thick, 0.07"-0.08" wide copper ribbon. Internal diameter of winding = 0.25"; winding length approx. 0.65". Tapped at 1-1 2 turns from C_1 end of winding.

L_2: Same as L_1 except winding length approx. 0.7"; no tap.

Fig.1 - 200-MHz power gain and noise-figure test circuit for type 3N159.

3N187

Silicon Dual Insulated-Gate Field-Effect Transistor

N-Channel Depletion Type

With Integrated Gate-Protection Circuits

For Military and Industrial Applications up to 300 MHz

Device Features

- Back-to-back diodes protect each gate against handling and in-circuit transients
- *High forward transconductance* – g_{fs} = 12,000 μmho (typ.)
- High unneutralized RF power gain ~ G_{ps} = 18 dB(typ.) at 200 MHz
- Low VHF noise figure – 3.5 dB(typ.) at 200 MHz

RCA-3N187 is an n-channel silicon, depletion type, dual insulated-gate field-effect transistor.

Special back-to-back diodes are diffused directly into the MOS▲ pellet and are electrically connected between each insulated gate and the FET's source. The diodes effectively bypass any voltage transients which exceed approximately ±10 volts. This protects the gates against damage in all normal handling and usage.

A feature of the back-to-back diode configuration is that it allows the 3N187 to retain the wide input signal dynamic range inherent in the MOSFET. In addition, the junction capacitance of these diodes adds little to the total capacitance shunting the signal gate.

The excellent overall performance characteristics of the RCA-3N187 make it useful for a wide variety of rf-amplifier applications at frequencies up to 300 MHz. The two serially-connected channels with independent control gates make possible a greater dynamic range and lower cross-modulation than is normally achieved using devices having only a single control element.

The two-gate arrangement of the 3N187 also makes possible a desirable reduction in feedback capacitance by operating in the common-source configuration and ac-grounding Gate No. 2. The reduced capacitance allows operation at maximum gain *without neutralization;* and, of special importance in rf-amplifiers, it reduces local oscillator feedthrough to the antenna.

The 3N187 is hermetically sealed in the metal JEDEC TO-72 package.

▲ Metal-Oxide-Semiconductor

Maximum Ratings,

Absolute-Maximum Values, at T_A = 25°C

DRAIN-TO-SOURCE VOLTAGE, V_{DS}	-0.2 to +20	V
GATE No. 1-TO-SOURCE VOLTAGE, V_{G1S}:		
Continuous (dc)	-6 to +3	V
Peak ac	-6 to +6	V
GATE No. 2-TO-SOURCE VOLTAGE, V_{G2S}:		
Continuous (dc)	-6 to 30% of V_{DS}	V
Peak ac	-6 to +6	V
* DRAIN-TO-GATE VOLTAGE, V_{DG1} OR V_{DG2}	+20	V
* DRAIN CURRENT, I_D	50	mA
* TRANSISTOR DISSIPATION P_T:		
At ambient temperatures up to 25°C	330	mW
At ambient temperatures above 25°C	derate linearly at 2.2 mW/°C	
* AMBIENT TEMPERATURE RANGE:		
Storage and Operating	-65 to +175	°C
* LEAD TEMPERATURE (During Soldering):		
At distances ≥ 1/32 inch from seating surface for 10 seconds max.	265	°C

* In accordance with JEDEC Registration Data Format JS-9 RDF-19A

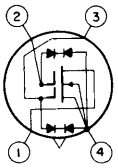
Applications

- RF amplifier, mixer, and IF amplifier in military, and industrial communications equipment
- Aircraft and marine vehicular receivers
- CATV and MATV equipment
- Telemetry and multiplex equipment

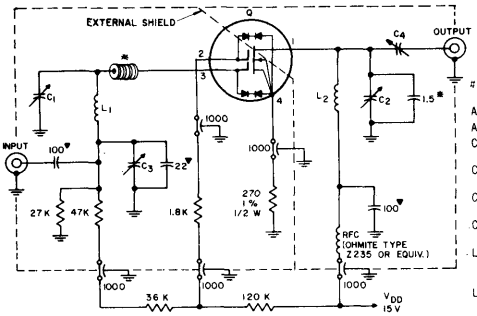
Performance Features

- Superior cross-modulation performance and greater dynamic range than bipolar or single-gate FET's
- Wide dynamic range permits large-signal handling before overload
- Virtually no agc power required
- Greatly reduces spurious responses in FM receivers

TERMINAL DIAGRAM



LEAD 1-DRAIN
LEAD 2-GATE No. 2
LEAD 3-GATE No. 1
LEAD 4-SOURCE, SUBSTRATE AND CASE



Ferrite bead (4): Pyroferric Co. "Carbonyl J" 0.09 in. OD; 0.03 in. ID; 0.063 in. thickness. Q = 3N187

All resistors in ohms

All capacitors in pF

C_1: 1.8–8.7 pF variable air capacitor: E.F. Johnson Type 160-104, or equivalent.

C_2: 1.5–5 pF variable air capacitor: E.F. Johnson Type 160-102, or equivalent.

C_3: 1–10 pF piston-type variable air capacitor: JFD Type VAM-010; Johanson Type 4335, or equivalent.

C_4: 0.8–4.5 pF piston type variable air capacitor: Erie 560-013 or equivalent.

L_1: 4 turns silver-plated 0.02-in. thick, 0.075-0.085-in. wide, copper ribbon. Internal diameter of winding = 0.25 in, winding length approx. 0.08 in.

L_2: 4½ turns silver-plated 0.02-in thick, 0.085-0.095-in. wide, 5/16-in. ID. Coil ≈.90 in. long.

* Tubular ceramic.

Disc ceramic.

Fig. 1 - 200-MHz Power gain and noise-figure test circuit

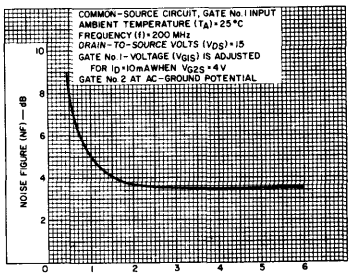


Fig. 2 - NF vs. V_{G2S}

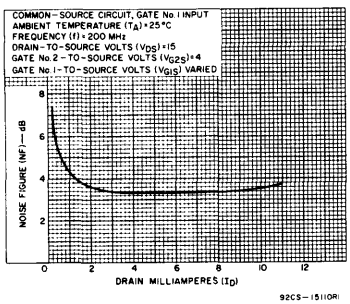


Fig. 3 - NF vs. I_D

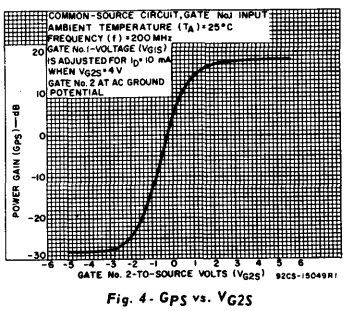


Fig. 4 - G_{PS} vs. V_{G2S}

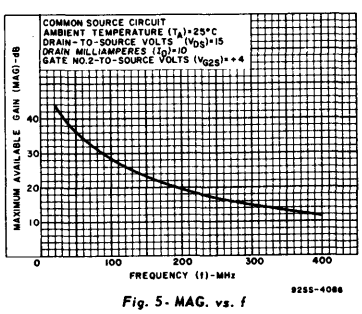


Fig. 5 - MAG. vs. f

ELECTRICAL CHARACTERISTICS, at T_A = 25° C unless otherwise specified

CHARACTERISTICS	SYMBOL	TEST CONDITIONS	LIMITS Min.	Typ.	Max.	UNITS
* Gate No. 1-to-Source Cutoff Voltage	$V_{G1S(off)}$	V_{DS} = +15 V, I_D = 50 μA, V_{G2S} = +4 V	-0.5	-2	-4	V
* Gate No. 2-to-Source Cutoff Voltage	$V_{G2S(off)}$	V_{DS} = +15 V, I_D = 50 μA, V_{G1S} = 0	-0.5	-2	-4	V
* Gate No. 1-Terminal Forward Current	I_{G1SSF}	V_{G1S} = +1 V, V_{G2S} = V_{DS} = 0, T_A = 25° C	–	–	50	nA
		T_A = 100° C	–	–	5	μA
* Gate No. 1-Terminal Reverse Current	I_{G1SSR}	V_{G1S} = -6 V, V_{G2S} = V_{DS} = 0, T_A = 25° C	–	–	50	nA
		T_A = 100° C	–	–	5	μA
* Gate No. 2-Terminal Forward Current	I_{G2SSF}	V_{G2S} = +6 V, V_{G1S} = V_{DS} = 0, T_A = 25° C	–	–	50	nA
		T_A = 100° C	–	–	5	μA
* Gate No. 2-Terminal Reverse Current	I_{G2SSR}	V_{G2S} = -6 V, V_{G1S} = V_{DS} = 0, T_A = 25° C	–	–	50	nA
		T_A = 100° C	–	–	5	μA
* Zero-Bias Drain Current	I_{DS}	V_{DS} = +15 V, V_{G2S} = +4 V, V_{G1S} = 0	5	15	30	mA
Forward Transconductance (Gate No. 1-to-Drain)	g_{fs}	V_{DS} = +15 V, I_D = 10 mA, V_{G2S} = +4 V, f = 1 kHz	7000	12,000	18,000	μmho
* Small-Signal, Short-Circuit Input Capacitance†	C_{iss}	V_{DS} = +15 V, I_D = 10 mA, V_{G2S} = +4 V, f = 1 MHz	4.0	6.0	8.5	pF
* Small-Signal, Short-Circuit, Reverse Transfer Capacitance (Drain-to-Gate No. 1) ♦	C_{rss}		0.005	0.02	0.03	pF
* Small-Signal, Short-Circuit Output Capacitanc	C_{oss}		–	2.0	–	pF
Power Gain (see Fig. 1)	G_{PS}	V_{DS} = +15 V, I_D = 10 mA, V_{G2S} = +4 V, f = 200 MHz	16	18	22	dB
Maximum Available Power Gain	MAG		–	20	–	dB
Maximum Usable Power Gain (unneutralized)	MUG		–	20▲	–	dB
Noise Figure (see Fig. 1)	NF		–	3.5	4.5	dB
* Magnitude of Forward Transadmittance	$\|Y_{fs}\|$		–	12,000	–	μmho
* Phase Angle of Forward Transadmittance	θ		–	-35	–	Degrees
Magnitude of Reverse Transadmittance	$\|Y_{rs}\|$		–	25	–	μmho
Angle of Reverse Transadmittance	θ_{rs}		–	-25	–	Degrees
* Input Resistance	r_{iss}		–	1.0	–	kΩ
* Output Resistance	r_{oss}		–	2.8	–	kΩ
Gate-to-Source Forward Breakdown Voltage: Gate No. 1	$V_{(BR)G1SSF}$	I_{G1SSF} = I_{G2SSF} = 100 μA	6.5	10	–	V
Gate No. 2	$V_{(BR)G2SSF}$					
Gate-to-Source Reverse Breakdown Voltage: Gate No. 1	$V_{(BR)G1SSR}$	I_{G1SSR} = I_{G2SSR} = -100 μA	-6.5	-10	–	V
Gate No. 2	$V_{(BR)G2SSR}$					

▲ Limited only by practical design considerations.

† Capacitance between Gate No. 1 and all other terminals

♦ Three-terminal measurement with Gate No. 2 and Source returned to ground terminal.

* In accordance with JEDEC Registration Data Format JS-9 RDF-19A

OPERATING CONSIDERATIONS

The flexible leads of the 3N187 are usually soldered to the circuit elements. As in the case of any high-frequency semiconductor device, the tips of soldering irons MUST be grounded.

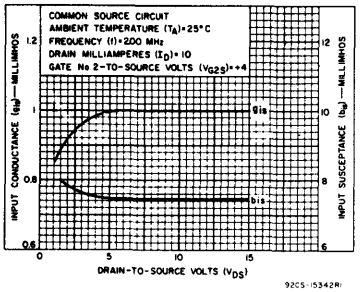


Fig. 8- y_{is} vs. V_{DS}

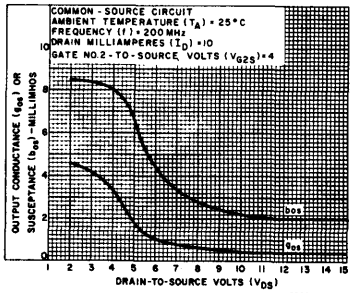


Fig. 9- y_{os} vs. V_{DS}

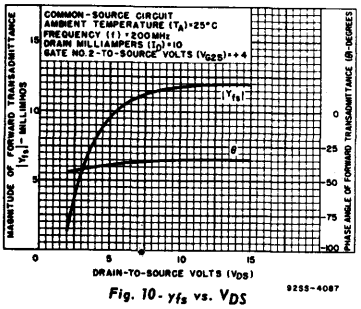


Fig. 10- y_{fs} vs. V_{DS}

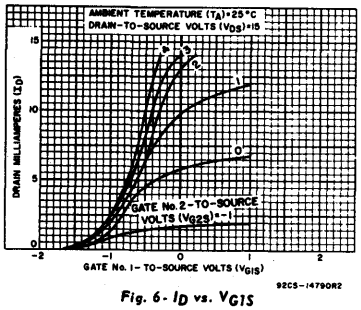


Fig. 6- I_D vs. V_{G1S}

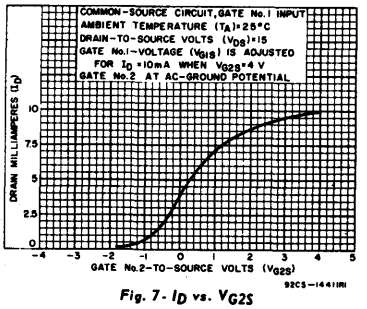


Fig. 7- I_D vs. V_{G2S}

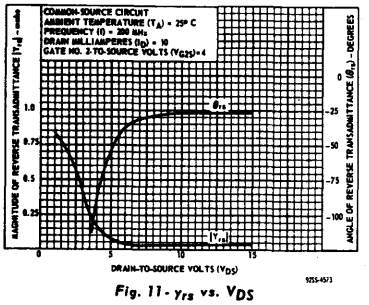


Fig. 11- y_{rs} vs. V_{DS}

3N187

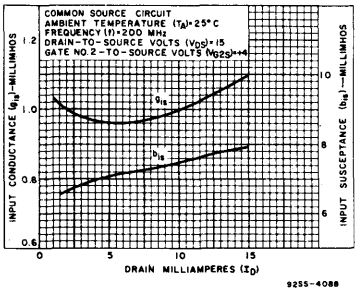


Fig. 12 - y_{is} vs. I_D

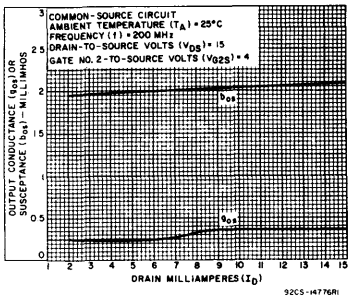


Fig. 13 - y_{os} vs. I_D

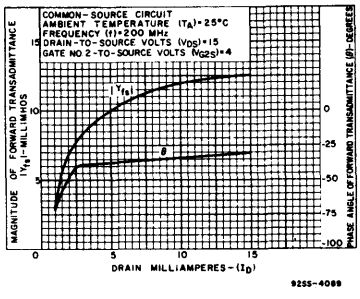


Fig. 14 - y_{fs} vs. I_D

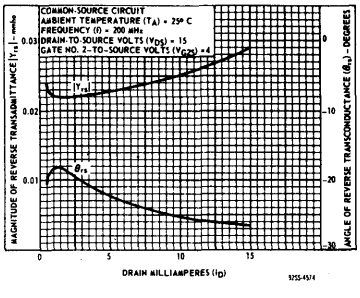


Fig. 15 - y_{rs} vs. I_D

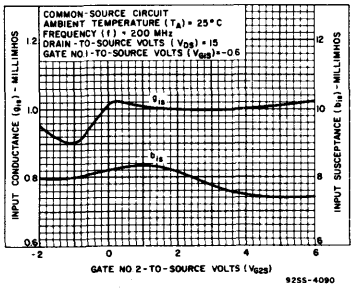


Fig. 16 - y_{is} vs. V_{G2S}

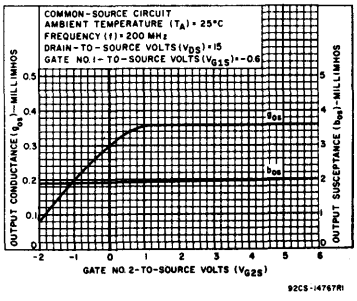


Fig. 17 - y_{os} vs. V_{G2S}

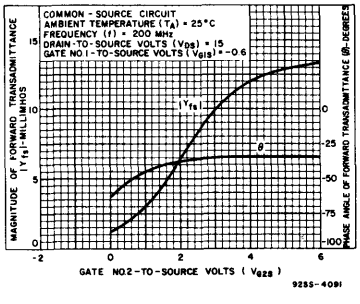


Fig. 18 - y_{fs} vs. V_{G2S}

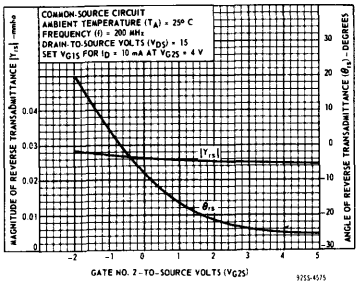


Fig. 19 - y_{rs} vs. V_{G2S}

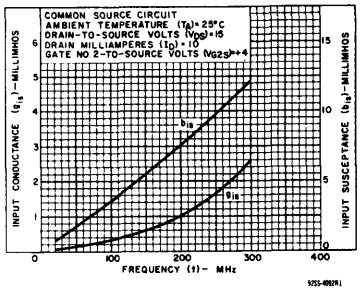


Fig. 20 - y_{is} vs. frequency

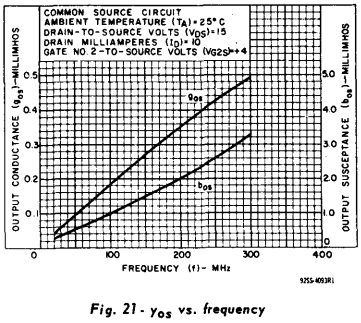


Fig. 21 - y_{os} vs. frequency

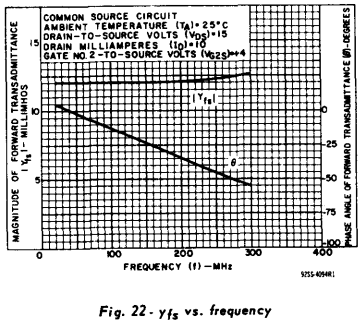


Fig. 22 - y_{fs} vs. frequency

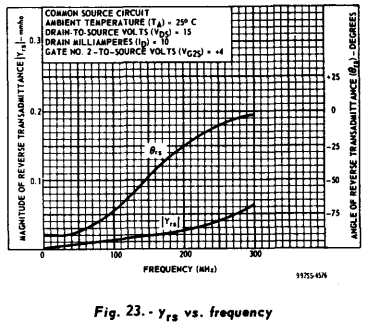


Fig. 23 - y_{rs} vs. frequency

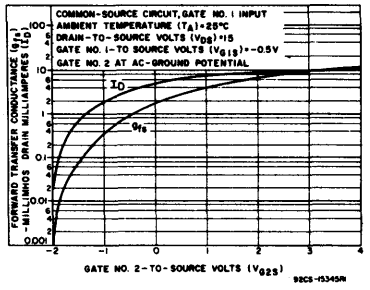


Fig. 24 - g_{fs} and I_D vs. V_{G2S}

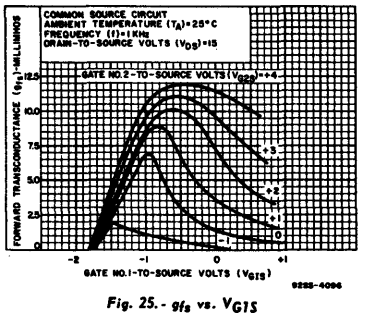


Fig. 25. - g_{fs} vs. V_{G1S}

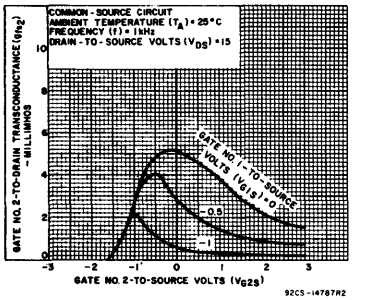


Fig. 26 - g_{fs_2} vs. V_{G2S}

3N200

Silicon Dual Insulated-Gate Field-Effect Transistor

N-Channel Depletion Types

With Integrated Gate-Protection Circuits

For Military and Industrial Applications up to 500 MHz

RCA-3N200 is an n-channel silicon, depletion type, dual insulated-gate field-effect transistor.

Special back-to-back diodes are diffused directly into the MOS pellet and are electrically connected between each insulated gate and the FET's source. The diodes effectively bypass any voltage transients which exceed approximately ±10 volts. This protects the gates against damage in all normal handling and usage.

A feature of the back-to-back diode configuration is that it allows the 3N200 to retain the wide input signal dynamic range inherent in the MOSFET. In addition, the low junction capacitance of these diodes adds little to the total capacitance shunting the signal gate.

The excellent overall performance characteristics of the RCA-3N200 make it useful for a wide variety of rf-amplifier applications at frequencies up to 500 MHz. The two serially-connected channels with independent control gates make possible a greater dynamic range and lower cross-modulation than is normally achieved using devices having only a single control element.

The two-gate arrangement of the 3N200 also makes possible a desirable reduction in feedback capacitance by operating in the common-source configuration and ac-grounding Gate No. 2. The reduced capacitance allows operation at maximum gain *without neutralization;* and, of special importance in rf-amplifiers, it reduces local oscillator feedthrough to the antenna.

The 3N200 is hermetically sealed in the metal JEDEC TO-72 package.

Maximum Ratings, *Absolute-Maximum Values, at T_A = 25°C*

DRAIN-TO-SOURCE VOLTAGE, V_{DS}	-0.2 to +20	V
GATE No.1-TO-SOURCE VOLTAGE, V_{G1S}:		
Continuous (dc)	-6 to +3	V
Peak ac	-6 to +6	V
GATE No.2-TO-SOURCE VOLTAGE, V_{G2S}:		
Continuous (dc)	-6 to 30% of V_{DS}	V
Peak ac	-6 to +6	V
* DRAIN-TO-GATE VOLTAGE, V_{DG1} OR V_{DG2}	+20	V
* DRAIN CURRENT, I_D	50	mA
* TRANSISTOR DISSIPATION, P_T:		
At ambient temperatures up to 25°C	330	mW
At ambient temperatures above 25°C	derate linearly at 2.2 mW/°C	
* AMBIENT TEMPERATURE RANGE:		
Storage and Operating	-65 to +175	°C
* LEAD TEMPERATURE (During soldering):		
At distances ≥ 1/32 inch from seating surface for 10 seconds max.	265	°C

*In accordance with JEDEC registration data format (JS-9 RDF-19A)

Performance Features

- Superior cross-modulation performance and greater dynamic range than bipolar or single-gate FET s
- Wide dynamic range permits large-signal handling before overload
- Dual-gate permits simplified agc circuitry
- Virtually no agc power required
- Greatly reduces spurious responses in FM receivers

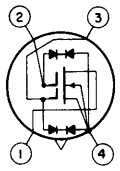
Device Features

- Back-to-back diodes protect each gate against handling and in-circuit transients
- High forward transconductance — g_{fs} = 15,000 μmho (typ.)
- High unneutralized RF power gain — G_{ps} = 12.5 dB (typ.) at 400 MHz; 19 dB (typ.) at 200 MHz
- Low VHF noise figure — 3.9 dB (typ.) at 400 MHz; 3.0 dB (typ.) at 200 MHz

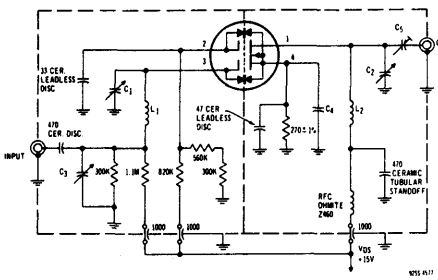
Applications

- RF amplifier, mixer, and IF amplifier in military and industrial communications equipment
- Aircraft and marine vehicular receivers
- CATV and MATV equipment
- Telemetry and multiplex equipment

TERMINAL DIAGRAM



LEAD 1-DRAIN
LEAD 2-GATE No. 2
LEAD 3-GATE No. 1
LEAD 4-SOURCE, SUBSTRATE AND CASE



All resistances in ohms
All capacitances in pF

C_1, C_2: 1.3-5.4 pF variable air capacitor: Hammerland Mac 5 type or equivalent

C_3: 1.9-13.8 pF variable air capacitor: Hammerland Mac 15 type or equivalent

C_4: Approx. 300 pF-capacitance formed between socket cover & chassis

C_5: 0.8-4.5 pF piston type variable air capacitor: Erie 560-013 or equivalent

L_1,L_2: Inductance to tune circuit

Fig. 1 - 400 MHz power gain and noise figure test circuit

y and s Parameters vs. Frequency

TEST CONDITIONS: Drain-to-Source Volts (V_{DS}) = 15, Drain Milliamperes (I_D) = 10, Gate No. 2-to-Source Volts (V_{G2S}) = 4

CHARACTERISTICS	SYMBOL	FREQUENCY (MHz)					UNITS
		100	200	300	400	500	
Maximum Available Power Gain	MAG	32	24	17.5	13	10	dB
Maximum Usable Power Gain (Unneutralized)*	MUG	32	24	17.5	13	10	dB
Y Parameters							
Input Conductance	g_{is}	0.25	0.8	2.0	3.6	6.2	mmho
Input Susceptance	b_{is}	3.4	5.8	8.5	11.2	15.5	mmho
Magnitude of Forward Transadmittance	$\|y_{fs}\|$	15.3	15.3	15.4	15.5	16.3	mmho
Angle of Forward Transadmittance	$\angle y_{fs}$	-15	-25	-35	-47	-60	degrees
Output Conductance	g_{os}	0.15	0.3	0.5	0.8	1.1	mmho
Output Susceptance	b_{os}	1.5	2.7	3.6	4.25	5.0	mmho
Magnitude of Reverse Transadmittance	$\|y_{rs}\|$	0.012	0.025	0.06	0.14	0.26	mmho
Angle of Reverse Transadmittance	$\angle y_{rs}$	-60	-25	0	14	20	degrees
S Parameters							
Magnitude of Input Reflection Coeff.	$\|s_{is}\|$	0.97	0.90	0.84	0.78	0.70	
Angle of Input Reflection Coeff.	$\angle s_{is}$	-20	-32	-55	-68	-82	degrees
Magnitude of Forward Transmission Coeff.	$\|s_{fs}\|$	1.50	1.40	1.25	1.1	0.9	
Angle of Forward Transmission Coeff.	$\angle s_{fs}$	153	133	112	90	70	degrees
Mangitude of Output Reflection Coeff.	$\|s_{os}\|$	0.985	0.95	0.93	0.92	0.91	
Angle of Output Reflection Coeff.	$\angle s_{os}$	-7.5	-16	-22	-28	-34	degrees
Magnitude of Reverse Transmission Coeff.	$\|s_{rs}\|$	0.001	0.0025	0.005	0.010	0.0165	
Angle of Reverse Transmission Coeff.	$\angle s_{rs}$	100	125	141	150	142	degrees

*Limited only by practical design considerations

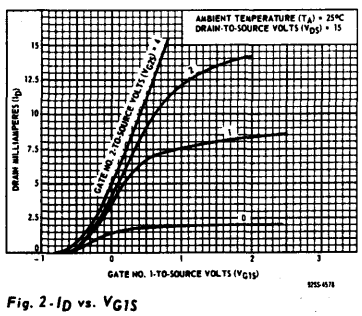


Fig. 2 - I_D vs. V_{G1S}

ELECTRICAL CHARACTERISTICS at T_A = 25°C unless otherwise specified		SYMBOLS	TEST CONDITIONS		LIMITS Min.	Typ.	Max.	UNITS
* Gate No. 1-to-Source Cutoff Voltage		$V_{G1S(off)}$	V_{DS} = +15 V, I_D = 50 μA; V_{G2S} = +4 V		-0.1	-1	-3	V
* Gate No. 2-to-Source Cutoff Voltage		$V_{G2S(off)}$	V_{DS} = +15 V, I_D = 50 μA; V_{G1S} = 0		-0.1	-1	-3	V
* Gate No. 1-Terminal Forward Current		I_{G1SSF}	V_{G1S} = +1 V; V_{G2S} = V_{DS} = 0	T_A = 25°C	-	-	50	nA
				T_A = 100°C	-	-	5	μA
* Gate No. 1-Terminal Reverse Current		I_{G1SSR}	V_{G1S} = -6 V; V_{G2S} = V_{DS} = 0	T_A = 25°C	-	-	50	nA
				T_A = 100°C	-	-	5	μA
* Gate No. 2-Terminal Forward Current		I_{G2SSF}	V_{G2S} = +6 V; V_{G1S} = V_{DS} = 0	T_A = 25°C	-	-	50	nA
				T_A = 100°C	-	-	5	μA
* Gate No. 2-Terminal Reverse Current		I_{G2SSR}	V_{G2S} = -6 V; V_{G1S} = V_{DS} = 0	T_A = 25°C	-	-	50	nA
				T_A = 100°C	-	-	5	μA
* Zero-Bias Drain Current		I_{DS}	V_{DS} = +15 V, V_{G1S} = 0; V_{G2S} = +4 V		0.5	5.0	12	mA
* Forward Transconductance (Gate No. 1-to-Drain)		g_{fs}	V_{DS} = +15 V; I_D = 10 mA; V_{G2S} = +4 V	f = 1 kHz	10,000	15,000	20,000	μmho
Small-Signal, Short-Circuit Input Capacitance†		C_{iss}		f = 1 MHz	4.0	6.0	8.5	pF
* Small-Signal, Short-Circuit, Reverse Transfer Capacitance (Drain-to-Gate-No. 1)♦		C_{rss}			0.005	0.02	0.03	pF
Small-Signal, Short-Circuit Output Capacitance		C_{oss}			-	2.0	-	pF
* Power Gain (see Fig. 1)		G_{PS}		f = 400 MHz	10	12.5	-	dB
Noise Figure (see Fig. 1)		NF			-	3.9	6.0	dB
* Bandwidth		BW			28	-	38	MHz
* Gate-to-Source Forward Breakdown Voltage	Gate No. 1	$V_{(BR)G1SSF}$	I_{G1SSF} = I_{G2SSF} = 100 μA	V_{G2S} = V_{DS} = 0	6.5	-	13	V
	Gate No. 2	$V_{(BR)G2SSF}$		V_{G1S} = V_{DS} = 0				
* Gate-to-Source Reverse Breakdown Voltage	Gate No. 1	$V_{(BR)G1SSR}$	I_{G1SSR} = I_{G2SSR} = 100 μA	V_{G2S} = V_{DS} = 0	-6.5	-	-13	V
	Gate No. 2	$V_{(BR)G2SSR}$		V_{G1S} = V_{DS} = 0				

*In accordance with JEDEC registration data format (JS-9 RDF-19A)

†Capacitance between Gate No. 1 and all other terminals.

♦Three-terminal measurement with Gate No. 2 and Source returned to guard terminal.

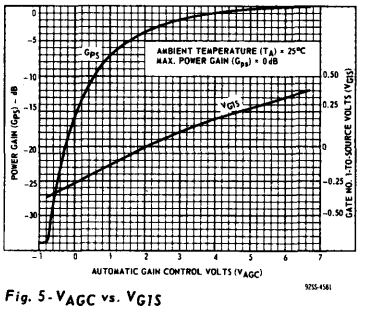


Fig. 5 - V_{AGC} vs. V_{G1S}

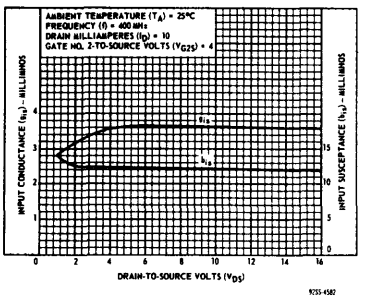


Fig. 6 - y_{is} vs. V_{DS}

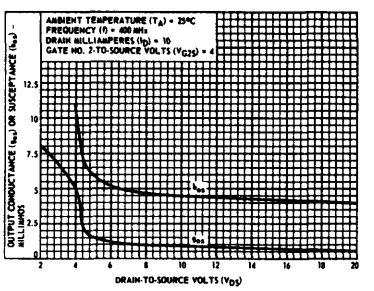


Fig. 7 - y_{os} vs. V_{DS}

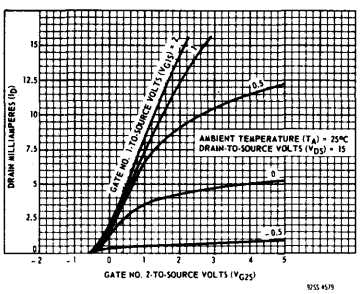


Fig. 3 - I_D vs. V_{G2S}

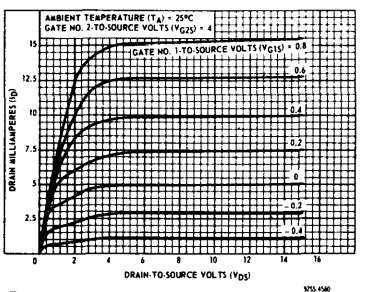


Fig. 4 - I_D vs. V_{DS}

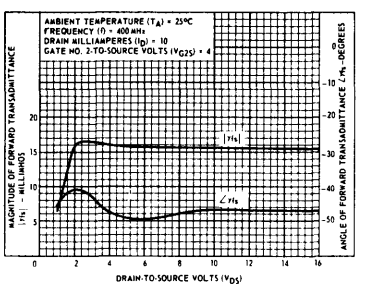


Fig. 8 - y_{fs} vs. V_{DS}

3N200

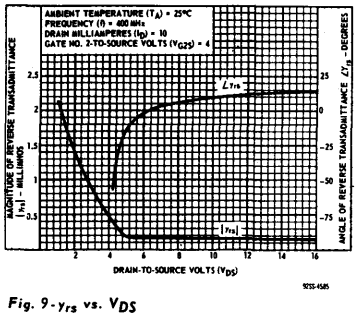


Fig. 9 - y_{rs} vs. V_{DS}

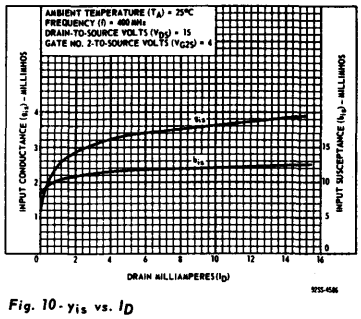


Fig. 10 - y_{is} vs. I_D

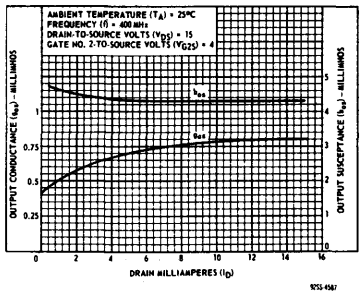


Fig. 11 - y_{os} vs. I_D

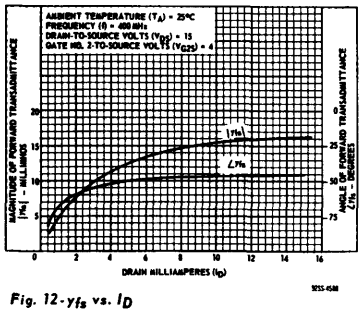


Fig. 12 - y_{fs} vs. I_D

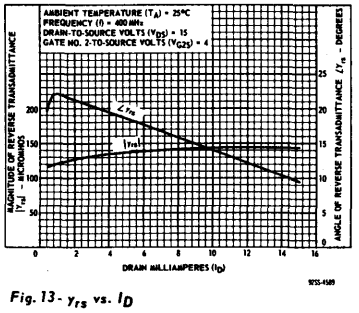


Fig. 13 - y_{rs} vs. I_D

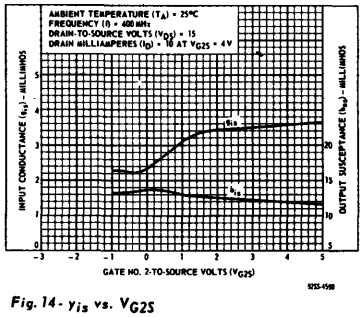


Fig. 14 - y_{is} vs. V_{G2S}

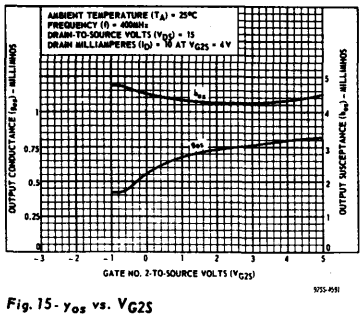


Fig. 15 - y_{os} vs. V_{G2S}

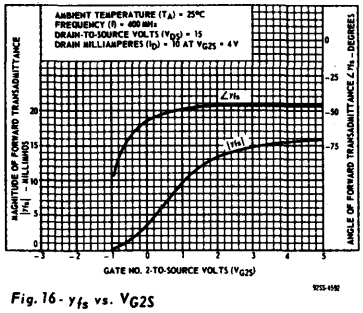


Fig. 16 - y_{fs} vs. V_{G2S}

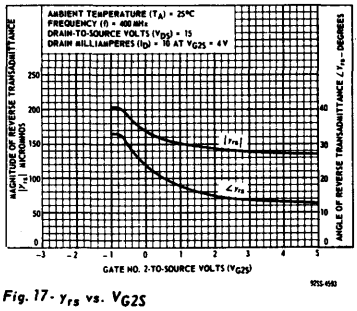


Fig. 17 - y_{rs} vs. V_{G2S}

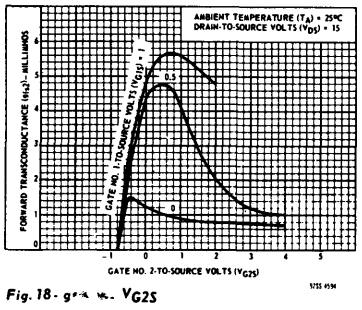


Fig. 18 - g_{fs} vs. V_{G2S}

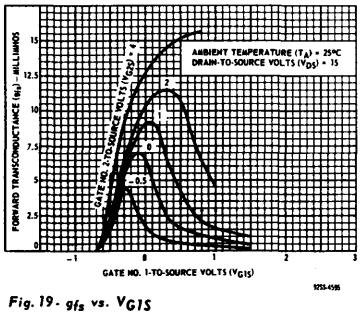


Fig. 19 - g_{fs} vs. V_{G1S}

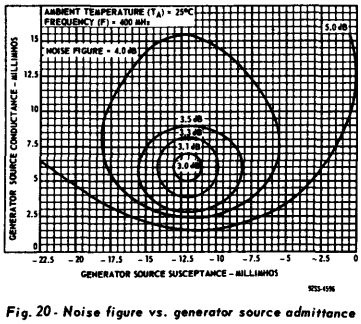


Fig. 20 - Noise figure vs. generator source admittance

Silicon Dual-Insulated-Gate Field-Effect Transistors

With Integrated Gate-Protection Circuits
For VHF TV Applications

3N204 – RF Amplifier
3N205 – Mixer
3N206 – TV IF Amplifier

The RCA-3N204, 3N205, and 3N206 are n-channel silicon, depletion type, dual-insulated gate, field-effect transistors intended for vhf TV applications. Integrated back-to- back diodes protect the gates from excessive input voltages.

The 3N204 is intended for use in vhf rf amplifiers and delivers linear, low-noise amplification. Its extremely low feedback capacitance allows high-gain stable operation without neutralization. The 3N205 is specified for low noise vhf mixer applications. The 3N206 is intended for use in tuned high-frequency amplifiers such as TV if strips.

Features:

- Low C_{rss} – 0.03 pF max.
- High $|Y_{fs}|$ – 14 mmho typ. for 3N204 and 3N205
- Integrated gate-protection diodes

MAXIMUM RATINGS,
Absolute Maximum Values at $T_A = 25^oC$

* DRAIN-TO-GATE No.1 VOLTAGE	30	V
* DRAIN-TO-GATE No.2 VOLTAGE	30	V
* DRAIN-TO-SOURCE VOLTAGE	25	V
* GATE No.1-TERMINAL FORWARD CURRENT▲	10	mA
* GATE No.2-TERMINAL FORWARD CURRENT▲	10	mA
* GATE No.1-TERMINAL REVERSE CURRENT	–10	mA
* GATE No.2-TERMINAL REVERSE CURRENT	–10	mA
* CONTINUOUS DRAIN CURRENT	50	mA
* DEVICE DISSIPATION:		
Up to $T_A = 25^oC$	360	mW
Above $T_A = 25^oC$ derate linearly	2.4	mW/°C
Up to $T_C = 25^oC$	1.2	W
Above $T_C = 25^oC$ derate linearly	8	mW/°C
* AMBIENT TEMPERATURE RANGE:		
Operating	–65 to +175	°C
Storage	–65 to +200	°C
* LEAD TEMPERATURE (DURING SOLDERING):		
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max.	+300	°C

▲ Forward gate-terminal current is the current into a gate terminal with a forward-gate-to-source voltage applied. This voltage is of such polarity that an increase in its magnitude causes the channel resistance to decrease.
* In accordance with JEDEC registration data format (JS-9 RDF-198)

OPERATING CHARACTERISTICS at $T_A = 25^oC$

CHARACTERISTIC	TEST CONDITIONS	LIMITS Min.	Typ.	Max.	UNITS
3N204					
* Common-Source Spot Noise Figure, NF	V_{DD}=18 V, V_{GG}=7 V, f = 200 MHz, See Fig.13	–	–	3.5	dB
* Small-Signal Common-Source Insertion Power Gain, G_{ps}		20	–	28	dB
* Bandwidth, BW		7	–	12	MHz
* Gain-Control Gate-Supply Voltage, $V_{GG(GC)}$	V_{DD}=18 V, ΔG_{ps}=–30dB,[1] f=200 MHz, See Fig. 13	0	–	–2	V
* Common-Source Spot Noise Figure, NF	V_D=15 V, V_{G2S} = 4 V, f = 450 MHz, I_D = 10 mA, See Figs. 15 and 16	–	–	5	dB
* Small-Signal Common Source Insertion Power Gain, G_{ps}		14	–	–	dB
3N205					
* Small-Signal Conversion Power Gain, G_{ps} (conv)	V_{DD}=18 V, f_{LO}=245 MHz,[3] f_{RF}=200 MHz, See Fig.17	17	–	28	dB
* Bandwidth, BW		4	–	7	MHz
3N206					
* Common-Source Spot Noise Figure, NF	V_{DD}=24 V, V_{GG}=6 V, f=45 MHz, See Fig. 14	–	–	4	dB
* Small-Signal Common-Source Insertion Power Gain, G_{ps}		25	–	35	dB
* Bandwidth, BW		3	–	6	MHz
* Gain-Control Gate-Supply Voltage, $V_{GG(GC)}$	V_{DD}=24 V, ΔG_{ps}=–30dB,[2] f=45 MHz, See Fig. 14	–1.6	–	0.6	V

*In accordance with JEDEC registration data format (JS-9 RDF-198).
1. ΔG_{ps} is defined as the change in G_{ps} from the value at V_{GG} = 7V.
2. ΔG_{ps} is defined as the change in G_{ps} from the value at V_{GG} = 6V.
3. Amplitude at input from local oscillator is 3 V RMS.

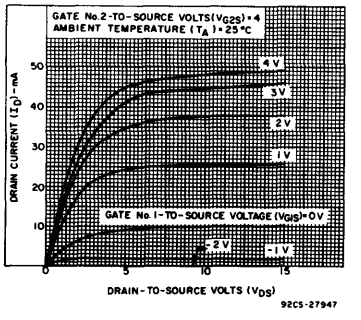


Fig.1 – Drain current vs. drain-to-source volts (pulse-tested with pulse duration = 300 μs, duty cycle ⩽2%).

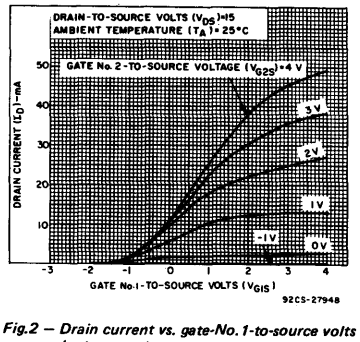


Fig.2 – Drain current vs. gate-No.1-to-source volts (pulse-tested with pulse duration = 300 μs, duty cycle ⩽ 2%).

3N204, 3N205, 3N206

ELECTRICAL CHARACTERISTICS, At T_A = 25°C (unless otherwise specified)

	CHARACTERISTIC	TEST CONDITIONS		LIMITS MIN.	LIMITS MAX.	UNITS
*	Drain-to-Source Breakdown Voltage, $V_{(BR)DS}$	I_D = 10μA, $V_{G1S}=V_{G2S}$ = −5V		25	–	V
*	Gate No.1-to-Source Forward Breakdown Voltage, $V_{(BR)G1SSF}$[1]	I_{G1}=10mA, $V_{G2S}=V_{DS}$=0		6	30	V
*	Gate No.1-to-Source Reverse Breakdown Voltage, $V_{(BR)G1SSR}$[1]	I_{G1}= −10mA, $V_{G2S}=V_{DS}$=0		−6	−30	V
*	Gate No.2-to-Source Forward Breakdown Voltage, $V_{(BR)G2SSF}$[1]	I_{G2}=10mA, $V_{G1S}=V_{DS}$=0		6	30	V
*	Gate No.2-to-Source Reverse Breakdown Voltage, $V_{(BR)G2SSR}$[1]	I_{G2}= −10mA, $V_{G1S}=V_{DS}$=0		−6	−30	V
*	Gate No.1-Terminal Forward Current, I_{G1SSF}	V_{G1S}=5V, $V_{G2S}=V_{DS}$=0		–	10	nA
*	Gate No.1-Terminal Reverse Current, I_{G1SSR}	V_{G1S}= −5V, $V_{G2S}=V_{DS}$=0	T_A=25°C	–	−10	nA
			T_A=150°C	–	−10	μA
*	Gate No.2-Terminal Forward Current, I_{G2SSF}	V_{G2S}=5V, $V_{G1S}=V_{DS}$=0		–	10	nA
*	Gate No.2-Terminal Reverse Current, I_{G2SSR}	V_{G2S}= −5V, $V_{G1S}=V_{DS}$=0	T_A=25°C	–	−10	nA
			T_A=150°C	–	−10	μA
*	Zero-Gate No.1-Voltage Drain Current, I_{DS}[2]	V_{DS}=15V, V_{GIS}=0, V_{G2S}=4V	3N204	6	30	mA
			3N205	6	30	
			3N206	3	15	
*	Gate No.1-to-Source Cutoff Voltage, $V_{G1S(off)}$	V_{DS}=15V, V_{G2S}=4V, I_D=20μA		−0.5	−4	V
*	Gate No.2-to-Source Cutoff Voltage, $V_{G2S(off)}$	V_{DS}=15V, V_{G1S}=0, I_D=20μA		−0.2	−4	V
*	Small-Signal Common-Source Forward Transfer Admittance, $\lvert y_{fs}\rvert$[3]	V_{DS}=15V, V_{G1S}=0, V_{G2S}=4V, f=1 kHz	3N204	10	22	mmho
			3N205	10	22	
			3N206	7	17	
*	Small-Signal Common-Source Reverse Transfer Capacitance, C_{rss}	V_{DS}=15V, V_{G2S}=4V, I_D = 10 mA, f = 1 MHz		0.005	0.03	pF

*In accordance with JEDEC registration data format (JS-9 RDF-19B).

1. All gate breakdown voltages are measured while the device is conducting rated gate current. This ensures that the gate-voltage-limiting network is functioning properly.
2. This characteristic must be measured using pulse techniques (t_W = 300μs, duty cycle ≤ 2%).
3. This characteristic must be measured with bias voltages applied for less than 5 seconds to avoid overheating. The signal is applied to gate No.1 with gate No.2 at ac ground.

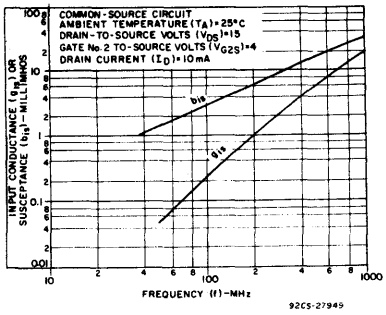


Fig.3 – Y_{is} vs. f

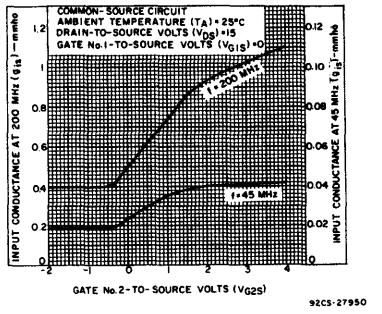


Fig.4 – Y_{is} vs. V_{G2S}

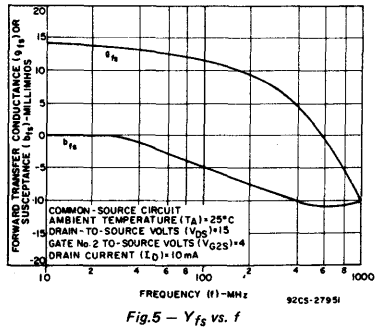


Fig.5 – Y_{fs} vs. f

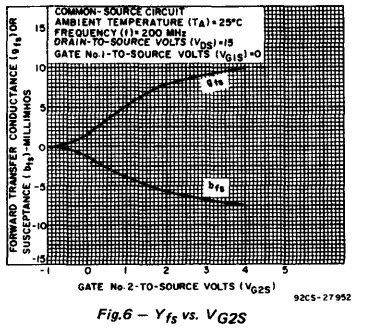


Fig.6 – Y_{fs} vs. V_{G2S}

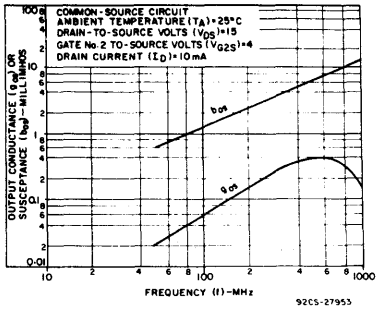


Fig.7 – Y_{os} vs. f

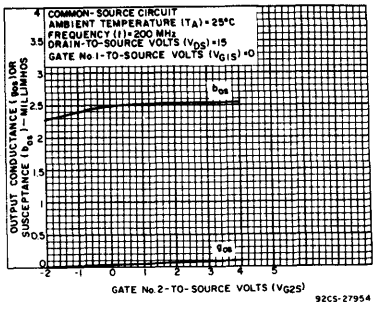


Fig.8 – Y_{os} vs. V_{G2S}

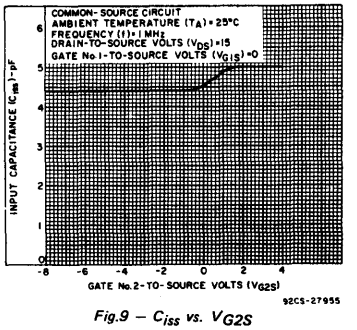


Fig.9 – C_{iss} vs. V_{G2S}

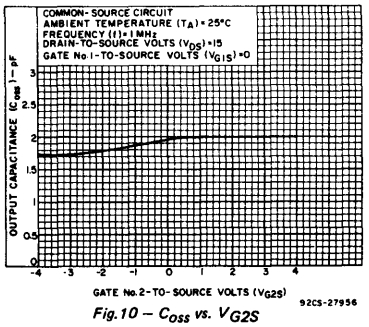
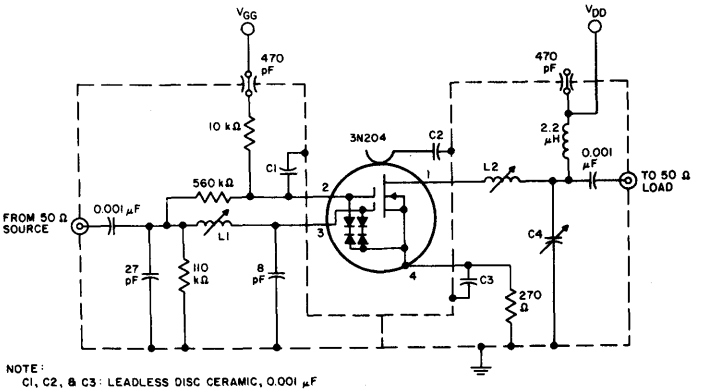


Fig.10 – C_{oss} vs. V_{G2S}



NOTE:
C1, C2, & C3: LEADLESS DISC CERAMIC, 0.001 μF
C4: ARCO 462, 5-80 pF, OR EQUIVALENT
L1: 3 TURNS No. 18 WIRE, 3/16 INCH-DIA. ALUMINUM SLUG
L2: 9 TURNS No. 20 WIRE, 3/16 INCH-DIA. ALUMINUM SLUG

92CS-27960

Fig.12 – 200-MHz power gain, gain-control voltage, and noise-figure test circuit for 3N204.*

* In accordance with JEDEC registration data format (JS-9 RDF-19B).

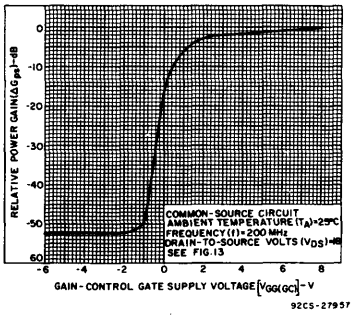


Fig.11 – ΔG_{ps} vs. $V_{GG(GC)}$

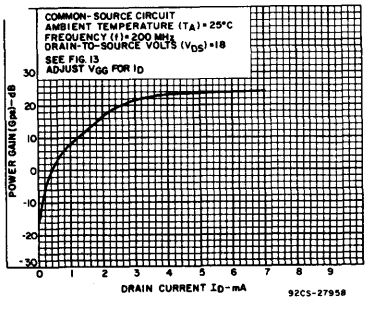
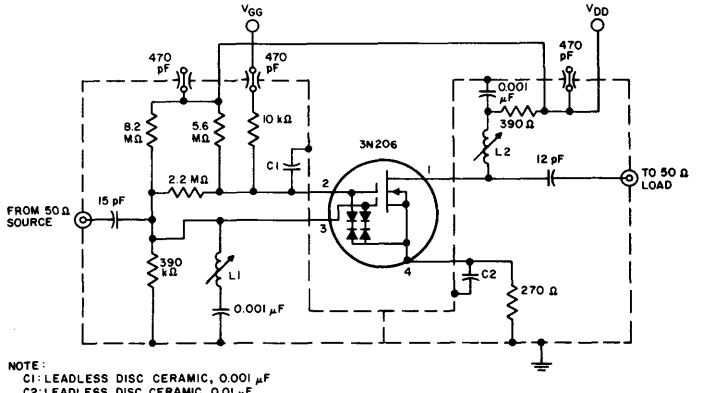


Fig.13 – G_{ps} vs. I_D

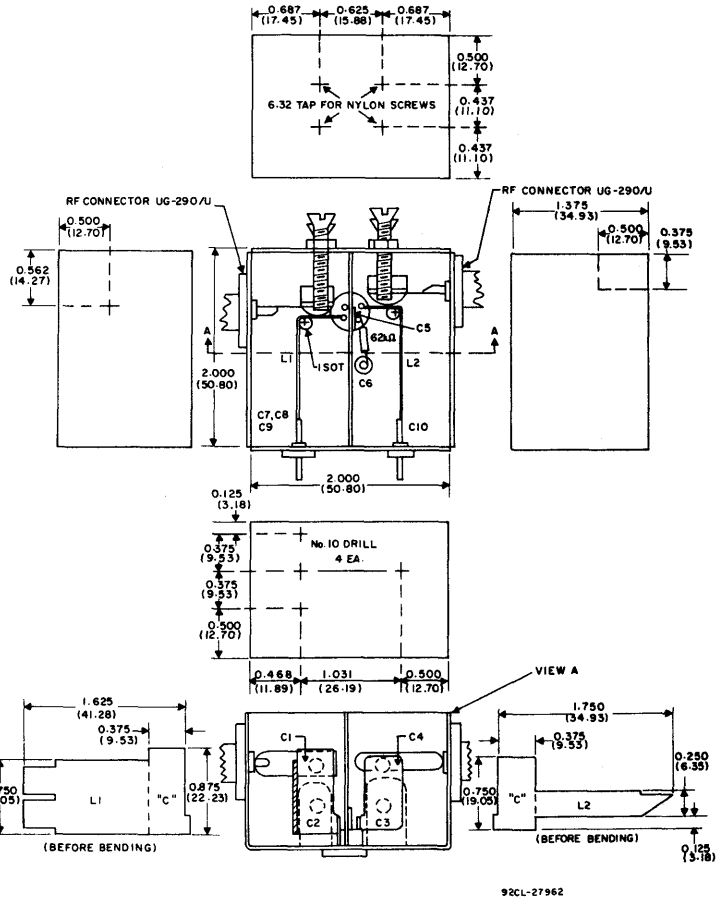


NOTE:
C1: LEADLESS DISC CERAMIC, 0.001 μF
C2: LEADLESS DISC CERAMIC, 0.01 μF
L1: 8 TURNS No. 28 WIRE, 5/32 INCH-DIA. FORM, TYPE "J" SLUG
L2: 9 TURNS No. 28 WIRE, 5/32 INCH-DIA. FORM, TYPE "J" SLUG

92CM-27959

Fig.14 – –45-MHz power-gain and noise-figure test circuit for 3N206.*

* In accordance with JEDEC registration data format (JS-9 RDF-19B).



NOTES:

A. Dimensions in parentheses are in millimeters and are derived from the basic inch dimensions, as indicated.

B. The removable top of test fixture is not shown.

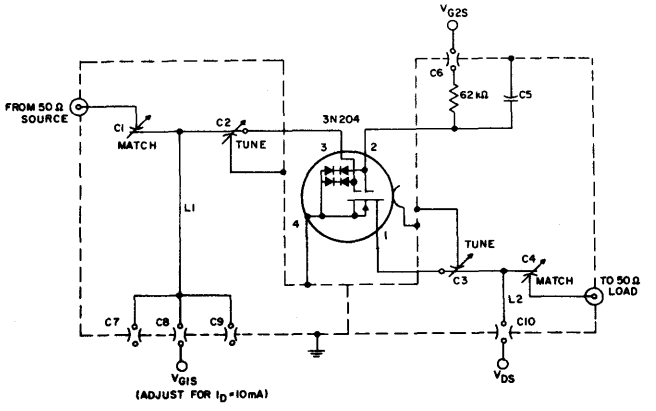
C. For clarity, the 62 kΩ resistor, the source and gate-2 socket pins, and insulating stand-off terminals (ISOT) soldered into the fold of L1 and L2 respectively for mechanical support, are not shown in view A.

D. C1 and C2 (C3 and C4) consist of shim brass and the "C" portion of L1 (L2) separated by air and the mylar tape covering the "C" portion of L1 (L2).

E. The four views surrounding the center view are as they would appear before the metal is bent up to form the sides.

Fig. 15 — —450 MHz power-gain and noise-figure test fixture.*

* In accordance with JEDEC registration data format (JS-9 RDF-19B).

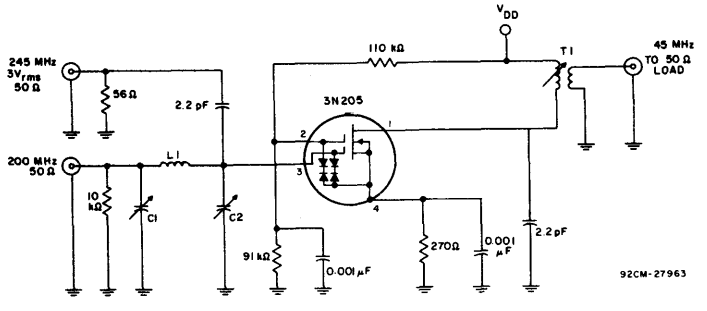


NOTE:
FOR TEST FIXTURE, SEE PICTORAL DRAWING IN FIGURE 16
C1 THRU C4: SEE FIGURE 16, NOTE D
C5: 0.001 μF LEADLESS DISC CAPACITOR
C6 THRU C10: ALLEN-BRADLEY FSAU 0.001 μF FEED-THROUGH CAPACITORS, OR EQUIVALENT
L1 & L2: SEE FIGURE 16

92CM-27961

Fig. 16 – –450-MHz power-gain and noise-figure test circuit for 3N204.*

* In accordance with JEDEC registration data format (JS-9 RDF-19B).



92CM-27963

NOTE:
C1: ARCO 462, 5–80 pF, or EQUIVALENT
C2: ARCO 460, 1.5–15 pF, OR EQUIVALENT
L1: 4 TURNS No.14 WIRE, 1/4 INCH INSIDE DIA.

T1: PRI: 16 TURNS No.30 WIRE CLOSE WOUND ON 1/4 INCH DIA. FORM, TYPE "J" SLUG
SEC: 5 TURNS No.30 WIRE CENTERED OVER PRIMARY

Fig. 17 – –200 MHz-to-45-MHz circuit for conversion power gain for 3N205.*

* In accordance with JEDEC registration data format (JS-9 RDF-19B).

3N211, 3N212, 3N213

Silicon Dual-Insulated-Gate Field-Effect Transistors

N-Channel Depletion Types

With Integrated Gate-Protection Circuits
For VHF TV Applications

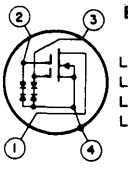
3N211 – RF Amplifiers
3N212 – Mixers
3N213 – TV IF Strips

Features:
- Low C_{rss} – 0.05 pF max.
- High $|Y_{fs}|$ – 30 mmho typ. for 3N211 and 3N212
- Integrated gate-protection diodes

The RCA-3N211, 3N212, and 3N213 are n-channel silicon, depletion type, dual-insulated gate, field-effect transistors intended for VHF TV applications. Integrated back-to-back diodes protect the gates from excessive input voltages.

The 3N211 is intended for use in VHF RF amplifiers and delivers linear, low-noise amplification. Its extremely low feedback capacitance allows high-gain stable operation without neutralization. The 3N212 is specified for low-noise VHF mixer applications. The 3N213 is intended for use in tuned high-frequency amplifiers such as TV IF strips.

TERMINAL DIAGRAM
Bottom View



LEAD 1 – DRAIN
LEAD 2 – GATE No.2
LEAD 3 – GATE No.1
LEAD 4 – SOURCE, SUBSTRATE AND CASE

MAXIMUM RATINGS,
Absolute Maximum Values at $T_A = 25°C$

	3N211, 3N212	3N213	
DRAIN-TO-GATE No.1 VOLTAGE	35	40	V
DRAIN-TO-GATE No.2 VOLTAGE	35	40	V
DRAIN-TO-SOURCE VOLTAGE	27	35	V
GATE No.1-TERMINAL FORWARD CURRENT*	10		mA
GATE No.2-TERMINAL FORWARD CURRENT*	10		mA
GATE No.1-TERMINAL REVERSE CURRENT	−10		mA
GATE No.2-TERMINAL REVERSE CURRENT	−10		mA
CONTINUOUS DRAIN CURRENT	50		mA
DEVICE DISSIPATION:			
Up to $T_A = 25°C$	360		mW
Above $T_A = 25°C$ derate linearly	2.4		mW/°C
Up to $T_C = 25°C$	1.2		mW
Above $T_C = 25°C$ derate linearly	8		mW/°C
AMBIENT TEMPERATURE RANGE:			
Operating	−65 to +175		°C
Storage	−65 to +200		°C
LEAD TEMPERATURE (DURING SOLDERING):			
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max.	+300		°C

* Forward gate-terminal current is the current into a gate terminal with a forward gate-to-source voltage applied. This voltage is of such polarity that an increase in its magnitude causes the channel resistance to decrease.

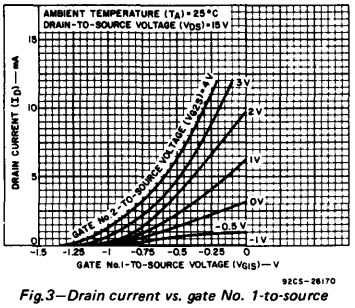


Fig.3—Drain current vs. gate No. 1-to-source voltage for all types.

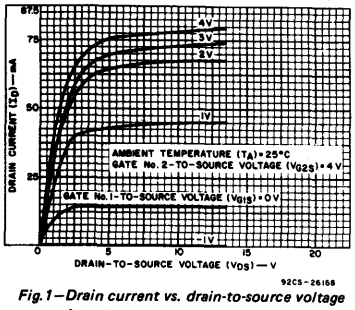


Fig.1—Drain current vs. drain-to-source voltage for all types.

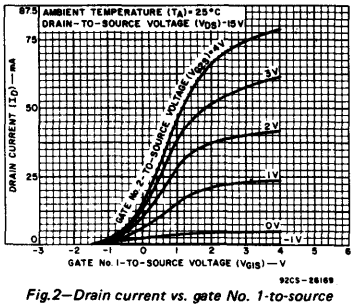


Fig.2—Drain current vs. gate No. 1-to-source voltage for all types.

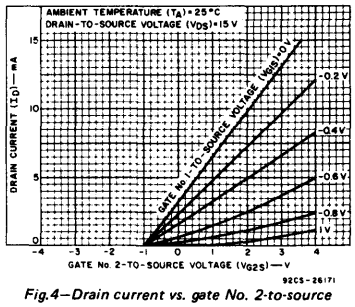


Fig.4—Drain current vs. gate No. 2-to-source voltage for all types.

(Figures 1 – 4 are pulse tested. Pulse duration = 300 μs, duty cycle <2%.)

ELECTRICAL CHARACTERISTICS, At T_A = 25°C (unless otherwise specified)

	CHARACTERISTIC	TEST CONDITIONS		LIMITS MIN.	LIMITS MAX.	UNITS
*	Drain-to-Source Breakdown Voltage, $V_{(BR)DS}$	I_D = 10μA, $V_{G1S}=V_{G2S}$= −4V	3N211	27	–	V
			3N212	27	–	
			3N213	35	–	
*	Gate No.1-to-Source Forward Breakdown Voltage, $V_{(BR)G1SSF}$[1]	I_{G1}=10mA, $V_{G2S}=V_{DS}$=0		6	–	V
*	Gate No.1-to-Source Reverse Breakdown Voltage, $V_{(BR)G1SSR}$[1]	I_{G1}= −10mA, $V_{G2S}=V_{DS}$=0		−6	–	V
*	Gate No.2-to-Source Forward Breakdown Voltage, $V_{(BR)G2SSF}$[1]	I_{G2}=10mA, $V_{G1S}=V_{DS}$=0		6	–	V
*	Gate No.2-to-Source Reverse Breakdown Voltage, $V_{(BR)G2SSR}$[1]	I_{G2}= −10mA, $V_{G1S}=V_{DS}$=0		−6	–	V
*	Gate No.1-Terminal Forward Current, I_{G1SSF}	V_{G1S}=5V, $V_{G2S}=V_{DS}$=0		–	10	nA
*	Gate No.1-Terminal Reverse Current, I_{G1SSR}	V_{G1S}= −5V, $V_{G2S}=V_{DS}$=0	T_A=25°C	–	−10	nA
			T_A=150°C	–	−10	μA
*	Gate No.2-Terminal Forward Current, I_{G2SSF}	V_{G2S}=5V, $V_{G1S}=V_{DS}$=0		–	10	nA
*	Gate No.2-Terminal Reverse Current, I_{G2SSR}	V_{G2S}= −5V, $V_{G1S}=V_{DS}$=0	T_A=25°C	–	−10	nA
			T_A=150°C	–	−10	μA
*	Zero-Gate No.1-Voltage Drain Current, I_{DS}[2]	V_{DS}=15V, V_{G1S}=0, V_{G2S}=4V		6	40	mA
*	Gate No.1-to-Source Cutoff Voltage, $V_{G1S(off)}$	V_{DS}=15V, V_{G2S}=4V, I_D=20μA	3N211	−0.5	−5.5	V
			3N212	−0.5	−4	
			3N213	−0.5	−5.5	
*	Gate No.2-to-Source Cutoff Voltage, $V_{G2S(off)}$	V_{DS}=15V, V_{G1S}=0, I_D=20μA	3N211	−0.2	−2.5	V
			3N212	−0.2	−4	
			3N213	−0.2	−4	
*	Small-Signal Common-Source Forward Transfer Admittance, $\lvert Y_{fs}\rvert$[3]	V_{DS}=15V, V_{G1S}=0, V_{G2S}=4V, f=1 kHz	3N211	17	40	mmho
			3N212	17	40	
			3N213	15	35	
*	Small-Signal Common-Source Reverse Transfer Capacitance, C_{rss}	V_{DS}=15V, V_{G2S}=4V, I_D=1mA, f=1MHz		0.005	0.05	pF

*In accordance with JEDEC registration data format (JS-9 RDF-19B).

1. All gate breakdown voltages are measured while the device is conducting rated gate current. This ensures that the gate-voltage-limiting network is functioning properly.
2. This characteristic must be measured using pulse techniques (t_W = 300μs, duty cycle ≤ 2%).
3. This characteristic must be measured with bias voltages applied for less than 5 seconds to avoid overheating. The signal is applied to gate No.1 with gate No.2 at ac ground.

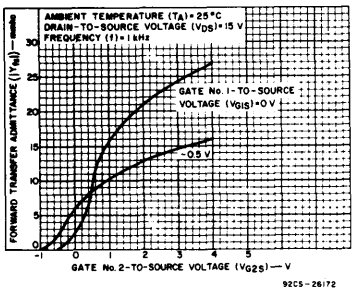


Fig. 5 – $|Y_{fs}|$ vs. V_{G2S} for 3N211 and 3N212.

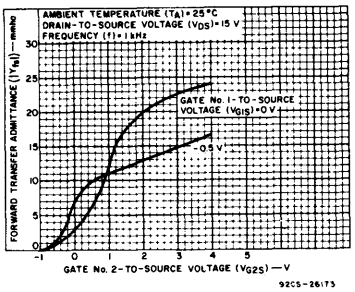


Fig. 6 – $|Y_{fs}|$ vs. V_{G2S} for 3N213.

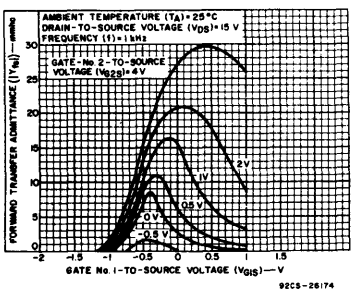


Fig. 7 – $|Y_{fs}|$ vs. V_{G1S} for 3N211, and 3N212.

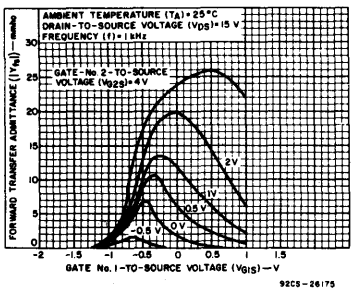
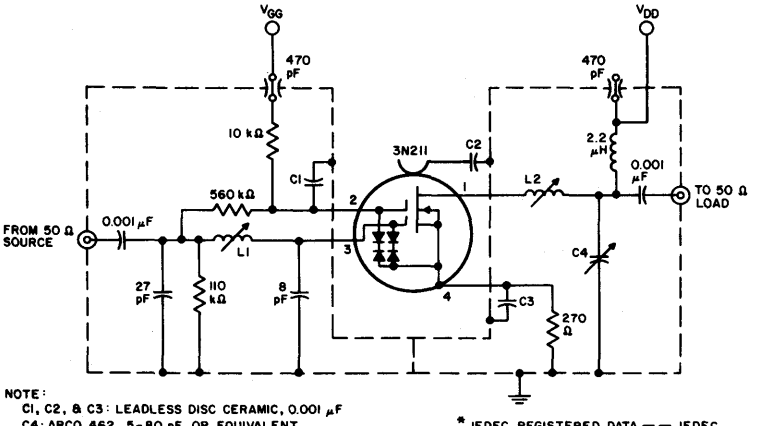


Fig. 8 – $|Y_{fs}|$ vs. V_{G1S} for 3N213.

OPERATING CHARACTERISTICS at T_A = 25°C

	CHARACTERISTIC	TEST CONDITIONS	LIMITS Min.	LIMITS Typ.	LIMITS Max.	UNITS
	3N211					
*	Common-Source Spot Noise Figure, F	V_{DD}=18V, V_{GG}=7V, f = 200MHz, See Fig.9	–	–	3.5	dB
*	Small-Signal Common-Source Insertion Power Gain, G_{ps}		24	–	35	dB
*	Bandwidth, B		5	–	12	MHz
*	Gain-Control Gate-Supply Voltage, $V_{GG(GC)}$	V_{DD}=18V, ΔG_{ps}= −30dB,[1] f=200MHz, See Fig.9	0	–	−2	V
*	Common-Source Spot Noise Figure, F	V_{DD}=24V, V_{GG}=6V, f=45MHz, See Fig.10	–	–	4	dB
*	Small-Signal Common-Source Insertion Power Gain, G_{ps}		29	–	37	dB
*	Bandwidth, B		3.5	–	6	MHz
*	Gain-Control Gate-Supply Voltage, $V_{GG(GC)}$	V_{DD}=24V, ΔG_{ps}= −30dB,[2] f=45MHz, See Fig.10	–	–	±1	V
	3N212					
*	Small-Signal Conversion Power Gain, G_{ps} (conv)	V_{DD}=18V, f_{LO}=245MHz,[3] f_{RF}=200MHz, See Fig.11	21	–	28	dB
*	Bandwidth, B		4	–	7	MHz
	3N213					
*	Common-Source Spot Noise Figure, F	V_{DD}=24V, V_{GG}=6V, f=45MHz, See Fig.9	–	–	4	dB
*	Small-Signal Common-Source Insertion Power Gain, G_{ps}		27	–	35	dB
*	Bandwidth, B		3.5	–	6	MHz
*	Gain-Control Gate-Supply Voltage, $V_{GG(GC)}$	V_{DD}=24V, ΔG_{ps}= −30dB,[2] f=45MHz, See Fig.9	–	–	±1	V

*In accordance with JEDEC registration data format (JS-9 RDF-198). 2. ΔG_{ps} is defined as the change in G_{ps} from the value at V_{GG} = 6V.
1. ΔG_{ps} is defined as the change in G_{ps} from the value at V_{GG} = 7V. 3. Amplitude at input from local oscillator is adjusted for maximum $G_{ps(conv)}$.

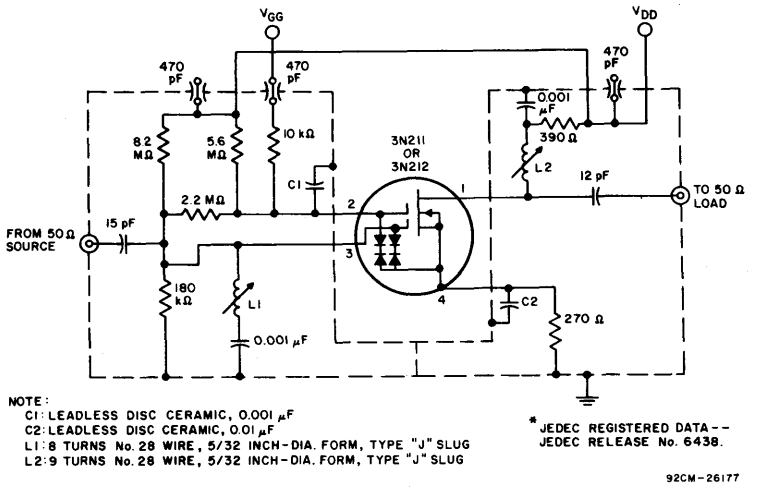


NOTE:
C1, C2, & C3: LEADLESS DISC CERAMIC, 0.001 μF
C4: ARCO 462, 5-80 pF, OR EQUIVALENT
L1: 3 TURNS No. 18 WIRE, 3/16 INCH-DIA. ALUMINUM SLUG
L2: 8 TURNS No. 20 WIRE, 3/16 INCH-DIA. ALUMINUM SLUG

*JEDEC REGISTERED DATA — — JEDEC RELEASE No. 6438.

92CM-26176

Fig.9—200 MHz power gain, gain control voltage, and noise figure test circuit for 3N211.*



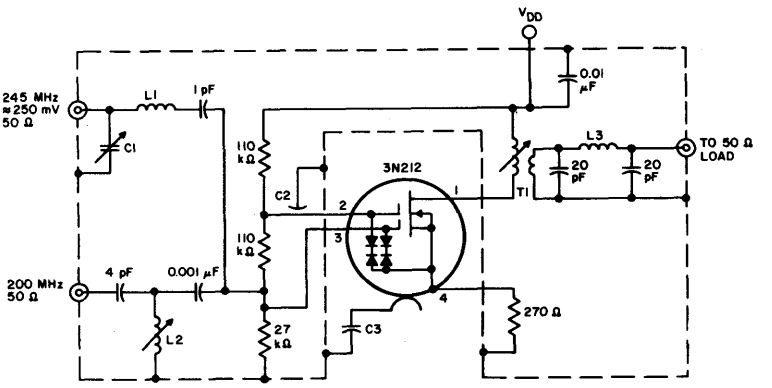
NOTE:
C1: LEADLESS DISC CERAMIC, 0.001 μF
C2: LEADLESS DISC CERAMIC, 0.01 μF
L1: 8 TURNS No. 28 WIRE, 5/32 INCH-DIA. FORM, TYPE "J" SLUG
L2: 9 TURNS No. 28 WIRE, 5/32 INCH-DIA. FORM, TYPE "J" SLUG

*JEDEC REGISTERED DATA -- JEDEC RELEASE No. 6438.

92CM-26177

Fig.10—45 MHz power gain and noise figure test circuit for 3N211 and 3N213.*

TEST CIRCUITS (CONT'D)



NOTE:
L1: 7 TURNS No. 34 WIRE, 1/4 INCH DIA. ALUMINUM SLUG
L2: 5 1/2 TURNS No. 20 WIRE, 1/4 INCH DIA. ALUMINUM SLUG
L3: 7 TURNS No. 24 WIRE, 1/4 INCH DIA. AIR CORE
T1: PRI: 25 TURNS No. 30 WIRE CLOSE WOUND ON 1/4 INCH DIA. FORM, TYPE "J" SLUG
SEC: 4 TURNS No. 30 WIRE CENTERED OVER PRIMARY
C1: ARCO TYPE 462, 5-80 pF
C2: 0.001 μF LEADLESS DISC
C3: 0.01 μF LEADLESS DISC

92CM-26178

*JEDEC REGISTERED DATA -- JEDEC RELEASE No. 6438.

Fig.11—200 MHz-to-45 MHz circuit for conversion power gain for 3N212.*

40467A

Silicon MOS Transistor N-Channel Depletion Type

For VHF Tuners and Other VHF Amplifier
Applications in Industrial & Commercial Electronic Equipment
Operating up to 220 MHz

Device Features:

- **Low feedback capacitance - C_{rss} = 0.25 pF typ.**
- **High forward transconductance - g_{fs} = 7500 μmho typ.**
- **High vhf power gain - G_{PS} = 16 dB typ at 200 MHz**
- **Low vhf noise figure - NF = 3.5 dB typ at 200 MHz**
- **Exceptionally good cross-modulation characteristics**

Performance Features:

- **Large dynamic range**
- **Greatly reduced spurious responses**
- **Permits use of vacuum-tube biasing techniques**
- ***Excellent thermal stability***
- **Superior cross-modulation performance and greater dynamic range than bipolar transistors**

RCA-40467A is an n-channel depletion-type silicon insulated-gate field-effect transistor utilizing the MOS■ construction. It is intended primarily for vhf-amplifier applications in industrial and commercial electronic equipment.

The 40467A is useful in vhf applications requiring devices capable of providing high useful power gains at frequencies up to approximately 220 MHz.

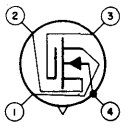
The 40467A features high forward transconductance, high dc gate-to-source resistance, and low feedback capacitance. Because of the improved transfer characteristic and increased dynamic range, the 40467A provides substantially better cross-modulation performance in linear-amplifier applications than conventional (bipolar) transistors and is free from diode-current loading, a problem that exists in junction type FET s. This device is hermetically sealed in the TO-72 metal case and utilizes full-gate construction.

■ Metal-Oxide Semiconductor

Maximum Ratings, *Absolute-Maximum Values at T_A=25°C*

DRAIN-TO-SOURCE VOLTAGE, V_{DS} ..... +20 V
DRAIN-TO-GATE VOLTAGE, V_{DG} ..... +20 V
GATE-TO-SOURCE VOLTAGE, V_{GS}:
CONTINUOUS (dc) ..... +1, −8 V
PEAK ac ..... +15 V
DRAIN CURRENT, I_D ..... 50 mA
TRANSISTOR DISSIPATION:
At ambient temperatures up to 25°C ..... 330 mW
above 25°C ..... derate at 2.2 mW/°C
AMBIENT TEMPERATURE RANGE:
Storage ..... −65 to +175°C
Operating ..... −65 to +175°C
LEAD TEMPERATURE (During Soldering):
At distances not closer than 1/32 inch to seating surface for 10 seconds maximum ..... 265°C

TERMINAL DIAGRAM



1 - Drain
2 - Source
3 - Insulated Gate
4 - Bulk (Substrate) and Case

ELECTRICAL CHARACTERISTICS AT T_C = 25°C WITH BULK (SUBSTRATE) CONNECTED TO SOURCE

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS			LIMITS			UNITS
		FREQUENCY f	DC DRAIN-TO-SOURCE VOLTAGE V_{DS}	DC DRAIN CURRENT I_D	RCA 40467A			
		MHz	V	mA	Min	Typ.	Max.	
Gate-to-Source Cutoff Voltage	V_{GS}(off)		12	0.1	-	-	-8	V
Gate Leakage Current	I_{GSS}		0	V_{GS} = +1V	-	-	1	nA
			0	V_{GS} = -8V	-	-	1	nA
Zero-Bias Drain Current	I_{DSS}		15	V_{GS} = 0	5	15	30	mA
Small-Signal, Short-Circuit Forward Transconductance	g_{fs}	1 KHz	15	5	4000	7500	-	μmho
Small-Signal, Short-Circuit Reverse-Transfer Capacitance (Drain-to-Gate)	C_{rss}	1	15	5	0.12	0.25	0.35	pF
Small Signal Short-Circuit Input Capacitance	C_{iss}	1	15	5	-	5.5	-	pF
Input Admittance	Y_{is}	Common Source Configuration f = 200 mHz V_{DS} = 15V I_D = 5 mA			-	0.4 + j7.3	-	
Forward Transfer Admittance	Y_{fs}				-	7 - j2	-	
Output Admittance	Y_{os}				-	0.28 + j1.8	-	
Maximum Available Power Gain	MAG	200	15	5	-	21	-	dB
Maximum Usable Power Gain (unneutralized)	MUG	200	15	5	-	12	-	dB
Maximum Usable Power Gain (neutralization)	MUG	200	15	5	12	16	-	dB
Noise Figure	NF	200	15	5	-	3.5	5	dB

For characteristics curves, refer to types 3N128 and 3N143.

MOS Silicon Transistors

N-Channel Depletion Types

For RF Amplifier and Mixer Applications in FM and AM/FM Receivers

RCA-40468A and 40559A are silicon insulated-gate field-effect transistors of the n-channel depletion type utilizing the MOS* construction. They are intended primarily for use as the rf amplifier and mixer, respectively, in FM receivers covering the 88 to 108 MHz band, but can be used for general amplifier applications at frequencies up to 125 MHz. For circuit design and typical performance data refer to RCA Application Note AN3535 "An FM Tuner Using Single-Gate MOS Field-Effect Transistors as RF Amplifier and Mixer".

The wide dynamic range of these transistors reduces cross-modulation effects in AM receivers and minimizes the generation of spurious responses in FM receivers.

Operating as a neutralized amplifier at 100 MHz, the 40468A can provide a power gain of 17 dB (typ.). A power gain of 14 dB (typ.) can be realized without neutralization.

* Metal-Oxide-Semiconductor.

Performance Features:

- reduced spurious responses in FM tuners
- reverse bias on substrate improves linearity
- reduced cross-modulation effects in AM receivers

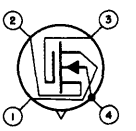
Maximum Ratings, *Absolute-Maximum Values at* $T_A = 25^oC$:

DRAIN-TO-SOURCE VOLTAGE, V_{DS}	+20	V
DRAIN-TO-GATE VOLTAGE, V_{DG}	+20	V
GATE-TO-SOURCE VOLTAGE, V_{GS}:		
CONTINUOUS (dc)	+1, -8	V
PEAK ac	±15	V
DRAIN CURRENT, I_D	25	mA
TRANSISTOR DISSIPATION:		
At ambient temperatures up to 25°C	330	mW
At ambient temperatures above 25°C	derate at 2.2	mW/°C
AMBIENT TEMPERATURE RANGE:		
Storage	-65 to +175	°C
Operating	-65 to +175	°C
LEAD TEMPERATURE (During Soldering):		
At distances not closer than 1/32 inch to seating surface for 10 seconds maximum	265	°C

Device Features:

- high forward transconductance - - g_{fs} = 7500 μmho typ. for 40468A
- low feedback capacitance - - C_{rss} = 0.35 pF max. for 40468A, 0.38 pF max. for 40559A
- high useful power gains - - neutralized - 17 dB typ., unneutralized - 14 dB typ.
- hermetically sealed in TO-72 metal package

TERMINAL DIAGRAM



LEAD 1 - DRAIN
LEAD 2 - SOURCE
LEAD 3 - INSULATED GATE
LEAD 4 - BULK (SUBSTRATE) AND CASE

ELECTRICAL CHARACTERISTICS, at T_A = 25°C
With Bulk (Substrate) Connected to Source Unless Otherwise Specified

Characteristics	Symbols	TEST CONDITIONS: Frequency f	DC Drain-to-Source V_{DS}	DC Drain Current I_D	LIMITS: RCA-40468A RF Amplifier Min.	Typ.	Max.	RCA-40559A Mixer Min.	Typ.	Max.	Units
		MHz	V	mA							
Drain-to-Source Cutoff Current	$I_{D(off)}$	-	12	V_{GS} = -8V	-	-	100	-	-	500	μA
Gate Leakage Current	I_{GSS}	-	0	V_{GS} = -8V	-	-	1	-	-	1	nA
		-	0	V_{GS} = +1V	-	-	1	-	-	1	nA
Zero-Bias Drain Current	I_{DSS}	-	15	V_{GS} = 0	5	15	30	5	15	30	mA
Small-Signal, Short-Circuit Forward Transconductance	g_{fs}	1 kHz	15	5	-	7500	-	-	-	-	μmho
Small-Signal, Short-Circuit Reverse-Transfer Capacitance (Drain-to-Gate)	C_{rss}	1	15	5	-	0.25	0.35	-	0.25	0.38	pF
Input Capacitance	C_{iss}	1	15	5	-	5.5	-	-	5.5	-	pF
Admittance	-	RF | Mixer		RF | Mixer	-			-			-
Input Admittance	Y_{is}	100 MHz	15	5 | 3	0.155 + j 3.45			0.14 + j 3.38			mmho
Forward Transfer Admittance	Y_{fs}	100 MHz	15	5 | 3	7.4 + j 0.9			-			mmho
Output Admittance	Y_{os}	100 MHz | 10.7 MHz	15	5 | 3	0.21 + j 0.9			0.076 + j 0.153			mmho
Forward Conversion Transconductance	$g_{fs(c)}$	1 kHz	15	3	-	-	-	-	2800*	-	μmho
Maximum Available Power Gain	MAG	100	15	5	-	26	-	-	-	-	dB
Maximum Usable Power Gain (Unneutralized)	MUG	100	15	5	-	14	-	-	-	-	dB
Maximum Usable Power Gain (Neutralized)	MUG	100	15	5	14	17	-	-	-	-	dB
Maximum Available Conversion Gain	MAG_C	f_{in} = 100, f_{out} = 10.7	15	3	-	-	-	-	22	-	dB
Noise Figure	NF	100	15	5	-	3.5	5	-	-	-	dB

* Bulk (Substrate)-to-Source Volts (V_{BS}) = -3.

For characteristics curves, refer to types 3N128 and 3N143.

SILICON DUAL INSULATED-GATE FIELD-EFFECT TRANSISTORS

N-Channel Depletion Types
For VHF TV Receiver Applications

RCA 40600, 40601, and 40602 are n-channel depletion type, dual-insulated-gate, field-effect transistors utilizing the MOS construction. These devices have characteristics which make them highly desirable for rf-amplifier applications (40600), mixer applications (40601), and first-if-amplifier applications (40602) in vhf TV receivers and other types of commercial equipment operating at frequencies up to approximately 250 MHz.

These transistors feature a series arrangement of two separate channels, each channel having an independent control gate. In amplifier applications the 40600 and 40602 with their wide dynamic range provide substantially better cross-modulation performance than is obtainable with bipolar or single-gate field-effect transistors. In mixer applications the 40601 provides excellent isolation between the oscillator and rf signals because each of the two signal frequencies being mixed has its own control element. The wide dynamic range of the 40601 minimizes cross-modulation which is generally encountered in mixer stages.

Provision of two insulated gates also results in extremely low feedback capacitances (0.02 pF typ.), a feature which enables the 40600 and 40602 to provide high maximum useable power gains in unneutralized circuits – for example, 20 dB at 200 MHz typ. for the 40600, and 35 dB typ. at 44 MHz for the 40602. The gain of the rf and if stages can be controlled by applying agc voltage to gate No.2 and agc delay is easily obtained. Virtually no agc power is required for full gain reduction.

Types 40600, 40601, and 40602 are hermetically sealed in metal JEDEC TO-72 packages.

APPLICATIONS

- **VHF TV Receiver**
  - **40600 for rf amplifier applications**
  - **40601 for mixer applications**
  - **40602 for first-if-amplifier applications**

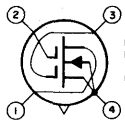
PERFORMANCE FEATURES

- **superior cross-modulation performance and greater dynamic range than bipolar and single-gate field-effect transistors**
- **permits use of vacuum-tube biasing techniques**
- **excellent thermal stability**

DEVICE FEATURES

- **extremely low feedback capacitance**
  C_{rss} = 0.02 pF typ.
- **high power gain**
  MUG_u = 20 dB typ. for 40600
  MAG = 35 dB typ. for 40602
  MAG_c = 14 dB typ. for 40601

TERMINAL DIAGRAM



Maximum Ratings, *Absolute-Maximum Values at T_A = 25°C:*

DRAIN-TO-SOURCE VOLTAGE, V_{DS} ..... 0 to +20 V

GATE No.1-TO-SOURCE VOLTAGE, V_{G1S}:
- Continuous (dc) ..... +1 to -8 V
- Peak ac ..... +20 to -8 V

GATE No.2-TO-SOURCE VOLTAGE, V_{G2S}:
- Continuous (dc) ..... -8 to 40% of V_{DS} V
- Peak ac ..... -8 to +20 V

DRAIN-TO-GATE VOLTAGE, V_{DG1} or V_{DG2} ..... +20 V

DRAIN CURRENT, I_D (Pulsed):
- Pulse duration ≤ 20 ms, duty factor ≤ 0.15 ..... 50 mA

TRANSISTOR DISSIPATION, P_T:
- At ambient temperatures up to 25°C ..... 400 mW
- above 25°C ..... derate linearly at 2.67 mW/°C

AMBIENT TEMPERATURE RANGE:
- Storage and Operating ..... -65 to +175 °C

LEAD TEMPERATURE (During soldering):
- At distances ≥ 1/32" from seating surface for 10 seconds max. ..... 265 °C

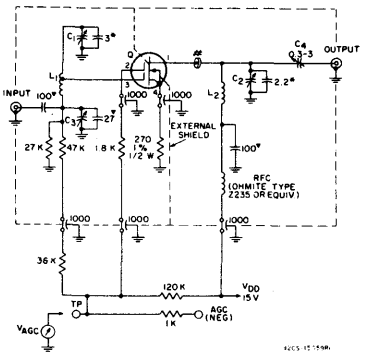
ELECTRICAL CHARACTERISTICS, at T_A = 25°C

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	LIMITS 40600, 40601, 40602			UNITS
			Min.	Typ.	Max.	
Gate No.1-to-Source Cutoff Voltage	$V_{G1S(off)}$	V_{DS} = +15V, I_D = 200 μA, V_{G2S} = +4V	-	-2	-	V
Gate No.2-to-Source Cutoff Voltage	$V_{G2S(off)}$	V_{DS} = +15V, I_D = 200 μA, V_{G1S} = 0	-	-2	-	V
Gate No.1 Leakage Current	I_{G1SS}	V_{G1S} = -20V, V_{G2S} = 0, V_{DS} = 0	-	-	1	nA
Gate No.2 Leakage Current	I_{G2SS}	V_{G2S} = -20V, V_{G1S} = 0, V_{DS} = 0	-	-	1	nA
Drain Current	I_{DSS}	V_{DS} = +13V, V_{G1S} = 0, V_{G2S} = +4V	-	18	-	mA
Forward Transconductance	g_{fs}	V_{DS} = +13V, I_D = 10 mA, V_{G2S} = +4V, f = 1 kHz	-	10000	-	μmho

TYPICAL PERFORMANCE CHARACTERISTICS, at T_A = 25°C

CHARACTERISTICS	SYMBOLS	40600 RF AMPLIFIER f = 200 MHz	40602 IF AMPLIFIER f = 44 MHz	40601 MIXER f = 200 MHz	UNITS
		V_{G1S} is adjusted for I_D = 10 mA, Gate No.2 at AC ground potential, V_{DS} = 13V, V_{G2S} = +4V		Local-oscillator injection Voltage on Gate No.2 = 750 mV, V_{DS} = 15V, V_{G2S} = +0.6V, V_{G1S} = 0.75V	
Small-Signal, Short Circuit Reverse-Transfer Capacitance (Drain-to-Gate No.1) at f = 1 MHz	C_{rss}	0.02 typ. 0.03 max.	0.02 typ. 0.03 max.	0.02 typ. 0.03 max.	pF
Output Capacitance	C_{oss}	2.2	2.2	2.2 at f = 44 MHz	pF
Input Capacitance	C_{iss}	5.5	5.5	5.5	pF
Input Resistance	r_{iss}	1.2	10	1.2	KΩ
Output Resistance	r_{oss}	2.8	12	12 at f = 44 MHz	KΩ
Magnitude of Forward Transadmittance	$\|Y_{fs}\|$	11000	11000	2700*	μmho
Phase Angle of Forward Transadmittance	$\angle\theta$	-46	-11	-	degrees
Maximum Available Power Gain	MAG	20	35	14**	dB
Maximum Usable Power Gain (Unneutralized)	MUG_u	20▲	1 Stage 28 2 Stages 26 3 Stages 24	-	dB
Power Gain See Fig.1 for measurement circuit	G_{PS}	17.5	-	-	dB
Noise Figure	NF	5 max.	-	-	dB

* Magnitude of forward conversion transadmittance ** Maximum available conversion gain ▲ Limited by practical design considerations



* Tubular ceramic.

▼ Disk ceramic.

\# Ferrite bead (½ used); Indiana General No. H1742C-(A-147) or F1157-1-H, or equivalent.

C_1, C_2: 1.5-5 pF variable air capacitor: E. F. Johnson Type 160-1C?, or equivalent.

C_3: 1-10 pF piston-type variable air capacitor: JFD Type VAM-010, Johanson Type 4335, or equivalent.

C_4: 0.3-3 pF piston-type variable air capacitor: Roanwell Type MH-13, or equivalent.

L_1: 5 turns silver-plated 0.02" thick, 0.07"- 0.08" wide copper ribbon. Internal diameter of winding = 0.25"; winding length approx. 0.65". Tapped at 1-1/2 turns from C_1 end of winding.

L_2: Same as L_1 except winding length approx. 0.7"; no tap.

Fig.1 - 200 MHz Power Gain and Noise Figure Test Circuit for 40600 and 40602

For characteristics curves, refer to type 3N140.

SILICON DUAL INSULATED-GATE FIELD-EFFECT TRANSISTORS

N-Channel Depletion Types
For FM Tuner Applications

RCA 40603 and 40604 are n-channel silicon, depletion type, dual insulated-gate, field-effect transistors utilizing the MOS construction.

These devices have exceptional characteristics for rf-amplifier (40603) and mixer applications (40604) in FM tuners and other commercial equipment operating at frequencies up to approximately 150 MHz. These transistors feature a series arrangement of two separate channels, each channel having an independent control gate. For amplifier applications the 40603 with its wide dynamic range provides substantially better cross-modulation performance and relative freedom from spurious responses than is obtainable with bipolar or single-gate field-effect transistors. The mixing function performed by the 40604 is unique in that the signal applied to gate No.2 is used to modulate the input-gate (gate No.1) transfer characteristic. This technique is superior to conventional "square law" mixing, which can only be accomplished in the non-linear region of the device transfer characteristic.

Because of the low feedback capacitance (0.02 typ. pF) the 40603 can provide a power gain of 25 dB (typ.) at 100 MHz in an unneutralized amplifier circuit.

The gain of the rf stage can be controlled by applying agc voltage to gate No.2. Virtually no agc power is required for full gain reduction.

The 40603 and 40604 are hermetically sealed in JEDEC TO-72 packages.

Maximum Ratings, *Absolute-Maximum Values at* T_A = 25°C:

DRAIN-TO-SOURCE VOLTAGE, V_{DS}	0 to +20	V
GATE No.1-TO-SOURCE VOLTAGE, V_{G1S}:		
Continuous (dc)	-8 to +1	V
Peak ac	-8 to +20	V
GATE No.2-TO-SOURCE VOLTAGE, V_{G2S}:		
Continuous (dc)	-8 to 40% of V_{DS}	V
Peak ac	-8 to +20	V
DRAIN-TO-GATE VOLTAGE, V_{DG1} or V_{DG2}	+20	V
DRAIN CURRENT, I_D (Pulsed): Pulse duration ≤ 20 ms, duty factor ≤ 0.15	50	mA
TRANSISTOR DISSIPATION, P_T:		
At ambient temperatures up to 25°C	400	mW
At ambient temperatures above 25°C	derate linearly at 2.67 mW/°C	
AMBIENT TEMPERATURE RANGE:		
Storage and Operating	-65 to +175	°C
LEAD TEMPERATURE (During soldering):		
At distances ≥ 1/32" from seating surface for 10 seconds max.	265	°C

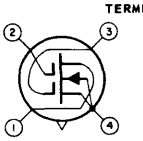
PERFORMANCE FEATURES

- large dynamic range permits large-signal handling before overload
- dual gates allow product mixing with extremely low harmonic generation
- greatly reduces spurious responses in FM receivers
- permits use of vacuum-tube biasing techniques
- excellent thermal stability
- superior cross-modulation performance and greater dynamic range than bipolar and single-gate field-effect transistors

DEVICE FEATURES

- extremely low feedback capacitance
  C_{rss} = 0.02 pF typ.
- high unneutralized RF power gain
  MUG = 25 dB (typ.) for 40603
- low noise figure
  NF = 2.5 dB typ. for 40603

TERMINAL DIAGRAM



Lead 1 — Drain
Lead 2 — Gate No. 2
Lead 3 — Gate No. 1
Lead 4 — Source, Substrate and Case

ELECTRICAL CHARACTERISTICS, at T_A = 25°C

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	LIMITS 40603 RF AMPLIFIER Typ.	40603 RF AMPLIFIER Max.	40604 MIXER Typ.	40604 MIXER Max.	UNITS
Gate No.1-to-Source Cutoff Voltage	$V_{G1S(off)}$	V_{DS} = +15 V, I_D = 200 μA, V_{G2S} = +4 V	-2	--	-2	--	V
Gate No.2-to-Source Cutoff Voltage	$V_{G2S(off)}$	V_{DS} = +15 V, I_D = 200 μA, V_{G1S} = 0	-2	--	-2	--	V
Gate No.1 Leakage Current	I_{G1SS}	V_{G1S} = -20 V, V_{G2S} = 0, V_{DS} = 0	--	1	--	1	nA
Gate No.2 Leakage Current	I_{G2SS}	V_{G2S} = -20 V, V_{G1S} = 0, V_{DS} = 0	--	1	--	1	nA
Zero-Bias-Voltage Drain Current	I_{DSS}	V_{G2S} = +4 V, V_{G1S} = 0, V_{DS} = +13 V	18	--	18	--	mA
Small-Signal, Short-Circuit Reverse-Transfer Capacitance (Drain-to-Gate-No.1)	C_{rss}	V_{DS} = +13 V, I_D = 10 mA, f = 1 MHz, V_{G2S} = +4 V	0.02	0.03	0.02	0.03	pF
Input Capacitance	C_{iss}	V_{DS} = +13 V, I_D = 10 mA, V_{G2S} = +4 V, f = 1 MHz	5.5	--	5.5	--	pF
Output Capacitance	C_{oss}	V_{DS} = +13 V, I_D = 10 mA, V_{G2S} = +4 V, f = 100 MHz	2.1	--	2.3	--	pF
Input Resistance	r_{is}	V_{DS} = +13 V, I_D = 10 mA, V_{G2S} = +4 V, f = 100 MHz	3.5	--	3.5	--	kΩ
Output Resistance	r_{os}	V_{DS} = +13 V, I_D = 10 mA, V_{G2S} = +4 V; f = 100 MHz	4	--	--	--	kΩ
		V_{DS} = +13 V, I_D = 10 mA, V_{G2S} = +4 V; f = 10.7 MHz	--	--	20	--	kΩ
Forward Transconductance	g_{fs}	V_{DS} = +13 V, I_D = 10 mA, V_{G2S} = +4 V, f = 1 kHz	10,000	--	2800*	--	μmho
Maximum Available Power Gain	MAG	V_{DS} = +13 V, I_D = 10 mA, V_{G2S} = +4 V, f = 100 MHz, f_{out} for 40604 (mixer) = 10.7 MHz	26	--	21	--	dB
Maximum Usable Power Gain (Unneutralized)	MUG		25▲	-	--	--	dB
Noise Figure	NF		2.5	--	--	--	dB

* conversion transconductance
▲ or limited by design considerations

For characteristics curves, refer to type 3N140.

40673

SILICON DUAL INSULATED-GATE FIELD-EFFECT TRANSISTOR
N-Channel Depletion Type With Integrated Gate-Protection Circuits
For RF Amplifier Applications up to 400 MHz

RCA-40673 is an n-channel silicon, depletion type, dual insulated-gate field-effect transistor.

Special back-to-back diodes are diffused directly into the MOS* pellet and are electrically connected between each insulated gate and the FET's source. The diodes effectively bypass any voltage transients which exceed approximately ±10 volts. This protects the gates against damage in all normal handling and usage.

A feature of the back-to-back diode configuration is that it allows the 40673 to retain the wide input signal dynamic range inherent in the MOSFET. In addition, the low junction capacitance of these diodes adds little to the total capacitance shunting the signal gate.

The excellent overall performance characteristics of the RCA-40673 make it useful for a wide variety of rf-amplifier applications at frequencies up to 400 MHz. The two serially-connected channels with independent control gates make possible a greater dynamic range and lower cross-modulation than is normally achieved using devices having only a single control element.

The two gate arrangement of the 40673 also makes possible a desirable reduction in feedback capacitance by operating in the common-source configuration and ac-grounding Gate No. 2. The reduced capacitance allows operation at maximum gain *without neutralization;* and, of special importance in rf-amplifiers, it reduces local oscillator feedthrough to the antenna.

The 40673 is hermetically sealed in the metal JEDEC TO-72 package.

*Metal-Oxide-Semiconductor.

Maximum Ratings, *Absolute-Maximum Values, at* T_A = 25°C

DRAIN-TO-SOURCE VOLTAGE, V_{DS}	-0.2 to +20	V
GATE No.1-TO-SOURCE VOLTAGE, V_{G1S}:		
Continuous (dc)	-6 to +1	V
Peak ac	-6 to +6	V
GATE No.2-TO-SOURCE VOLTAGE, V_{G2S}:		
Continuous (dc)	-6 to 30% of V_{DS}	V
Peak ac	-6 to +6	V
DRAIN-TO-GATE VOLTAGE, V_{DG1} OR V_{DG2}	+20	V
DRAIN CURRENT, I_D	50	mA
TRANSISTOR DISSIPATION, P_T:		
At ambient temperatures up to 25°C	330	mW
At ambient temperatures above 25°C	derate linearly at 2.2 mW/°C	
AMBIENT TEMPERATURE RANGE:		
Storage and Operating	-65 to +175	°C
LEAD TEMPERATURE (During soldering):		
At distances ≥ 1/32 inch from seating surface for 10 seconds max.	265	°C

ELECTRICAL CHARACTERISTICS, at T_A = 25°C unless otherwise specified

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	LIMITS			UNITS
			Min.	Typ.	Max.	
Gate-No.1-to-Source Cutoff Voltage	$V_{G1S(off)}$	V_{DS} = +15V, I_D = 200μA, V_{G2S} = +4V	–	-2	-4	V
Gate-No.2-to-Source Cutoff Voltage	$V_{G2S(off)}$	V_{DS} = +15V, I_D = 200μA, V_{G1S} = 0	–	-2	-4	V
Gate-No.1-Leakage Current	I_{G1SS}	V_{G1S} = +1 or -6 V, V_{DS} = 0, V_{G2S} = 0	–	–	50	nA
Gate-No.2-Leakage Current	I_{G2SS}	V_{G2S} = ±6V, V_{DS} = 0, V_{G1S} = 0	–	–	50	nA
Zero-Bias Drain Current	I_{DSS}	V_{DS} = +15V, V_{G2S} = +4V, V_{G1S} = 0	5	15	35	mA
Forward Transconductance (Gate-No.1-to-Drain)	g_{fs}	V_{DS} = +15V, I_D = 10mA, V_{G2S} = +4V, f = 1kHz	–	12,000	–	μmho
Small-Signal, Short-Circuit Input Capacitance †	C_{iss}	V_{DS} = +15V, I_D = 10mA, V_{G2S} = +4V, f=1MHz	–	6	–	pF
Small-Signal, Short-Circuit, Reverse Transfer Capacitance (Drain-to-Gate No.1) ♦	C_{rss}		0.005	0.02	0.03	pF
Small-Signal, Short-Circuit Output Capacitance	C_{oss}		–	2.0	–	pF
Power Gain (see Fig. 1)	G_{PS}	V_{DS} = +15V, I_D = 10mA, V_{G2S} = +4V, f = 200 MHz	14	18	–	dB
Maximum Available Power Gain	MAG		–	20	–	dB
Maximum Usable Power Gain (unneutralized)	MUG		–	20*	–	dB
Noise Figure (see Fig. 1)	NF		–	3.5	6.0	dB
Magnitude of Forward Transadmittance	$\lvert Y_{fs} \rvert$		–	12,000	–	μmho
Phase Angle of Forward Transadmittance	θ		–	-35	–	degrees
Input Resistance	r_{iss}		–	1.0	–	kΩ
Output Resistance	r_{oss}		–	2.8	–	kΩ
Protective Diode Knee Voltage	V_{knee}	$I_{DIODE(REVERSE)}$ = ±100μA	–	±10	–	V

*Limited only by practical design considerations.

†Capacitance between Gate No. 1 and all other terminals

♦Three-terminal measurement with Gate No. 2 and Source returned to guard terminal.

For characteristics curves, refer to type 3N187.

APPLICATIONS

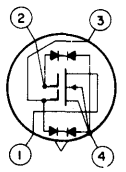
- RF amplifier, mixer, and IF amplifier in military, industrial, and consumer communications equipment
- aircraft and marine vehicular receivers
- CATV and MATV equipment
- telemetry and multiplex equipment

PERFORMANCE FEATURES

- superior cross-modulation performance and greater dynamic range than bipolar or single-gate FET s
- wide dynamic range permits large-signal handling before overload
- dual-gate permits simplified agc circuitry
- virtually no agc power required
- greatly reduces spurious responses in fm receivers
- permits use of vacuum-tube biasing techniques
- excellent thermal stability

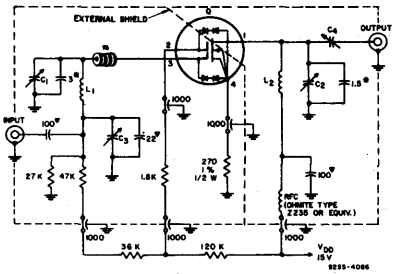
DEVICE FEATURES

- back-to-back diodes protect each gate against handling and in-circuit transients
- low gate leakage currents — I_{G1SS} & I_{G2SS} = 20 nA(max.) at T_A = 25°C
- high forward transconductance — g_{fs} = 12,000 μmho (typ.)
- high unneutralized RF power gain — G_{ps} = 18 dB(typ.) at 200 MHz
- low VHF noise figure — 3.5 dB(typ.) at 200 MHz



TERMINAL DIAGRAM

LEAD 1 - DRAIN
LEAD 2 - GATE No. 2
LEAD 3 - GATE No. 1
LEAD 4 - SOURCE, SUBSTRATE AND CASE



#Ferrite bead (4); Pyroferric Co. "Carbonyl J" 0.09 in. OD; 0.03 in. ID; 0.063 in. thickness.

All resistors in ohms

All capacitors in pF

Q = 40673
▼ Disc ceramic.
•Tubular ceramic.

C_1: 1.8 – 8.7 pF variable air capacitor: E.F. Johnson Type 160-104, or equivalent.

C_2: 1.5 – 5 pF variable air capacitor: E.F. Johnson Type 160-102, or equivalent.

C_3: 1 – 10 pF piston-type variable air capacitor: JFD Type VAM-010; Johanson Type 4335, or equivalent.

C_4: 0.8 – 4.5 pF piston type variable air capacitor: Erie 560-013 or equivalent.

L_1: 4 turns silver-plated 0.02-in. thick, 0.075-0.085-in. wide, copper ribbon. Internal diameter of winding = 0.25 in, winding length approx. 0.80 in.

L_2: 4½ turns silver-plated 0.02-in. thick, 0.085-0.095-in. wide, 5/16-in. ID. Coil ≈ .90 in. long.

Fig. 1. 200-MHz Power gain and noise-figure test circuit

Silicon Dual-Insulated-Gate Field-Effect Transistor

N-Channel Depletion Type

With Integrated Gate-Protection Circuits

For RF Amplifier Applications up to 250 MHz

RCA-40819 is an n-channel silicon, depletion type, dual insulated-gate field-effect transistor (FET).

The excellent overall performance characteristics of the RCA-40819 make it useful for a wide variety of rf-amplifier applications at frequencies up to 250 MHz. The two serially-connected channels with independent control gates make possible a greater dynamic range and lower cross-modulation than is normally achieved using devices having only a single control element.

The two-gate arrangement of the 40819 also makes possible a desirable reduction in feedback capacitance by operating in the common-source configuration and ac grounding Gate No.2. The reduced capacitance allows operation at maximum gain *without neutralization* and reduces local oscillator feedthrough to the antenna – features of special importance in rf and if amplifiers.

Special back-to-back diodes are diffused directly into the MOS pellet and are electrically connected between each insulated gate and the FET's source. The diodes effectively bypass any voltage transients which exceed approximately ±10 volts and protect the gates against damage in all normal handling and usage.

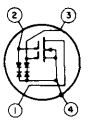
The back-to-back diode configuration permits the 40819 to retain the wide input signal dynamic range inherent in the MOSFET. In addition, the low junction capacitance of these diodes adds little to the total capacitance shunting the signal gate.

The 25-volt drain-to-source rating permits the use of higher voltage power supplies.

The 40819 is hermtically sealed in the metal JEDEC TO-72 package.

TERMINAL DIAGRAM

LEAD 1 - DRAIN
LEAD 2 - GATE No.2
LEAD 3 - GATE No.1
LEAD 4 - SOURCE, SUBSTRATE, AND CASE



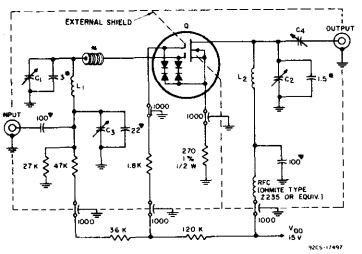
ELECTRICAL CHARACTERISTICS, at T_A = 25° C unless otherwise specified

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	LIMITS			UNITS
			Min.	Typ.	Max.	
Gate-No.1-to-Source Cutoff Voltage	$V_{G1S(off)}$	V_{DS} = +15 V, I_D = 200 μA, V_{G2S} = +4 V	–	–2	–4	V
Gate-No.2-to-Source Cutoff Voltage	$V_{G2S(off)}$	V_{DS} = +15 V, I_D = 200 μA, V_{G1S} = 0	–	–2	–4	V
Gate-No.1-Leakage Current	I_{G1SS}	V_{G1S} = ± 6 V, V_{DS} = 0, V_{G2S} = 0	–	–	50	nA
Gate-No.2-Leakage Current	I_{G2SS}	V_{G2S} = ± 6 V, V_{DS} = 0, V_{G1S} = 0	–	–	50	nA
Zero-Bias Drain Current	I_{DSS}	V_{DS} = + 15 V, V_{G2S} = +4 V, V_{G1S} = 0	5	15	35	mA
Forward Transconductance (Gate-No.1-to-Drain)	g_{fs}	V_{DS} = +15 V, I_D = 10 mA, V_{G2S} = +4 V, f = 1 kHz	–	12,000	–	μmho
Small-Signal, Short-Circuit Input Capacitance†	C_{iss}	V_{DS} = +15 V, I_D = 10 mA, V_{G2S} = +4 V, f = 1 MHz	–	6	–	pF
Small-Signal, Short-Circuit, Reverse Transfer Capacitance (Drain-to-Gate No.1)♣	C_{rss}		0.005	0.02	0.03	pF
Small-Signal, Short-Circuit Output Capacitance	C_{oss}		–	2	–	pF
Power Gain (see Fig. 1)	G_{PS}	V_{DS} = +15 V, I_D = 10 mA, V_{G2S} = +4 V, f = 200 MHz	14	18	–	dB
Maximum Available Power Gain	MAG		–	20	–	dB
Maximum Usable Power Gain (unneutralized)	MUG		–	20*	–	dB
Noise Figure (see Fig. 1)	NF		–	3.5	6.0	dB
Magnitude of Forward Transadmittance	$\|Y_{fs}\|$		–	12,000	–	μmho
Phase Angle of Forward Transadmittance	θ		–	–35	–	degrees
Input Resistance	r_{iss}		–	1	–	kΩ
Output Resistance	r_{oss}		–	2.8	–	kΩ
Protective Diode Knee Voltage	V_{knee}	I_{diode} (reverse) = ±100 μA	–	±10	–	V

* Limited only by practical design considerations.

† Capacitance between Gate No.1 and all other terminals.

♣ Three-terminal measurement with Gate No.2 and Source returned to guard terminal.



#Ferrite bead (4); Pyroferric Co. "Carbonyl J" 0.09 in OD; 0.03 in ID; 0.063 in thickness.

Q = 40673
▼Disc ceramic.
* Tubular ceramic.

All resistors in ohms

All capacitors in pF

C_1: 1.8 – 8.7 pF variable air capacitor: E. F. Johnson Type 160-104, or equivalent.

C_2*: 1.5 – 5 pF variable air capacitor: E. F. Johnson Type 160-102, or equivalent.

C_3: 1 – 10 pF piston-type variable air capacitor: JFD Type VAM-010; Johanson Type 4335, or equivalent.

C_4: 0.8 – 4.5 pF piston type variable air capacitor: Erie 560-013 or equivalent.

L_1: 4 turns silver-plated 0.02-in thick, 0.075-0.085 in wide, copper ribbon. Internal diameter of winding = 0.25 in, winding length approx. 0.80 in.

L_2: 4-1/2 turns silver-plated 0.02 in thick, 0.085-0.095 in wide, 5/16-in; ID Coil = .90 in long.

Fig. 1. 200 MHz power gain and noise figure test circuit

For characteristics curves, refer to type 3N187.

Device Features

- **back-to-back diodes protect each gate against handling and in-circuit transients**
- **high forward transconductance: g_{fs} = 12,000 μmho (typ.)**
- **high unneutralized RF power gain: G_{ps} = 18 dB (typ.) at 200 MHz**
- **low VHF noise figure: 3.5 dB (typ.) at 200 MHz**
- **low gate leakage currents: I_{G1SS} & I_{G2SS} = 50 nA at T_A = 25° C**
- **increased drain-to-source voltage rating: V_{DS} = –0.2 to +25 V**

Performance Features

- **superior cross-modulation performance and greater dynamic range than bipolar or single-gate FET s**
- **wide dynamic range permits large-signal handling before overload**
- **virtually no agc power required**
- **greatly reduces spurious responses in FM receivers**
- **dual gate permits simplified AGC circuitry**

Applications

- **RF amplifier, mixer, and IF amplifier in military, industrial, and consumer communications equipment**
- **aircraft and marine vehicular receivers**
- **CATV and MATV equipment**
- **telemetry and multiplex equipment**

Absolute Maximum Values, at T_A = 25°C:

Drain-to-Source Voltage, V_{DS}	–0.2 to +25	V
Gate Terminal Current, I_{G1S} or I_{G2S}	±100	μA
Drain-to-Gate Voltage, V_{DG1} or V_{DG2}	+31	V
Drain Current, I_D	50	mA
Transistor Dissipation, P_T:		
At T_A up to 25°C	330	mW
At T_A above 25°C	derate linearly 2.2 mW/°C	
Ambient Temperature Range:		
Operating and Storage	–65 to +175	°C
Lead Temperature (During Soldering):		
At distances 1/32 in from seating surface for 10 s max.	265	°C

Maximum Ratings

Continuous Working Voltage#, at T_A = 25°C:

Gate No.1-to-Source Voltage, V_{G1S}	–6 to +3	V
Gate No.2-to-Source Voltage, V_{G2S}	–6 to +6 or 40% of V_{DS} (whichever value is less)	V
Drain-to-Gate Voltage, V_{DG1} or V_{DG2}	+25	V

Continuous Working Voltage Ratings must be observed to maintain device characteristics. These ratings are based on long-term continuous voltage operation but may be exceeded for short durations (e.g. testing of device characteristics), provided the absolute Maximum Ratings are not exceeded.

40820, 40821

Silicon Dual-Insulated-Gate Field-Effect Transistors

N-Channel Depletion Types

With Integrated Gate-Protection Circuits
For VHF-TV Tuner Applications

40820 – RF Amplifier **40821** – Mixer

Device Features

- back-to-back diodes protect each gate against handling and in-circuit transients
- high forward transconductance: g_{fs} = 12,000 μmho (typ.)
- high unneutralized RF power gain: G_{ps} = 17 dB (typ.) at 200 MHz (40820)
- low VHF noise figure: 3.5 dB (typ.) at 200 MHz (40820)
- low gate leakage currents: I_{G1SS} & I_{G2SS} = 50 nA

RCA-40820 and 40821 are n-channel silicon, depletion type, dual-insulated-gate, MOS▲ field-effect transistors for RF amplifier (40820) and mixer (40821) applications in VHF-TV receivers and other commercial equipment operating at frequencies up to 250 MHz.

These devices designed for VHF performance, provide excellent power gain, low-noise figures and have wide dynamic range. The dual-gate feature offers good cross-modulation performance over the AGC range and reduces feedback capacitance by shielding Gate No. 1 from the drain. The very low feedback capacitance also eliminates the need for circuit neutralization and reduces local oscillator feed-through to the antenna.

Virtually no AGC power is required because of the high gate input resistance of the MOS FET types. Automatic AGC delay can be achieved with a very slight change in the input impedance by the application of AGC voltage to Gate No. 2.

▲ Metal-Oxide-Semiconductor.

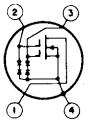
Performance Features

- superior cross-modulation performance and greater dynamic range than bipolar or single-gate FET s
- wide dynamic range permits large-signal handling before overload
- virtually no agc power required
- dual gate permits simplified AGC circuitry

The dual-gate arrangement also makes it possible to isolate the local oscillator signal from the incoming signal by applying each signal to a separate gate.

Integral back-to-back diodes protect the gates against damage in normal handling and usage by limiting transient voltages that exceed ±10 volts. The 40820 and 40821 are hermetically sealed in metal JEDEC TO-72 packages.

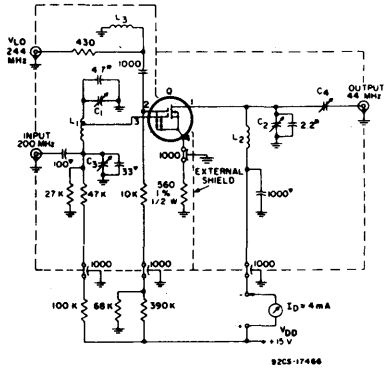
TERMINAL DIAGRAM



LEAD 1 – DRAIN
LEAD 2 – GATE No.2
LEAD 3 – GATE No.1
LEAD 4 – SOURCE, SUBSTRATE, AND CASE

Maximum Ratings *Continuous Working Voltage#, at T_A = 25°C:*	**40820**	**40821**	
Gate No. 1-to-Source Voltage, V_{G1S}	−6 to +3	−4.5 to +3	V
Gate No. 2-to-Source Voltage, V_{G2S}	-6 to +6 or 40% of V_{DS} (whichever value is less)	-4.5 to +4.5 or -4.5 to 40% of V_{DS} (whichever value is less)	V
Drain-to-Gate Voltage, V_{DG1} or V_{DG2}	+20	+20	V
Absolute Maximum Values, at T_A = 25°C:			
Drain-to-Source Voltage, V_{DS}	−0.2 to +20	−0.2 to +20	V
Gate Terminal Current, I_{G1S} or I_{G2S}	±100	±100	μA
Drain-to-Gate Voltage, V_{DG1} or V_{DG2}	+26	+24.5	V
Drain Current, I_D	50	50	mA
Transistor Dissipation:			
At T_A up to 25°C	330	330	mW
At T_A above 25°C	derate linearly 2.2 mW/°C		
Ambient Temperature Range:			
Operating and Storage	−65 to +175	−65 to +175	°C
Lead Temperature (During Soldering):			
At distances 1/32 in from seating surface for 10 s max.	265	265	°C

Continuous Working Voltage Ratings must be observed to maintain device characteristics. These ratings are based on long-term continuous voltage operation but may be exceeded for short durations (e.g. testing of device characteristics), provided the *Absolute Maximum Ratings* are not exceeded.



Q = 40821
▼Disc. ceramic.
* Tubular ceramic.
All resistors in ohms
All capacitors in pF

C_1, C_2: 1.5–5 pF variable air capacitor: E.F. Johnson Type 160-102 or equivalent.

C_3: 1-10 pF piston-type variable air capacitor: JFD Type VAM-010, Johanson Type 4335, or equivalent.

C_4: 0.9–7 pF compression-type capacitor: ARCO 400 or equivalent.

L_1: 5 turns silver-plated 0.02" thick, 0.07"–0.08" wide copper ribbon. Internal diameter of winding = 0.25"; winding length approx. 0.65". Tapped at 1-1/2 turns from C_1 end of winding.

L_2: Ohmite Z-235 RF choke or equivalent

L_3: J. W. Miller Co. #4580 0.1 μH RF choke or equivalent.

Note: If 50Ω meter is used in place of sweep detector, a low pass filter must be provided to eliminate local oscillator voltage from load.

Fig.1 – Conversion power gain test circuit for type 40821.

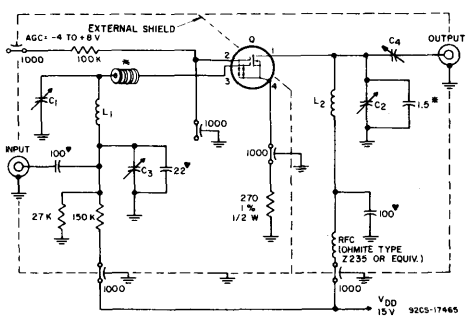
For characteristics curves, refer to type 3N187.

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS		LIMITS 40820 Min.	40820 Typ.	40820 Max.	40821 Min.	40821 Typ.	40821 Max.	UNITS
Gate No. 1 to Source Cutoff Voltage	$V_{G1S(off)}$	V_{DS}= +15V, I_D=200μA, V_{G2S}= +4V		-	-1	-3	-	-1	-3	V
Gate No. 2 to Source Cutoff Voltage	$V_{G2S(off)}$	V_{DS}= +15V, I_D=200μA, V_{G1S}=0		-	-1	-3	-	-1	-3	V
Gate to Source Forward Breakdown Voltage Gate No. 1	$V_{(BR)G1SSF}$	I_{G1SSF} I_{G2SSF} 100 μA	V_{G2S} V_{DS} 0	-	9	-	-	11	-	V
Gate No. 2	$V_{(BR)G2SSF}$		V_{G1S} V_{DS} 0	-	9	-	-	11	-	V
Gate to Source Reverse Breakdown Voltage Gate No. 1	$V_{(BR)G1SSR}$	I_{G1SSR} I_{G2SSR} 100 μA	V_{G2S} V_{DS} 0	-	9	-	-	11	-	V
Gate No. 2	$V_{(BR)G2SSR}$		V_{G1S} V_{DS} 0	-	9	-	-	11	-	V
Gate No. 1 Terminal Forward Current	I_{G1SSF}	V_{DS} V_{G2S} 0	V_{G1S} 6 V	-	-	50	-	-	-	nA
			V_{G1S} 4.5 V	-	-	-	-	-	50	nA
Gate No. 1 Terminal Reverse Current	I_{G1SSR}	V_{DS} V_{G2S} 0	V_{G1S} 6 V	-	-	50	-	-	-	nA
			V_{G1S} 4.5 V	-	-	-	-	-	50	nA
Gate No. 2 Terminal Forward Current	I_{G2SSF}	V_{DS} V_{G1S} 0	V_{G2S} 6 V	-	-	50	-	-	-	nA
			V_{G2S} 4.5 V	-	-	-	-	-	50	nA
Gate No. 2 Terminal Reverse Current	I_{G2SSR}	V_{DS} V_{G1S} 0	V_{G2S} -6 V	-	-	50	-	-	-	nA
			V_{G2S} 4.5 V	-	-	-	-	-	50	nA
Zero-Bias Drain Current	I_{DS}	V_{DS} +15 V, V_{G1S} 0, V_{G2S} +4 V		0.5	8	15	0.5	8	20	mA
Forward Transconductance (Gate No. 1-to-Drain)	g_{fs}	V_{DS} +15 V I_D 10 mA V_{G2S} +4 V	f 1 kHz	-	12000	-	-	12000	-	μmho
Small Signal, Short Circuit Input Capacitance♦	C_{iss}		f 1 MHz	-	6	8.5	-	6	9	pF
Small Signal, Short Circuit, Reverse Transfer Capacitance (Drain to Gate No. 1)♦	C_{rss}			0.005	0.02	0.03	0.005	0.02	0.04	pF
Small Signal, Short Circuit Output Capacitance	C_{oss}			-	2	-	-	2	-	pF
Power Gain (see Fig. 6)	G_{PS}		f 200 MHz	14	17	-	-	-	-	dB
Noise Figure (see Fig. 6)	NF			-	4.5	6	-	-	-	dB
Conversion Gain	$G_{PS(C)}$		f 200/44 MHz	-	-	-	11	-	-	dB

♦ Capacitance between Gate No. 1 and all other terminals

♠ Three terminal measurement with Gate No. 2 and Source returned to guard terminal.



=Ferrite bead (4); Pyroferric Co. "Carbonyl J" 0.09 in OD; 0.03 in ID. 0.063 in. thickness.

Q = 40820
▼ Disc ceramic.
* Tubular ceramic

All resistors in ohms

All capacitors in pF

C_1: 1.8 – 8.7 pF variable air capacitor: E. F. Johnson Type 160-104, or equivalent.

C_2: 1.5 – 5 pF variable air capacitor: E. F. Johnson Type 160-102, or equivalent.

C_3: 1 – 10 pF piston-type variable air capacitor: JFD Type VAM-010; Johanson Type 4335, or equivalent.

C_4: 0.8 – 4.5 pF piston type variable air capacitor: Erie 560-013 or equivalent.

L_1: 4 turns silver-plated 0.02-in. thick, 0.075-0.085 in. wide, copper ribbon. Internal diameter of winding = 0.25 in, winding length approx. 0.80 in.

L_2: 4-1/2 turns silver-plated 0.02 in. thick, 0.085-0.095 in wide, 5/16-in. ID Coil ≈ 0.90 in. long.

Fig.2 – 200 MHz power gain and noise figure test circuit for type 40820.

Table 1 – y parameters vs. frequency

CHARACTERISTICS	SYMBOL	FREQUENCY (MHz) 50	100	200	250	UNITS
Y Parameters						
Input Conductance	g_{is}	0.08	0.33	1.0	1.6	mmho
Input Susceptance	b_{is}	1.8	3.6	7.5	9.8	mmho
Magnitude Forward Transadmittance	$\lvert y_{fs}\rvert$	12	12	12	12.3	mmho
Angle of Forward Transadmittance	$\angle y_{fs}$	-2	-13	-35	-45	degrees
Output Conductance	g_{os}	0.10	0.18	0.36	0.42	mmho
Output Susceptance	b_{os}	0.5	1.0	2.0	2.6	mmho
Magnitude of Reverse Transadmittance	$\lvert y_{rs}\rvert$	8	12	25	40	μmho
Angle of Reverse Transadmittance	$\angle y_{rs}$	-88	-73	-25	-10	degrees

40822-40823

Silicon Dual-Insulated-Gate Field-Effect Transistors

N-Channel Depletion Types

With Integrated Gate-Protection Circuits

For FM Tuner Applications

40822 – RF Amplifier **40823** – Mixer

RCA-40822 and 40823 are n-channel silicon, depletion type, dual-insulated-gate, field-effect transistors for RF amplifier (40822) and mixer (40823) applications in FM receivers and other commercial equipment operating at frequencies up to 150 MHz.

These devices designed for VHF performance, provide excellent power gain, low-noise figures and have wide dynamic range. The dual-gate feature offers good cross-modulation performance over the AGC range and reduces feedback capacitance by shielding Gate No. 1 from the drain. The very low feedback capacitance also eliminates the need for circuit neutralization and reduces local oscillator feed-through to the antenna.

Virtually no power is required in AGC utilizing the 40822 and 40823. In addition, these devices minimize input impedance variations and automatically achieve AGC delay when AGC is applied to Gate No. 2. The dual-gate arrangement also makes it possible to isolate the local oscillator signal from the incoming signal by applying each signal to a specific gate.

Back-to-back diodes, diffused directly into the MOS pellet, protect the gates against damage in normal handling and usage by limiting transient voltages that exceed +10 volts. The 40822 and 40823 are hermetically sealed in metal JEDEC TO-72 packages.

Performance Features

- superior cross-modulation performance and greater dynamic range than bipolar or single-gate FET's
- wide dynamic range permits large-signal handling before overload
- virtually no agc power required
- greatly reduces spurious responses in FM receivers
- dual gate permits simplified AGC circuitry

Device Features

- back-to-back diodes protect each gate against handling and in-circuit transients
- high forward transconductance: g_{fs} = 12,000 μmho (typ.)
- high unneutralized RF power gain: G_{ps} = 24 dB (typ.) at 100 MHz (40822)
- low VHF noise figure: 2 dB (typ.) at 100 MHz (40822)
- low gate leakage currents: I_{G1SS} & I_{G2SS} = 50 nA at T_A = 25°C

Maximum Ratings

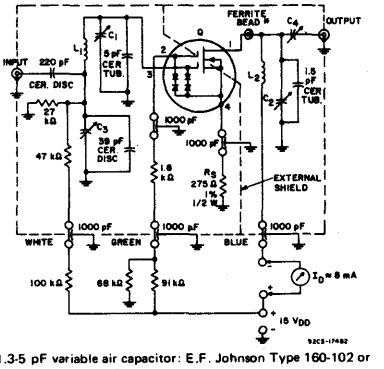
	40822	40823	
Continuous Working Voltage#, at T_A = 25°C:			
Gate No. 1-to-Source Voltage, V_{G1S}	−6 to +3	−4.5 to +3	V
Gate No. 2-to-Source Voltage, V_{G2S}	-6 to +6 or 40% of V_{DS} (whichever value is less)	-4.5 to +4.5 or 40% of V_{DS} (whichever value is less)	V
Drain-to-Gate Voltage, V_{DG1} or V_{DG2}	+20	+20	V
Absolute Maximum Values, at T_A = 25°C:			
Drain-to-Source Voltage, V_{DS}	−0.2 to +18	−0.2 to +18	V
Gate Terminal Current, I_{G1S} or I_{G2S}	±100	±100	μA
Drain-to-Gate Voltage, V_{DG1} or V_{DG2}	+24	+22.5	V
Drain Current, I_D	50	50	mA
Transistor Dissipation:			
At T_A up to 25°C	330	330	mW
At T_A above 25°C	derate linearly 2.2 mW/°C		
Ambient Temperature Range:			
Operating and Storage	−65 to +175	−65 to +175	°C
Lead Temperature (During Soldering):			
At distances 1/32 in from seating surface for 10 s max.	265	265	°C

Continuous Working Voltage Ratings must be observed to maintain device characteristics. These ratings are based on long-term continuous voltage operation but may be exceeded for short durations (e.g. testing of device characteristics), provided the Absolute Maximum Ratings are not exceeded.

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS		LIMITS 40822 Min.	40822 Typ.	40822 Max.	40823 Min.	40823 Typ.	40823 Max.	UNITS
Gate No. 1-to-Source Cutoff Voltage	$V_{G1S(off)}$	V_{DS}= +15V, I_D=200μA, V_{G2S}= +4V		–	−2	−4	–	−2	−4	V
Gate No. 2-to-Source Cutoff Voltage	$V_{G2S(off)}$	V_{DS}= +15V, I_D=200μA, V_{G1S}=0		–	−2	−4	–	−2	−4	V
Gate-to-Source Forward Breakdown Voltage: Gate No. 1	$V_{(BR)G1SSF}$	I_{G1SSF} = I_{G2SSF} = 100 μA	V_{G2S} = V_{DS} = 0	–	9	–	–	11	–	V
Gate No. 2	$V_{(BR)G2SSF}$		V_{G1S} = V_{DS} = 0	–	9	–	–	11	–	V
Gate-to-Source Reverse Breakdown Voltage: Gate No. 1	$V_{(BR)G1SSR}$	I_{G1SSR} = I_{G2SSR} = 100 μA	V_{G2S} = V_{DS} = 0	–	9	–	–	11	–	V
Gate No. 2	$V_{(BR)G2SSR}$		V_{G1S} = V_{DS} = 0	–	9	–	–	11	–	V
Gate No. 1-Terminal Forward Current	I_{G1SSF}	V_{DS} = V_{G2S} = 0	V_{G1S} = 6 V	–	–	50	–	–	–	nA
			V_{G1S} = 4.5 V	–	–	–	–	–	50	nA
Gate No. 1-Terminal Reverse Current	I_{G1SSR}	V_{DS} = V_{G2S} = 0	V_{G1S} = −6 V	–	–	50	–	–	–	nA
			V_{G1S} = −4.5 V	–	–	–	–	–	50	nA
Gate No. 2-Terminal Forward Current	I_{G2SSF}	V_{DS} = V_{G1S} = 0	V_{G2S} = 6 V	–	–	50	–	–	–	nA
			V_{G2S} = 4.5 V	–	–	–	–	–	50	nA
Gate No. 2-Terminal Reverse Current	I_{G2SSR}	V_{DS} = V_{G1S} = 0	V_{G2S} = −6 V	–	–	50	–	–	–	nA
			V_{G2S} = −4.5 V	–	–	–	–	–	50	nA
Zero-Bias Drain Current	I_{DS}	V_{DS} = +15 V, V_{G1S} = 0, V_{G2S} = +4 V		5	15	30	5	15	35	mA
Forward Transconductance (Gate No. 1-to-Drain)	g_{fs}	V_{DS} = +15 V, I_D = 10 mA, V_{G2S} = +4 V	f = 1 kHz	–	12000	–	–	12000	–	μmho
Small-Signal, Short-Circuit Input Capacitance†	C_{iss}		f = 1 MHz	–	6.5	9.5	–	6.5	10	pF
Small-Signal, Short-Circuit, Reverse Transfer Capacitance (Drain-to-Gate-No. 1)♦	C_{rss}			0.005	0.020	0.030	0.005	0.025	0.045	pF
Small-Signal, Short-Circuit Output Capacitance	C_{oss}			–	2	–	–	2	–	pF
Power Gain (see Fig. 5)	G_{PS}			19	24	–	–	–	–	dB
Noise Figure (see Fig. 5)	NF		f = 100 MHz	–	2	3.5	–	–	–	dB
Conversion Gain	$G_{PS(C)}$		f = 100 to 10.7 MHz	–	–	–	14	18	–	dB

† Capacitance between Gate No. 1 and all other terminals.

♦ Three-terminal measurement with Gate No. 2 and Source returned to guard terminal.



1.3-5 pF variable air capacitor: E.F. Johnson Type 160-102 or equivalent.

2.7-19.6 pF variable air capacitor: E.F. Johnson Type 160-110 or equivalent.

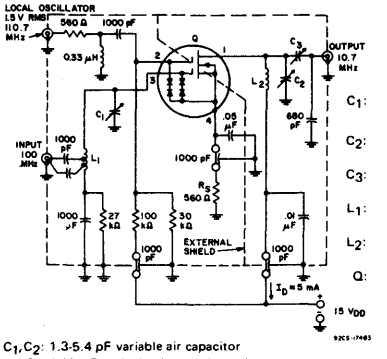
80 pF max. compression-type capacitor: Arco 405 or equivalent

8 turns No. 22 wire on 1/4" diameter air core. One turn spacing between windings. Tapped at one turn from low end.

37 turns No. 34 wire on 3/16" diameter air core. Unloaded Q = 63

40823.

Fig.1 - 100/10.7-MHz conversion power gain test circuit for type 40823.



C_1, C_2: 1.3-5.4 pF variable air capacitor

C_3: 1-10 pF variable air capacitor, piston type: Johanson Co., No. 4335

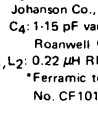
C_4: 1-15 pF variable air capacitor, precision piston type: Roanwell Corp. SG11129/AG

L_1, L_2: 0.22 μH RF choke (7T): Miller, No. 4584

*Ferramic toroid (1/2 used): Indiana General, No. CF101-(0-6)

Fig.2 - 100-MHz power gain and noise figure test circuit for type 40822.

TERMINAL DIAGRAM



LEAD 1 – DRAIN
LEAD 2 – GATE No.2
LEAD 3 – GATE No.1
LEAD 4 – SOURCE, SUBSTRATE AND CASE

For characteristics curves, refer to type 3N187.

Silicon Dual-Insulated Gate Field-Effect Transistor

N-Channel Depletion Type

With Integrated Gate-Protection Circuits

General-Purpose Economy Type for Applications from DC to 500 MHz

RCA-40841 is an n-channel silicon, depletion type, dual-insulated gate, field-effect transistor intended for general-purpose applications from DC to frequencies up to 500 MHz.

This MOS/FET provides excellent power gain, linear-circuit operation and has a wide dynamic operating range. Its square-law characteristics result in low cross-modulation performance over the AGC range. Its dual-gate construction reduces feedback capacitance by shielding Gate No. 1 from the drain, and makes it possible to isolate the local oscillator signal from the incoming signal by applying the two signals to separate gates. The very low feedback capacitance of this device eliminates the need for neutralization in circuits using the dual-gate configuration. Use of the device in the RF input stage of a receiver reduces local oscillator feed-through to the antenna. The 40841 requires negligible AGC power, provides automatic delay when AGC is applied to Gate No. 2, and exhibits slight input impedance variations during AGC functioning. The device has exceptionally high input impedance, an attribute for timing-circuit design.

Back-to-back diodes are fabricated on the same monolithic silicon pellet as the MOS/FET to protect the gates against damage due to electrostatic charges frequently encountered during normal handling. These back-to-back diodes also function as "transient trappers" by limiting in-circuit transient voltages that exceed ±10 volts.

Maximum ratings and electrical characteristics are included in the data for operation of the 40841 as the equivalent of a single-gate device. For single-gate operation, connect Gate No. 1 (Term. 2) to Gate No. 2 (Term. 3), as shown in the Terminal Diagrams on Page 2. The 40841 MOS/FET is hermetically sealed in the metal JEDEC TO-72 package.

Device Features:

- back-to-back diodes protect gate insulation against damage due to static charges frequently encountered during handling
- high forward transconductance: g_{fs} = 12,000 μmho (typ.)
- high power gain: G_{ps} = 32 dB (typ.) at 44 MHz
- gate leakage currents: I_{G1SS} and I_{G2SS} = 60 nA (max.) at T_A = 25°C
- high input impedance
- excellent thermal stability

Performance Features:

- superior cross-modulation performance and greater dynamic range than bipolar and junction-gate FETs
- wide dynamic range permits large-signal handling before overloading
- virtually no agc power required
- greatly reduced spurious responses in AM and FM receivers
- dual-gate configuration permits simplified AGC circuitry
- operates at frequencies to 500 MHz without neutralization in circuits utilizing the dual-gate configuration
- operates up to UHF with low-noise performance

The following dual-gate MOS/FET types are specified for applications requiring premimum-grade performance: 3N200, 3N187, 40673, 40819, 40820, 40821, 40822, and 40823.

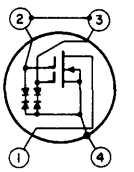
Detailed information, utilizing RCA dual-gate protected MOS/FETs in RF applications, is given in the following RCA Application Notes: AN-4431 "RF Applications of the Dual-Gate MOS/FET up to 500 MHz" and AN-4018 "Design of Gate-Protected MOS Field-Effect Transistors".

Applications:

- DC amplifiers
- RF amplifiers
- mixers
- IF amplifiers
- video amplifiers
- differential amplifiers
- frequency multipliers
- phase splitters
- industrial timers – long time delays
- thyristor trigger circuits
- choppers
- voltage-controlled attenuators
- constant-current source
- voltage regulators
- telemetry & multiplex
- servo amplifiers
- proximity switches

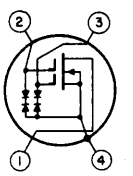
TERMINAL DIAGRAMS

SINGLE-GATE CONFIGURATION



LEAD 1–DRAIN
LEADS–2 AND 3–GATE
LEAD 4–SOURCE, SUBSTRATE AND CASE

DUAL-GATE CONFIGURATION



LEAD 1–DRAIN
LEAD 2–GATE No.2
LEAD 3–GATE No.1
LEAD 4–SOURCE SUBSTRATE AND CASE

Maximum Ratings

	Dual-Gate Configuration	Single-Gate Configuration	
Absolute Maximum Values, at T_A = 25°C:			
Drain-to-Source Voltage, V_{DS}	−0.2 to +18	−0.2 to +18	V
Gate Terminal Current, I_{G1S} or I_{G2S}	±100	−	μA
Gate Terminal Current, I_{GS}	−	±100	μA
Drain-to-Gate Voltage, V_{DG1} or V_{DG2}	+24	−	V
Drain-to-Gate Voltage, V_{DG}	−	+24	V
Drain Current, I_D	50	50	mA
Transistor Dissipation:			
At T_A up to 25°C	330	330	mW
At T_A above 25°C	derate linearly 2.2 mW/°C		
Ambient Temperature Range:			
Operating and Storage	−65 to +175	−65 to +175	°C
Lead Temperature (During Soldering):			
At distances 1/32 in from seating surface for 10 s max.	265	265	°C
Continuous Working Voltage#, at T_A = 25°C:			
Gate No. 1-to-Source Voltage, V_{G1S}	−4.5 to +3	−	V
Gate No. 2-to-Source Voltage, V_{G2S}	-4.5 to +4.5 or 40% of V_{DS} (whichever value is less)	−	V
Gate-to-Source Voltage, V_{GS}	−	−4.5 to +3	V
Drain-to-Gate Voltage, V_{DG1} or V_{DG2}	+20	−	V
Drain-to-Gate Voltage, V_{DG}	−	+20	V

#Continuous Working Voltage Ratings must be observed to maintain device characteristics. These ratings are based on long-term continuous voltage operation but may be exceeded for short durations (e.g. testing of device characteristics), provided the Absolute Maximum Ratings are not exceeded.

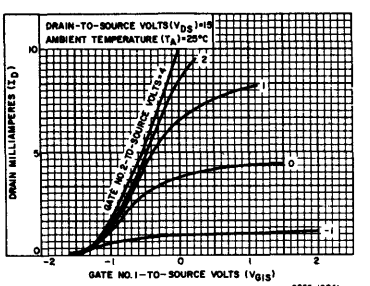


Fig.1–I_D vs. V_{G1S}.

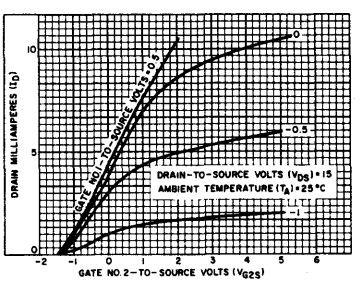


Fig.2–I_D vs. V_{G2S}.

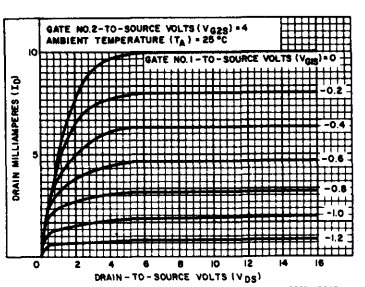


Fig.3–I_D vs. V_{DS}.

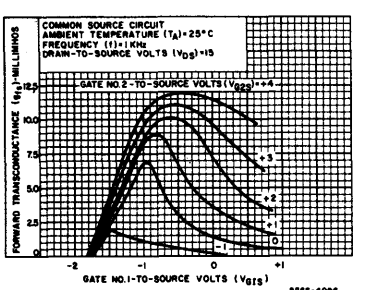


Fig.4–g_{fs} vs. V_{G1S}.

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS		LIMITS CONFIGURATION DUAL-GATE MIN.	DUAL-GATE TYP.	DUAL-GATE MAX.	SINGLE-GATE MIN.	SINGLE-GATE TYP.	SINGLE-GATE MAX.	UNITS
Gate-to-Source Cutoff Voltage:										
Dual-Gate (No. 1)	$V_{G1S(off)}$	V_{DS} = +15V, I_D = 200μA, V_{G2S} = +4V		–	–2	–	–	–	–	V
Dual-Gate (No. 2)	$V_{G2S(off)}$	V_{DS} = +15V, I_D = 200μA, V_{G1S} = 0		–	–2	–	–	–	–	V
Single-Gate	$V_{GS(off)}$	V_{DS} = +15V, I_D = 200μA		–	–	–	–	–1.6	–	V
Gate-to-Source Forward Breakdown Voltage:		I_{G1SSF} =								
Dual-Gate (No. 1)	$V_{(BR)G1SSF}$	I_{G2SSF} =	V_{G2S} = V_{DS} = 0	–	9	–	–	–	–	V
Dual-Gate (No. 2)	$V_{(BR)G2SSF}$	100μA	V_{G1S} = V_{DS} = 0	–	9	–	–	–	–	V
Single-Gate	$V_{(BR)GSSF}$	I_{GSSF} = 100μA, V_{DS} = 0		–	–	–	–	9	–	V
Gate-to-Source Reverse Breakdown Voltage:		I_{G1SSR} =								
Dual-Gate (No. 1)	$V_{(BR)G1SSR}$	I_{G2SSR} =	V_{G2S} = V_{DS} = 0	–	9	–	–	–	–	V
Dual-Gate (No. 2)	$V_{(BR)G2SSR}$	100μA	V_{G1S} = V_{DS} = 0	–	9	–	–	–	–	V
Single-Gate	$V_{(BR)GSSR}$	I_{GSSR} = 100μA, V_{DS} = 0		–	–	–	–	9	–	V
Gate Terminal Forward Current:										
Dual-Gate (No. 1)	I_{G1SSF}	V_{DS} = V_{G2S} = 0, V_{G1S} = 6 V		–	–	60	–	–	–	nA
Dual Gate (No. 2)	I_{G2SSF}	V_{DS} = V_{G1S} = 0, V_{G2S} = 6 V		–	–	60	–	–	–	nA
Single-Gate	I_{GSSF}	V_{DS} = 0, V_{GS} = 6 V		–	–	–	–	–	120	nA
Gate Terminal Reverse Current:										
Dual-Gate (No. 1)	I_{G1SSR}	V_{DS} = V_{G2S} = 0, V_{G1S} = –6 V		–	–	60	–	–	–	nA
Dual-Gate (No. 2)	I_{G2SSR}	V_{DS} = V_{G1S} = 0, V_{G2S} = –6 V		–	–	60	–	–	–	nA
Single-Gate	I_{GSSR}	V_{DS} = 0, V_{GS} = –6 V		–	–	–	–	–	120	nA
Zero-Bias Drain Current:										
Dual-Gate	I_{DS}	V_{DS} = +15 V, V_{G1S} = 0, V_{G2S} = +4 V		–	10	–	–	–	–	mA
Single-Gate	I_{DSS}	V_{DS} = +15 V, V_{GS} = 0		–	–	–	–	3.7	–	mA
Forward Transconductance (Gate-to-Drain)										
Dual-Gate	g_{fs}		1 kHz	–	12000	–	–	–	–	μmho
Single-Gate	g_{fs}			–	–	–	–	7000	–	μmho
Small-Signal, Short-Circuit Input Capacitance†	C_{iss}			–	6.5	–	–	11	–	pF
Small-Signal, Short-Circuit, Reverse Transfer Capacitance (Drain-to-Gate-No. 1)♣	C_{rss}	V_{DS} = +15 V, I_D = 10 mA, [Dual-Gate only] V_{G2S} = +4 V	f = 1 MHz	–	0.02	–	–	0.54	–	pF
Small-Signal, Short-Circuit Output Capacitance	C_{oss}			–	2	–	–	2	–	pF
Audio Spot Noise Figure*										
Dual-Gate	NF		f = 1 kHz	–	0.46	–	–	–	–	dB
Single-Gate	NF		f = 1 kHz	–	–	–	–	0.29	–	dB
Power Gain	G_{ps}		44 MHz	–	32	–	–	–	–	dB
Conversion Gain	$G_{ps(C)}$			–	24	–	–	–	–	dB

† Capacitance between Gate No. 1 and all other terminals (Dual-Gate), Gate and all other terminals (Single-Gate)

♣ Three-terminal measurement with Gate No. 2 and Source returned to guard terminal (Dual-Gate)

* Noise Figure = $10 \log_{10}\left[1 + \frac{e_n^2}{4\,KT\,BW\,R_g}\right]$ where K = 1.38×10^{-23}; T = Temperature in °Kelvin; BW = Bandwidth in Hz; R_g = Generator resistance

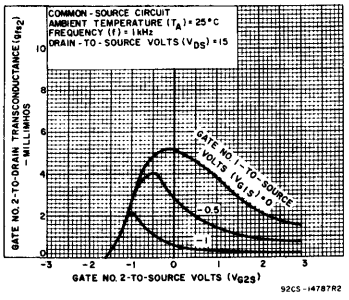


Fig.5—g_{fs2} vs. V_{G2S}.

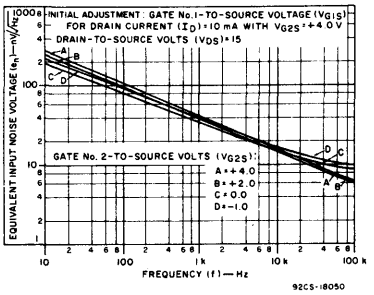


Fig.6—e_n vs. f.

TYPICAL CHARACTERISTICS FOR 40841 IN SINGLE-GATE CONFIGURATION
(Terminals 2 and 3 tied together to comprise effective single-gate)

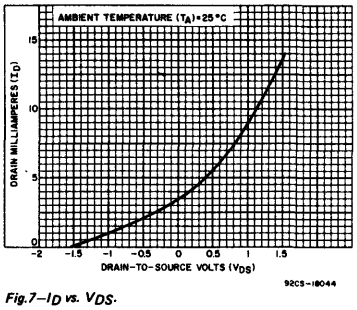


Fig.7—I_D vs. V_{DS}.

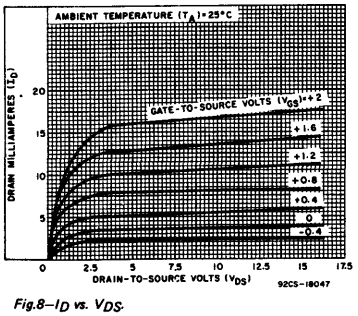


Fig.8—I_D vs. V_{DS}.

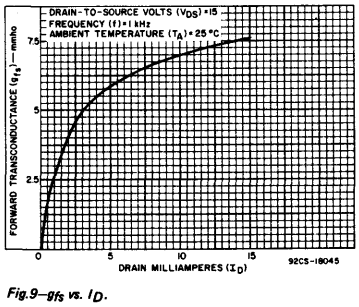


Fig.9—g_{fs} vs. I_D.

TYPICAL CHARACTERISTICS FOR 40841 IN SINGLE-GATE & DUAL-GATE CONFIGURATION

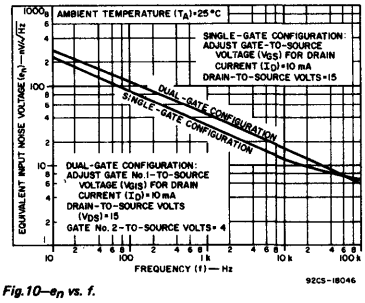


Fig.10—e_n vs. f.

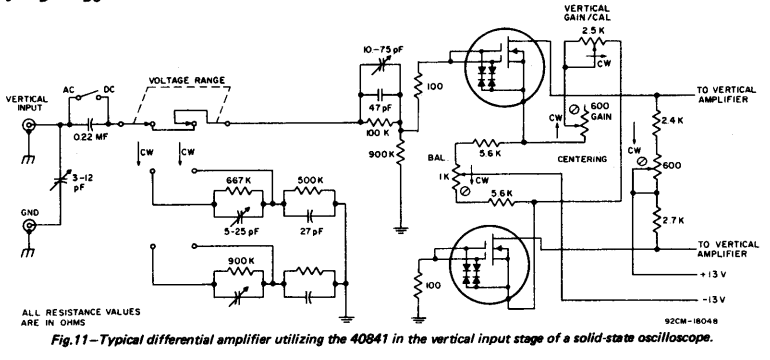
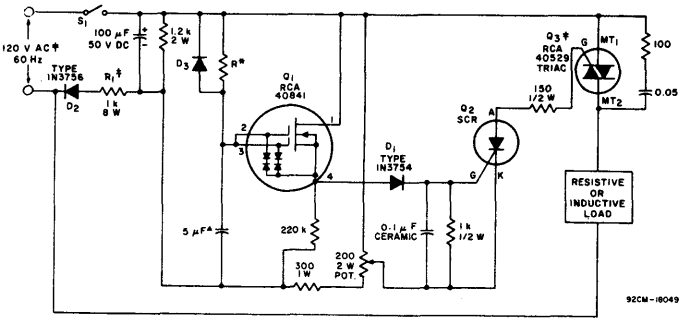


Fig.11—Typical differential amplifier utilizing the 40841 in the vertical input stage of a solid-state oscilloscope.

SOLID-STATE TIMER FOR INDUSTRIAL APPLICATIONS



▲ Cornell-Dubilier Electronics—Type MMW or equivalent.

* R controls duration of time delay. At R = 60 MΩ up to 5-minute delay (IRC resistor, Type CGH or equivalent)

‡ This circuit can also be used at supply voltages of 240 V AC and 24V AC (60Hz) by changing the values of R1 and Q3.

TIMING CIRCUIT CHARACTERISTICS

T_A = −25°C to +60°C

Accuracy: ±10% (over temperature)

Repeatability: ±3% (at 25°C)

Reset Time: Less than 150 ms

Q2: V_{DRM} = 60V, I_{GT} = 200μA, I_T = 0.8A

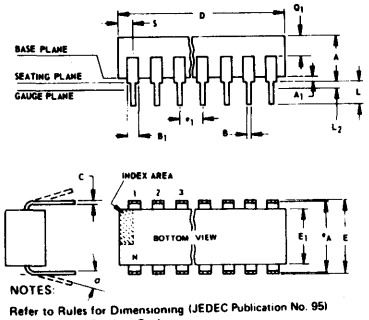
D3: I_R = 1 nA, V_R = 60V

Fig.12—Typical timing circuit utilizing the 40841 in a single-gate configuration.

Dimensional Outlines

Dimensional Outlines

CERAMIC DUAL-IN-LINE PACKAGES



Refer to Rules for Dimensioning (JEDEC Publication No. 95) for Axial Lead Product Outlines.

1. When this device is supplied solder-dipped, the maximum lead thickness (narrow portion) will not exceed 0.013".
2. Leads within 0.005" (0.12 mm) radius of True Position (TP) at guage plane with maximum material condition and unit installed.
3. e_A applies in zone L_2 when unit installed.
4. α applies to spread leads prior to installation.
5. N is the maximum quantity of lead positions.
6. N_1 is the quantity of allowable missing leads.

(D) Suffix (JEDEC MO-001-AD) 14-Lead

SYMBOL	INCHES		NOTE	MILLIMETERS	
	MIN.	MAX.		MIN.	MAX.
A	0.120	0.160		3.05	4.06
A_1	0.020	0.065		0.51	1.65
B	0.014	0.020		0.356	0.508
B_1	0.050	0.065		1.27	1.65
C	0.008	0.012	1	0.204	0.304
D	0.745	0.770		18.93	19.55
E	0.300	0.325		7.62	8.25
E_1	0.240	0.260		6.10	6.60
e_1	0.100 TP		2	2.54 TP	
e_A	0.300 TP		2, 3	7.62 TP	
L	0.125	0.150		3.18	3.81
L_2	0.000	0.030		0.000	0.76
α	0°	15°	4	0°	15°
N	14		5	14	
N_1	0		6	0	
Q_1	0.050	0.085		1.27	2.15
S	0.065	0.090		1.66	2.28

92SS-4411R2

(D) Suffix (JEDEC MO-001-AE) 16-Lead

SYMBOL	INCHES		NOTE	MILLIMETERS	
	MIN.	MAX.		MIN.	MAX.
A	0.120	0.160		3.05	4.06
A_1	0.020	0.065		0.51	1.65
B	0.014	0.020		0.356	0.508
B_1	0.035	0.065		0.89	1.65
C	0.008	0.012	1	0.204	0.304
D	0.745	0.785		18.93	19.93
E	0.300	0.325		7.62	8.25
E_1	0.240	0.260		6.10	6.60
e_1	0.100 TP		2	2.54 TP	
e_A	0.300 TP		2, 3	7.62 TP	
L	0.125	0.150		3.18	3.81
L_2	0.000	0.030		0.000	0.76
α	0°	15°	4	0°	15°
N	16		5	16	
N_1	0		6	0	
Q_1	0.050	0.085		1.27	2.15
S	0.015	0.060		0.39	1.52

92SS-4286R5

DUAL-IN-LINE PLASTIC AND FRIT-SEAL CERAMIC PACKAGES

(E) and (G) Suffixes (JEDEC MO-001-AN) 8-Lead Plastic (Mini-DIP)

SYMBOL	INCHES		NOTE	MILLIMETERS	
	MIN.	MAX.		MIN.	MAX.
A	0.155	0.200		3.94	5.08
A_1	0.020	0.050		0.508	1.27
B	0.014	0.020		0.356	0.508
B_1	0.035	0.065		0.889	1.65
C	0.008	0.012	1	0.203	0.304
D	0.370	0.400		9.40	10.16
E	0.300	0.325		7.62	8.25
E_1	0.240	0.260		6.10	6.60
e_1	0.100 TP		2	2.54 TP	
e_A	0.300 TP		2, 3	7.62 TP	
L	0.125	0.150		3.18	3.81
L_2	0.000	0.030		0.000	0.762
α	0	15	4	0	15°
N	8		5	8	
N_1	0		6	0	
Q_1	0.040	0.075		1.02	1.90
S	0.015	0.060		0.381	1.52

92CS-24026R1

(E, (F) and (G) Suffixes (JEDEC MO-001-AB) 14-Lead

SYMBOL	INCHES		NOTE	MILLIMETERS	
	MIN.	MAX.		MIN.	MAX.
A	0.155	0.200		3.94	5.08
A_1	0.020	0.050		0.51	1.27
B	0.014	0.020		0.356	0.508
B_1	0.050	0.065		1.27	1.65
C	0.008	0.012	1	0.204	0.304
D	0.745	0.770		18.93	19.55
E	0.300	0.325		7.62	8.25
E_1	0.240	0.260		6.10	6.60
e_1	0.100 TP		2	2.54 TP	
e_A	0.300 TP		2, 3	7.62 TP	
L	0.125	0.150		3.18	3.81
L_2	0.000	0.030		0.000	0.76
α	0°	15°	4	0°	15°
N	14		5	14	
N_1	0		6	0	
Q_1	0.040	0.075		1.02	1.90
S	0.065	0.090		1.66	2.28

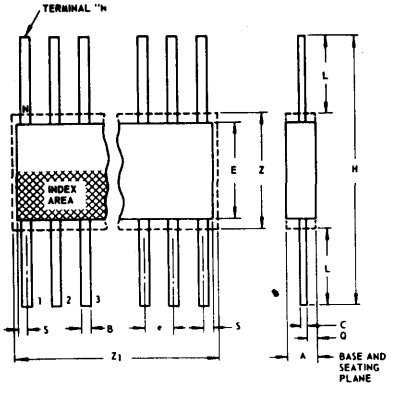
92SS-4296R3

(E), (F), and (G) Suffixes (JEDEC MO-001-AC) 16-Lead

SYMBOL	INCHES		NOTE	MILLIMETERS	
	MIN.	MAX.		MIN.	MAX.
A	0.155	0.200		3.94	5.08
A_1	0.020	0.050		0.51	1.27
B	0.014	0.020		0.356	0.508
B_1	0.035	0.065		0.89	1.65
C	0.008	0.012	1	0.204	0.304
D	0.745	0.785		18.93	19.93
E	0.300	0.325		7.62	8.25
E_1	0.240	0.260		6.10	6.60
e_1	0.100 TP		2	2.54 TP	
e_A	0.300 TP		2, 3	7.62 TP	
L	0.125	0.150		3.18	3.81
L_2	0.000	0.030		0.000	0.76
α	0°	15°	4	0°	15°
N	16		5	16	
N_1	0		6	0	
Q_1	0.040	0.075		1.02	1.90
S	0.015	0.060		0.39	1.52

92CM-15967R4

CERAMIC FLAT PACKS



NOTES:
1. Refer to Rules for Dimensioning (JEDEC Publication No. 95) for Axial Lead Product Outlines.
2. Leads within .005" (.12 mm) radius of True Position (TP) at maximum material condition.
3. N is the *maximum* quantity of lead positions.
4. Z and Z_1 determine a zone within which all body and lead irregularities lie.

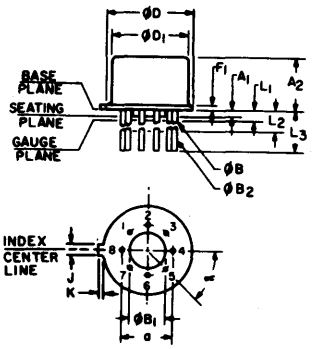
(K) Suffix (JEDEC MO-004-AF) 14-Lead

SYMBOL	INCHES		NOTE	MILLIMETERS	
	MIN.	MAX.		MIN.	MAX.
A	0.008	0.100		0.21	2.54
B	0.015	0.019	1	0.381	0.482
C	0.003	0.006	1	0.077	0.152
e	0.050 TP		2	1.27 TP	
E	0.200	0.300		5.1	7.6
H	0.600	1.000		15.3	25.4
L	0.150	0.350		3.9	8.8
N	14		3	14	
Q	0.005	0.050		0.13	1.27
S	0.000	0.050		0.00	1.27
Z	0.300		4	7.62	
Z_1	0.400		4	10.16	

92SS-4300R3

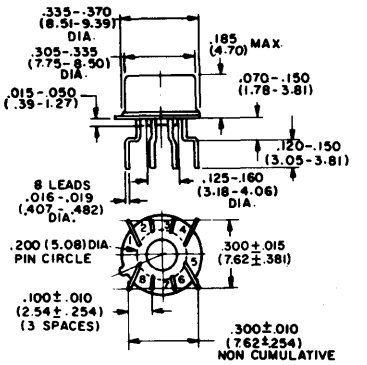
TO–5 STYLE PACKAGES

(T) Suffix (JEDEC MO-002-AL) 8-Lead TO-5 Style



SYMBOL	INCHES		NOTE	MILLIMETERS	
	MIN.	MAX.		MIN.	MAX.
a	0.200 TP		2	5.88 TP	
A_1	0.010	0.050		0.26	1.27
A_2	0.165	0.185		4.20	4.69
øB	0.016	0.019	3	0.407	0.482
$øB_1$	0.125	0.160		3.18	4.06
$øB_2$	0.016	0.021	3	0.407	0.533
øD	0.335	0.370		8.51	9.39
$øD_1$	0.305	0.335		7.75	8.50
F_1	0.020	0.040		0.51	1.01
I	0.028	0.034		0.712	0.863
k	0.029	0.045	4	0.74	1.14
L_1	0.000	0.050	3	0.00	1.27
L_2	0.250	0.500	3	6.4	12.7
L_3	0.500	0.562	3	12.7	14.27
α	45° TP			45° TP	
N	8		6	8	
N_1	3		5	3	

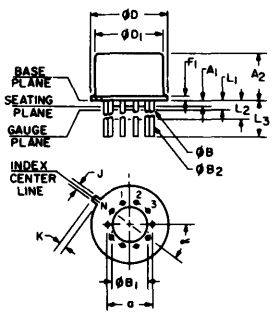
(S) Suffix 8-Lead TO-5 Style with Dual-In-Line Formed Leads (DIL-CAN)



NOTES

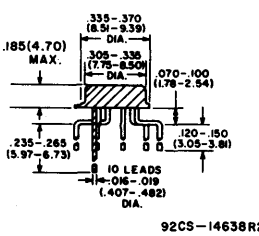
1. Refer to JEDEC Publication No 95 for Rules for Dimensioning Axial Lead Product Outlines.
2. Leads at gauge plane within 0.007" (0.178 mm) radium of True Position (TP) at maximum material condition.
3. øB applies between L_1 and L_2. $øB_2$ applies between L_2 and 0.500" (12.70 mm) from seating plane. Diameter is uncontrolled in L_1 and beyond 0.500" (12.70 mm).
4. Measure from Max. øD.
5. N_1 is the quantity of allowable missing leads.
6. N is the maximum quantity of lead positions.

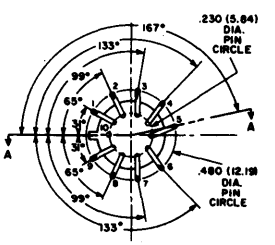
(T) Suffix (JEDEC MO-006-AF) 10-Lead TO-5 Style



SYMBOL	INCHES		NOTE	MILLIMETERS	
	MIN.	MAX.		MIN.	MAX.
a	0.230 TP		2	5.84 TP	
A_1	0	0		0	0
A_2	0.165	0.185		4.19	4.70
øB	0.016	0.019	3	0.407	0.482
$øB_1$	0	0		0	0
$øB_2$	0.016	0.021	3	0.407	0.533
øD	0.335	0.370		8.51	9.39
$øD_1$	0.305	0.335		7.75	8.50
F_1	0.020	0.040		0.51	1.01
j	0.028	0.034		0.712	0.863
k	0.029	0.045	4	0.74	1.14
L_1	0.000	0.050	3	0.00	1.27
L_2	0.250	0.500	3	6.4	12.7
L_3	0.500	0.562	3	12.7	14.27
α	36° TP			36° TP	
N	10		6	10	
N_1	1		5	1	

(V) Suffix 10 Formed Leads Radially Arranged TO-5 Type





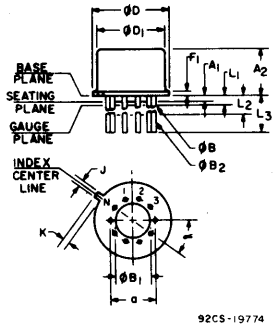
NOTES:

1. Refer to Rules for Dimensioning Axial Lead Product Outlines.
2. Leads at gauge plane within 0.007" (0.178 mm) radius of True Position (TP) at maximum material condition.
3. øB applies between L_1 and L_2. $øB_2$ applies between L_2 and 0.500" (12.70 mm) from seating plane. Diameter is uncontrolled in L_1 and beyond 0.500" (12.70 mm).
4. Measure from Max. øD.
5. N_1 is the quantity of allowable missing leads.
6. N is the maximum quantity of lead positions.

Dimensional Outlines

TO–5 STYLE PACKAGE (Cont'd)

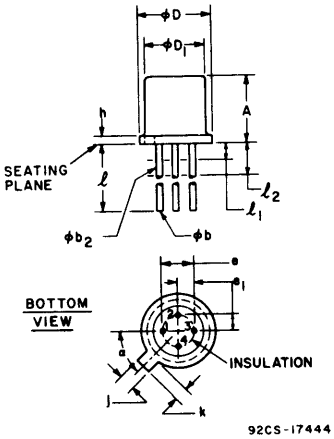
(T) Suffix(JEDEC MO-006-AG)12-Lead TO-5 Style



SYMBOL	INCHES		NOTE	MILLIMETERS	
	MIN.	MAX.		MIN.	MAX.
a	0.230		2	5.84 TP	
A_1	0	0		0	0
A_2	0.165	0.185		4.19	4.70
ϕB	0.016	0.019	3	0.407	0.482
ϕB_1	0	0		0	0
ϕB_2	0.016	0.021	3	0.407	0.533
ϕD	0.335	0.370		8.51	9.39
ϕD_1	0.305	0.335		7.75	8.50
F_1	0.020	0.040		0.51	1.01
j	0.028	0.034		0.712	0.863
k	0.029	0.045	4	0.74	1.14
L_1	0.000	0.050	3	0.00	1.27
L_2	0.250	0.500	3	6.4	12.7
L_3	0.500	0.562	3	12.7	14.27
α	30° TP			30° TP	
N	12		6	12	
N_1	1		5	1	

NOTES:
1. Refer to Rules for Dimensioning Axial Lead Product Outlines.
2. Leads at gauge plane within 0.007" (0.178 mm) radius of True Position (TP) at maximum material condition.
3. ϕB applies between L_1 and L_2. ϕB_2 applies between L_2 and 0.500" (12.70 mm) from seating plane. Diameter is uncontrolled in L_1 and beyond 0.500" (12.70 mm).
4. Measure from Max. ϕD.
5. N_1 is the quantity of allowable missing leads.
6. N is the maximum quantity of lead positions.

JEDEC TO-72 PACKAGE



SYMBOL	INCHES		MILLIMETERS		NOTES
	MIN	MAX	MIN	MAX	
A	0.170	0.210	4.32	5.33	
ϕb	0.016	0.021	0.406	0.533	2
ϕb_2	0.016	0.019	0.406	0.483	2
ϕD	0.209	0.230	5.31	5.84	
ϕD_1	0.178	0.195	4.52	4.95	
e	0.100 T.P.		2.54 T.P.		4
e_1	0.050 T.P.		1.27 T.P.		4
h		0.030		0.762	
l	0.036	0.046	0.914	1.17	
k	0.028	0.048	0.711	1.22	3
l	0.500		12.70		2
l_1		0.050		1.27	2
l_2	0.250		6.35		2
α	45° T.P.		45° T.P.		4, 6

Note 1 (Four leads) Maximum number leads omitted in this outline, "none" (0). The number and position of leads actually present are indicated in the product registration. Outline designation determined by the location and minimum angular or linear spacing of any two adjacent leads.

Note 2 (All leads) ϕb_2 applies between l_1 and l_2. ϕb applies between l_2 and .500" (12.70 mm) from seating plane. Diameter is uncontrolled in l_1 and beyond .500" (12.70 mm) from seating plane.

Note 3 Measured from maximum diameter of the product.

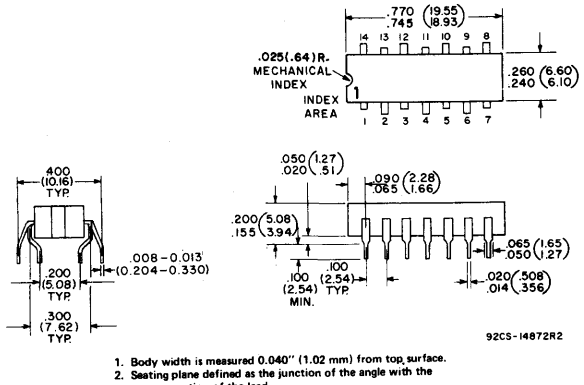
Note 4 Leads having maximum diameter .019" (.483 mm) measured in gaging plane .054" (1.37 mm) + .001" (.025 mm) - .000" (.000 mm) below the seating plane of the product shall be within .007" (.178 mm) of their true position relative to a maximum width tab.

Note 5 The product may be measured by direct methods or by gage.

Note 6 Tab centerline.

QUAD-IN-LINE PLASTIC PACKAGES

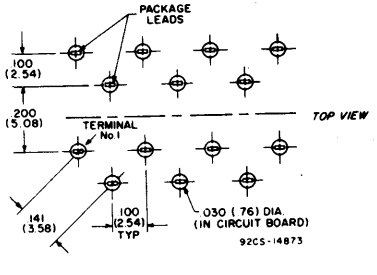
(Q) Suffix 14-Lead Staggered



1. Body width is measured 0.040" (1.02 mm) from top surface.
2. Seating plane defined as the junction of the angle with the narrow portion of the lead.

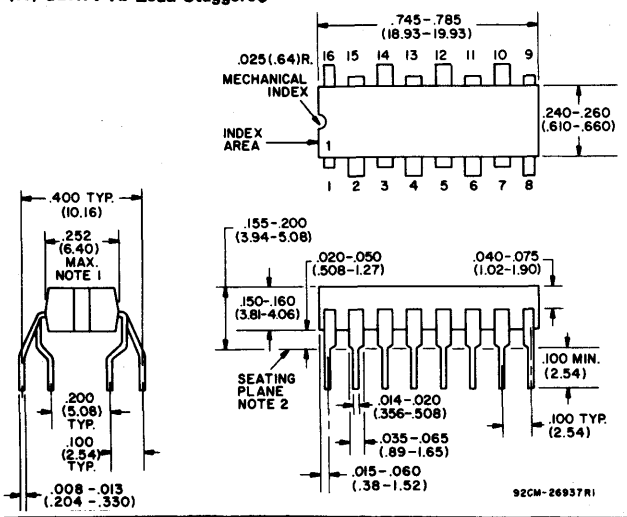
Dimensions in parentheses are millimeter equivalents of the basic inch dimensions.

Recommended Mounting – Hole Dimensions and Spacing

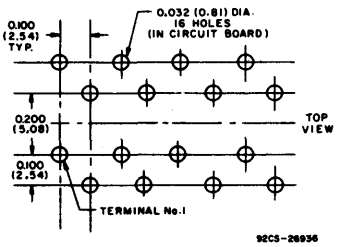


(W) Suffix 16-Lead Staggered

QUAD IN-LINE PACKAGES (Cont'd)



Recommended Mounting – Hole Dimensions and Spacing

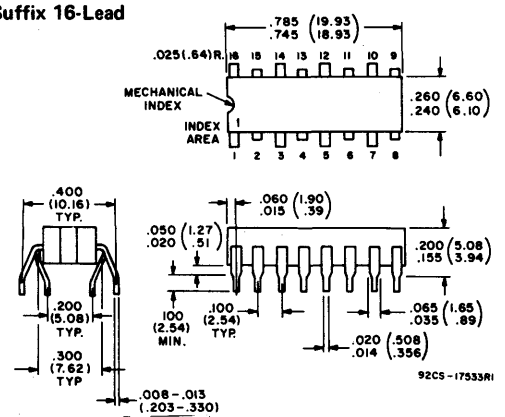


NOTES:

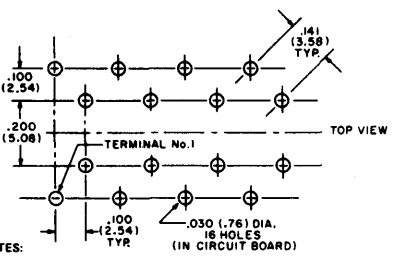
1. Body width is measured 0.040" (1.02 mm) from top surface.
2. Seating plane defined as the junction of the angle with the narrow portion of the lead.

Dimensions in parentheses are millimeter equivalents of the basic inch dimensions.

(Q) Suffix 16-Lead



Recommended Mounting – Hole Dimensions and Spacing

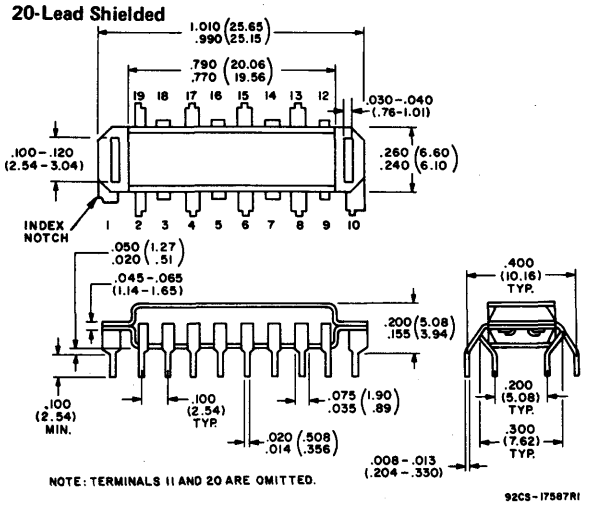


NOTES:

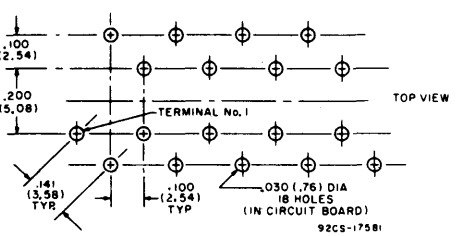
1. Body width is measured 0.040" (1.02 mm) from top surface.
2. Seating plane defined as the junction of the angle with the narrow portion of the lead.

Dimensions in parentheses are millimeter equivalents of the basic inch dimensions.

20-Lead Shielded



Recommended Mounting – Hole Dimensions and Spacing



NOTES:

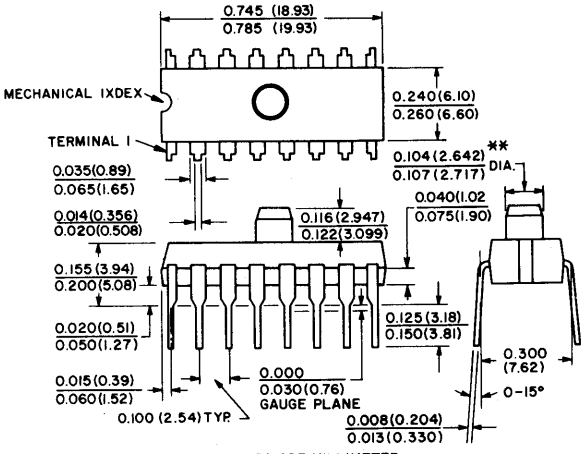
1. Body width is measured 0.040" (1.02 mm) from top surface.
2. Seating plane defined as the junction of the angle with the narrow portion of the lead.

Dimensions in parentheses are millimeter equivalents of the basic inch dimensions.

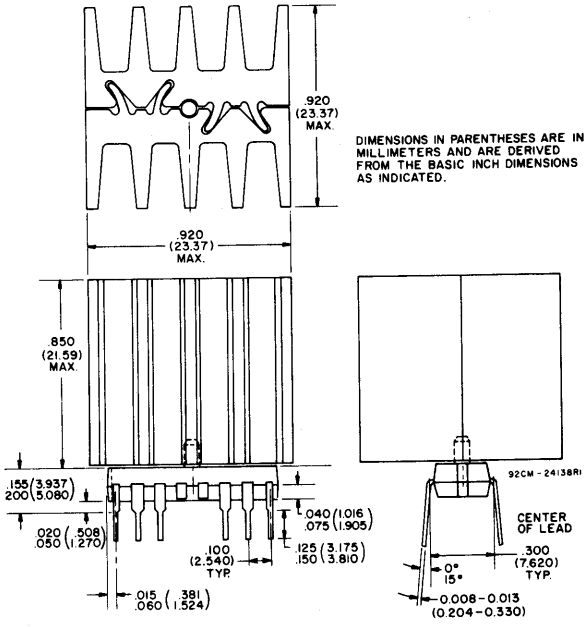
Dimensional Outlines

DUAL-IN-LINE AND QUAD-IN-LINE PLASTIC PACKAGES
(Power Stud and Heat-Sink Types)

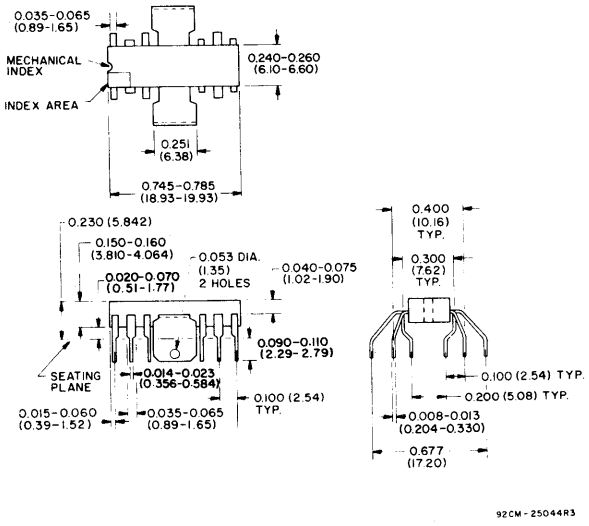
(E) Suffix
16-Lead "Power-Stud" Package



(EM) Suffix
Modified 16-Lead with Integral Heat Sink

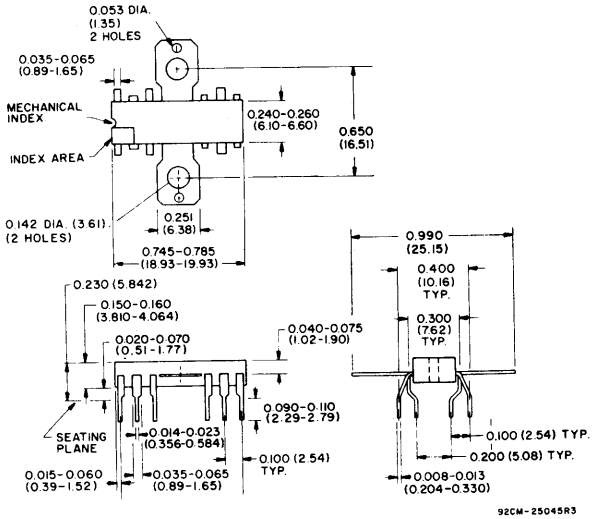


(Q) Suffix
Modified 16-Lead with Integral Bent Down Wing-Tab Heat Sink



DIMENSIONS IN PARENTHESES ARE MILLIMETER EQUIVALENTS OF THE BASIC INCH DIMENSIONS

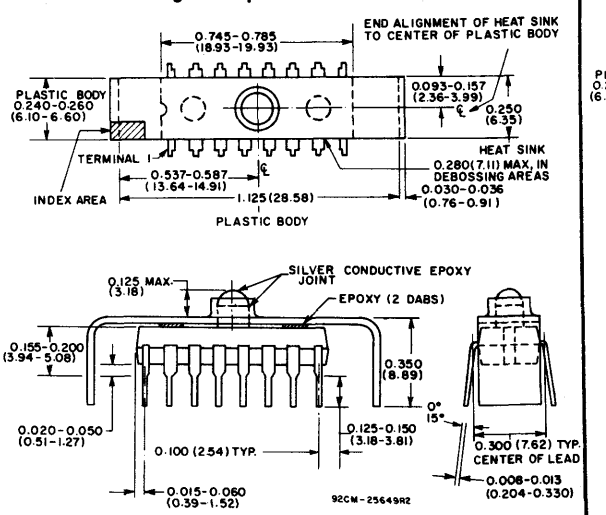
(QM) Suffix
Modified 16-Lead with Integral Flat Wing-Tab Heat Sink



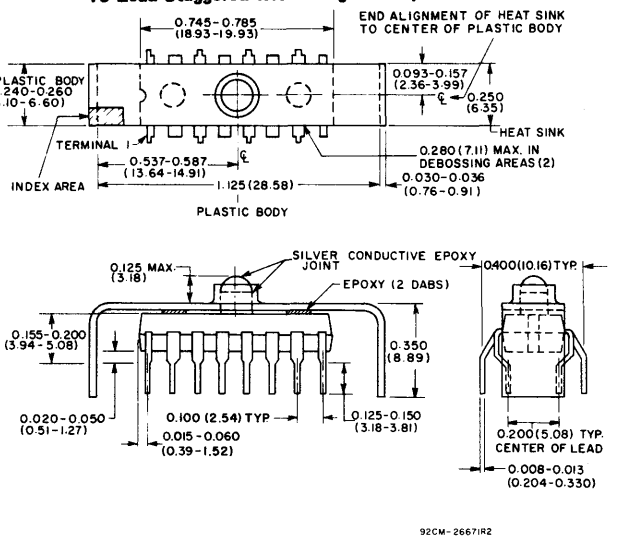
DIMENSIONS IN PARENTHESES ARE MILLIMETER EQUIVALENTS OF THE BASIC INCH DIMENSIONS

DUAL-IN-LINE AND QUAD-IN-LINE PLASTIC PACKAGES
(Power Stud and Heat-Sink Types)

(EM) Suffix
16-Lead with Integral Strap Heat Sink

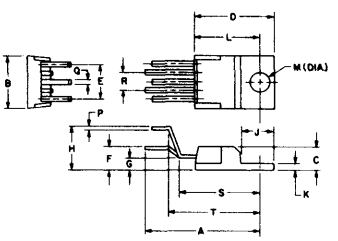


(QM) Suffix
16-Lead Staggered with Integral Strap Heat Sink



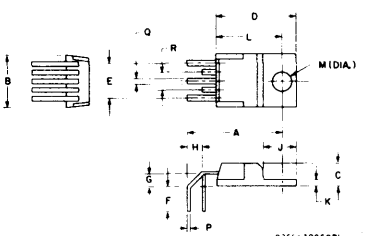
TO-220-STYLE (VERSA-V) PLASTIC PACKAGE

VERTICAL MOUNT



SYMBOL	INCHES		MILLIMETERS	
	MIN.	MAX.	MIN.	MAX.
A	0.876	0.896	22.25	22.75
B	0.396	0.408	10.06	10.36
C	0.173	0.182	4.395	4.622
D	0.604	0.619	15.35	15.72
E	0.263	0.273	6.681	6.934
F	0.168	0.188	4.268	4.775
G	0.100	0.104	2.540	2.641
H	0.320	0.340	8.128	8.638
J	0.246	0.254	6.249	6.451
K	0.046	0.054	1.169	1.371
L	0.496	0.508	12.60	12.90
M	0.140	0.150	3.556	3.810
N	5		5	
P	0.015	0.020	0.381	0.406
Q	0.033	0.040	0.839	1.016
R	0.129	0.139	3.277	3.530
S	0.600	0.630	15.24	16.00
T	0.680	0.710	17.27	18.03

HORIZONTAL MOUNT (M Suffix)



SYMBOL	INCHES		MILLIMETERS	
	MIN.	MAX.	MIN.	MAX.
A	0.726	0.746	18.44	18.94
B	0.396	0.408	10.06	10.36
C	0.173	0.182	4.395	4.622
D	0.604	0.619	15.35	15.72
E	0.263	0.273	6.681	6.934
F	0.221	0.251	5.614	6.375
G	0.100	0.104	2.540	2.641
H	0.143	0.163	3.633	4.140
J	0.246	0.254	6.249	6.451
K	0.046	0.054	1.169	1.371
L	0.496	0.508	12.60	12.90
M	0.140	0.150	3.556	3.810
N	5		5	
P	0.015	0.020	0.381	0.406
Q	0.033	0.040	0.839	1.016
R	0.129	0.139	3.277	3.530

Application Notes

AN-4431

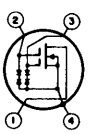
RF Applications of the Dual-Gate MOS FET up to 500 MHz

by L. S. Baar

The RCA dual-gate protected, metal-oxide silicon, field-effect transistor (MOS FET) is especially useful for high-frequency applications in RF amplifier circuits. The dual-gate feature permits the design of simple AGC circuitry requiring very low power. The integrated diodes protect the gates against damage due to static discharge that may develop during handling and usage. This Note describes the use of the RCA-3N200 dual-gate MOS FET in RF applications. The 3N200 has good power gain and a low noise factor at frequencies up to 500 MHz, offers especially good cross-modulation performance, and has a wide dynamic range; its low-feedback capacitance provides stable performance without neutralization.

Gate-Protection Diodes

Fig. 1 shows the terminal diagram for the 3N200. Gate No. 1 is the input signal electrode and Gate No. 2 is normally used to obtain gain control. The back-to-back diodes are connected from each of the gates to the source terminal, lead No. 4. If short duration pulses greater than ±10 volts, generated for example by static discharge, are inadvertently applied to either gate, the protective diodes limit these voltages and shunt the current to the source terminal. Thus the gates, under normal operating conditions, are protected against the effects of overload voltages.[1]



LEAD 1 - DRAIN
LEAD 2 - GATE NO. 2
LEAD 3 - GATE NO. 1
LEAD 4 - SOURCE, SUBSTRATE, AND CASE

Fig. 1 – Terminal diagram for the 3N200.

Operating Conditions

Typical two-port characteristics at 400 MHz including both "y" and "s" parameters, are given for the 3N200 in the RCA technical bulletin, File No. 437. This note makes use of the "y" parameters; however, designers who prefer the alternate method can, by parallel analysis, make use of the "s" parameters.

A recommended operating drain current (I_D) for the 3N200 is approximately 10 milliamperes with Gate No. 2 sufficiently forward biased such that a change in the bias voltage does not greatly affect the drain current. An adequate Gate No. 2-to-source voltage (V_{G2S}) is approximately +4 volts. The forward transadmittance (y_{fs}) increases with drain current, but saturates at higher current levels. The increase in RF performance at drain currents above 10 milliamperes is achieved only with less efficient use of input power.

To establish the optimum operating conditions for a type, consideration must be given to the range of variations in characteristics values encountered in production quantities of the type.[2] One important measure of type variation is the range of zero bias drain current (I_{DS}). The current range given in the 3N200 technical bulletin for I_{DS} is from 0.5 mA to 12 mA. A fixed bias condition intended to center the range of drain current at the desired level, still will produce an operating drain current range of 11.5 milliamperes with a resultant wide range of forward transconductance (g_{fs}). The drain current can be regulated by applying dc feedback with a bypassed source resistor (R_S). A good approximation of R_S (where $I_{DQ} \geq I_{DS}/2$) can be calculated by the use of the following formula*, assuming that V_{G1S} vs. I_{DS} is linear over the current range under consideration:

$$R_S \approx \left(\frac{1}{g_{fs}(\text{min.})}\right)\left(\frac{\Delta I_{DS}}{\Delta I_{DQ}} - 1\right) \qquad \text{Eq. 1}$$

*See Appendix

where:

ΔI_{DS} is the current range given in the 3N200 technical bulletin

ΔI_{DQ} is the desired range of operating current

g_{fs}(min.) is the minimum forward transconductance at 1000 Hz

With the value of R_S established, then the Gate-No. 1 Voltage (V_{G1}) can be calculated from the equation

$$V_{G1} = V_{G1S} + I_{DQ} R_S \qquad \text{Eq. 2}$$

where V_{G1S} is estimated by:

$$V_{G1S} \approx \frac{I_{DQ} - I_{DS}}{g_{fs}(\text{avg.})} \qquad \text{Eq. 3}$$

where:

g_{fs}(avg.) is the average forward transconductance

To establish the Gate-No. 2 Voltage (V_{G2}), follow the same procedure described for calculating the Gate-No. 1 Voltage, except that a fixed V_{G2S} of approximately 4 volts is adequate.

If gain control is desired, apply a negative-going voltage to Gate No. 2. Because Gate No. 2 has little control in the voltage range of +2 to +5 volts, this characteristic may be used to effect AGC delay of the device in order to maintain the low noise figure until the RF signal is out of the noise-level range.

Stability Considerations

Typical "y" parameter data as a function of frequency are given in Table 1. Maximum available gain (MAG) calculated from these data are also included to indicate ideal gain performance (i.e., $y_{rs} = 0$). The ability of the MOS FET to approach these gain levels depends on the device maintaining stable performance at the required operating frequency.

There are several methods which may be used to test for gain vs. stability. One of these methods, the Linvill Criteria (C), is defined by the equation:

$$C = \frac{y_{rs}\, y_{fs}}{2g_{is}\, g_{os} - R_e\,(y_{rs}\, y_{fs})} \qquad \text{Eq. 4}$$

A value for C which is less than 1 indicates unconditional stability. Applying the 400-MHz values taken from Table 1 to the Linvill Criteria yields a value of C = 0.615; substantially less than the value indicating unconditional stability.

The following equation for Maximum Usable Gain (MUG)[3] is:

$$MUG = \frac{2K\,\left|y_{fs}\right|}{\left|y_{rs}\right|(1 + \cos\theta)} \qquad \text{Eq. 5}$$

where:

$\theta = \angle y_{fs} + \angle y_{rs}$

K = skew factor

$\angle y_{rs}$ = angle of reverse transadmittance

$\angle y_{fs}$ = angle of forward transadmittance

The skew factor, introduced in this equation, is a safety measure that establishes an arbitrary degree of skewing in the frequency response which may be introduced by regeneration. A value of 0.2 for K has been established on the basis of past experience. The value of MUG calculated at 400 MHz is 13.8 dB. This value of MUG is greater than the value of MAG, again indicating unconditional stability, since MAG, ignoring inherent feedback, is the conjugately matched gain. Therefore, neutralization or circuit loading is not required to insure stable performance, and the gain can approach MAG, limited only by circuit losses.

Reverse transadmittance (y_{rs}) is composed of several components, but the major ones are feedback capacitance (C_{rss}) and source-lead inductance (L_S). Therefore, care must be exercised in the application of the y_{rs} values, shown in Table 1, at the upper end of the usable frequency range. The 3N200 utilizes a JEDEC TO-72 package that has 4 leads. The data in Table 1 was compiled with the use of a socket which contacts the leads of the 3N200 as close as possible to the bottom of the package as specified by the JEDEC Standard Proposal SP-1028 "Measurement of VHF-UHF "y" Parameters". The leads are shielded from each other to eliminate stray capacitance between the leads, but some lead inductance is inevitable. If the device is soldered directly to the circuit components using commercial production techniques rather than by precise laboratory methods, then additional source lead inductance can be expected. Also, some additional capacitive coupling may result if the input and output circuits are not completely isolated from each other.

CHARACTERISTICS	SYMBOL	FREQUENCY (MHz) 100	200	300	400	500	UNITS
y Parameters							
Input Conductance	g_{is}	0.25	0.8	2.0	3.6	6.2	mmho
Input Susceptance	b_{is}	3.4	5.8	8.5	11.2	15.5	mmho
Magnitude of Forward Transadmittance	$\lvert y_{fs} \rvert$	15.3	15.3	15.4	15.5	16.3	mmho
Angle of Forward Transadmittance	$\angle y_{fs}$	-15.0	-25.0	-35.0	-47.0	-60.0	degrees
Output Conductance	g_{os}	0.15	0.3	0.5	0.8	1.1	mmho
Output Susceptance	b_{os}	1.5	2.7	3.6	4.25	5.4	mmho
Magnitude of Reverse Transadmittance	$\lvert y_{rs} \rvert$	0.012	0.025	0.06	0.14	0.26	mmho
Angle of Reverse Transadmittance	$\angle y_{rs}$	-60.0	-25.0	0	14.0	20.0	degrees
Maximum Available Gain	MAG	32.0	24.0	17.5	13	10.0	dB

Table 1 – "y" Parameters from 100 to 500 MHz

Because the published y_{rs} value for the 3N200 is very small, the circuit y_{rs} values may differ significantly from the y_{rs} values shown in Table 1 and hence, may result in an unstable operating condition. It is impossible to provide data for all possible mounting combinations, therefore, a recommended mounting arrangement is shown in Fig. 2. The source and substrate in the TO-72 package of the 3N200 are internally connected to lead No. 4 and the case. The source-lead inductance can be reduced, if the case is used as the source connection. Fig. 2 illustrates a partial component layout in which the case is held by a clamp or other fingered device. The clamp is soldered to a feedthrough capacitor to provide an effective, very-low inductance bypass to RF signals. This mounting arrangement still permits the use of a source resistor for DC stability, and enables the case to provide isolation between the input and output circuit in addition to the isolation afforded by the shield.

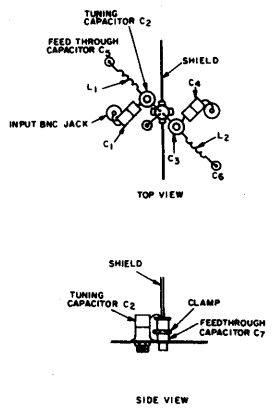


Fig. 2 – Partial component layout of 400-MHz amplifier circuit

The reduction of source-lead inductance provides in addition to greater stability, a lower input and output conductance. Table 2 shows the differences in "y" parameter values at 400 MHz when measured with the source connection made to lead No. 4 (in accordance with the published data for the 3N200) and when measured with the case connected directly to the ground plane of the test jig. The magnitude of reverse transadmittance is halved with a significant change in its phase angle. The input conductance is reduced by 30%, and the output conductance is reduced by 13%. A recalculation of the expressions for MAG, MUG, and Linvill Criteria (C) shows a significant improvement in gain and circuit stability.

While it is difficult to provide accurate information on the effects of shielding between the input and output circuits, its effect can be demonstrated when all other feedback components have been reduced to negligible values. The circuit, shown in Fig. 3 (for component layout see Fig. 2), was measured both with and without a shield. The maximum gain, without the shield, averaged 0.8 dB lower than with the use of the shield.

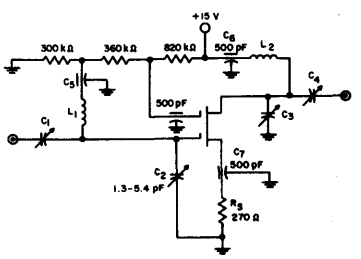


Fig. 3 – 400-MHz amplifier circuit

When receiver sensitivity is an important consideration in the design of an RF amplifier, a compromise must be made in the circuit power gain to achieve a lower noise factor. A contour plot of noise figure as a function of generator source admittance is shown in Fig. 4. Each contour is a plot of noise figure as a function of the generator source conductance and susceptance. Data for the noise figure were obtained from a test amplifier designed with very low feedback. Even though the area of very low-noise figure in the curves in Fig. 4 cover a broad range of source admittance, impedance-matching for maximum power gain could result in

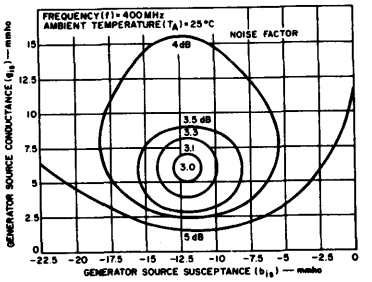


Fig. 4 – Noise factor vs. generator source (input) admittance (Y_{is})

a relatively poor noise figure. As shown in Table 2, the input conductance (g_{is}) with the case grounded is 2.5 mmho. With the reactive portion tuned out, the noise factor at power matched conditions is almost 1 dB higher than the optimum noise figure. However, matching to 5.0 mmho results in a near optimum noise factor with a loss of only 0.5 dB in gain. In addition, impedance matching to high conductance also benefits crossmodulation performance, as will be discussed in a later section.

Gate Protection Diodes

The diodes incorporated into RCA dual-gate MOS FETs, for gate protection, have been designed to minimize RF loading on the input circuits. The small amount of RF loading results in only a fraction of a dB loss in power gain and a negligible increase in the noise figure. The advantages of diode protection, greatly outweigh the slight loss in power gain, especially in an RF amplifier intended for the input stage of a receiver.

In addition to the protection afforded in normal handling, the diodes also provide in-circuit protection against events such as: static discharge due to contact with the antenna, delay in transmit-receive switching, or connection of an antenna with an accumulated charge to the receiver.

Crossmodulation

Crossmodulation is an important consideration because it is an inherent device characteristic where circuit considerations are secondary. Crossmodulation is the transfer of modulation from an undesired signal on a desired signal caused by the non-linear characteristics of a device.

Crossmodulation is proportional to the third-order term of the expansion of the I_D - V_{GS} curve. It is normally specified as the undesired signal voltage required to produce a crossmodulation factor of 0.01. The crossmodulation factor is defined as the percent modulation on a desired carrier by the modulated undesired signal divided by the percent modulation of the undesired signal.[4]

Inspection of the I_D - V_{G1S} curve of Fig. 5 offers an insight to the possible crossmodulation as a function of gain-reduction performance. When both channels of the 3N200 are fully conducting current, as shown by the V_{G2S} = 4-volt curve, the device approximately follows a square-law characteristic. If the I_D - V_{G1S} curve was ideal, the third-order term would be zero; but in practical cases, the third-order term and crossmodulation have some low values.

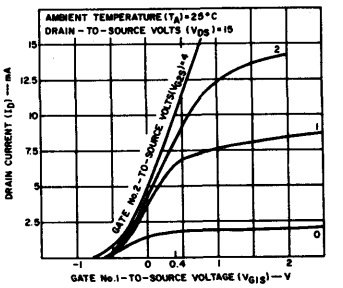


Fig. 5 – Drain current (I_D) vs. gate No. 1-to-source voltage (V_{G1S})

CHARACTERISTICS	SYMBOL	FREQUENCY (f) = 400 MHz		UNITS
		Normal Connection	Case Grounded	
Maximum Available Power Gain	MAG	13.0	15.7	dB
Maximum Usable Power Gain (unneutralized)	MUG	13.8	19.4	dB
Linvill Stability Factor, C	C	0.615	0.335	mmho
"y" Parameters				
Input Conductance	g_{is}	3.6	2.5	mmho
Input Susceptance	b_{is}	11.2	11.7	mmho
Magnitude of Forward Transadmittance	$\lvert y_{fs} \rvert$	15.5	15.5	mmho
Angle of Forward Transadmittance	$\angle y_{fs}$	-47.0	-40.0	degrees
Output Conductance	g_{os}	0.8	0.65	mmho
Output Susceptance	b_{os}	4.25	4.25	mmho
Magnitude of Reverse Transadmittance	$\lvert y_{rs} \rvert$	0.14	0.07	mmho
Angle of Reverse Transadmittance	$\angle y_{rs}$	14.0	49.0	degrees

Table 2 – "y" Parameters at 400 MHz with source connection to lead No. 4 and with case connected to ground plane of test jig

When the gain is reduced, by the application of bias to Gate No. 2, the square-law characteristic changes to a curve with a knee. Sharp curvatures usually result in larger high-order terms and poorer crossmodulation performance can be expected at lower gain conditions. If in Fig. 6, Circuit A, we assume a fixed bias (V_{G1S}) of approximately +0.4 volt, then the expected variation in crossmodulation is determined at the points where the ordinate at V_{G1S} = +0.4 volt crosses the curves. Crossmodulation performance at values of V_{G2S} = +4 volts to cutoff is as follows: good (low crossmodulation) at +4 volts, poorer at +2 volts, poorest at +1 volt, and again improves from zero volts to cutoff.

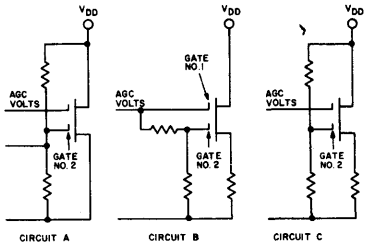


Fig. 6 – Biasing circuits using the 3N200

Curve A, Fig. 7 shows a curve of the undesired signal with a crossmodulation factor of 0.01 as a function of gain reduction. The curve indicates performance is poorest when gain reduction is in the 3- to 15-dB region; this region represents a Gate No. 2-voltage range of approximately 0.5 volt to 2 volts. The exception to the poor crossmodulation performance in this range is the sharp peak which occurs at the 5-dB level and is due to a curve inversion that takes place just prior to the knee. Beyond the 15-dB level, crossmodulation generally shows an improvement.

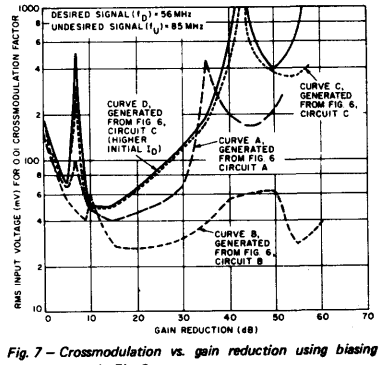


Fig. 7 – Crossmodulation vs. gain reduction using biasing circuits shown in Fig. 6

If Gate No. 1 is also reverse biased in conjunction with Gate No. 2 in the manner shown in Fig. 6, Circuit B, then the overall performance is poorer because the Gate No. 1 voltage will tend to follow the knee of each curve. This occurrence is evident in Fig. 7, Curve B. If Gate No. 1 is biased as shown in Fig. 6, Circuit C, the Gate No. 1-to-Source voltage intercepts the Gate No. 2 curves where the curvature is less severe, indicating as shown by Fig. 7, Curve C an improvement in crossmodulation performance. A further slight improvement is possible by the use of a higher initial operating drain current, which effectively moves the intercepts to the right on each curve. This improvement is indicated in Fig. 7, Curve D.

The curves in Fig. 7 establish that the biasing arrangement which provides optimum crossmodulation performance is the one in which Gate No. 1 forward bias increases as Gate No. 2 controls the gain. This biasing arrangement is easily accomplished by the use of a fixed Gate No. 1 voltage and a source resistor. As the Gate No. 2 bias voltage reduces the drain current, there is also a decrease in source voltage and an increase in the Gate No. 1-to-Source voltage. The gate-to-source voltage ratings must not be exceeded under any circumstances.

Summary

An RF amplifier, ideally, should provide high gain, a low-noise figure, and low crossmodulation. The 3N200 offers a good compromise in providing these three features. As indicated in the section on "Stability Considerations" a mismatch at the circuit input to a higher conductance level, provides an improved noise figure. The same mismatch condition also improves crossmodulation performance. The input signal at the gate of the device, when mismatched as indicated above, is lower than if it is power matched. The same ratio applies to any undesired signal and, thus, reduces the possibility of crossmodulation interference.

Appendix

The drain current of a device is established by the relationship

$$I_D = g_{fs} V_{G1S} + I_{DS}$$

where:

I_{DS} = drain current

at:

$V_{G1S} = 0$, $V_{G2S} = +4$ volts.

If a source resistor is used, as shown in Fig. A1, the gate No. 1-to-source voltage is

$$V_{G1S} = V_{G1} - I_D R_S$$

then

$$I_D = g_{fs} (V_{G1} - I_D R_S) + I_{DS} \quad \text{or}$$

$$I_D = \frac{g_{fs} V_{G1}}{1 + g_{fs} R_S} + \frac{I_{DS}}{1 + g_{fs} R_S}$$

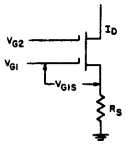


Fig. A1 – Bias circuit using the 3N200

The typical curves in Fig. A2 show drain current vs. Gate No. 1-to-Source Voltage as a function of I_{DS} level. These curves are almost linear when the typical operating drain current is in the 10-milliampere region. For the remainder of the analysis a linear relationship will be assumed for the required range of quiescent current. The assumption of linearity dictates that g_{fs} is a constant.

The required range of drain current is $I_{D2} - I_{D1}$

where:

$$I_{D2} = \frac{g_{fs} V_{G1}}{1 + g_{fs} R_S} + \frac{I_{DS}\,(\text{max.})}{1 + g_{fs} R_S}$$

$$I_{D1} = \frac{g_{fs} V_{G1}}{1 + g_{fs} R_S} + \frac{I_{DS}\,(\text{min.})}{1 + g_{fs} R_S}$$

$$\Delta I_D = I_{D2} - I_{D1} = \frac{I_{DS}\,(\text{max.}) - I_{DS}\,(\text{min.})}{1 + g_{fs} R_S} = \frac{\Delta I_{DS}}{1 + g_{fs} R_S}$$

Solving the above equation for R_S gives

$$R_S = \frac{(\Delta I_{DS}/\Delta I_D) - 1}{g_{fs}}$$

where:

g_{fs} is equal to the expected minimum value at the required I_D

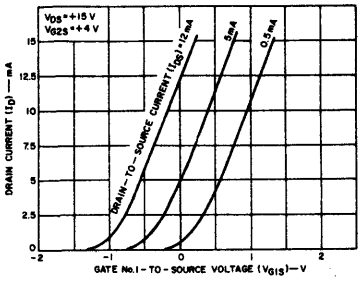


Fig. A2 – Drain current vs. gate No. 1-to-source voltage

References:

1. L. A. Jacobus and S. Reich, "Design of Gate-Protected MOS Field-Effect Transistors", RCA Application Note AN-4018
2. S. Reich, "Field-Effect Transistor Biasing Techniques", EEE, Sept. 1970
3. R. A. Santilli, "RF and IF Amplifier Design Considerations", IEEE Transactions on Broadcast and TV Receivers, Nov. 1967
4. H. Thanos, "Crossmodulation on Transistorized TV Tuners", IEEE Transactions on Broadcast and TV Receivers, Vol. 9 No. 3, Nov. 1963

Some Applications of a Programmable Power/Switch Amplifier

by L. R. Campbell and H. A. Wittlinger

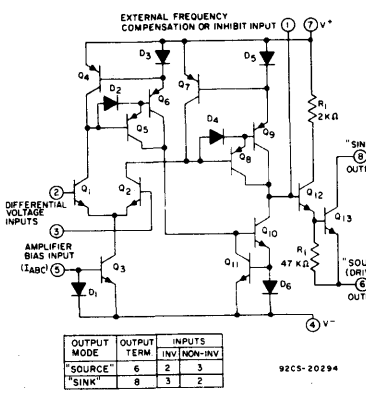
The RCA-CA3094 unique monolithic programmable power switch/amplifier IC consists of a high-gain preamplifier driving a power-output amplifier stage. It can deliver average power of 3 watts or peak power of 10 watts to an external load, and can be operated from either a single or dual power supply. This Note briefly describes the characteristics of the CA3094, and illustrates its use in the following circuit applications:

- Class A instrumentations and power amplifiers
- Class A driver-amplifier for complementary power transistors
- Wide-frequency-range power multivibrators
- Current- or voltage-controlled oscillators
- Comparators (threshold detectors)
- Voltage regulators
- Analog timers (long time delays)
- Alarm systems
- Motor-speed controllers
- Thyristor-firing circuits
- Battery-charger regulator circuits
- Ground-fault-interrupter circuits

Circuit Description

The CA3094 series of devices offers a unique combination of circuit flexibility and power-handling capability. Although these monolithic IC's dissipate only a few microwatts when quiescent, they have a high current-output capability (100 milliamperes average, 300 milliamperes peak) in the active state, and the premium-grade devices can operate at supply voltages up to 44 volts.

Fig. 1 shows a schematic diagram of the CA3094. The portion of the circuit preceding transistors Q_{12} and Q_{13} is the preamplifier section and is generically similar to that of the RCA-CA3080 Operational Transconductance Amplifier (OTA).[1,2] The CA3094 circuits can be gain-programmed by either digital and/or analog signals applied to a separate Amplifier-Bias-Current (I_{ABC}) terminal (No. 5 in Fig. 1) to control circuit sensitivity. Response of the amplifier is essentially linear as a function of the current at terminal 5. This additional signal input "port" provides added flexibility in many applications. Thus, the output of the amplifier is a function of input signals applied differentially at terminals 2 and 3 and/or in a single-ended configuration at terminal 5. The output portion of the monolithic circuit in the CA3094 consists of a Darlington-connected transistor pair with access provided to both the collector and emitter terminals to provide capability to "sink" and/or "source" current.



OUTPUT MODE	OUTPUT TERM	INPUTS INV	INPUTS NON-INV
"SOURCE"	6	2	3
"SINK"	8	3	2

Fig.1–CA3094 circuit schematic diagram.

The CA3094 series of circuits consists of six types that differ only in voltage-handling capability and package options, as shown below; other electrical characteristics are identical.

Package Options	Maximum Voltage Rating
CA3094S; CA3094T	24 V
CA3094AS; CA3094AT	36 V
CA3094BS; CA3094BT	44 V

The suffix "S" indicates circuits packaged in TO-5 enclosures with leads formed to an 8-lead dual-in-line configuration (0.1" pin spacing). The suffix "T" indicates circuits packaged in 8-lead TO-5 enclosures with straight leads. The generic CA3094 type designation is used throughout this Note.

Class A Instrumentation Amplifiers

One of the more difficult instrumentation problems frequently encountered is the conversion of a differential input signal to a single-ended output signal. Although this conversion can be accomplished in a straightforward design through the use of classical op-amps, the stringent matching requirements of resistor ratios in feedback networks make the conversion particularly difficult from a practical standpoint. Because the gain of the preamplifier section in the CA3094 can be defined as the product of the transconductance and the load resistance ($g_m R_L$), feedback is not needed to obtain predictable open-loop gain performance. Fig. 2 shows the CA3094 in this basic type of circuit.

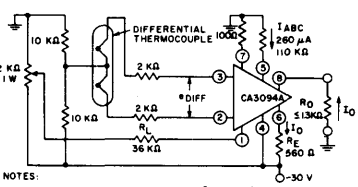


Fig.2–Open-loop instrumentation amplifier with differential input and single-ended output.

The gain of the preamplifier section (to terminal No. 1) is $g_m R_L = (5 \times 10^{-3})(36 \times 10^3) = 180$. The transconductance g_m is a function of the current into terminal No. 5, I_{ABC}, the amplifier-bias-current. In this circuit an I_{ABC} of 260 microamperes results in a g_m of 5 millimhos. The operating point of the output stage is controlled by the 2-kilohm potentiometer. With no differential input signal ($e_{diff} = 0$), this potentiometer is adjusted to obtain a quiescent output current I_O of 12 milliamperes. This output current is established by the 560-ohm emitter resistor, R_E, as follows:

$$I_O \approx \frac{(g_m R_L)(e_{diff})}{R_E}$$

Under the conditions described, an input swing e_{diff} of ±26 millivolts produces a variation in the output current I_O of ±8.35 milliamperes. The nominal quiescent output voltage is 12 milliamperes times 560 ohms or 6.7 volts. This output level drifts approximately −4 millivolts, or −0.0595 per cent, for each °C change in temperature. Output drift is caused by temperature-induced variations in the base-emitter voltage of the two output transistors, Q_{12} and Q_{13}.

Fig. 3 shows the CA3094 used in conjunction with a resistive-bridge input network; and Fig. 4 shows a single-supply amplifier for thermocouple signals. The RC networks* connected between terminals 1 and 4 in Figs. 3 and 4 provide compensation to assure stable operation.

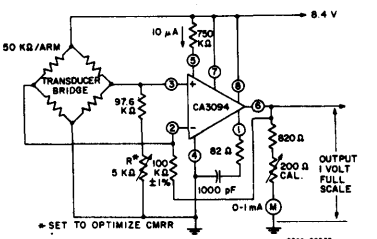


Fig.3–Single-supply differential-bridge amplifier.

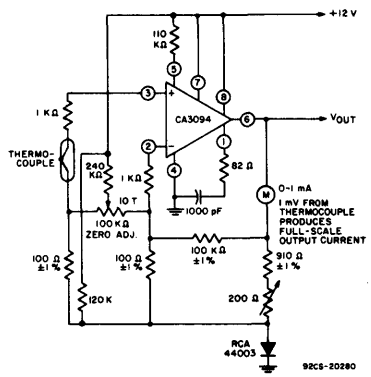


Fig.4–Single-supply amplifier for thermocouple signals.

Class A Power Amplifiers

The CA3094 is attractive for power-amplifier service because the output transistor can control current up to 100 milliamperes (300 milliamperes peak), the premium devices (CA3094B) can operate at supply voltages up to 44 volts, and the TO-5 package can dissipate power up to 1.6 watts when equipped with a suitable heat sink that limits the case temperature to 55°C.

*The components of the RC network are chosen so that

$$\frac{1}{2\pi RC} \approx 2 \text{ MHz}.$$

Fig. 5 shows a Class A amplifier circuit using the CA3094A that is capable of delivering 280 milliwatts to a 350-ohm resistive load. This circuit has a voltage gain of 60 dB and a

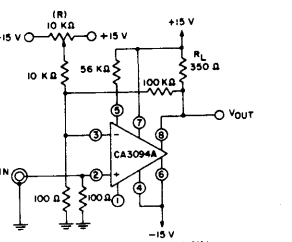


Fig.5–Class-A amplifier – 280-mW capability into a resistive load.

3-dB bandwidth of about 50 kHz. Operation is stable without the use of a phase-compensation network. Potentiometer R is used to establish the quiescent operating point for class A operation.

The circuit of Fig. 6 illustrates the use of the CA3094 in a class A power-amplifier circuit driving a transformer-coupled load. With dual power supplies of +7.5 volts and −7.5 volts, a

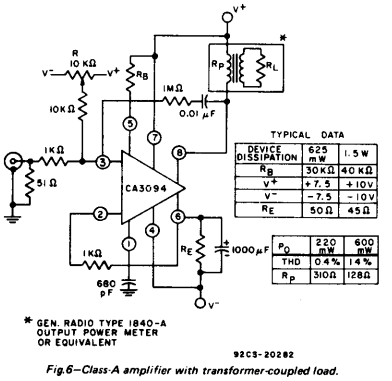


Fig.6—Class-A amplifier with transformer-coupled load.

base resistor R_B of 30 kilohms, and an emitter resistor R_E of 50 ohms, CA3094 dissipation is typically 625 milliwatts. With supplies of +10 volts and −10 volts, R_B of 40 kilohms, and R_E of 45 ohms, the dissipation is 1.5 watts. Total harmonic distortion is 0.4 per cent at a power-output level of 220 milliwatts with a reflected load resistance R_P of 310 ohms, and is 1.4 per cent for an output of 600 milliwatts with an R_P of 128 ohms. The setting of potentiometer R establishes the quiescent operating point for class A operation. The 1-kilohm resistor connected between terminals 6 and 2 provides dc feedback to stabilize the collector current of the output transistor. The ac gain is established by the ratio of the 1-megohm resistor connected between terminals 8 and 3 and the 1-kilohm resistor connected to terminal 3. Phase compensation is provided by the 680-picofarad capacitor connected to terminal 1.

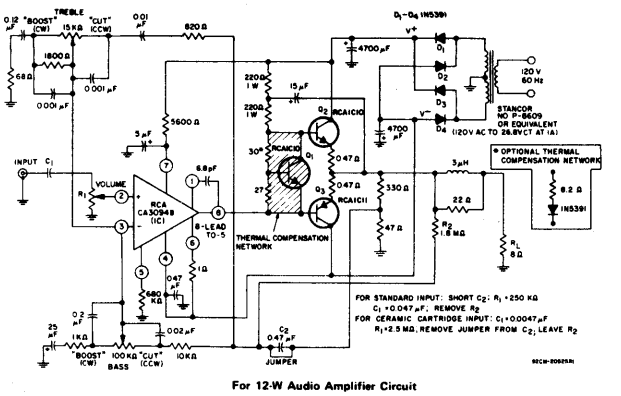
Class A Driver-Amplifier for Complementary Power Transistors

The CA3094 configuration and characteristics are ideal for driving complementary power-output transistors;[3] a typical circuit is shown in Fig. 7. This circuit can provide 12 watts of audio power output into an 8-ohm load with intermodulation distortion (IMD) of 0.2 per cent when 60-Hz and 2-kHz signals are mixed in a 4:1 ratio. Intermodulation distortion is shown as a function of power output in Fig. 8.

The large amount of loop gain and the flexibility of feedback arrangements with the CA3094 make it possible to incorporate the tone controls into a feedback network that is closed around the entire amplifier system. The tone controls in the circuit of Fig. 7 are part of the feedback network connected from the amplifier output (junction of the 330- and 47-ohm resistors driven by the emitters of Q_2 and Q_3) to terminal 3 of the CA3094. Fig. 9 shows voltage gain as a function of frequency with tone controls adjusted for "flat" response and for responses at the extremes of tone-control rotation. The use of tone controls incorporated in the feedback network results in excellent signal-to-noise ratio. Hum and noise are typically 700 microvolts (83 dB down) at the output.

In addition to the savings resulting from reduced parts count and circuit size, the use of the CA3094 leads to further savings in the power-supply system. Typical values of power-supply rejection and common-mode rejection are 90 dB and 100 dB, respectively. An amplifier with 40-dB gain and 90-dB power-supply rejection would require a 31-millivolt power-supply ripple to produce one millivolt of hum at the output. Therefore, no filtering is required other than that provided by the energy-storage capacitors at the output of the rectifier system shown in Fig. 7.

For applications in which the operating temperature range is limited (e.g., consumer service) the thermal compensation network (shaded area) can be replaced by a more economical configuration consisting of a resistor-diode combination (8.2 ohms and 1N5391) as shown in Fig. 7.



For 12-W Audio Amplifier Circuit

Power Output (8Ω load, Tone Control set at "Flat")		
Music (at 5% THD, regulated supply)	15	W
Continuous (at 0.2% IMD, 60 Hz & 2 kHz mixed in a 4:1 ratio, unregulated supply) See Fig. 8	12	W
Total Harmonic Distoration		
At 1 W, unregulated supply	0.05	%
At 12 W, unregulated supply	0.57	%
Voltage Gain	40	dB
Hum and Noise (Below continuous Power Output)	83	dB
Input Resistance	250	kΩ
Tone Control Range	See Fig. 9	

Fig.7—12-watt amplifier circuit featuring true complementary-symmetry output stage with CA3094 in driver stage.

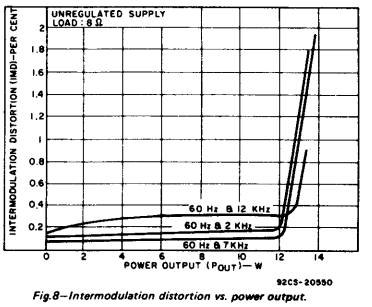


Fig.8—Intermodulation distortion vs. power output.

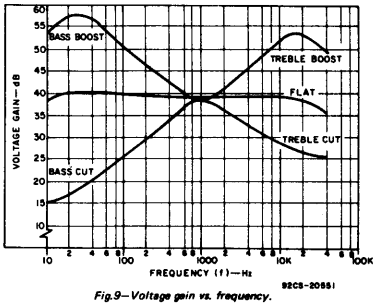


Fig.9—Voltage gain vs. frequency.

Power Multivibrators (Astable and Monostable)

The CA3094 is suitable for use in power multivibrators because its high-current output transistor can drive low-impedance circuits while the input circuitry and the frequency-determining elements are operating at micropower levels. A typical example of an astable multivibrator using the CA3094 with a dual power supply is shown in Fig. 10. The output frequency f_{OUT} is determined as follows:

$$f_{OUT} = \frac{1}{2RC\ln[(2R1/R2)+1]}$$

If R2 is equal to 3.08 R1, then f_{OUT} is simply the reciprocal of RC.

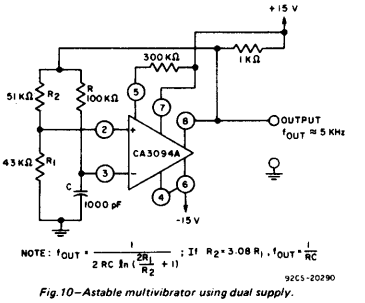


Fig.10—Astable multivibrator using dual supply.

Fig. 11 is a single-supply astable multivibrator circuit which illustrates the use of the CA3094 for flashing an incandescent lamp. With the component values shown, this circuit produces one flash per second with a 25-per-cent "on"-time while delivering output current in excess of 100 milliamperes. During

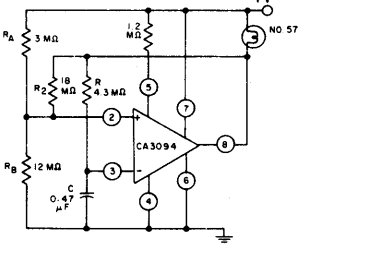


Fig.11—Astable multivibrator using single supply.

the 75-per-cent "off"-time it idles with micropower consumption. The flashing rate can be maintained within ±2 per cent of the nominal value over a battery voltage range from 6 to 15 volts and a temperature excursion from 0 to 70°C. The CA3094 series of circuits can supply peak-power output in

excess of 10 watts when used in this type of circuit. The frequency of oscillation f_{OSC} is determined by the resistor ratios, as follows:

$$f_{OSC} = \frac{1}{2RC\ln[(2R_1/R_2)+1]}$$

where

$$R1 = \frac{R_A R_B}{R_A + R_B}$$

Provisions can easily be made in the circuit of Fig. 11 to vary the multivibrator pulse length while maintaining an essentially constant pulse repetition rate. The circuit shown in Fig. 12 incorporates a potentiometer Rp for varying the width of pulses generated by the astable multivibrator to drive a light-emitting diode (LED).

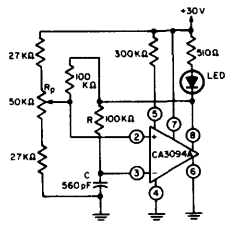


Fig.12–Astable power multivibrator with provisions for varying duty cycle.

Fig. 13 shows a circuit incorporating independent controls (R_{ON} and R_{OFF}) to establish the "on" and "off" periods of the current supplied to the LED. The network between points "A" and "B" is analogous in function to that of the 100-kilohm resistor R in Fig. 12.

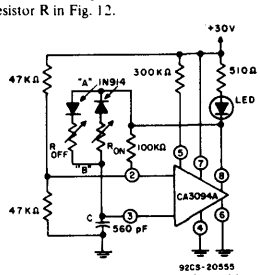


Fig.13–Astable power multivibrator with provisions for independent control of LED "on-off" periods.

The CA3094 is also suitable for use in monostable multivibrators, as shown in Fig. 14. In essence, this circuit is a pulse counter in which the duration of the output pulses is independent of trigger-pulse duration. The meter reading is a function of the pulse repetition rate which can be monitored with the speaker.

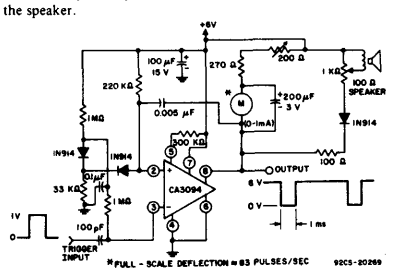


Fig.14–Power monostable multivibrator.

Current- or Voltage-Controlled Oscillators

Because the transconductance of the CA3094 varies linearly as a function of the amplifier bias current (I_{ABC}) supplied to terminal 5, the design of a current- or voltage-controlled oscillator is straightforward, as shown in Fig. 15. Fig. 16 and 17 show oscillator frequency as a function of I_{ABC} for a current-controlled oscillator for two different values of capacitor C in Fig. 15. The addition of an appropriate resistor (R) in series with terminal 5 in Fig. 15 converts the circuit into a voltage-controlled oscillator. Linearity with respect to either current or voltage control is within 1 per cent over the middle half of the characteristics. However, variation in the symmetry of the output pulses as a function of frequency is an inherent characteristic of the circuit in Fig. 15, and leads to distortion when this circuit is used to drive the phase detector in phase-locked-loop applications. This type of distortion can be eliminated by interposing an appropriate flip-flop between the output of the oscillator and the phase-locked discriminator circuits.

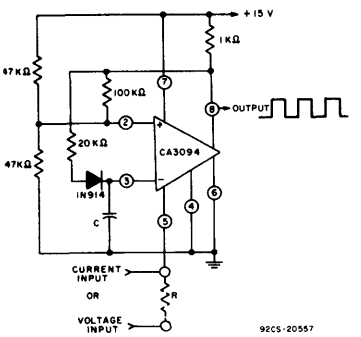


Fig.15–Current- or voltage-controlled oscillator.

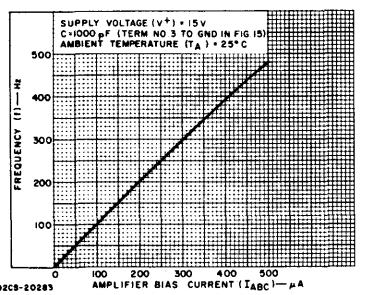


Fig.16–Frequency as a function of I_{ABC} for C=1000 pF for circuit in Fig. 15.

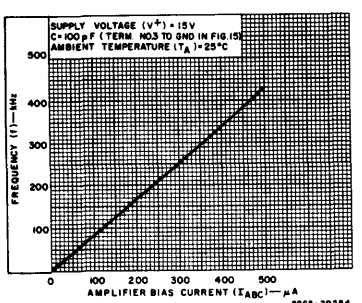


Fig.17–Frequency as a function of I_{ABC} for C=100 pF for circuit in Fig. 15.

Comparators (Threshold Detectors)

Comparator circuits are easily implemented with the CA3094, as shown by the circuits in Fig. 18. The circuit of Fig. 18(a) is arranged for dual-supply operation; the input voltage exceeds the positive threshold, the output voltage swings essentially to the negative supply-voltage rail (it is assumed that there is negligible resistive loading on the output terminal). An input voltage that exceeds the negative threshold value results in a positive voltage output essentially equal to the positive supply voltage. The circuit in Fig. 18(b), connected for single-supply operation, functions similarly.

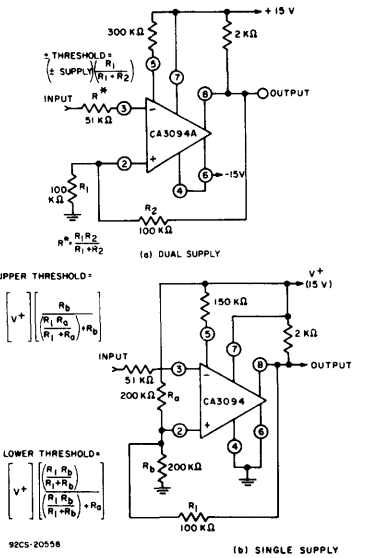


Fig.18–Comparators (threshold detectors) – dual- and single-supply types.

Fig. 19 shows a dual-limit threshold detector circuit in which the high-level limit is established by potentiometer R1

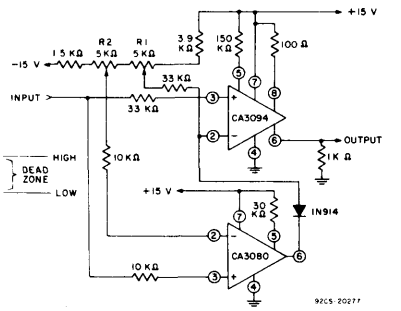


Fig.19–Dual-limit threshold detector.

and the low-level limit is set by potentiometer R2 to actuate the CA3080 low-limit detector.[1,2] A positive output signal is delivered by the CA3094 whenever the input signal exceeds either the high-limit or the low-limit values established by the appropriate potentiometer settings. This output voltage is approximately 12 volts with the circuit shown.

The high current-handling capability of the CA3094 makes it useful in Schmitt power-trigger circuits such as that shown in Fig. 20. In this circuit, a relay coil is switched whenever the

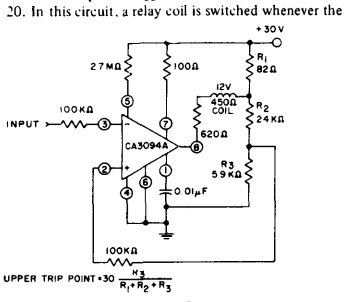


Fig.20–Precision Schmitt power-trigger circuit.

input signal traverses a prescribed upper or lower trip point, as defined by the following expressions:

$$\text{Upper Trip Point} = 30\left(\frac{R3}{R1 + R2 + R3}\right)$$

$$\text{Lower Trip Point} \cong (30 - 0.026R1)\frac{R3}{R2 + R3}$$

The circuit is applicable, for example, to automatic ranging. With the values shown in Fig. 20, the relay coil is energized when the input exceeds approximately 5.9 volts and remains energized until the input signal drops below approximately 5.5 volts.

Power-Supply Regulators

The CA3094 is an ideal companion device to the CA3085 series regulator circuits[4] in dual-voltage tracking regulators that handle currents up to 100 milliamperes. In the circuit of Fig. 21, the magnitude of the regulated positive voltage provided by the CA3085A is adjusted by potentiometer R. A sample of this positive regulated voltage supplies the power for the CA3094A negative regulator and also supplies a reference voltage to its terminal 3 to provide tracking. This circuit provides a maximum line regulation equal to 0.075 per cent per volt of input voltage change and a maximum load regulation of 0.075 per cent of the output voltage.

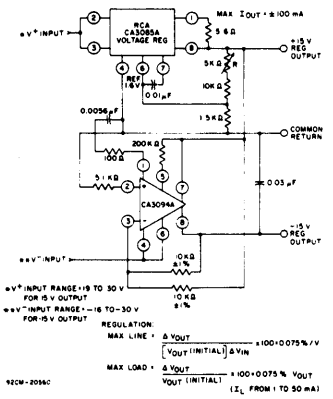


Fig.21—Dual-voltage tracking regulator.

Fig. 22 shows a regulated high-voltage supply similar to the type used to supply power for Geiger-Mueller tubes. The CA3094, used as an oscillator, drives a step-up transformer which develops suitable high voltages for rectification in the RCA-44007 diode network. A sample of the regulated output voltage is fed to the CA3080A operational transconductance amplifier through the 198-megohm and 910-kilohm divider to control the pulse repetition rate of the CA3094. Adjustment of potentiometer R determines the magnitude of the regulated output voltage. Regulation of the desired output voltage is maintained within one per cent despite load-current variations of 5 to 26 microamperes. The dc-to-dc conversion efficiency is about 48 per cent.

Timers

The programmability feature inherent in the CA3094 (and operational transconductance amplifiers in general) simplifies the design of presettable timers such as the one shown in Fig. 23. Long timing intervals (e.g., up to 4 hours) are achieved by discharging a timing capacitor C_1 into the signal-input terminal (e.g., No. 3) of the CA3094. This discharge current is controlled precisely by the magnitude of the amplifier bias current I_{ABC} programmed into terminal 5 through a resistor selected by switch S2. Operation of the circuit is initiated by charging capacitor C_1 through the momentary closing of switch S_1. Capacitor C_1 starts discharging and continues discharging until voltage E_1 is less than voltage E_2. The differential input transistors in the CA3094 then change state, and terminal 2 draws sufficient current to reverse the polarity of the output voltage (terminal 6). Thus, the CA3094 not only has provision for readily presetting the time delay, but also provides significant output current to drive control devices such as thyristors. Resistor R_5 limits the initial charging current for C_1. Resistor R_7 establishes a minimum voltage of at least 1 volt at terminal 2 to insure operation within the common-mode-input range of the device. The diode limits the maximum differential input voltage to 5 volts. Gross changes in time-range selection are made with switch S2, and vernier trimming adjustments are made with potentiometer R_6.

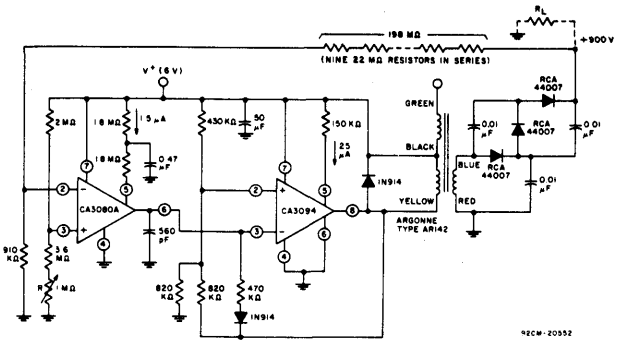


Fig.22—Regulated high-voltage supply.

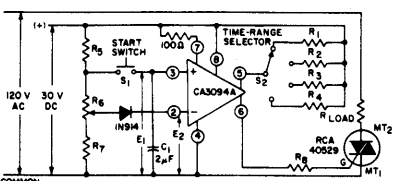


Fig.23—Presettable analog timer.

In some timer applications, such as that shown in Fig. 24, a meter readout of the elapsed time is desirable. This circuit uses the CA3094 and the CA3083 transistor array[5] to control the meter and a load-switching triac. The timing cycle starts with the momentary closing of the start switch to charge capacitor C_1 to an initial voltage determined by the 50-kilohm vernier timing adjustment. During the timing cycle, capacitor C_1 is discharged by the input bias current at terminal 3, which is a function of the resistor value R_1 chosen by the time-range selection switch. During the timing cycle the output of the CA3094, which is also the collector voltage of Q_1, is "high". The base drive for Q_1 is supplied from the positive supply through a 91-kilohm resistor. The emitter of Q_1, through the 75-ohm resistor, supplies gate-trigger current to the triac. Diode-connected transistors Q_4 and Q_5 are connected so that transistor Q_1 acts as a constant-current source to drive the triac. As capacitor C_1 discharges, the CA3094 output voltage at terminal 6 decreases until it becomes less than the V_{CEsat} of Q_1. At this point the flow of drive current to the triac ceases and the timing cycle is ended. The 20-kilohm resistor between terminals 2 and 6 of the CA3094 is a feedback resistor. Diode-connected transistors Q_2 and Q_3 and their associated networks serve to compensate for nonlinearities in the discharge-circuit network by bleeding corrective current into the 20-kilohm feedback resistor. Thus, current flow in the meter is essentially linear with respect to the timing period. The time periods as a function of R_1 are indicated on the Time-Range Selection Switch in Fig. 24.

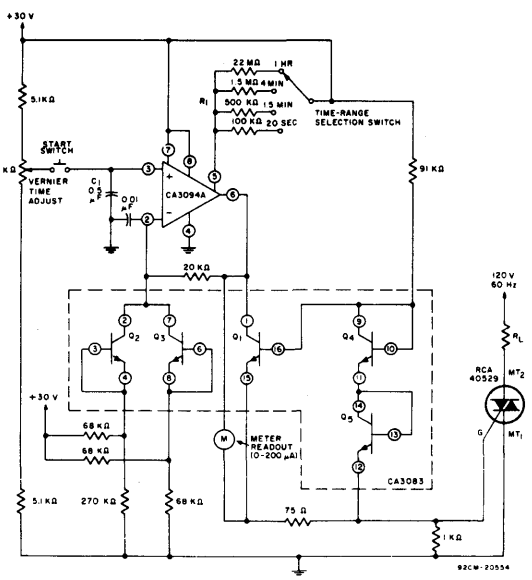


Fig.24—Presettable timer with linear readout.

Alarm Circuit

Fig. 25 shows an alarm circuit utilizing two "sensor" lines. In the "no-alarm" state, the potential at terminal 2 is lower than the potential at terminal 3, and terminal 5 (I_{ABC}) is driven with sufficient current through resistor R_5 to keep the output voltage "high". If either "sensor" line is opened, shorted to ground, or shorted to the other sensor line, the output goes "low" and activates some type of alarm system. The back-to-back diodes connected between terminals 2 and 3 protect the CA3094 input terminals against excessive differential voltages.

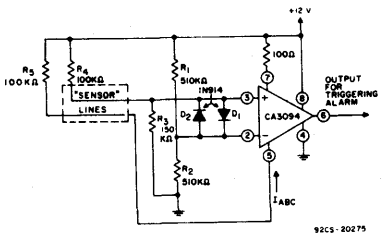


Fig.25—Alarm system.

Motor-Speed Controller System

Fig. 26 illustrates the use of the CA3094 in a motor-speed controller system. Circuitry associated with rectifiers D_1 and D_2 comprises a full-wave rectifier which develops a train of half-sinusoid voltage pulses to power the dc motor. The motor speed depends on the peak value of the half-sinusoids and the period of time (during each half-cycle) the SCR is conductive.

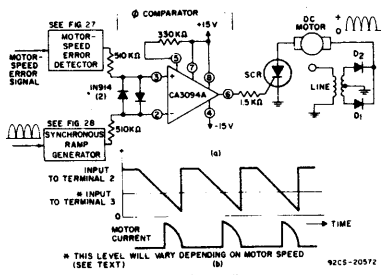


Fig.26—Motor-speed controller system.

The SCR conduction, in turn, is controlled by the time duration of the positive signal supplied to the SCR by the phase comparator. The magnitude of the positive dc voltage supplied to terminal 3 of the phase comparator depends on motor-speed error as detected by a circuit such as that shown in Fig. 27. This dc voltage is compared to that of a fixed-amplitude ramp wave generated synchronously with the ac-line-voltage frequency. The comparator output at terminal 6 is "high" (to trigger the SCR into conduction) during the period when the ramp potential is less than that of the error voltage on terminal 3. The motor-current conduction period is increased as the error voltage at terminal 3 is increased in the positive direction. Motor-speed accuracy of ±1 per cent is easily obtained with this system.

Motor-Speed Error Detector. Fig. 27(a) shows a motor-speed error detector suitable for use with the circuit of Fig. 26. A CA3080 operational transconductance amplifier is used as a voltage comparator. The reference for the comparator is established by setting the potentiometer R so that the voltage at terminal 3 is more positive than that at terminal 2 when the motor speed is too low. An error voltage E_1 is derived from a tachometer driven by the motor. When the motor speed is too low, the voltage at terminal 2 of the voltage comparator is less positive than that at terminal 3, and the output voltage at terminal 6 goes "high". When the motor speed is too high, the opposite input conditions exist, and the output voltage at terminal 6 goes "low". Fig. 27(b) also shows these conditions graphically, with a linear transition region between the "high" and "low" output levels. This linear transition region is known as "proportional bandwidth". The slope of this region is determined by the proportional bandwidth control to establish the error-correction response time.

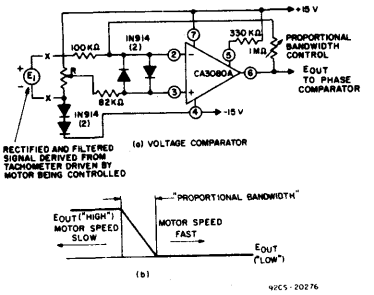


Fig. 27—Motor speed error detector.

Synchronous Ramp Generator. Fig. 28 shows a schematic diagram and signal waveforms for a synchronous ramp generator suitable for use with the motor-controller circuit of Fig. 26. Terminal 3 is biased at approximately +2.7 volts (above the negative supply voltage). The input signal E_{IN} at terminal 2 is a sample of the half-sinusoids (at line frequency) used to power the motor in Fig. 26. A synchronous ramp signal is produced by using the CA3094 to charge and discharge capacitor C_1 in response to the synchronous toggling of E_{IN}.

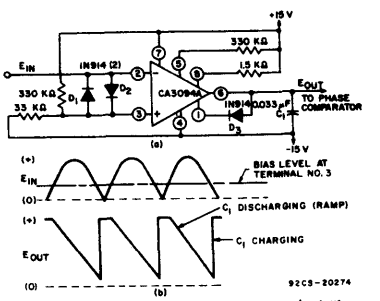


Fig.28—Synchronous ramp generator with input and output waveforms.

The charging current for C_1 is supplied by terminal 6. When terminal 2 swings more positive than terminal 3, transistors Q_{12} and Q_{13} in the CA3094 (Fig. 1) lose their base drive and become non-conductive. Under these conditions, C_1 discharges linearly through the external diode D_3 and the Q_{10}, D_6 path in the CA3094 to produce the ramp wave. The E_{out} signal is supplied to the phase comparator in Fig. 26.

Thyristor Firing Circuits

Temperature Controller. In the temperature control system shown in Fig. 29, the differential input of the CA3094 is connected across a bridge circuit comprised of a PTC (positive-temperature-coefficient) temperature sensor, two 75-kilohm resistors, and an arm containing the temperature set control. When the temperature is "low", the resistance of the PTC-type sensor is also low; therefore, terminal 3 is more positive than terminal 2 and an output current from terminal 6 of the CA3094 drives the triac into conduction. When the temperature is "high", the input conditions are reversed and the triac is cut off. Feedback from terminal 8 provides hysteresis to the control point to prevent rapid cycling of the system. The 1.5-kilohm resistor between terminal 8 and the positive supply limits the triac gate current and develops the voltage for the hysteresis feedback. The excellent power-supply-rejection and common-mode-rejection ratios of the CA3094 permit accurate repeatability of control despite appreciable power-supply ripple. The circuit of Fig. 29 is equally suitable for use with NTC (negative-temperature-coefficient) sensors provided the positions of the sensor and the associated resistor R are interchanged in the circuit. The diodes connected back-to-back across the input terminals of the CA3094 protect the device against excessive differential input signals.

Thyristor Control from AC-Bridge Sensor. Fig. 30 shows a line-operated thyristor-firing circuit controlled by a CA3094 that operates from an ac-bridge sensor. This circuit is particularly suited to certain classes of sensors that cannot be operated from dc. The CA3094 is inoperative when the high side of the ac line is negative because there is no I_{ABC} supply to terminal 5. When the sensor bridge is unbalanced so that terminal 2 is more positive than terminal 3, the output stage of the CA3094 is cut off when the ac line swings positive, and the output level at terminal 8 of the CA3094 goes "high". Current from the line flows through the 1N3193 diode to charge the 100-microfarad reservoir capacitor, and also provides current to drive the triac into conduction. During the succeeding negative swing of the ac line, there is sufficient remanent energy in the reservoir capacitor to maintain conduction in the triac.

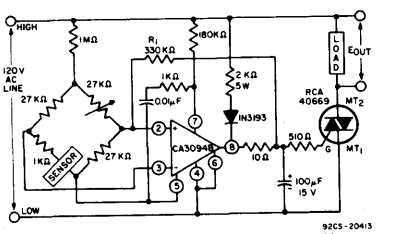


Fig.30—Line-operated thyristor-firing circuit controlled by ac-bridge sensor.

When the bridge is unbalanced in the opposite direction so that terminal 3 is more positive than terminal 2, the output of the CA3094 at terminal 8 is driven sufficiently "low" to "sink" the current supplied through the 1N3193 diode so that the triac gate cannot be triggered. Resistor R_1 supplies the hysteresis feedback to prevent rapid cycling between turn-on and turn-off.

Battery-Charger Regulator Circuit

The circuit for a battery-charger regulator circuit using the CA3094 is shown in Fig. 31. This circuit accurately limits the peak output voltage to 14 volts, as established by the zener

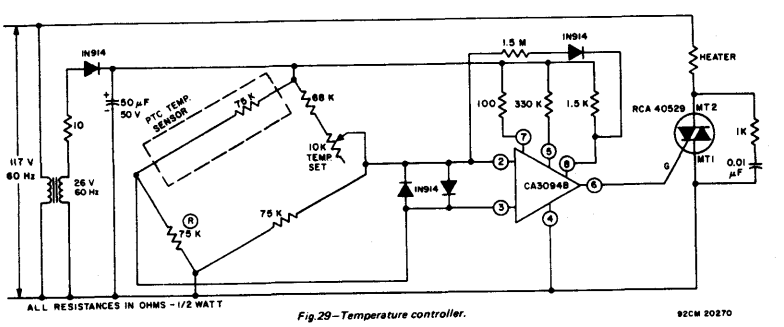


Fig.29—Temperature controller.

diode connected across terminals 3 and 4. When the output voltage rises slightly above 14 volts, signal feedback through a 100-kilohm resistor to terminal 2 reduces the current drive supplied to the 2N3054 pass transistor from terminal 6 of the CA3094. An incandescent lamp serves as the indicator of charging-current flow. Adequate limiting provisions protect the circuit against damage under load-short conditions. The advantage of this circuit over certain other types of regulator circuits is that the reference voltage supply doesn't drain the battery when the power supply is disconnected. This feature is important in portable service applications, such as in a trailer where a battery is kept "on-charge" when the trailer is parked and power is provided from an ac line.

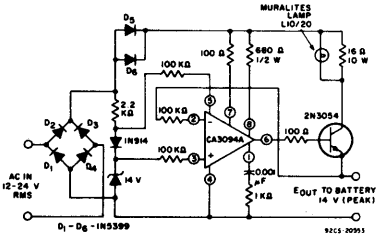
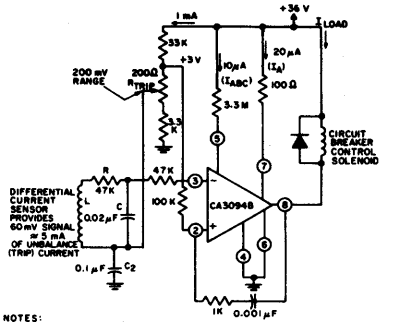


Fig.31–Battery-charger regulator circuit.

Ground-Fault Interrupters (GFI)

Ground-fault-interrupter systems are used to continuously monitor the balance of current between the high and neutral lines of power-distribution networks. Power is interrupted whenever the unbalance exceeds a preset value (e.g., 5 milliamperes). An unbalance of current can occur when, for example, defective insulation in the high side of the line permits leakage of current to an earth ground. GFI systems can be used to reduce the danger of electrocution from accidental contact with a "high" line because the unbalance caused by the leakage of current from the "high" line through a human body to ground results in an interruption of current flow.

The CA3094 is ideally suited for GFI applications because it can be operated from a single supply, has adequate sensitivity, and can drive a relay or thyristor directly to effect power interruption. Fig. 32 shows a typical GFI circuit. Vernier adjustment of the trip point is made by the R_{TRIP} potentiometer. When the differential current sensor supplies a signal that exceeds the selected trip-point voltage level (e.g., 60 millivolts), the CA3094 is toggled "on" and terminal 8 goes "low" to energize the circuit-breaker trip coil. Under quiescent conditions, the entire circuit consumes approximately 1 milliampere. The resistor R, connected to one leg of the current sensor, provides current limiting to protect the CA3094 against voltage spikes as large as 100 volts. Fig. 32 also shows the pertinent waveform for the GFI circuit.



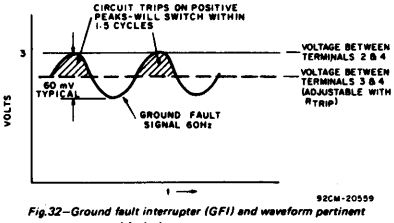


Fig.32–Ground fault interrupter (GFI) and waveform pertinent to ground fault detector.

Because hazards of severe electrical shock are a potential danger to the individual user in the event of malfunctions in GFI apparatus, it is mandatory that the highest standards of good engineering practice be employed in designing equipment for this service. Every consideration in design and application must be given to the potentially serious consequences of component malfunction in such equipment. Use of "reliability-through-redundancy" concepts and so-called "fail-safe" features is encouraged.

Acknowledgments

The authors thank A. Sheng and R. Baird for their assistance in designing some of the circuits described in this Note.

References

1. RCA Published Data for CA3080 and CA3080A, File No. 475.
2. Applications of the CA3080 and CA3080A High Performance Operational Transconductance Amplifiers, RCA Application Note ICAN-6668.
3. L. Kaplan and H. A. Wittlinger, "An IC Operational-Transconductance-Amplifier (OTA) with Power Capability". Paper originally presented at the IEEE Chicago Spring Conference on Broadcast and TV Receivers, June 1972. Reprinted by the RCA Solid State Division as Publication No. ST-6077.
4. RCA Published Data for CA3085, File No. 491.
5. RCA Published Data for CA3083, File No. 481.

An IC Operational-Transconductance-Amplifier (OTA) With Power Capability

by L. Kaplan and H. Wittlinger

In 1969, RCA introduced the first triple operational-transconductance-amplifier or OTA. The wide acceptance of this new circuit concept prompted the development of the single, highly linear operational-transconductance-amplifier, the CA3080. Because of its extremely linear transconductance characteristics with respect to amplifier bias current, the CA3080 gained wide acceptance as a gain-control block. The CA3094 improved on the performance of the CA3080 through the addition of a pair of transistors; these transistors extended the current-carrying capability to 300 milliamperes, peak. This new device, the CA3094, is useful in an extremely broad range of circuits in consumer and industrial applications; this paper describes only a few of the many consumer applications.

WHAT IS AN OTA?

The OTA, operational-transconductance-amplifier, concept is as basic as the transistor; once understood, it will broaden the designer's horizons to new boundaries and make realizable designs that were previously unobtainable. Fig. 1 shows an equivalent diagram of the OTA. The differential input circuit is the same as that found on many modern operational amplifiers. The remainder of the OTA is composed of current mirrors as shown in Fig. 2. The geometry of these mirrors is such that the current gain is unity. Thus, by highly degenerating the current mirrors, the output current is precisely defined by the differential-input amplifier. Fig. 3 shows the output-current transfer-characteristic of the amplifier. The shape of this characteristic remains

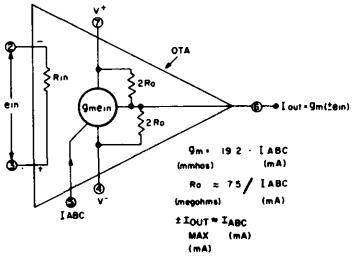


Fig. 1– Equivalent diagram of the OTA.

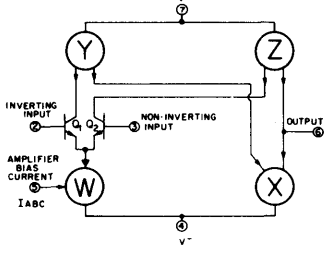


Fig. 2– Current mirrors W, X, Y, and Z used in the OTA.

constant and is independent of supply voltage. Only the maximum current is modified by the bias current.

The major controlling factor in the OTA is the input amplifier bias current I_{ABC}; as explained in Fig. 1, the total output current and gm are controlled by this current. In addition, the input bias current, input resistance, total supply current, and output resistance are all proportional to this amplifier bias current. These factors provide the key to the performance of this most flexible device, an idealized

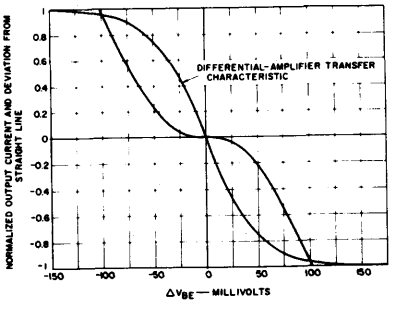


Fig. 3– The output-current transfer-characteristic of the OTA is the same as that of an idealized differential amplifier.

differential amplifier, i.e., a circuit in which differential input to single-ended output conversion can be realized. With this knowledge of the basics of the OTA, it is possible to explore some of the applications of the device.

DC Gain Control

The methods of providing dc gain-control functions are numerous. Each has its advantage – simplicity, low cost, high level control, low distortion. Many manufacturers who have nothing better to offer propose the use of a four-quadrant multiplier. This is analogous to using an elephant to carry a twig. It may be elegant but it takes a lot to keep it going! When operated in the gain-control mode, one input of the standard transconductance multiplier is offset so that only one half of the differential input is used; thus, one-half of the multiplier is being thrown away.

The OTA, while providing excellent linear amplifier characteristics, does provide a simple means of gain control. For this application the OTA may be considered the realization of the ideal differential amplifier in which the full differential amplifier gm is converted to a single-ended output. Because the differential amplifier is ideal, its gm is directly proportional to the operating current of the differential-amplifier; in the OTA the maximum output current is equal to the amplifier bias current I_{ABC}. Thus, by varying the amplifier bias current, the amplifier gain may be varied: $A = Gm\ R_L$ where R_L is the output load resistance. Fig. 4 shows the basic configuration of the OTA dc gain-control circuit.[1]

As long as the differential input signal to the OTA remains under 50-millivolts peak-to-peak, the deviation from a linear transfer will remain under 5 percent. Of course, the total harmonic distortion will be considerably less than this value. Signal excursions beyond this point only result in an undesired "compressed" output. The reason for this compression can be seen in the transfer characteristic of the differential amplifier in Fig. 3. Also shown in Fig. 3 is a curve depicting the departure from a linear line of this transfer characteristic.

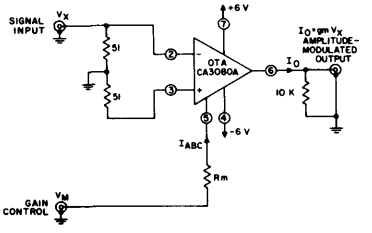


Fig. 4– Basic configuration of the OTA dc gain-control circuit.

The actual performance of the circuit shown in Fig. 4 is plotted in Fig. 5. Both signal-to-noise ratio and total harmonic distortion are shown as a function of signal input. Figs. 5(b) and (c) show how the signal-handling capability of the circuit is extended through the connection of diodes on

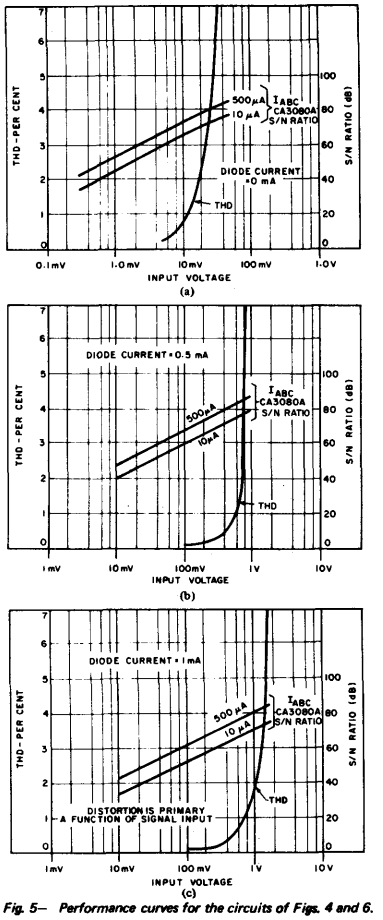


Fig. 5– Performance curves for the circuits of Figs. 4 and 6.

the input as shown in Fig. 6.[2] Fig. 7 shows total system gain as a function of amplifier bias current for several values of diode current. Fig. 8 shows an oscilloscope photograph of the CA3080 transfer characteristic as applied to the circuit of Fig. 4. The oscilloscope photograph of Fig. 9 was obtained with the circuit shown in Fig. 6. Note the improvement in

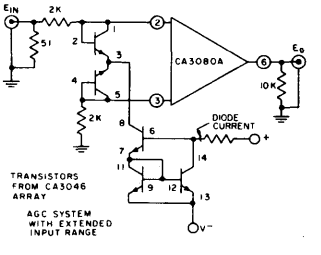


Fig. 6– A circuit showing how the signal-handling capability of the circuit of Fig. 4 can be extended through the connection of diodes on the input.

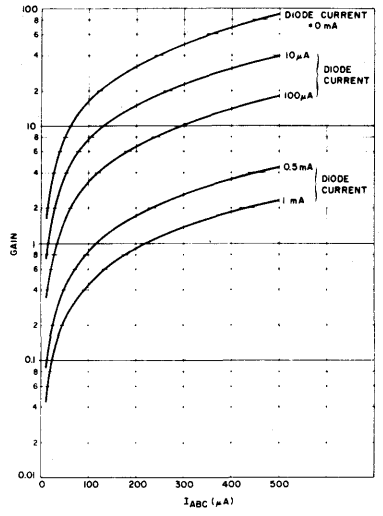


Fig. 7– Total system gain as a function of amplifier bias current for several values of diode current.

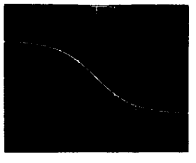


Fig. 8– Oscilloscope photograph of the CA3080 transfer characteristic as applied to the circuit of Fig. 4.

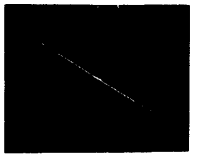


Fig. 9– Oscilloscope photograph of the CA3080 transfer characteristic as applied to the circuit of Fig. 6.

linearity of the transfer characteristic. Reduced input impedance does result from this shunt connection. Similar techniques could be used on the OTA output, but then the output signal would be reduced and the correction circuitry further removed from the source of non-linearity. It must be emphasized that the input circuitry is differential.

Simplified Differential-Input to Single-Ended Output Conversion

One of the more exacting configurations for operational amplifiers is the differential-to-single-ended conversion circuit. Fig. 10 shows some of the basic circuits that are usually employed. The ratios of the resistors must be precisely matched to assure the desired common-mode rejection. Fig. 11 shows another system using the CA3080 to obtain this conversion without the use of precision resistors. Differential input signals must be kept under ±26-millivolts for better than 5-percent non-linearity. The common-mode range is that of the CA3080 differential amplifier. In addition, the gain characteristic follows the standard differential-amplifier Gm-temperature coefficient of -0.3%/°C. Although the system of Fig. 11 does not provide the precise gain control obtained with the standard operational-amplifier approach, it does provide a good simple compromise suitable for many differential transducer-amplifier applications.

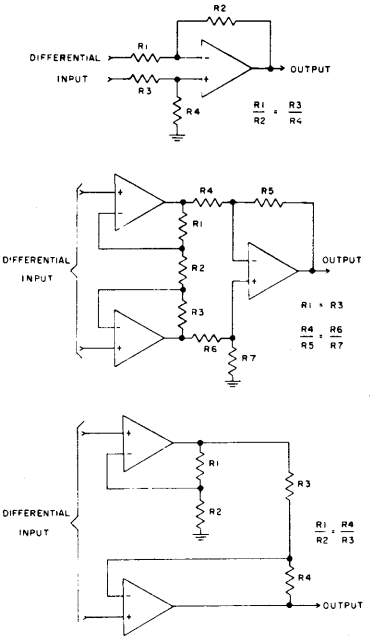


Fig. 10– Some typical differential-to-single-ended conversion circuits.

THE CA3094

The RCA CA3094 offers a unique combination of characteristics that suit it ideally to use as a programmable gain block for audio power amplifiers. It is a transconductance amplifier in which gain and open-loop bandwidth can be controlled between wide limits. The device has a large reserve of output-current capability, and breakdown and power-dissipation ratings sufficiently high to allow it to drive a complementary pair of transistors. For example, a 12-watt power-amplifier stage (8-ohm load) can be driven with peak currents of 35 milliamperes (assuming a minimum output-transistor beta of 50) and supply voltages of ±18 volts. In this application, the RCA CA3094A is operated substantially below its supply-voltage rating of 44-volts max. and its dissipation rating of 1.6-watts max. Also in this application, a high value of open-loop gain suggests the possibility of precise adjustment of frequency-response characteristics by adjustment of impedances in the feedback networks.

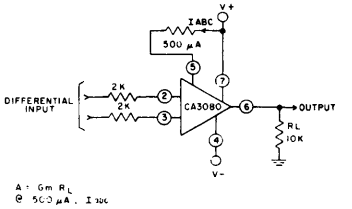


Fig. 11– A differential-to-single-ended conversion circuit without precision resistors.

Implicit Tone Controls

In addition to low distortion, the large amount of loop gain and flexibility of feedback arrangements available when using the CA3094 make it possible to incorporate the tone controls into the feedback network that surrounds the entire amplifier system. Consider the gain requirements of a phonograph playback system that uses a typical high-quality magnetic cartridge.[3] A desirable system gain would result in from 2 to 5 watts of output at a recorded velocity of 1 cm/s. Magnetic pickups have outputs typically ranging from 4 to 10 millivolts at 5 cm/s. To get the desired output, the total system needs about 72 dB of voltage gain at the reference frequency.

Fig. 12 is a block diagram of a system that uses a passive or "losser"-type of tone-control circuit that is inserted ahead of the gain control. Fig. 13 shows a system in which the tone controls are implicit in the feedback circuits of the power amplifier. Both systems assume the same noise input voltage at the equalizer and main-amplifier inputs. The feedback system shows a small improvement (3.8 dB) in signal-to-noise ratio at maximum gain but a dramatic improvement (20 dB) at the zero gain position. For purposes of comparison, the assumption is made that the tone controls are set "flat" in both cases.

Cost Advantages

In addition to the savings resulting from reduced parts count and circuit size, the use of the CA3094 leads to further savings in the power-supply system. Typical values of power-supply rejection and common-mode rejection are 90 and 100 dB, respectively. An amplifier with 40 dB of gain and 90 dB of power-supply rejection would require 316 millivolts of power-supply ripple to produce one millivolt of hum at the output. Thus, no further filtering is required other than that given by the energy-storage capacitor at the output of the rectifier system.

POWER AMPLIFIER USING THE CA3094

A complete power amplifier using the CA3094 and three additional transistors is shown schematically in Fig. 14. The amplifier is shown in a single-channel configuration, but power-supply values are designed to support a minimum of two channels. The output section comprises Q1 and Q2, complementary epitaxial units connected in the familiar "bootstrap" arrangement. Capacitor C3 provides added base drive for Q1 during positive excursions of the output. The circuit can be operated from a single power supply as well as from a split supply as shown in Fig. 15. The changes required for 14.4-volt operation with a 3.2-ohm speaker are also indicated in the diagram.

The amplifier may also be modified to accept input from ceramic phonograph cartridges. For standard inputs (equalizer preamplifiers, tuners, etc.) C1 is 0.047, R1 is 250 kilohms, and R2 and C2 are omitted. For ceramic-cartridge inputs, C1 is 0.0047, R1 is 2.5 megohms, and the jumper across C2 is removed.

Output Biasing

Instead of the usual two-diode arrangement for establishing idling currents in Q1 and Q2, a "V_{be} Multiplier", transistor Q3, is used. This method of biasing establishes the voltage between the base of Q1 and the base of Q2 at a constant multiple of the base-to-emitter voltage of a single transistor while maintaining a low variational impedance between its collector and emitter (see Appendix A). If transistor Q3 is mounted in intimate thermal contact with the output units, the operating temperature of the heat sink forces the V_{be} of Q3 up and down inversely with heat-sink temperature. The voltage bias between the bases of Q1 and Q2 varies inversely with heat-sink temperature and tends to keep the idling current in Q1 and Q2 constant.

A bias arrangement that can be accomplished at lower cost than those already described replaces the V_{be} multiplier with a 1N5391 diode in series with an 8.2-ohm resistor. This arrangement does not provide the degree of bias stability of the V_{be} multiplier, but is adequate for many applications.

Tone-Controls

The tone controls, the essential elements of the feedback system, are located in two sets of parallel paths. The bass

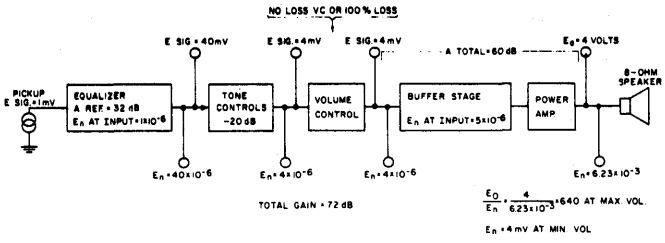


Fig. 12– Block diagram of a system using a "losser"-type tone-control circuit.

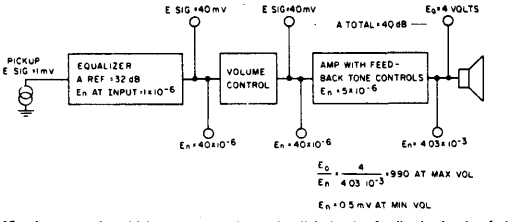


Fig. 13– A system in which tone controls are implicit in the feedback circuit of the power amplifier.

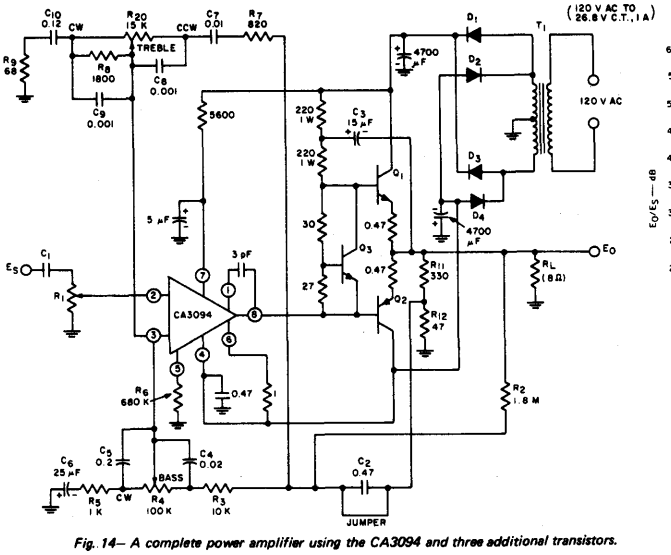


Fig. 14– A complete power amplifier using the CA3094 and three additional transistors.

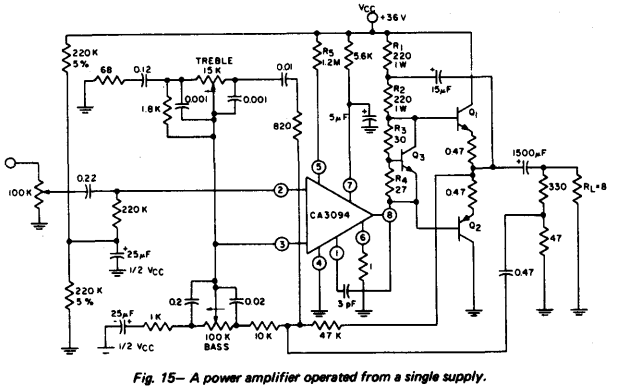


Fig. 15– A power amplifier operated from a single supply.

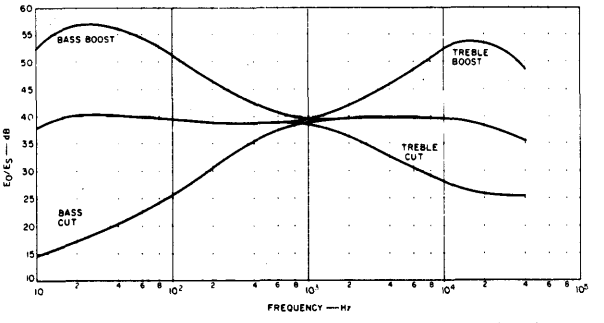


Fig. 16– The measured response of the amplifier at extremes of tone-control rotation.

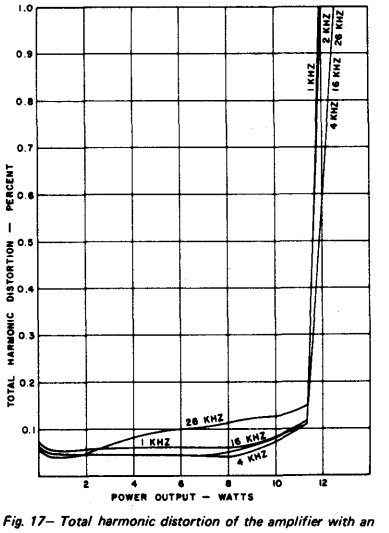


Fig. 17– Total harmonic distortion of the amplifier with an unregulated power supply.

network includes R3, R4, R5, C4, and C5. C6 blocks the dc from the feedback network so that the dc gain from input to the feedback takeoff point is unity. The residual dc-output-voltage at the speaker terminals is then $I_{ABC} R_1 \frac{R_{11} + R_{12}}{R_{12}}$ where R_1 is the source resistance. The input bias current is then $\frac{I_{ABC}}{2\beta} = -\frac{(V_{CC} - V_{be})}{2\beta R6}$. The treble network consists of R7, R8, R9, R10, C7, C8, C9, and C10. Resistors R7 and R9 limit the maximum available cut and boost, respectively. The boost limit is useful in curtailing heating due to finite turn-off time in the output units. The limit is also desirable when there are tape recorders nearby. The cut limit aids the stability of the amplifier by cutting the loop gain at higher frequencies where phase shifts become significant.

In cases in which absolute stability under all load conditions is required, it may be necessary to insert a small inductor in the output lead to isolate the circuit from capacitive loads. A 3-microhenry inductor (1 ampere) in parallel with a 22-ohm resistor is adequate. The derivation of circuit constants is shown in Appendix B. Curves of control action versus electrical rotation are also given.

Performance

Fig. 16 is a plot of the measured response of the complete amplifier at the extremes of tone-control rotation. A comparison of Fig. 16 with the computed curves of Fig. B4 (Appendix B) shows good agreement. The total harmonic distortion of the amplifier with an unregulated power supply is shown in Fig. 17; IM distortion is plotted in Fig. 18. Hum and noise are typically 700 microvolts at the output, or 83-dB down.

COMPANION RIAA PREAMPLIFIER

Many available preamplifiers are capable of providing the drive for the power amplifier of Fig. 14. Yet the unique characteristics of the amplifier – its power supply, input impedance, and gain – make possible the design of an RIAA

preamplifier that can exploit these qualities. Since the input impedance of the amplifier is essentially equal to the value of the volume-control resistance (250 kilohms), the preamplifier need not have high output-current capability. Because the gain of the power amplifier is high (40 dB) the preamplifier gain only has to be approximately 30 dB at the reference frequency (1 kHz) to provide optimum system gain.

Fig. 19 shows the schematic diagram of a CA3080 preamplifier. The CA3080, a low-cost OTA, provides sufficient open-loop gain for all the bass boost necessary in RIAA compensation. For example, a gm of 10,000 micromhos with a load resistance of 250 kilohms provides an

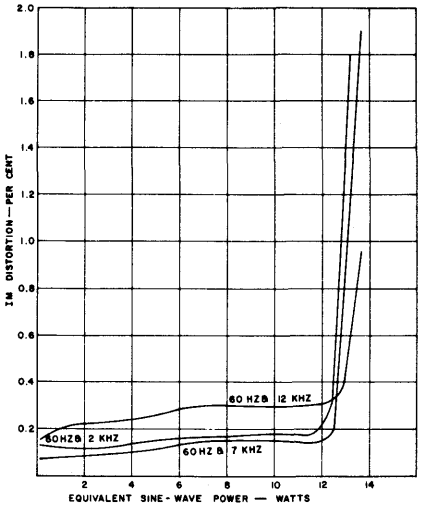


Fig. 18– IM distortion of the amplifier with an unregulated supply.

open-loop gain of 68 dB, thus allowing at least 18 dB of loop gain at the lowest frequency. The CA3080 can be operated from the same power supply as the main amplifier with only minimal decoupling because of the high power-supply rejection inherent in the device circuitry. In addition, the high voltage-swing capability at the output enables the CA3080 preamplifier to handle badly over-modulated (over-cut) recordings without overloading. The accuracy of equalization is within ±1 dB of the RIAA curve, and distortion is virtually unmeasurable by classical methods. Overload occurs at an output of 7.5 volts, which allows for undistorted inputs of up to 186 millivolts (260 millivolts peak).

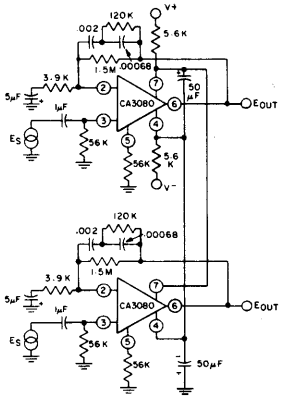


Fig. 19– A CA3080 preamplifier.

APPENDIX A – V_{be} MULTIPLIER

The equivalent circuit for the V_{be} multiplier is shown in Fig. A1. The voltage E_1 is given by:

$$E_1 = \frac{R1 I}{\beta+1} + V_{be}\left[1 + \frac{R1}{R2\,(\beta+1)}\right] \qquad (A1)$$

The value of V_{be} is itself dependent on the emitter current of the transistor, which is, in turn, dependent on the input current I since:

$$I_e = I - \frac{V_{be}}{R2} \qquad (A2)$$

The derivative of Eq. A1 with respect to I yields the incremental impedance of the V_{be} multiplier:

$$\frac{dE_1}{dI} = Z = \frac{R1}{\beta+1} + \left[1 + \frac{BR1}{(\beta+1)\,R2}\right]\left[\frac{K_3 R2}{R2 I_e + K_3}\right] \qquad (A3)$$

where K_3 is a constant of the transistor Q1 and can be found from:

$$V_{be} = K_3 \ln I_e - K_2 \qquad (A4)$$

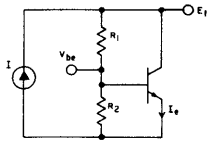


Fig. A1– Equivalent circuit for the V_{be} multiplier.

Eq. A4 is but another form of the diode equation:[4]

$$I_e = I_s\left(e^{\frac{qV_{be}}{KT}} - 1\right) \qquad (A5)$$

Using the values shown in Fig. 14 plus data on the 2N5494 (a typical transistor that could be used in the circuit), the dynamic impedance of the circuit at a total current of 40 milliamperes is found to be 4.6 ohms. In the actual design of the V_{be} multiplier, the value of IR2 must be greater than V_{be} or the transistor will never become forward biased.

APPENDIX B – TONE CONTROLS

Fig. B1 shows four operational-amplifier circuit configurations and the gain expressions for each. The asymptotic low-frequency gain is obtained by letting S approach zero in each case:

Bass Boost: $A_{Low} = \frac{R1+R2+R3}{R2}$

Bass Cut: $A_{Low} = \frac{R1+R2+R3}{R2+R3}$

Treble Boost: $A_{Low} = \frac{C1+C4}{C4}$

Treble Cut: $A_{Low} = \frac{C1+C4}{C4}$

The asymptotic high-frequency gain is obtained by letting S increase without limit in each expression:

Bass boost: $A_{High} = \frac{R1+R2}{R2}$

Bass cut: $A_{High} = \frac{R1+R2}{R2}$

Treble boost: $A_{High} = 1 + C1\left(\frac{C3+C4}{C3C4}\right)$

Treble cut: $A_{High} = \frac{C2 + \frac{C1C4}{C1+C4}}{C1+C2}$

Note that the expressions for high-frequency gain are identical for both bass circuits, while the expressions for low-frequency gain are identical for the treble circuits.

Fig. B2 shows cut and boost bass and treble controls that have the characteristics of the circuits of Fig. B1. The value R_{EFF} in the treble controls of Fig. B1 is derived from the parallel combination of R1 and R2 of Fig. B2 when the control is rotated to its maximum counterclockwise position. When the control is rotated to its maximum clockwise position, the value is equal to R1.

To compute the circuit constants, it is necessary to decide in advance the amounts of boost and cut desired. The gain expressions of Fig. B1 indicate that the slope of the amplitude versus frequency curve in each case will be 6 dB per octave (20 dB per decade). If the ratios of boosted and cut gain are set at 10, i.e.:

Bass circuit: $A_{Low(Boost)} = 10\,A_{Mid}$

$A_{Low\,(Cut)} = \frac{A_{Mid}}{10}$

Treble Circuit: $A_{High\,(Boost)} = 10\,A_{Mid}$

$A_{High(Cut)} = \frac{10\,A_{Mid}}{10}$

then the following relationships result:

Bass circuit: $R1 = 10\,R2$

$R3 = 99\,R2$

Treble circuit: $C1 = 10\,C4$

$C2 = \frac{10\,C4}{99}$

The unaffected portion of the gain (A high for the bass control and A low for the treble control) is 11 in each case.

To make the controls work symmetrically, the low- and high-frequency break points must be equal for both boost and cut.
Thus:

Bass Control: $\frac{C1\,R3\,(R1+R2)}{R1+R2+R3} = \frac{C2\,R2R3}{R2+R3}$

and $C1\,R3 = \frac{C2\,R3\,(R1+R2)}{R1+R2+R3}$

since $R3 \simeq R2+R3$, $C2 = 10C1$

Treble Control: $R1\,\frac{(C1C4+C3C4+C1C3)}{C1+C4}$

$= \frac{R1R2}{R1+R2}\,(C1+C2)$

and $R2C3 = \left(\frac{R1R2}{R1+R2}\right)\frac{(C1C4+C2C4+C1C2)}{(C1+C4)}$

since $C1 \simeq 100C2$, $C2 = C3$ and $C1 = 10C4$, $R1 = 9R2$

To make the controls work in the circuit of Fig. 14, breaks were set at 1000 Hz:

for the base control $0.1C1R3 = \frac{1}{2\pi \times 1000}$

and for the treble control $R1C3 = \frac{1}{2\pi \times 1000}$

Response and Control Rotation

In a practical design, it is desirable to make "flat" response correspond to the 50-percent rotation position of the control, and to have an aural sensation of smooth variation of response on either side of the mechanical center. It is easy to show that the "flat" position of the bass control occurs when the wiper arm is advanced to

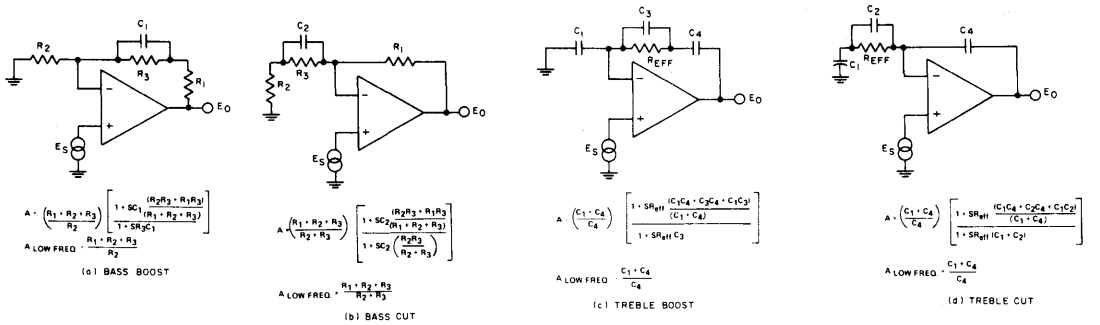


Fig. B1– Four operational-amplifier circuit configurations and the gain expressions for each.

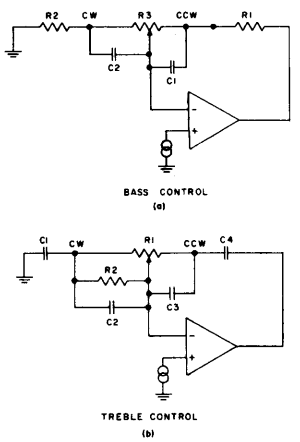


Fig. B2– Cut and boost bass and treble controls that have the characteristics of the circuits of Fig. B1.

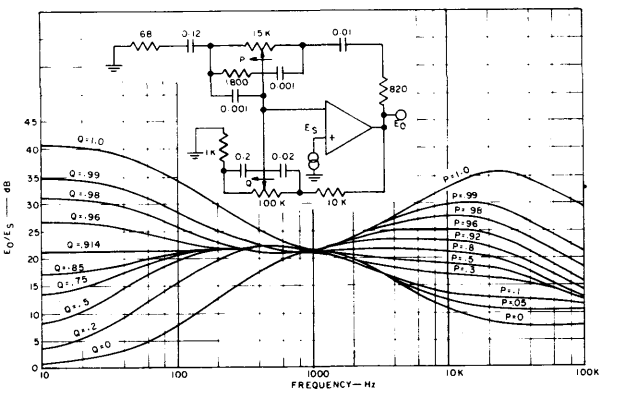


Fig. B3– A plot of the response of the circuit of Fig. 14 with bass and treble tone controls combined at various settings of both controls.

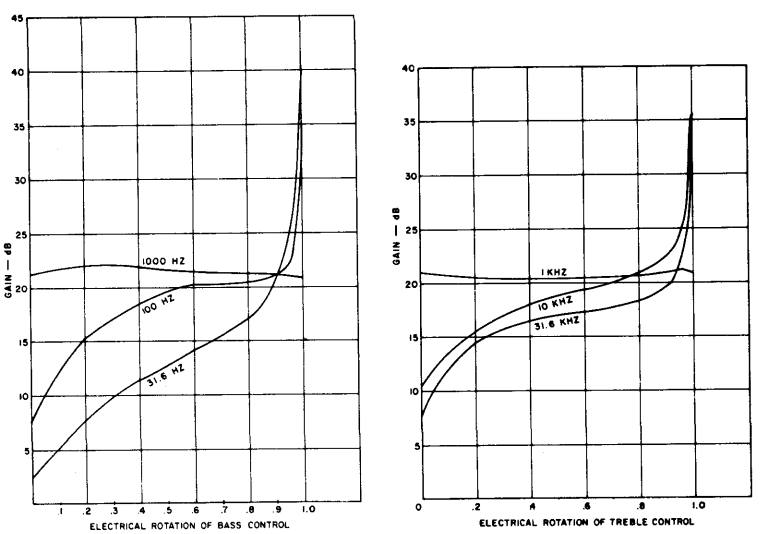


Fig. B4– The information of Fig. B3 plotted as a function of electrical rotation.

91-percent of its total resistance. The amplitude response of the treble control is, however, never completely "flat"; a computer was used to generate response curves as controls were varied.

Fig. B3 is a plot of the response with bass and treble tone controls combined at various settings of both controls. The values shown are the practical ones used in the actual design. Fig. B4 shows the information of Fig. B3 replotted as a function of electrical rotation. The ideal taper for each control would be the complement of the 100-Hz plot for the bass control and the 10-kHz response for the treble control. The mechanical center should occur at the crossover point in each case.

References*

1. "Applications of the CA3080 and CA3080A High-Performance Operational Transconductance Amplifiers," H. A. Wittlinger, RCA Application Note ICAN-6668
2. "A New Wide-band Amplifier Technique," B. Gilbert, IEEE Journal of Solid State Circuits, Vol. SC-3, No. 4, December, 1968.
3. "Trackability," James A. Kogar, *Audio*, December 1966
4. *RCA Linear Integrated Circuits Manual*, RCA Technical Series IC-42

*RCA publications available through RCA Solid State Division, Box 3200, Somerville, N.J., 08876.

Applications of the CA3085-Series Monolithic IC Voltage Regulators

by A. C. N. Sheng and L. R. Avery

The RCA-CA3085, CA3085A, and CA3085B monolithic IC's are positive-voltage regulators capable of providing output currents up to 100 milliamperes over the temperature range from −55° to +125°C. They are supplied in 8-lead TO-5 type packages; their characteristics and ratings are given in RCA Data File No. 491. The following tabulation shows some key characteristics and salient differences between devices in the CA3085 Series.

Type	$V_{IN}(V_I)$ Range V	$V_{OUT}(V_O)$ Range V	Max. $I_{OUT}(I_O)$ mA	Max. Load Regulation % V_O
CA3085	7.5–30	1.8–26	12*	0.1
CA3085A	7.5–40	1.7–36	100	0.15
CA3085B	7.5–50	1.7–46	100	0.15

* This value may be extended to 100 mA; however, regulation is not specified beyond 12 mA.

In addition to these differences, the range of some specified performance parameters is more tightly controlled in the CA3085B than in the CA3085A, and more in the CA3085A than in the CA3085.

This Note describes the basic circuit of the CA3085-series devices and some typical applications that include a high-current regulator, constant-current regulators, a switching regulator, a negative-voltage regulator, a dual-tracking regulator, high-voltage regulators, and various methods of providing current limiting. A circuit in which the CA3085 is used as a general-purpose amplifier is also shown.

CIRCUIT DESCRIPTION

The block diagram of the CA3085-series circuits is shown in Fig. 1. Fundamentally, the circuit consists of a frequency-compensated error-amplifier which compares an internally generated reference voltage with a sample of the output voltage and controls a series-pass amplifier to regulate the output. The starting circuit assures stable latch-in of the voltage-reference circuitry. The current-limiting portion of the circuit is an optional feature that protects the IC in the event of overload.

Terminal 5 provides a source of stable reference voltage for auxiliary use; a current of about 250 microamperes can be supplied to an external circuit without significantly disturbing reference-voltage stability. If necessary, filtering of the inherent noise of the reference-voltage circuit can be accomplished by connecting a suitable bypass capacitor between terminals 5 and 4.

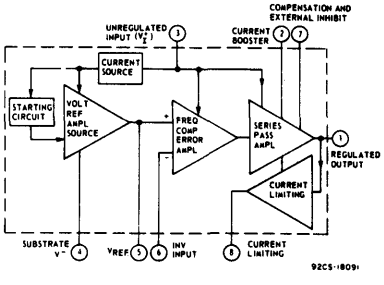


Fig. 1– Block diagram of CA3085 series.

Terminal 6 (the "inverting input" in accordance with operational-amplifier terminology) is the input through which a sample of the regulated output voltage is applied.

The collector of the series-pass output transistor is brought out separately at terminal 2 ("current booster") to provide base drive for an external p-n-p transistor; this approach is one method of regulating currents greater than 100 milliamperes.

Because the voltage regulator is essentially an operational amplifier having considerable feedback, frequency compensation is necessary in some circuits to prevent oscillations. Terminal 7 is provided for external frequency compensation; it can also be used to "inhibit" (strobe, squelch, pulse, key) the operation of the series-pass amplifier.

Brief Description of CA3085 Schematic Diagram

The schematic diagram of the CA3085-series circuits is shown in Fig. 2. The left-hand section includes the starting circuit, the voltage-reference circuit, and the constant-current circuit. The center section is basically an elementary operational amplifier which serves as the voltage-error amplifier. It controls the series-pass Darlington pair (Q13, Q14) shown in the right-hand section. When controlled by an appropriate external sensing network, transistor Q15 serves to provide protective current-limiting characteristics by diverting base drive from the series-pass circuit. For operation at the highest current levels, terminals 2 and 3 are tied together to eliminate the voltage drop which would otherwise be developed across resistor R5.

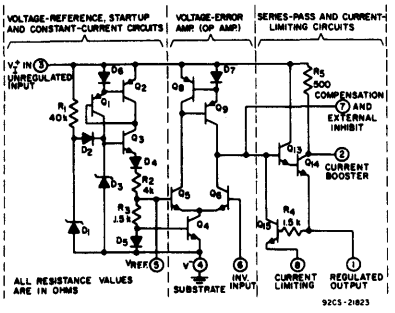


Fig. 2– Schematic diagram of CA3085 series.

Voltage-Reference Circuits

The basic voltage-reference element used in the CA3085 is zener diode D3. It provides a nominal reference voltage of 5.5 volts and exhibits a positive temperature coefficient of approximately 2.5 millivolts/°C. If this reference voltage were used directly in conjunction with the error-amplifier (Q5, Q6, etc.), the IC would exhibit two major undesirable characteristics: (1) its performance with temperature variations would be poor, and (2) its use as a regulator would be restricted to circuits in which the minimum regulated output voltages are in excess of 5.5 volts. Consequently, it is necessary to provide means of compensating for the positive temperature coefficient of D3 and at the same time provide for obtaining a stable source of lower reference voltage. Both temperature compensation and the reduction of the reference voltage are accomplished by means of the series divider network consisting of the base-emitter junction of Q3, diode D4, resistors R2 and R3, and diode 5.

The voltage developed across D3 drives the divider network and a voltage of approximately 4 volts is developed between the cathode of D4 and the cathode of D5 (terminal 4). The current through this divider network is held nearly constant with temperature because of the combined temperature coefficients of the zener diode (D3), Q3 base-emitter junction, D4 D5, and the resistors R2 and R3. This constant current through the diode D5 and the resistor R3 produces a voltage drop between terminals 4 and 5 that results in the reference voltage (≈ 1.6 volts) having an effective temperature coefficient of about 0.0035 per cent/°C.

The reference diode D3 receives a current of approximately 620 microamperes from a constant-current circuit consisting of Q3 and the current-mirror* D6, Q1, and Q2. Current to startup the constant-current source initially is provided by auxiliary zener diode D1 and R1. Diode D2 blocks current from the R1-D1 source after latch-in of the constant-current source establishes a stable reference potential, and thereby prevents modulation of the reference voltage by ripple voltage on the unregulated input voltage.

Voltage-Error Amplifier

Transistors Q5 and Q6 comprise the basic differential amplifier that is used as a voltage-error amplifier to compare the stable reference voltage applied at the base of Q5 with a sample of the regulator output voltage applied at terminal 6. The D5-Q4 combination is a current-mirror which maintains essentially constant-current flow to Q5 and Q6 despite variations in the unregulated input voltage. The Q8, Q9, and D7 network provides a "mirrored" active collector load for Q5 and Q6 and also provides a variable single-ended drive to the Q13 and Q14 series-pass transistors in accordance with the difference signal developed between the bases of Q5 and Q6. The open-loop gain of the error-amplifier is greater than 1000.

Series-Pass and Current-Limiting Circuits

In the normal mode of operation, or in the current-boost mode when terminals 2 and 3 are tied together, the Darlington pair Q13-Q14 performs the basic series-pass regulating function between the unregulated input voltage and the regulated output voltage at terminal 1. In the current-limiting mode transistor Q15 provides current-limiting to protect the CA3085 and/or limit the load current. To provide current-limiting protection, a resistor (e.g., 5 ohms) is connected between terminals 1 and 8; terminal 8 becomes the source of regulated output voltage. As the voltage drop across this resistor increases, base drive is supplied to transistor Q15 so that it becomes increasingly conductive and diverts base drive from the Q13-Q14 pass transistor to reduce output current accordingly. Resistor R4 is provided to protect Q15 against overdrive by limiting its base current under transient and load-short conditions.

Because the CA3085 regulator is essentially an op-amp having considerable feedback, frequency compensation may be required to prevent oscillations. Stability must also be maintained despite line and load transients, even during operation into reactive loads (e.g., filter capacitors). Provisions are included in the CA3085 so that a small-value capacitor may be connected between terminals 6 and 7 to compensate the regulator, when necessary, by "rolling-off" the amplifier frequency-response. Terminal 7 is also used to externally "inhibit" operation of the CA3085 by diverting base current supplied to Q13-Q14, thereby permitting the use of keying, strobing, programming, and/or auxiliary overload-protection circuits.

* The fundamentals of current-mirror theory are reviewed in the Appendix of Application Note ICAN-6668.

APPLICATIONS

A Simple Voltage Regulator

Fig. 3 shows the schematic diagram of a simple regulated power supply using the CA3085. The ac supply voltage is stepped down by T1, full-wave rectified by the diode bridge circuit, and smoothed by the large electrolytic capacitor C1 to provide unregulated dc to the CA3085 regulator circuit. Frequency compensation of the error-amplifier is provided by capacitor C2. Capacitor C3 bypasses residual noise in the reference-voltage source, and thus decreases the incremental noise-voltage in the regulator circuit output.

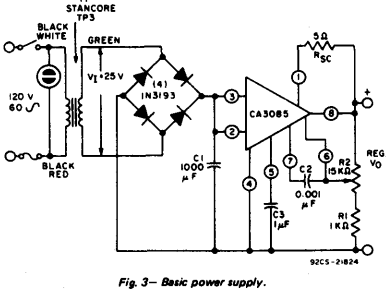


Fig. 3– Basic power supply.

Because the open-loop gain of the error-amplifier is very high (greater than 1000), the output voltage may be directly calculated from the following expression:

$$V_O = \frac{(R2 + R1)}{R1} V_{ref} \qquad (1)$$

In the circuit shown in Fig. 3, the output voltage can be adjusted from 1.8 volts to 20 volts by varying R2. The maximum output current is determined by R_{SC}; load-regulation charac-

teristics for various values of R_{SC} are shown in Fig. 4.

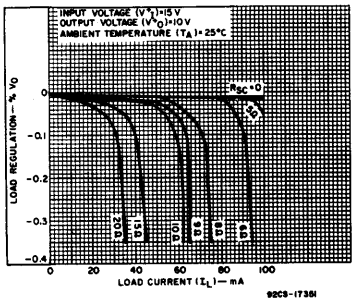


Fig. 4– Load regulation characteristics for circuit of Fig. 3.

When this circuit is used to provide high output currents at low output voltages, care must be exercised to avoid excessive IC dissipation. In the circuit of Fig. 3, this dissipation current can be accomplished by increasing the primary-to-secondary transformer ratio (a reduction in V_I) or by using a dropping resistor between the rectifier and the CA3085 regulator. Fig. 5 gives data on dissipation limitation (V_I-V_O vs. I_O) for CA3085-series circuits.

The short-circuit current is determined as follows:

$$I_{SC} = \frac{V_{BE}}{R_{SC}} \approx \frac{0.7}{R_{SC}} \text{ amperes} \qquad (2)$$

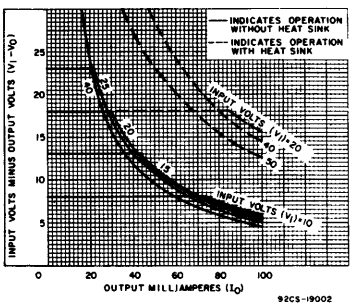


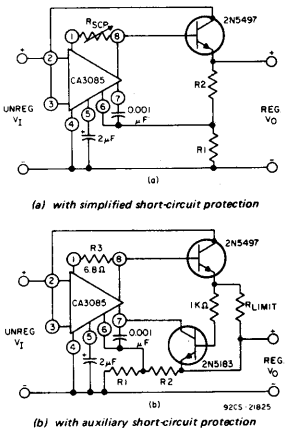
Fig. 5– Dissipation limitation ($V_I - V_O$ vs. I_O) for CA3085 series circuits.

The line- and load-regulation characteristics for the circuit shown in Fig. 3 are approximately 0.05 per cent of the output voltage.

High-Current Voltage Regulator

When regulated voltages at currents greater than 100 milliamperes are required, the CA3085 can be used in conjunction with an external n-p-n pass-transistor as shown in the circuits of Fig. 6. In these circuits the output current available from the regulator is increased in accordance with the h_{FE} of the external n-p-n pass-transistor. Output currents up to 8 amperes can be regulated with these circuits. A Darlington power transistor can be substituted for the 2N5497 transistor when currents greater than 8 amperes are to be regulated.

A simplified method of short-circuit protection is used in connection with the circuit of Fig. 6(a). The variable resistor R_{SCP} serves two purposes: (1) it can be adjusted to optimize the base drive requirements (h_{FE}) of the particular 2N5497 transistor being used, and (2) in the event of a short-circuit in the regulated output voltage the base drive current in the 2N5497 will increase, thereby increasing the voltage drop across R_{SCP}. As this voltage-drop increases the short-circuit protection system within the CA3085 correspondingly reduces the output current available at terminal 8, as described previously. It should be noted that the degree of short-circuit protection depends on the value of R_{SCP}, i.e., design compromise is required in choosing the value of R_{SCP} to provide the desired base drive for the 2N5497 while maintaining the desired short-circuit protection. Fig. 6(b) shows an alternate circuit in which an additional transistor (2N5183) and two resistors have been added as an auxiliary short-circuit protection feature. Resistor R3 is used to establish the desired base drive for the 2N5497, as described above. Resistor R_{limit} now controls the short-circuit output current because, in the event of a short-circuit, the voltage drop developed across its terminals increases sufficiently to increase the base drive to the 2N5183 transistor. This increase in base drive results in reduced output from the CA3085 because collector current flow in the 2N5183 diverts base drive from the Darlington output stage of the CA3085 (see Fig. 2) through terminal 7. The load regulation of this circuit is typically 0.025 per cent with 0 to 3-ampere load-current variation; line regulation is typically 0.025 per cent/volt change in input voltage.



(a) with simplified short-circuit protection

(b) with auxiliary short-circuit protection

Fig. 6– High-current voltage regulator using n-p-n pass transistor.

Voltage Regulator with Low V_I-V_O Difference

In the voltage regulators described in the previous section, it is necessary to maintain a minimum difference of about 4 volts between the input and output voltages. In some applications this requirement is prohibitive. The circuit shown in Fig. 7 can deliver an output current in the order of 2 amperes with a V_I-V_O difference of only one volt.

It employs a single external p-n-p transistor having its base and emitter connected to terminals 2 and 3, respectively, of the CA3085. In this circuit, the emitter of the output transistor (Q14 in Fig. 2) in the CA3085 is returned to the negative supply rail through an external resistor (R_{SCP}) and two series-connected diodes (D1, D2). These forward-biased diodes maintain Q6 in the CA3085 within linear-mode operation. The choice of resistors R1 and R2 is made in accordance with Eq. (1). Adequate frequency compensation for this circuit is provided by the 0.01-microfarad capacitor connected between terminal 7 of the CA3085 and the negative supply rail.

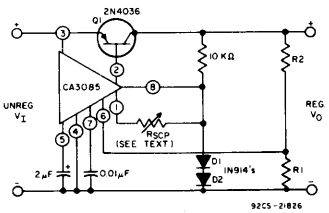


Fig. 7– Voltage regulator for low $V_I - V_O$ difference.

Fig. 8, which shows the output impedance of the circuit of Fig. 7 as a function of frequency, illustrates the excellent ripple-rejection characteristics of this circuit at frequencies below 1 kHz. Lower output impedances at the higher frequencies can be provided by connecting an appropriate capacitor across the output voltage terminals. The addition of a capacitor will, however, degrade the ability of the system to react to transient-load conditions.

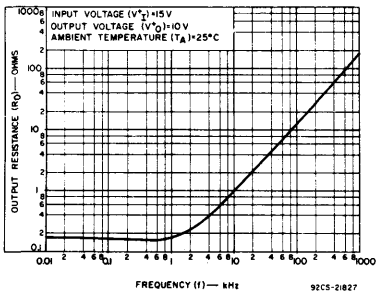


Fig. 8– Output resistance vs. frequency for circuit of Fig. 7.

High-Voltage Regulator

Fig. 9 shows a circuit that uses the CA3085 as a voltage-reference and regulator control device for high-voltage power supplies in which the voltages to be regulated are well above the input-voltage ratings of the CA3085-series circuits. The external transistors Q1 and Q2 require voltage ratings in excess of the maximum input voltage to be regulated. Series-pass transistor Q2 is controlled by the collector current of Q1, which in turn is controlled by the normally regulated current output supplied by the CA3085. The input voltage for the CA3085 regulator at terminal 3 is supplied through dropping resistor R3 and the clamping zener diode D1. The values for resistor R1 and R2 are determined in accordance with Eq. (1).

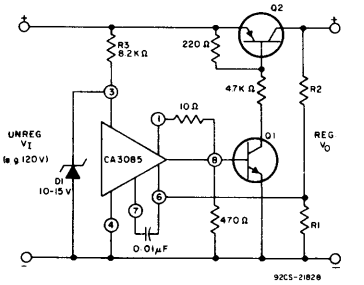


Fig. 9– High-voltage regulator.

Negative-Voltage Regulator

The CA3085 is used as a negative-supply voltage regulator in the circuit shown in Fig. 10. Transistor Q3 is the series-pass transistor. It should be noted that the CA3085 is effectively connected across the load-side of the regulated system. Diode D1 is used initially in a "circuit-starter" function; transistor Q2 "latches" D1 out of its starter-circuit function so that the CA3085 can assume its role in controlling the pass-transistor Q3 by means of Q1.

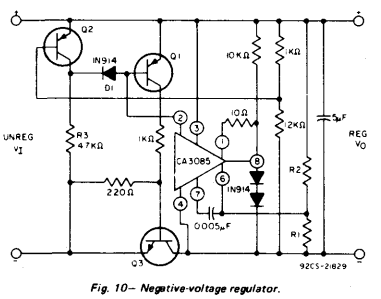


Fig. 10– Negative-voltage regulator.

Operation of the circuit is as follows: current through R3 and D1 provides base drive for Q1, which in turn provides

base drive for the pass-transistor Q3. By this means operating potential for the CA3085 is developed between the collector of Q3 (terminal 4 of the CA3085) and the positive supply-rail (terminal 3 of the CA3085). When the output voltage has risen sufficiently to maintain operation of the CA3085 (approx. 7.5 volts), transistor Q2 is driven into conduction by the base drive supplied from the 1 kilohm-12 kilohm voltage divider. As Q2 becomes conductive, it diverts the base drive being supplied to Q1 through the R3-D1 path, and diode D1 ceases to conduct. Under these conditions, base-current drive to Q1 through terminal 2 of the CA3085 regulates the base drive to Q3. Values of R1 and R2 are determined in accordance with Eq. (1).

The circuit shown in Fig. 11 is similar to that of Fig. 10, except for the addition of a constant-current limiting circuit

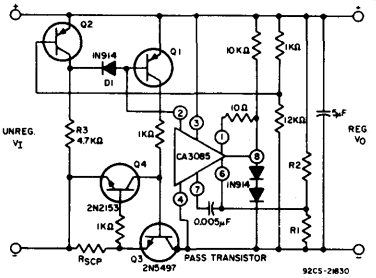


Fig. 11– Negative-voltage regulator with constant-current limiting circuit.

consisting of transistor Q4, a 1-kilohm resistor, and resistor R_{SCP}. When the load current increases above a particular design value, the corresponding increase in the voltage drop across resistor R_{SCP} provides additional base drive to transistor Q4. Thus, as transistor Q4 becomes increasingly conductive, its collector current diverts sufficient base drive from Q3 to limit the current in the pass transistor feeding the regulated load. With the types of transistors shown in Figs. 10 and 11, maximum currents in the order of 5 amperes can be regulated.

High-Output-Current Voltage Regulator With "Foldback" Current-Limiting (Also known as "Switch-Back" Current-Limiting)

In high-current voltage regulators employing constant-current limiting (e.g., Figs. 6 and 7), it is possible to develop excessive dissipation in the series-pass transistor when a short-circuit develops across the output terminals. This situation can be avoided by the use of the "foldback" current-limiting circuitry as shown in Fig. 12. In this circuit, terminal 8 of the

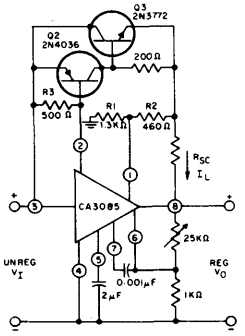


Fig. 12– High-output-current voltage regulator with "foldback" current limiting.

CA3085 senses the output voltage, and terminal 1 is tied to a tap on a voltage-divider network connected between the emitter of the pass-transistor (Q3) and ground. The current-foldback trip-point is established by the value of resistor R_{SC}.

The protective tripping action is accomplished by forward-biasing Q15 in the CA3085 (see Fig. 2). Conditions for tripping-circuit operation are defined by the following expressions:

$$V_{BE(Q15)} = (\text{voltage at terminal 1}) - (\text{output voltage})$$

$$= \left[(V_O + I_L R_{SC})\frac{R1}{R1 + R2}\right] - V_O \qquad (3)$$

If $\frac{R1}{R1 + R2} = K$, then

$$V_{BE(Q15)} = (V_O + I_L R_{SC})\,K - V_O = KV_O + KI_L R_{SC} - V_O$$

and therefore

$$R_{SC} = \frac{V_O + V_{BE(Q15)} - KV_O}{KI_L} \qquad (4)$$

Under load short-circuit conditions, terminal 8 is forced to ground potential and current flows from the emitter of Q14 in the CA3085, establishing terminal 1 at one V_{BE}-drop [≅ 0.7 V] above ground and Q15 in a partially conducting state. The current through Q14 necessary to establish this one-V_{BE} condition is the sum of currents flowing to ground through R1 and [R2 + R_{SC}]. Normally R_{SC} is much smaller than R2 and can be ignored; therefore, the equivalent resistance R_{eq} to ground is the parallel combination of R1 and R2. The Q14 current is then given by:

$$I_{Q14} = \frac{V_{BE(Q15)}}{R_{eq}} = \frac{V_{BE(Q15)}}{\frac{R1R2}{R1 + R2}} = \frac{0.7\ [\,1.3 + 0.46\,]}{1.3 \times 0.46} = 2.06 \text{ milliamperes}$$

This current provides a voltage between terminals 2 and 3 as follows:

$$V_{2-3} = I_{Q14} \times 250 \text{ ohms} = 2.06 \times 10^{-3} \times 250 = 0.515 \text{ volt}$$

The effective resistance between terminals 2 and 3 is 250 ohms because the external 500-ohm resistor R3 is in parallel with the internal 500-ohm resistor R5. It should be understood that the V_{2-3} potential of 0.515 volt is insufficient to maintain the external p-n-p transistor Q2 in conduction, and, therefore, Q3 has no base drive. Thus the output current is reduced to zero by the protective circuitry. Fig. 13 shows the foldback characteristic typical of the circuit of Fig. 12.

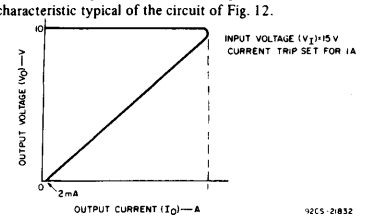


Fig. 13– Typical "foldback" current-limiting characteristic for circuit of Fig. 12.

An alternative method of providing "foldback" current-limiting is shown in Fig. 14. The operation of this circuit is similar to that of Fig. 12 except that the foldback-control transistor Q2 is external to the CA3085 to permit added flexibility in protection-circuit design.

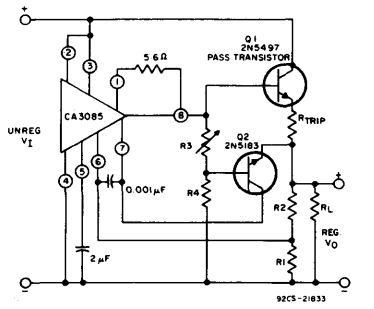


Fig. 14– High-output-current voltage regulator using auxiliary transistor to provide "foldback" current limiting.

Under low load conditions Q2 is effectively reverse-biased by a small amount, depending upon the values of R3 and R4. As the load current increases the voltage drop across R_{trip} increases, thereby raising the voltage at the base of Q1, and Q2 starts to conduct. As Q2 becomes increasingly conductive it diverts base current from transistors Q13 and Q14 in the CA3085, and thus reduces base drive to the external pass-transistor Q1 with a consequent reduction in the output voltage. The point at which current-limiting occurs, I_{trip}, is calculated as follows:

$V_{BE(Q1)}$ = voltage at terminal 8 – V_O (assuming a low value for R_{trip})

$V_{BE(Q2)}$ = voltage at terminal 8 $\left(\frac{R4}{R3 + R4}\right) - V_O$

$$= \left[V_O + I_L R_{trip} + V_{BE(Q1)}\right]\left[\frac{R4}{R3 + R4}\right] - V_O$$

if $K = \frac{R4}{R3 + R4}$, then the trip current is given by:

$$I_{trip} = \frac{V_{BE(Q2)} - K\left[V_O + V_{BE(Q1)}\right] + V_O}{KR_{trip}} \qquad (7)$$

In the circuit in Fig. 12 the load current goes to zero when a short circuit occurs. In the circuit of Fig. 14 the load current is significantly reduced but does not go to zero. The value for I_{SC} is computed as follows:

$$V_{BE(Q2)} + \left[\frac{V_{BE(Q2)}}{R2} + I_{B(Q2)}\right] R1 = V_{BE(Q1)} + I_{SC}R_{trip}$$

$$I_{SC} = \frac{V_{BE(Q2)} + \left[\frac{V_{BE(Q2)}}{R2} + I_{B(Q2)}\right] R1 - V_{BE(Q1)}}{R_{trip}} \qquad (8)$$

Fig. 15 shows that the transfer characteristic of the load current is essentially linear between the "trip-point" and the "short-circuit" point.

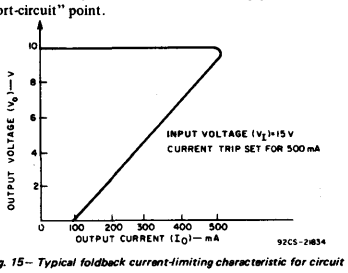


Fig. 15– Typical foldback current-limiting characteristic for circuit of Fig. 14.

High-Voltage Regulator Employing Current "Snap-Back" Protection

In high-voltage regulators (e.g., see Fig. 9), "foldback" current-limiting cannot be used safely because the high voltage across the pass transistor can cause second breakdown despite the reduction in current flow. To adequately protect the pass transistor in this type of high-voltage regulator, the so-called "snap-back" method of current limiting can be employed to reduce the current to zero in a few microseconds, and thus prevent second-breakdown destruction of the device.

The circuit diagram of a high-voltage regulator employing current "snap-back" protection is shown in Fig. 16. The basic regulator circuit is similar to that shown in Fig. 9. The additional circuitry in the circuit of Fig. 16 quickly interrupts base drive to the pass transistor in event of load fault. The point of current-trip is established as follows:

$$I_{trip} = \frac{V_{BE(Q1)}}{R_{SC}} \qquad (9)$$

Thus, when a sufficient voltage drop is developed across R_{sc}, transistor Q1 becomes conductive and current flows into the base of Q2 so that it also becomes conductive. Transistor Q3, in turn, is driven into conduction, thereby latching the Q2-Q3 combination (basic SCR action) so that it diverts (through terminal 7) base drive from the output stage (Q13, Q14) in the CA3085. By this means, base drive is diverted from Q4 and the pass transistor Q5. To restore regulator operation, normally closed switch S1 is momentarily opened and unlatches Q2-Q3.

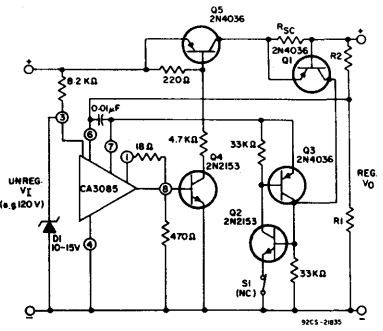


Fig. 16– High-voltage regulator incorporating current "snap-back" protection.

Switching Regulator

When large input-to-output voltage differences are necessary, the regulators described above are inefficient because they dissipate significant power in the series-pass transistor. Under these conditions, high-efficiency operation can be achieved by using a switching-type regulator of the generic type shown in Fig. 17(a). Transistor Q1 acts as a keyed switch and operates in

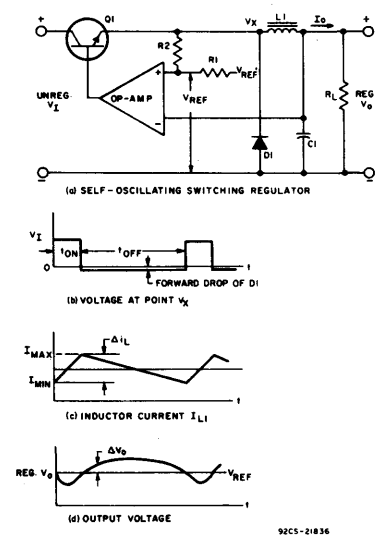


Fig. 17– Switching regulator and associated waveforms.

either a saturated or cut-off condition to minimize dissipation. When transistor Q1 is conductive, diode D1 is reverse-biased and current in the inductance L1 increases in accordance with the following relationship:

$$i_L = \frac{1}{L}\int_{t_o}^{t_1} V dt \qquad (10)$$

where V is the voltage across the inductance L1. The current through the inductance charges the capacitor C1 and supplies current to the load. The output voltage rises until it slightly exceeds the reference voltage V_{ref}. At this point the op-amp removes base drive to Q1 and the unregulated input voltage V_I is "switched off". The energy stored in the inductor L1 now causes the voltage at V_X to swing in the negative direction and current flows through diode D1, while continuing to supply current into the load R_L. As the current in the inductor falls below the load current, the capacitor C1 begins to discharge and V_O decreases. When V_O falls slightly below the value of V_{ref}, the op-amp turns on Q1 and the cycle is repeated. It should be apparent that the output voltage oscillates about V_{ref} with an amplitude determined by R1 and R2. Actually, the value of V_{ref} varies from being slightly more positive than V_{ref}' when Q1 is conducting, to being slightly more negative than V_{ref}' when D1 is conducting. The voltage and current waveforms are shown in Fig. 17(b), (c), and (d).

Design Example: The following specifications are used in decomputations for a switching regulator:

V_I = 30 V, V_O = 5 V, I_O = 500 mA,
switching frequency = 20 kHz,
output ripple = 100 mV.

If it is assumed that transistor Q1 is in steady-state saturated operation with a low voltage-drop, the current in the inductor is given by Eq. 10, as follows:

$$i_L = \frac{1}{L}\int_{t_o}^{t_1} V dt = \left(\frac{V_I - V_O}{L1}\right) t_{on} \qquad (11)$$

When transistor Q1 is off, the current in the inductor is given by:

$$i_L \cong \frac{(V_O + V_{D1})\ t_{off}}{L1} \qquad (12)$$

From Eq. 11,

$$L_1 = \frac{(V_I - V_O)}{i_L} \cdot \frac{1}{f} \cdot \frac{V_O}{V_I} \qquad (13)$$

If i_{max} is 1.3 I_L, then during t_{on} the current in the inductor (i_L) will be 0.5 A x 1.3 = 0.65 A; therefore, Δi_L = 0.15 A
Substitution in Eq. 13 yields

$$L_1 = \frac{(30 - 5)}{0.15} \cdot \frac{1}{(20 \times 10^3)} \cdot \frac{5}{30} = 1.4 \text{ mH} \qquad (14)$$

Current discharge from the capacitor C1 is given by:

$$i_c = C\frac{dv}{dt} \qquad (15)$$

Thus, $\Delta i_c = C\frac{\Delta v}{\Delta t}$, or $C = \frac{\Delta i_c\ \Delta t}{\Delta v}$

Since $i_c = i_L$ and $\Delta t = t_{off}$, then

$$C = \frac{\Delta i_L\ t_{off}}{\Delta v}$$

Substitution for the value of i_L from Eq. 13 yields

$$C = \frac{\left(\frac{V_I - V_O}{L1}\right) \cdot \frac{1}{f} \cdot \left(\frac{V_O}{V_I}\right) \cdot t_{off}}{\Delta v} \qquad (16)$$

The total period $T = t_{off} + t_{on}$, and $T = \frac{1}{f}$. Therefore,

$$t_{off} = \frac{1}{f} - t_{on} \qquad (17)$$

For optimum efficiency t_{on} should be

$$\cong \left(\frac{V_O}{V_I}\right) T \cong \left(\frac{V_O}{V_I}\right) \frac{1}{f} \qquad (18)$$

Substitution for t_{on} in Eq. 18 yields

$$t_{off} = \frac{1}{f} - \left(\frac{V_O}{V_I}\right)\frac{1}{f} = \frac{1}{f}\left(1 - \frac{V_O}{V_I}\right) \qquad (19)$$

Substitution for t_{off} in Eq. 16 yields

$$C = \frac{\frac{(V_I - V_O)}{L_1} \cdot \frac{1}{f} \cdot \frac{V_O}{V_I} \cdot \frac{1}{f} \cdot \left(1 - \frac{V_O}{V_I}\right)}{\Delta v}$$

Substitution of numerical values in Eq. 20 produces the following value for C:

$$C = \frac{\frac{30 - 5}{1.4 \times 10^{-3}} \cdot \frac{1}{20 \times 10^3} \cdot \frac{5}{30} \cdot \frac{1}{20 \times 10^3} \cdot \left(1 - \frac{5}{30}\right)}{10^{-1}} = 63\ \mu F$$

A switching-regulator circuit using the CA3085 is shown in Fig. 18. The values of L and C (1.5 millihenries and 50 microfarads, respectively) are commercially available components having values approximately equal to the computed values in the previous design example.

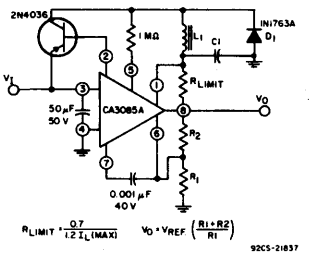


Fig. 18– Typical switching regulator circuit.

Current Regulators

The CA3085 series of voltage regulators can be used to provide a constant source or sink current. A regulated-current supply capable of delivering up to 100 milliamperes is shown in Fig. 19(a). The regulated load current is controlled by R1 because the current flowing through this resistor must establish a voltage difference between terminals 6 and 4 that is equal to the internal reference voltage developed between terminals 5 and 4. The actual regulated current, reg I_L, is the sum of the quiescent regulator current and the current through R1, i.e.,

$$\text{reg } I_L = I_{quiescent} + I_{R1}$$

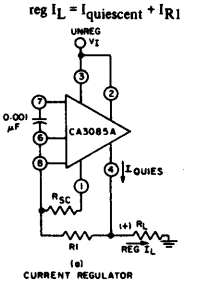


Fig. 19– Constant current regulators.

Fig. 19(b) shows a high-current regulator using the CA3085 in conjunction with an external n-p-n transistor to regulate currents up to 3 amperes. In this circuit the quiescent regulator current does not flow through the load and the output current can be directly programmed by R1, i.e.,

$$\text{Reg } I_L = \frac{V_{ref}}{R1}$$

With this regulator currents between 1 milliampere and 3 amperes can be programmed directly. At currents below 1 milliampere inaccuracies may occur as a result of leakage in the external transistor.

A Dual-Tracking Voltage Regulator

A dual-tracking voltage regulator using a CA3085 and a CA3094A* is shown in Fig. 20. The CA3094A is basically an op-amp capable of supplying 100 milliamperes of output current.

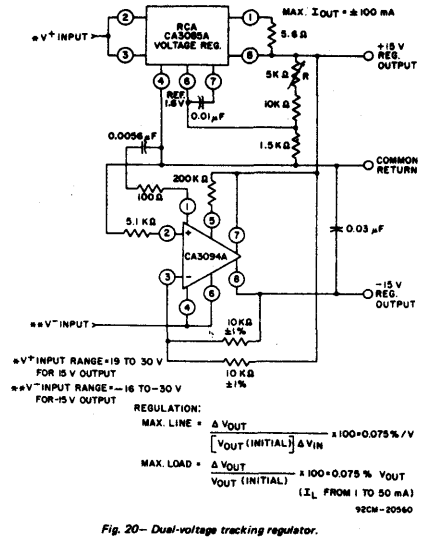


Fig. 20– Dual-voltage tracking regulator.

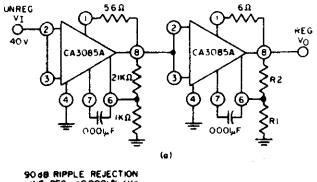
The positive output voltage is regulated by a CA3085 operating in a configuration essentially similar to that described in connection with Fig. 3. Resistor R is used as a vernier adjustment of output voltage. The negative output voltage is regulated by the CA3094A, which is "slaved" to the regulated positive voltage supplied by the CA3085. It should be noted that the non-inverting input of the CA3094A and the negative supply terminal of the CA3085 are connected to a common ground reference. The "slaving" potential for the CA3094A is derived from an accurate 1:1 voltage-divider network comprised of two 10-kilohm resistors connected between the +15-volt and −15-volt output terminals. The junction of these two resistors is connected to the inverting input of the CA3094A. The voltage at this junction is compared with the voltage at the non-inverting input, and the CA3094A then automatically adjusts the output current at the negative terminal to maintain a negative regulated output voltage essentially equal to the regulated positive output voltage. Typical performance data for this circuit are shown in Fig. 20.

* Specifications for the CA3094A appear in RCA Data File No. 598 and application information is presented in ICAN-6048.

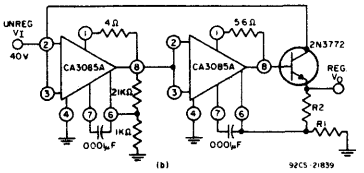
The basic circuit of Fig. 20 can be modified to regulate dissimilar positive and negative voltages (e.g., +15 V, −5 V) by appropriate selection of resistor ratios in the voltage-divider network discussed previously. As an example, to provide tracking of the +15 V and −5 V regulated voltages with the circuit of Fig. 20, it is only necessary to replace the 10-kilohm resistor connected between terminals 3 and 8 of the CA3094A with a 3.3-kilohm resistor.

Regulators With High Ripple Rejection

When the reference-voltage source in the CA3085 is adequately filtered, the typical ripple rejection provided by the circuit is 56 dB. It is possible to achieve higher ripple-rejection performance by cascading two stages of the CA3085, as shown in Fig. 21. The voltage-regulator circuit in Fig. 21(a) provides 90 dB of ripple rejection. The output voltage is adjustable over the range from 1.8 to 30 volts by appropriate adjustment of resistors R1 and R2. Higher regulated output currents up to 1 ampere can be obtained with this circuit by adding an external n-p-n transistor as shown in Fig. 21(b).



(a) voltage regulator with high ripple rejection



(b) high-current voltage regulator with high ripple rejection

Fig. 21– Regulators with high ripple rejection.

The CA3085 As A Power Source For Sensors

Certain types of sensor applications require a regulated power source. Additionally, low-impedance sensors can consume significant power. An example of a circuit with these requirements, in which a CA3085 provides regulated power for a low-impedance sensor and the CA3059* zero-voltage switch, is shown in Fig. 22. Terminal 12 on the CA3059 provides the ac trigger-signal which actuates the zero-voltage switch synchronously with the power line to control the load-switching triac.

* Technical specifications on the CA3059 zero-voltage switch appear in RCA Data File No. 490, and related application information is provided in Application Notes ICAN-6158 and ICAN-6268.

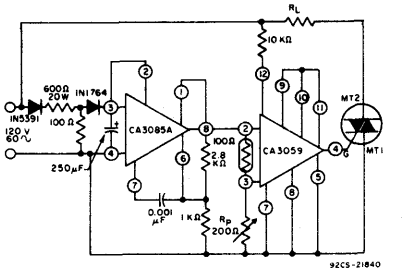


Fig. 22– Voltage regulator for sensor and zero-voltage switch.

The CA3085 As A General-Purpose Amplifier

As described above, the CA3085 series regulators contain a high-gain linear amplifier having a current-output capability up to 100 milliamperes. The premium type (CA3085B) can operate at supply voltages up to 50 volts. When equipped with an appropriate radiator or heat-sink, the TO-5 package of these devices can dissipate up to 1.6 watts at 55°C. A very stable internal voltage-reference source is used to bias the high-gain amplifier and/or provide an external voltage-reference despite extreme temperature or supply-voltage variations. These factors, plus economics, prompt consideration of this circuit for general-purpose uses, such as amplifiers, relay controls, signal-lamp controls, and thyristor firing.

As an example, Fig. 23 shows the application of the CA3085 in a general-purpose amplifier. Under the conditions shown,

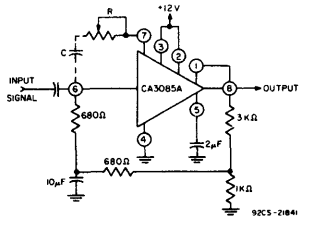


Fig. 23– General purpose amplifier using CA3085A.

the circuit has a typical gain of 70 dB with a flat response to at least 100 kHz without the RC network connected between terminals 6 and 7. The RC network is useful as a tone control or to "roll-off" the amplifier response for other reasons. Current limiting is not used in this circuit. The network connected between terminals 8 and 6 provides both dc and ac feedback. This circuit is also applicable for directly driving an external discrete n-p-n power transistor.

Features and Applications of RCA Integrated-Circuit Zero-Voltage Switches (CA3058, CA3059, and CA3079)

by A.C.N. Sheng, G.J. Granieri, J. Yellin, and T. McNulty

RCA-CA3058, CA3059 and CA3079 zero-voltage switches are monolithic integrated circuits designed primarily for use as trigger circuits for thyristors in many highly diverse ac power-control and power-switching applications. These integrated-circuit switches operate from an ac input voltage of 24, 120, 208 to 230, or 277 volts at 50, 60, or 400 Hz.

The CA3059 and CA3079 are supplied in a 14-terminal dual-in-line plastic package. The CA3058 is supplied in a 14-terminal dual-in-line ceramic package. The electrical and physical characteristics of each type are detailed in RCA Data Bulletin File No. 490.

RCA zero-voltage switches (ZVS) are particularly well suited for use as thyristor trigger circuits. These switches trigger the thyristors at zero-voltage points in the supply-voltage cycle. Consequently, transient load-current surges and radio-frequency interference (RFI) are substantially reduced. In addition, use of the zero-voltage switches also reduces the rate of change of on-state current (di/dt) in the thyristor being triggered, an important consideration in the operation of thyristors. These switches can be adapted for use in a variety of control functions by use of an internal differential comparator to detect the difference between two externally developed voltages. In addition, the availability of numerous terminal connections to internal circuit points greatly increases circuit flexibility and further expands the types of ac power-control applications to which these integrated circuits may be adapted. The excellent versatility of the zero-voltage switches is demonstrated by the fact that these circuits have been used to provide transient-free temperature control in self-cleaning ovens, to control gun-muzzle temperature in low-temperature environments, to provide sequential switching of heating elements in warm-air furnaces, to switch traffic signal lights at street intersections, and to effect other widely different ac power-control functions.

FUNCTIONAL DESCRIPTION

RCA zero-voltage switches are multistage circuits that employ a diode limiter, a zero-crossing (threshold) detector, an on-off sensing amplifier (differential comparator), and a Darlington output driver (thyristor gating circuit) to provide the basic switching action. The dc operating voltages for these stages is provided by an internal power supply that has sufficient current capability to drive external circuit elements, such as transistors and other integrated circuits. An important feature of the zero-voltage switches is that the output trigger pulses can be applied directly to the gate of a triac or a silicon controlled rectifier (SCR). The CA3058 and CA3059 also feature an interlock (protection) circuit that inhibits the application of these pulses to the thyristor in the event that the external sensor should be inadvertently opened or shorted. An external inhibit connection (terminal No. 1) is also available so that an external signal can be used to inhibit the output drive. This feature is not included in the CA3079; otherwise, the three integrated-circuit zero-voltage switches are electrically identical.

Over-all Circuit Operation

Fig. 1 shows the functional interrelation of the zero-voltage switch, the external sensor, the thyristor being triggered, and the load elements in an on-off type of ac power-control system. As shown, each of the zero-voltage switches incorporates four functional blocks as follows:

(1) Limiter-Power Supply – Permits operation directly from an ac line.

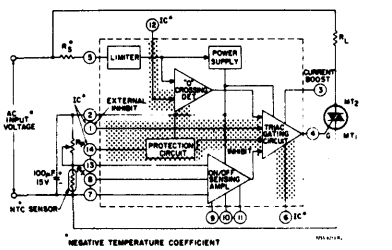
(2) Differential On/Off Sensing Amplifier – Tests the condition of external sensors or command signals. Hysteresis or proportional-control capability may easily be implemented in this section.

(3) Zero-Crossing Detector – Synchronizes the output pulses of the circuit at the time when the ac cycle is at a zero-voltage point and thereby eliminates radio-frequency inteference (RFI) when used with resistive loads.

(4) Triac Gating Circuit – Provides high-current pulses to the gate of the power-controlling thyristor.

In addition, the CA3058 and CA3059 provide the following important auxiliary functions (shown in Fig. 1):

(1) A built-in protection circuit that may be actuated to remove drive from the triac if the sensor opens or shorts.



AC Input Voltage (50/60 or 400 Hz) V AC	Input Series Resistor (R_S) k Ω	Dissipation Rating for R_S W
24	2	0.5
120	10	2
208/230	20	4
277	25	5

Fig. 1 – Functional block diagrams of the zero-voltage switches CA3058, CA3059, and CA3079.

(2) Thyristor firing may be inhibited through the action of an internal diode gate connected to terminal 1.

(3) High-power dc-comparator operation is provided by overriding the action of the zero-crossing detector. This override is accomplished by connecting terminal 12 to terminal 7. Gate current to the thyristor is continuous when terminal 13 is positive with respect to terminal 9.

Fig. 2 shows the detailed circuit diagram for the integrated-circuit zero-voltage switches. (The diagrams shown in Figs. 1 and 2 are representative of all three RCA zero-voltage switches, i.e., the CA3058, CA3059, and CA3079; the shaded areas indicate the circuitry that is not included in the CA3079.)

The limiter stage of the zero-voltage switch clips the incoming ac line voltage to approximately ±8 volts. This signal is then applied to the zero-voltage-crossing detector, which generates an output pulse each time the line voltage passes through zero. The limiter output is also applied to a rectifying diode and an external capacitor, C_F, that comprise the dc power supply. The power supply provides approximately 6 volts as the V_{CC} supply to the other stages of the zero-voltage switch. The on-off sensing amplifier is basically a differential comparator. The thyristor gating circuit contains a driver for direct triac triggering. The gating circuit is enabled when all the inputs are at a "high" voltage, i.e., the line voltage must be approximately zero volts, the sensing-amplifier output must be "high," the external voltage to terminal 1 must be a logical "0", and, for the CA3058 and CA3059, the output of the fail-safe circuit must be "high." Under these conditions, the thyristor (triac or SCR) is triggered when the line voltage is essentially zero volts.

Thyristor Triggering Circuits

The diodes D_1 and D_2 in Fig. 2 form a symmetrical clamp that limits the voltages on the chip to ±8 volts; the diodes D_7 and D_{13} form a half-wave rectifier that develops a positive voltage on the external storage capacitor, C_F.

The output pulses used to trigger the power-switching thyristor are actually developed by the zero-crossing detector and the thyristor gating circuit. The zero-crossing detector consists of diodes D_3 through D_6, transistor Q_1, and the associated resistors shown in Fig. 2. Transistors Q_1 and Q_6 through Q_9 and the associated resistors comprise the thyristor gating circuit and output driver. These circuits generate the output pulses when the ac input is at a zero-voltage point so that RFI is virtually eliminated when the zero-voltage switch and thyristor are used with resistive loads.

The operation of the zero-crossing detector and thyristor gating circuit can be explained more easily if the on state (i.e., the operating state in which current is being delivered to the thyristor gate through terminal 4) is considered as the operating condition of the gating circuit. Other circuit elements in the zero-voltage switch inhibit the gating circuit unless certain conditions are met, as explained later.

In the on state of the thyristor gating circuit, transistors Q_8 and Q_9 are conducting, transistor Q_7 is off, and transistor Q_6 is on. Any action that turns on transistor Q_7 removes the drive from transistor Q_8 and thereby turns off the thyristor. Transistor Q_7 may be turned on directly by application of a minimum of ±1.2 volts at 10 microamperes to the external-inhibit input, terminal 1. (If a voltage of more than

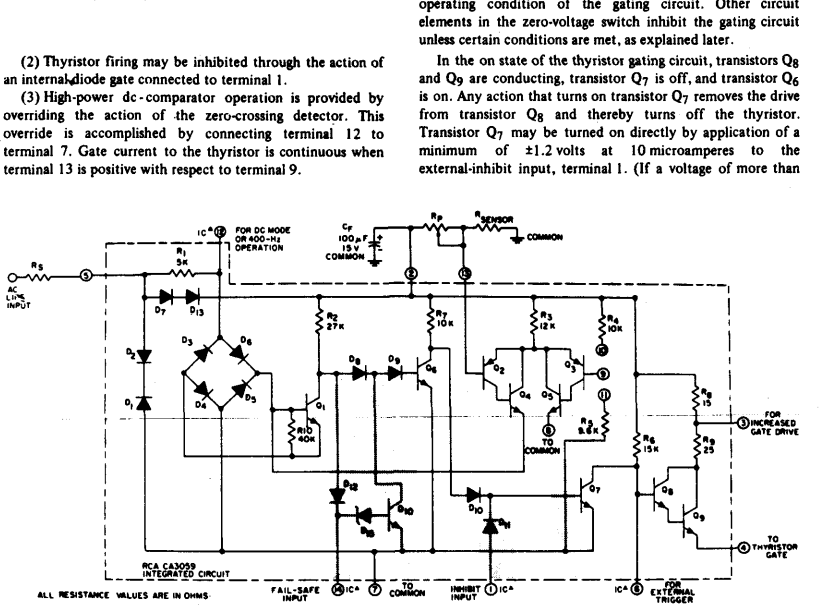


Fig. 2 – Schematic diagram of zero-voltage switches CA3058, CA3059, and CA3079.

1.5 volts is available, an external resistance must be added in series with terminal 1 to limit the current to 1 milliampere.) Diode D_{10} isolates the base of transistor Q_7 from other signals when an external-inhibit signal is applied so that this signal is the highest priority command for normal operation. (Although grounding of terminal 6 creates a higher-priority inhibit function, this level is not compatible with normal DTL or TTL logic levels.) Transistor Q_7 may also be activated by turning off transistor Q_6 to allow current flow from the power supply through resistor R_7 and diode D_{10} into the base of Q_7. Transistor Q_6 is normally maintained in conduction by current that flows into its base through resistor R_2 and diodes D_8 and D_9 when transistor Q_1 is off.

Transistor Q_1 is a portion of the zero-crossing detector. When the voltage at terminal 5 is greater than +3 volts, current can flow through resistor R_1, diode D_6, the base-to-emitter junction of transistor Q_1, and diode D_4 to terminal 7 to turn on Q_1. This action inhibits the delivery of a gate-drive output signal at terminal 4. For negative voltages at terminal 5 that have magnitudes greater than 3 volts, the current flows through diode D_5, the emitter-to-base junction of transistor Q_1, diode D_3, and resistor R_1, and again turns on transistor Q_1. Transistor Q_1 is off only when the voltage at terminal 5 is less than the threshold voltage of approximately ±2 volts. When the integrated-circuit zero-voltage switch is connected as shown in Fig. 1, therefore, the output is a narrow pulse which is approximately centered about the zero-voltage time in the cycle, as shown in Fig. 3. In some applications, however,

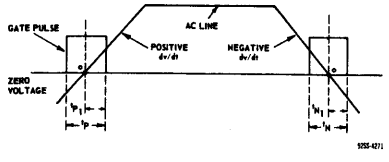


Fig. 3 – Waveform showing output-pulse duration of the zero-voltage switch.

particularly those that use either slightly inductive or low-power loads, the thyristor load current does not reach the latching-current value* by the end of this pulse. An external capacitor C_X connected between terminal 5 and 7, as shown in Fig. 4, can be used to delay the pulse to accommodate such loads. The amount of pulse stretching and delay is shown in Figs. 5(a) and 5(b).

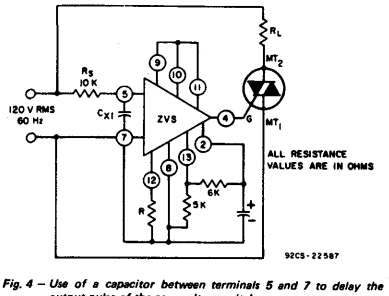


Fig. 4 – Use of a capacitor between terminals 5 and 7 to delay the output pulse of the zero-voltage switch.

Continuous gate current can be obtained if terminal 12 is connected to terminal 7 to disable the zero-crossing detector. In this mode, transistor Q_1 is always off. This mode of operation is useful when comparator operation is desired or when inductive loads must be switched. (If the capacitance in the load circuit is low, most RFI is eliminated.) Care must be taken to avoid overloading of the internal power supply in this mode. A sensitive-gate thyristor should be used, and a resistor should be placed between terminal 4 and the gate of the thyristor to limit the current, as pointed out later under **Special Application Considerations.**

* The latching current is the minimum current required to sustain conduction immediately after the thyristor is switched from the off to the on state and the gate signal is removed.

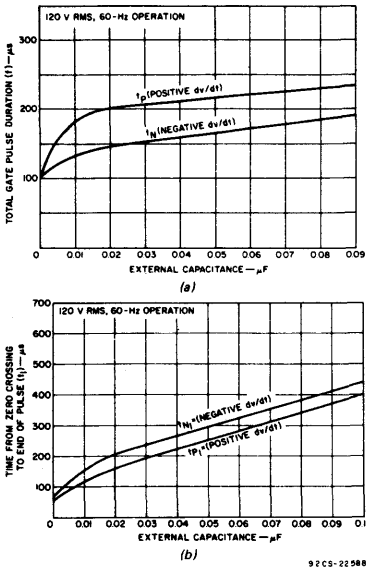
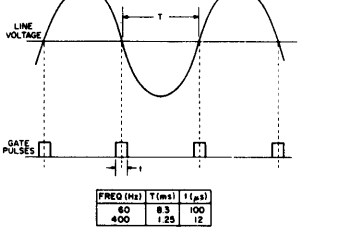


Fig. 5 – Curves showing effect of external capacitance on (a) the total output-pulse duration, and (b) the time from zero crossing to the end of the pulse.

Fig. 6 indicates the timing relationship between the line voltage and the zero-voltage- switch output pulses. At 60 Hz, the pulse is typically 100 microseconds wide; at 400 Hz, the pulse width is typically 12 microseconds. In the basic circuit shown, when the dc logic signal is "high", the output is disabled; when it is "low", the gate pulses are enabled.



FREQ (Hz)	T (ms)	t (μs)
60	8.3	100
400	1.25	12

Fig. 6 – Timing relationship between the output pulses of the RCA zero-voltage switch and the ac line voltage.

On-Off Sensing Amplifier

The discussion thus far has considered only cases in which pulses are present all the time or not at all. The differential sense amplifier consisting of transistors Q_2, Q_3, Q_4, and Q_5 (shown in Fig. 2) makes the zero-voltage switch a flexible power-control circuit. The transistor pairs Q_2-Q_4 and Q_3-Q_5 form a high-beta composite p-n-p transistors in which the emitters of transistors Q_4 and Q_5 act as the collectors of the composite devices. These two composite transistors are connected as a differential amplifier with resistor R_3 acting as a constant-current source. The relative current flow in the two "collectors" is a function of the difference in voltage between the bases of transistors Q_2 and Q_3. Therefore, when terminal 13 is more positive than terminal 9, little or no current flows in the "collector" of the transistor pair Q_2-Q_4. When terminal 13 is negative with respect to terminal 9, most of the current flows through that path, and none in terminal 8. When current flows in the transistor pair Q_2-Q_4, the path is from the supply through R_3, through the transistor pair Q_2-Q_4, through the base-emitter junction of transistor Q_1, and finally through the diode D_4 to terminal 7. Therefore, when V_{13} is equal to or more negative than V_9, transistor Q_1 is on, and the output is inhibited.

In the circuit shown in Fig. 1, the voltage at terminal 9 is derived from the supply by connection of terminals 10 and 11 to form a precision voltage divider. This divider forms one side of a transducer bridge, and the potentiometer R_p and the negative-temperature-coefficient (NTC) sensor form the other side. At low temperatures, the high resistance of the sensor causes terminal 13 to be positive with respect to terminal 9 so that the thyristor fires on every half-cycle, and power is applied to the load. As the temperature increases, the sensor resistance decreases until a balance is reached, and V_{13} approaches V_9. At this point, the transistor pair Q_2-Q_4 turns on and inhibits any further pulses. The controlled temperature is adjusted by variation of the value of the potentiometer R_p. For cooling service, either the positions of R_p and the sensor may be reversed or terminals 9 and 13 may be interchanged.

The low bias current of the sensing amplifier permits operation with sensor impedances of up to 0.1 megohm at balance without introduction of substantial error (i.e., greater than 5 per cent). The error may be reduced if the internal bridge elements, resistors R_4 and R_5, are not used, but are replaced with resistances which equal the sensor impedance. The minimum value of sensor impedance is restricted by the current drain on the internal power supply. Operation of the zero-voltage switch with low-impedance sensors is discussed later under **Special Application Considerations.** The voltage applied to terminal 13 must be greater than 1.8 volts at all times to assure proper operation.

Protection Circuit

A special feature of the CA3058 and CA3059 zero-voltage switches is the inclusion of an interlock type of circuit. This circuit removes power from the load by interrupting the thyristor gate drive if the sensor either shorts or opens. However, use of this circuit places certain constraints upon the user. Specifically, effective protection-circuit operation is dependent upon the following conditions:

(1) The circuit configuration of Fig. 1 is used, with an internal supply, no external load on the supply, and terminal 14 connected to terminal 13.

(2) The value of potentiometer R_p and of the sensor resistance must be between 2000 ohms and 0.1 megohm.

(3) The ratio of sensor resistance and R_p must be greater than 0.33 and less than 3.0 for all normal conditions. (If either of these ratios is not met with an unmodified sensor, a series resistor or a shunt resistor must be added to avoid undesired activation of the circuit.)

The protective feature may be applied to other systems when operation of the circuit is understood. The protection circuit consists of diodes D_{12} and D_{15} and transistor Q_{10}. Diode D_{12} activates the protection circuit if the sensor shown in Fig. 1 shorts or its resistance drops too low in value, as follows: Transistor Q_6 is on during an output pulse so that the junction of diodes D_8 and D_{12} is 3 diode drops (approximately 2 volts) above terminal 7. As long as V_{14} is more positive or only 0.15 volt negative with respect to that point, diode D_{12} does not conduct, and the circuit operates normally. If the voltage at terminal 14 drops to 1 volt, the anode of diode D_8 can have a potential of only 1.6 to 1.7 volts, and current does not flow through diodes D_8 and D_9 and transistor Q_6. The thyristor then turns off.

The actual threshold is approximately 1.2 volts at room temperature, but decreases 4 millivolts per degree C at higher temperatures. As the sensor resistance increases, the voltage at terminal 14 rises toward the supply voltage. At a voltage of approximately 6 volts, the zener diode D_{15} breaks down and turns on transistor Q_{10}, which then turns off transistor Q_6 and the thyristor. If the supply voltage is not at least 0.2 volt

more positive than the breakdown voltage of diode D_{15}, activation of the protection circuit is not possible. For this reason, loading the internal supply may cause this circuit to malfunction, as may selection of the wrong external supply voltage. Fig. 7 shows a guide for the proper operation of the protection circuit when an external supply is used with a typical integrated-circuit zero-voltage switch.

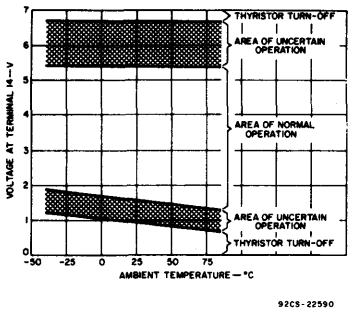


Fig. 7 – Operating regions for built-in protection circuits of a typical zero-voltage switch.

SPECIAL APPLICATION CONSIDERATIONS

As pointed out previously, the RCA integrated-circuit zero-voltage switches (CA3058, CA3059, and CA3079) are exceptionally versatile units that can be adapted for use in a wide-variety of power-control applications. Full advantage of this versatility can be realized, however, only if the user has a basic understanding of several fundamental considerations that apply to certain types of applications of the zero-voltage switches.

Operating-Power Options

Power to the zero-voltage switch may be derived directly from the ac line, as shown in Fig. 1, or from an external dc power supply connected between terminals 2 and 7, as shown in Fig. 8. When the zero-voltage switch is operated directly from the ac line, a dropping resistor R_S of 5,000 to 10,000 ohms must be connected in series with terminal 5 to limit the current in the switch circuit. The optimum value for this resistor is a function of the average current drawn from the internal dc power supply, either by external circuit elements or by the thyristor trigger circuits, as shown in Fig. 9. The chart shown in Fig. 1 indicates the value and dissipation rating of the resistor R_S for ac line voltages of 24, 120, 208 to 230, and 277 volts.

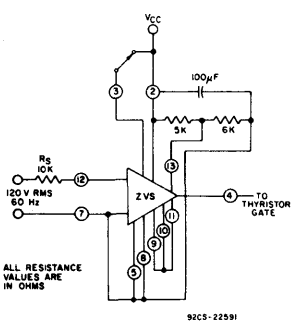


Fig. 8 – Operation of the zero-voltage switch from an external dc power supply connected between terminals 2 and 7.

Half-Cycling Effect

The method by which the zero-voltage switch senses the zero crossing of the ac power results in a half-cycling phenomenon at the control point. Fig. 10 illustrates this phenomenon. The zero-voltage switch senses the zero-voltage

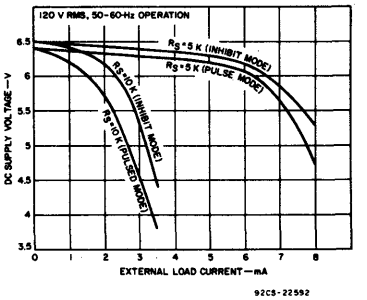


Fig. 9 – DC supply voltage as a function of external load current for several values of dropping resistance R_S.

crossing every half-cycle, and an output, for example pulse No. 4, is produced to indicate the zero crossing. During the remaining 8.3 milliseconds, however, the differential amplifier in the zero-voltage switch may change state and inhibit any further output pulses. The uncertainity region of the differential amplifier, therefore, prevents pulse No. 5 from triggering the triac during the negative excursion of the ac line voltage.

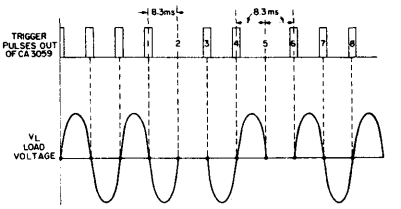


Fig. 10 – Half-cycling phenomenon in the zero-voltage switch.

When a sensor with low sensitivity is used in the circuit, the zero-voltage switch is very likely to operate in the linear mode. In this mode, the output trigger current may be sufficient to trigger the triac on the positive-going cycle, but insufficient to trigger the device on the negative-going cycle of the triac supply voltage. This effect introduces a half-cycling phenomenon, i.e., the triac is turned on during the positive half-cycle and turned off during the negative half-cycle.

Several techniques may be used to cope with the half-cycling phenomenon. If the user can tolerate some hystersis in the control, then positive feedback can be added around the differential amplifier. Fig. 11 illustrates this technique. The tabular data in the figure lists the recommended values of resistors R_1 and R_2 for different sensor impedances at the control point.

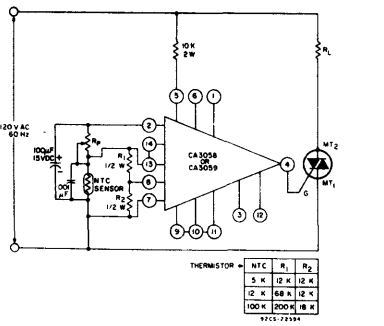


Fig. 11 – CA3058 or CA3059 on-off controller with hysteresis.

If a significant amount (greater than ±10%) of controlled hysteresis is required, then the circuit shown in Fig. 12 may be employed. In this configuration, external transistor Q_1 can be used to provide an auxiliary timed-delay function.

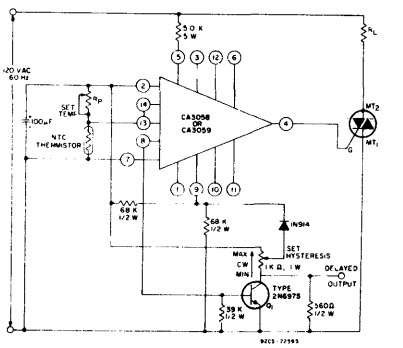


Fig. 12 – CA3058 or CA3059 on-off controller with controlled hysteresis.

For applications that require complete elimination of half-cycling without the addition of hysteresis, the circuit shown in Fig. 13 may be employed. This circuit uses a CA3098E integrated-circuit programmable comparator with a zero-voltage switch. A block diagram of CA3098E is shown in Fig. 14. Because the CA3098E contains an integral flip-flop, its output will be in either a "0" or "1" state. Consequently the zero-voltage switch cannot operate in the linear mode, and spurious half-cycling operation is prevented. When the signal-input voltage at terminal 8 of the CA3098E is equal to or less than the "low" reference voltage (LR), current flows from the power supply through resistor R_1 and R_2, and a logic "0" is

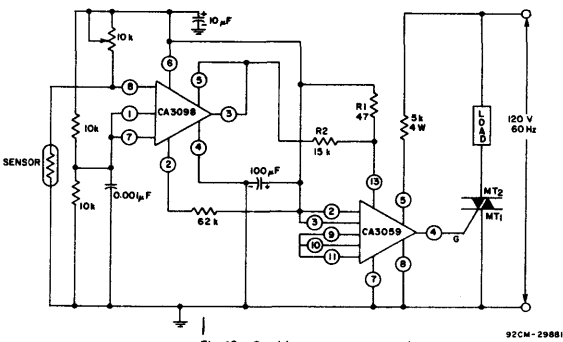


Fig. 13 – Sensitive temperature control.

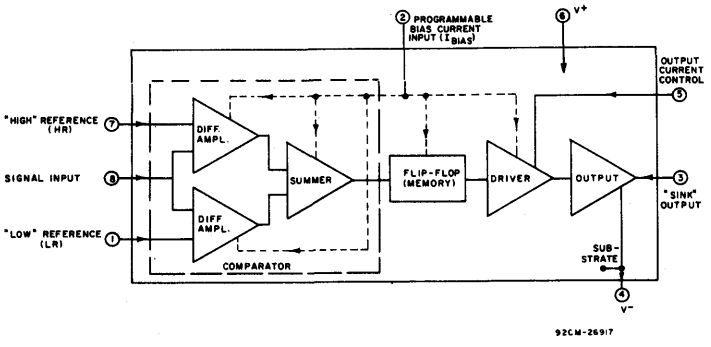


Fig. 14 – Block diagram of CA3098 programmable Schmitt trigger.

applied to terminal 13 of the zero-voltage switch. This condition turns off the triac. The triac remains off until the signal-input voltage rises to or exceeds the "high" reference voltage (HR), thereby effecting a change in the state of the flip-flop so that a logic "1" is applied to terminal 13 of the zero-voltage switch, and triggers the triac on.

"Proportional Control" Systems

The on-off nature of the control shown in Fig. 1 causes some overshoot that leads to a definite steady-state error. The addition of hysteresis adds further to this error factor. However, the connections shown in Fig. 15(a) can be used to add proportional control to the system. In this circuit, the sense amplifier is connected as a free-running multivibrator. At balance, the voltage at terminal 13 is much less than the voltage at terminal 9. The output will be inhibited at all times until the voltage at terminal 13 rises to the design differential voltage between terminals 13 and 9; then proportional control resumes. The voltage at terminal 13 is as shown in Fig. 15(b). When this voltage is more positive than the threshold, power is

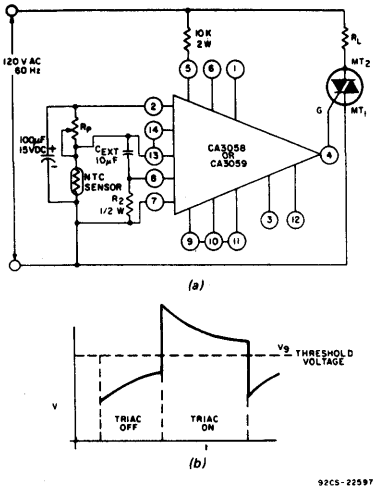


Fig. 15 – Use of the CA3058 or CA3059 in a typical heating control with proportional control: (a) schematic diagram, and (b) waveform of voltage at terminal 13.

applied to the load so that the duty cycle is approximately 50 per cent. With a 0.1 megohm sensor and values of R_P = 0.1 megohm, R_2 = 10,000 ohms, and C_{EXT} = 10 microfarads, a period greater than 3 seconds is achieved. This period should be much shorter than the thermal time constant of the system. A change in the value of any of these elements changes the period, as shown in Fig. 16. As the resistance of the sensor changes, the voltage on terminal 13 moves relative to V_9. A cooling sensor moves V_{13} in a positive direction. The triac is on for a larger portion of the pulse cycle and increases the average power to the load.

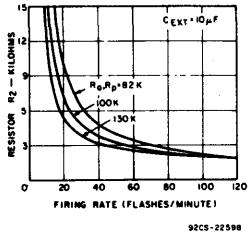


Fig. 16 – Effect of variations in time-constant elements on period.

As in the case of the hysteresis circuitry described earlier, some special applications may require more sophisticated systems to achieve either very precise regions of control or very long periods.

Zero-voltage switching control can be extended to applications in which it is desirable to have constant control of the temperature and a minimization of system hysteresis. A closed-loop top-burner control in which the temperature of the cooking utensil is sensed and maintained at a particular value is a good example of such an application; the circuit for this control is shown in Fig. 17. In this circuit, a unijunction

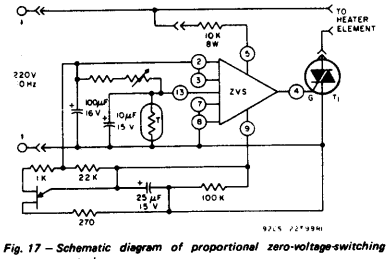


Fig. 17 – Schematic diagram of proportional zero-voltage-switching control.

oscillator is outboarded from the basic control by means of the internal power supply of the zero-voltage switch. The output of this ramp generator is applied to terminal 9 of the zero-voltage switch and establishes a varied reference to the differential amplifier. Therefore, gate pulses are applied to the triac whenever the voltage at terminal 13 is greater than the voltage at terminal 9. A varying duty cycle is established in which the load is predominantly on with a cold sensor and predominantly off with a hot sensor. For precise temperature regulation, the time base of the ramp should be shorter than the thermal time constant of the system but longer than the period of the 60-Hz line. Fig. 18, which contains various waveforms for the system of Fig. 17, indicates that a typical variance of ±0.5°C might be expected at the sensor contact to the utensil. Overshoot of the set temperature is minimized with this approach, and scorching of any type is minimized.

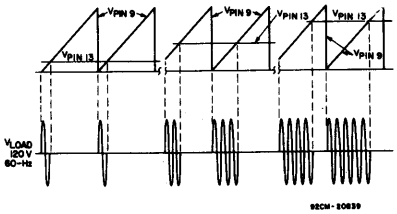


Fig. 18 – Waveforms for the circuit of Fig. 17.

Effect of Thyristor Load Characteristics

The zero-voltage switch is designed primarily to gate a thyristor that switches a resistive load. Because the output pulse supplied by the switch is of short duration, the latching current of the triac becomes a significant factor in determining whether other types of loads can be switched. (The latching-current value determines whether the triac will remain in conduction after the gate pulse is removed.) Provisions are included in the zero-voltage switch to accommodate inductive loads and low-power loads. For example, for loads that are less than approximately 4 amperes rms or that are slightly inductive, it is possible to retard the output pulse with respect to the zero-voltage crossing by insertion of the capacitor C_X from terminal 5 to terminal 7. The insertion of capacitor C_X permits switching of triac loads that have a slight inductive component and that are greater than approximately 200 watts (for operation from an ac line voltage of 120 volts rms). However, for loads less than 200 watts (for example, 70 watts), it is recommended that the user employ the T2300B* sensitive-gate triac with the zero-voltage switch because of the low latching-current requirement of this triac.

For loads that have a low power factor, such as a solenoid valve, the user may operate the zero-voltage switch in the dc mode. In this mode, terminal 12 is connected to terminal 7, and the zero-crossing detector is inhibited. Whether a "high" or "low" voltage is produced at terminal 4 is then dependent only upon the state of the differential comparator within the integrated-circuit zero-voltage switch, and not upon the zero crossing of the incoming line voltage. Of course, in this mode of operation, the zero-voltage switch no longer operates as a zero-voltage switch. However, for many applications that involve the switching of low-current inductive loads, the amount of RFI generated can frequently be tolerated.

For switching of high-current inductive loads, which must be turned on at zero line current, the triggering technique employed in the dual-output over-under temperature controller and the transient-free switch controller described subsequently in this Note is recommended.

Switching of Inductive Loads

For proper driving of a thyristor in full-cycle operation, gate drive must be applied soon after the voltage across the device reverses. When resistive loads are used, this reversal occurs as the line voltage reverses. With loads of other power factors, however, it occurs as the current through the load becomes zero and reverses.

There are several methods for switching an inductive load at the proper time. If the power factor of the load is high (i.e., if the load is only slightly inductive), the pulse may be delayed by addition of a suitable capacitor between terminals 5 and 7, as described previously. For highly inductive loads, however, this method is not suitable, and different techniques must be used.

If gate current is continuous, the triac automatically commutates because drive is always present when the voltage reverses. This mode is established by connection of terminals 7 and 12. The zero-crossing detector is then disabled so that current is supplied to the triac gate whenever called for by the sensing amplifier. Although the RFI-eliminating function of the zero-voltage switch is inhibited when the zero-crossing detector is disabled, there is no problem if the load is highly inductive because the current in the load cannot change abruptly.

* Formerly RCA 40526

Circuits that use a sensitive-gate triac to shift the firing point of the power triac by approximately 90 degrees have been designed. If the primary load is inductive, this phase shift corresponds to firing at zero current in the load. However, changes in the power factor of the load or tolerances of components will cause errors in this firing time.

The circuit shown in Fig. 19 uses a CA3086 integrated-circuit transistor array to detect the absence of load current by sensing the voltage across the triac. The internal zero-crossing detector is disabled by connection of terminal 12 to terminal 7, and control of the output is made through the external inhibit input, terminal 1. The circuit permits an output only when the voltage at point A exceeds two V_{BE} drops, or 1.3 volts. When A is positive, transistors Q_3 and Q_4 conduct and reduce the voltage at terminal 1 below the inhibit state. When A is negative, transistors Q_1 and Q_2 conduct. When the voltage at point A is less than ±1.3 volts, neither of the transistor pairs conducts; terminal 1 is then pulled positive by the current in resistor R_3, and the output in inhibited.

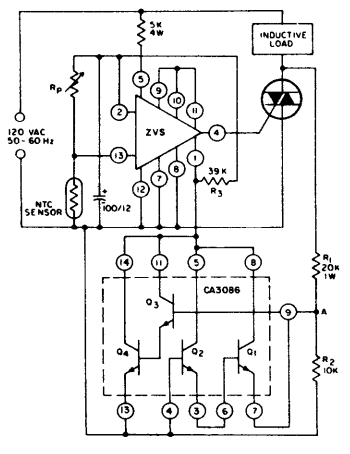


Fig. 19 – Use of the CA3058 or CA3059 together with CA3086 for switching inductive loads.

The circuit shown in Fig. 19 forms a pulse of gate current and can supply high peak drive to power traics with low average current drain on the internal supply. The gate pulse will always last just long enough to latch the thyristor so that there is no problem with delaying the pulse to an optimum time. As in other circuits of this type, RFI results if the load is not suitably inductive because the zero-crossing detector is disabled and initial turn-on occurs at random.

The gate pulse forms because the voltage at point A when the thyristor is on is less than 1.3 volts: therefore, the output of the zero-voltage switch is inhibited, as described above. The resistor divider R_1 and R_2 should be selected to assure this condition. When the triac is on, the voltage at point A is approximately one-third of the instantaneous on-state voltage (v_T) of the thyristor. For most RCA thyristors, v_T (max) is less than 2 volts, and the divider shown is a conservative one. When the load current passes through zero, the triac commutates and turns off. Because the circuit is still being driven by the line voltage, the current in the load attempts to reverse, and voltage increases rapidly across the "turned-off" triac. When this voltage exceeds 4 volts, one portion of the CA3086 conducts and removes the inhibit signal to permit application of gate drive. Turning the triac on causes the voltage across it to drop and thus ends the gate pulse. If the latching current has not been attained, another gate pulse forms, but no discontinuity in the load current occurs.

Provision of Negative Gate Current

Triacs trigger with optimum sensitivity when the polarity of the gate voltage and the voltage at the main terminal 2 are similar (I^+ and II^- modes). Sensitivity is degraded when the polarities are opposite (I^- and III^+ modes). Although RCA triacs are designed and specified to have the same sensitivity in both I^- and III^+ modes, some other types have very poor sensitivity in the III^+ condition. Because the zero-voltage switch supplies positive gate pulses, it may not directly drive some higher-current triacs of these other types.

The circuit shown in Fig. 20(a) uses the negative-going voltage at terminal 3 of the zero-voltage switch to supply a negative gate pulse through a capacitor. The curve in Fig. 20(b) shows the approximate peak gate current as a function of gate voltage V_G. Pulse width is approximately 80 microseconds.

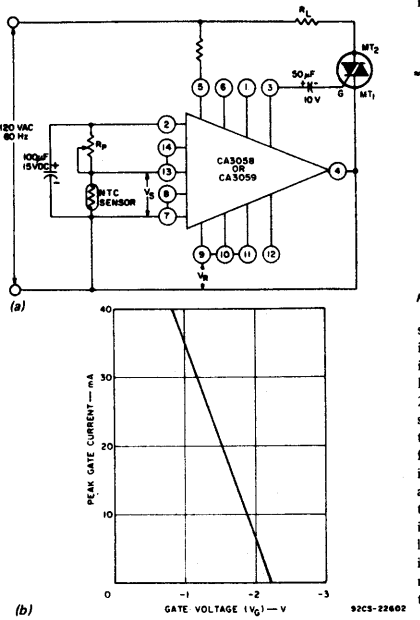


Fig. 20 – Use of the CA3058 or CA3059 to provide negative gate pulses: (a) schematic diagram; (b) peak gate current (at terminal 3) as a function of gate voltage.

Operation with Low-Impedance Sensors

Although the zero-voltage switch can operate satisfactorily with a wide range of sensors, sensitivity is reduced when sensors with impedances greater than 20,000 ohms are used. Typical sensitivity is one per cent for a 5000-ohm sensor and increases to three per cent for a 0.1-megohm sensor.

Low-impedance sensors present a different problem. The sensor bridge is connected across the internal power supply and causes a current drain. A 5000-ohm sensor with its associated 5000-ohm series resistor draws less than 1 milliampere. On the other hand, a 300-ohm sensor draws a current of 8 to 10 milliampers from the power supply.

Fig. 21 shows the 600-ohm load line of a 300-ohm sensor on a redrawn power-supply regulation curve for the zero-voltage switch. When a 10,000-ohm series resistor is used, the voltage across the circuit is less than 3 volts and both sensitivity and output current are significantly reduced. When a 5000-ohm series resistor is used, the supply voltage is nearly 5 volts, and operation is approximately normal. For more consistent operation, however, a 4000-ohm series resistor is recommended.

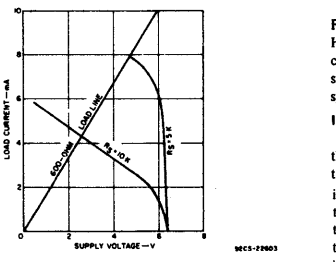


Fig. 21 – Power-supply regulation of the CA3058 or CA3059 with a 300-ohm sensor (600-ohm load) for two values of series resistor.

Although positive-temperature-coefficient (PTC) sensors rated at 5 kilohms are available, the existing sensors in ovens are usually of a much lower value. The circuit shown in Fig. 22 is offered to accommodate these inexpensive metal-wound sensors. A schematic diagram of the RCA CA3080 integrated-circuit operational transconductance amplifier used in Fig. 22, is shown in Fig. 23. With an amplifier bias current, I_{ABC}, of 100 microamperes, a forward transconductance of 2 milliohms is achieved in this configuration. The CA3080 switches when the voltage at terminal 2 exceeds the voltage at terminal 3. This action allows the sink current, I_s, to flow from terminal 13 of the zero-voltage switch (the input impedance to terminal 13 of the zero-voltage switch is approximately 50 kilohms); gate pulses are no longer applied to the triac because Q_2 of the zero-voltage switch is on. Hence, if the PTC sensor is cold, i.e., in the low resistance state, the load is energized. When the temperature of the PTC sensor increases to the desired temperature, the sensor enters the high resistance state, the voltage on terminal 2 becomes greater than that on terminal 3, and the triac switches the load off.

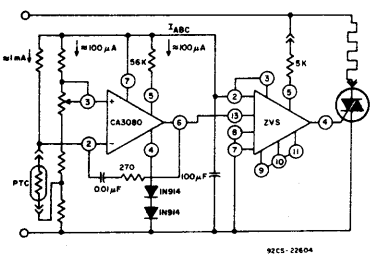


Fig. 22 – Schematic diagram of circuit for use with low-resistance sensor.

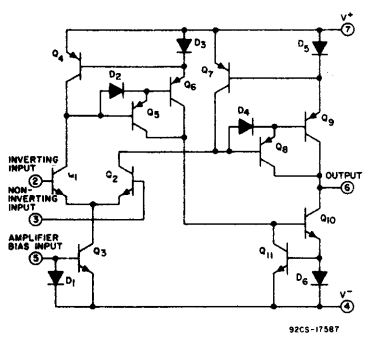


Fig. 23 – Schematic diagram of the CA3080.

Further cycling depends on the voltage across the sensor. Hence, very low values of sensor and potentiometer resistance can be used in conjunction with the zero-voltage switch power supply without causing adverse loading effects and impairing system performance.

Interfacing Techniques

Fig. 24 shows a system diagram that illustrates the role of the zero-voltage switch and thyristor as an interface between the logic circuitry and the load. There are several basic interfacing techniques. Fig. 25(a) shows the **direct input** technique. When the logic output transistor is switched from the on state (saturated) to the off state, the load will be turned on at the next zero-voltage crossing by means of the interfacing zero-voltage switch and the triac. When the logic output transistor is switched back to the on state, zero-crossing pulses from the zero-voltage switch to the triac

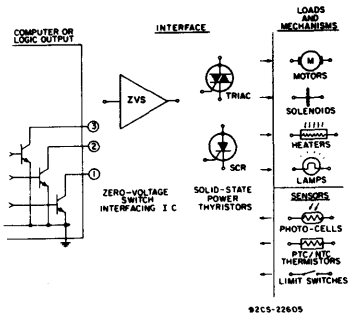


Fig. 24 – The zero-voltage switch and thyristor as an interface.

gate will immediately cease. Therefore, the load will be turned off when the triac commutates off as the sine-wave load current goes through zero. In this manner, both the turn-on and turn-off conditions for the load are controlled.

When electrical isolation between the logic circuit and the load is necessary, the **isolated-input** technique shown in Fig. 25(b) is used. In the technique shown, optical coupling is used to achieve the necessary isolation. The logic output transistor switches the light-source portion of the isolator. The light-sensor portion changes from a high impedance to a low impedance when the logic output transistor is switched from

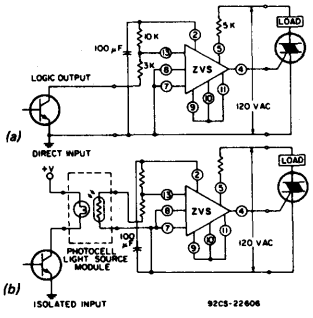


Fig. 25 – Basic interfacing techniques: (a) direct input; (b) isolated input.

off to on. The light sensor is connected to the differential amplifier input of the zero-voltage switch, which senses the change of impedance at a threshold level and switches the load on as in Fig. 25(a).

Sensor Isolation

In many applications, electrical isolation of the sensor from the ac input line is desirable. Several isolation techniques are shown in Figs. 26, 27, and 28.

Transformer Isolation – In Fig. 26, a pulse transformer is used to provide electrical isolation of the sensor from incoming ac power lines. The pulse transformer T_1 isolates the sensor from terminal No. 1 of the triac Y_1, and transformer T_2 isolates the CA3058 or CA3059 from the power lines. Capacitor C_1 shifts the phase of the output pulse at terminal No. 4 in order to retard the gate pulse delivered to triac Y_1 to compensate for the small phase-shift introduced by transformer T_1.

Many applications require line isolation but not zero-voltage switching. A line-isolated temperature controller for use with inductive or resistive loads that does not include zero-voltage switching is shown in Fig. 27.

In temperature monitoring or control applications the sensor may be a temperature-dependent element such as a resistor, thermistor, or diode. The load may be a lamp, bell, horn, recorder or other appropriate device connected in a feedback relationship to the sensor.

For the purpose of the following explanation, assume that the sensor is a resistor having a negative temperature coefficient

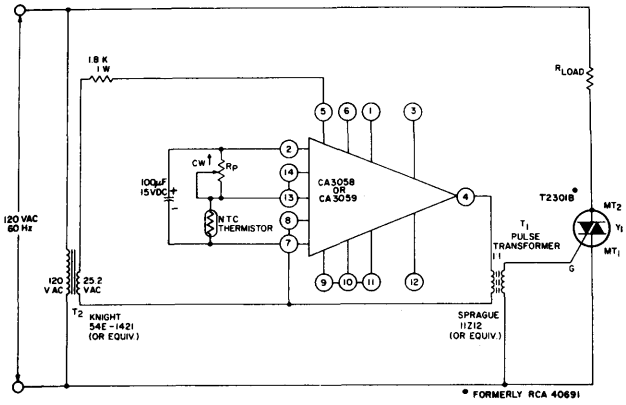


Fig. 26 – Zero-voltage switch, on-off controller with an isolated sensor.

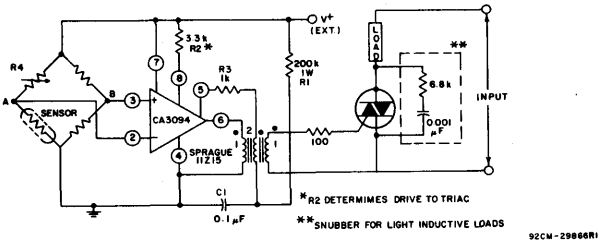


Fig. 27 – A line-isolated temperature controller for use with inductive or resistive loads; this controller does not include zero-voltage switching.

and that the load is a heater thermally coupled to the sensor, the object being to maintain the thermal-coupling medium at a desired reference temperature. Assume initially that the temperature at the coupling medium is low.

The operating potentials applied to the bridge circuit produce a common-mode potential, V_{CM}, at the input terminals of the CA3094. Assuming the bridge to have been initially balanced (by adjustment of R4), the potential at point A will increase when temperature is low since it was assumed that the sensor has a negative temperature coefficient. The potential at the noninverting terminal, being greater than that at the inverting terminal at the amplifier, causes the multivibrator to oscillate at approximately 10 kHz. The oscillations are transformer-coupled through a current-limiting resistor to the gate of the thyristor, and trigger it into conduction.

When the thyristor conducts, the load receives ac input power, which tends to increase the temperature of the sensor. This temperature increase decreases the potential at point A to a value below that at point B and the multivibrator is disabled, which action, in turn, turns off the thyristor. The temperature is thus controlled in an on-off fashion.

Capacitor C_1 is used to provide a low impedance path to ground for feedback-induced signals at terminal No. 5 while blocking the direct current bias provided by resistor R1. Resistor R2 provides current limiting. Resistor R3 limits the secondary current of the transformer to prevent excessive current flow to the control terminal of the CA3094.

Photocoupler Isolation – In Fig. 28, a photocoupler provides electrical isolation of the sensor logic from the incoming ac power lines. When a logic "1" is applied at the input of the photocoupler, the triac controlling the load will be turned on whenever the line voltage passes through zero. When a logic "0" is applied to the photocoupler, the triac will turn off and remain off until a logic "1" appears at the input of the photocoupler.

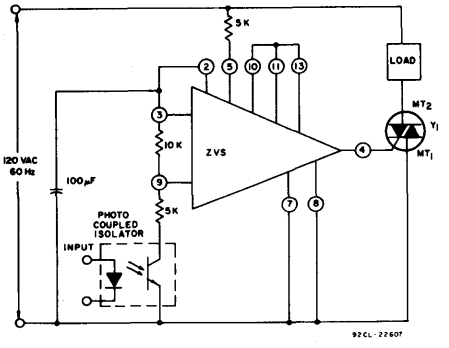


Fig. 28 – Zero-voltage switch, on-off controller with photocoupler.

TEMPERATURE CONTROLLERS

Fig. 29 shows a triac used in an on-off temperature-controller configuration. The triac is turned on at zero voltage whenever the voltage V_s exceeds the reference

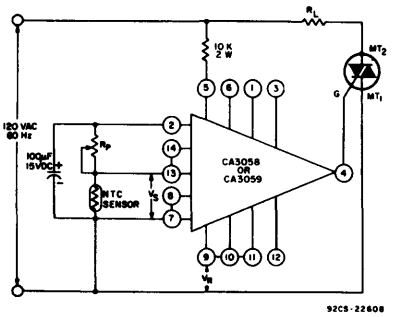


Fig. 29 – CA3058 or CA3059 on-off temperature controller.

voltage V_r. The transfer characteristic of this system, shown in Fig. 30(a), indicates significant thermal overshoots and undershoots, a well-known characteristic of such a system. The differential or hysteresis of this system, however, can be further increased, if desired, by the addition of positive feedback.

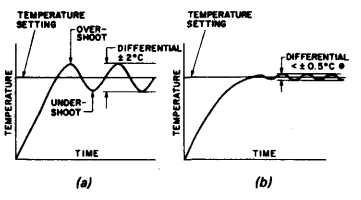


Fig. 30 – Transfer characteristics of (a) on-off and (b) proportional control systems.

For precise temperature-control applications, the proportional-control technique with synchronous switching is employed. The transfer curve for this type of controller is shown in Fig. 30(b). In this case, the duty cycle of the power supplied to the load is varied with the demand for heat required and the thermal time constant (inertia) of the system. For example, when the temperature setting is increased in an on-off type of controller, full power (100 per cent duty cycle) is supplied to the system. This effect results in significant temperature excursions because there is no anticipatory circuit to reduce the power gradually before the actual set temperature is achieved. However, in a proportional control technique, less power is supplied to the load (reduced duty cycle) as the error signal is reduced (sensed temperature approaches the set temperature).

Before such a system is implemented, a time base is chosen so that the on-time of the triac is varied within this time base. The ratio of the on-to-off time of the triac within this time interval depends on the thermal time constant of the system and the selected temperature setting. Fig. 31 illustrates the principle of proportional control. For this operation, power is supplied to the load until the ramp voltage reaches a value greater than the dc control signal supplied to the opposite side of the differential amplifier. The triac then remains off for the remainder of the time-base period. As a result, power is "proportioned" to the load in a direct relation to the heat demanded by the system.

For this application, a simple ramp generator can be realized with a minimum number of active and passive components. A ramp having good linearity is not required for proportional operation because of the nonlinearity of the

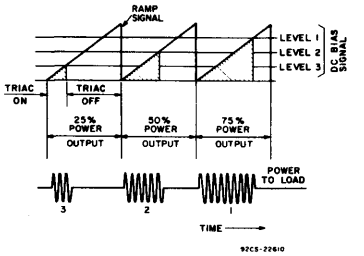


Fig. 31 – Principles of proportional control.

thermal system and the closed-loop type of control. In the circuit shown in Fig. 32, the ramp voltage is generated when the capacitor C_1 charges through resistors R_0 and R_1. The time base of the ramp is determined by resistors R_2 and R_3, capacitor C_2, and the breakover voltage of the D3202U* diac.

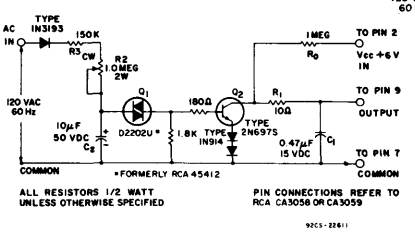


Fig. 32 – Ramp generator.

When the voltage across C_2 reaches approximately 32 volts, the diac switches and turns on the 2N697S transistor and 1N914 diodes. The capacitor C_1 then discharges through the collector-to-emitter junction of the transistor. This discharge time is the retrace or flyback time of the ramp. The circuit shown can generate ramp times ranging from 0.3 to 2.0 seconds through adjustment of R_2. For precise temperature regulation, the time base of the ramp should be shorter than the thermal time constant of the system, but long with respect to the period of the 60-Hz line voltage. Fig. 33 shows a triac connected for the proportional mode.

lag the incoming line voltage. The motors, however, are switched by the triacs at zero current, as shown in Fig. 34(b).

The problem of driving inductive loads such as these motors by the narrow pulses generated by the zero-voltage switch is solved by use of the sensitive-gate RCA-40526 triac. The high sensitivity of this device (3 milliamperes maximum) and low latching current (approximately 9 milliamperes) permit synchronous operation of the temperature-controller circuit. In Fig. 34(a), it is apparent that, though the gate pulse V_g of triac Y_1 has elapsed, triac Y_2 is switched on by the current through R_{L1}. The low latching current of the RCA-40526 triac results in dissipation of only 2 watts in R_{L1}, as opposed to 10 to 20 watts when devices that have high latching currents are used.

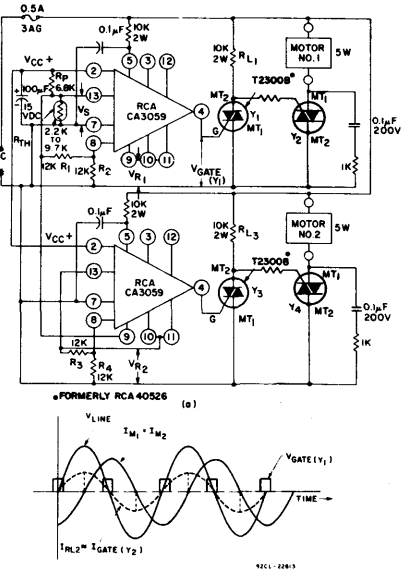


Fig. 34 – Dual output, over-under temperature controller (a) circuit, (b) voltage and current waveforms.

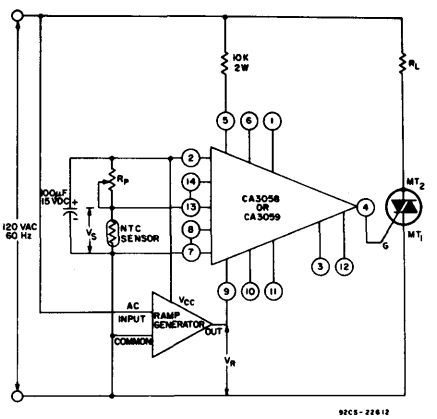


Fig. 33 – CA3058 or CA3059 proportional temperature controller.

Fig. 34(a) shows a dual-output temperature controller that drives two triacs. When the voltage V_s developed across the temperature-sensing network exceeds the reference voltage V_{R1}, motor No. 1 turns on. When the voltage across the network drops below the reference voltage V_{R2}, motor No. 2 turns on. Because the motors are inductive, the currents I_{M1}

Electric-Heat Application

For electric-heating applications, the RCA-2N5444 40-ampere triac and the zero-voltage switch constitute an optimum pair. Such a combination provides synchronous switching and effectively replaces the heavy-duty contactors which easily degrade as a result of pitting and wearout from

* Formerly RCA 45412

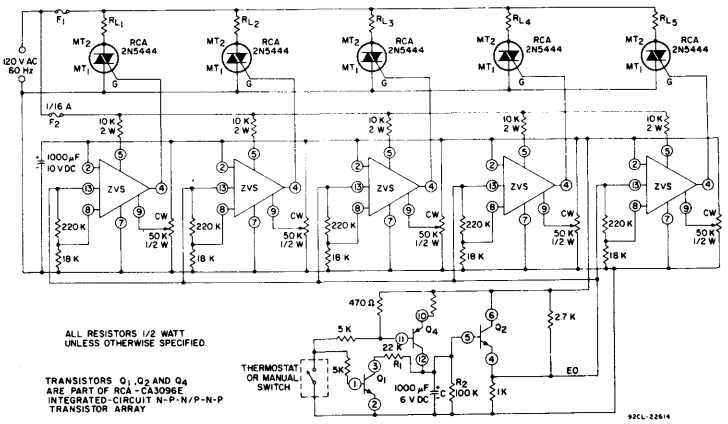


Fig. 35 – Synchronous-switching heat-staging controller using a series of zero-voltage switches.

the switching transients. The salient features of the 2N5444 40-ampere triac are as follows:

(1) 300-ampere single-surge capability (for operation at 60-Hz),

(2) a typical gate sensitivity of 20 milliamperes in the $I(^+)$ and $III(^+)$ modes,

(3) low on-state voltage of 1.5 volts maximum at 40 amperes, and

(4) available V_{DROM} equal to 600 volts.

Fig. 35 shows the circuit diagram of a synchronous-switching heat-staging controller that is used for electric heating systems. Loads as heavy as 5 kilowatts are switched sequentially at zero voltage to eliminate RFI and prevent a dip in line voltage that would occur if the full 25 kilowatts were to be switched simultaneously.

Transistor Q_1 and Q_4 are used as a constant-current source to charge capacitor C in a linear manner. Transistor Q_2 acts as a buffer stage. When the thermostat is closed, a ramp voltage is provided at output E_o. At approximately 3-second intervals, each 5-kilowatt heating element is switched onto the power system by its respective triac. When there is no further demand for heat, the thermostat opens, and capacitor C discharges through R_1 and R_2 to cause each triac to turn off in the reverse heating sequence. It should be noted that some half-cycling occurs before the heating element is switched fully on. This condition can be attributed to the inherent dissymmetry of the triac and is further aggravated by the slow-rising ramp voltage applied to one of the inputs. The timing diagram in Fig. 36 shows the turn-on and turn-off sequence of the heating system being controlled.

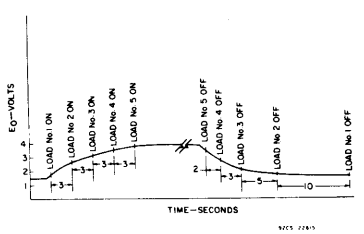


Fig. 36 – Ramp-voltage waveform for the heat-staging controller.

Seemingly, the basic method shown in Fig. 35 could be modified to provide proportional control in which the number of heating elements switched into the system, under any given thermal load, would be a function of the BTU's required by the system or the temperature differential between an indoor and outdoor sensor within the total system environment. That is, the closing of the thermostat would not switch in all the heating elements within a short time interval, which inevitably results in undesired temperature excursions, but would switch in only the number of heating elements required to satisfy the actual heat load.

Oven/Broiler Control

Zero-voltage switching is demonstrated in the oven control circuit shown in Fig. 37. In this circuit, a sensor element is included in the oven to provide a closed-loop system for accurate control of the oven temperature.

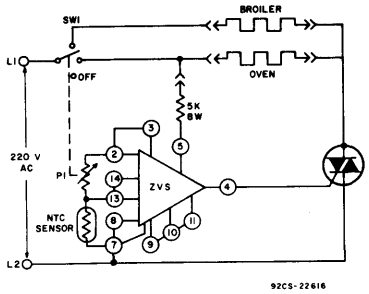


Fig. 37 – Schematic diagram of basic oven control.

As shown in Fig. 37, the temperature of the oven can be adjusted by means of potentiometer R_1, which acts, together with the sensor, as a voltage divider at terminal 13. The voltage at terminal 13 is compared to the fixed bias at terminal 9 which is set by internal resistors R_4 and R_5. When the oven is cold and the resistance of the sensor is high, transistors Q_2 and Q_4 are off, a pulse of gate current is applied to the triac, and heat is applied to the oven. Conversely, as the desired temperature is reached, the bias at terminal 13 turns the triac off. The closed-loop feature then cycles the oven element on and off to maintain the desired temperature to approximately ±2°C of the set value. Also, as has been noted, external resistors between terminals 13 and 8, and 7 and 8, can be used to vary this temperature and provide hysteresis. In Fig. 11, a circuit that provides approximately 10-per-cent hysteresis is demonstrated.

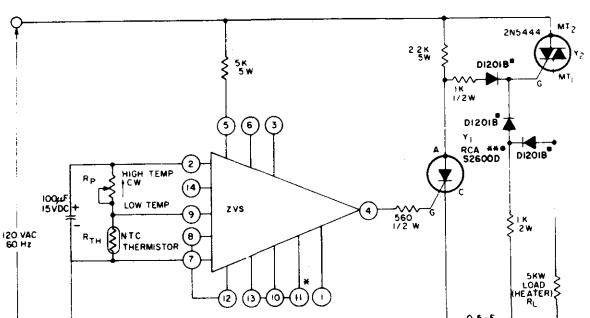
In addition to allowing the selection of a hysteresis value, the flexibility of the control circuit permits incorporation of other features. A PTC sensor is readily used by interchanging terminals 9 and 13 of the circuit shown in Fig. 37 and substituting the PTC for the NTC sensor. In both cases, the sensor element is directly returned to the system ground or common, as is often desired. Terminal 9 can be connected by external resistors to provide for a variety of biasing, e.g., to match a lower-resistance sensor for which the switching-point voltage has been reduced to maintain the same sensor current.

To accommodate the self-cleaning feature, external switching, which enables both broiler and oven units to be paralleled, can easily be incorporated in the design. Of course, the potentiometer must be capable of a setting such that the sensor, which must be characterized for the high, self-clean temperature, can monitor and establish control of the high-temperature, self-clean mode. The ease with which this self-clean mode can be added makes the over-all solid-state systems cost-competitive with electromechanical systems of comparable capability. In addition, the system incorporates solid-state reliability while being neater, more easily calibrated, and containing less-costly system wiring.

Integral-Cycle Temperature Controller (No half-cycling)

If a temperature controller which is completely devoid of half-cycling and hysteresis is required, then the circuit shown in Fig. 38 may be used. This type of circuit is essential for applications in which half-cycling and the resultant dc component could cause overheating of a power transformer on the utility lines.

In the integral-cycle controller, when the temperature being controlled is low, the resistance of the thermistor is high, and an output signal at terminal 4 of zero volts is obtained. The SCR (Y_1), therefore, is turned off. The triac (Y_2) is then triggered directly from the line on positive cycles of the ac voltage. When Y_2 is triggered and supplies power to the load R_L, capacitor C is charged to the peak of the input voltage.



* FOR PROPORTIONAL OPERATION OPEN TERMINALS 10, 11, AND 13, AND CONNECT POSITIVE RAMP VOLTAGE TO TERMINAL 13

** SELECTED FOR I_{GT} = 6 mA MAXIMUM.

■ FORMERLY RCA 44003

● FORMERLY RCA 40655

Fig. 38 – Integral-cycle temperature controller in which half-cycling effect is eliminated.

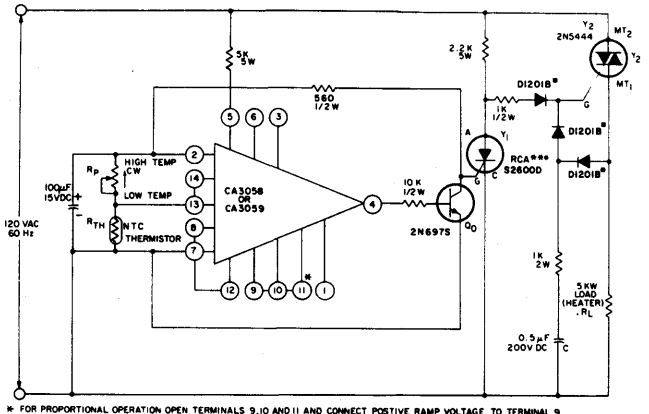


Fig. 39 – CA3058 or CA3059 integral-cycle temperature controller that features a protection circuit and no half-cycling effect.

When the ac line swings negative, capacitor C discharges through the triac gate to trigger the triac on the negative half-cycle. The diode-resistor-capacitor "slaving network" triggers the triac on negative half-cycle to provide only integral cycles of ac power to the load.

When the temperature being controlled reaches the desired value, as determined by the thermistor, then a positive voltage level appears at terminal 4 of the zero-voltage switch. The SCR then starts to conduct at the beginning of the positive input cycle to shunt the trigger current away from the gate of the triac. The triac is then turned off. The cycle repeats when the SCR is again turned OFF by the zero-voltage switch.

The circuit shown in Fig. 39 is similar to the configuration in Fig. 38 except that the protection circuit incorporated in the zero-voltage switch can be used. In this new circuit, the NTC sensor is connected between terminals 7 and 13, and transistor Q_0 inverts the signal output at terminal 4 to nullify the phase reversal introduced by the SCR (Y_1). The internal power supply of the zero-voltage switch supplies bias current to transistor Q_0.

Of course, the circuit shown in Fig. 39 can readily be converted to a **true proportional integral-cycle temperature controller** simply by connection of a positive-going ramp voltage to terminal 9 (with terminals 10 and 11 open), as previously discussed in this Note.

Thermocouple Temperature Control

Fig. 40 shows the CA3080A operating as a pre-amplifier for the zero-voltage switch to form a zero-voltage switching circuit for use with thermocouple sensors.

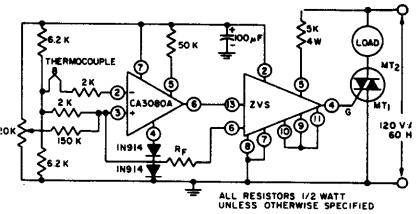


Fig. 40 – Thermocouple temperature control with zero-voltage switching.

Thermocouple Temperature Control with Zero-Voltage Load Switching

Fig. 41 shows the circuit diagram of a thermocouple temperature control system using zero-voltage load switching. It should be noted that one terminal of the thermocouple is connected to one leg of the supply line. Consequently, the thermocouple can be "ground-referenced", provided the appropriate leg of the ac line is maintained at ground. The comparator, A_1 (a CA3130), is powered from a 6.4-volt source of potential provided by the zero-voltage-switch (ZVS) circuit (a CA3079). The ZVS, in turn, is powered off-line through a series-dropping resistor R6. Terminal 4 of the ZVS provides trigger-pulses to the gate of the load-switching triac in response to an appropriate control signal at terminal 9.

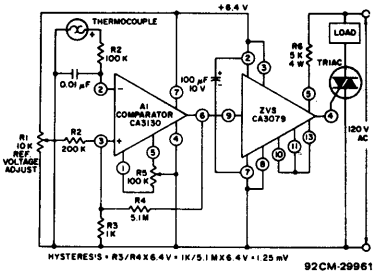


Fig. 41 – Thermocouple temperature control with zero-voltage switching.

The CA3130 is an ideal choice for the type of comparator circuit shown in Fig. 41 because it can "compare" low voltages (such as those generated by a thermocouple) in the proximity of the negative supply rail. Adjustment of potentiometer R1 drives the voltage-divider network R3, R4 so that reference voltages over the range of 0 to 20 millivolts can be applied to noninverting terminal 3 of the comparator. Whenever the voltage developed by the thermocouple at terminal 2 is more positive than the reference voltage applied at terminal 3, the comparator output is toggled so as to sink current from terminal 9 of the ZVS; gate pulses are then no longer applied to the triac. As shown in Fig. 41, the circuit is provided with a control-point "hysteresis" of 1.25 millivolts.

Nulling of the comparator is performed by means of the following procedure: Set R1 at the low end of its range and short the thermocouple output signal appropriately. If the triac is in the conductive mode under these conditions, adjust nulling potentiometer R5 to the point at which triac conduction is interrupted. On the other hand, if the triac is in the nonconductive mode under the conditions above, adjust R5 to the point at which triac conduction commences. The thermocouple output signal should then be unshorted, and R1 can be set to the voltage threshold desired for control-circuit operation.

MACHINE CONTROL AND AUTOMATION

The earlier section on interfacing techniques indicated several techniques of controlling ac loads through a logic system. Many types of automatic equipment are not complex enough or large enough to justify the cost of a flexible logic system. A special circuit, designed only to meet the control requirements of a particular machine, may prove more economical. For example, consider the simple machine shown in Fig. 42; for each revolution of the motor, the belt is advanced a prescribed distance, and the strip is then punched. The machine also has variable speed capability.

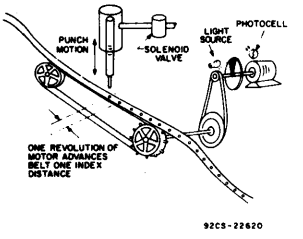


Fig. 42 – Step-and-punch machine.

The typical electromechanical control circuit for such a machine might consist of a mechanical cambank driven by a separate variable speed motor, a time delay relay, and a few logic and power relays. Assuming use of industrial-grade controls, the control system could get quite costly and large. Of greater importance is the necessity to eliminate transients generated each time a relay or switch energizes and deenergizes the solenoid and motor. Fig. 43 shows such transients, which might not affect the operation of this machine, but could affect the more sensitive solid-state equipment operating in the area.

A more desirable system would use triacs and zero-voltage switching to incorporate the following advantages:

a. Increased reliability and long life inherent in solid-state devices as opposed to moving parts and contacts associated with relays.

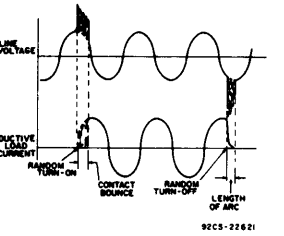


Fig. 43 – Transients generated by relay-contact bounce and non-zero turn-off of inductive load.

b. Minimized generation of EMI/RFI using zero-voltage switching techniques in conjunction with thyristors.
c. Elimination of high-voltage transients generated by relay-contact bounce and contacts breaking inductive loads, as shown in Fig. 42.
d. Compactness of the control system.

The entire control system could be on one printed-circuit board, and an over-all cost advantage would be achieved. Fig. 44 is a timing diagram for the proposed solid-state

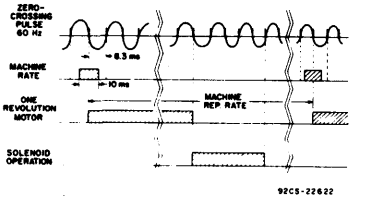


Fig. 44 – Timing diagram for proposed solid-state machine control.

machine control, and Fig. 45 is the corresponding control schematic. A variable-speed machine repetition rate pulse is set up using either a unijunction oscillator or a transistor astable multivibrator in conjunction with a 10-millisecond one-shot multivibrator. The first zero-voltage switch in Fig. 45 is used to synchronize the entire system to zero-voltage crossing. Its output is inverted to simplify adaptation to the rest of the circuit. The center zero-voltage switch is used as an interface for the photo-cell, to control one revolution of the motor. The gate drive to the motor triac is continuous dc, starting at zero voltage crossing. The motor is initiated when both the machine rate pulse and the zero-voltage sync are at low voltage. The bottom zero-voltage switch acts as a time-delay for pulsing the solenoid. The inhibit input, terminal 1, is used to assure that the solenoid will not be operated while the motor is running. The time delay can be adjusted by varying the reference level (50K potentiometer) at terminal 13 relative to the capacitor charging to that level on terminal 9. The capacitor is reset by the SCR during the motor operation. The gate drive to the solenoid triac is direct current. Direct current is used to trigger both the motor and solenoid triacs because it is the most desirable means of switching a triac into an inductive load. The output of the zero-voltage switch will be continuous dc by connecting terminal 12 to common. The output under dc operation should be limited to 20 milliamperes. The motor

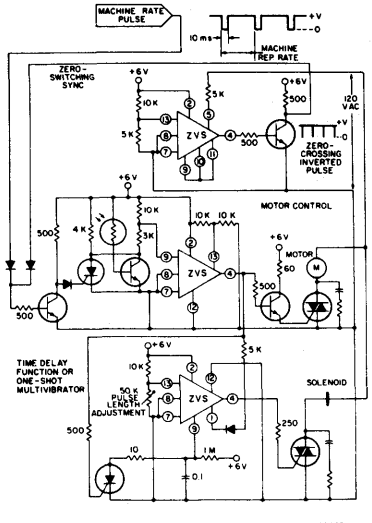


Fig. 45 – Schematic of proposed solid-state machine control.

triac is synchronized to zero crossing because it is a high-crurent inductive load and there is a chance of generating RFI. The solenoid is a very low current inductive load, so there would be little chance of generating RFI: therefore, the initial triac turn-on can be random, which simplifies the circuitry.

This example shows the versatility and advantages of the RCA zero-voltage switch used in conjunction with triacs as interfacing and control elements for machine control.

400-Hz TRIAC APPLICATIONS

The increased complexity of aircraft control systems, and the need for greater reliability than electromechanical switching can offer, has led to the use of solid-state power switching in aircraft. Because 400-Hz power is used almost universally in aircraft systems, RCA offers a complete line of triacs rated for 400-Hz applications. Use of the RCA zero-voltage switch in conjunction with these 400-Hz triacs results in a minimum of RFI, which is especially important in aircraft.

Areas of application for 400-Hz triacs in aircraft include:

a. Heater controls for food-warming ovens and for windshield defrosters.
b. Lighting controls for instrument panels and cabin illumination
c. Motor controls
d. Solenoid controls
e. Power-supply switches

Lamp dimming is a simple triac application that demonstrates an advantage of 400-Hz power over 60-Hz power. Fig. 46 shows the adjustment of lamp intensity by phase control of the 60-Hz line voltage. RFI is generated by the step functions of power each half cycle, requiring extensive filtering. Fig. 47 shows a means of controlling power to the lamp by the zero-voltage-switching technique. Use of 400-Hz power makes possible the elimination of complete or half cycles within a period (typically 17.5 milliseconds)

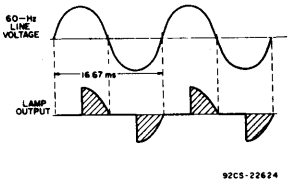


Fig. 46 – Waveforms for 60-Hz phase-controlled lamp dimmer.

without noticeable flicker. Fourteen different levels of lamp intensity can be obtained in this manner. A line-synced ramp is set up with the desired period and applied to terminal No. 9 of the differential amplifier within the zero-voltage switch, as shown in Fig. 48. The other side of the differential amplifier (terminal No. 13) uses a variable reference level, set by the 50K potentiometer. A change of the potentiometer setting changes the lamp intensity.

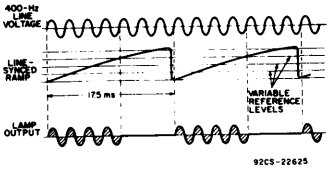


Fig. 47 – Waveforms for 400-Hz zero-voltage-switched lamp dimmer.

In 400-Hz applications it may be necessary to widen and shift the zero-voltage switch output pulse (which is typically 12 microseconds wide and centered on zero voltage crossing), to assure that sufficient latching current is available. The 4K resistor (terminal No. 12 to common) and the 0.015-microfarad capacitor (terminal No. 5 to common) are used for this adjustment.

SOLID-STATE TRAFFIC FLASHER

Another application which illustrates the versatility of the zero-voltage switch, when used with RCA thyristors, involves switching traffic-control lamps. In this type of application, it is essential that a triac withstand a current surge of the lamp load

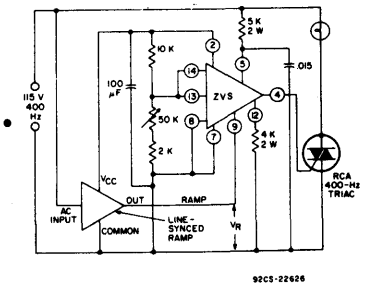


Fig. 48 – Circuit diagram for 400-Hz zero-voltage-switched lamp dimmer.

on a continuous basis. This surge results from the difference between the cold and hot resistance of the tungsten filament. If it is assumed that triac turn-on is at 90 degrees from the zero-voltage crossing, the first current-surge peak is approximately ten times the peak steady-state value or fifteen times the steady-state rms value. The second current-surge peak is approximately four times the steady-state rms value.

When the triac randomly switches the lamp, the rate of current rise di/dt is limited only by the source inductance. The triac di/dt rating may be exceeded in some power systems. In many cases, exceeding the rating results in excessive current concentrations in a small area of the device which may produce a hot spot and lead to device failure. Critical applications of this nature require adequate drive to the triac gate for fast turn-on. In this case, some inductance may be required in the load circuit to reduce the initial magnitude of the load current when the triac is passing through the active region. Another method may be used which involves the switching of the triac at zero line voltage. This method involves the supply of pulses to the triac gate only during the presence of zero voltage on the ac line.

Fig. 49 shows a circuit in which the lamp loads are switched at zero line voltage. This approach reduces the initial di/dt, decreases the required triac surge-current ratings, increases the operating lamp life, and eliminates RFI problems. This circuit consists of two triacs, a flip-flop (FF-1), the zero-voltage switch, and a diac pulse generator. The flashing rate in this circuit is controlled by potentiometer R, which provides between 10 and 120 flashes per minute. The state of FF-1 determines the triggering of triacs Y_1 or Y_2 by the output pulses at terminal 4 generated by the zero-crossing circuit. **Transistors Q_1 and Q_2 inhibit these pulses to the gates of the triacs until the triacs turn on by the logical "1" (V_{CC} high) state of the flip-flop.**

The arrangement described can also be used for a synchronous, sequential traffic-controller system by addition of one triac, one gating transistor, a "divide-by-three" logic circuit, and modification in the design of the diac pulse generator. Such a system can control the familiar red, amber, and green traffic signals that are found at many intersections.

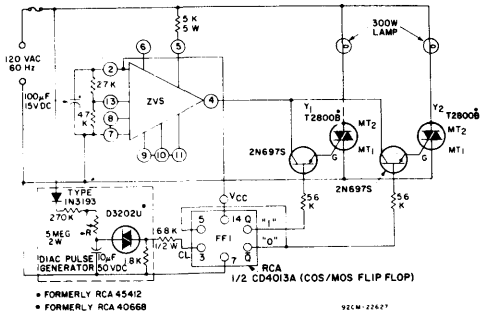


Fig. 49 – Synchronous-switching traffic flasher.

SYNCHRONOUS LIGHT FLASHER

Fig. 50 shows a simplified version of the synchronous-switching traffic light flasher shown in Fig. 49.

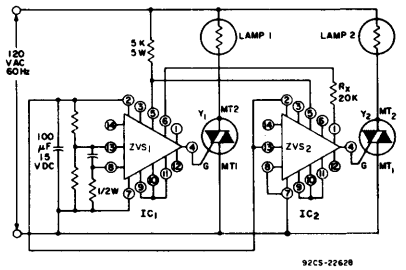


Fig. 50 – Synchronous light flasher.

Flash rate is set by use of the curve shown in Fig. 16. If a more precise flash rate is required, the ramp generator described previously may be used. In this circuit, ZVS_1 is the master control unit and ZVS_2 is slaved to the output of ZVS_1 through its inhibit terminal (terminal 1). When power is applied to lamp No. 1, the voltage of terminal 6 on ZVS_1 is high and ZVS_2 is inhibited by the current in R_X. When lamp No. 1 is off, ZVS_2 is not inhibited, and triac Y_2 can fire. The power supplies operate in parallel. The on-off sensing amplifier in ZVS_2 is not used.

TRANSIENT-FREE SWITCH CONTROLLERS

The zero-voltage switch can be used as a simple solid-state switching device that permits ac currents to be turned on or off with a minimum of electrical transients and circuit noise.

The circuit shown in Fig. 51 is connected so that, after the control terminal 14 is opened, the electronic logic waits until the power-line voltage reaches a zero crossing before power is applied to the load Z_L. Conversely, when the control terminals are shorted, the load current continues until it reaches a zero crossing. This circuit can switch a load at zero current whether it is resistive or inductive.

The circuit shown in Fig. 52 is connected to provide the opposite control logic to that of the circuit shown in Fig. 51. That is, when the switch is closed, power is supplied to the load, and when the switch is opened, power is removed from the load.

In both configurations, the maximum rms load current that can be switched depends on the rating of triac Y_2. If Y_2 is an RCA-2N5444 triac, an rms current of 40 amperes can be switched.

DIFFERENTIAL COMPARATOR FOR INDUSTRIAL USE

Differential comparators have found widespread use as limit detectors which compare two analog input signals and provide a go/no-go, logic "one" or logic "zero" output, depending upon the relative magnitudes of these signals. Because the signals are often at very low voltage levels and very accurate discrimination is normally required between them, differential comparators in many cases employ differential amplifiers as a basic building block. However, in many industrial control applications, a high-performance differential comparator is not required. That is, high resolution, fast switching speed, and similar features are not essential. The zero-voltage switch is ideally suited for use in such applications. Connection of terminal 12 to terminal 7 inhibits the zero-voltage threshold detector of the zero-voltage switch, and the circuit becomes a differential comparator.

Fig. 53 shows the circuit arrangement for use of the zero-voltage switch as a differential comparator. In this application, no external dc supply is required, as is the case with most commercially available integrated-circuit comparators; of course, the output-current capability of the zero-voltage switch is reduced because the circuit is operating in the dc mode. The 1000-ohm resistor R_G, connected between terminal 4 and the gate of the triac, limits the output current to approximately 3 milliamperes.

When the zero-voltage switch is connected in the dc mode, the drive current for terminal 4 can be determined from a curve of the external load current as a function of dc voltage

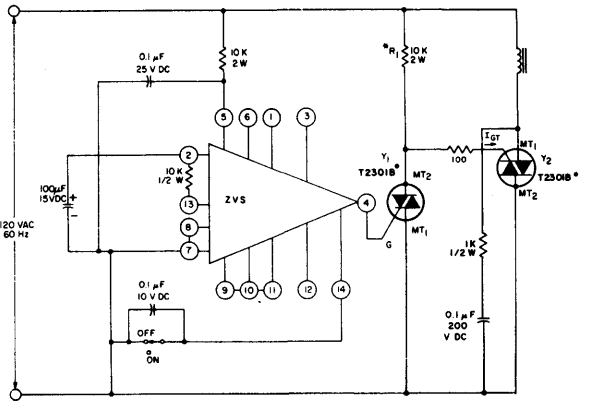


Fig. 51 – Zero-voltage switch transient-free switch controller in which power is supplied to the load when the switch is open.

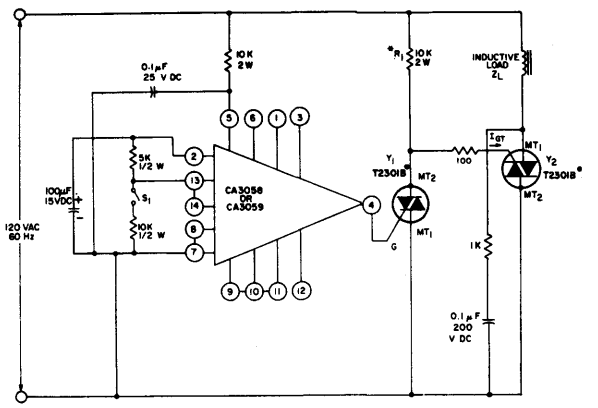


Fig. 52 – Zero-voltage switch transient-free switch controller in which power is applied to the load when the switch is closed.

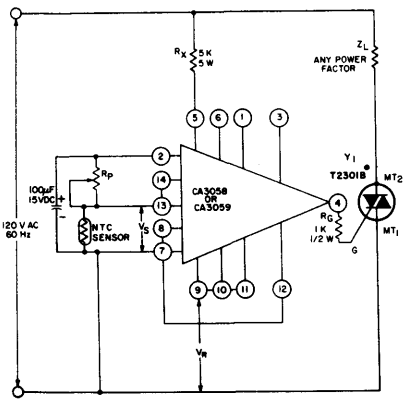


Fig. 53 – Differential comparator using the CA3058 or CA3059 integrated-circuit zero-voltage switch.

from terminals 2 and 7. This curve is shown in the technical bulletin for RCA integrated-circuit zero-voltage switches, File No. 490. Of course, if additional output current is required, an external dc supply may be connected between terminals 2 and 7, and resistor R_X (shown in Fig. 53) may be removed.

The chart below compares some of the operating characteristics of the zero-voltage switch, when used as a comparator, with a typical high-performance commercially available integrated-circuit differential comparator.

Parameters	Zero-Voltage Switch (Typical Values)	Typical Integrated-Circuit Comparator (710)
Sensitivity	30 mV	2 mV
Switching speed (rise time)	> 20 μs	90 ns
Output drive capability	*4.5 V at ⩽ 4 mA	3.2 V at ⩽ 5.0 mA

* Refer to Fig. 20; R_X equals 5000 ohms.

POWER ONE-SHOT CONTROL

Fig. 54 shows a circuit which triggers a triac for one complete half-cycle of either the positive or negative alternation of the ac line voltage. In this circuit, triggering is initiated by the push button PB-1, which produces triggering of the triac near zero voltage even though the button is randomly depressed during the ac cycle. The triac does not trigger again until the button is released and again depressed. This type of logic is required for the solenoid drive of electrically operated stapling guns, impulse hammers, and the like, where load-current flow is required for only one complete half-cycle. Such logic can also be adapted to keyboard consoles in which contact bounce produces transmission of erroneous information.

In the circuit of Fig. 54, before the button is depressed, both flip-flop outputs are in the "zero" state. Transistor Q_G is biased on by the output of flip-flop FF-1. The differential comparator which is part of the zero-voltage switch is initially

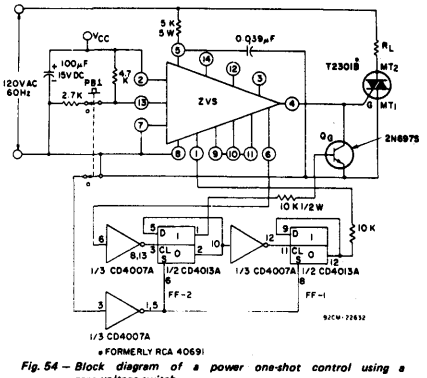


Fig. 54 – Block diagram of a power one-shot control using a zero-voltage switch.

biased to inhibit output pulses. When the push button is depressed, pulses are generated, but the state of Q_G determines the requirement for their supply to the triac gate. The first pulse generated serves as a "framing pulse" and does not trigger the triac but toggles FF-1. Transistor Q_G is then turned off. The second pulse triggers the triac and FF-1 which, in turn, toggles the second flip-flop FF-2. The output of FF-2 turns on transistor Q_7, as shown in Fig. 55, which inhibits all further output pulses. When the pushbutton is released, the circuit resets itself until the process is repeated with the button. Fig. 56 shows the timing diagram for the described operating sequence.

PHASE CONTROL CIRCUIT

Fig. 57 shows a circuit using a CA3058 or CA3059 zero-voltage switch together with two CA3086 integrated-circuit transistor arrays to form a phase-control circuit. This circuit is specifically designed for speed control of ac induction motors, but may also be used as a light dimmer.

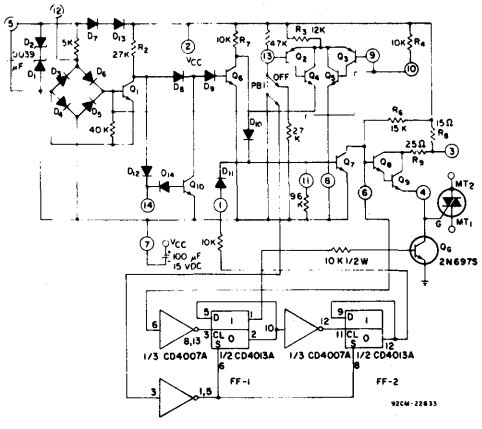


Fig. 55 – Circuit diagram for the power one-shot control.

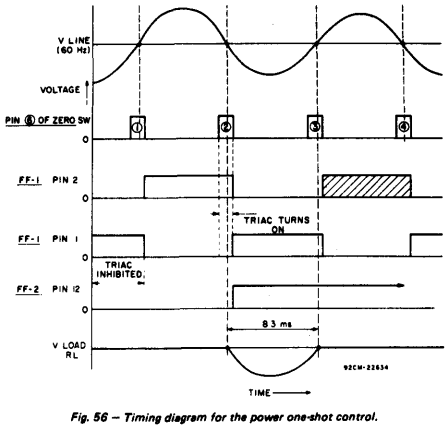


Fig. 56 – Timing diagram for the power one-shot control.

The circuit, which can be operated from a line frequency of 50-Hz to 400-Hz, consists of a zero-voltage detector, a line-synchronized ramp generator, a zero-current detector, and a line-derived control circuit (i.e., the zero-voltage switch). The zero-voltage detector (part of CA3086 No. 1) and the ramp generator (CA3086 No. 2) provide a line-synchronized ramp-voltage output to terminal 13 of the zero-voltage switch. The ramp voltage, which has a starting voltage of 1.8 volts, starts to rise after the line voltage passes the zero point. The ramp generator has an oscillation frequency of twice the incoming line frequency. The slope of the ramp voltage can be adjusted by variation of the resistance of the 1-megohm ramp-control potentiometer. The output phase can be controlled easily to provide 180° firing of the triac by programming the voltage at terminal 9 of the zero-voltage switch. The basic operation of the zero-voltage switch driving a thyristor with an inductive load was explained previously in the discussion on switching of inductive loads.

ON/OFF TOUCH SWITCH

The on/off touch switch shown in Fig. 58 uses the CA3240E to sense small currents flowing between two contact points on a touch plate consisting of a PC board metallization "grid". When the on plate is touched, current flows between the two halves of the grid, causing a positive shift in the output voltage (terminal 7) of the CA3240E. These positive transitions are fed into the CA3059, which is used as a latching circuit and zero-crossing triac driver. When a positive pulse occurs at terminal No. 7 of the CA3240E, the triac is turned on and held on by the CA3059 and associated positive feedback circuitry (51-kilohm resistor and 36-kilohm/42-kilohm voltage divider). When the pulse occurs at terminal No. 1, the triac is turned off and held off in a similar manner. Note that power for the CA3240E is derived from the CA3059 internal power supply. The advantage of using the CA3240E in this circuit is that it can sense the small currents associated with skin conduction while maintaining sufficiently high circuit impedance to protect against electrical shock.

TRIAC POWER CONTROLS FOR THREE-PHASE SYSTEMS

This section describes recommended configurations for power-control circuits intended for use with both inductive and resistive balanced three-phase loads. The specific design requirements for each type of loading condition are discussed.

In the power-control circuits described, the integrated-circuit zero-voltage switch is used as the trigger circuit for the power triacs. The following conditions are also imposed in the design of the triac control circuits:

1. The load should be connected in a three-wire configuration with the triacs placed external to the load; eiter delta or wye arrangements may be used. Four-wire loads in wye configurations can be handled as three independent single-phase systems. Delta configurations in which a triac is connected within each phase rather than in the incoming lines can also be handled as three independent single-phase systems.
2. Only one logic command signal is available for the

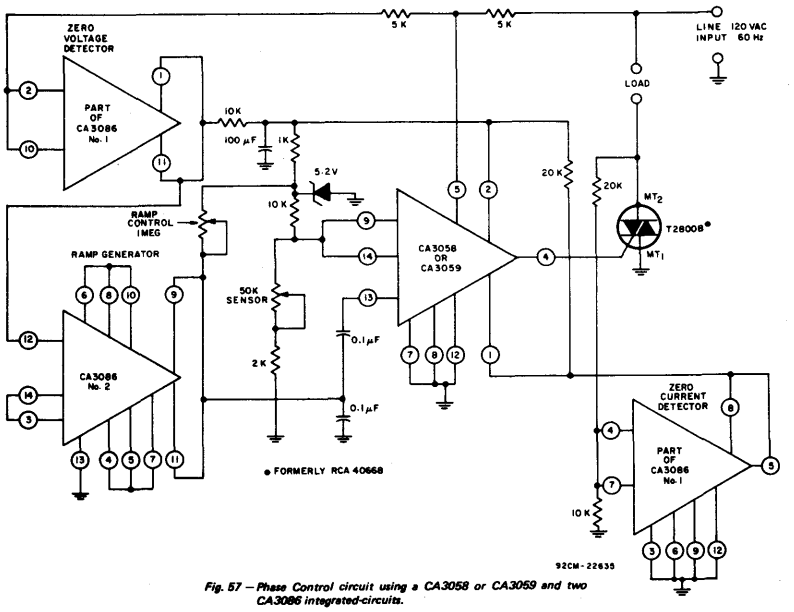


Fig. 57 – Phase Control circuit using a CA3058 or CA3059 and two CA3086 integrated-circuits.

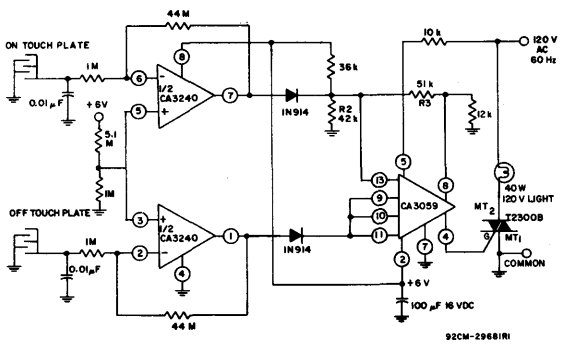


Fig. 58 – On-off touch switch.

control circuits. This signal must be electrically isolated from the three-phase power system.

3. Three separate triac gating signals are required.
4. For operation with resistive loads, the zero-voltage switching technique should be used to minimize any radio-frequency interference (RFI) that may be generated.

Isolation of DC Logic Circuitry

As explained earlier under **Special Application Considerations**, isolation of the dc logic circuitry* from the ac line, the triac, and the load circuit is often desirable even in many single-phase power-control applications. In control circuits for polyphase power systems, however, this type of isolation is essential, because the common point of the dc logic circuitry cannot be referenced to a common line in all phases.

In the three-phase circuits described in this section, photo-optic techniques (i.e., photo-coupled isolators) are used to provide the electrical isolation of the dc logic command signal from the ac circuits and the load. The photo-coupled isolators consist of an infrared light-emitting diode aimed at a silicon photo transistor, coupled in a common package. The light-emitting diode is the input section, and the photo transistor is the output section. The two components provide a voltage isolation typically of 1500 volts. Other isolation techniques, such as pulse transformers, magnetoresistors, or reed relays, can also be used with some circuit modifications.

Resistive Loads

Fig. 59 illustrates the basic phase relationships of a balanced three-phase resistive load, such as may be used in heater applications, in which the application of load power is controlled by zero-voltage switching. The following conditions are inherent in this type of application:

1. The phases are 120 degrees apart; consequently, all three phases cannot be switched on simultaneously at zero voltage.
2. A single phase of a wye configuration type of three-wire system cannot be turned on.
3. Two phases must be turned on for initial starting of the system. These two phases form a single-phase circuit which is out of phase with both of its component phases. The single-phase circuit leads one phase by 30 degrees and lags the other phase by 30 degrees.

These conditions indicate that in order to maintain a system in which no appreciable RFI is generated by the switching action from initial starting through the steady-state operating condition, the system must first be turned on, by zero-voltage switching, as a single-phase circuit and then must revert to synchronous three-phase operation.

Fig. 60 shows a simplified circuit configuration of a three-phase heater control that employs zero-voltage synchronous switching in the steady-state operating condition, with random starting. In this system, the logic command to turn on the system is given when heat is required, and the command to turn off the system is given when heat is not required. Time proportioning heat control is also possible through the use of logic commands.

The three photo-coupled inputs to the three zero-voltage switches change state simultaneously in response to a "logic command". The zero-voltage switches then provide a positive pulse, approximately 100 microseconds in duration, only at a zero-voltage crossing relative to their particular phase. A balanced three-phase sensing circuit is set up with the three zero-voltage switches each connected to a particular phase on their common side (terminal 7) and referenced at their high side (terminal 5), through the current-limiting resistors R4, R5, and R6, to an established artificial neutral point. This artificial neutral point is electrically equivalent to the inaccessible neutral point of the wye type of three-wire load and, therefore, is used to establish the desired phase relationships. The same artificial neutral point is also used to establish the proper phase relationships for a delta type of three-wire load. Because only one triac is pulsed on at a time, the diodes (D1, D2, and D3) are necessary to trigger the opposite-polarity triac, and, in this way, to assure initial latching-on of the system. The three resistors (R1, R2, and R3) are used for current limiting of the gate drive when the opposite-polarity triac is triggered on by the line voltage.

In critical applications that require suppression of all generated RFI, the circuit shown in Fig. 61 may be used. In addition to synchronous steady-state operating conditions, this circuit also incorporates a zero-voltage starting circuit. The start-up condition is zero-voltage synchronized to a single-phase, 2-wire, line-to-line circuit, comprised of phases A and B. The logic command engages the single-phase start-up zero-voltage switch and three-phase photo-coupled isolators OC13, OC14, OC15 through the photo-coupled

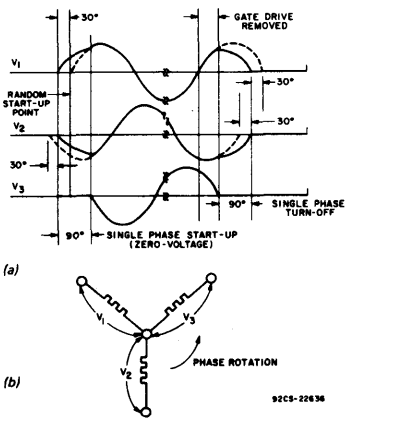


Fig. 59 – Voltage phase relationship for a three-phase resistive load when the application of load power is controlled by zero-voltage switching: (a) voltage waveforms, (b) load-circuit orientation of voltages. (The dashed lines indicate the normal relationship of the phases under steady-state conditions. The deviation at start-up and turn-off should be noted.)

* The dc logic circuitry provides the low-level electrical signal that dictates the state of the load. For temperature controls, the dc logic circuitry includes a temperature sensor for feedback. The RCA integrated-circuit zero-voltage switch, when operated in the dc mode with some additional circuitry, can replace the dc logic circuitry for temperature controls.

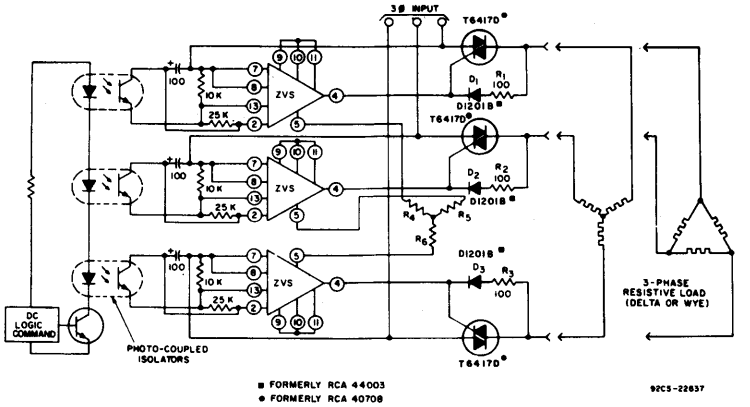


Fig. 60 – Simplified diagram of a three-phase heater control that employs zero-voltage synchronous switching in the steady-state operating conditions.

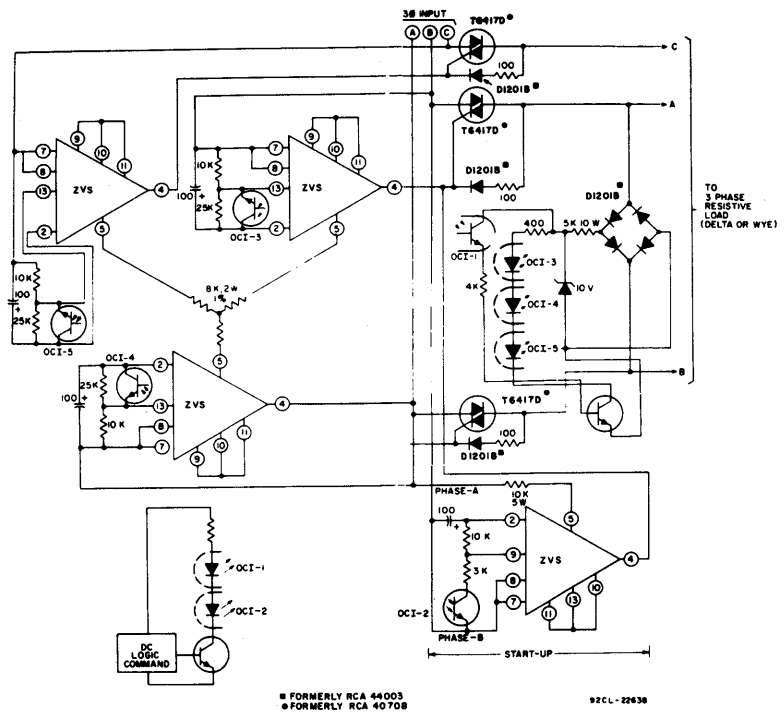


Fig. 61 – Three-phase power control that employs zero-voltage synchronous switching both for steady-state operation and for starting.

isolators OC11 and OC12. The single-phase zero-voltage switch, which is synchronized to phases A and B, starts the system at zero voltage. As soon as start-up is accomplished, the three photo-coupled isolators OC13, OC14, and OC15 take control, and three-phase synchronization begins. When the "logic command" is turned off, all control is ended, and the triacs automatically turn off when the sine-wave current decreases to zero. Once the first phase turns off, the other two will turn off simultaneously, 90° later, as a single-phase line-to-line circuit, as is apparent from Fig. 59.

Inductive Loads

For inductive loads, zero-voltage turn-on is not generally required because the inductive current cannot increase instantaneously; therefore, the amount of RFI generated is usually negligible. Also, because of the lagging nature of the inductive current, the triacs cannot be pulse-fired at zero voltage. There are several ways in which the zero-voltage switch may be interfaced to a triac for inductive-load applications. The most direct approach is to use the zero-voltage switch in the dc mode, i.e., to provide a continuous dc output instead of pulses at points of zero-voltage crossing. This mode of operation is accomplished by connection of terminal 12 to terminal 7, as shown in Fig. 62. The output of the zero-voltage switch should also be limited to approximately 5 milliamperes in the dc mode by the 750-ohm series resistor. Use of a triac such as the T2301D* is recommended for this application. Terminal 3 is connected to terminal 2 to limit the steady-state power dissipation within the zero-voltage switch. For most three-phase inductive load applications, the current-handling capability of the 40692 triac (2.5 amperes) is not sufficient. Therefore, the 40692 is used as a trigger triac to turn on any other currently available power triac that may be used. The trigger triac is used only to provide trigger pulses to the gate of the power triac (one pulse per half cycle); the power dissipation in this device, therefore, will be minimal.

Simplified circuits using pulse transformers and reed relays will also work quite satisfactorily in this type of application. The RC networks across the three power triacs are used for suppression of the commutating dv/dt when the circuit operates into inductive loads.

The specific integrated-circuits, triacs, SCR's, and rectifiers included in circuit diagrams shown in this Application Note are listed below. Additional information on these devices can be obtained by requesting the applicable RCA data-bulletin file number.

Type No.	File No.
CA3058, CA3059, and CA3079	490
CA3099E	620
CA3086	483
CA3080	475
CD4007A, CD4013A	479
2N5444	456
T2800B (40668)	364
T2300B (40526)	470
T2301B (40691), T2301D (40692)	431
T64170 (40708)	406
S2600D (40655)	496
D1201B (44003)	495
D3202U (45412)	577

Note: Numbers in parenthesis (e.g. 40668) are former RCA type numbers.

* Formerly RCA 40692

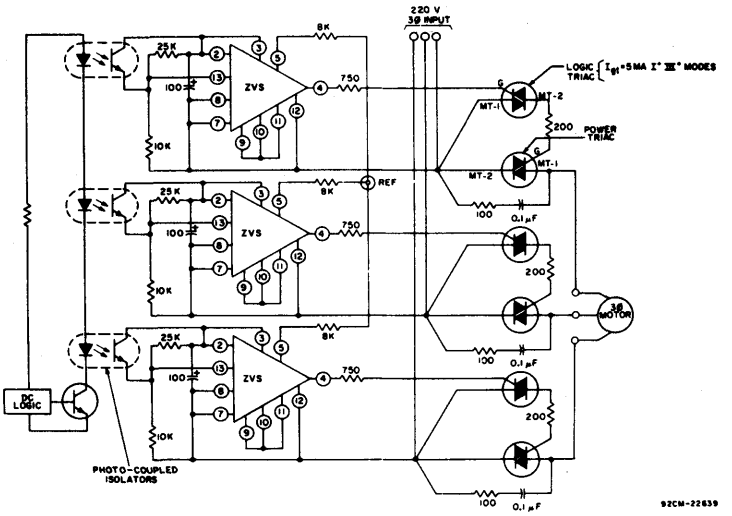


Fig. 62 – Triac three-phase control circuit for an inductive load, i.e., three-phase motor.

ICAN-6222

DESIGNING WITH AN IC TRANSISTOR ARRAY CONTAINING MATCHED SUPER-BETA TRANSISTORS

by T. J. Robe

Many small-signal, low-frequency, and video-frequency applications require an amplifying device with a very high input impedance, low input-bias current, and low-noise characteristics that can maintain low dc offset voltage and drift. Examples of such applications include:

Small-signal instrumentation - chart recorders, meters, etc.
Pre-amps for op-amps.
Magnetic tape-head and phono-cartridge amplifiers.
Opto-electric amplifiers.
Medical electronics.

Devices suitable for such applications are also generally applicable in circuits having long time constants, such as timers, integrators, monostable oscillators, comparators, and low-frequency oscillators. Matched super-beta transistors are well suited for use in these applications.

Super-beta transistors are similar to conventional bipolar transistors except that they have betas in the range of 1000 to 5000; the beta range of a conventional bipolar transistor is from 50 to 400. Since super-beta transistors are commonly used in high-source-impedance applications that require high input impedance and low-noise characteristics, it is equally important that they exhibit super-beta performance at operating currents of a few microamperes. On the other hand, to achieve broadband characteristics in video-amplifier applications, super-beta performance must be maintained with collector currents of 1 milliampere or more. This Note describes the RCA-CA3095 super-beta transistor array and discusses its operation in some typical applications.

A TRANSISTOR-ARRAY IC CONTAINING SUPER-BETA TRANSISTORS

Fig.1 shows the schematic diagram of the RCA-CA3095E, a monolithic IC[1] containing an array of n-p-n transistors of which Q_1 and Q_2 have super-beta characteristics. Q_1 and Q_2 are connected, in conjunction with transistors Q_3 and Q_4, in a differential-cascode-amplifier configuration. Since the super-beta transistors in this IC have a collector-emitter breakdown voltage $V_{(BR)CEO}$ of about 2 volts, it is necessary to limit the collector-emitter voltage accordingly. Limiting is accomplished



Fig. 1 – Schematic diagram of the CA3095E.

by means of the voltage-limiting network composed of D_1, D_2 and Q_5; these components are included on the CA3095E chip. With this arrangement, the voltage applied to the bases of Q_1 and Q_2 can be varied over a wide common-mode voltage range while the collector-emitter voltages are restrained within a maximum range of about 1.5 volts. The collectors of transistors Q_3 and Q_4 have a minimum collector-to-emitter breakdown voltage of 35 volts. The salient characteristics of the differential-cascode amplifier (Q_1 through Q_4) is the "super-high" ratio between the collector currents (at terminals 10 and 6) and the base currents (into terminals 9 and 7, respectively). The beta of the circuit is typically in the range of 1000 to 5000 at collector currents ranging from less than 1 microampere to more than 1 milliampere. As a consequence, amplifier circuits can be designed to operate with extremely small input base-bias currents. Implicit in this performance are high input impedance, low noise, and low dc offset-error effects.

Transistors Q_5, Q_6, and Q_7 are conventional n-p-n types with betas in the range of 150 to 400 at collector currents in the range of 1 microampere to 10 milliamperes. These transistors also have a minimum collector-emitter breakdown voltage $V_{(BR)CEO}$ of 35 volts.

Operating the Super-Beta, Differential-Cascode Amplifier

Application of the differential-cascode amplifier in the CA3095E is similar to that of the classical differential-cascode amplifier; differential input signals are applied at terminals 7 and 9 with balanced collector loads connected from terminals 6 and 10 to the positive supply voltage. The common emitter connection, terminal 8, may be made directly or through a "current source" (e.g., a transistor or resistor) to the negative supply voltage. The circuit in Fig. 2 illustrates the use of "mirrored" transistors (Q_7, Q_8) as a constant-current source to provide high emitter impedance for the differential-cascode amplifier. As an alternative, a resistor may be used as a "current source" (as illustrated by the circuit in Figs. 8, 9, and 11). The IC substrate (terminal 5) is usually connected directly to the negative supply terminal, as shown in Fig. 2, because it must be maintained at the most negative potential of all elements on the CA3095E chip.

The only additional requirement for CA3095E operation is for bias current into terminal 11 to forward-bias the network composed of D_1, D_2 and Q5, and to supply base-bias current for transistors Q_3 and Q_4. This base-bias current can be provided by connecting a dropping resistor between terminal 11 and the positive supply voltage; this arrangement is illustrated by the use of resistor R_{BIAS} in Fig. 2. As an alternative, this current can be supplied from the positive supply to terminal 11 through a p-n-p constant-current-source transistor to maximize common-mode and power-supply rejection characteristics. In most applications, however, such a

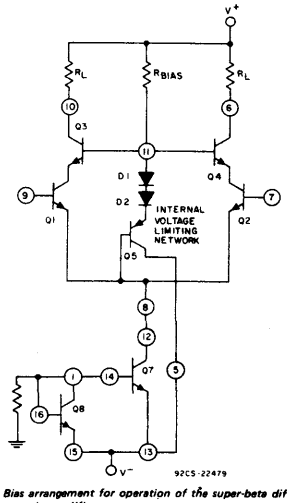


Fig. 2 – Bias arrangement for operation of the super-beta differential-cascode amplifier.

constant-current-source arrangement is not necessary because transistor Q_5 conducts most of the D_1, D_2 signal current to the IC substrate connected to terminal 5. As a general rule, the current supplied to terminal 11 should be approximately 4 to 10 per cent of the current drawn from terminal 8. The input signals to the super-beta transistors (terminals 7 and 9) should not be permitted to swing more than 6 volts below the voltage at terminal 11 to avoid exceeding the V_{CBO} rating of super-beta transistors Q_1 and Q_2. This factor is normally a design consideration only when one or both of the input-stage transistors is to be biased off.

Low-Frequency Operation

When an amplifier is to operate at very low frequencies, or as a dc amplifier, the signal source must be directly coupled to the amplifier input. This coupling requires the use of an amplifying device with a very low input dc offset error and low offset-error drift with temperature variations. A matched differential-cascode amplifier, like the one used in the CA3095E, is particularly well suited to this requirement, not only because of its low input offset voltage (V_{IO} = 1 mV, typical), but also because of its low input offset current (I_O = 4 nA, typical, at I_C = 100 μA). When the input signal is provided from a high-impedance source (R_s), both of these characteristics assume importance because the total effective input-offset-voltage error is the sum of their effects:

$$\text{Total Offset Error} = V_{IO} + I_{IO}R_s$$

The differential input impedance in megohms (Z_{id} in Fig. 3) of an amplifier operating at low frequencies is given by:

$$Z_{id} = \frac{26\text{ mV X the number of p-n junctions in the input stage}}{I_{IB}\text{ (in nA)}}\ \text{M}\Omega$$

where I_{IB} is the input-stage base-bias current. Consequently, the input bias current (I_{IB}) must be quite low if a high input impedance is to be established. The characteristics of the super-beta transistors in the CA3095E are well suited for use

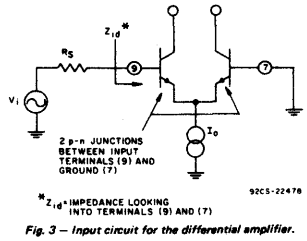


Fig. 3 – Input circuit for the differential amplifier.

in dc amplifiers requiring high input impedance. For example, with the super-beta transistors operating at input-stage emitter currents of 1 microampere and an H_{FE} of 2000, the base-bias current is only 0.5 nanoampere. Under these conditions,

$$Z_{id} = \frac{26\text{ mV X }2}{0.5\text{ nA}} = 104\text{ megohms}$$

Impedance levels of this order can also be realized by using negative feedback in connection with devices having higher input bias currents, as illustrated by the circuit shown in Fig. 4. In this arrangement, the use of the feedback network effectively multiplies the differential input impedance. Unfortunately, this arrangement does not avoid the input-offset-voltage effect resulting from the flow of unequal currents through the signal-source resistance (R_s) and the equivalent resistance of the feedback network; i.e., $R_i//R_f$. Consequently, the advantage to be gained by using super-beta transistors is apparent.

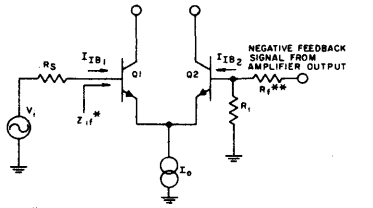


Fig. 4 – Differential input with provisions for feedback.

Considerations in Low-Noise Performance

Fig.5 shows the schematic diagram of a noise model useful in a review of the considerations pertinent to optimizing low-noise performance in amplifier operation.

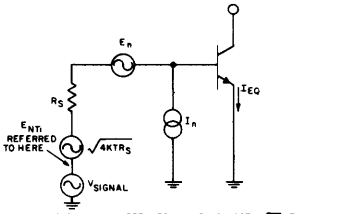


Fig. 5 – Sources of noise in the transistor-amplifier stage.

This model illustrates that consideration must be given to three major sources of noise:

1. Noise contributed by the "thermal-noise" voltage developed across the signal-source resistance, R_s. The magnitude of this voltage in $V/\sqrt{Hz}$ is approximately equal to $\sqrt{4KTR_s}$ for a 1-cycle bandwidth, where k is Boltzmann's constant (1.38×10^{-23} joule/°K), T is the temperature in degrees Kelvin, and R_s is the source resistance in ohms.
2. The noise voltage, E_n, resulting from the combined effects of shot noise due to emitter current flow and thermal noise due to transistor base resistance. These effects add in rms fashion to give a total E_n equal to $(E_{shot}^2 + 4KTrb')^{1/2}$. The shot-noise component, E_{shot}, is inversely proportional to the square root of I_{EQ}, and has a value

$$E_{shot} = \frac{14.2 \times 10^{-12}}{\sqrt{I_{EQ}}} (V/Hz.)$$

   In super-beta transistors, the base resistance component of E_n tends to dominate, particularly at currents greater than 10 microamperes. In addition, this component of E_n has been experimentally found to be inversely related to operating current. Therefore, the total value of E_n is inversely related to operating current I_{EQ}. For example, the CA3095E has a total 1-kHz E_n of approximately 15 nV/$\sqrt{Hz}$ at a collector current of 5 microamperes and approximately 8 nV/$\sqrt{Hz}$ at 50 microamperes.
3. The noise current, I_n, resulting from the combined "shot noise" generated by the flow of base current and the 1/f noise generated in the transistor. The magnitude of I_n is approximately proportional to $\sqrt{I_{IB}}$, where I_{IB} is the base current. The value of I_n is typically 0.12 pA/$\sqrt{Hz}$ at f = 10 Hz when the super-beta differential-cascode amplifier in the CA3095E is operating at I_{EQ} = 5 μA. I_n decreases to approximately 0.03 pA$\sqrt{Hz}$ at f = 1 kHz.

When each input terminal in a differential amplifier is driven from a source resistance (R_s), the total noise voltage (referred to the input, see Fig. 5) per unit bandwidth is given by:

$$E_{nti}\ (\text{in } V/\sqrt{Hz}) = \sqrt{2KTR_s + 2(\overline{I_nR_s})^2 + (E_n)^2}$$

When amplifiers are driven from low source impedances, E_n is the predominant factor in noise contributions, whereas the effect of I_n predominates when input signals are supplied from high source impedances. Consequently, since the CA3095E operates with very high beta at very low operating currents, it has exceptionally low values of I_n, and is an excellent choice to amplify signals from high source resistances when low amplifier noise contribution is desired. Additionally, the incidence of "popcorn" (burst) noise[2] is low in the CA3095E, a characteristic which further enhances its suitability for use in amplifying signals supplied from high-impedance sources. Figs. 6 and 7 show typical data on I_n and E_n characteristics, respectively, as a function of frequency, for the super-beta transistors in the CA3095E.

Because the operating current of the super-beta transistors in the CA3095E is adjustable over a wide range, the circuit designer can optimize the operating current for maximum

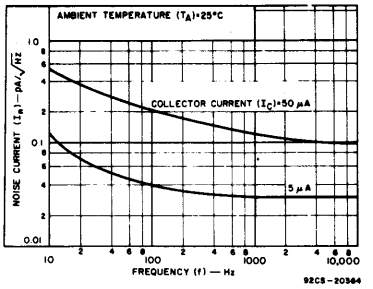


Fig. 6 – Noise current I_n as a function of frequency f for each super-beta cascode-amplifier transistor pair (Q_1 - Q_3 and Q_2 - Q_4).

signal-to-noise ratio at a particular frequency and source resistance. This adjustment is accomplished by selecting an operating point for which E_n is approximately equal to $\sqrt{2}\,I_n\,R_s$. For example, the optimum operating collector currents in the differential-cascode amplifier are about 5 microamperes when the amplifier is to be driven from two 300-kilohm source resistors. For operation from higher source resistances, the currents should be proportionately lower, and vice versa. Operating currents in the range from 0.1 to 1.0 milliampere are recommended when the amplifier is to be operated as a low-noise video amplifier. At these current levels, the gain-bandwidth product (f_T) is increased significantly with respect to low collector current operation.

ILLUSTRATIVE CIRCUIT APPLICATIONS

Like other RCA transistor-array IC's, the CA3095E offers the circuit designer a class of solid-state devices featuring matched electrical and thermal characteristics, compactness, ease of physical handling, economy, and versatility of use. The

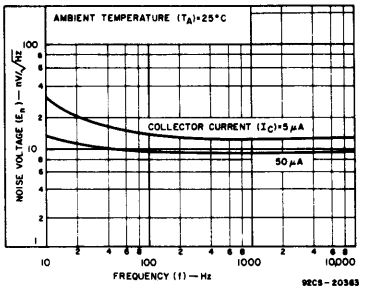


Fig.7 – Noise voltage E_n as a function of frequency f for each super-beta cascode-amplifier transistor pair (Q_1–Q_3 and Q_2–Q_4).

CA3095E is an electronic "building block" which permits the designer to optimize performance of a particular circuit for gain, noise, power consumption, bandwidth, and/or other specific considerations. Some typical circuit applications of the CA3095E are described below.

High-Input-Resistance Low-Noise Amplifier

The CA3095E contains all the transistors necessary for the construction of a low-noise, feedback amplifier having a high input resistance ($R_{IN} \cong 20$ MΩ) and a 3-dB bandwidth of about 50 kHz. In the circuit shown in Fig. 8, voltage gain is provided by a cascade of two stages, the differential-cascode stage (Q_1, Q_3 - Q_2, Q_4) and the differential stage (Q_7, Q_8). Transistor Q_6 is an interstage emitter-follower. The voltage gain of the amplifier (approximately 30 dB with the circuit values shown) is essentially established by the ratio of R_8 to the parallel combination of R_5 and R_6. The R_8, C_2 network couples feedback around the entire amplifier. Capacitor C_4 provides stabilizing compensation. The output-voltage swing (E_O) is typically 3 volts, peak-to-peak. Typical noise-figure data are shown in Fig. 8. Power consumption of the amplifier

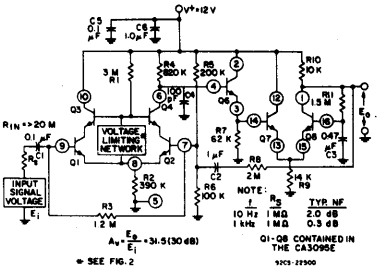


Fig. 8 – High-input-resistance, low-noise amplifier circuit.

is typically about 750 microamperes at a supply voltage of 12 volts, although the current in transistors Q_1 and Q_2 is less than 5 microamperes.

Low-Noise Video Amplifier

The circuit shown in Fig. 9 illustrates the use of super-beta transistors in the input stage of a video amplifier. The circuit is capable of delivering 4 volts, peak-to-peak, of output signal with a typical gain of 33 dB across a bandwidth from dc to 10 MHz (3-dB point). In this application, each super-beta transistor is biased for operation at about 400 microamperes to achieve wideband operation. The super-beta transistor characteristics minimize the contributions to noise generated by noise current (I_n) in the input stage. The equivalent input-noise-voltage-vs-frequency characteristics for the entire amplifier circuit are shown in Fig.10. Transistors Q_1 through Q_4 are connected as an emitter-coupled pair of cascode amplifiers with a single-ended load resistor, R_3, to drive a discrete transistor Q-PNP. This combination provides sufficient current gain to drive Q_6, the voltage-gain-stage transistor, with load resistor R_8. Resistor R_7 provides a path for dc and ac feedback around this stage. Transistor Q_8 is an emitter-follower output stage. The typical current drain of the amplifier is approximately 8 milliamperes at a total supply voltage of 10 volts.

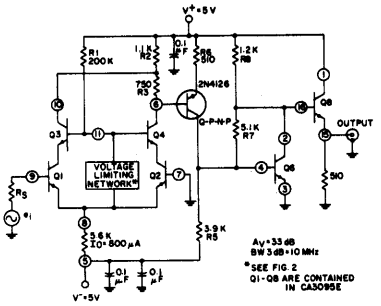


Fig. 9 – Video amplifier.

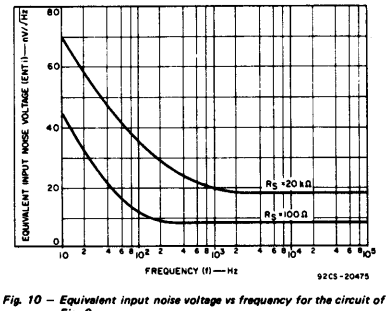


Fig. 10 – Equivalent input noise voltage vs frequency for the circuit of Fig. 9.

Long-Delay Monostable Multivibrator

Super-beta transistors are useful in the design of long-delay, monostable multivibrator circuits, as illustrated in the circuit of Fig. 11. Basically, the circuit is a differential-cascode amplifier biased so that, in the quiescent state, the current path through transistors Q_1 and Q_3 is cut off, and the path through transistors Q_2 and Q_4 is conductive. This arrangement is accomplished by biasing the base of transistor Q_2 so that it

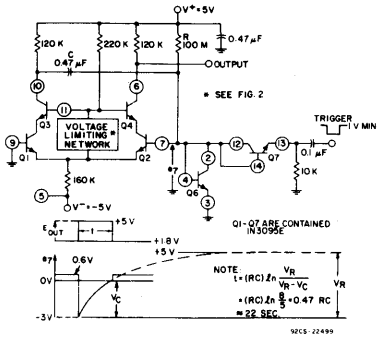


Fig. 11 – Long-delay monostable-multivibrator circuit.

is one step of V_{be} (potential $e_7 \cong 0.6$ V) above the base of transistor Q_1, a condition established by the diode-connected transistor Q_6 in shunt with the base of Q_2. The timing cycle is initiated by the application of a negative-going trigger pulse at terminal 13, which is coupled to the base of Q_2 through diode-connected transistor Q_7. Q_2 is cut off and the output voltage rises to essentially V+. Since Q_1 and Q_2 are emitter-coupled, Q_1 is turned on by the trigger pulse, and the potential at terminal 10 drops rapidly; this drop pulls the left-side electrode of capacitor C toward ground. The right-side electrode of capacitor C is connected to the positive 5-volt supply terminal through resistor R; this connection permits the capacitor to be charged in the exponential characteristic of an RC network and, primarily, determines the delay time. Eventually, voltage e_7 rises sufficiently to again switch Q_2 on (and Q1 off) so that the output voltage drops to its quiescent level (approximately +1.8 V) to signify the end of the timing period. The trigger-voltage pulse should be at least -1 volt in amplitude to assure positive switching.

Low-Input-Bias Current Comparator

The circuit shown in Fig. 12 employs the super-beta and conventional transistors in the CA3095E in a comparator circuit that requires an input signal of only 1.5 nanoamperes at threshold to produce toggling. The output of the circuit can interface directly with COS/MOS logic circuits. Transistor pairs $Q_1 - Q_3$ and $Q_2 - Q_4$ are connected in the differential-cascode arrangement described above. Transistor Q_6 is a programmable constant-current source (i.e., capable of being keyed, gated, clocked, etc.) for the differential-cascode pair. Transistors Q_7 and Q_8 are a differential pair used to provide sufficient gain for the control of the external discrete transistor Q-PNP. The reference voltage for the comparator is applied at terminal 7; voltages in the range from 1.5 to 6.0 volts are suitable for satisfactory circuit operation.

Analog Timer for Long Delays

The very low input-bias-current characteristic of the super-beta transistors in the CA3095E is very desirable in the design of an analog timer for long time delays; the circuit shown in Fig. 13 is illustrative, and functions in a manner quite similar to that of the circuit of Fig. 12. Time delay can be varied in accordance with the expression shown in Fig. 13.

The timing cycle is initiated by momentarily closing the push-button switch to discharge timing capacitor C. At this instant, transistors Q_1 and Q_3 are non-conductive and Q_2 and Q_4 are conductive; this arrangement prevents conduction in transistors Q_8 and Q-PNP. Consequently, the output is essentially zero volts. Timing capacitor C is charged exponentially through resistor R until the voltage across the capacitor is sufficiently large to toggle transistor Q_1 and Q_3 into conduction (and Q_2 and Q_4 into non-conduction). Transistors Q_7 and Q_8 are also switched so as to drive Q-PNP into conduction and produce a "high" output signal (approximately 9.7 volts). With the values shown, the time delay can be varied over the range from about 90 to 725 seconds, depending on the setting of the 100-kilohm potentiometer. The leakage-current loading on timing capacitor C due to transistor Q_1 is quite small, and is only of second-order importance in determining timing accuracy.

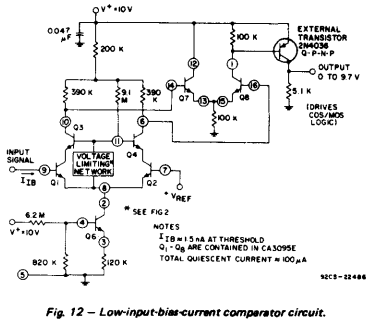


Fig. 12 – Low-input-bias-current comparator circuit.

Super-Beta Transistors in Preamplifier Applications

The wide operating-current range and circuit-connection flexibility available in the super-beta transistors contained in the CA3095E offer numerous advantages for applications in preamplifiers. For example, these transistors can be simply connected as a preamplifier for many of the common, economy-type op-amps (e.g., CA741, CA748)[3] The combination of the CA3095E and one of these op-amps provides an op-amp with the superior input characteristics offered by some of the high-priced op-amps that use super-beta transistors in their input stages. The CA3095E in conjunction with "commodity-class" op-amps can provide an over-all circuit exhibiting orders-of-magnitude improvement over the op-amp itself in terms of input impedance, noise, and the effects of error currents. Several circuit combinations of this type are described below.

Mention has already been made of the low-noise performance which can be achieved with the super-beta transistors in the CA3095E. This attribute is a requisite for preamplifiers operating with low-level signal outputs from sources such as tape heads and magnetic phonograph cartridges. The circuits described below illustrate the simplicity with which super-beta transistor circuits using the CA3095E can be equalized to meet the requirements for NAB playback and RIAA phonograph-record reproduction.

Unity-Gain Preamplifier. The circuit in Fig. 14 illustrates a unity-gain preamplifier using the transistors in the CA3095E to drive a CA741 op-amp. This circuit boosts the input impedance, Z_{id}, of the op-amp to the order of 20 megohms, typical. Transistors Q_1 - Q_3 and Q_2 - Q_4 operate as a differential-cascode amplifier with transistor Q_8 as their constant-current source. Transistors Q_6 and Q_7 are diode-connected to establish dc levels which are appropriate for direct connection to the CA741 input terminals. No additional external compensation is required with this circuit because the unity voltage gain provided by the preamplifier precedes the internally compensated CA741 op-amp. The resultant offset voltage of the combination circuit is the algebraic sum of the offsets due to Q_1, Q_7 vs Q_2, Q_6, and the offset due to the CA741. The resultant offset can be nulled at the normal nulling terminals on the CA741. This circuit is ideal for amplification of signals emanating from sources with very high impedances.

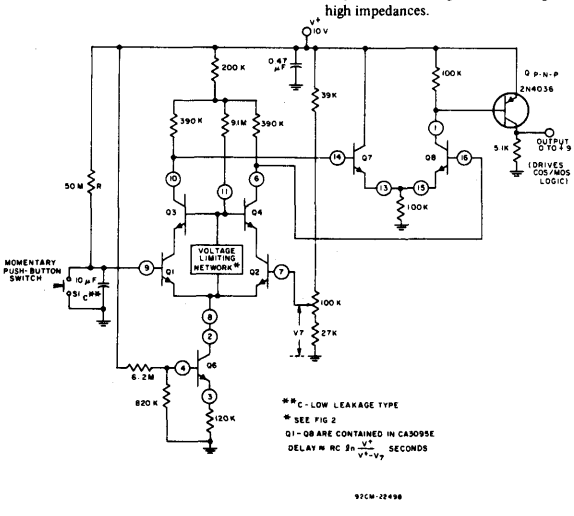


Fig. 13 – Analog timer for long delays.

High-Gain Preamplifier The circuit in Fig. 15 shows a high-gain preamplifier using the transistors in the CA3095E to drive a CA748 op-amp. Transistors Q_1 - Q_3, and Q_2 - Q_4 operate as a differential-cascode amplifier with transistor Q_8 as their constant-current source. Transistor Q_7 is diode-connected to drop the dc common-mode voltage at the input of the CA741 to within its linear operating range.

This circuit boosts the input impedance, Z_{id}, of the op-amp to the order of 20 megohms, typical. It also can capitalize on the low-noise operational capability of the CA3095E, and reduces the noise (and offset-voltage) contributions of the CA748 by an increment equal to the gain of the preamplifier, typically about 28 dB with the circuit constants shown. In this case, external compensation of the CA748 may be necessary when the over-all op-amp is connected with feedback.

The approximate value of the compensation capacitor required can be computed by use of the following relationship:

$$C_C \text{ (in pF)} = \frac{30 \text{ X the voltage gain of the preamp}}{\text{closed-loop voltage gain of the composite op-amp}}$$

The circuits in Fig. 16 illustrate applications utilizing the superior input characteristics of super-beta op-amps. A circuit such as the one shown in Fig. 16(a) is useful in applications where input impedances in the order of 150 megohms are required at frequencies up to 5 kHz. Since input bias currents are required to flow through 10-megohm resistors in this circuit, it is mandatory that the input stage exhibit both low bias current and very low input offset current. (The capacitive

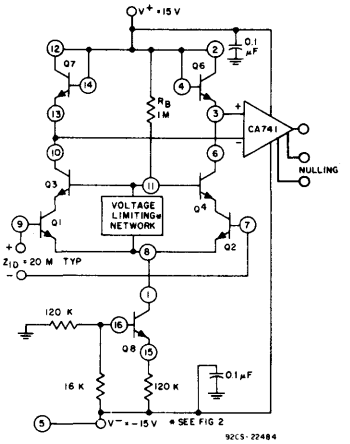


Fig. 14 – Op-amp with unity gain preamplifier.

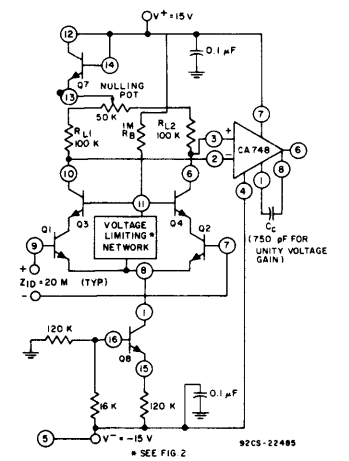


Fig. 15 – Op-amp with high-gain preamplifier.

reactance of the coupling capacitor C must be much lower than 5 megohms to achieve the high-input-impedance characteristic described above.) The low-drift, long-time-constant integrator circuit shown in Fig. 16(b) is another excellent application for the CA3095E super-beta transistor array. Because exceedingly low input bias currents permit the use of a high-value integrating resistor, R, without introducing substantial error, long-time-constant integration can be accomplished. The low input-offset-voltage drift characteristic of the super-beta transistors also contributes to low-error performance. Further reductions in error effects, particularly with temperature variation, can be achieved by using a temperature compensated bias-current source (e.g., R_1–R_3, D_1). In the circuit of Fig.15, such an arrangement could be implemented by using Q_6 in a diode connected fashion to serve as D_1. With such an arrangement, temperature-compensated bias current is applied to the inverting input terminal.

High-Input-Impedance DC-Voltmeter Circuit

The combination of a preamplifier circuit using the CA3095E in conjunction with a CA748 op-amp as described above is adaptable to dc-voltmeter circuits requiring high input impedance, as illustrated by the circuit of Fig. 17. An appropriate resistor-divider network is provided to develop a dc input signal at terminal 9 of the CA3095E with transistors Q_1 · Q_3 and Q_2 · Q_4 connected in the differential-cascode

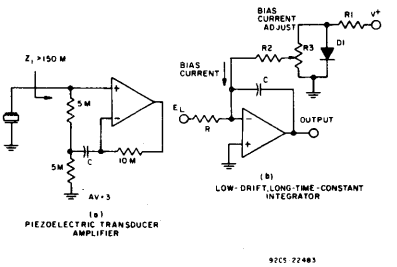


Fig. 16 – Typical super-beta op-amp applications: (a) piezoelectric transducer amplifier (b) low-drift, long-time-constant integrator.

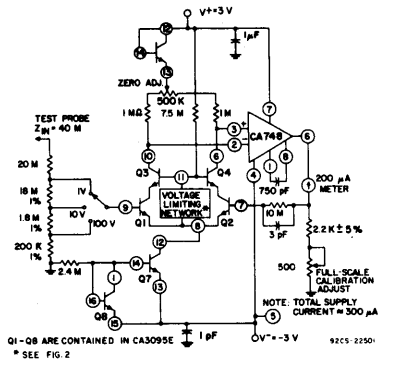


Fig. 17 – Typical high-input-impedance dc-voltmeter circuit.

arrangement. Biasing and dc feedback are applied at terminal 7 of the CA3095E through a 10-megohm resistor. The CA748 op-amp drives a 200-microampere meter calibrated in terms of the voltages to be measured. A full-scale reading occurs when the voltage applied to pin 9 is 500 millivolts dc. The entire circuit is nulled with the 500-kilohm zero-adjustment potentiometer. The total power-supply requirement is only 6 volts with a supply current of only 300 microamperes; this requirement can be met with batteries. The input impedance of this simple circuit is approximately 40 megohms on all scales.

Preamplifier for Tape-Head Signals

The exceptional low-noise characteristics of the CA3095E make it suitable for preamplifier service in professional-grade tape-playback systems. A typical circuit with equalization for NAB standards (7.5 in/s) is shown in Fig. 18. Transistors Q_1 and Q_3 are cascode-connected as the input stage, and transistor Q_6 is connected as a common-emitter post-amplifier. Transistors Q_2 and Q_4 are non-conductive because the emitter-base junction in Q_2 and the base-collector junction in Q_4 are shunted by external wiring. Equalization for the NAB tape-playback, frequency-response characteristics is provided by the R_1, C_1, C_2 network connected in the ac feedback path; DC feedback stabilization is provided by the path through resistor R_2. The amplifier has an over-all gain of about 37 db at 1 kHz, and can deliver output voltages in the order of 25 volts, peak-to-peak. The circuit configuration of Fig.18 is preferred to the differential-amplifier configuration because it limits the input-noise contribution to that of a single transistor (e.g., Q_1).

Preamplifier for Signals from Magnetic-Phonograph Cartridges

The exceptional low-noise characteristics of the CA3095E are also of great use in preamplifier service in equipment used to reproduce signals from magnetic phonograph cartridges. A typical circuit for this application with equalization for RIAA

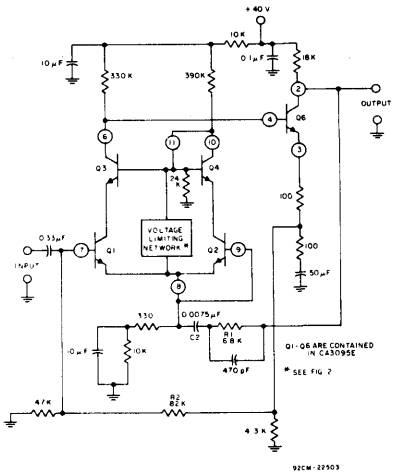


Fig. 18 – Tape play-back preamplifier equalized for NAB standards (7.5 in/s).

playback standards is shown in Fig. 19. Transistors Q_1 and Q_3 are cascode-connected as the input stage, and transistor Q_6 is connected as a common-emitter post-amplifier. Transistor Q_2 and Q_4 are non-conductive because the emitter-base junction in Q_2 and the base-collector junction in Q_4 are shunted by external wiring. Equalization for the RIAA phonograph-frequency-response characteristics is provided by the R_1, C_1 network connected in the ac feedback path. DC feedback stabilization is provided by the path through resistor R_2. The amplifier has an over-all gain of about 40 dB at 1 Hz, and can deliver output voltages in the order of 25 volts, peak-to-peak. The dynamic range of these circuits is typically about 95 dB with the gains indicated.

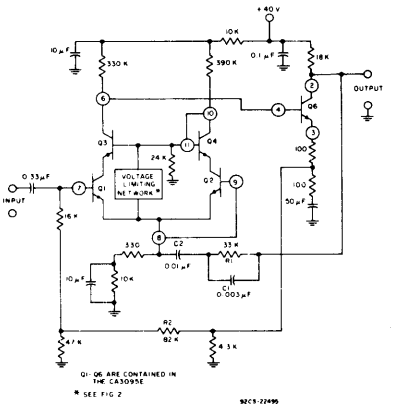


Fig. 19 – Preamplifier equalized for RIAA standards applicable to magnetic phonograph catridges.

ACKNOWLEDGMENT

The circuits of Figs. 18 and 19 were designed by L.A. Kaplan.

REFERENCES

1. "Super-Beta Transistor Array," CA3095E, RCA Data File No. 591.
2. "Measurement of Burst ("Popcorn") Noise in Linear Integrated Circuits," T. J. Robe, RCA Application Note ICAN-6732.
3. "Operational Amplifiers," RCA Data File No. 531.